SHOULDER & ELBOW — REVISION NOTES

– MCQ-focused · Orthobullets + OB/TO question explanations + TO lecture notes · 61 chapters (1 from TO notes) –

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Shoulder & Elbow

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Every chapter, same order: 1 Basics → 2 Mechanism → 3 Presentation → 4 Diagnosis & Classification → 5 Management → 6 Complications → X vs Y → How it's asked

Always included where sources have them: criteria (diagnostic, operative) · genes & lab markers · key radiology findings

Key: TO = Target Ortho fact (must-know) · ↑ increased · ↓ decreased · → leads to / then · ✗ Wrong options = MCQ distractors · ✓ correct answer · "TO — verify" = TO figure that differs from standard teaching

– – Active space – –

AC ARTHRITIS

– Shoulder & Elbow · 1 / 60 –

1 Basics

Epidemiology

Frequency	**most common cause of AC joint pain**
Demographics	↑ with age; can start by **2nd decade**
Classic patient	**weight-lifters** + overhead throwing athletes (constant heavy overhead activity)
Risk factors	trauma · post-traumatic (**clavicle fracture, AC instability**) · **distal clavicle osteolysis** · inflammatory (RA) · post-infectious (septic arthritis)

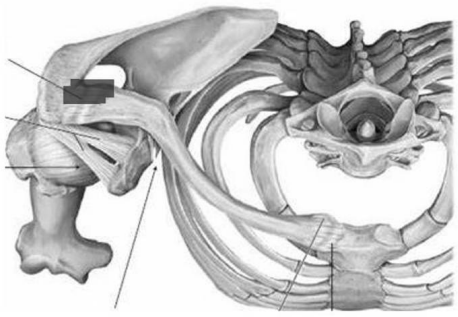
Pathophysiology

Mechanism	**large axial loads through a small contact area → repetitive microtrauma** → OA (**same mechanism as distal clavicle osteolysis**)
Aggravators	articular disc degeneration + oblique joint surface orientation → ↑ contact forces

Anatomy – AC joint & ligaments

AC joint **diarthrodial**; scapula ↔ clavicle; **fibrocartilaginous intra-articular disc** (analogous to knee meniscus); small surface, high axial + rotational loads

Ligament	Stability provided	Injury →
AC ligaments	**Anterior–posterior (horizontal)**	AP instability
Posterior + superior AC ligaments	**Most important** AC stabilisers	**Posterior instability** of distal clavicle
Anterior + superior AC ligaments	Anterior	**Anterior instability** (one OB explanation) · **no instability** (another OB explanation) – read the options
CC ligaments (conoid + trapezoid)	**Superior–inferior (vertical)**	**Superior instability**



Axial view – AC joint ligaments; posterior + superior AC ligaments highlighted

Not AC stabilisers	**coracohumeral ligament** (rotator interval) · **coracoacromial ligament** (anterior acromion; released in subacromial decompression) · deltotrapezial fascia (only elevated in open DCE)
Rare instabilities	**inferior** (CC ligaments resist superior displacement) · **rotational** (rotation is normal baseline load)

3 Presentation

Symptoms	activity-related **superior shoulder pain**; overhead activity, **pressing (bench press, push-up)**, lying on affected side
Palpation	**direct AC joint tenderness**; prominent distal clavicle (osteophytes)
Provocative	**cross-body adduction test** (most classic) · **O'Brien active compression** (90° forward flexion) → AC pain



Cross-body adduction test

4 Diagnosis & Classification

Key radiology findings

Modality	Finding
X-ray – Zanca view (15° cephalic tilt; 10–15° in TO/OB)	**Best view**; osteophytes, **joint space narrowing**, **distal clavicle osteolysis**
X-ray caveat	Changes often present **without symptoms** (poor clinical correlation)
MRI	↑ signal + **oedema** in AC joint; shows associated cuff / LHB pathology

– – Active space – –



Zanca view – AC osteophytes and joint space narrowing

Diagnostic injection

AC joint injection (corticosteroid / local anaesthetic, e.g. bupivacaine)	**diagnostic + therapeutic**; relief confirms AC source
Accuracy	joint hard to access, **often missed → ultrasound ↑ accuracy**; most get **no long-term relief**

5 Management

- **Symptomatic AC arthritis**
  - First line → **activity modification (avoid pressing) + PT (shoulder girdle) + NSAIDs**
  - Persistent → **corticosteroid injection (US-guided)**
  - Failed nonop, severe symptoms → **distal clavicle excision (Mumford)**
    - Arthroscopic → **resect 0.5–1 cm, evaluate + treat GH pathology**
    - Open → **meticulous deltotrapezial fascia repair**

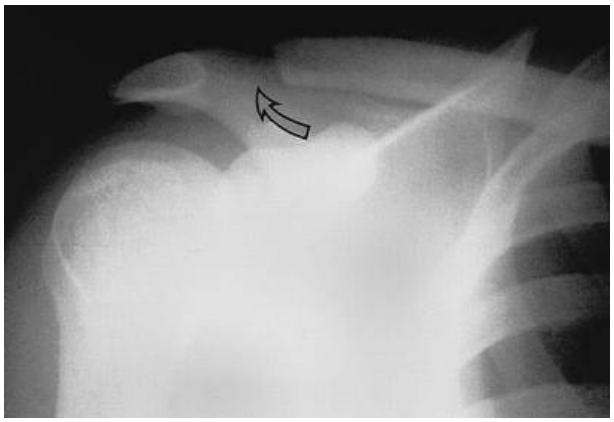
Distal clavicle excision – preserving posterosuperior capsule

Mumford procedure distal clavicle resection, arthroscopic (burr) or open; indication = **severe symptoms, failed nonop**

Resection amount	Source
0.5–1.0 cm (lateral clavicle)	Orthobullets, OB
4–8 mm may suffice arthroscopically (preserves superior AC ligaments → less bone than open)	OB (Nuber)
>1–1.5 cm = aggressive → **instability**	Orthobullets, OB

Structure to preserve	**posterosuperior joint capsule / posterior + superior AC ligaments** → prevents **AP (posterior) instability**
Over-resection	AC or CC ligament damage → instability; even correct length + posterosuperior damage → AP instability

Under-resection persistent pain



Arthroscopic distal clavicle excision

Arthroscopic vs open distal clavicle excision

Outcomes	similar; choice = surgeon preference
Only proven arthroscopic advantage	**evaluation of the glenohumeral joint** + treat associated pathology (**rotator cuff, LHB, labrum / SLAP**)
Missed pathology with open	**15 missed SLAP lesions** in failed open DCE (Berg) → do **diagnostic arthroscopy even with open DCE**
Not proven arthroscopic advantages	lower complication rate · lower infection rate · ↓ surgical time · preservation of CC or inferior AC ligaments
Other arthroscopic pluses (OB)	direct view of clavicle end, less soft-tissue violation, shorter rehab, quicker return
Open DCE	requires **meticulous deltotrapezial fascia repair**

6 Complications

Complication	Cause
AP instability	**Over-resection >1–1.5 cm** · sacrificing **posterior + superior AC ligaments**
Superior–inferior instability	Iatrogenic; resection compromising **CC ligaments**
Persistent pain	**Incomplete resection** – most commonly **posterior-superior distal clavicle**
Heterotopic ossification	
Stiffness, infection	Causes of failed DCE (OB)
Deltoid dehiscence	Inadequate **deltotrapezial fascia repair** after **open** DCE

Failed DCE management conservative, revision, or **ligament reconstruction**

– – Active space – –

X vs Y

	AC ligaments	CC ligaments
Components	Superior, posterior, anterior, inferior	**Conoid + trapezoid**
Plane	**Anterior–posterior**	**Superior–inferior**
Injured by	Aggressive DCE / debridement	Over-aggressive resection (iatrogenic)
Instability	**Posterior** (post + sup injured)	**Superior**

	Arthroscopic DCE	Open DCE
GH evaluation	**Yes**	No (add scope)
Key closure step	–	**Deltotrapezial fascia repair**
Specific risk	Under-resection (visualisation)	**Deltoid dehiscence**, missed SLAP

How it's asked

"Weightlifter, superior shoulder pain with bench press — best X-ray view?"	→ **Zanca (15° cephalic tilt)**
"Weightlifter, persistent popping after arthroscopic DCE, posterosuperior ligaments damaged — instability direction?"	→ **posterior** ✗ Wrong options: anterior · inferior · rotational · superior = CC ligaments
"Structure most important to preserve during arthroscopic DCE?"	→ **posterosuperior joint capsule** ✗ Wrong options: coracohumeral · coracoacromial · anterosuperior capsule · deltotrapezial insertion
"Only documented benefit of arthroscopic over open DCE?"	→ **ability to evaluate the glenohumeral joint** ✗ Wrong options: lower complications · ↓ surgical time · CC / inferior AC ligament preservation
"Persistent pain after DCE — commonest cause?"	→ **incomplete resection of posterior-superior distal clavicle**
"Deltoid weakness after open DCE?"	→ **inadequate deltotrapezial fascia repair**
"How much clavicle to resect arthroscopically?"	→ **0.5–1 cm** (4–8 mm may suffice; >1–1.5 cm → instability)

ACROMIOCLAVICULAR JOINT

– Shoulder & Elbow · 2 / 60 –

1 Basics

Osteology & articular disc

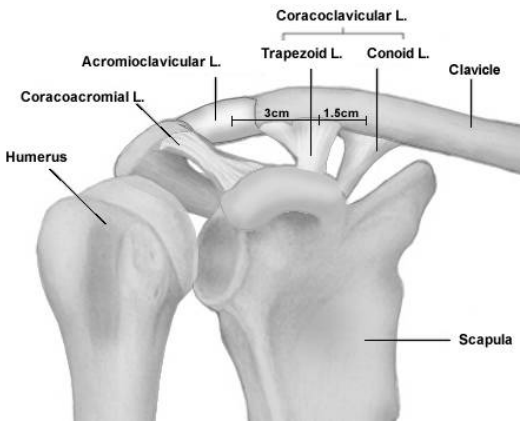
Joint type	**diarthrodial**; capsule + intra-articular synovium + articular cartilage interface
Intra-articular disc	**fibrocartilaginous**; usually **meniscoid-like** · complete disc **very rare** (TO); variable size/shape
Cartilage (TO)	hyaline → **fibrocartilage**: **acromial side by age 17** · **clavicular side by age 24**

Motion

Clavicle	rotates **40–50° posteriorly** with shoulder elevation
Through AC joint	only **8°** of rotation; remainder = **scapular rotation + SC joint motion**
Joint itself	**gliding motions only**; most motion from the bones, not through the joint

Stability – AC vs CC ligaments

Restraint	Direction	Key facts
AC ligaments / capsule (superior, inferior, anterior, posterior)	**Horizontal (AP)**	**Superior strongest**, then **posterior**; constrain posterior clavicular rotation; small-displacement vertical stability (TO)
Trapezoid (CC; **anterolateral**)	Horizontal + vertical loads	Inserts **3 cm** from clavicle end; **>3× thicker at clavicular end** (TO); origin **posterior half of coracoid dorsum** (TO)
Conoid (CC; **posteromedial**)	**Vertical**	Inserts **4.5 cm** from clavicle end, **posterior border**; **more important** CC ligament; clavicular insertion **~2× wider/ thicker** than coracoid → **inverted cone** (TO); origin more posterior on **coracoid base** (TO)
Deltotrapezial fascia, capsule, deltoid, trapezius	Additional	Secondary stabilisers



Coracoclavicular ligaments – trapezoid and conoid

– – Active space – –

Normal CC distance	**11–13 mm** (superior coracoid → inferior clavicle)

Biomechanics of the AC ligaments (TO)

AP vs vertical (TO)	AC capsule + ligaments give **3× more control in AP** than vertical direction
Thickness (TO)	**1–5 mm**; inner osseous insertion **3–4 mm from joint line**, peripheral fibres farther
Superior ligament	**most robust**; principal horizontal stabiliser; **~50%** of AP strength (OB); **thickest posteriorly**
Posterior ligament	adds **~25%** of AP strength (OB)
Two-bundle model (TO)	**superoposterior bundle** resists **posterior** clavicular translation · thin **anteroinferior bundle** limits **anterior** displacement

5 Management

Distal clavicle excision – posterosuperior AC ligaments

Mumford procedure	(arthroscopic) distal clavicle resection for **AC arthrosis**; resection length **debated**
Must preserve	**superior + posterior AC ligaments / capsule** → prevents post-op **AP (posterior) instability**
Over-resection sign	popping + pain + **mild posterior instability**, no deformity on X-ray (e.g. 8 mm excised in weightlifter) → **posterior + superior AC ligaments compromised**
✗ Wrong option: anterior + superior AC	opposite answer; these resist **anterior** translation
✗ Conoid / trapezoid	**vertical** restraints; injured in **grade III or higher** separations → superior clavicular migration
✗ Wrong option: coracohumeral ligament	not an AC stabiliser

AC separation – reduction & stabilisation

AC separation	Feature	Treatment
Type III	–	**Controversial**
Type V	Clavicle elevated **100–300%** + **extensive soft-tissue stripping**	**Surgical AC reduction + stabilisation**

– – Active space – –



Type V AC separation – marked clavicular prominence

✗ Type V distractors sling · NSAIDs + rapid return · arthroscopic partial claviculectomy · acromionectomy

X vs Y

	AC ligaments	CC ligaments (conoid + trapezoid)
Plane	**Horizontal / AP**	**Vertical**
Strongest component	**Superior** (then posterior)	**Conoid** (more important)
Threatened by	**Distal clavicle excision** (too much)	**Grade ≥III** AC separation
Failure →	**Posterior instability**, popping	**Superior clavicle migration**

	Trapezoid	Conoid
Position	**Anterolateral**	**Posteromedial**
Distance from clavicle end	**3 cm**	**4.5 cm**
Shape (TO)	>3× thicker at clavicular end	**Inverted cone**
Role	Horizontal + vertical	**Vertical** (primary)

How it's asked

"Structure responsible for AP stability of AC joint?"	→ **posterosuperior part of capsule** (TO) ✗ Wrong options: anteroinferior capsule · trapezoid · conoid
"Arthroscopic distal clavicle excision – preserve what to prevent AP instability?"	→ **superior + posterior AC capsule** ✗ Wrong options: trapezoid · conoid · coracohumeral
"Popping + posterior clavicle instability after Mumford – what was compromised?"	→ **posterior + superior AC ligaments** ✗ Wrong options: anterior + superior
"Quarterback, clavicle tenting under skin (type V) – management?"	→ **AC reduction + stabilisation** ✗ Wrong options: sling · NSAIDs · claviculectomy · acromionectomy
"Primary vertical stabiliser of AC joint?"	→ **conoid ligament**

– – Active space – –

"Strongest AC ligament?"	→ **superior** (~50% AP strength), then posterior (~25%)

– – Active space – –

ACROMIOCLAVICULAR JOINT INJURY

– Shoulder & Elbow · 3 / 60 –

1 Basics

Epidemiology

Frequency	**9% of shoulder girdle injuries · 40–50% of shoulder injuries in sports** (OB)
Demographics	♂ **>** ♀, **athletes**, young
AKA	"shoulder separation" = disruption of AC ± CC ligaments

Anatomy

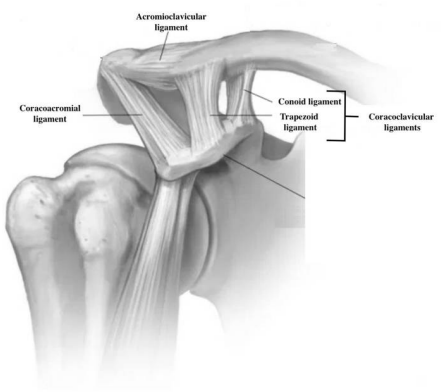
AC joint	**diarthrodial**; medial acromion + lateral clavicle; oblique surface
Intra-articular disc	**fibrocartilaginous** (≈ knee meniscus); involutes, **gone by age 40**
Motion	mainly **gliding**; clavicle rotates **40–50° posteriorly** with elevation but only **~8° through AC joint** (synchronous scapuloclavicular motion)
Dynamic stabilisers	**anterior deltoid + trapezius** (deltotrapezial fascia)

AC vs CC ligament function (horizontal vs vertical stability)

Ligament	Controls	Key fact
AC ligaments (sup, inf, ant, post)	**Horizontal = anterior–posterior** stability	**Posterior + superior (posterosuperior) most important** → preserve in distal clavicle resection
Conoid (CC)	**Vertical = superior–inferior**	**Most important for vertical stability**; **conoid tubercle, posteromedial**; inserts **4.5 cm** medial to lateral clavicle edge
Trapezoid (CC)	Vertical	**Trapezoid tubercle, anterolateral**; inserts **3 cm** medial to lateral edge
Coracoacromial (CA)	–	**Spared** in AC injuries

✗ Ligament traps ✗ Wrong options: anterosuperior AC = vertical · conoid = horizontal · **inferior AC** = no significant horizontal role · trapezoid "most posteromedial" (that is conoid) · **anterior + inferior AC** most important · trapezoid at 4.5 cm

– – Active space – –



AC, CA and CC (conoid, trapezoid) ligaments

Normal CC distance	**11–13 mm** (top of coracoid → undersurface of clavicle)
AC joint type (TO)	**atypical synovial** with fibrocartilaginous disc · **AC ligament = horizontal**, **CC = vertical** stability · conoid = medial/central, trapezoid = lateral · conoid–trapezoid gap **~1.5 cm** · CC distance **>25% increase = disruption** (TO)

2 Mechanism

Mechanism	**direct blow to the shoulder** (fall onto **adducted** shoulder); bike falls, contact sport

3 Presentation

Symptoms	pain over AC joint ± referred to **trapezius**; visible step-off
Exam	AC/lateral clavicle tenderness; abnormal contour vs other side
Stability testing	**horizontal (AP) = AC ligaments** → cross-body adduction; **vertical (sup–inf) = CC ligaments**
Horizontal instability (ISAKOS IIIB)	may need more aggressive treatment
Type IV clue	**skin tenting + posterior fullness**
Type V clue	**severe shoulder droop, not corrected by shrug**
Type VI clue	rare, associated injuries, **paraesthesias**

Provocative tests

Test	Technique	Positive =
O'Brien (active compression)	**Resisted flexion at 90° flexion, 10° adduction, full IR**	**Superficial pain → AC joint · deep pain → SLAP**
Cross-body adduction	Arm adducted across chest	AC joint pain
✗ Wrong option: Jobe / empty can	Resisted abduction 90° scapular plane, pronated	Supraspinatus tear
✗ Wrong option: apprehension	Supine abduction + ER	Anterior instability
✗ Wrong option: belly press	Palm on abdomen, resisted IR	Subscapularis tear
✗ Wrong option: Neer	Passive forward elevation in IR	Subacromial impingement

Persistent AC pain after nonop type III (6 wk) **O'Brien / cross-body adduction** → if pain + instability → **AC + CC reconstruction**

4 Diagnosis & Classification

Key radiology findings

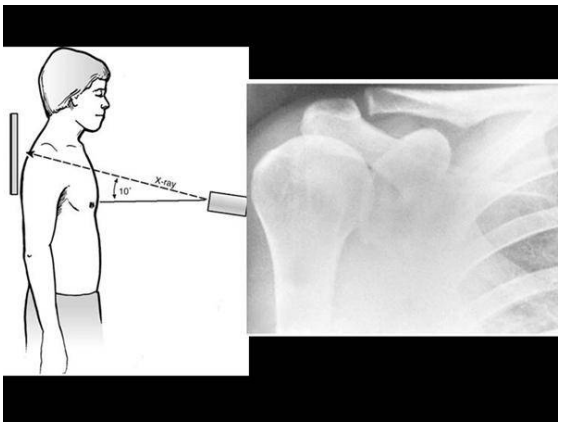
View	Technique	Use
Bilateral AP AC joints	**1/3 penetration** of standard shoulder AP	Compare **CC distance** with contralateral (III vs V)
Axillary lateral	–	**Required to diagnose type IV (posterior)**
Zanca	**10–15° cephalad tilt, 50% penetrance**	**Most accurate view of AC joint**
Cross-body adduction (Basmania)	Scapular Y with cross-body adduction stress	Horizontal stress view
Alexander	Scapular Y centred on AC joint, **15° cranial tilt**	**Posterior displacement** (also superior)
Weighted stress views	–	Rarely used; II vs III
CT	–	Degree/direction of displacement; often unnecessary
✗ Wrong option: **Serendipity**	**40° cephalic tilt**	**SC joint**
✗ Wrong option: Stryker notch · West Point · supraspinatus outlet · Velpeau	–	Not for AC joint

Penetration AC joint needs **1/3–1/2** of glenohumeral exposure → standard AP **over-penetrates (dark)** AC joint

Zanca tilt **10–15°** (Orthobullets, OB) · **15°** (one OB explanation)

TO views AP routine · **15° cephalic tilt (Zanca)** clears thoracic shadow · **stress view with 5 kg hanging** → subtle AC separation · axillary lateral → anterior/posterior displacement

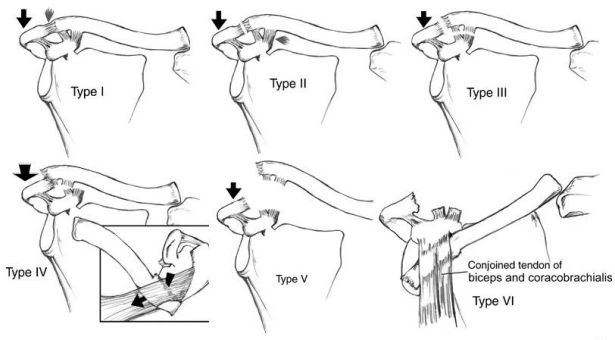
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Zanca view technique

Rockwood classification & ligament injury pattern

Type	AC lig	CC lig	Exam	Radiograph	Reducible	Treatment
I	Sprain	Normal	Tender, **no instability**	Normal	Yes	Sling
II	**Torn**	**Sprain**	**Horizontal instability**	AC disrupted; **CC ↑ < 25%**	Yes	Sling
III	Torn	**Torn**	–	**CC ↑ 25–100%**	Yes	**Controversial**
IIIA (ISAKOS)	Torn	Torn	**Vertical instability, no horizontal**	CC ↑ 25–100%	Yes	–
IIIB (ISAKOS)	Torn	Torn	**Vertical + horizontal instability**	CC ↑ 25–100%	Yes	**May need more aggressive**
IV	Torn	Torn	**Skin tenting, posterior fullness**	Clavicle **posterior through trapezius** on **axillary**	**No**	Surgery
V	Torn	Torn	**Severe droop**, not improved by shrug	**CC ↑ > 100%**	**No**	Surgery
VI	Torn	Torn	Rare; associated injuries; paraesthesias	**Inferior** – **subacromial or subcoracoid**, tethered by **conjoint tendon**	**No**	Surgery



Rockwood classification types I–VI

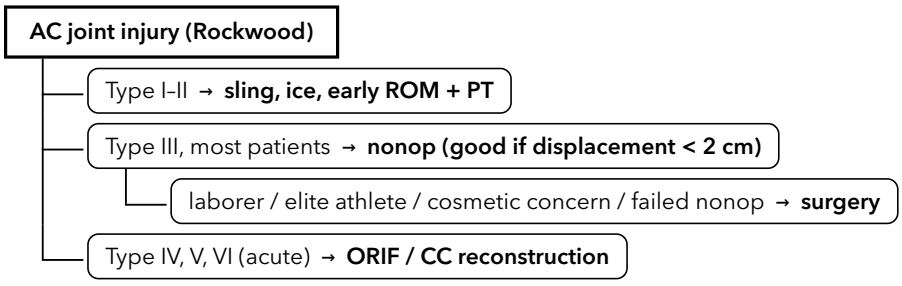
Type III CC threshold	**25–100%** (Orthobullets, OB) · **20–100%** (one OB explanation) — read the options
Type I trap	written in one OB explanation as "AC sprain + CC sprain, no instability" (Orthobullets: CC normal)
Types III–VI	**both AC + CC torn**; IV–VI = **irreducible**
Type V irreducible	clavicle **buttonholed through deltotrapezial fascia** (most common culprit)
✗ Interposed-structure traps	conjoint tendon (coracoid tip) · CC ligaments (ruptured, not incarcerated) · CA ligament
Type IV and trapezius	clavicle **pierces trapezial fascia** → muscle innervated by **CN XI (spinal accessory)**
✗ Nerve traps	axillary (deltoid, teres minor) · CN VII (face, platysma) · pectoral nerves · suprascapular (SS, IS)
ISAKOS consensus	Rockwood misses nuances → added **biomechanical stability (IIIA vs IIIB)**

Differential (mimics)

Mimic	Distinguishing feature
Base of coracoid fracture	Superiorly displaced clavicle but **normal CC distance** (CC ligs intact)
Distal clavicle fracture Neer 2A	Ligaments stay on distal fragment; medial fragment displaces
Distal clavicle fracture IIB (OB)	Fracture **between conoid + trapezoid** → **only conoid** disrupted; distal fragment CC distance intact
Paediatric distal / medial clavicle physeal injury	Physis

AC separation vs IIB clavicle fracture	AC separation = **conoid + trapezoid** both injured (CC ↑ 25–100%)

5 Management



Low-grade injuries – sling & early motion

TO nonop	sling **3–4 wk** → early motion · **persistent pain at 2–3 months → distal clavicle excision (1–2.5 cm)** · type III **controversial**: nonop generally, **operative for young / athletes / manual workers** · IV–VI operative
Nonop regimen	brief sling (**2–4 wk** for I–III, OB), ice, activity modification, **early ROM**, PT; ± steroid injection
Timeline (nonop)	functional motion by **6 wk** · normal activity at **12 wk**
20% ↑ CC distance	= **type II → sling + early PT** ✗ Wrong options: CC reconstruction · AC capsule repair · retrograde pinning · distal clavicle excision

– – Active space – –

58-yo accountant, non-dominant, II vs III, no AP translation	**sling + early ROM** ✗ Wrong options: Weaver-Dunn · CC recon
Nonop complications	AC arthritis, chronic subluxation/instability

Operative criteria

Indication	Detail
Acute type IV, V, VI	Recent studies: no functional difference op vs nonop even for high grade (IV–V may be treated nonop, OB)
Acute type III	**Laborers, elite athletes, cosmetic concerns**
Chronic type III	**Failed nonop**
Contraindications	Poor rehab compliance; skin problems over approach

Timing	acute stabilisation usually **within first 2 wk** (OB) · surgery **after 6 wk failed nonop = same outcome as immediate** (Orthobullets)
Old teaching	acute → ORIF, chronic → CC reconstruction (no longer supported)

Type III injuries – operative vs nonoperative

Outcome	Operative	Nonoperative
Complications	**Higher** (hardware commonest)	Lower
Return to work/sport	**Slower**	**Faster**
Early function (6 wk, 3 mo)	Worse	**Better**
Function / PROs / QoL at 1 yr	**Equal**	**Equal**
Return to athletics at 1 yr	Equal	Equal
Cosmesis	**Better** (only consistent benefit)	Poorer
Radiographic alignment	**Better** (hook plate, COTS RCT)	Worse
Disability at 2 yr	**Both improve significantly**	**Both improve significantly**
ROM / strength / endurance	No difference	No difference
Secondary surgery	Higher (hardware removal)	Lower
Cost	Higher	Lower

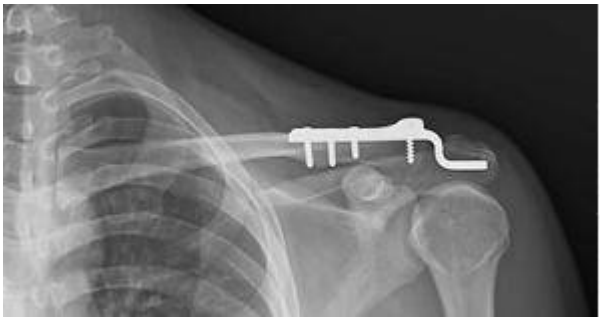
Early-score wording	nonop "higher DASH at 6 wk & 3 mo" (Orthobullets; ↑ DASH = worse) · nonop **better** early scores/function at 6 wk (OB) – read the options
Spencer systematic review	surgery **not better**, **higher complication rate**, **longer recovery**
Canadian (COTS) RCT	hook plate: **better radiographic alignment**, **not clinically superior**; nonop better early scores

Surgical techniques

– – Active space – –

Technique	Pros	Cons / complications
CC screw (Bosworth) – clavicle → coracoid, sup → inf	Rigid	Out of favour; screw too long → critical structures; **routine removal 8–12 wk** (breakage from clavicle–scapula motion); **failure at coracoid thread purchase**
CC suture fixation – around/through clavicle + around coracoid base (or anchors)	**No hardware failure/migration**	Weaker than screw; **careful passage under coracoid** (NV structures); **suture erosion → distal clavicle fracture**
AC pin (Phemister) – wire across AC joint	Percutaneous	**High pin migration** (catastrophic); rarely done
Hook plate	Rigid	**2nd surgery** for removal; **acromial erosion**, hook pullout
Modified Weaver-Dunn – distal clavicle excision + CA ligament transfer to clavicle	–	**CA ligament only 20% as strong** as CC; failure without internal fixation
Free tendon graft CC reconstruction – figure-of-8 around coracoid, clavicle tunnels	**Closest to native CC strength**	Graft morbidity; needs internal fixation backup
Cortical flip button (Dog Bone) ± arthroscopy	–	–

Grafts autograft **palmaris longus, semitendinosus** · allograft **tibialis anterior**



Hook plate fixation

Post-op rehab sling **6 wk, no ROM** · full activity at **6 months**

Technique (TO)	Concept
Phemister K-wire	**Obsolete** (migration)
Hook plate	Hook **under posterior acromion** (anterior placement → impingement) · **remove 8–10 wk**
Bosworth screw	CC screw · **remove 8–10 wk** (vs 8–12 wk above)
TightRope / Endobutton	FiberWire suspensory CC fixation keeping **~1.2 cm** CC distance
Suture anchor	Coracoid + clavicle tunnels at **3 cm (trapezoid) + 4.5 cm (conoid)**
Weaver-Dunn	CA ligament transfer + distal clavicle excision – chronic
Hamstring autograft / Mazzocca	**Anatomic conoid + trapezoid reconstruction → gold standard for chronic** (TO)

– – Active space – –

AC joint cyst (Geyser sign) (TO)

Entity (TO) **synovial cyst of AC joint**; uncommon; elderly; chronic shoulder pain, ↓ ROM, large lump, cosmetic concern

Type (TO)	Cause	Treatment (TO)
Type 1	**AC joint chronic arthropathy**	**Excision + lateral clavicle resection**
Type 2 (**most common**)	**Chronic rotator cuff tear**	Mild/moderate pain + reparable cuff → **distal clavicle resection + cuff repair** · severe pain + irreparable cuff (elderly) → **shoulder arthroplasty (RSA > hemi)**

Mechanism (TO) chronic cuff tear → **superior humeral head migration** → erodes **inferior AC capsule** → fluid escapes into AC joint + subcutaneous tissue

Geyser sign (TO) fluid jet through AC joint on **MRI / arthrogram / USG**; MRI reliably shows cyst type

(✗ Wrong options: **Distractors (TO)** – AC exostosis) · SLAP type 4 · biceps sheath GCT



Geyser sign on MRI

6 Complications

Complication	Fact
Residual AC pain	**30–50%**
AC arthritis	**More common after surgery** than nonop (nonop: no articular contact → no arthritic symptoms; surgery: cartilage injury + incongruity)
Hardware failure	CC screw breakage/pullout **at coracoid**
Pin migration	AC pinning (Phemister)
Coracoid fracture	**Coracoid tunnel drilling**
Distal clavicle fracture	Suture erosion
Acromial erosion	Hook plate
Neurological injury	**Musculocutaneous nerve** closest to coracoid (✗ Wrong options: axillary (runs at inferior subscapularis, distant))

Surgical complications (Weaver-Dunn AP translation, AC arthritis)

Failed modified Weaver-Dunn	persistent symptoms from **anterior–posterior (horizontal) translation** ✗ Wrong options: sup–inf translation · inadequate resection · over-reduction · SC instability
Prevention	**anatomic** repair/reconstruction **preserving distal clavicle** → less horizontal instability
CC screw statement traps	✗ Wrong options: "more migration than AC pinning" · "fails at clavicle" (coracoid) · "avoid removal" (remove at 8–12 wk) · "axillary nerve"

X vs Y

	Type III	Type V
CC distance ↑	**25–100%**	**> 100%**
Reducible	Yes	**No** (deltotrapezial fascia)
Shrug	Corrects	**Does not correct droop**
Treatment	Controversial / mostly nonop	Surgery

	Type IV	Type VI
Direction	**Posterior** (through trapezius)	**Inferior** (subacromial/subcoracoid)
View	**Axillary lateral** (Alexander)	AP
Clue	Posterior fullness, skin tenting	Rare, paraesthesias, associated injuries

	Zanca	Serendipity
Joint	**AC**	**SC**
Tilt	**10–15° cephalad**	**40° cephalic**

How it's asked

"Type V on X-ray – ligaments torn?"	→ **AC + CC** ✗ Wrong options: SC · CA
"CC distance 13 mm contralateral, type III – which labelled ligaments torn?"	→ **AC + conoid + trapezoid** ✗ Wrong options: CA ligament (spared)
"Type II AC injury?"	→ **AC torn, CC sprain, horizontal instability** ✗ Wrong options: no instability = type I · CC torn = III
"IIB distal clavicle fracture vs AC separation?"	→ **fracture = conoid only; AC separation = conoid + trapezoid**
"Posterior clavicle displacement after fall over handlebars?"	→ **type IV**
"Type V, irreducible, skin tenting – interposed structure?"	→ **deltotrapezial fascia** ✗ Wrong options: conjoint · CC · CA

– – Active space – –

"Type IV, posterior fullness — nerve to muscle involved?"	→ **CN XI** (trapezius)
"Most important for horizontal stability?"	→ **posterosuperior AC ligament**
"Most important for vertical stability?"	→ **conoid**
"Trapezoid inserts how far from lateral clavicle?"	→ **3 cm** (conoid 4.5 cm)
"Best extra view for AC joint?"	→ **Zanca** ✗ Wrong options: Stryker · West Point · outlet · Velpeau · Serendipity = SC
"Type III — surgery vs nonop?"	→ **higher complication rate** with surgery
"Type III non-athlete — only benefit of fixation?"	→ **improved cosmesis**
"Type III, 19-yo wants surgery — true?"	→ **surgery ↑ complications + nonop faster return** ✗ Wrong options: better function
"Type III nonop vs op?"	→ **nonop better function at 6 wk**; equal at 1 yr
"Hook plate RCT — true?"	→ **both groups improve significantly by 2 yr** ✗ Wrong options: op faster return · op worse cosmesis · equal alignment
"Type III nonop, still painful at 6 wk — exam?"	→ **O'Brien: resisted flexion at 90° flexion, 10° adduction, full IR**
"20% ↑ CC distance, symptomatic — treatment?"	→ **sling + early PT**
"58-yo accountant, non-dominant AC separation, no AP translation?"	→ **sling + early ROM**
"Persistent symptoms after modified Weaver-Dunn?"	→ **anterior–posterior translation**
"CC screw fixation — true post-op?"	→ **AC arthritis more likely than nonop**
"Elderly, large shoulder lump, MRI Geyser sign?"	→ **AC joint cyst** (TO)
"Commonest AC cyst type?"	→ **type 2 — chronic cuff tear** (TO)
"Geyser sign, severe pain, irreparable cuff, elderly?"	→ **arthroplasty (RSA > hemi)** (TO)
"Correct hook-plate position?"	→ **hook under the posterior acromion** (anterior → impingement) (TO)
"Gold standard for chronic AC dislocation?"	→ **anatomic CC tendon-graft reconstruction** (TO)

ADHESIVE CAPSULITIS (FROZEN SHOULDER)

– Shoulder & Elbow · 4 / 60 –

1 Basics

Epidemiology

Prevalence (TO)	**2–5%** of population · incidence in **diabetics 10–36%**
Demographics	♀ **>** ♂, **40–60 y**; **nondominant** arm most often; **minorities > Caucasians** ✗ Wrong options: Caucasian ↑ risk
Bilateral	**age <50** and **diabetes** → ↑ risk of bilateral disease; **sequential, not simultaneous**

Risk factors & associations (diabetes, thyroid disease)

Risk factors (TO)	**female**, **age >40**, **preceding trauma**, **HLA-B27 positivity**, **prolonged immobilisation** of GH joint
Associations (TO)	**diabetes**, **thyroid disease**, **cerebrovascular disease**, **coronary artery disease**, **autoimmune disease**, **Dupuytren's**
Other associations	atherosclerotic disease, **cervical disc disease**, **history of MI**, **Crohn's**, **breast cancer treatment / axillary dissection**, previous surgery
Diabetes (both types)	**most common association**; stiffness may be **first manifestation of DM** → work up · often **bilateral** · **worse outcomes regardless of treatment** (nonop + surgical)
Insulin-dependent (TO)	**more severe** limitation + **more resistant** to nonop than non-insulin diabetics
Risk ↑ within diabetics	**older age (>50)** · **duration of DM** · **autonomic neuropathy** · **history of MI**
Not risk factors in diabetics	**type of diabetes** (↑ in **both type I and II**) · diabetic regimen (insulin vs non-insulin) · glycaemic control
Thyroid	autoimmune aetiology; **hypothyroidism** (↑ TSH) classic
✗ Distractors	**menopause** (no association) · **steroid use** (steroids treat it) · epilepsy · renal disease · prior dislocations

– – Active space – –

Genes & lab markers

Test	Finding in AC	✗ Distractor meaning
TSH	↑ (hypothyroidism)	↑ **free T4** = hyperthyroidism (wrong option)
HbA1c (hemoglobin A1C)	↑ (diabetes)	–
HLA-B27 (TO)	Positivity = risk factor	–
Uric acid	Normal	↑ = gout
ESR / CRP	Normal	↑ = infection / inflammatory arthropathy
RF / anti-CCP	Normal	↑ = RA
ANA	Normal	+ = lupus

Workup metabolic panel + endocrine labs (**TSH, HbA1c**) before choosing treatment

Pathophysiology – fibroblastic proliferation & reduced capsular volume

Forms **primary/idiopathic** (most) · **post-traumatic** (proximal humerus fracture, immobilisation) · **post-surgical** (RCR, axillary dissection)

Pathoanatomy inflammation (subsynovial layer, TO) → **fibroblastic proliferation** of capsule → thickening, fibrosis, adherence of capsule to itself + humerus → **mechanical block** · **↓ intracapsular volume**

Cell type **fibroblasts / myofibroblasts**
✗ Wrong options: adipocytes · granulocytes · lymphocytes · monocytes

Collagen abundant **type III collagen**

Essential lesion **coracohumeral ligament (CHL)** + **rotator interval capsule**

Why ER lost first (TO) changes start in **subsynovial layer of capsule in rotator interval**

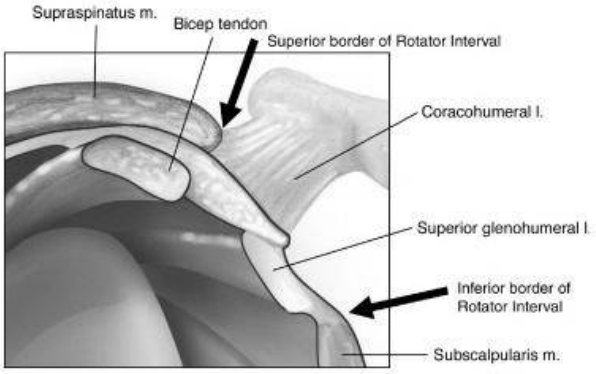
✗ Wrong options (other processes) isolated posterior capsular tightness = **GIRD** · ↓ blood supply = **AVN** · degenerative tendon tear = **cuff tear** (active loss only)

Rotator interval anatomy

Boundary	Structure
Superior	**Anterior edge of supraspinatus**
Inferior	**Superior edge of subscapularis**
Medial (base)	**Coracoid process**
Lateral (apex)	**Transverse humeral ligament**
Contents	**LHB tendon, SGHL, CHL, RI capsule**

✗ RI traps LHB = content · AIGHL + **axillary pouch** lie **inferior** to RI · MGHL is not the inferior border (subscapularis is)
✗ Wrong options: superior border

RI clinical **closed** for instability · **contracted** in AC → **released**

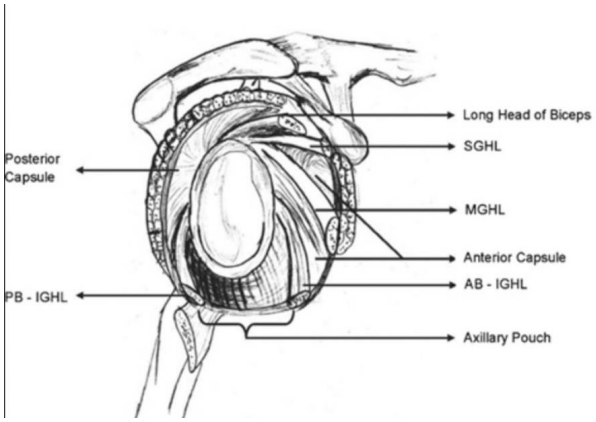


Rotator interval borders and contents

Capsuloligamentous restraints

Structure	Restraint	Release effect
RI (SGHL + CHL + capsule)	**Inferior translation + ER in adduction**	↑ **ER at side** + inferior translation
SGHL	**Inferior translation at 0° abduction**	–
MGHL	**Anterior (± posterior) translation at ~45° abduction** + ER	↑ anterior translation at 45°
AB-IGHL	**Anterior + inferior translation at 90° abduction + max ER**	↑ anteroinferior translation at 90°
Posterior capsule	IR + **cross-body adduction**	↑ **IR + cross-body adduction**

Normal role check reins at extremes of motion; IGHL complex = anterior band + axillary fold + posterior band



Capsular structures from glenoid side

3 Presentation

Clinical diagnosis – loss of active and passive motion

Symptoms **insidious** diffuse pain **preceding** loss of motion · **night pain / difficulty sleeping** · ADL difficulty (reaching overhead)

Exam **symmetric loss of both active AND passive ROM** · **tethered endpoint** · pain throughout arc or at terminal motion (stage-dependent)

– – Active space – –

First movement lost (TO) **external rotation**
✗ Wrong options: IR · flexion · abduction

Most common + most sensitive **passive ER deficit vs contralateral** shoulder

Provocative tests impingement, biceps, SLAP often **positive** (false); cuff testing limited by stiffness

Diagnosis **clinical** (normal X-ray + painful active + passive loss); MRI not necessary

✗ Wrong option: exam pattern	Diagnosis
Active < passive (e.g. active FF 100°, passive 150°)	**Weakness** – cuff tear
Pain 0–30° active abduction, full passive	Supraspinatus tendinosis/tear
Empty can + pain/weakness, full passive	Rotator cuff tear
Speed's +, full ROM	Biceps tendinopathy
Isolated IR loss (posterior capsule)	GIRD

Clinical stages

Stage	Features	Duration
1 Freezing / painful	Gradual diffuse pain at rest + motion, night pain	**6 wk – 9 mo**
2 Frozen / stiff	Pain lessens; ↓ ROM affecting ADLs	**4 – 9 mo or more**
3 Thawing / resolution	Pain gone; gradual return of motion (less than normal)	**5 – 26 mo**

Total duration self-limited; average **3–6 mo**, up to **18 mo – 2 y** (OB) – read the options

4 Diagnosis & Classification

Key radiology findings

X-ray views **AP in neutral rotation**, **scapular-Y**, **axillary lateral** (± AP in IR/ER) · **mandatory** to exclude **OA, posterior dislocation, hardware**

X-ray finding normal or **disuse osteopenia**; no high-riding head

MRI (± arthrography) not needed for diagnosis; excludes other pathology · **loss of axillary recess** (capsular contracture) · thickened CHL

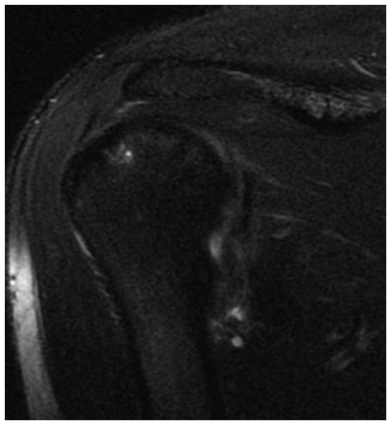
MR arthrogram **↓ intracapsular volume**

MRI sign of AC (TO)	Value
Axillary recess thickening	**≥1.3 cm** (TO)
Anterior capsule thickening	**>3.5 mm** + hyperintensity
Synovial thickness	**>4 mm**
RI	Soft tissue thickening + signal change; scarring
Biceps anchor	Encased by abnormal soft tissue
CHL	**Thickened**
Subcoracoid triangle sign	**Obliteration of fat** in subcoracoid space
Inferior capsular fold	**Obliterated**; capsule hypervascular

NOT seen (TO) **widening** of axillary pouch (pouch is contracted/obliterated)

CHL on MRI thickened (Orthobullets) · **not better visualised** than normal (OB wrong option) — read the options

Incidental partial cuff tear common in **asymptomatic >60 y** → still treat AC with stretching



Coronal MRI, loss of axillary recess

✗ MRI distractor	Diagnosis
Fluid extravasation down humerus	**HAGL**
Hypertrophied synovium, papillary projections	**PVNS**
Massive retracted tear	Cuff tear

Arthroscopic stages

Stage	Finding
1	**Patchy, fibrinous synovitis**
2	**Capsular contraction + fibrinous adhesions**
3	Increasing contraction, **synovitis resolving**
4	**Severe contraction**

5 Management

Adhesive capsulitis

- Early (freezing) → **reassurance + gentle stretching ± IA steroid + NSAIDs**
  (treat diabetes / thyroid)
- Failed nonop (3–6 mo) → **MUA and/or arthroscopic capsular release**
  - diabetic / post cuff or labral repair → **release (MUA 50% fail)**
- Residual IR / cross-body loss → **add posterior capsular release**

Nonoperative management – stretching, injection & natural history

First-line **PT + NSAIDs ± IA steroid injection** ± heat/cryotherapy; often effective, **majority** succeed

– – Active space – –

PT program	**gentle, pain-free stretching + moist heat**; supervised; **3–6 mo**; stress **motion, not strengthening**
Home exercise program	**as helpful as organised therapy**
IA steroid	↓ inflammation → allows stretching; **earlier** return of ROM, **no long-term difference** (OB, TO) · stage II → **IA steroid + stretching**
PT detail (TO)	stretching then strengthening = mainstay **regardless of stage**; **avoid abduction initially** (impingement)

(✗ Wrong options: **Wrong early** – **sling** (↑ contracture, prolonged recovery)) · **immediate aggressive therapy** (counterproductive unless pain controlled) · **PRP** (no evidence) · MUA / arthroscopy before conservative trial · cuff strengthening

Distension arthrography	rarely performed
Natural history	**self-limited**; most resolve without surgery
Most common outcome after nonop stretching	**decreased range of motion vs contralateral shoulder** ✗ Wrong options: limiting pain · recurrence (low) · surgery (rare, **>90% satisfied**)

Stretching exercise mechanics

ER wand exercise (arm adducted, elbow flexed)	stretches **anterior-superior capsule, SGHL, CHL, IGHL** · **LEAST stretch on posterior capsule**
Most commonly contracted	**anterior-superior capsule (RI)** → limits **ER in adduction**

Operative criteria

Factor	Threshold / rule
Failed therapy before release	**3 mo** (Orthobullets) · **3–6 mo** (OB) · **≥6 mo** (OB) · **4–6 mo** compliant therapy (TO) – read the options
Indication	**Progressive loss of motion** despite prolonged PT
MUA contraindication	**Diabetics (50% failure)** · **after RCR or labral repair** · freezing phase (controversial)
Arthroscopy (TO)	Reserve for **recalcitrant** cases (✗ Wrong options: RCT superiority over MUA)

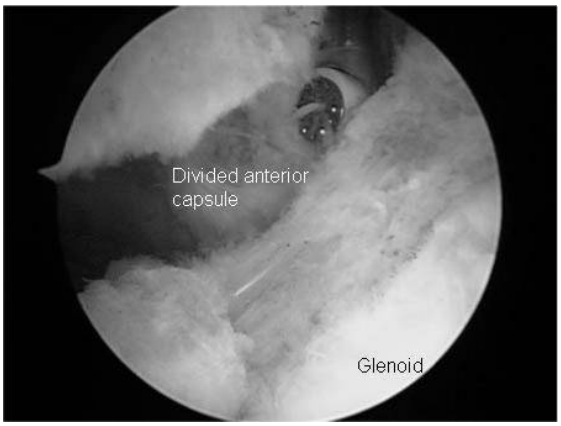
Manipulation under anaesthesia

Technique	**document ROM** first · **full muscle paralysis** · indwelling regional catheter · **steady force** · **early PT**
FEAR sequence (TO)	**F**lexion, **E**xtension, **A**bduction, adduction, external + internal **R**otation
Post-MUA (TO)	**20 ml 0.5% bupivacaine + corticosteroid** IA
Results (TO)	**up to 94%** subjective relief, sustained long term

Arthroscopic capsular release – structures & outcomes

Portals	slightly **higher** than normal (contracted capsule)
Rotator interval release	from **anterior biceps to superior subscapularis** → CHL visualised + released → restores **ER at side**

Released	**SGHL, MGHL, IGHL, CHL** (circumferential) · **not coracoacromial ligament** (= landmark for complete RI release)
Posterior capsular release	↑ **IR + cross-body adduction**
Subacromial	**bursectomy + adhesion release**; **no acromioplasty**
Sequence	MUA before or after release; TO: release before manipulation allows inspection + precise capsulotomy; **anterior release always ± posterior** (posterior: early benefit, similar outcome)
Advantage over open	**spares subscapularis**; releases intra-articular + subacromial adhesions
Not done	completion + repair of small partial tear (that is for **>6 mm or 50%**) · **capsular plication** (instability) · **distal clavicle resection** (AC arthritis)
Outcome	gains in ROM + function **maintained at long-term follow-up**



Arthroscopic anterior capsular release

6 Complications

Complication	Detail / prevention
Residual / recurrent stiffness	**Most common** after release → **immediate ROM + PT**, **no sling**, ± CPM, regional analgesia
Axillary nerve injury	**Inferior capsular release** → stay **close to glenoid rim**; rare
Infection, instability, impingement	Less common
MUA (overzealous, osteoporotic bone)	**Proximal humerus fracture**, dislocation, **cuff + labral tears**, **brachial plexus palsy**

Prognosis	self-limited · **worse in diabetics**

– – Active space – –

X vs Y

	Adhesive capsulitis	Cuff tear	GIRD	GH OA / posterior dislocation
Active ROM	↓	↓	↓ IR	↓
Passive ROM	**↓ (= active)**	**Normal**	↓ IR only	↓
First / key loss	**ER**	Strength (weakness)	**IR** (posterior capsule)	–
X-ray	Normal / osteopenia	± high-riding head	Normal	**Diagnostic** (why X-ray is mandatory)

✗ Fibroblastic distractor	Diagnosis
>50 y, giant cell arteritis, bilateral shoulder stiffness	**PMR**
Podagra, acute shoulder pain	**Gout** (crystal)
Ulnar drift	**RA**
+ANA	**Lupus**

How it's asked

"Known risk factor for adhesive capsulitis?"	**→ diabetes mellitus** ✗ Wrong options: menopause · steroid use · renal disease · epilepsy
"Risk factor in diabetics EXCEPT"	**→ type of diabetes** ✗ Wrong options: age >50, duration, MI, autonomic neuropathy
"55 F, stiff shoulder, no PMH – lab most likely abnormal?"	**→ ↑ TSH** ✗ Wrong options: uric acid, ESR, anti-CCP, dislocations
"Basic blood work – which elevated?"	**→ HbA1c** ✗ Wrong options: free T4, RF, CRP, lactic acid
"True statement about AC?"	**→ age <50 + diabetes → bilateral disease** ✗ Wrong options: Caucasian · CHL better seen · release usually needed · flexion lost first
"NOT seen in adhesive capsulitis (MRI)?"	**→ widening and thickening of axillary pouch** (TO)
"MR arthrogram in diabetic stiff shoulder?"	**→ ↓ intracapsular volume** ✗ Wrong options: PVNS · HAGL extravasation

"Fibroblastic proliferative process — which patient?"	→ **50 F hypothyroid, loss of active + passive motion** ✗ Wrong options: PMR, gout, RA, lupus
"Pathophysiology + other finding?"	→ **fibroblastic proliferation of capsule; HbA1c 11.7%** ✗ Wrong options: posterior capsule = GIRD
"Cell line responsible?"	→ **fibroblasts**
"First movement hampered?"	→ **ER** (TO)
"Uncontrolled diabetic, 3 mo shoulder pain — exam?"	→ **symmetric loss of active + passive ROM**
"Most common outcome after stretching program?"	→ **↓ ROM vs contralateral**
"Early stiffening stage, 1 mo — treatment?"	→ **reassurance + gentle stretching** ✗ Wrong options: aggressive PT, PRP, MUA, arthroscopy
"Diabetic, 10 wk, greatest ER loss — treatment?"	→ **PT + medical management** ✗ Wrong options: cuff repair · posterior release · RSA
"Recent-onset AC, 2 wk — next step?"	→ **home exercise program** ✗ Wrong options: sling
"4 mo diffuse pain, stage II — management?"	→ **IA corticosteroid + stretching** ✗ Wrong options: sling · aggressive stretching
"Global stiffness + MRI partial supraspinatus tear, >60 y?"	→ **stretching regimen** ✗ Wrong options: strengthening · acromioplasty / repair
"Mitigate most common complication of arthroscopic release?"	→ **immediate ROM + PT** (stiffness)
"Release increases inferior translation + ER in adduction?"	→ **rotator interval (A)** ✗ Wrong options: MGHL · AIGHL
"Release improves cross-body adduction?"	→ **posterior capsule (H)**
"Failed nonop, 2 mm articular tear — surgery includes?"	→ **rotator interval release** ✗ Wrong options: tear completion, plication, SAD, DCR
"NOT released in AC arthroscopic release?"	→ **coracoacromial ligament**
"Expectation counselling before release?"	→ **improvements maintained long term** ✗ Wrong options: axillary nerve most common · posterior capsule most common stiffness · stiffness rare
"Arthroscopic release structure (figure)?"	→ **rotator interval (D)** ✗ Wrong options: subscapularis, supraspinatus, biceps, infraspinatus

– – Active space – –

"Rotator interval superior border?"	→ **anterior edge of supraspinatus** ✗ Wrong options: LHB · AIGHL · MGHL · axillary pouch
"ER wand exercise — LEAST stretch?"	→ **posterior capsule**

AVASCULAR NECROSIS OF THE SHOULDER

– Shoulder & Elbow · 5 / 60 –

1 Basics

Pathogenesis & diagnosis (corticosteroid risk)

Pathogenesis interrupted blood supply → **cellular death of subchondral bone** → resorption + remodelling → **subchondral collapse** → incongruity → arthritis

✗ Pathogenesis distractors cell-mediated synovial hypertrophy = **RA** · humoral response after infection, HLA-B27 = **Reiter's** · hyperuricaemic crystals = **gout** · bacterial PMN cartilage destruction = **septic arthritis**

Most common cause **corticosteroids** (asthma, Crohn's flares, SLE, leukaemia chemotherapy, graft-versus-host disease); also **sickle cell**, chronic **alcohol**

Young patients high-dose steroids in chemotherapy regimens → ↑ demand for arthroplasty in the young

vs hip humeral head AVN **less common than hip**; same pathophysiology (genetic predisposition + risk factors such as steroids); treatment similar to hip, based on **subchondral collapse**

Workup proximal humerus AVN on X-ray → also get **hip radiographs**

ASEPTIC mnemonic	Cause
A	Alcohol, AIDS
S	**Steroids (most common)**, Sickle cell, SLE
E	Erlenmeyer flask (**Gaucher**)
P	Pancreatitis
T	Trauma
I	Idiopathic / Infection
C	Caisson's (the bends)

Gaucher (wrong option) **large bony infarcts** + severe pain; less common cause than chronic steroids

Post-traumatic AVN

Proximal humerus injury	AVN rate
4-part fracture-dislocation	**~100%**
Displaced 4-part	~45%
3-part	~14%
Valgus-impacted 4-part	**~11%**

– – Active space – –

Anatomy – humeral head blood supply

Anterior humeral circumflex	**ascending branch** runs **lateral to LHB tendon in bicipital groove → arcuate artery** = intraosseous continuation, penetrates head · **35%** ✗ Wrong options: injure when plating
Posterior humeral circumflex	**main supply** per most current literature · **65%**

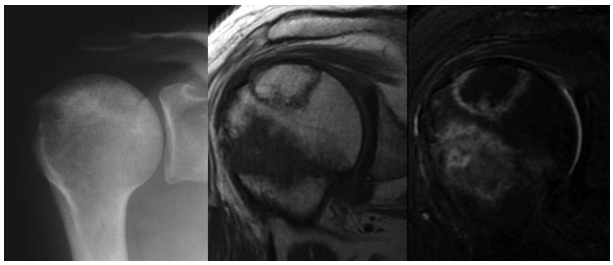
3 Presentation

Symptoms	**insidious** shoulder pain, often no inciting event · ↓ motion · crepitus · weakness
Exam	limited ROM (pain throughout arc) · crepitus · cuff + deltoid weakness (cuff strength may be normal)

4 Diagnosis & Classification

Key radiology findings

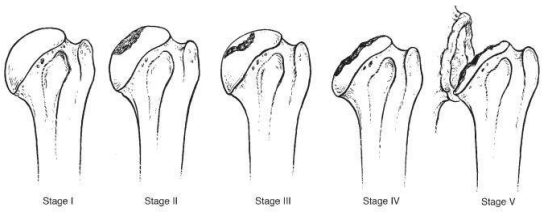
X-ray	5 views; best seen on **neutral rotation AP** · **normal at onset** · osteolytic lesion first at **superior-middle humeral head** (most common initial site)
Stage signs (X-ray)	**"snowcap" sign** = sclerosis, contour preserved (II) · **crescent sign** = subchondral fracture (III) · flattening/collapse, loss of sphericity (IV) · glenoid changes (V)
MRI	**preferred** modality · **~100% sensitive** · early = **↓ subchondral signal on T1 and T2** (oedema) · variegated pattern · metaphyseal oedema common/normal



Cruess stage II: snowcap sclerosis with preserved contour (X-ray, T1, T2 fat-sat)

Cruess classification

Stage	Radiographic findings	Treatment
I	**Normal X-ray, MRI changes only**	Core decompression
II	**Sclerosis** (wedged, mottled), osteopenia; snowcap	Core decompression
III	**Crescent sign** = subchondral fracture	Resurfacing or hemiarthroplasty
IV	**Flattening and collapse** (glenoid spared)	Hemiarthroplasty (or resurfacing)
V	Degenerative changes **extend to glenoid**	**TSA**



Cruess stages I–V

Ficat-Arlet hip staging adapted to the shoulder (used in core decompression series)

5 Management

Humeral head AVN

- First line → **analgesia, activity modification (no overhead/manual labour), PT**
- Precollapse (Cruess I–II) → **core decompression ± arthroscopy**
- Collapse, glenoid + cuff intact (III–IV) → **hemiarthroplasty**
  - focal III defect + enough epiphyseal bone → **resurfacing**
- Glenoid involved (V) → **TSA**
- Older, cuff pathology → **reverse TSA**

Operative criteria

Procedure	Indication	Not when
Core decompression (+ arthroscopy to confirm cartilage)	**Precollapse Cruess I–II**	Subchondral fracture/flattening (post-collapse = **contraindicated**)
Resurfacing hemiarthroplasty	**Stage III focal chondral defect + sufficient epiphyseal bone**; least bone removal	Large defect · global head involvement
Hemiarthroplasty	**III–IV**: collapse, **glenoid intact, cuff intact**; young → preserves glenoid bone stock for later TSA	**Irreparable cuff tear** · glenoid changes
TSA	**Stage V** (glenoid involved)	Young + intact glenoid (glenoid loosening)
Reverse TSA	Older + cuff pathology	**Intact cuff**, young

– – Active space – –

Treatment by stage – core decompression vs resurfacing/hemiarthroplasty

Core decompression ↓ subchondral marrow pressure → **neovascularisation** → slows progression · ± bone graft into void · percutaneous small-diameter technique (Harreld: no arthroplasty at 32 mo, 95% improved)

Core decompression (LaPorte, ~10 y; Ficat stage)	Success
I	**94%**
II	**88%**
III	70%
IV	14%

Stage I–II outcome good/excellent **88–100%** at 5–10 y → **~90% successful long term** · better in **non-steroid** than steroid cases
✗ Wrong options: 50% satisfied · most need arthroplasty by 10 y · unpredictable · not indicated

Conflict on core decompression effective for I–II (Mont, LaPorte; LaPorte recommends I–III; Harreld tries it with arthroscopy in III) · role in precollapse "**highly debated**" and drilling did **not** alter progression (L'insalata) · Sarris: I–II **conservative**, surgery when persistently symptomatic + collapse — read the options

Arthroscopic debridement / release no sustained benefit; AVN keeps progressing; role unknown

Most reliable, durable treatment after collapse **prosthetic arthroplasty → hemiarthroplasty**
✗ Wrong options: arthroscopic debridement · arthrodesis · resection arthroplasty · cortisone injection

Arthrodesis indications brachial plexus injury · infantile paralysis · deltoid + cuff insufficiency · chronic infection · failed revision arthroplasty · refractory instability · tumour resection bone loss

Stemless resurfacing preserves metaphyseal bone · Levy & Copeland 285 cementless · Hawi (Eclipse) 9 y, no loosening

"Ream and run" HA + glenoid reaming → unnecessary with intact glenoid cartilage

Arthroplasty outcomes HA + TSA both safe/effective · 10-y survivorship **HA 88.9%, TSA 80%**; **TSA better internal rotation** (Wang) · RSA ≈ TSA when cuff compromised · AVN arthroplasty has **↑ complications** vs other diagnoses (bone quality, comorbidity)

Prognosis related to **stage**; stage + lesion size predict progression

6 Complications

Hemiarthroplasty **glenoid cartilage wear** over time → 45% unsatisfactory (Smith, steroid AVN); still used in young active III–IV

Glenoid component in young **loosening** risk → avoid TSA/RTSA when glenoid spared

X vs Y

	AVN	Calcific tendinitis	OA	Gaucher
Onset	Insidious, steroid user	**Severe acute pain**	Long-standing symptoms	Severe pain
Imaging	**Subchondral** ↓ T1/T2 signal, collapse, fragmentation	**Hypointense deposit in cuff soft tissue**	No subchondral collapse/ fragmentation	**Large bony infarcts**

Stage	Core decompression	Hemiarthroplasty	TSA
Cruess	I–II (precollapse)	III–IV (collapse, glenoid spared)	V (glenoid)
Cuff	–	Must be intact	Intact
Key trap	Not after collapse	Not with irreparable cuff or glenoid cysts	Not in young with intact glenoid

How it's asked

"Asthmatic, 20 y steroids, humeral head collapse — pathogenesis?"	→ **cellular death of subchondral bone after vascular interruption** ✗ Wrong options: RA · Reiter's · gout · septic
"Crohn's on prednisone, atraumatic pain, MRI subchondral lesion — diagnosis?"	→ **osteonecrosis** ✗ Wrong options: Gaucher · OA · cuff tendinopathy · calcific tendinitis
"Steroid user, snowcap sclerosis, contour preserved — surgery?"	→ **core decompression** ✗ Wrong options: HA · RTSA with intact cuff · resurfacing · TSA
"Leukaemia + high-dose cortisone, collapse — most reliable pain relief?"	→ **hemiarthroplasty** ✗ Wrong options: debridement · arthrodesis · resection · injection
"39F, collapse + loss of sphericity, glenoid spared, failed PT — procedure?"	→ **resurfacing hemiarthroplasty** ✗ Wrong options: arthroscopic release · core decompression post-collapse · TSA/RTSA glenoid loosening
"GVHD steroid AVN — best case for hemiarthroplasty?"	→ **subchondral fracture + intact cuff** ✗ Wrong options: sclerosis = decompression · irreparable cuff · glenoid cysts = TSA
"SLE patient had HA — preop imaging?"	→ **isolated humeral head collapse** (Cruess IV) ✗ Wrong options: glenoid changes = V · osteopenia / MRI-only / mottled sclerosis = I–II
"Leukaemia, flattening + collapse, glenoid intact, cuff normal?"	→ **hemiarthroplasty** ✗ Wrong options: TSA · core decompression · ream and run · RTSA

– – Active space – –

"Alcoholic, Cruess II – core decompression outcome?"	**→ ~90% successful long term**
"Humeral head AVN on X-ray – what else to image?"	**→ hips**

BICEPS SUBLUXATION

– Shoulder & Elbow · 6 / 60 –

1 Basics

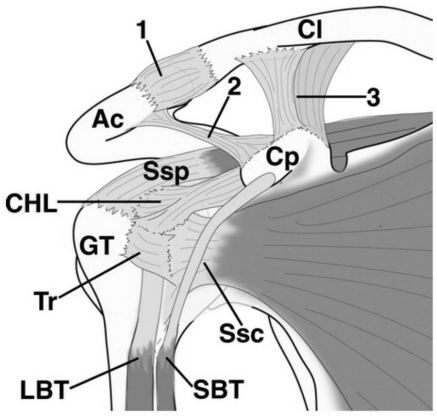
Association with subscapularis tears

Definition	LHBT subluxes/dislocates **medially** out of the bicipital groove → recognised cause of **anterior shoulder pain**
Most common associated pathology	**subscapularis tear** (even **partial**); biceps subluxation is **almost always** associated with a subscapularis tear
Most important medial restraint	**subscapularis tendon** = most important restraint to medial subluxation/ dislocation of LHBT
Walch series	dislocated LHBT associated with partial/complete **subscapularis tear in 96%**
Other cause	**disruption of the biceps sling (pulley)**
Not the cause	**subacromial impingement** does not cause LHBT instability · isolated degenerative tendinosis (instability is from the subscapularis tear) · SLAP

Biceps pulley stabilisers (coracohumeral & transverse humeral ligaments)

Origin	LHBT arises from **supraglenoid tubercle + superior labrum** → intra-articular → bicipital groove
Function	**dynamic stabiliser**; shoulder flexion, abduction

Biceps sling / pulley (intra-articular)	Rotator interval tissue?
Coracohumeral ligament (CHL)	**Yes**
Superior glenohumeral ligament (SGHL)	**Yes**
Subscapularis fibres	Contributes to sling
Supraspinatus (anterior fibres)	Contributes to sling



Biceps sling: CHL, SGHL, subscapularis, supraspinatus over LHBT

Transverse humeral ligament (THL)	stabilises LHBT in the groove **at the level of the greater + lesser tuberosities** (distal to pulley); **not rotator interval tissue**

– – Active space – –

THL (TO) old concept of THL as the LHB stabiliser is **obsolete**; **medial pulley = subscapularis + SGHL + CHL** prevents medial subluxation — read the options

Comma sign (TO) in **subscapularis tears**: torn CHL + SGHL + subscapularis pulled together into a comma shape → **pulley lesion**

Habermeyer pulley lesion (TO)	Injury
I	**SGHL** → anterior LHB instability
II	SGHL + **anterior supraspinatus** tear
III	SGHL + **cranial subscapularis** tear (**comma sign**)
IV	SGHL + anterior supraspinatus + cranial subscapularis → **anteroposterior instability**

Bicipital groove length 81–87 mm · width 7.7–11.6 mm · depth 3.7–6 mm · **bottleneck ~2/3 down** from top → THL may constrain groove → anterior pain; tenodesis **below the groove** may be preferred

✗ Distractor	Why wrong
Coracoacromial ligament (CAL)	Not rotator interval; no LHBT stability at tuberosities
IGHL	Not rotator interval; not an LHBT stabiliser
AIGHL	Not associated with biceps subluxation (injured in anterior dislocation)
Superior labrum / infraspinatus / posterior labrum	Not associated with **medial** biceps subluxation

Imaging finding = injured structure MRI medial biceps subluxation → **CHL disruption** (sling)

Anterosuperior cuff tear sling disrupted → LHBT subluxes medially → **signals subscapularis tear** (often missed on MRI)

3 Presentation

Symptoms **anterior** shoulder pain ± **clicking**

Yergason test anterior pain with **resisted supination**, arm at side, elbow flexed 90°

Speed test anterior pain with **resisted forward flexion**, shoulder 90° flexed, elbow extended, palm up

Palpable click with **abduction + external rotation** as tendon subluxes/dislocates out of groove

Clinical clues (TO) anterior pain · clicking/snapping · MRI medial subluxation/dislocation · **often missed** unless subscapularis/cuff evaluated

Always test subscapularis strength **lift-off** (Gerber), belly press, push-off

Lift-off test (Gerber) back of hand held off mid-lumbar spine in IR; hand falls or patient extends elbow/shoulder to compensate → **subscapularis tear**

– – Active space – –

✗ Exam distractor	Tests
Drop arm	**Supraspinatus**
External rotation lag	**Infraspinatus**
Hornblower's	**Teres minor**
Kim test	**Posteroinferior labral tear**

4 Diagnosis & Classification

Key radiology findings

MRI ↑ **T2 signal** + LHBT **displaced out of bicipital groove** (axial: on lesser tuberosity → fully medial); coincides with **subscapularis tear**

MRI limitation misses subscapularis tears unless tear extends **≥½ cranio-caudal** length → small tears missed → test lift-off

LHBT subluxation on MRI as predictor of full-thickness subscapularis tear (Shi) sensitivity **82%** · specificity **80%** · PPV **35%** · **NPV 97%** → value lies in **NPV** (no subluxation → full-thickness subscapularis tear unlikely); correlates with tear severity

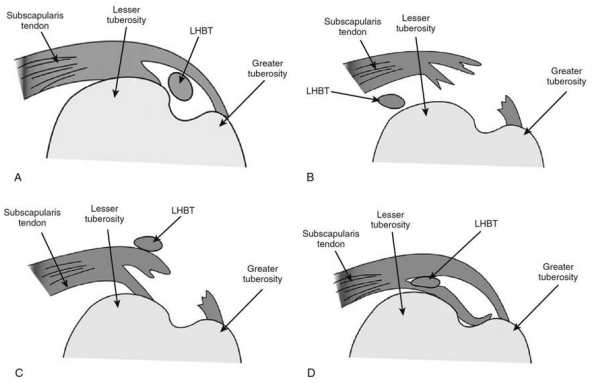
Ultrasound **dynamic test** of biceps instability

Arthroscopy ↓ **biceps–glenoid angle** associated with subluxation (**87°** vs **90°**; Koh) → statistically significant but **not clinically relevant**



Axial MRI: LHBT dislocated medial to empty groove

– – Active space – –



LHBT position relative to subscapularis and tuberosities

5 Management

Biceps subluxation

- Initial → **NSAIDs + PT + steroid injection**
  (injection near, not into, the tendon)
- Refractory → **arthroscopic/open tenotomy or tenodesis**
  - subscapularis tear → **+ subscapularis repair**

Nonoperative

First line	**NSAIDs, PT strengthening, activity modification, steroid injection** (adjacent to tendon, **not into it**)

Operative

Indication	**refractory** symptoms
Options	arthroscopic vs open **biceps tenotomy vs tenodesis** ± **subscapularis repair**; instability can be tested intra-op
Acute traumatic medial LHBT instability (Maier)	open subscapularis repair + LHBT stabilisation = equal function to tenotomy/tenodesis, **better cosmesis, less cramping**
THL release	routine release at tenotomy/tenodesis may ↓ post-op anterior pain

Biceps tenotomy for irreparable cuff tear

Indication	large/massive **irreparable** cuff tear, **pain** main complaint, **good active elevation** (e.g. 120°), **normal articular surfaces → biceps tenotomy**

Alternative	When
Latissimus dorsi transfer	Severe **ER deficit** + deficient **teres minor**
Reverse TSA	**Cuff tear arthropathy + pseudoparalysis**
✗ Wrong option: SLAP repair	Contraindicated in this age group

X vs Y

	Biceps sling (pulley)	Transverse humeral ligament
Level	Intra-articular, rotator interval	Bicipital groove at **tuberosities**
Components	CHL, SGHL + subscapularis, supraspinatus	Single ligament, superficial roof of groove
Rotator interval tissue	**CHL + SGHL = yes**	**No**

	Tenotomy for massive cuff tear	Latissimus transfer	Reverse TSA
Key feature	Pain, elevation preserved, no arthritis	ER deficit, teres minor deficient	Arthropathy + pseudoparalysis

How it's asked

"62F, anterior pain, + Yergason/Speed, MRI LHBT on lesser tuberosity then dislocated — true statement?"	→ **biceps pathology due to partial subscapularis tear** ✗ Wrong options: SLAP repair · isolated tendinosis · impingement
"MR arthrogram, biceps out of groove — arthroscopy will show?"	→ **subscapularis tear + biceps subluxation** ✗ Wrong options: THL disruption · absent CHL · short head rupture
"Skier, MRI dislocated biceps tendon — most important exam?"	→ **lift-off test** ✗ Wrong options: drop arm · ER lag · Hornblower · Kim
"Post-dislocation, MRI medial biceps subluxation — injured structure?"	→ **coracohumeral ligament** ✗ Wrong options: AIGHL · superior labrum · infraspinatus · posterior labrum
"Stabilises LHBT at tuberosities, not rotator interval?"	→ **transverse humeral ligament** ✗ Wrong options: CHL · SGHL = interval · CAL · IGHL
"Main diagnostic value of LHBT subluxation on MRI?"	→ **negative predictive value (97%)**
"Massive irreparable cuff tear, FE 120°, normal joint, MRI biceps subluxed?"	→ **biceps tenotomy** ✗ Wrong options: latissimus transfer · reverse TSA
"Most important medial restraint of LHBT?"	→ **subscapularis tendon**
"Comma sign at arthroscopy indicates?"	→ **subscapularis tear with biceps pulley lesion** (TO)
"Habermeyer type III pulley lesion?"	→ **SGHL + cranial subscapularis tear** (TO)

– – Active space – –

BICEPS TENDONITIS

– Shoulder & Elbow · 7 / 60 –

1 Basics

Pathoanatomy

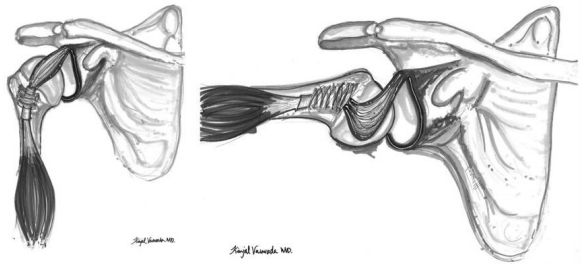
Nature **tendinosis** (degenerative) > true inflammation; inflammatory, traumatic or degenerative causes

Associations usually **secondary** to other shoulder pathology: **subacromial impingement**, **bicipital groove stenosis**, **rotator cuff tears – esp. subscapularis**

Tendonitis group (TO)	Cause	Treatment
Young active (throwers, swimmers, gymnasts)	Overuse of intra-articular LHB; isolated or with **internal impingement**	Rest, NSAIDs, injection · **tenodesis if pain >3 months**
Elderly / cuff disease	Subacromial impingement, osteophytes; often **supraspinatus tear**	Address impingement + cuff repair + **tenotomy** (tenodesis if active)

Rule (TO) always assess the biceps in cuff disease; treating only the cuff may leave pain

Hourglass biceps chronic inflammation → **hypertrophied intra-articular tendon** → **incarcerated in joint** during motion, esp. **elevation**



Hourglass biceps incarcerated in the joint

Long head of biceps anatomy – labral attachment, rotator interval & pulley

Origin (TO) **supraglenoid tubercle + superior labrum**; most commonly **50% labrum / 50% tubercle**

Insertion with short head → **radial tuberosity** · action: **elbow flexion + forearm supination** · **no shoulder ROM/strength contribution**

Synovial status (TO) **intra-articular but extrasynovial**; synovial sheath communicates with GH joint → extends into groove

"Policeman" and "thief" (TO) LHB orients the arthroscopist yet its lesions are subtle, chronic · **short head = function, long head = pathology** · **~95%** of biceps pathology involves the LHB

Length (TO) intra-articular portion **~3.5 cm** · origin → musculotendinous junction **~9 cm**

– – Active space – –

LHB zone (TO)	Region
Zone 1	Traditional bicipital groove
Zone 2	**Fibro-osseous tunnel** (subscapularis fibres form part) → **hidden pathology**
Zone 3	Below groove (musculotendinous junction)

Function (TO) elbow flexion + **supination (2× the supinator with elbow extended)** · assists **abduction of externally rotated arm** · **humeral head depressor** (balances superior migration in cuff tears) – contrasts with "no shoulder ROM/strength contribution" above – read the options

Tuoheti classification – LHB attachment on labrum (TO)

Type	Attachment	Frequency
1	Completely posterior	–
2	**Predominantly posterior**	**Most common**
3	Anterior + posterior	–
4	Anterior	**Least common**

Rotator interval (TO)

Feature	Structure
Superior border	**Supraspinatus**
Inferior border	**Subscapularis**
Medial edge (base)	**Coracoid process**
Ceiling	Joint capsule
Floor	Articular surface of humeral head
Reinforced	**CHL laterally**, **SGHL medially**
Contents	**LHB** (on the base, covered by capsule), CHL, SGHL, RI capsule

✗ Boundary trap (TO) **CHL is NOT a boundary** (content/reinforcement)

Shape (TO) triangular **space** (3-D), not 2-D area

Jost 4 layers (TO) 1 superficial CHL fibres (coracoid → greater + lesser tuberosities) · 2 CHL blended with cuff tendons · 3 deep CHL fibres · 4 **SGHL + capsule** mesh

CHL (TO) arises **outside joint**, lateral base of **coracoid** → inserts **greater + lesser tuberosities** → **spans bicipital groove** · superficial layer covers articular surface of supra/infraspinatus · **deep layer thicker** → greater tuberosity

– – Active space – –

Biceps pulley & stabilisers (TO)

Biceps pulley (TO) **SGHL + CHL** fuse along lateral half of RI with **subscapularis + supraspinatus** fibres → cradle intra-articular LHB

LHB portion	Stabilisers (TO)
Intra-articular	**Labrum + capsuloligamentous structures**, **CHL**, **SGHL**, **supraspinatus**, **subscapularis**
Extra-articular	**Transverse humeral ligament**, **pectoralis major** muscle/tendon, **bicipital groove**

✗ Stabiliser traps (TO) **transverse humeral ligament = extra-articular** (✗ Wrong options: intra-articular) · CHL/SGHL = **intra-articular** stabilisers (✗ Wrong options: extra-articular)

Rotator cable vs crescent (TO)

	Rotator cable	Rotator crescent
Structure	**Thick bundle** of fibres	**Thin, crescent-shaped sheet** of cuff
Location	Avascular zone of CHL; **perpendicular to supraspinatus fibres**	**Distal** supraspinatus + infraspinatus insertions
Relationship	Proximal margin of crescent	Bounded proximally by cable

"Suspension bridge" (TO) cable + crescent span supraspinatus + infraspinatus insertions · swapping the two definitions = classic trap

3 Presentation

Symptoms **anterior shoulder pain** ± radiating down biceps; mimics cuff/impingement pain

Groove tenderness over **bicipital groove**; worse with arm **internally rotated 10°**

Test	Manoeuvre	Positive
Speed	Forward elevate against resistance, **elbow extended + forearm supinated**	Pain in bicipital groove; **also + in SLAP**
Yergason	Active **supination against resistance**, **elbow 90°**, forearm **pronated**	Pain in bicipital groove
Popeye	Distal bulge of biceps muscle	**Rupture**

Rupture (TO)	Site	Patient	Sign	Treatment
Proximal	**Top of intertubercular sulcus (commonest)** > musculotendinous junction > supraglenoid origin (rare)	Older, degenerative	**Popeye** (bulge in lower arm)	Tenotomy / tenodesis
Distal	Radial tuberosity	Young, trauma	**Reverse Popeye** (bulge near shoulder)	Repair to tuberosity

Rupture weakness (TO) **supination > flexion**

Bicipital groove palpation (TO) arm in **~30° ER**, elbow 90° → marks groove



Popeye deformity

4 Diagnosis & Classification

Key radiology findings

US	**thickened tendon** within bicipital groove
MRI (TO)	normal LHB = **"coffee bean"** on axial · tendonitis = thick/irregular + peritendinous fluid · rupture = **empty groove**, retracted tendon · degenerative = frayed, intermediate signal, may sublux medially · US = dynamic adjunct
MRI	evaluates **concurrent pathology** (subscapularis, cuff) · **thickening + tenosynovitis** of proximal LHB · **↑ T2 signal around tendon** · longitudinal split tears

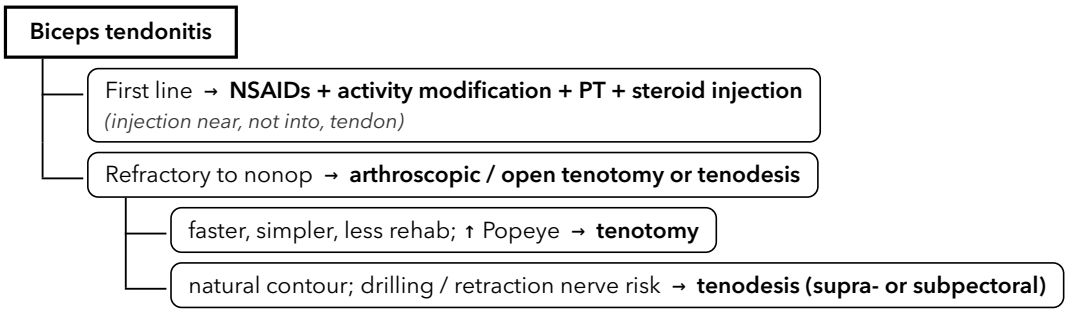


Axial MRI, fluid around LHB in groove

Diagnostic US-guided injection	groove steroid → temporary relief supports LHB as pain generator (e.g. 90% relief × 4 wk)

– – Active space – –

5 Management



Nonoperative

Regimen	NSAIDs, **PT strengthening**, activity modification, **steroid injection** (in proximity, **not into tendon**)

Tenodesis vs tenotomy

Only proven difference	**Popeye (cosmetic) deformity ↑ with tenotomy**
No difference	**strength**, **pain (VAS)**, **ASES / patient-reported outcomes**, **shoulder + elbow ROM**, **supination** (level 1 evidence)
Cramping	tenodesis **possibly ↓ arm cramping** (surgeon belief) · **not shown in high-level studies** – read the options
Tenotomy pros	faster, simpler, **less rehab**, **faster return** to activity
Tenodesis pros	natural muscle **contour**; but extra risks: nerve injury (**bicortical drilling**, **musculocutaneous retraction**)
Tenotomy downsides (TO)	Popeye **up to 20%** · **~20% flexion** and **20–30% supination** strength loss (TO) · no strength difference (level 1, Orthobullets/OB) – read the options
Patient choice (TO)	tenotomy: **elderly, low-demand, obese, non-compliant**, cosmesis unimportant · tenodesis: **young, active, cosmetic-conscious, manual workers** · **<40 y → tenodesis, >40 y → tenotomy** (TO)
Superior migration (TO)	LHB depresses head → **avoid tenotomy alone** if superior migration
Bite study (OB)	tenotomy vs suprapectoral tenodesis with cuff repair: differences small; non-inferiority **statistically inconclusive** on Constant-Murley at 1 y

Tenodesis technique – suprapectoral vs subpectoral, arthroscopic vs open

	Arthroscopic suprapectoral	**Open subpectoral**
Fixation site	**Proximal** in groove	**Distal** groove, at **pectoralis major insertion**
Rationale	–	Removes tendon from groove (eliminates **groove pain**)
Residual / groove pain	**Similar**	**Similar**
Popeye, ROM, strength, PROMs, revision	**Similar** (revision very low)	**Similar**
Nerve injury	Lower (**unicortical** drilling)	**↑ transient neuropraxia** · **axillary n.** (bicortical drilling) · **medial NV** structures (retraction)

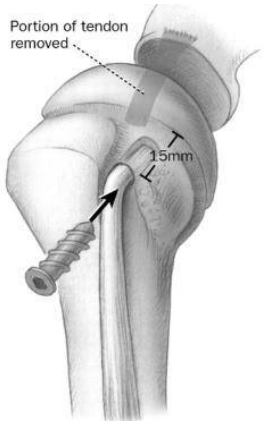
– – Active space – –

Tenodesis level (TO)	Pros	Cons
Top of groove (suprapectoral, arthroscopic)	Safe, good haemostasis, revision possible	**Higher revision** – pathology may persist in tunnel
In groove (zone 2)	Addresses **hidden tunnel pathology**, decompresses sheath	Demanding, **near axillary nerve**
Subpectoral (open)	Easy, **lowest recurrence**, excises diseased sheath	**Humeral stress riser**, stiffness

Philosophy (TO) "**once you go high you can always go low; once you go low you can't go back**" · **preferred site = subpectoral** (TO)

Complications overall complication rates **similar** (OB) · **subpectoral ↑ transient neuropraxia** (OB) – read the options

Groove pain Orthobullets lists tenodesis → possible "**groove pain**"; recent prospective studies → **no difference** supra vs subpectoral



Subpectoral tenodesis

Postoperative function after tenodesis

Rehab full active shoulder ROM; **avoid active supination with elbow 90° + resisted elbow flexion** until healed

Long-term **no change in shoulder or elbow strength**
✗ Wrong options: ↓ elbow flexion · ↓ supination · ↓ shoulder abduction/elevation

6 Complications

Procedure	Complication
Tenotomy	**Popeye deformity** · ? cramping
Tenodesis	Groove pain · **nerve injury** (axillary, musculocutaneous, medial NV) · fixation failure

– – Active space – –

X vs Y

	Speed	Yergason
Elbow	**Extended**	**Flexed 90°**
Forearm	Supinated	Pronated → **resisted supination**
Resisted motion	**Forward elevation**	**Supination**
Also + in	**SLAP**	–

	Tenotomy	Tenodesis
Popeye	↑	↓
Strength / pain / PROMs / ROM	Same	Same
Rehab / return	**Faster**	Slower
Nerve risk	–	Drilling / retraction

How it's asked

"Tenodesis vs tenotomy – possibly lower incidence of?"	→ **arm cramping** ✗ Wrong options: elbow flexion weakness · stiffness · shoulder weakness
"True comparing tenodesis to tenotomy?"	→ **↓ Popeye deformity with tenodesis** ✗ Wrong options: strength, pain, supination, ROM differences
"Primary outcome difference tenodesis vs tenotomy?"	→ **cosmetic deformity** ✗ Wrong options: ASES, ROM, strength, VAS
"Photographer worried about strength + deformity (figure D)?"	→ **tenotomy ↑ cosmetic deformity, not inferior in PROMs** ✗ Wrong options: tenodesis superior
"Suprapectoral vs subpectoral – true?"	→ **similar residual anterior shoulder pain** ✗ Wrong options: ROM, cosmesis, nerve, PROMs differences
"Subpectoral differs from suprapectoral in?"	→ **higher rate of nerve injury** ✗ Wrong options: PROMs, strength, revision, ROM
"Arthroscopic suprapectoral vs prior open subpectoral – counselling?"	→ **equivalent patient-reported outcomes**
"Post-op avoid active supination at 90° – long-term function?"	→ **no change in shoulder or elbow strength**
"Rotator interval boundaries EXCEPT"	→ **CHL** (TO) ✗ Wrong options: coracoid, subscapularis, supraspinatus

"True about rotator interval structures?"	→ **intra-articular LHB is extrasynovial** (TO) ✗ Wrong options: cable/crescent swapped · CHL/SGHL extra-articular
"NOT a stabiliser of intra-articular LHB?"	→ **transverse humeral ligament** (TO)
"Tuoheti — most common LHB attachment?"	→ **predominantly posterior** (TO)
"Speed test also positive in?"	→ **SLAP lesion**
"Most common site of proximal LHB rupture?"	→ **top of the intertubercular sulcus** (TO)
"Preferred tenodesis site with lowest recurrence?"	→ **subpectoral** (TO)

– – Active space – –

– – Active space – –

BRACHIAL NEURITIS (PARSONAGE-TURNER SYNDROME)

– Shoulder & Elbow · 8 / 60 –

1 Basics

Epidemiology

Incidence	**1.6–3/100,000/y** (Orthobullets text) · **1–30/100,000/y** (Orthobullets INA vs HNA table); underdiagnosed
Demographics	♂ **>** ♀ (1.5:1 to 11.5:1) · **middle age (4th decade)**, 20–60 y, mean 41 (range 3–81)
Side	unilateral · **bilateral 10–30%** (16% simultaneous)
Synonyms	**neuralgic amyotrophy** (NA) · idiopathic neuralgic amyotrophy (**INA**) · Parsonage-Turner syndrome (**PTS**) · acute brachial neuritis/plexitis · idiopathic brachial plexus neuropathy

Nerves involved

Distribution	any brachial plexus nerve/branch · **outside plexus in 17%** · usually **upper plexus + >1 nerve**
Most common nerves	**long thoracic** (→ **medial scapular winging**) · **suprascapular** · **axillary** · **musculocutaneous** · radial (Orthobullets, OB)
Others	cervical roots, **AIN**, PIN, LABC
Muscles	deltoid, supraspinatus, infraspinatus, **serratus anterior**, biceps



Medial scapular winging (long thoracic nerve) in brachial neuritis

Pathophysiology

Aetiology	unknown; multifactorial → **inflammatory response in brachial plexus**
Autoimmune	↑ blastogenic lymphocyte activity vs plexus tissue · ↑ **complement-fixing antibodies to peripheral nerve myelin**
Biomechanical	**mobile upper trunk** → wear of **blood–nerve barrier** → immune factors reach nerve (strenuous exercise, manual labour)
Operative finding	**hourglass-like constrictions** of nerves, **no external compression**

Risk factors / antecedent events

Trigger	Frequency	Examples
Viral infection (most common)	**25–55%**	EBV, VZV, Coxsackie B, **parvovirus B19**, CMV, mumps, smallpox, HIV; "flu-like illness", URTI
Bacterial infection	–	Leptospira, TB, Yersinia, Salmonella, Borrelia
Immunisation	**15%**	Tetanus, hepatitis B; **influenza vaccine** (OB stem)
Perioperative / peripartum	**14%**	Elective surgery
Strenuous exercise	**8%**	Cross-training, heavy upper-body work
Drugs	–	Abacavir, streptokinase, **heroin**, infliximab, **interferon**
Iatrogenic	–	**Interscalene block**, surgery, LP, irradiation; burns

Viral illness figure — **25–55%** (Orthobullets, OB) · **25–50%** (one OB explanation) – read the options

Genes & lab markers

HNA gene — **autosomal dominant** · **SEPT9 (septin 9)** on **chromosome 17q24** (cytoskeletal protein in glial cells) · ~200 families worldwide

Routine labs — **CBC + ESR usually normal**

Inconsistent labs — ↑ liver enzymes · **+ antiganglioside antibodies** · + ANA

CSF — usually normal; ± mildly ↑ protein, slight pleocytosis, oligoclonal bands

3 Presentation

Diagnosis – acute pain followed by weakness

Phase	Features	Duration
I – Pain	**Sudden, severe, unrelenting shoulder pain** (90%); radiates to proximal arm/neck; **wakes from sleep**; burning; often atraumatic	**Days–weeks** (♂ 45 d vs ♀ 23 d)
II – Painless flaccid paralysis	Weakness as pain subsides: within **24 h (33%) to 4 wk (80%)**; **patchy paresis** (muscles of same nerve unequally affected)	–
III – Recovery	Slow, steady motor return; length ∝ length of pain phase	**6–18 months**

Key exam pattern — **inability to actively abduct** with **full passive motion** · **normal cervical spine exam** (–ve Spurling)

Pain behaviour — **not worsened by motion or palpation** · **not relieved by subacromial lidocaine** (neuropathic)

Motor — **severe weakness of ER + abduction** (SS, IS, deltoid) · hypotonia, **areflexia** (LMN) · **atrophy**, fasciculations · **medial winging** (serratus anterior)

Sensory — changes in **78%**: paraesthesia/hypoaesthesia over deltoid, lateral arm, radial forearm; often unnoticed · sensation may be **normal** (OB stems)

Autonomic (15%) — trophic skin, temperature dysregulation, ↑ sweating, altered nail/hair

– – Active space – –

Upper trunk pattern pain → weakness + atrophy + scapular winging after viral illness = **upper trunk neuralgic amyotrophy**
✗ Wrong options: isolated LTN palsy (doesn't explain whole picture)

4 Diagnosis & Classification

Diagnosis **clinical**; often a **diagnosis of exclusion**; tests rule out mimics; **EMG confirms**

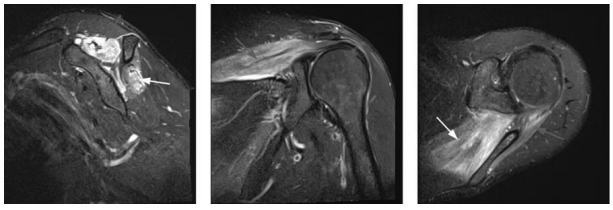
Key radiology findings

X-ray shoulder / C-spine **normal**; rule out **calcific tendinitis** (also severe incapacitating pain)

Chest X-ray baseline to exclude compressive **Pancoast tumour** (OB)

MRI timing	Sequence	Finding
Early	**T2 / STIR**	**Diffuse ↑ signal (oedema)** in muscles of plexus (SS, IS, deltoid, teres minor most common)
Late	**T1**	**Fatty infiltration + atrophy**

MRI also excludes cuff tear (intact cuff) · SSN notch/spinoglenoid **cyst** · brain MS plaques · cervical disc



T2 MRI – diffuse muscle oedema in brachial neuritis

EMG findings

EMG compatible findings in **96%**; confirms diagnosis + **monitors recovery** (serial)

Timing	EMG
Early – **3–4 wk** (Orthobullets) · **first 3 wk** (OB)	**Acute denervation**: **positive sharp waves + fibrillation potentials** in **both peripheral nerve AND nerve root distributions**
Late – **3–4 months** (Orthobullets) · **as early as 6 wk** (OB)	Chronic denervation + **early reinnervation**: **polyphasic MUPs** (= **resolving** PTS)

Sensory NCS less useful than EMG · **↓ amplitude (axonal loss)**, **preserved conduction velocity + distal latency** · may be normal (doesn't exclude)

✗ EMG distractor	Diagnosis
Normal EMG	PTS shows denervation
Fibrillations from **spinoglenoid notch** compression	SSN compression (cyst on MRI)
Sharp waves + fibs **deltoid + biceps only**	**C5 root** compression
Polyphasic MUPs, early reinnervation	**Resolving** PTS (not early)

Classification – INA vs HNA

Factor	INA (idiopathic)	HNA (hereditary)
Incidence	1–30/100,000/y	Very rare (~200 families)
Gene	–	**Septin 9 (chr 17q24)**, **AD**
Age at onset	Middle age (20–60)	**Young** (childhood/adolescence; mean 28)
Recurrence	**Rare** (~1.5 episodes)	**Frequent** (3.5 episodes)
Appearance	Normal	**Dysmorphic**: short stature, **hypotelorism**, cleft palate, facial asymmetry, skin folds
Nerves outside plexus	Uncommon (**17%**)	Common (**56%**): **lower cranial nerves** (VII, IX, X, XI, XII), lumbosacral plexus, phrenic, recurrent laryngeal
Severity / outcome	Better	More severe paresis, worse outcome

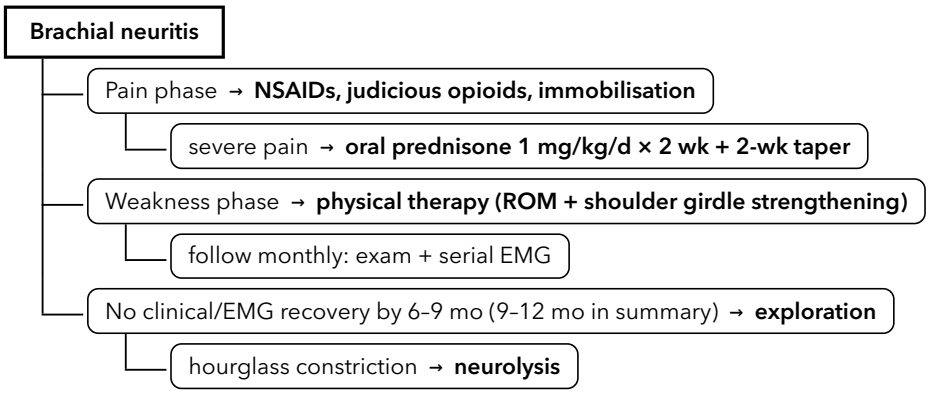
Differential diagnosis

Mimic	Distinguishing feature
Cervical radiculopathy (C5/C6–7, disc)	Single root dermatome/myotome; pain from neck; **↑ with movement**; abnormal neck exam/Spurling (normal neck exam here makes it unlikely)
Rotator cuff tear / impingement	Pain persists with weakness; impingement signs; **relieved by subacromial lidocaine**; MRI shows tear; does not explain winging
Calcific tendinitis	Calcification on X-ray; no acute severe weakness
Adhesive capsulitis (frozen shoulder)	**Loss of passive + active** motion
Entrapment neuropathy (SSN)	**Single nerve** distribution on EMG (SS + IS only)
Poliomyelitis	No (excluded by acute severe weakness pattern, OB)
ALS	**UMN + LMN** signs, **painless, no sensory loss**, cranial nerve signs
Idiopathic hypertrophic brachial neuritis	**Painless**; plexus hypertrophy on MRI; **demyelinating** NCS (slow velocity, prolonged latency)
Multiple sclerosis	Brain MRI demyelination

Exclusion logic (OB) loss of **active** motion excludes radiculopathy + impingement · **normal MRI** excludes cuff tear · **full passive motion** excludes adhesive capsulitis · normal C-spine exam excludes disc disease

– – Active space – –

5 Management



Nonoperative

Mainstay	**observation + pain control**; self-limiting
Corticosteroids	early, severe pain; **2 wk prednisone 1 mg/kg/d + 2 wk taper** · faster pain resolution, **no change in progression/prognosis** · OB: may shorten symptoms/accelerate recovery
Physical therapy	once pain abated; ROM + strengthening; reduce traction on nerves · reverses atrophy · **observation alone = observation + PT at 1 y**
Steroid window (TO)	**high-dose corticosteroids if diagnosed early (within 72 h)**, rarely possible · some spontaneous recovery usual, **full recovery not assured**

Operative criteria

Indication	Threshold
Exploration / neurolysis / graft / transfer	**No regeneration on exam + EMG by 6–9 months** (Orthobullets treatment) · **9–12 months** (Orthobullets summary) – read the options

Procedure	Note
Neurolysis	**Superior** to neurorrhaphy + grafting; LTN microneurolysis, ulnar transposition, Guyon/radial tunnel/carpal tunnel release
Neurorrhaphy / grafting	Excise diseased segment
Nerve transfer	–
Split pectoralis major transfer	**Serratus anterior paralysis** (winging)

✗ Wrong options: decompression / arthroscopic repair	not indicated (no compressive lesion, cuff intact)

Prognosis

Overall **good**; **gradual return of strength without long-term functional deficit**; slow · residual weakness may last months–years

Time	Recovery
1 month	**66%** some motor recovery
1 year	**35%** recovered (excellent 36%)
2 years	Excellent **75%**
3 years	**~90%** full strength, no pain (89%)
Up to 8 years	Full strength in some

Poor prognosis **female** · **lower trunk** involvement (upper trunk = best) · **persistent pain + no motor recovery by 3 months** · **hereditary** cases

No effect **age**

Recurrence **rare** in idiopathic (common in HNA)

X vs Y

Prognosis statement (OB distractor)	Disease
Gradual return of strength, no long-term deficit	✓ **PTS**
Decompression improves strength	**SSN notch cyst**
Immunomodulators ↓ relapses	**Multiple sclerosis**
Arthroscopic repair best for active patient	**Rotator cuff tear**
Progressive weakness → **respiratory failure**	**ALS** (pure motor; no ↓ sensory amplitudes)

	Brachial neuritis	Cervical radiculopathy	Rotator cuff tear
Pain onset	Acute severe → resolves before weakness	Neck → arm	Persists with weakness
Pain with motion	**No**	**Yes**	Yes
Subacromial lidocaine	No relief	–	Relief
Distribution	**Multiple roots + peripheral nerves**, patchy	One root	Cuff only, no winging
EMG	**Root + peripheral nerve** denervation	Root	Normal

How it's asked

"Acute atraumatic burning shoulder pain, can't abduct, full passive ROM, normal C-spine + MRI – diagnosis?"

→ **brachial neuritis**
✗ Wrong options: C6–7 radiculopathy · impingement · cuff tear · adhesive capsulitis

– – Active space – –

"Flu-like illness 1 wk earlier, then burning pain → weakness, normal X-rays?"	→ **brachial neuritis** ✗ Wrong options: calcific tendinitis · poliomyelitis · cervical disc disease · impingement
"URTI, 3 mo pain + weakness, atrophy, medial winging, flaccid — diagnosis?"	→ **upper trunk neuralgic amyotrophy** ✗ Wrong options: isolated LTN palsy · ALS · frozen shoulder · cuff tear
"Post-influenza-vaccine pain → weakness, T2 oedema SS/IS — expected EMG?"	→ **acute denervation in both peripheral nerve + nerve root distributions with sharp waves + fibrillations**
"EMG showing polyphasic MUPs / early reinnervation?"	→ **resolving PTS** (late)
"Sharp waves + fibs in deltoid + biceps only?"	→ **C5 root compression**
"Cross-training athlete, viral illness, pain → weakness, ↓ sensory amplitudes, normal brain/C-spine MRI — prognosis?"	→ **gradual return of strength without long-term deficits** ✗ Wrong options: decompression · immunomodulators · arthroscopic repair · respiratory failure
"Most common antecedent event?"	→ **viral illness (25–55%)**
"Most commonly involved nerve / cause of winging?"	→ **long thoracic nerve**
"Gene in hereditary neuralgic amyotrophy?"	→ **SEPT9, chromosome 17q24, AD**
"Poor prognostic factors?"	→ **female, lower trunk, no recovery by 3 months, hereditary** ✗ Wrong options: age
"Indication for surgery?"	→ **no EMG/clinical recovery by 6–9 (9–12) months → neurolysis**

CALCIFIC TENDONITIS

– Shoulder & Elbow · 9 / 60 –

1 Basics

Epidemiology

Definition	calcification + tendon degeneration **near rotator cuff insertion** → pain + ↓ ROM; **unknown etiology**
Age	**30–60 y**
Sex	♀ > ♂
Location	**supraspinatus** most common · supraspinatus and/or **infraspinatus** (OB)
Risk factors	endocrine: **diabetes**, **hypothyroidism**

Genes & lab markers

Crystal	**calcium hydroxyapatite** (Orthobullets, OB) · "calcium carbonate apatite" (one OB explanation) – read the options
Histology	hydroxyapatite deposits surrounded by **fibroblasts, macrophages, giant cells, lymphocytes**
Molecular pathways	**osteopontin** · **cathepsin K** · **transglutaminase 2**
✗ Wrong option: calcium pyrophosphate	**positively birefringent rhomboid/rod** crystals → CPPD / **pseudogout**
✗ Wrong option: calcium oxalate	**kidney stones**, urine; excess in **ethylene glycol (antifreeze)** poisoning → renal failure
Process	**active, cell-mediated** ✗ Wrong options: purely degenerative; Uhthoff & Loehr

Phases of disease (resorptive phase pain)

Stage (Uhthoff)	Phase	Pathology	Pain
Precalcific	–	**Fibrocartilaginous metaplasia**; **tenocytes → chondrocytes**	**Pain-free**
Calcific	**Formative**	**Cell-mediated** calcific deposits	±
	Resting	**No inflammation / vascular infiltration**	±
	Resorptive	**Phagocytic resorption + vascular infiltration**	**Most painful**
Postcalcific	–	Calcium resorbed + **tissue remodelling**	–

✗ Resorptive distractors	"vascular infiltration, painless" (it is painful) · lowest inflammation (= resting) · cell-mediated deposits (= formative) · fibrocartilaginous metaplasia (= precalcific)

– – Active space – –

Fibrocartilaginous metaplasia	only in **calcific tendonitis** among shoulder pathologies ✗ Wrong options: cuff arthropathy · biceps tenosynovitis · chronic cuff tear · neuropathic arthropathy
Surgery timing	avoid surgery in **resorptive** phase (Uhthoff & Loehr) · surgery for refractory cases in **formative** phase

3 Presentation

History	similar to **subacromial impingement**
Symptoms	**atraumatic pain** (worst in resorptive phase) · **catching**, crepitus · mechanical block · pain with overhead activity / active elevation
Exam	supraspinatus fossa **atrophy** · **↓ active ROM** · scapular dyskinesia · ± ↓ cuff strength · positive **impingement signs**

4 Diagnosis & Classification

Key radiology findings

Modality	Fact
Radiographs	**Gold standard / mainstay**; monitor progression; location, density, extent, delineation
AP, supraspinatus outlet, axillary	**Supraspinatus** calcification
Internal rotation view	**Infraspinatus + teres minor**
External rotation view	**Subscapularis**
Deposit site	**1–1.5 cm** from supraspinatus insertion (Orthobullets) · **1–2 cm**, zone of ↓ vascularity, can be multifocal (OB) — read the options
CT	Rarely required; 3D anatomy
MRI	**Low signal on all sequences** (globular, T1 + STIR/T2 hypointense); limited diagnostic use; refractory pain → concomitant cuff tear
Ultrasound	Deposits **hyperechoic**; quantify extent; **guides needle lavage**
MR arthrography alone	Poor sensitivity + inter-rater agreement (Zubler)

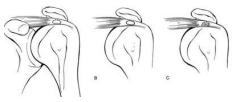


Calcific deposit over supraspinatus insertion

✗ Wrong options (next tests after failed barbotage) bone scan · MR arthrogram · EMG (no added value)

Imaging ✗ distractors no acromial spur (✗ Wrong options: impingement) · no tear (✗ Wrong options: partial/full-thickness tear) · **adhesive capsulitis = clinical diagnosis**, no consistent X-ray findings

Gartner & Heyer classification

Type	Appearance	Phase	Image
I	**Well circumscribed, dense**	**Formative**	
II	Soft contour/dense **or** sharp/transparent	–	
III	**Translucent, cloudy**, no clear circumscription	**Resorptive**	

Radiographic types (one OB explanation) Type I = **fluffy, fleecy, poorly defined** → acute pain, formative phase · Type II = **discrete homogeneous, well defined** → subacute/chronic – conflicts with Gartner (Orthobullets: type I well circumscribed) – read the options

Mole et al. classification

Type	Appearance
A	Dense, **homogeneous**, sharp contours
B	Dense, **segmented**, sharp contours
C	**Heterogeneous, soft** contours
D	**Dystrophic** calcification at cuff **insertion**

5 Management

Calcific tendonitis

- All phases, first line → **NSAIDs + PT (stretch/strengthen, moist heat) ± steroid injection**
- Refractory, formative/resting phase → **adjunct ESWT**
- Persistent, resorptive phase → **US-guided needle lavage / barbotage**
- Progressive, refractory, ↓ ADLs → **arthroscopic / mini-open debridement of deposit**
  - remaining cuff defect → **± rotator cuff repair (controversial)**

Treatment – conservative, barbotage & arthroscopic debridement

Nonop outcome symptoms resolve **60–70% at 6 months**; most improve spontaneously

Failure more likely **bilateral or large** deposits · deposit under **anterior third of acromion** · deposit **medial to acromion**

Steroid injection common but **controversial**; variable relief

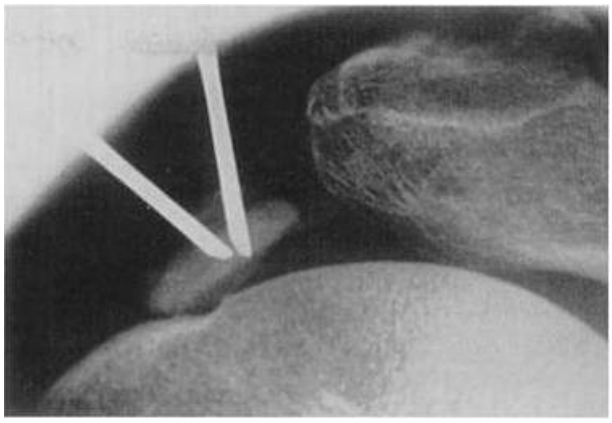
– – Active space – –

Formative phase, 4-month mild pain
moist heat + stretching first
✗ Wrong options: aspiration (deposit **granular** → hard to aspirate) · ESWT (role poorly defined, OB) · **low-dose radiation** (malignant transformation risk) · arthroscopic debridement (only after failed nonop)

ESWT
adjunct; refractory **formative + resting** phases; **dose-dependent**: high-energy > low-energy for outcome scores + resorption, but also > procedural pain + local reaction (ecchymosis)

Needle lavage
2 needles (inflow saline ± anaesthetic + aspiration outflow); indicated for **resorptive phase**; better outcomes **Gartner II/III vs I**

Barbotage
needle breaks up deposit → **corticosteroid injection**

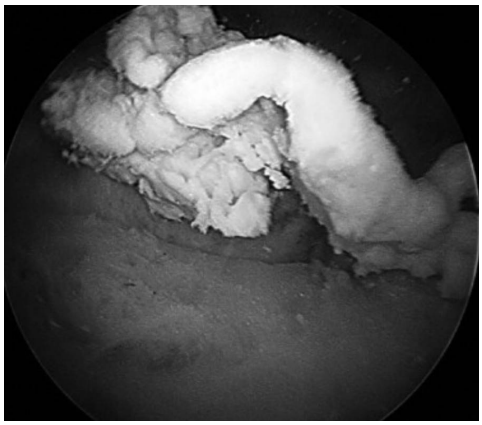


US-guided two-needle lavage

Operative criteria

Indication for surgical decompression of deposit
Progression of symptoms
Refractory to nonoperative treatment (incl. failed barbotage)
Interference with ADLs / work

Surgery
arthroscopic or mini-open removal (shaver or spinal needle) ± SAD ± RCR; **paste-like ("toothpaste") calcium** expressed from cuff



Arthroscopic paste-like calcium expressed from cuff

After failed barbotage
arthroscopic debridement
✗ Wrong options: isolated RCR without debridement (won't resolve symptoms)

RCR after debridement — **controversial**; depends on remaining tissue integrity + defect size · debridement without RCR safe: **3.7%** cuff tears at 7 y (El Shewy)

Deposit with osseous involvement — same results as tendon-only (Seyahi)

Debridement ± subacromial decompression

SAD + debridement vs debridement alone	Result
Time to pain-free activity	**↑ (longer)** with SAD (Marder)
Pain at midterm	No difference
ROM	No difference
Recurrence	No difference
Cuff tears	**↑ in operated shoulder vs contralateral** regardless (Balke, 6 y)

Debridement + SAD — 91% good/satisfactory at 26 mo (Ark); SAD gives **no apparent benefit**

Return to work — longer with SAD and/or RCR

6 Complications

List — **recurrence** · **persistent pain** · **stiffness** (operative risk) · **iatrogenic cuff injury** · ↑ cuff tears after surgery

X vs Y

	Formative phase	Resting phase	Resorptive phase
Process	Cell-mediated deposition	No inflammation / vascular infiltration	Phagocytosis + vascular infiltration
Pain	±	±	**Most severe**
Deposit (Gartner)	**Type I** dense, well circumscribed	–	**Type III** cloudy, translucent
Treatment niche	**ESWT**; conservative; aspiration difficult (granular)	**ESWT**	**Needle lavage**; avoid surgery

	Hydroxyapatite	CPP	Calcium oxalate
Disease	**Calcific tendonitis**	**Pseudogout / CPPD**	Kidney stones, ethylene glycol
Crystal	Deposits ringed by fibroblasts, macrophages, giant cells	**Positively birefringent rhomboid/rod**	Urine crystals
Colchicine	**no role**	–	–

– – Active space – –

How it's asked

"Calcific deposit on X-ray — structure abnormal at arthroscopy?"	→ **supraspinatus** ✗ Wrong options: infraspinatus · labrum · biceps · GH cartilage
"35-y ♀, insidious pain, low-signal globule in supraspinatus on MRI?"	→ **calcific tendinitis** ✗ Wrong options: impingement · adhesive capsulitis · cuff tear
"Fibrocartilaginous metaplasia — which image?"	→ **AP X-ray with supraspinatus calcification** ✗ Wrong options: cuff arthropathy · biceps tenosynovitis · neuropathic arthropathy
"Best description of resorptive phase?"	→ **highest pain levels** ✗ Wrong options: painless vascular infiltration
"Formative-phase deposit, mild 4-month pain — initial treatment?"	→ **moist heat + stretching** ✗ Wrong options: aspiration · ESWT · radiation · arthroscopy
"Debrided material pathology + next step?"	→ **calcium hydroxyapatite; debridement ± cuff repair** ✗ Wrong options: CPP · oxalate · IV colchicine
"Failed PT, NSAIDs and US barbotage — next?"	→ **arthroscopic debridement** ✗ Wrong options: isolated RCR · bone scan · MR arthrogram · EMG
"Adding SAD to debridement — effect?"	→ **longer time to pain-free activity** ✗ Wrong options: better pain · ROM · fewer tears · ↓ recurrence
"Most painful phase?"	→ **resorptive**
"View for subscapularis calcification?"	→ **external rotation AP**

DISTAL BICEPS AVULSION

– Shoulder & Elbow · 10 / 60 –

1 Basics

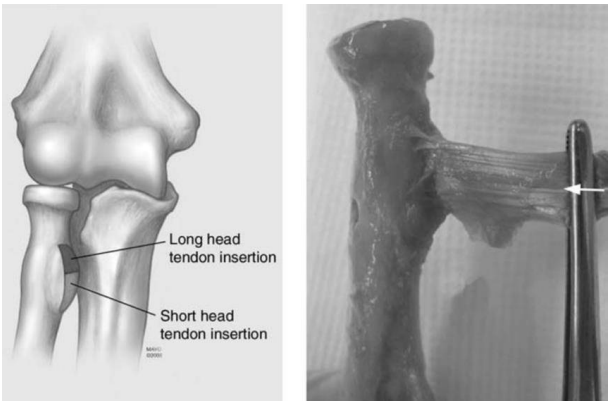
Epidemiology

Frequency	rare; **~10% of all biceps ruptures**
Demographics	**dominant elbow (86%)** of **men (93%)** in their **40s** (30–50 y)
Risk factors	**anabolic steroids** · **smoking (7.5× risk)** · hypovascularity · intrinsic degeneration · mechanical impingement in space available for tendon
Patterns	complete avulsion · **partial** (on **radial side** of tuberosity footprint) · **intersubstance muscle transection** (rope wrapped round arm, **tug-of-war**)
Rare association	**median nerve compression**

Footprint anatomy

Insertion **radial (bicipital) tuberosity**; innervation **musculocutaneous only**
✗ Wrong options: dual musculocutaneous + radial

	Short head	Long head
Origin	**Coracoid**	Supraglenoid tubercle / superior labrum
Footprint	**Distal + anterior** (thin sliver)	**Proximal + posterior** (oval; **apex** of tuberosity)
Better at	**Elbow flexion**	**Supination** (furthest from axis of rotation)



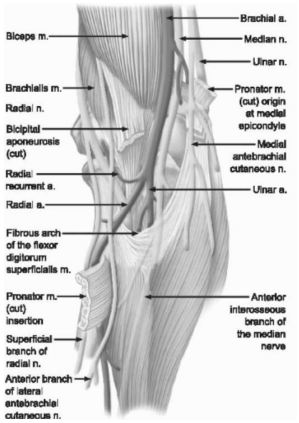
Long head inserts proximal, short head distal on radial tuberosity

Biceps function	**primary supinator**; lesser elbow flexor · **brachialis = primary elbow flexor** (inserts **coronoid + ulnar tuberosity**, not radial tuberosity)
Annular ligament	anterior radial head, PRUJ stability; does not attach to radial tuberosity
Lacertus fibrosus (bicipital aponeurosis)	from **medial side of distal short head**, distal to elbow crease → obliquely distal + medial across cubital fossa → protects **brachial artery + median nerve** → continuous with flexor-origin fascia; **may be mistaken for intact tendon**; intact lacertus → ↓ deficit in low-demand patient

– – Active space – –

Antecubital fossa

Medial → lateral	**median nerve** (most medial) · **brachial artery** · **biceps tendon** · **radial nerve** (most lateral)
Borders	**pronator teres** ulnarly · **brachioradialis** radially
Radial recurrent artery	**superficial** to biceps tendon, crosses near tuberosity
Superficial radial nerve	**lateral (radial) + superficial** to biceps tendon; radial nerve bifurcates (PIN + SRN) **at level of antecubital fossa / radiocapitellar joint**
LABCN	terminal sensory branch of **musculocutaneous (lateral cord)** → exits **between biceps + brachialis** lateral to musculotendinous junction → pierces fascia, **lateral to biceps tendon, deep to cephalic vein** → superficial along **brachioradialis** → volar-lateral forearm sensation



Antecubital fossa: radial recurrent artery, SRN, LABCN

Nerve course (✗ Wrong options: distractors for LABCN)	Nerve
Branches 5–8 cm distal to lateral epicondyle, **between 2 heads of PT**, along volar FDP → pronator quadratus	**AIN**
From radial n. at radiocapitellar joint, **through supinator (arcade of Frohse)**, around **radial neck** → dorsal wrist capsule	**PIN**
Deep to brachioradialis, lateral to radial artery, pierces fascia **7–9 cm proximal to wrist** → snuffbox, dorsoradial hand	**SRN**
With brachial artery between **PT + biceps tendon**, between **FDS + FDP**, emerges between FDS + FPL → carpal tunnel	**Median**

Radial nerve motor order (Abrams)	BR → ECRL → supinator → ECRB → EDC → ECU → EDQ → APL → EPL → EPB → **EIP last**

2 Mechanism

Mechanism – eccentric contraction

Eccentric contraction	**muscle lengthens during contraction**; flexed elbow forced into extension ("**flexed elbow unacceptably challenged**")
Contributors	**vascular watershed** · mechanical attrition (abrasion during pronosupination)

Partial tear mechanism pronation → **sharp radial margin of tuberosity impinges tendon** → released in supination; rupture through pre-existing degeneration on **radial aspect**

✗ Wrong option: contraction type	Definition
Isometric	Constant **length**
Isokinetic	Constant **speed**
Isotonic	Constant **force**
Concentric	Muscle **shortens** (biceps curl)

3 Presentation

History painful **"pop"** as elbow eccentrically loaded (catching falling object, lifting furniture/engine, deadlift, curls)

Hallmark **weakness + pain in supination** > flexion

Inspection **"reverse Popeye"** (proximal retraction of muscle belly), contour change, **antecubital / medial ecchymosis**, **palpable defect**

Complete vs partial tendon **absent** in complete · **palpable but painful** in partial (otherwise similar picture)

Diagnosis – hook test, MRI, partial tears

Hook test (O'Driscoll) (TO) patient looks at palm, **shoulder elevated, elbow 90°, forearm supinated** → intact tendon = cord-like structure finger hooks around; torn → **brachialis is flat** → nothing to hook

Hook technique index finger hooks **lateral edge** of tendon, **lateral → medial**; intact/partial → finger 1 cm beneath tendon

Accuracy **sensitivity + specificity 100%** → **most sensitive exam** for complete rupture

False "normal" hook **partial tear** · **intact lacertus fibrosus** (hooking **medial → lateral** catches lacertus) · underlying **brachialis**

Partial tears hook test **normal but painful**; hook test not useful for partial tears



Hook test: finger hooks lateral edge of tendon

Ruland biceps squeeze test elbow **60–80° flexion**, forearm slightly **pronated**, squeeze biceps belly → **no supination = positive** (like Thompson); **sensitivity 96%**

Other exam (OB) **biceps crease interval test**, **passive forearm pronation** test

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✗ Wrong option: milking manoeuvre	tests **UCL**, not biceps

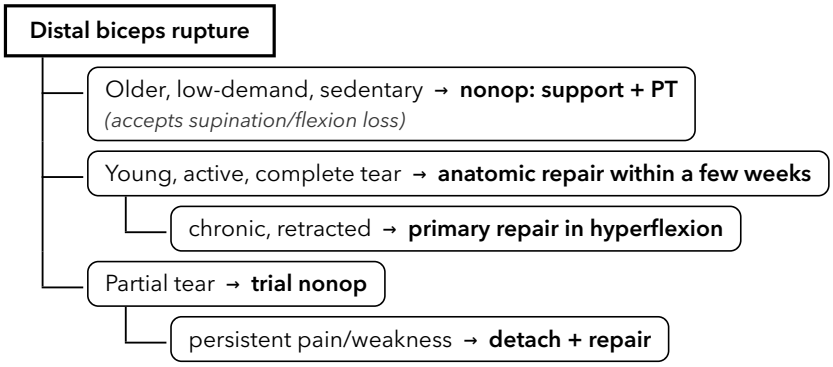
4 Diagnosis & Classification

Key radiology findings

X-ray	usually **normal**; occasionally **fleck/avulsion** from radial tuberosity
MRI	distinguishes **complete vs partial**, muscle vs tendon, **degree of retraction**; **FABS position** (elbow **flexion**, shoulder **abduction**, forearm **supination**) ↑ sensitivity
MRI when?	**needed for partial tear** (palpable but painful tendon) · not always needed for complete tear with conclusive exam
Partial tear MRI	↑ signal + ↑ **diameter** of tendon · **avulsive marrow oedema** in tuberosity · **bicipitoradial bursa distension**
Ultrasound	can confirm diagnosis in office

Clinical finding	Next step
Abnormal hook, retraction, ecchymosis	Clinical diagnosis → repair
Palpable tendon, painful hook	**MRI** (✗ Wrong options: CT · immobilise 3 wk · exploration · early PT)
Partial tear, failed ≥6 mo nonop	**Detachment + repair** of biceps tendon

5 Management



Nonoperative outcome – loss of supination strength

Greatest loss **forearm supination**
✗ Wrong options: elbow flexion (brachialis compensates; flexion returns near normal)

Nonop deficit	Orthobullets	OB explanations
Sustained supination (endurance)	**50%**	50%
Supination strength	**40%**	40% (up to 40%)
Flexion strength	**30%**	**30%** · **10%** (one OB) · **~40% sustained flexion** (one OB)
Grip	15%	–

Flexion loss value **30%** (Orthobullets, most OB) · **10%** (one OB) · **40% sustained** (one OB) – read the options

Also with nonop **delayed return to activity**
✗ Wrong options: chronic elbow pain · instability (terrible triad / anteromedial coronoid) · lateral pain on resisted wrist extension (lateral epicondylitis)

Activity hardest **using a screwdriver** (supination)

✗ Wrong option: activity affected	Injury
Pushing up from chair	**LUCL** tear
Bench press	**Pectoralis major** tear
Wrist extension	ECRB/ECRL (common extensor)
Elbow extension	**Triceps** tear

Nonop EXCEPT **LABCN neuritis** (exclusively surgical)

Nonop indications **older, low-demand, sedentary**, willing to sacrifice function · not healthy middle-aged with dominant-arm tear

Operative criteria

Indication	Detail
Young, healthy, active	Don't want to sacrifice function (**main aim = restore supination strength**)
Partial tear failing nonop	Persistent pain + weakness
Subacute / chronic	**Direct repair** (no allograft) usually possible; may need **hyperflexion**
Timing	**Within a few weeks**; delay → harder primary repair, **more extensile approach**

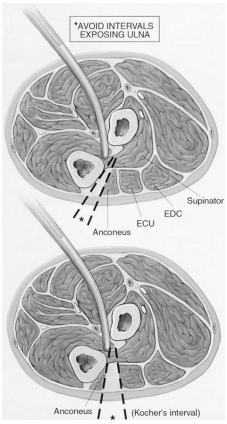
Not surgery indications cosmesis · preventing elbow degeneration

Repair outcome anatomic repairs **high success**; **loss of terminal extension** common after repair
✗ Wrong options: high re-rupture

– – Active space – –

Surgical anatomy – BR/PT interval & recurrent radial artery

	Anterior single-incision	Dual-incision (Boyd-Anderson)
Rationale	Developed to ↓ **HO + synostosis** of 2-incision	Developed to avoid **radial nerve / PIN** injury
Interval	**Brachioradialis (radial n.) / pronator teres (median n.)** – not PT/FCR	Same anterior + **posterolateral: ECU / EDC**
✗ Wrong interval	–	**Kocher (ECU/anconeus)**, **anconeus/ulna** → expose ulna → HO/synostosis
Forearm	**Maximal supination** protects PIN (✗ Wrong options: pronation)	Hemostat through IO space along **medial border of tuberosity** → pierces anconeus → tents skin
LABCN injury	**Higher**	Lower
HO / synostosis	Lower (↓ periosteal stripping) – **not eliminated**	**Higher**
Repair	–	**More anatomic** (better footprint exposure; ulnar placement)
Supination strength	Similar	Similar or slightly better



Avoid intervals exposing the ulna (ECU/anconeus, Kocher)

Single-incision steps limited antecubital incision → BR lateral, PT medial → **identify LABCN** (exits between biceps + brachialis) and retract laterally → **recurrent radial vessels** coagulated/ligated or retracted → protect PIN
✗ Wrong options: forceful lateral retraction, keep supinated

Recurrent radial artery **crosses field immediately deep to BR/PT interval, near tuberosity**; large, may bleed profusely → **ligate** if cannot retract → "flash of bright red blood" with deep retractor

✗ Artery distractor	Location
Anterior interosseous a.	From ulnar a., along **IOM**
Anterior recurrent ulnar a.	From ulnar a., proximal + superficial to brachialis insertion
Recurrent interosseous a.	From posterior interosseous a., **posterior forearm**
Inferior ulnar collateral a.	From brachial a., **medial elbow**
Ulnar a.	Far medial, between **FDS + FDP**

Approach vs effectiveness single vs two-incision: **no proven difference** in effectiveness

Post-op immobilise **110° flexion + moderate supination**

Fixation techniques

Technique	Approach	Key step
Bone tunnel	2-incision	Acorn reamer slot; **2–3 × 2-mm holes 1 cm apart** through far (lateral) radius; sutures tied over bone bridge
Suture anchors	Single	2 anchors; **distal anchor tied first** (brings tendon out to length) → maximises contact
Intraosseous (interference) screw	Single	Whip-stitch sutures through bioabsorbable tenodesis screw
Suspensory cortical button	Single	Ream anterior cortex, smaller hole far cortex, **flip button** on far cortex, tension-slide tendon in

Load	Force
Elbow 90°, no load	**50 N** (repair must withstand)
Elbow 90°, 1 kg load	112 N
Tendon rupture	200 N

Strength order **suture button (400 N) > suture anchor (380) > bone tunnel (310) > interference screw (230)**

Strongest **button + interference screw combination** · **weakest** – **interference screw alone**
✗ Wrong options: bone tunnels weakest · button alone strongest

Repair technique – anatomic (ulnar) footprint & chronic retracted tears

Insertion point repair to **ulnar aspect of tuberosity → ↑ supination lever arm (windlass effect)**; not more distal (off tuberosity); no flexion advantage

Chronic retracted (e.g. 5 wk, 6 cm) **primary fixation in hyperflexion is well tolerated**: **no flexion contracture / ROM loss**

Allograft augmentation only if tendon **cannot reach tuberosity even in hyperflexion**; less consistent results; not "necessary" for retraction

Chronic traps ✗ Wrong options: two-incision required (may need extensile approach) · add interference screw to button if tendon reaches · nonop equivalent

6 Complications

LABCN injury – most common complication (single-incision)

LABCN neuropraxia **most common complication overall** and **most common nerve** injured, either approach; **more common with single-incision**

Cause **aggressive / prolonged retraction**; not related to fixation type (anchor/screw/button) or timing
✗ Wrong options: direct laceration

Deficit numbness **volar-lateral (radial) forearm**

– – Active space – –

Origin — **lateral cord → musculocutaneous → LABCN** (labelled musculocutaneous on plexus diagrams)

LABCN incidence	Source
9%	Orthobullets, OB
5–7%	OB (Cohen)
Nearly one-quarter	OB

Resolution — **3–6 mo** (Orthobullets, most OB) · **by 3 mo** (one OB) · by 6 mo, **up to 2 y** (one OB) – read the options; **observe → gradual resolution**; permanent deficit/neuroma rare (neuroma excision + tension-free repair, traction resection, transposition)

PIN injury – two-incision technique & forearm positioning

PIN — **most commonly injured motor nerve** (**1–2%**; 5% Cohen); resolves 3–6 mo

Mechanism — **radially based retractors** around radial neck · prolonged traction · **Beath pin out dorsal forearm** · **entrapment by cortical button** on posterior radius

Protect PIN — **maximal supination** + limit forceful lateral retraction

PIN palsy — **weak wrist extension with radial deviation** (ECRL intact – radial n. proximal; ECU out – PIN) · weak thumb / **index finger extension**

PIN risk by approach — **single-incision higher** (Orthobullets, OB) · **two-incision higher** (other OB explanations) · **similar** (recent meta-analysis, OB) – read the options

Complication rates (OB)

Minor (~20%)	Rate	Major (~4%)	Rate
LABCN neuropraxia	**9%**	**PIN neuropraxia**	**1–2%**
HO	3–4%	Re-rupture	1–2%
SRN palsy	**2–3%**	Deep infection	1%
Superficial infection	1%	**Radioulnar synostosis**	1%
Stiffness	1%	Ulnar / AIN injury	**<0.1%**

Nerve order — LABCN > SRN > PIN > median > AIN (OB)

2nd most common nerve — **SRN** = 2nd most common **cutaneous/sensory** (Orthobullets, OB) · **PIN** = 2nd most common nerve (one OB) – read the options

SRN injury — numbness **dorsoradial hand / dorsal thumb**; lateral + superficial to tendon

Re-rupture — **<4–5%** (OB) · **1–2%** (OB) → weak supination > flexion

HO — when **IOM + ulnar periosteum disrupted**; risk of **2-incision**; often asymptomatic; some advise prophylaxis

Synostosis — loss of pronation/supination; avoid exposing **ulnar periosteum** and dissection between radius + ulna; **more with 2-incision**; ↓ with intermuscular (ECU/EDC) interval

Proximal radius fracture — large tunnels or tunnels **proximal to tuberosity**

Suture rupture — bone tunnel method

Expanding hematoma brachial artery laceration / radial recurrent bleeding — very uncommon

– – Active space – –

Post-op deficit	Nerve
Lateral volar forearm numbness	**LABCN**
Dorsal thumb / dorsoradial hand numbness	**SRN**
Weak wrist / thumb / index extension	**PIN**
Medial volar forearm numbness	MABCN
Can't flex middle finger	Median
Can't flex thumb IP	AIN
Weak finger abduction/adduction	Ulnar

X vs Y

Proximal long-head biceps rupture

	Proximal LHB rupture	Distal biceps rupture
Deformity	**Popeye** deformity	**Reverse Popeye** (belly retracts proximally)
Hook test	–	**Abnormal** (no cord)
Main deficit	–	**Supination**

✗ Photo distractors brachial plexus palsy · pectoralis major tear · normal biceps

	Complete tear	Partial tear
Hook test	Abnormal	**Normal but painful**
Tendon	Absent	Palpable
Location	Off tuberosity	**Radial side** of footprint
MRI	Not always needed	**Needed**
Treatment	Repair (active)	Trial nonop → repair if fails

	LABCN	PIN
Type	Sensory (most common overall)	**Motor** (most common motor)
Approach	**Single-incision**	Disputed (see above)
Protection	Gentle retraction, retract laterally	**Supination**, no radial retractors

How it's asked

"Nonop distal biceps rupture – greatest loss?"	→ **forearm supination** ✗ Wrong options: elbow flexion (brachialis)
"Main indication for repair in a 45-year-old coach?"	→ **restore supination strength** ✗ Wrong options: cosmesis · prevent degeneration

– – Active space – –

"Nonop management results in all EXCEPT"	→ **lateral antebrachial neuritis**
"Partial tear MRI, nonop — outcome?"	→ **decreased supination strength** ✗ Wrong options: instability · chronic pain
"Activity hardest after nonop?"	→ **using a screwdriver** ✗ Wrong options: bench press = pec · chair push-up = LUCL
"Hook test diagnoses?"	→ **biceps tear** (TO)
"Partial tear, normal hook, failed 6 mo nonop — next?"	→ **detach + repair biceps** ✗ Wrong options: radial tunnel release · EMG
"Palpable tendon, painful hook — next step?"	→ **MRI scan** ✗ Wrong options: CT · immobilise + re-examine
"Partial tears typically occur in which part of the footprint?"	→ **radial** side
"True of the injured structure (MRI distal biceps)?"	→ **inability to hook a cord-like structure is most sensitive** ✗ Wrong options: LH distal · dual innervation · primary flexor · high re-rupture
"Orange area: distal + anterior on tuberosity?"	→ **short head of biceps** ✗ Wrong options: long head = proximal/posterior · lacertus = medial
"Healthy 44-year-old, positive hook, weak supination — management?"	→ **distal biceps repair** ✗ Wrong options: sling + PT · allograft · UCL reconstruction
"Contraction causing rupture?"	→ **muscle lengthens during contraction** (eccentric)
"Nerve injured in superficial dissection, single anterior incision?"	→ **LABCN**
"Most likely deficit after two-incision repair?"	→ **numbness lateral volar forearm** (LABCN)
"Most common complication after button repair?"	→ **LABCN irritation** ✗ Wrong options: re-rupture · synostosis · PIN
"True about surgical treatment of distal biceps rupture?"	→ **most common complication = injury to terminal branch of musculocutaneous nerve** ✗ Wrong options: button alone strongest · bone tunnels weakest
"Course of nerve in commonest neurologic complication?"	→ **pierces biceps fascia, lateral to tendon, deep to cephalic vein, superficial along BR**

"Burning radial forearm pain 3 mo after single-incision reconstruction — sequela?"	→ **gradual resolution**
"True of LABCN injury?"	→ **highest risk after single-incision** ✗ Wrong options: laceration · permanent · less common than PIN
"Sensory nerve from lateral cord injured — true?"	→ **more common single- than double-incision** ✗ Wrong options: resolves 9–12 mo · second most common
"Commonest nerve injured — brachial plexus origin?"	→ **lateral cord**
"Labelled plexus: branch of which nerve most at risk, single incision?"	→ **musculocutaneous (A)**
"Altered lateral forearm sensation after repair — contributing factor?"	→ **single-incision approach** ✗ Wrong options: fixation type · timing
"Dorsoradial hand numbness after single-incision repair — nerve relation to tendon?"	→ **lateral (radial)** (SRN)
"True of complications of distal biceps repair?"	→ **LABCN = most common complication of single-incision** ✗ Wrong options: ECU/EDC ↑ HO · screw highest load · synostosis eliminated
"Most common sensory + motor nerve deficits?"	→ **lateral forearm numbness + weak wrist extension** (LABCN + PIN)
"Arm kept maximally supinated during footprint prep protects?"	→ **PIN → wrist extension weakness**
"Most common complication, findings?"	→ **volar/radial forearm numbness**
"True of single-incision technique?"	→ **higher rate of sensory nerve injury** ✗ Wrong options: PT/FCR interval · pronation · more anatomic
"True of management (positive hook)?"	→ **single-incision uses BR/PT interval** ✗ Wrong options: anconeus/ulna interval · more flexion than supination loss
"Vessel most at risk in volar approach?"	→ **recurrent radial artery**
"Deep retractor in BR/PT interval, bright red blood?"	→ **recurrent radial artery** ✗ Wrong options: inferior ulnar collateral · recurrent interosseous
"5 weeks, 6 cm retraction — true?"	→ **primary fixation in hyperflexion well tolerated** ✗ Wrong options: allograft necessary · two-incision required
"Resident's plan: button on radial side of tuberosity — improve by?"	→ **move insertion more ulnar** ✗ Wrong options: more distal · anchors
"More ulnar insertion improves?"	→ **forearm supination**

– – Active space – –

"Photo best showing long head biceps rupture?"

→ **Popeye deformity of upper arm**
✗ Wrong options: abnormal hook = distal

– – Active space – –

DISTAL CLAVICLE OSTEOLYSIS

– Shoulder & Elbow · 11 / 60 –

1 Basics

Epidemiology

Demographics	**20s**, mostly ♂; **weightlifters** (bench press, push-ups), overhead athletes, wrestlers
Risk factor	history of **trauma** (acute injury) or **repetitive microtrauma**
Adolescents (OB)	mean age 15.9 y in one series; overhead sports / weightlifting

Anatomy

Clavicle	S-shaped; **last bone to fuse** (medial physis fuses early 20s)
AC joint	**diarthrodial** with **fibrocartilage meniscus**

Ligament	Stability
AC ligaments (esp. posterosuperior capsule)	**Horizontal**
CC ligaments	**Vertical**

2 Mechanism

Pathoanatomy	**repetitive stress + micro-fractures** of distal clavicle → **osteopenia + bone resorption**
Load	bench press / manoeuvres with **elbows posterior to torso** → traction through AC joint

3 Presentation

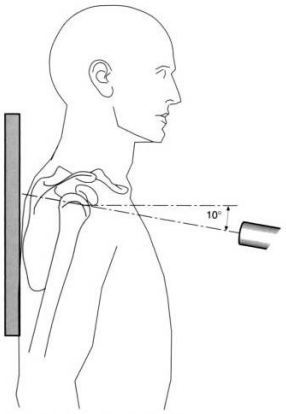
Pain	**distal clavicle / anterosuperior shoulder**; **insidious**; ↑ with repetitive loading (bench press, push-ups); mimics AC arthritis
Exam	**point tenderness at AC joint / distal clavicle** · **pain with cross-body adduction** · pain without weakness on resisted forward flexion + adduction · **full ROM**

4 Diagnosis & Classification

Key radiology findings

Views	**AP clavicle** + **Zanca view (15° cephalad tilt)** (Orthobullets) · **10–15°** (TO, OB) – read the options

– – Active space – –



Zanca view technique

X-ray (distal clavicle only, not acromion)	**cysts**, **osteopenia**, **resorption + erosion**, **tapering** of distal clavicle, **AC joint widening**



Distal clavicle osteolysis, AP

MRI	**↑ T2 signal + bone marrow oedema** in distal clavicle / AC joint (fluid-sensitive)
Bone scan	**↑ uptake** distal clavicle; **earlier than X-ray changes**
Follow-up MRI (OB)	mean **5 mm AC widening**

5 Management

- **Distal clavicle osteolysis**
  - First line → **rest + NSAIDs + PT + modify lifting technique**
    - persistent → **corticosteroid injection (diagnostic + therapeutic; US-guided)**
  - Failed nonop → **distal clavicle excision (open or arthroscopic)**
    - address cuff + LHB pathology at same time

Nonoperative

Activity modification	avoid aggravating lifts; **narrow hand grip**; **stop bar descent 4–6 cm above chest**; avoid elbows behind torso
Steroid injection	**diagnostic + therapeutic**; **more accurate with US**
Success (OB)	**93%** respond to conservative treatment (adolescent series)

Distal clavicle excision

Indication	persistent symptoms after **failed nonop** (rest, PT, cortisone)
Mumford (arthroscopic)	resect **only 0.5–1 cm**; **too much → AC instability**
Open vs arthroscopic	surgeon preference; arthroscopic **comparable pain relief**, ↓ morbidity, fewer restrictions, **quicker return**, better cosmesis, **evaluates GH joint** + treats concomitant pathology
Open	meticulous **deltoid-trapezial fascia** repair
Results (OB)	Auge: **10 weightlifters**, limited arthroscopic resection → **return to sport 3.2 d**, weight training **9.1 d** · Zawadsky: 22 excellent, 16 good, **3 failures – all traumatic aetiology**



After arthroscopic distal clavicle excision

6 Complications

Horizontal instability	**violating posterosuperior capsule / AC ligaments** during excision
AC instability	**over-resection** (beyond 0.5–1 cm)

X vs Y

	Distal clavicle osteolysis	AC separation	SLAP / biceps (wrestler)	Cuff tear
Key clue	AC tenderness, cross-body pain, **no weakness**, full ROM	**Instability**	Common in wrestlers, but exam ≠ AC	**Weakness**
Imaging	Lysis, cysts, tapering; marrow oedema	–	Labral pathology	Tendon tear
Surgery	**Distal clavicle excision**	**CC ligament reconstruction ± DCE**	SLAP repair / tenodesis	Cuff repair

– – Active space – –

How it's asked

"Bodybuilder, cross-body pain, AC tenderness, X-ray lysis — most successful treatment?"	→ **arthroscopic resection of the distal clavicle** ✗ Wrong options: GH injection · periscapular strengthening · labral repair · capsular release
"Wrestler, failed rest + PT + cortisone, AC tenderness, MRI marrow oedema — next step?"	→ **distal clavicle excision** ✗ Wrong options: SLAP repair · subpectoral tenodesis · CC reconstruction
"Best early imaging before X-ray changes?"	→ **bone scan**
"Complication of excessive / posterosuperior capsular resection?"	→ **horizontal AC instability**
"How much clavicle to resect?"	→ **0.5–1 cm**

– – Active space – –

ELBOW ANATOMY & BIOMECHANICS

– Shoulder & Elbow · 12 / 60 –

1 Basics

Joints & function

3 articulations	**ulnohumeral** (hinge) · **radiocapitellar / radiohumeral** (pivot) · **proximal radioulnar**
Function	lever arm positioning the hand · fulcrum for forearm lever

Range of motion & load sharing

Parameter	Value
Total flexion–extension	**0–150°**
Functional flexion–extension	**30–130°**
Functional pronation/supination	**50° each**
Carrying angle (valgus)	♂ **5–10°** · ♀ **10–15°** (anatomy note) · ~11° ♂ / ~13° ♀ (exam note) · 6–11° (UCL note); ↓ with flexion
Axial load, extended elbow	**40% ulnohumeral · 60% radiohumeral**

Osteology

Spiral groove	posterior humeral shaft; **radial nerve**; **~13 cm** proximal to trochlear articular surface
Distal flare	medial + lateral epicondyles; flare = half of elbow joint · **medial + lateral columns**
Trochlea	spool-shaped, **medial** · **capitellum** – lateral
Sublime tubercle	medial **coronoid** process; distal attachment of **anterior bundle of MCL**

– – Active space – –

Arthrology

Feature	Value
Articular surface **anterior tilt**	**~30°** relative to humeral shaft
Valgus	**6°**
Rotation	**5° internal**
Axis of flexion	Centre of **trochlea + capitellum**; passes through **anteroinferior medial epicondyle**
Radial head cartilage	Covers **~240°**; **lateral 120° bare** (safe zone for radial head fixation)
Capsule	**Max distension 70–80° flexion**; anterior capsule inserts **6 mm distal to coronoid tip → coronoid intra-articular**
Coronoid fossa	Receives coronoid tip in deep flexion
Coronoid tip	**Buttress against posterior dislocation**

Muscles of the elbow

Flexors **biceps** (radial tuberosity) · **brachialis** (inserts **11 mm distal to coronoid tip**) · brachioradialis

Extensor triceps

Collateral ligaments – MCL anterior bundle & LCL complex

Stabiliser tier	Structures
Primary static (3)	**Ulnohumeral joint (coronoid) · MCL (anterior bundle) · LCL complex**
Secondary static	**Radiocapitellar joint · capsule · flexor + extensor origins**
Dynamic	Muscles crossing elbow: **anconeus, brachialis, triceps, biceps** → compression

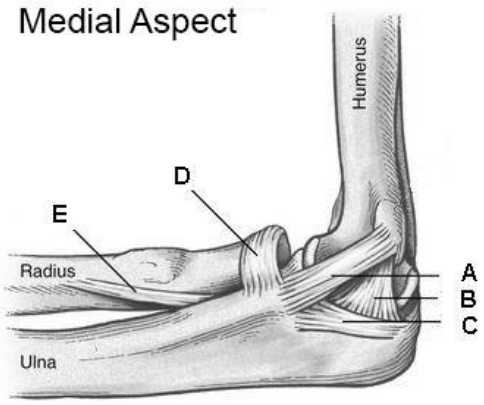
Coronoid loss of **≥50% height** → instability

MCL anterior, posterior, transverse bundles; resists **valgus + distraction**; origin **anteroinferior medial epicondyle** → insertion **sublime tubercle**

MCL bundle	Role
Anterior	**Most important valgus restraint** (prime medial stabiliser); esp. in flexion (Morrey)
Posterior (posterior oblique)	**Floor of cubital tunnel · primary valgus restraint in maximal flexion** · secondary medial stabiliser · tension ↑ **60–120° flexion** → contracture **limits flexion**
Transverse	**No role** in valgus stability or ROM

2nd most important valgus restraint **radial head**
✗ Wrong options: ulnotrochlear articulation

– – Active space – –

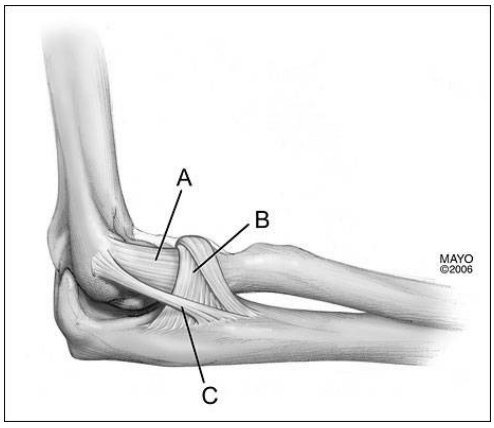


MCL – A anterior bundle, B posterior oblique band, C transverse, D annular, E oblique cord

LCL complex — **RCL** · **LUCL** · **accessory collateral ligament** · **annular ligament**; whole complex = **capsular thickening**; arises from **isometric point on lateral capitellum**

LCL component	Key fact
RCL	**Originates at axis of rotation for flexion**
LUCL	**Primary restraint to varus + external rotation** through full arc; origin **posterior lateral epicondyle** → insertion **crista supinatoris** (✗ Wrong options: sublime tubercle)
Accessory collateral	± substantial lateral stability (with RCL)
Annular	**PRUJ** stability

LCL repair — optimal stability with **appropriately tensioned** repair



LCL complex – A RCL, B annular, C LUCL

Not LCL complex — ulnohumeral joint (primary but not LCL) · radiocapitellar joint (secondary) · anterior bundle UCL · **transverse ligament (MCL complex)**

LUCL injury — abnormal **supination of ulna** on humerus → radial head subluxes **posteriorly** → **PLRI** (external rotatory subluxation of ulna + valgus on trochlea)

LCL disruption — **uniform** in dislocations + fracture-dislocations; commonest pattern = **avulsion from distal humerus** (McKee)

Radiocapitellar joint — important valgus constraint; radial head = **~30% of valgus stability**, most at **0–30° flexion / pronation**

– – Active space – –

Capsule greatest stability contribution **in extension** · anterior capsule resists varus-valgus **in extension, not flexion** (Morrey)

Stability (TO)	Primary	Secondary
Valgus	**MCL anterior bundle**	Radial head, flexor-pronator mass
Varus	**Coronoid process** (TO)	LUCL, extensor mass
Posterolateral rotatory (PLRI)	**LUCL**	Radial head, MCL
Posteromedial rotatory (PMRI)	Coronoid (anteromedial facet)	MCL (TO table)

Dynamic (TO) **flexor-pronator mass resists valgus** · **extensor-supinator mass resists varus**

Coronoid (TO) anterior buttress against posterior dislocation · fracture **>50% height → instability** · fractured in **10–15%** of dislocations

Radial head safe zone (TO) **"90° arc between radial styloid and Lister's tubercle" (Caputo)** · "posterolateral 80° non-articular; 280° articular" (other TO) · lateral **120°** bare (above) – read the options

Horii cycle (TO) FOOSH valgus + axial + supination: **LUCL → anterior/posterior capsule → MCL** · **reverse Horii**: varus + pronation → **PMRI**, coronoid anteromedial facet + LCL

Nerves of the elbow

Nerve	Origin	At the elbow
Musculocutaneous	Lateral cord	Between **biceps + brachialis** (supplies both); exits laterally distal to biceps tendon → **LABC deep to cephalic vein**
Radial	Posterior cord	**Triangular interval** (teres major, long head triceps, humerus) → spiral groove **13 cm** above trochlea → pierces **lateral IM septum 7.5 cm** above trochlea (middle/distal third) → between **brachialis + brachioradialis** → superficial to capsule at radiocapitellar joint
Median	Medial + lateral cords	With brachial artery (lateral → medial); superficial to brachialis; **no branches in arm**
Ulnar	Medial cord	Medial to brachial artery → pierces medial IM septum (**arcade of Struthers**) → cubital tunnel; **no branches in arm**; first FCU branch **distal to elbow**

Blood supply

Brachial artery medial in arm → enters cubital fossa → splits into **radial + ulnar** arteries

Cubital fossa contents lateral → medial: **biceps tendon, brachial artery, median nerve** · borders: **brachioradialis** (lateral), **pronator teres** (medial), distal humerus (proximal)

Branches superior/inferior ulnar collateral · nutrient/muscular · supratrochlear

Elbow imaging – X-ray & MRI (TO)

Condyles (TO) **lateral condyle (larger)** carries **capitellum + lateral epicondyle** · medial = medial epicondyle + trochlea → "medial condyle fracture" is really a **medial epicondyle or trochlear** injury

– – Active space – –

Fossae (TO) — **coronoid fossa (anterior)** receives coronoid in flexion · **olecranon fossa (posterior)** receives olecranon in extension

CRITOE (TO) — capitellum **1** · radial head **3** · internal (medial) epicondyle **5** · trochlea **7** · olecranon **9** · external epicondyle **11** (or 2-4-6-8-10-12) — **order matters** · **trochlea visible but medial epicondyle not in place → avulsion with entrapment**

X-ray sign (TO)	Meaning
Anterior "sail" / visible posterior fat pad	**Haemarthrosis**
Fat–fluid level	**Lipohaemarthrosis → intra-articular fracture**
Anterior humeral line	Normally through **middle third of capitellum**; posterior displacement → anterior part / misses
Small lateral flake (child)	May hide **lateral condyle fracture** into cartilage → **MRI**
Wider physis vs other side	**Epicondyle avulsion** (adolescent athlete)

Occult radial head/ neck fracture (TO) — after any elbow trauma: **AP + lateral + external oblique** · **Coyle view** = lateral with beam **45°** → radial head free of ulna · negative X-ray + suspicion (esp. children) → **screening MRI** (also coronoid tip fracture)

CT (TO) — little role except **deformity**

MRI technique (TO) — **prone "superman" position**, elbow in magnet centre (large FOV with chest → respiratory artefact) · **distal biceps protocol = 2 scans** (shoulder–arm supine, arm–elbow prone) · coronal T1 + PD FS, optional **GRE** (trabeculae, faint calcification), sagittal PD FS, axial PD ± FS

MRI structure (TO)	Normal / pitfall
UCL	Anterior band (key) · posterior band = **cubital tunnel floor** · transverse rarely seen · **grey fan-shaped humeral end = normal**; dark cord at sublime tubercle
RCL	Cord-like; small radial sulcus normal
LUCL	Lateral epicondyle → behind radius → ulna; **often hard to see even when intact**
Radiocapitellar plica	Normal synovial fold
Capitellar pseudodefect	**Normal posterior** capitellar groove — **not OCD**
Ulnar "bare area"	Normal cartilage gap in trochlear notch
Radial head fovea	Central cartilage thinning can be normal
Annular ligament	From ulnar tubercle around radial head

– – Active space – –

Distal biceps on MRI (TO) short + long heads stay **separate to the radial tuberosity** · lacertus runs medially · don't confuse **brachial artery (flow void)**, veins or median nerve with the tendon · rupture = fluid in biceps–brachialis plane, no insertion → **measure retraction**

MRI case (TO)	Finding
Medial elbow pain	Oedematous common flexor origin + faint calcification on GRE = **calcific tendinosis**
Pain on supination	Thick distal biceps + fluid = **radiobicipital bursitis / tendinosis**
Lateral elbow pain	**Empty common extensor origin**, retracted extensors, ligaments intact
Young athlete	**Capitellar OCD = ANTERIOR** cartilage/subchondral irregularity (pseudodefect = posterior)

2 Mechanism

Axis of forearm rotation

Flexion–extension axis centre of trochlea; line through isometric points of capitellum + trochlea

Pronation–supination axis **centre of radial head → fovea of ulnar head** (capitellum → radial head → distal ulna); forms a **cone**

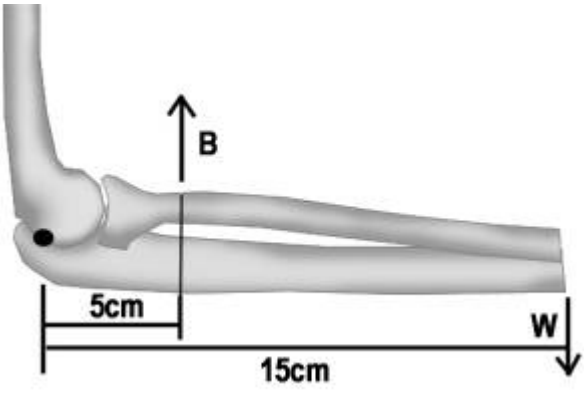
✗ Axis distractors radial head–radial styloid · radial head–ulnar styloid · coronoid–sigmoid notch · coronoid–radial styloid

Forearm rotation radiocapitellar + PRUJ + DRUJ + IOM + forearm muscles · **pronators**: pronator teres + quadratus · **supinators**: biceps + supinator

Biomechanics calculation – moment arms & anchor strength

Joint reaction force large; **short, inefficient lever arms** (biceps inserts near COR) → degenerative change · static loads ≈ body weight · **dynamic loads > body weight**

Free body ΣM = 0 → **B × 5 cm = W × 15 cm → B = 3W**



Free body diagram – biceps 5 cm, weight 15 cm

Anchor problem arm 20 N at 15 cm, biceps 5 cm → biceps force **60 N** × safety factor 3 = **180 N** minimum pullout

3 Presentation

– – Active space – –

Olecranon bursitis (student's elbow)

Stem (TO) direct blow posterior elbow + **olecranon spur** → continued sport → **painless swelling with effusion** over elbow

Answer (TO) key **not provided by source**; module label = **olecranon bursitis (student's elbow)** · options also list miner's elbow, little leaguer's elbow, golfer's elbow — read the options

5 Management

Surgical approaches to the elbow (TO)

Group	Approach (TO)	Plane / key step
Posterior	**Campbell triceps split**	15 cm midline incision; triceps split to bone; lateral half with anconeus, medial half with flexor fascia
Posterior	**Campbell aponeurosis tongue**	Tongue of triceps aponeurosis reflected proximally → **V-Y lengthening** for **extension contracture**
Posterior	**Wadsworth** (extensile posterolateral)	Triceps tongue + **anconeus elevated from lateral epicondyle** → radial head + posterior humerus
Posterior	**Olecranon osteotomy (MacAusland, Müller)**	**~2 cm from tip** · best articular exposure · fixed TBW (K-wire TBW commonest), TBW over screw, plate
Posterior	**Bryan–Morrey**	Triceps + forearm fascia + periosteum + anconeus reflected **medial → lateral** as one sleeve → **only posterior approach keeping extensor mechanism in continuity** · triceps must be repaired
Lateral	**Kaplan**	**ECRB ↔ EDC** · close to PIN → forearm pronated · mainly PIN release
Lateral	**Column** (extensile proximal lateral)	**BR + ECRL elevated off lateral supracondylar ridge** · extensile
Lateral	**Kocher**	**Anconeus ↔ ECU** · **most common lateral approach** · **cannot extend distally (PIN)**; annular ligament = distal limit
Medial	**Molesworth / Campbell**	**Medial epicondylectomy**, flexor origin reflected with fragment; ulnar nerve isolated
Medial	**Hotchkiss "over the top"**	**Split flexor-pronator mass between PT + FCU**, brachialis reflected → **coronoid** · protect **MABCN**
Anterior	**Henry**	**Mobile wad ↔ biceps tendon**; supinator elevated from radius

![Cross-section map of approaches around the elbow](../../TO notes/attachments/TOn-notes-approaches-around-elbow-chronic-elbow-and-forearm-inst-p02-f004.jpg)

Olecranon osteotomy (TO)	Feature
Transverse (MacAusland)	**Most commonly performed**
Müller	Only triceps-insertion part cut; residual ulna blocks view → rarely used
Chevron	V-cut → **rotational control + larger healing area**

– – Active space – –

Lateral boundaries (TO)	every lateral approach lies between **radial nerve (anterior)** and **LUCL (posterior)** · **pronate forearm** → PIN moves anterior, away from dissection (Kaplan + Kocher)
Radial nerve course (TO)	pierces lateral IM septum at **middle/distal third junction** → brachialis–ECRL → brachialis–BR → winds around proximal radius **between two heads of supinator** (PIN)
Medial approach indications (TO)	**coronoid fixation when intact radial head blocks lateral view** · medial osteophyte excision
Triceps-split/ reflecting approaches (TO)	poor view of intra-articular distal humerus

Posterior bundle of MCL – release for flexion

Release for flexion loss	**posterior (oblique) band of MCL + posterior capsule**; flexion 89° → **127°** (Wada; medial incision, ulnar nerve freed)
Limited extension	**anterior capsulectomy**
✗ Release distractors	**anterior band MCL** (→ valgus instability) · transverse band (no ROM gain) · **annular ligament** (never) · oblique cord of IOM

Arthrodesis position

Situation	Position
Unilateral	**90° flexion**, **0–7° valgus**
Bilateral	One at **110°** (feeding) · other at **65°** (perineal hygiene)

Diagnostic injection

Soft spot	triangle of **lateral epicondyle + olecranon + radial head** → intra-articular injection

Complex instability (OB)

Simple dislocation	closed reduction + short immobilisation; recurrent instability **uncommon**
Complex / persistent instability	fix or replace bony elements + ligament repair ± **hinged fixation**

X vs Y

	MCL (anterior bundle)	LCL complex (LUCL)
Resists	**Valgus** + distraction	**Varus + external rotation**
Origin	**Anteroinferior medial epicondyle**	**Posterior lateral epicondyle** / isometric point capitellum
Insertion	**Sublime tubercle**	**Crista supinatoris**
Failure	Valgus instability	**PLRI**
Backup	**Radial head** (2nd valgus restraint)	–

– – Active space – –

	Posterior bundle MCL	Anterior bundle MCL
Valgus role	Primary only in **max flexion**	**Most important overall**
Release for stiffness?	**Yes — restores flexion**	**No —** valgus instability
Cubital tunnel	**Floor**	—

How it's asked

"Primary stabiliser against valgus (labelled figure)?"	→ **anterior bundle MCL** ✗ Wrong options: posterior bundle = secondary · transverse · annular · biceps
"Structure A on lateral ligament figure?"	→ **radial collateral ligament** (B annular, C LUCL; transverse = MCL complex)
"Sublime tubercle is the insertion of?"	→ **anterior bundle MCL**
"True statement about elbow stability?"	→ **RCL originates at axis of flexion** ✗ Wrong options: LUCL on sublime tubercle · anterior bundle primary in max flexion · ulnotrochlear 2nd valgus restraint
"Varus + external rotation instability — which primary stabilisers incompetent?"	→ **LUCL, accessory collateral, annular, RCL** ✗ Wrong options: ulnohumeral · radiocapitellar · transverse ligament
"Release to restore flexion?"	→ **posterior oblique band MCL (B)** ✗ Wrong options: anterior band · annular
"Axis of forearm rotation?"	→ **radial head → ulnar head**
"Anchor pullout, 20 N at 15 cm, biceps 5 cm, safety 3?"	→ **180N**
"Posterior elbow blow, olecranon spur, painless effusion?"	→ **student's elbow** (olecranon bursitis; key not provided, TO)
"Optimal unilateral arthrodesis?"	→ **90° flexion**
"Only posterior approach preserving extensor mechanism continuity?"	→ **Bryan–Morrey** (TO)
"Kocher lateral approach interval?"	→ **anconeus and ECU** (TO) ✗ Wrong options: ECRB–EDC = Kaplan · mobile wad–biceps = Henry
"Safe zone for radial head fixation?"	→ **lateral 120° non-articular**
"Primary varus stabiliser of the elbow?"	→ **coronoid process** (TO)
"Posterior capitellar groove on sagittal MRI?"	→ **normal pseudodefect, not OCD** (OCD is anterior) (TO)
"View for occult radial head fracture?"	→ **external oblique / Coyle (45°) view** (TO)

– – Active space – –

ELBOW ARTHRITIS

– Shoulder & Elbow · 13 / 60 –

1 Basics

Types & epidemiology

3 types **osteoarthritis (primary)** · **post-traumatic** · **inflammatory**

	Primary OA	Post-traumatic	Inflammatory (RA)
Frequency	**Rare** (symptomatic ~**2%**)	**2nd most common**	**Most common / most prevalent** elbow arthritis (historically)
Patient	**Middle-aged ♂ laborer**, **♂:♀ 4:1**, 20–70 y (**avg 50**), **dominant arm**	**Younger** than other types	Older ♀; elbow involved in **20–50%** of RA
Cause	Strenuous manual labour	Non-op radial head fracture, (fracture-)dislocation, traumatic instability	RA, psoriatic, SLE, PVNS
Joint	**Radiocapitellar** preferentially; **ulnohumeral relatively spared**	Whole joint or focal ulnohumeral/radiocapitellar	Global; symmetric narrowing
Hallmark	**Osteophytes, capsular contracture, loose bodies**	Incongruity → altered load; stiffness, instability, malunion/ nonunion	Synovitis, **osteopenia**, ligament attenuation, instability

Secondary OA causes post-traumatic · **OCD** · **synovial osteochondromatosis** · **MUCL insufficiency / valgus extension overload**

Genes & lab markers

RA pathology **pannus → hyperplastic synovium → collagen + proteoglycan degradation** → chronic synovitis, ligament attenuation, **periarticular osteopenia**, contracture

RA trap synovial proliferation from pannus, **not rheumatoid factor mutation**

RA pathoanatomy fixed flexion contracture · instability · **ulnar** (or less often radial) neuropathy · erosions · cysts · deformity · progressive bone resorption

Stabilisers (relevance to surgery)

Primary **anterior band MCL** (**anterior oblique fibres** most important; valgus + distraction) · **LCL** · olecranon–coronoid–trochlea congruity

Secondary **radial head** (most important secondary; **30%** valgus stability; **0–30° flexion + pronation**) · capsule (**distraction in full extension**) · anconeus + lateral capsule (varus)

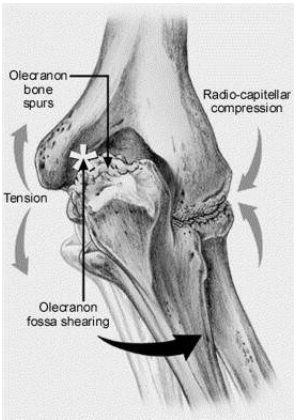
3 Presentation

OA symptoms	pain at **end range** · **loss of terminal extension** · painful **locking/catching** (loose bodies) · night pain unusual ✗ Wrong options: mid-range
OA exam	forearm rotation preserved early · **ulnar neuropathy in up to 50%**
RA	**hand + wrist precede elbow** · fixed flexion contracture · ligamentous incompetence · ± ulnar neuropathy · rheumatoid nodule · **evaluate C-spine in all RA**

4 Diagnosis & Classification

Key radiology findings

Modality	Fact
X-ray	**AP + lateral**; **C-spine X-rays for RA before surgery**
OA osteophytes	**Coronoid process + fossa** · **radial head + fossa** · **olecranon tip + posteromedial olecranon fossa**
Joint space	Narrowing; **ulnohumeral relatively preserved** in OA
Loose bodies	**Underestimated on plain X-ray**
RA	**Periarticular erosions + cysts**, osteopenia; **osteophytes not prominent** (✗ Wrong options: characteristic)
CT	Surgical planning; better defines **osteophytes + loose bodies**



Olecranon osteophyte impinging in olecranon fossa

Larsen grading (RA)

Stage	Radiographs
1	Soft tissues involved; **near-normal** radiographs
2	**Periarticular erosions + mild cartilage loss**; soft-tissue swelling, osteopenia
3	**Marked joint space narrowing**
4	Advanced erosions **penetrating subchondral plate**
5	Advanced damage, **loss of articular contour**

– – Active space – –

TEA threshold **Larsen 3–5** in patients >65 y

5 Management

Elbow arthritis – failed NSAIDs, injection, splint, activity modification

- Mild–moderate OA, arc >90°, painful end-range, mechanical symptoms → **arthroscopic OCA**
- Severe contracture (e.g. 40–90°), prior ulnar transposition → **open: O-K or column procedure**
- Young (<65), high demand, END-STAGE (OA/RA/PTA), stable → **interposition arthroplasty**
  - isolated radiocapitellar → **anconeus interposition / radial head replacement**
- >65, low demand, severe (Larsen 3–5) → **semiconstrained TEA**

Primary OA – osteocapsular arthroplasty (arthroscopic vs open) & patient selection

Nonop NSAIDs · cortisone injection · resting splints · activity modification → **mild–moderate** symptoms

OCA removal of osteophytes (coronoid, olecranon, fossae) + loose bodies + **capsular release**

Best OCA candidate **<65 y**, active, **arc >90°**, **mild–moderate** OA, **painless mid-arc + painful end-range**

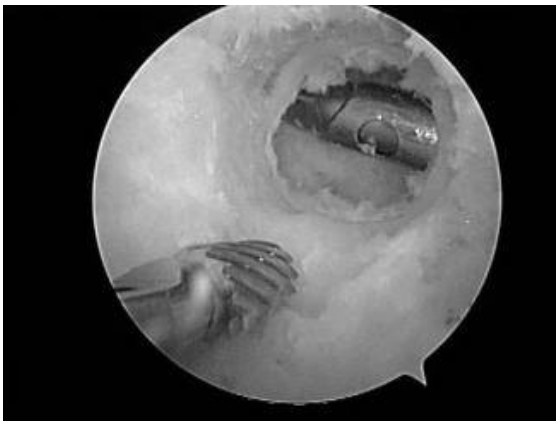
Arthroscopic debridement / release	
Indications	**Mechanical symptoms (loose bodies)** · capsular contracture · **bony block** to motion · **>90° motion** preferred
Contraindications (relative)	**Prior ulnar nerve transposition** (nerve location unknown) · **severe contracture / arthrofibrosis** · prior extensive open procedures
Not a CI	Prior **olecranon osteotomy** · ulnar nerve **subluxation** (✗ Wrong options: absolute CI; identify pre-op, esp. prior submuscular/intramuscular transposition)
Complications	**Neurologic** (**ulnar** palsy commonest, **radial** 2nd – anterolateral portal) · **synovial fistula** · recurrent stiffness

Arthroscopic vs open OCA	**Arthroscopic**
Post-op pain	↓
HO / ankylosis	↓ (✗ Wrong options: equal)
Access	**Easy anterior + posterior**, better cartilage view
Reliability	**Pain relief** more reliable (open → more predictable **ROM** gain)

O'Driscoll & Blonna 8 steps posterior view → remove debris + fat pad → osteophytes → posterior capsule release → anterior view → strip anterior capsule → coronoid osteophytes/reshape → excise anterior capsule; **4.8% transient nerve injury** (502 pts)

Elbow arthroscopy indications septic I&D · synovectomy (inflammatory) · debridement + loose body removal (OA) · contracture release · OCD capitellum · lateral epicondylitis release

Advanced radiocapitellar OA	consider **radial head replacement**
Avoid radial head resection in laborer	↑ load on arthritic ulnohumeral joint; doesn't fix ROM or end-range impingement
No PT / dynamic splints	no role initially in advanced painful arthritis with **fixed** contracture · no static splinting when arc functional + complaint is pain
Fusion	severe functional loss; rare (unmanageable **sepsis**, **Charcot**)



Outerbridge–Kashiwagi olecranon fossa fenestration

Outerbridge–Kashiwagi (ulnohumeral arthroplasty)	open **posterior** approach; **burr fenestration of olecranon fossa** → release posterior impingement + anterior access; younger pts with ↓ ROM; **decompress ulnar nerve if pre-op flexion contracture**; failure = missed anterior / peripheral (medial + lateral olecranon) osteophytes
Column procedure	medial or lateral open capsular release + bony resection for extrinsic contracture; **lateral most common** (access to all 3 joints) via lateral supracondylar ridge · anterior arthrotomy **ECRL/common extensor interval, stay anterior to LUCL** · posterior: elevate triceps · **go medial to gain flexion** (excise **posterior band MCL**)
Open release	elderly regain motion **more reliably than <20 y** (Strubel & Cohen)

Ulnar nerve – neuritis & decompression when gaining flexion

Risk	severely limited pre-op motion: **flexion <100°** · **extensor lag / max extension >60°** → **post-op ulnar nerve dysfunction**
Rule	any motion-restoring procedure (OCA, O-K, column, interposition) in such patients → **concomitant ulnar nerve decompression ± transposition**
Not radial neuritis	radial tension ↓ with gained extension
Not column procedure add-ons	release anterior band MUCL (primary valgus) · release LUCL (primary varus) · radial head resection (rotation full + painless) · **TFL interposition** (reserved **<50 y**)

Young post-traumatic arthritis – interposition arthroplasty

Indication	**young, high-demand, END-STAGE** OA / RA / **post-traumatic** who would otherwise get TEA if older; **no lifting restrictions**; needs **preserved bone stock + inherent stability**
Technique	burr distal humerus + proximal ulna congruent → graft over distal humerus (**Achilles allograft**, **autogenous TFL**, dermal/fascial) → ligament repair → **hinged distracting ex-fix** (offloads graft, early ROM, protects repairs)

– – Active space – –

Advantage can **convert to TEA** later (successful; Blaine)

Interposition contraindication	Type
Elbow instability / chronic instability without bone stock for ligament reconstruction	**Absolute**
Active infection	**Absolute**
Elderly **>60 y** (TEA candidates)	Relative
Bone loss: **destructive JRA**, **distal humerus osteonecrosis**	→ prosthetic replacement
Not CI	Young age · post-traumatic · inflammatory · previous debridement

Anconeus interposition arthritis **isolated to radiocapitellar** joint

Failed interposition (68 y, pain + instability) **convert to TEA**
✗ Wrong options: revision interposition · hemiarthroplasty (ulnar side arthritic) · arthrodesis

Rheumatoid elbow – TEA & synovectomy

Synovectomy open or arthroscopic; **young RA, joint space preserved**; most improve

Radial head resection + synovectomy only if synovitis **encircles radial head/neck** + painful crepitus on rotation + **PIN signs** (can't extend thumb)

Complication of radial head resection **rapid progression of ulnohumeral arthritis**
✗ Wrong options: loss of pronation/supination · longitudinal instability with intact IOM · ulnar palsy (less likely)

End-stage RA **semiconstrained TEA** (concomitant ligamentous instability)
✗ Wrong options: constrained (**high early loosening**) · arthroscopy (incomplete, unpredictable)

TEA outcome RA vs OA **equal or better** for RA
✗ Wrong options: poorer

Operative criteria – TEA

TEA	
Indications	**>65 y**, severe arthritis (**Larsen 3–5**) · complex distal humerus fracture, elderly, poor bone · distal humerus non/malunion, elderly low demand · post-traumatic arthritis (lower demand) · **failed interposition**
Contraindications	**Active <65 y** · **infection** · **Charcot joint**
Restriction	**Lifelong 5–10 lb lifting** limit

TEA design	Feature
Unconstrained	Needs **competent collaterals**, soft-tissue envelope, good bone
Semiconstrained	**"Sloppy hinge"** → some varus-valgus + rotational laxity → ↓ bone-cement stress → ↓ loosening; **best longevity + function**
Constrained	**High early loosening**

Elbow arthrodesis position (TO)

Arthrodesis	Position
Unilateral (TO)	**90° flexion + neutral to slight valgus** (✗ Wrong options: varus · 110°)
Bilateral (TO)	One **110°** (feeding) · other **65°** (perineal hygiene)

6 Complications

Total elbow arthroplasty complications

Rate
up to **43%** (Orthobullets) · up to **25%** (OB) – read the options

Bushing wear
most common mode of failure of semiconstrained TEA at midterm, esp. **post-traumatic indication** (younger, non-adherence to restrictions, unstable injury, deformity, contracture)
✗ Wrong options: gender · smoking · cement · age



Bushing wear in semiconstrained TEA

Infection / wound risk factors
prior elbow surgery · prior infection (**S. epidermidis**) · psychiatric comorbidity · **RA** · wound drainage · reoperation · poor skin (**long-term steroids**); two-stage revision → poor survival

Aseptic loosening
linked implants · **post-traumatic** OA

Instability
unlinked implants

Others
ulnar neuritis (iatrogenic or after motion restored without decompression) · **triceps avulsion/insufficiency** · fracture · mechanical failure

– – Active space – –

X vs Y

	Arthroscopic OCA	Interposition arthroplasty	TEA
Stage	**Mild–moderate**	**End-stage**	**End-stage**
Age / demand	Active, <65	**Young, high demand**	**>65, low demand**
Motion	Arc **>90°**	Any; needs stability	Any
Key CI	**Prior ulnar transposition**, severe contracture	**Instability**, infection	**Active <65**, infection, **Charcot**
Restriction	–	**None**	**5–10 lb lifetime**

	Elbow OA	Elbow RA
X-ray	**Osteophytes**, loose bodies, ulnohumeral spared	**Erosions, cysts, osteopenia**, symmetric narrowing
Symptom	**End-range** pain	Diffuse pain, effusion
Instability	No	**Ligament attenuation**
End-stage op	OCA / interposition / TEA	**Semiconstrained TEA**; synovectomy early

How it's asked

"Plumber, advanced OA, lacks 45° extension, flexes to 110° – initial treatment?"	→ **osteophyte resection + capsular release** ✗ Wrong options: PT/dynamic splints · TEA · fusion · radial head resection
"Carpenter, mid-range locking, arc 35–130°, failed injection?"	→ **arthroscopic capsular release + loose body removal + osteophyte decompression** ✗ Wrong options: interposition · radial head arthroplasty · TEA
"Firefighter 58, arc 30–130°, pain – most predictable pain relief?"	→ **arthroscopic debridement** ✗ Wrong options: open debridement = ROM · static splint · interposition
"True of arthroscopic vs open OCA?"	→ **less post-op pain** ✗ Wrong options: equal HO · limited access · radial palsy commonest · subluxation absolute CI
"Relative CI to arthroscopic debridement in post-traumatic OA?"	→ **prior ulnar nerve transposition** ✗ Wrong options: loose bodies · contracture · bony block · olecranon osteotomy
"Who benefits most from arthroscopic OCA?"	→ **50-y ♂, painful limited end-range** ✗ Wrong options: RA destruction · end-stage · elderly pain throughout

"Best OCA candidate?"	→ **51-y mailman, mild OA, arc 30–135°** ✗ Wrong options: 40–90° contracture → open · prior transposition · end-stage → interposition
"OCA for 50–90° arc — specific complication?"	→ **ulnar neuritis** ✗ Wrong options: radial neuritis · implant wear
"Lateral column procedure, pre-op arc 30–90° — add?"	→ **ulnar nerve decompression ± transposition** ✗ Wrong options: release anterior MUCL · LUCL · radial head resection · TFL
"Distraction interposition arthroplasty best for?"	→ **41-y laborer, post-traumatic arthritis** ✗ Wrong options: destructive JRA · osteonecrosis · 70-y RA
"Failed interposition at 68 y with pain + instability?"	→ **convert to TEA**
"Contraindication to interposition arthroplasty?"	→ **elbow instability** ✗ Wrong options: young age · PTA · inflammatory · prior debridement
"40-y inflammatory end-stage, stable, arc 75°?"	→ **Achilles interposition** ✗ Wrong options: anconeus = isolated radiocapitellar · TEA
"Burred surfaces + Achilles allograft + hinged ex-fix — best candidate?"	→ **20-y severe post-traumatic arthritis** ✗ Wrong options: active infection · chronic instability · elderly → TEA
"61-y RA, end-stage destruction?"	→ **semiconstrained TEA** ✗ Wrong options: constrained (early loosening)
"RA elbow pathology — true?"	→ **proteoglycan + collagen destruction** ✗ Wrong options: osteophytes · worse TEA results · unconstrained · RF mutation
"RA synovectomy + radial head resection — likely complication?"	→ **rapid ulnohumeral arthritis**
"Bushing wear at 6 y — risk factor?"	→ **indication for TEA (post-traumatic)** ✗ Wrong options: gender · smoking · cement · age
"Optimal unilateral elbow arthrodesis?"	→ **90° flexion + slight valgus** (TO)

– – Active space – –

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ELBOW ARTHROSCOPY: INDICATIONS & APPROACH

– Shoulder & Elbow · 14 / 60 –

1 Basics

Indications

Source	Indications
Orthobullets	**Loose body removal**, osteophyte debridement, **synovectomy**, **capsular release (stiffness)**, **OCD capitellum**, lateral epicondylitis, **septic arthritis** debridement
TO	Loose bodies · **OCD capitellum** · osteochondral lesions radial head · osteophytes humerus/olecranon · **RA (partial synovectomy)** · stiff elbow (debridement, lysis of adhesions) · **tennis elbow release** · evaluation of painful elbow · **inflamed olecranon bursa** · **intra-articular fractures**

Osteocapsular arthroplasty — young patient with degenerative elbow where **TEA inappropriate**, or significant contracture

Advantages — better articular visualisation, ↓ post-op pain, **faster recovery**

Disadvantages — technically demanding; **high risk of NV injury** (proximity to joint)

Contraindications (prior ulnar nerve transposition)

Source	Contraindications
Orthobullets	Prior trauma, surgical scarring, **previous ulnar nerve transposition**
TO	**Bony ankylosis** · **severe fibrous ankylosis** (scope can't enter safely) · **active infection** · surgery altering anatomy e.g. **anterior ulnar nerve transposition (relative)**
OB	**Ulnar neuropathy with prior submuscular transposition** = contraindication

Not contraindications (OB) — **Panner disease** · loose body ulnohumeral joint · **s/p radial head ORIF** · elbow stiffness · **osteoarthritis**

Ulnar nerve subluxation — **not an absolute contraindication**; identify pre-op (esp. prior submuscular/intramuscular transposition)

Capsular anatomy & distension (TO)

Capsule (TO) — attached to articular margins; fibres blend with **annular ligament**; anterior → above coronoid + radial fossae; posterior → above olecranon fossa

Tension (TO) — taut **anteriorly in extension**, **posteriorly in flexion**

Maximum distension (TO) — **70–80° flexion**; capacity **25–30 mL**
✗ Wrong options: 90–120° · 10° · 120–150°

5 Management

Positioning & joint distension (TO)

Positions	**supine** (suspended), **prone**, **lateral decubitus**
Most commonly used (TO)	**lateral**, elbow hanging freely on support (**flexed**)
All positions = elbow flexed (TO)	**elbow extended, prone** = not a position
Anaesthesia	**general** (muscle relaxation, tolerates prone/lateral) · regional: **no immediate post-op nerve check**, poor tolerance of position

Portal technique

Distend first	**fully distend joint through lateral soft spot** → NV structures move away from trocar
Nick and spread	skin only with blade, **hemostat spread** to capsule, **blunt trocar**
Elbow position	**anterior portals at 90° flexion** · posterior portals in some extension
Kit	standard **30° scope** · tourniquet · **solid cannulas** → maintain distension, ↓ extravasation ✗ Wrong options: trephinated
Landmarks	olecranon, both epicondyles, radiocapitellar joint, **ulnar nerve**; **mark before insufflation** (distension distorts)
Order	distend → viewing portal (medial or lateral) → working portal under **direct vision + needle localisation**
First portal	**anterolateral** after distension (TO, OB) · establishing a **medial portal first** (before swelling) may be advantageous (Orthobullets) — read the options
Viewing vs working (TO)	**anterolateral = viewing**; **proximal anteromedial = most common working portal** + most commonly used medial portal
Posteromedial portal	**avoided** (lies over **ulnar nerve**)

Factors that ↓ NV injury (TO)

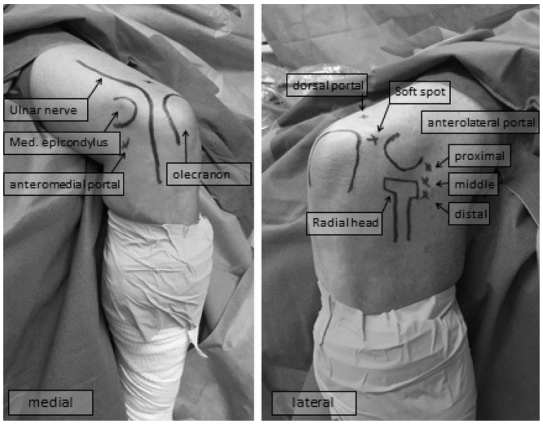
✓ **Manoeuvre (TO)**	**Protects**
90° flexion + joint distension	Radial nerve moves anteriorly
Pronation	**PIN**
Keep lateral portal more proximal	**Radial nerve / PIN**
Avoid deep knife; hemostat + blunt trocar	Lateral + posterior antebrachial cutaneous
Draw landmarks, **palpate ulnar nerve**	Ulnar
Low-flow infusion 40–45 mL/min or gravity (✗ Wrong options: pressurised)	Compartment syndrome
Instrument tips always in view; no suction near nerves	All

NOT a protective manoeuvre (TO)	**more distal lateral portal** (risk ↑ with distal placement)

– – Active space – –

Portals & nerves at risk

Portal	Location	Use	Nerves at risk
Proximal anterolateral	**1–2 cm proximal, 1 cm anterior** to lateral epicondyle	Views anteromedial joint; **safest anterolateral** (TO)	**Radial** (↓ risk more proximally)
Mid anterolateral (TO)	Just proximal, ~1 cm anterior to radiocapitellar joint	–	–
Distal anterolateral	**1–3 cm distal, 1 cm anterior** to lateral epicondyle; anterior to radial head/neck	1st portal supine; radial head, medial elbow, coronoid, trochlea, brachialis insertion, coronoid fossa	**PIN** + **lateral antebrachial cutaneous**; **radial at maximum risk** (TO) → rarely used
Direct lateral (soft spot)	Triangle **olecranon – radial head – lateral epicondyle**	**Initial distension**; posterior compartment (capitellum, radial head, radioulnar joint)	**Relatively safe**; LABC
Anteromedial (direct medial)	**2 cm anterior + 2 cm distal** to medial epicondyle; through flexor-pronator	Augments proximal AM; medial recess; place under direct vision	**MABC + median** (+ basilic vein, TO)
Proximal anteromedial (superomedial)	**2 cm proximal** to medial epicondyle, **anterior to IM septum**	Entire anterior compartment, radial head, capitellum, coronoid, trochlea; **most common working portal** (TO)	**MABC**; ulnar (7 mm, posterior to septum) + median (OB, TO)
Straight posterior (transtriceps)	**3 cm proximal** to olecranon, midline through triceps	Partial extension; **olecranon osteophytes**, posteromedial loose bodies	**Posterior antebrachial cutaneous**; ulnar **15–25 mm** away (OB)
Posterolateral	**2–3 cm proximal** to olecranon, **just lateral to triceps** (centre of anconeus triangle)	**20–30° flexion** (relax triceps); **best posterior compartment**, radiocapitellar (**OCD capitellum**), olecranon fossa	PABC, medial brachial cutaneous, ulnar (Orthobullets) · PABC + **lateral brachial cutaneous** (TO)
✗ Wrong option: **posteromedial**	Medial to triceps	**Not used**	**Ulnar nerve** – unacceptably high

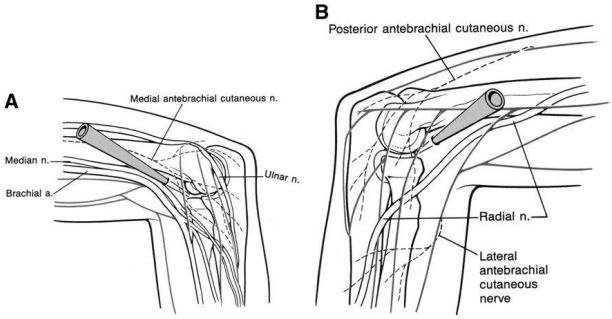


Standard elbow arthroscopy portals marked

Proximal anterolateral location	**1–2 cm proximal** (Orthobullets) · **2 cm proximal** (TO, OB) · **3 cm proximal** (one OB explanation) — read the options
Distal anterolateral location	**1–3 cm distal** (Orthobullets) · **2–3 / 3 cm distal** (TO) · **1–2 cm distal** (OB)
Straight posterior location	**3 cm proximal** (Orthobullets, OB) · **2–3 cm proximal, 2 cm medial to posterolateral** (TO)
Posterior compartment workhorses	**straight posterior + posterolateral**, both **3 cm proximal to olecranon tip** ✗ Wrong options: any posteromedial portal

Nerve distances (OB)

Portal	Nerve – distance
Anterolateral (joint line)	**Radial 4.8 mm** · PABC 12.6 mm
Proximal anterolateral	**Radial 4–7 mm** · PIN 1–14 mm
Anteromedial	**MABC 1 mm · median 7 mm** (one OB) · MABC 8.9 · median 12.9 · ulnar 22.1 mm (another OB)
Proximal anteromedial	MABC 1 mm · **median 12 mm** · ulnar 7 mm (posterior to septum)
Posterolateral	Radial 4.8 mm (3–8) (OB)
Straight posterior	Ulnar **~20 mm (15–25)**



Nerves at risk from medial (A) and lateral (B) portals

Radial nerve & anterior capsulectomy

Radial nerve course	on the **anterior capsule at mid-capitellum** (OB), runs longitudinally **medial to capitellum** → at risk in capsular excision; seen near probe when viewing radiocapitellar joint from **anteromedial** portal
Protective layer	**brachialis (~4 mm)** between capsule and radial nerve **proximal to radial neck** ✗ Wrong options: anconeus · brachioradialis · biceps · pronator teres
Danger zone	distal capsule **over radial head/neck** → transect capsule **≥2–3 cm proximal to radial head**

- - Active space - -

6 Complications

Nerve injury

Rate **nerve palsy 1–5%**; neurologic = **most common** complication, mostly **transient**

Greatest risk **rheumatoid arthritis**, **elbow contracture**

Ranking	Answer
Most common complication / nerve (TO, OB)	**Transient (sensory) ulnar nerve palsy**
2nd	**Radial nerve palsy** (Orthobullets, OB) · **persistent wound drainage**, then transient sensory radial palsy (TO)
Most common permanent nerve palsy (TO)	**PIN**, then ulnar
Least involved nerve (TO)	**Median**

Mechanism **direct** (trocars, instruments, no blunt dissection → neuromas) · **indirect** (**compartment syndrome** from aggressive distension/extravasation; **local anaesthetic extravasation** – transient)

Deficit → nerve → portal

Deficit	Nerve	Portal
Wrist drop: no wrist + MCP extension	**Radial nerve proper**	**Proximal anterolateral**
Weak wrist extension + **ulnar deviation** (lost ECRL)	Radial nerve proper	Proximal anterolateral
No finger (MCP) extension, partial wrist extension; can't extend thumb IP; wrist extends in **radial deviation**; no paraesthesia	**PIN** (ECRL spared)	**(Distal) anterolateral**
Paraesthesia dorsal 5th digit, intrinsic weakness, ↓ finger ab/adduction	**Ulnar**	**Posteromedial**
Paraesthesia **medial volar forearm**	MABC	Anteromedial
Can't flex wrist	Median	Anteromedial
Sensory deficit only	Superficial radial	–

PIN palsy EPL (thumb IP extension) + finger extensors; **ECRL** innervated **proximal to bifurcation** → spared

✗ PIN traps index abduction = **1st DI (ulnar)** · APB = **median** (APL = PIN) · "wrist extension + radial deviation lost" = radial nerve proper

– – Active space – –



PIN palsy after anterolateral portal: finger drop, partial wrist extension

Other complications

HO / ankylosis	**less than open**; minimise bleeding
Infection	**sinus tract (posterolateral portal)**; ↑ risk with **intra-articular steroid at time of arthroscopy**; possibly ↑ with **inflammatory arthritis**

Infection prevention – avoid intra-articular steroid

✓ **Prevent infection**	**avoid intra-op steroid injection**
✗ **Traps**	avoid posterior midline portals · avoid scope in OA · absorbable subcuticular suture only (no difference) · restrict ROM 3–4 wk (↑ stiffness)

X vs Y

	Proximal anterolateral	**Distal anterolateral**
Location	1–2 cm **proximal**, 1 cm anterior to LE	1–3 cm **distal**, 1 cm anterior to LE
Nerve	**Radial nerve proper** (wrist drop)	**PIN** (finger + thumb IP extension), LABC
Safety	**Safer** → more commonly used	Riskier → rarely used

	Anteromedial	**Proximal anteromedial**
Location	2 cm anterior + 2 cm distal to ME	2 cm proximal to ME, anterior to septum
Nerves	**MABC (1 mm) + median (7 mm)**	MABC (1 mm), ulnar (7 mm), median (12 mm)
Role	Augment PAMP, medial recess	**Most common working portal**

How it's asked

"First portal in elbow arthroscopy?"	→ **anterolateral** (TO)
"Most common nerve involved?"	→ **ulnar** (TO)
"Most common complication?"	→ **transient sensory ulnar nerve palsy** (TO)
"Most common permanent nerve palsy?"	→ **PIN** (TO)
"Least involved nerve?"	→ **median** (TO)

– – Active space – –

"NOT a manoeuvre to avoid nerve injury?"	→ **more distal lateral portal** (TO) ✗ Wrong options: pronation + 90° · distension · low flow
"Most commonly used working portal?"	→ **proximal anteromedial** (TO)
"NOT a position for elbow arthroscopy?"	→ **elbow extended, prone** (TO)
"Maximum elbow distension at?"	→ **70–80° flexion** (TO)
"Portal generally not used?"	→ **posteromedial** (ulnar nerve)
"Shaver through anterolateral portal, finger drop — structure?"	→ **PIN**
"Structure nearest probe, radiocapitellar joint from anteromedial portal?"	→ **radial nerve**
"Greatest risk with proximal anterolateral portal?"	→ **radial nerve**
"Two safest posterior compartment portals?"	→ **straight posterior + posterolateral, 3 cm proximal to olecranon**
"Portal 2 cm proximal, 1 cm anterior to LE — deficit if nerve injured?"	→ **loss of wrist + MCP extension**
"Can't extend thumb IP post-op — portal?"	→ **anterolateral** ✗ Wrong options: direct lateral = LABC · posterolateral = PABC
"Portal 2 cm anterior + 2 cm distal to ME — risk?"	→ **MABC + median**
"AL, AM and posteromedial portals used — likely deficit?"	→ **paraesthesia dorsal 5th digit** (ulnar)
"Protects radial nerve in anterior capsulectomy?"	→ **brachialis**
"Prior radial head ORIF, AL portal — iatrogenic finding?"	→ **can't extend thumb IP** ✗ Wrong options: wrist extension + radial deviation = radial proper
"Contraindication to elbow arthroscopy?"	→ **prior submuscular ulnar nerve transposition** ✗ Wrong options: Panner · loose body · stiffness · s/p radial head ORIF
"Prevent infection after elbow arthroscopy?"	→ **avoid intra-op steroid injection**

– – Active space – –

ELBOW PHYSICAL EXAM

– Shoulder & Elbow · 15 / 60 –

1 Basics

Exam sequence & normal values

Sequence inspection → palpation → ROM → stability → motor → sensory → vascular → provocative tests

Parameter	Normal	Functional
Carrying angle (full extension)	**~11° ♂ · ~13° ♀** (exam note) · ♂ 5–10° / ♀ 10–15° (anatomy note); ↑ in throwers	–
Flexion–extension	0–140°	**30–130°**
Pronation	75°	**50°**
Supination	85°	**50°**

Pronation–supination test with shoulders fully adducted + elbow at 90°; compare both sides, active + passive

End point **soft = effusion / capsular tightness** · **firm = mechanical block** (loose body, fracture, osteophyte)

Throwers loss of full extension even **without pathology**

Motor – muscle & root

Movement	Muscle	Root
Elbow flexion, **full supination**	Brachialis + biceps	C5–C6
Elbow flexion, **neutral** (thumb to ceiling)	**Brachioradialis**	C6
Elbow extension	Triceps	C7
Pronation	Flexor-pronator mass	C7–C8
Supination	Biceps (primarily)	C6
Wrist extension	ECRL, ECRB, ECU	C6–C8
Wrist flexion	FCR, FCU	C6–C8
Finger/thumb extension	EDC, EPL	C7–C8
Finger/thumb flexion	FDS, FDP, FPL	C7–C8
Intrinsics	Lumbricals, interossei	T1

Brachial artery anterior elbow, **medial to biceps tendon**; also check radial + ulnar pulses

– – Active space – –

3 Presentation

Inspection

Effusion elbow held at **70–80° flexion** (= max capsular distension) · fullness of **soft spot** (radial head + lateral epicondyle + olecranon)

Cubital tunnel **1st dorsal interosseous / 1st web atrophy** · ring/small clawing (more with **Guyon's** – unopposed FDP)

Distal biceps rupture **medial ecchymosis** + swelling · proximal retraction of belly = **reverse Popeye sign**

Palpation – point of maximal tenderness

Condition	Tender point
MCL injury	**MCL origin, just inferior to medial epicondyle**; palpate at **50–70° flexion** (flexor-pronator mass moves anterior)
Valgus extension overload	**Posteromedial olecranon**
Lateral epicondylitis	ECRB insertion, **few mm distal to tip of lateral epicondyle**
Radial tunnel syndrome	**3–5 cm distal to lateral epicondyle** (more distal than tennis elbow)
Medial epicondylitis	**5–10 mm distal + anterior** to medial epicondyle; swelling/warmth if inflamed

Radial head best palpated while rotating forearm pronation → supination

Ulnar nerve subluxation over medial epicondyle, extension → flexion · hypermobility in **33% of adults**
✗ Wrong options: necessarily CuTS

Snapping medial triceps snaps over medial epicondyle on **resisted extension from full flexion** (differentiate from ulnar nerve)

4 Diagnosis & Classification

Stability – MCL (valgus)

Valgus stress test flex **20–30°** (**unlocks olecranon**) + **externally rotate humerus** + valgus → tests **MCL**

Test	Technique	Positive	Accuracy
Milking maneuver	**Pull patient's thumb**, forearm **supinated**, elbow **90°**; supine or seated/standing	Apprehension, instability or **pain at MCL origin**	Sens **87.5%**, **NPV 100%**
Moving valgus stress	Same position + valgus while ranging full arc; shoulder **fully ER**	Pain at MCL origin **between 70° and 120°** (early acceleration 70° · late cocking 120°)	**Sens 100%, spec 75%**

– – Active space – –



Milking maneuver

Valgus extension overload — **pain with forced elbow extension** · valgus in terminal extension reproduces pain · **varus loading ↓ pain**

Stability – LCL / posterolateral rotatory instability

Test	Technique	Positive / accuracy
Lateral pivot-shift	Supine, arm overhead, shoulder fully ER, **forearm supinated + valgus**, extension → flexion	Pain/apprehension or **clunk at 40°** (radiocapitellar dislocated); 2nd clunk with ↑ flexion (triceps reduces radial head); **more reliable under anaesthesia**
PLR drawer	Supine, elbow **40°**, forearm supinated, AP force over lateral proximal forearm	Apprehension or **skin dimple** (radial head posterior subluxation)
Chair push-up	Push up from chair, forearm **supinated**	Unable / apprehension; sens **87.5%** (**100% with prone push-up**)
Prone push-up	Push-up, forearm supinated	Unable; sens 87.5% (100% with chair test)
Table-top relocation	Press-up on table edge, elbow lateral, forearm supinated; 3 parts	Pain in parts 1 + 3, **relief with thumb on radial head (part 2)** = PLRI

Table-top (wrong option) — pain in **all 3 parts** = intra-articular **radial head fracture**

– – Active space – –

Provocative tests – nerve

Condition	Test / sign	Finding
Cubital tunnel	**Froment**	Thumb **IP flexion by FPL (AIN)** on key pinch → compensates adductor pollicis (ulnar)
Cubital tunnel	**Jeanne**	Thumb **MCP hyperextension** + adduction by **EPL (radial)** on key pinch
Cubital tunnel	**Wartenberg**	Persistent **small finger abduction** + extension; weak intrinsics + **unopposed EDM**
Cubital tunnel	**Masse**	Flat palmar arch, lost ulnar hand elevation (weak **ODQ**, ↓ small MCP flexion)
Cubital tunnel	**Tinel**	Over cubital tunnel, **elbow extended**
Cubital tunnel	**Elbow flexion test**	Symptoms with flexion **>60 s**
Radial tunnel	**Resisted long finger extension** · **resisted supination** (elbow + wrist extended) · **passive pronation + wrist flexion**	Pain at radial tunnel (weakness from pain); supinator stretch → **250 mmHg** (normal 50)
PIN syndrome	Resisted supination ↑ pain · **normal tenodesis** (vs extensor tendon rupture)	Weak finger MCP extension; **wrist extends in radial deviation** (ECRL intact, ECU absent)
AIN syndrome	**OK sign** (FDP index + FPL)	**Can't make OK**; weak **PQ** = weak resisted pronation with **elbow maximally flexed**
AIN vs FPL rupture (RA)	**Tenodesis**	Intact tendons: passive wrist extension flexes thumb IP + index DIP

Cubital tunnel motor **weak pinch** (up to **70%** lost; adductor pollicis)

Pronator syndrome – site-specific tests

Pronator syndrome **Tinel proximal anterior forearm**, not at wrist · wrist-flexion provocation as in CTS

Resisted test	Compression site
Elbow flexion + forearm supination	**Bicipital aponeurosis (lacertus)**
Forearm pronation, elbow extended	**Two heads of pronator teres**
FDS contraction to middle finger	**FDS fibrous arch**

Sensory

Cutaneous nerves medial, lateral, posterior antebrachial cutaneous · ulnar · median · superficial radial

Cubital tunnel ↓ 2PD **small + ulnar ring** · **↓ dorsal ulnar hand** = proximal to wrist (dorsal branch arises **5 cm proximal** to wrist)

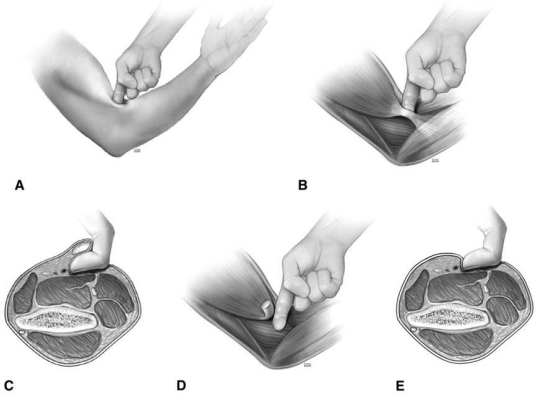
Pronator syndrome ↓ sensation in **palmar cutaneous branch** territory (arises **4–5 cm proximal** to carpal tunnel) · not in CTS

Provocative tests – tendon

Condition	Test	Key fact
Triceps rupture	**Modified Thompson squeeze**	Prone, forearm hanging; squeeze triceps → **no elbow extension** = complete tear (triceps + lateral expansion)
Distal biceps	**Hook test**	Elbow 90°, **full supination**, hook **lateral** edge; intact = finger **1 cm beneath tendon**; **sens + spec 100%**
Distal biceps	**Ruland biceps squeeze**	Elbow **60–80°**, slight pronation; squeeze belly → **no supination** = positive; **sens 96%**
Distal biceps	**Biceps crease interval (BCI)**	Crease → cusp distance **>6 cm or 1.2× contralateral**; sens 92%, spec 100%
Distal biceps	**Passive forearm pronation**	Belly moves proximal on supination, distal on pronation (intact)
Distal biceps	Strength	**Loss of supination > flexion**

Hook test false positives — **partial tear** · **intact lacertus fibrosus** · **underlying brachialis**

Biceps triad — hook + passive pronation + BCI in sequence → **100% sens + 100% spec** for complete rupture



Hook test



Biceps crease interval

Lateral epicondylitis — pain on resisted wrist extension (elbow extended, pronated) · **Maudsley = resisted middle finger extension** (ECRB) · maximal wrist flexion · passive wrist flexion in pronation

Chair test — elbow extended, forearm pronated, shoulder flexed, lift chair → lateral elbow pain = **lateral epicondylitis**

Medial epicondylitis — pain on **resisted pronation + wrist flexion** · **no instability/apprehension** on valgus stress or milking (vs MCL)

– – Active space – –

5 Management

Olecranon bursitis – aspiration & compression

Cause — **repeated minor trauma** (leaning on elbow, desk work) > single injury → inflamed lining secretes fluid



Olecranon bursitis

Aseptic olecranon bursitis

- First line → **nonoperative (more effective + safer than surgery)**
- Aspiration chosen → **posterolateral needle, aspirate alone**
  - then → **compressive wrap (bursal walls adhere)**

Aspiration ✓ — **alone** (no steroid) + **compressive dressing** → walls adhere, prevents reaccumulation

Approach — **posterolateral**
✗ Wrong options: posteromedial → ulnar nerve

Avoid steroid — **↑ septic bursitis** (local immunosuppression); no outcome benefit · **recurrence occurs with or without steroid**
✗ Wrong options: "steroid ↑ recurrence"

LUCL rupture — from repeated steroid injections for **lateral epicondylitis**, not bursa aspiration

Finding	Septic	Aseptic
Tenderness	88%	36%
Erythema/cellulitis	83%	27%
Warmth	84%	56%
Trauma / skin lesion	50%	25%
Fever	38%	**0%**

Septic vs aseptic — clinical + lab data **overlap** → hard to distinguish

X vs Y

	Lateral epicondylitis	Radial tunnel syndrome
Tender point	**Few mm distal** to lateral epicondyle (ECRB)	**3–5 cm distal**
Provocation	Maudsley, chair test, resisted wrist extension	Resisted long finger extension, resisted supination

	MCL injury	Medial epicondylitis
Tender point	**Just inferior** to medial epicondyle	**5–10 mm distal + anterior**
Milking / valgus stress	**Pain, apprehension, instability**	**No** instability or apprehension
Provocation	Moving valgus stress 70–120°	Resisted pronation + wrist flexion

	Pronator syndrome	Carpal tunnel syndrome
Tinel	**Proximal forearm**	Wrist
Palmar cutaneous branch sensation	↓	**Spared**

	Cubital tunnel	Guyon's canal
Clawing	Less	**More** (unopposed FDP)
Dorsal ulnar hand	↓	Spared

How it's asked

"Olecranon bursa aspiration – true statement?"	→ **compressive wrap after aspiration alone promotes bursal wall adherence** ✗ Wrong options: steroid ↑ recurrence · posteromedial · similar infection risk with steroid · LUCL rupture
"Steroid injected into olecranon bursa – risk?"	→ **↑ septic bursitis**
"Needle approach for olecranon bursa?"	→ **posterolateral** (avoid ulnar nerve)
"Position to palpate MCL origin?"	→ **50–70° flexion**, just inferior to medial epicondyle
"Elbow position to test valgus stability?"	→ **20–30° flexion** (unlocks olecranon), humerus ER
"Moving valgus stress test positive arc?"	→ **70–120°**
"Clunk at 40° on pivot-shift?"	→ **dislocated radiocapitellar joint** (PLRI)
"Table-top test, pain in all 3 parts?"	→ **radial head fracture** ✗ Wrong options: PLRI
"Hook test false positive?"	→ **partial tear / intact lacertus / brachialis**
"Tenderness 3–5 cm distal to lateral epicondyle?"	→ **radial tunnel syndrome**

– – Active space – –

"Elbow held at 70–80° flexion?"	→ **effusion** (max capsular volume)
"Wrist extends into radial deviation?"	→ **PIN palsy** (ECRL intact, ECU lost)
"Weak resisted pronation with elbow fully flexed?"	→ **pronator quadratus (AIN)**
"Resisted pronation, elbow extended reproduces median symptoms?"	→ **pronator teres heads**
"Numbness thenar eminence + forearm Tinel?"	→ **pronator syndrome** ✗ Wrong options: CTS (palmar cutaneous spared)
"Firm end point to flexion?"	→ **mechanical block** ✗ Wrong options: effusion = soft

– – Active space – –

ELBOW STIFFNESS AND CONTRACTURES

– Shoulder & Elbow · 16 / 60 –

1 Basics

Functional range of motion

Motion	Functional range
Flexion–extension	**30°–130° = 100° arc** (Orthobullets, OB) · **90° arc** (one OB explanation) – read the options
Pronation–supination	**50°–50°**

Tolerance — **30° loss of extension well tolerated · flexion loss causes more dysfunction than extension loss**

Causes

Aetiology — **OA**, **trauma (fractures)**, surgery, **CP**, **TBI**, **burns**, prolonged immobilisation, recurrent haemarthroses, congenital (**arthrogryposis**, **congenital radial head dislocation**)

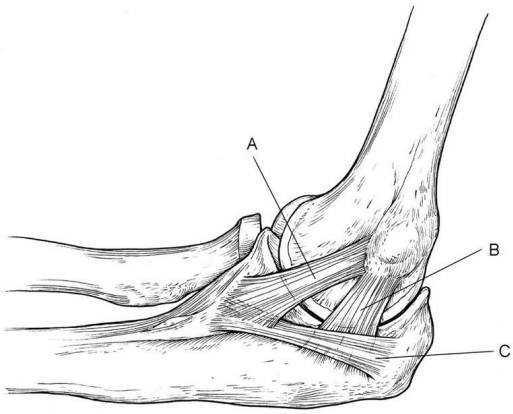
Type	Causes
Intrinsic (intra-articular)	Joint incongruity, synovitis, **loose bodies**, intra-articular fracture, malunion, OCD, post-traumatic arthritis (**coronoid + olecranon tip osteophytes**, radiocapitellar narrowing)
Extrinsic (extra-articular)	Burn **eschar**, **HO**, **capsular adhesions/contracture**, ligament contracture (**posterior oblique MCL**)
Mixed	Late effect of intrinsic → extrinsic stiffness

Elbow ligaments

Ligament	Key fact
Anterior bundle MUCL (anterior oblique)	**Isometric**; **primary valgus + distraction stabiliser**; highest tensile strength; **Tommy John** reconstruction · do not release → **valgus instability**
Posterior bundle MUCL (posterior oblique)	**Dorsal to axis of rotation**; lengthens **9 mm between 60° and 120°** flexion · contracture → **loss of flexion · release → ↑ flexion without valgus instability**
Transverse band MUCL	Release **does not improve ROM**
Radial collateral ligament	Do not release
LUCL	Do not release → **PLRI** (pivot shift)
Annular ligament	–

Anterior capsule contracture — **loss of extension**

– – Active space – –



Medial ligaments: A anterior band, B posterior band, C transverse

Ulnar nerve	close to joint → at risk when contracted / during flexion gain

3 Presentation

Pain	**mid-arc pain → intra-articular** pathology · **pain at extremes** → extrinsic contracture (bony impingement + soft-tissue stretch)
Function	↓ motion limits **ADLs**
Exam	**scars**, skin quality, inflammation · flexion/extension + pronation/supination vs contralateral · **median, radial, ulnar nerve** function
Flexion <90–100°	**posterior band of MCL contracted** → release (+ consider ulnar nerve decompression/transposition)

4 Diagnosis & Classification

Key radiology findings

X-ray	**AP, lateral, oblique**; **serial X-rays if HO**



Heterotopic ossification, elbow

CT	**loose bodies**, nonunion, joint incongruity, abnormal bony anatomy; HO planning
MRI	rarely indicated
HO maturity	**well-corticated margins + no change on serial X-rays** · **labs not needed** · matures **12–18 mo** (OB) → **no bone scan** years after injury

5 Management

- **Elbow stiffness**
  - Most / contracture <40° → **NSAIDs + supervised PT (active + passive ROM)**
    - failed PT + flexion contracture >30° or flexion <130° → **static progressive splint**
  - Plateau 4–6 mo, extrinsic, congruent joint → **capsular release ± posterior band MCL**
  - Intrinsic arthritis → **choose by extent + demand**
    - confined to olecranon fossa → **osteophyte excision + capsular release**
    - diffuse, young high-demand → **distraction interposition arthroplasty**
    - diffuse, elderly low-demand → **TEA**

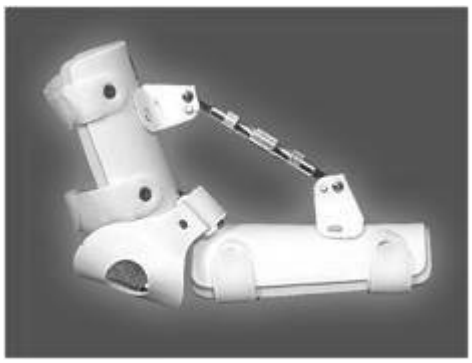
Static progressive splinting

Indications after **failed PT** with **flexion contracture >30°** or **flexion <130°**

Best initial option (OB) **supervised therapy + static progressive (turnbuckle) splinting** over **6 mo** (6–12 mo) → greatest ↑ **DASH** + functional ROM

Not indicated **mechanical block** (esp. pronation/supination) · **ankylosis** (extensive HO) · functional arc already (e.g. 10–140°) · routinely after arthroscopic loose body removal

✗ Alternative	Why wrong
Self-directed exercise	**Poor results**
Supervised therapy alone	First line, but response **variable, often unsatisfactory**
CPM	Limited role in **established** contractures; no end-range gain
Closed manipulation (MUA)	**Worsens stiffness**: tearing, **haemarthrosis**, more fibrosis



Turnbuckle static progressive splint

– – Active space – –

Operative criteria

Procedure	Indication
Capsular release ± posterior band MCL	**Extrinsic** contracture + **normal joint congruency** (most predictable) · arthritis (less predictable) · ankylosis → less predictable
Osteophyte excision (+ capsular release)	**Intrinsic** contracture, **arthritis confined to olecranon fossa**; bony block to terminal motion → remove **coronoid, coronoid fossa, olecranon, olecranon fossa**
Distraction interposition arthroplasty	Intrinsic, **diffuse arthritis**, **young high-demand**
TEA	Intrinsic, diffuse arthritis, **elderly low-demand** · high failure in young · **permanent 5-lb lifting limit**
Musculocutaneous neurectomy	**Neurogenic** contracture, **flexion deformity <90°**
Timing of release	**4–6 mo** post-injury/surgery once ROM **plateaued** despite splinting/therapy
HO excision	At **maturity**

Contraindications to release — **Charcot elbow**, **neurologic elbow disorder** · **poor skin** (relative → rotational flap)

Release – posterior bundle of MCL for flexion

Flexion gain — capsular release + **posterior oblique band of MUCL**
✗ Wrong options: anterior band · transverse · RCL · LUCL

Flexion <100° preop — consider **ulnar nerve transposition/decompression** (↓ nerve injury)

Arthroscopic vs open release (prior ulnar transposition, HO)

Arthroscopic — technically demanding · posterior: olecranon fossa/osteophyte debridement + posterior capsular release (**no suction medially** – ulnar nerve) · anterior: coronoid fossa debridement + anterior capsulotomy/capsulectomy

Nerves at risk with portals — **radial (most)** > ulnar > median (stiffness note) · overall most common = **transient ulnar**, most common permanent = **PIN** (TO, arthroscopy MCQs) – read the options

Relative contraindications to arthroscopic release — **prior ulnar nerve transposition** (variable nerve position) · **HO** · obesity · severe loss of pronation–supination · muscular contracture (**CP**)

Not contraindications age >40, gender, **osteophytes / arthritis** (unpredictable outcome only), heavy labour

Open approach	Key points
Lateral column (Morrey)	Lateral or posterior skin incision · elevate **ECRL + BR** off anterior supracondylar ridge, triceps posteriorly · brachialis off anterior capsule · **robust anterior access** → capsular release + **HO excision**; coronoid + olecranon tip/fossa, radiocapitellar joint
Medial "over the top" (Hotchkiss)	Extrinsic contracture, **MCL calcification**, **baseline ulnar nerve symptoms** · with ulnar nerve decompression/transposition · **release posterior band MCL** · work anterior to flexor-pronator mass
Combined medial + lateral	Single **posterior incision** → both columns

Rehabilitation & outcomes

Post-op **regional block**, **CPM through full ROM**, compressive dressing, active + active-assist ROM, extension / dynamic / static progressive splinting as needed

Compliance critical

Outcome variable; **retain ~two-thirds of motion gained** at surgery

6 Complications

Complication	Detail
Heterotopic ossification	Prophylaxis **low-dose radiation** or **indomethacin** · no radiation with **acute fracture** (nonunion risk)
Transient ulnar neuropraxia	After release (nerve close to contracted joint)
Ulnar nerve damage	Transpose if **preop flexion <100°**
Recurrent contracture	–

X vs Y

	Posterior band MUCL	Anterior band MUCL
Length change	**+9 mm (60°→120°)**	**Isometric**
Contracture →	**Loss of flexion**	–
Release →	**↑ flexion**, stable	**Valgus instability**
Other	–	Valgus restraint in throwing; Tommy John; strongest

	Arthroscopic release	Open release
Best for	Osteophytes, capsular contracture, loose bodies	**Extensive HO**, **prior ulnar transposition**, severe contracture
Risk	**Radial nerve** (portals) > ulnar > median	Ulnar neuropraxia / damage

– – Active space – –

How it's asked

"Static progressive turnbuckle splint most appropriate?"	→ **3 mo post distal humerus ORIF, arc 45–100°, plateaued** ✗ Wrong options: radial head block to supination · 10–140° arc · ankylosis with HO · after arthroscopic loose body removal
"2.5 mo after fracture-dislocation fixation, arc 75°, DASH 45 — best initial?"	→ **supervised therapy + static progressive splinting** ✗ Wrong options: self-directed · CPM · MUA
"True about posterior oblique MCL?"	→ **released to gain flexion in post-traumatic contracture** ✗ Wrong options: strongest · Tommy John · valgus restraint = anterior bundle · pivot shift = LUCL
"Lacks 40° extension, flexion to 80° — release capsule plus?"	→ **posterior band MCL (B)** ✗ Wrong options: anterior band → valgus instability · LUCL → PLRI
"Relative contraindication to arthroscopic release?"	→ **prior ulnar nerve transposition** ✗ Wrong options: age, gender, osteophytes, labour
"Terrible triad 5 y ago, arc 60–95°, extensive HO — treatment?"	→ **open capsular release + HO excision via lateral column approach** ✗ Wrong options: bone scan · arthroscopy · downsize radial head · interposition arthroplasty
"Nerve most at risk with elbow arthroscopy portals?"	→ **radial**
"Functional elbow ROM?"	→ **30–130° (100° arc)**; pronation/supination **50–50°**

GLENOHUMERAL ARTHRITIS (SHOULDER ARTHRITIS)

– Shoulder & Elbow · 17 / 60 –

1 Basics

Epidemiology

Incidence ↑ with age; mostly **>60 y** · ♀ **>** ♂

Dislocation as risk **56%** of primary anterior dislocators have arthrosis at **25 y**

Aetiology

Type	Key point
Primary OA	No known cause · irreversible cartilage loss + **hypertrophic subchondral bone** · flattened head, osteophytes, cysts, **posterior subluxation + posterior glenoid wear**
Post-traumatic	Healed articular incongruity (humeral fractures, chronic dislocations)
Dislocation arthropathy	↑ with **age >25 y** at dislocation + **posterior** dislocation · **not related to number of dislocations**
Post-capsulorrhaphy	Over-tightening (e.g. **Putti-Platt**) pushes head one way → eccentric wear; presents 5th decade
Cuff tear arthropathy	Torn cuff → head migrates up → abrasion against acromion
RA	Affects **90%** of RA patients · **central glenoid wear + medialisation** · cuff tear 25–50%
CPPD / gout	Pyrophosphate (pseudogout) / urate crystals
AVN	Proximal humerus fracture: **35% 3-part, 90% 4-part** (this topic's Orthobullets note) · chronic dislocation · cuff repair · steroid, EtOH, haemoglobinopathy, Gaucher
Chondrolysis	After arthroscopy: **radiofrequency**, **continuous intra-articular anaesthetic infusion**, bioabsorbable anchors, contrast · **fewer osteophytes** than OA
Neuropathic	Charcot

Cuff tears with OA **5–10%** (Orthobullets) · full-thickness supraspinatus **<10%** in primary OA (OB) · CTA excluded by definition
✗ Wrong options: 30–55% = asymptomatic >60 y population

Anatomy

Version glenoid **~3° retroverted** · humerus **20–30° retroverted**

Stabilisers static: GH ligaments, labrum · dynamic: cuff, rotator interval, biceps

2 Mechanism

Posterior wear + subluxation posterior capsule stretched · **anterior capsule + subscapularis contract → loss of ER**

– – Active space – –

Walch B vs C — **B**: posterior subluxation **causes** posterior wear · **C**: glenoid retroversion causes wear → subluxation later

3 Presentation

Symptoms — pain with motion, often none at rest · difficulty sleeping · **loss of ER** (anterior capsule contracture)

Exam — ↓ ER, FF, IR · crepitus · catching/squeaking · examine cuff carefully

Subscapularis tear — **weak belly-press** + **↑ passive ER** (90° at side)

Outcome scores — ASES · Constant · Oxford

4 Diagnosis & Classification

Key radiology findings

Views — **true AP (Grashey)** + AP + **axillary lateral** (axillary/CT to judge posterior glenoid wear)

Cause	X-ray hallmark
Primary OA	JSN, sclerosis, cysts, **inferior "goat's beard" osteophyte**, **posterior glenoid wear**, fixed posterior subluxation
Dislocation arthropathy	Large osteophytes, hardware
RA	**Osteopenia**, marginal erosions, **few osteophytes**, **central wear + medialisation**, ↓ acromiohumeral distance
CPPD	**Chondrocalcinosis**
Gout	Usually unremarkable; erosions after repeat attacks
AVN	Normal early → central head resorption → **crescent sign** → collapse
CTA	**Superior migration**, ↓ acromiohumeral interval, acromial erosion, superior glenoid loss, **"acetabularisation"** of CA arch, osteopenia



Primary OA with inferior goat's beard osteophyte

CT — **glenoid morphology** + templating · underestimates full-thickness RCT and fatty infiltration vs MRI

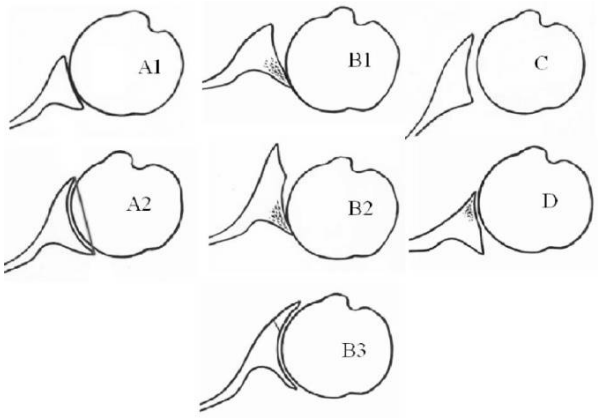
MRI — cuff assessment · **less accurate than CT** for glenoid type · **Goutallier IV** fatty infiltration = irreparable cuff

Glenoid wear patterns (Walch, posterior wear, medialisation in RA)

Most common OA wear — **posterior** (or central)
✗ Wrong options: anterior · superior · inferior

Inflammatory (RA/JRA) vs OA — **concentric erosion + medialisation of joint line** · severe erosion → glenoid resurfacing not always feasible → some favour HA
✗ Wrong options: inferior osteophyte · posterior subluxation · sclerotic glenoid · posterior wear = OA

Walch	Description
A (59%)	**Concentric**, head centred · **A1** minor central erosion · **A2** deep central erosion (rim line transects head)
B	**Biconcave**, asymmetric wear, **posterior subluxation** · **B0** pre-arthritic subluxation · **B1** (17%) posterior JSN, osteophytes, sclerosis, no bone loss · **B2** (15%) posterior rim erosion, **retroversion <25°** · **B3 monoconcave**, posterior wear, subluxation **≥70%** or retroversion **>15°**
C (9%)	**Dysplastic**, **retroversion >25°** · **C1** >25° regardless of erosion · **C2** biconcave + posterior bone loss + posterior translation · subtypes **"Lazy J"** + **"Delta"**
D	**Anteversion** or **anterior** subluxation (<40%)



Walch classification

B2 trap — biconcave + posterior subluxation + **10°** retroversion = B2 · same with 25° = **C2** · deep concentric = A2 · monoconcave + 70% subluxation = B3 · anteversion = D

B2 outcomes — poorer after arthroplasty · anatomic TSA → instability/early glenoid loosening if retroversion **>27°** or **80%** posterior subluxation → **reverse** often better in older/severe deformity

– – Active space – –

5 Management

Glenohumeral arthritis

- First line → **NSAIDs, PT (capsular stretching), activity modification**
- Second line → **intra-articular corticosteroid (HA/PRP/stem cell: limited evidence)**
- Young, high demand → **arthroscopic debridement / CAM (temporising)**
- Failed nonop, end stage → **arthroplasty by cuff status**
  - cuff intact (incl. partial tear) → **anatomic TSA**
  - cuff deficient / >80% subluxation / Walch D → **RSA**

Operative criteria

Procedure	Indications	Contraindications / cautions
Anatomic TSA	**Intact cuff** (partial tear OK), glenoid wear, posterior subluxation, failed nonop	**Deficient deltoid or cuff** · active infection · **Charcot** · young high-demand (glenoid loosening)
Hemiarthroplasty	Young · **RA with irreparable cuff / poor glenoid bone stock** · **AVN without glenoid involvement**	Early failure, poor pain/function in OA (not routine)
Ream-and-run	Young, **intact cuff, no inflammatory arthropathy**	–
RSA	**Irreparable/large cuff tear** · CTA · OA/RA with major glenoid pathology · age · **failed arthroplasty** · complex fracture	Higher complication rate than TSA
Arthroscopic debridement	Mild–moderate OA without structural change · loose bodies · small AVN lesions · synovial chondromatosis	Advanced degeneration
Arthrodesis	Paralysis · recurrent infection · poor deltoid/soft tissue · brachial plexus palsy · failed repair of instability	–

Anatomic TSA vs hemiarthroplasty – indications & cuff status

TSA vs HA
TSA: **better pain relief + function** · HA: **↑ revision rate**
✗ Wrong options: better pain · ↑ blood loss · ↑ instability · ↑ infection

End-stage OA, cuff strong
TSA = most predictable pain relief
✗ Wrong options: RSA unnecessary · arthroscopic osteophyte removal · interposition · HA

Partial articular cuff tear (35%)
good candidate for anatomic TSA (± repair)

Not anatomic TSA candidates
52 y powerlifter (glenoid loosening → HA) · prior aTSA with high-riding humerus (cuff tear → **convert to RSA**) · Walch B **85%** subluxation (RSA if **>80%**) · elderly **Walch D** (RSA)

Tight anterior structures
ER **<40–45°** under EUA → **subscapularis lengthening + anterior capsule release**

TSA outcomes — 10-y survival **92–95%** · complications: **glenoid/humeral loosening**, infection, fracture, nerve injury, cuff tear · **↑ PJI if within 3 months of arthroscopy** · vs RSA: **↑ IR** + **↑ glenoid loosening**

Reverse shoulder arthroplasty – cuff deficiency, post-instability arthropathy, RA

Irreparable cuff + OA (Goutallier IV) — NOT appropriate = **arthroscopic cuff repair** · appropriate = NSAIDs/injection, PT, activity modification, then HA or RSA after failed nonop

Elderly, can't elevate arm (CTA), failed nonop — **RSA** most reliable (Delta III: Constant +42, +65° elevation, **96%** minimal/no pain) · HA = historical standard, non-uniform results

OA + complete subscapularis tear — **RSA**
✗ Wrong options: anatomic TSA: subscap repair fails

Post-Bankart arthropathy + supraspinatus fatty atrophy + posterior subluxation/erosion — **RSA**
✗ Wrong options: HA: glenoid pain persists · TSA: incompetent cuff · arthroscopic release

RA + cuff tear — **RSA good outcomes**, even with **reparable** tear · options = TSA + cuff repair or RSA · **TSA > HA** in RA (pain, elevation, ROM) · **etanercept/DMARDs/prednisone: no effect** on RSA outcome · RSA complications in RA **≈ CTA** (not more) · no interposition arthroplasty in elderly

RSA outcomes — 10-y survival **~90–95%** · complications: **scapular notching**, infection, instability, nerve injury

Young patients – nonoperative care & arthroscopic debridement (CAM procedure)

Young OA first step — **NSAIDs, cortisone injection, PT / ROM stretching** before any surgery
✗ Wrong options: TSA · fusion · resurfacing · arthroscopy as first step

Arthroscopic debridement — **temporising** · improves pain + ROM · better with **age <50**, little JSN, concentric joint, added subacromial procedure · **~20% convert to arthroplasty within 1 y**
✗ Wrong options: 45%

Arthroscopic debridement predictor	Effect
Nonconcentric joint space / moderate–severe degeneration	**Poor**
Posterior glenoid wear, ↑ retroversion	**Poor** (↑ failure)
Bipolar (head + glenoid) OA	Poor
Preoperative stiffness, loose bodies	**Good**
Prior infection / instability without degeneration	No effect

Technique — treat **both glenoid and humeral head aggressively**
✗ Wrong options: ream glenoid + spare humerus

CAM procedure — young patient · GH debridement, chondroplasty, synovectomy, loose body removal, **humeral osteoplasty (goat's beard excision)**, capsular release, subacromial + subcoracoid decompression, **axillary nerve decompression**, biceps tenodesis

– – Active space – –

Inferior humeral osteophyte

axillary nerve entrapment → **teres minor** fatty infiltration
✗ Wrong options: infraspinatus/supraspinatus = suprascapular · subscapularis = upper + lower subscapular · teres major = lower subscapular ± thoracodorsal

6 Complications

Procedure	Typical complications
TSA	**Glenoid / humeral loosening**, infection, fracture, nerve injury, cuff tear (most common group)
RSA	**Scapular notching**, infection, instability, nerve; overall > TSA
HA	Glenoid wear/pain, **↑ revision**
Arthroscopy (pain pump, RF)	**Chondrolysis**

X vs Y

	Primary OA	RA	CTA
Glenoid wear	**Posterior**	**Central, medialisation**	**Superior**
Osteophytes	**Large, inferior**	**Few**	–
Head position	**Posterior subluxation**	Medialised	**Superior migration**
Bone	Sclerosis	**Osteopenia**, erosions	Osteopenia, acetabularisation
Cuff	Tear 5–10%	Tear 25–50%	Torn by definition
Arthroplasty	**TSA**	TSA (+ repair) or RSA; HA if no glenoid	**RSA**



RA: osteopenia, central wear, few osteophytes

	TSA	RSA
Design	Concave glenoid + convex humerus	**Convex glenoid (ball) + concave humerus**
Cuff	Must be intact	Deficient OK
10-y survival	92–95%	~90–95%
Distinct	↑ IR, ↑ glenoid loosening	Scapular notching, ↑ complications

How it's asked

"72 y, OA X-ray, neutral ER — where is glenoid wear?"	→ **posterior**
"Inflammatory vs osteoarthritis at arthroplasty — more likely?"	→ **medialisation of joint line**
"Walch B2 description?"	→ **biconcave, posterior subluxation + erosion, 10° retroversion** ✗ Wrong options: 25° = C2 · anteversion = D · deep central = A2 · monoconcave 70% = B3
"Chance of full-thickness supraspinatus tear at TSA for primary OA?"	→ **<10%**
"TSA vs HA for OA with intact cuff — HA gives?"	→ **↑ revision surgery**
"74 y, end-stage OA, posterior erosion, good cuff strength?"	→ **TSA**
"Best candidate for anatomic TSA?"	→ **60 y with 35% partial articular cuff tear**
"80 y, drop arm, Goutallier IV, GH arthritis — NOT appropriate?"	→ **arthroscopic cuff repair**
"75F, can't elevate arm, cuff arthropathy, most reliable surgery?"	→ **RSA** ✗ Wrong options: HA · resurfacing · debridement · fusion
"Old cuff repair, weak belly-press, ER 90°, end-stage OA?"	→ **RSA**
"Post-Bankart arthropathy, supraspinatus fatty atrophy, posterior erosion?"	→ **RSA**
"RA on etanercept, 2 cm cuff tear — true statement?"	→ **RSA has good outcomes in this cohort**
"Predictor of poor outcome after arthroscopic debridement?"	→ **nonconcentric joint space**
"40 y labourer, OA, no prior treatment?"	→ **NSAIDs, cortisone injection, PT**
"35 y weightlifter, new pain + stiffness, OA X-ray — next step?"	→ **anti-inflammatory drugs + range-of-motion stretching program** ✗ Wrong options: TSA · HA + interposition · resurfacing · debridement first

– – Active space – –

"Inferior humeral osteophyte neuropraxia — fatty infiltration of?"	→ **teres minor** (axillary nerve)
"True about arthroscopy for OA?"	→ **↑ failure with posterior glenoid wear** ✗ Wrong options: 45% conversion at 1 y · >50 y better · bipolar no effect · spare humerus

– – Active space – –

GLENOHUMERAL INTERNAL ROTATION DEFICIT (GIRD)

– Shoulder & Elbow · 18 / 60 –

1 Basics

Epidemiology

Who	**overhead athletes**, esp. **baseball pitchers**; also javelin
Definition	loss of glenohumeral **IR of throwing shoulder vs contralateral** · ↑ ER · ↓ total arc (symptomatic)
Course	deficit worsens with ↑ repetitions over a season + a career; may be **painless**

Definition & thrower adaptations (IR loss, total arc, posterior capsule)

Characteristic motion change	↓ **IR** in throwing arm · pitchers: ↓ **IR** + ↑ **ER** ✗ Wrong options: ER, abduction, adduction, forward elevation

Adaptation (dominant throwing shoulder)	Direction
Humeral head retroversion	↑ (✗ Wrong options: anteversion)
Glenoid retroversion	↑ (✗ Wrong options: anteversion)
Posterior capsule / PIGHL	**Tight + thickened**
Anterior capsule / AIGHL	**Lengthened, attenuated** (✗ Wrong options: stretch it)
ER at 90° + scapular plane	↑
IR	↓

Retroversion mechanism	humerus retroverted in utero → **de-rotates until maturity** → throwing stress on physis **halts de-rotation** → ↑ retroversion → more ER, less IR · bony changes in those who **began pitching before skeletal maturity**
Crockett CT	athletes: ↑ humeral head retroversion, ↑ glenoid retroversion, ↑ ER at 90° + scapular plane, ↓ IR · not in non-athletes
Why ER gain	**arc of acceleration** ↑ → **velocity** ↑ · short IR arc + tight posterior capsule → **deceleration harder** → overuse injury
Posterior capsule thickening	from **repetitive eccentric stress in follow-through**

Total arc of motion concept

Situation	Total arc (IR + ER)
Adaptive, asymptomatic thrower (ERG ≥ GIRD)	**Normal** vs other side; normal kinematics
Thrower vs **average population**	↑ **overall arc** (ER gain > IR loss)
Pathological GIRD (GIRD > ERG)	↓ vs contralateral → deranged kinematics

Total arc	should be **equal between shoulders** · ~**180°** (total motion concept)

– – Active space – –

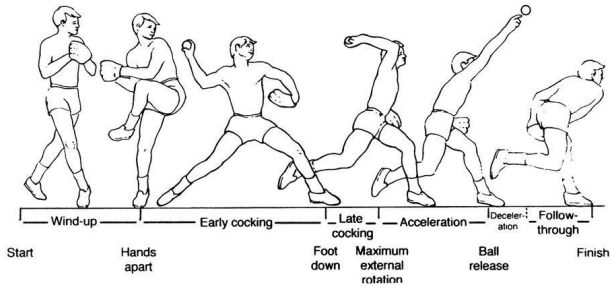
Source nuance	"tight posterior capsule with **normal** total arc" (OB) · "**↑** total arc vs average population" (OB) · "**↓** total arc" in symptomatic GIRD (Orthobullets, OB) – read the stem: adaptation vs pathology
Injury risk ✓	**↓ ER vs non-dominant arm** independently linked to injury (esp. with pre-existing IR deficit) ✗ Wrong options: ↑ ER (compensatory) · ↑ IR (uncommon) · ↑ horizontal adduction · ↓ flexion
Newer view	loss of total arc + **failure to gain ER** may matter more than IR loss alone
GIRD thresholds (TO thrower lecture)	GIRD = IR loss **≥20°** vs contralateral · **physiologic** = total rotational motion (TRM) equal · **pathologic** = **TRM loss >5°**

2 Mechanism

Cause	**repetitive throwing**; during **late cocking + early acceleration**
Capsular constraint mechanism	tight capsule → humeral head translates **OPPOSITE** to tight region

Tight region	Position	Head translation
Posterior capsule	Flexion	**Anterosuperior**
Posteroinferior capsule (PIGHL)	**ABER / cocking**	**Posterosuperior** ✓

Posterosuperior shift ↑ clearance for greater tuberosity but → internal impingement + **peel-back** of biceps anchor (biceps vector shifts posterior)

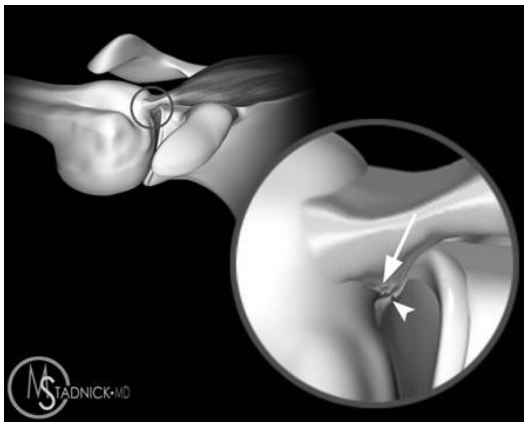


Throwing phases

Throwing phases	wind-up → early cocking → **late cocking (max ER)** → acceleration → deceleration → follow-through · "5 distinct phases" (OB)
Pathologic cascade	↓ velocity + command → posterior stiffness, **slow to loosen up** → posterior pain in late cocking / early acceleration (no mechanical symptoms) → **"SLAP event"** (posterosuperior labrum + biceps anchor fail in tension) → **surgery after SLAP event**; stretching works before it

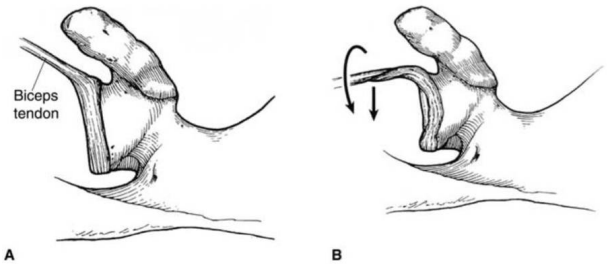
Internal impingement & associated lesions (SLAP, articular-sided cuff tears)

Most closely linked pathology	**internal impingement** ✗ Wrong options: subacromial · HAGL · Hill-Sachs · biceps tendinitis
Internal impingement	in **ABER (late cocking)** greater tuberosity + **posterosuperior glenoid** pinch **articular side of posterior supraspinatus + infraspinatus** and posterosuperior labrum



Internal impingement

Cuff tear in throwers	**articular-sided partial**, **SS–IS junction** (posterior SS + anterior IS) · (✗ Wrong options: bursal-sided · full-thickness (not described)) · general population = anterior supraspinatus
Tear mechanism	**compressive** (GT vs glenoid) + **tensile** (repetitive eccentric load, hyper-ER twisting)
Lesion most associated	**SLAP tear** (type II, posterior extension) ✗ Wrong options: Bankart · HAGL · anterior capsular contracture
SLAP risk	throwers with GIRD **25% more likely** to have SLAP · **peel-back** in late cocking



Peel-back mechanism

Cause of internal impingement	**posterior capsular contracture (GIRD)** or **anterior capsular laxity**
Chondral injury	**posterosuperior** ✗ Wrong options: anteroinferior = anterior instability
Associated conditions	GH instability · internal impingement · articular-sided partial cuff tears · SLAP · **ipsilateral elbow pain** (altered throwing)

SICK scapula

Letter	Meaning
S	Scapular malposition (hangs lower)
I	**Inferior medial border prominence**
C	**Coracoid pain** + malposition (pec minor contracture → tendinitis)
K	Dys**K**inesis of scapular movement

– – Active space – –

3 Presentation

Symptoms	vague / **posterior** shoulder pain · may be painless · **↓ velocity + command** · "dead arm" · arm heavy in later innings · slow to loosen up
Exam	**stabilise scapula** for true GH rotation · ↓ IR + ↑ ER at 90° abduction · **↑ sulcus sign** (stretched CHL + rotator interval)
Internal impingement exam	deep **posterior** pain at **90° abduction + max ER**; negative apprehension
Subacromial impingement (wrong option)	**anterolateral** pain with IR in abduction

4 Diagnosis & Classification

Diagnostic criteria

Criterion	Value
Clinically relevant GIRD	**IR loss >25°** vs contralateral (Orthobullets, OB) · **20–25°** (OB, Little Leaguer's) — read the options
Range reported	up to **50°**
Pathological	**GIRD > ERG** (↓ total arc)
No	**<15°** loss

Key radiology findings

X-ray	AP + lateral; **usually normal**
CT	**↑ glenoid retroversion**
MRI	**ABER view** shows associated lesions (articular-sided cuff tear, labrum) · often normal · not needed initially for isolated posterior tightness

Little Leaguer's shoulder (proximal humeral epiphysiolysis)



Little Leaguer's shoulder — widened physis

X-ray	**widened proximal humeral physis** + metaphyseal changes · **17% normal**
Risk	pitchers throwing **breaking pitches**, ignoring pitch counts/rest, year-round pitching

GIRD link GIRD exam findings at diagnosis → **increased likelihood of recurrence (3×)**, mean **6.2–7.6 months** · **pain after 2 months PT** = independent predictor of recurrence

Treatment stop throwing (**~3 months** no pitching) → PT (restore motion, kinetic chain) → return-to-throw · **no breaking balls until skeletal maturity**

5 Management

GIRD in thrower

- First line → **rest from throwing + posterior capsular stretching** *(sleeper + cross-body + periscapular/pec minor + cuff)*
- Fails extensive PT → **arthroscopic posteroinferior capsular release** *(some advocate anterior stabilisation)*
- After "SLAP event" → **surgery (SLAP repair)**

Treatment – posterior capsular stretching (sleeper/cross-body) & scapular work

Initial ✓ **PT directed at posteroinferior capsule stretching** + periscapular strengthening, **regardless of pathology** · then cuff + scapular stabilisation

Regimen rest from throwing + PT **6 months** (Orthobullets)

Emphasise **sleeper + cross-body stretches + periscapular strengthening**

Sleeper stretch lying on affected side, shoulder **90° abduction** (Orthobullets, OB) · **90° forward flexion** (OB), elbow 90°, push forearm to table, **scapula stabilised** → stretches **posterior capsule**
✗ Wrong options: anterior · anteroinferior · rotator interval · MGHL



Sleeper stretch

– – Active space – –

Stretch / exercise	Target	Restores IR?
Sleeper stretch	Posteroinferior capsule	**Yes**
Roll-over sleeper	Arm flexed **60°**, body rolled forward **30°**	Yes
Cross-body adduction	Posterior capsule	Yes
Doorway stretch	–	–
Pec minor stretch (rolled towel, push shoulders back)	SICK scapula, coracoid pain	No (adjunct)
Rows (retraction) · punches (protraction)	Scapula	No
Humeral head depression, ball on wall	Closed-chain cuff	No

Coracoid pain + GIRD — **pec minor stretching + sleeper stretches**
✗ Wrong options: weighted ball program · subacromial steroid · arthroscopy

Stretching protocol (TO) — posteroinferior capsule only (not global): **sleeper** (90°, scapula stabilised) · **cross-body adduction** · **doorway** (90° abduction, lean forward) · **roll-over** (50–60° flexion) · **5 reps × 30 s daily × 6 wk**

Outcomes — **~90%** respond (young throwers) · ~10% older throwers fail → arthroscopic release

Internal impingement + partial posterior SS tear — **NSAIDs + posterior capsular stretching + cuff strengthening** · refractory → SLAP repair / cuff debridement
✗ Wrong options: cuff repair

✗ Wrong initial option	Why
Arthroscopic posteroinferior release	Only after failed PT
Anterior(-inferior) capsular stretch / release	Anterior capsule already **lax/lengthened**
Selective ER stretching	ER already normal / ↑
Posterior labral repair + capsulorrhaphy	**Posterior instability** (linemen, weightlifters)
Capsulolabral imbrication / plication	**MDI**
Subacromial steroid	Pathology is **intra-articular**
Sling	Comfort only
Cessation 3–6 months alone	Does not address contracture; deconditioning
↑ deltoid / latissimus weights	↑ stiffness
Change pitch mix	Pitch selection not a risk factor for GIRD
MRI arthrogram first	Not needed for isolated tightness

Operative criteria

Procedure	Indication
Arthroscopic posteroinferior capsule release (± PIGHL)	**Failure of extensive PT** only
Anterior stabilisation	Alternative (controversial)
Repair thinned cuff	Significant thinning (transcuff or takedown)

Posterior capsule release vs anterior stabilization

– – Active space – –

Technique	electrocautery via **posterior portal**, camera **anterior** · release posteroinferior capsule + **posterior band IGHL** from **9 to 6 o'clock** at glenoid rim until cuff fibres seen · **shaver widens gap** (prevents recurrence) · gentle manipulation
Result	immediate gain **~65° IR**

X vs Y

	GIRD / internal impingement	**Subacromial impingement**	**Adhesive capsulitis**
Pain	**Deep posterior**, ABER	**Anterolateral**, IR in abduction	Global
Motion	↓ IR, ↑ ER	Normal	**Global active + passive ↓**
Cuff tear side	**Articular**	Bursal	–

	GIRD thrower	**Anterior instability**
Anterior capsule	Lax (microtrauma)	Torn / Bankart
Lesion	**SLAP**, articular cuff	**Bankart, HAGL**
Chondral injury	**Posterosuperior**	Anteroinferior

How it's asked

"Motion decreased in thrower's dominant arm?"	→ **internal rotation**
"Dominant thrower adaptations?"	→ **tight posterior capsule with normal total arc** ✗ Wrong options: humeral anteversion
"Pitcher, ↓ velocity, posterior pain, slow to loosen up – exam?"	→ **IR loss >25°**
"Side-lying stretch, arm 90°, push forearm to table – diagnosis + finding?"	→ **GIRD; >25° IR loss** ✗ Wrong options: adhesive capsulitis · <15°
"ROM finding most associated with injury risk?"	→ **decreased ER** vs non-dominant
"12-year-old, GIRD + widened physis – clinical course?"	→ **↑ recurrence** ✗ Wrong options: elbow pain unlikely · non-breaking pitches
"Pitcher, 30° IR loss vs average population?"	→ **thickened posterior capsule** ✗ Wrong options: ↓ total arc · humeral anteversion · anteroinferior chondral injury
"Sleeper + prone IR exercises – diagnosis?"	→ **GIRD**

– – Active space – –

"Pitcher, 30° IR loss, positive O'Brien, normal MRI – emphasise?"	**→ sleeper + cross-body + periscapular strengthening**
"Total arc 150°, scapula hangs low – initial management?"	**→ aggressive PT, posterior capsular stretching + scapular strengthening**
"IR stretch at 90° flexion, scapula stabilised – stretches?"	**→ posterior capsule**
"Pitching staff, mean GIRD 45° – intervention?"	**→ posterior capsular stretching**
"IR deficit 25°, negative O'Brien/Neer/Hawkins – next step?"	**→ posterior capsular stretching program** ✗ Wrong options: MRI · arthroscopy
"ER 110° vs 100°, IR 30° vs 70° – best treatment?"	**→ posterior capsular stretching** ✗ Wrong options: selective ER stretching · thermal shift
"Late-cocking posterior pain, IR 25° vs 60°, O'Brien painful – next?"	**→ posterior capsular stretching** ✗ Wrong options: SLAP repair · subacromial injection
"Home exercise to improve IR deficit?"	**→ sleeper stretch** ✗ Wrong options: rows · punches · towel pec stretch
"↓ IR + inferomedial scapular prominence + partial posterior SS tear – treatment?"	**→ NSAIDs + posterior capsular stretching + cuff strengthening**
"Pitcher, figure of GIRD – initial treatment?"	**→ PT for posteroinferior capsule stretching** ✗ Wrong options: anteroinferior stretch/release
"GIRD + coracoid tenderness, normal MRA – next?"	**→ pec minor + sleeper stretches**
"IR 5° vs 15°, ER 115° vs 90° – next?"	**→ posterior capsular stretches** ✗ Wrong options: 3–6 months rest · anterior stretches
"Humeral translation in cocking with posterior contracture?"	**→ posterosuperior**
"GIRD linked most closely to?"	**→ internal impingement**
"Range of motion in pitchers with internal impingement?"	**→ ↓ IR, ↑ ER**
"Javelin thrower, posterior pain at 90° abduction + max ER?"	**→ internal impingement**
"Arthroscopy: posterior SS + posterosuperior labral lesion – throwing phase?"	**→ late cocking**
"Lesion most associated with internal impingement?"	**→ SLAP tear** ✗ Wrong options: Bankart · HAGL · full-thickness tear
"Sleeper + cross-body stretches – associated EXCEPT?"	**→ bursal-sided partial cuff tear**
"Deep posterior pain at max ER, 30° less total arc – MRI?"	**→ articular-sided SS/IS partial tear**

"Dead arm, total arc −45° – arthroscopic finding?"	→ **SLAP lesion** ✗ Wrong options: glenoid anteversion · bursal tear	– – Active space – –

– – Active space – –

GLENOHUMERAL JOINT ANATOMY, STABILIZER, AND BIOMECHANICS

– Shoulder & Elbow · 19 / 60 –

1 Basics

Scapulohumeral rhythm

GH : ST ratio (TO)	**2:1** ✗ Wrong options: 1:2 · 2.5:1 · 3:1
First 30° elevation (TO)	largely **glenohumeral**; scapulothoracic small + inconsistent → after 30° both move **simultaneously**
180° abduction	**120° GH (⅔) + 60° scapulothoracic (⅓)**
Rhythm variation (TO thrower lecture)	ratio shifts toward **1:1 at ~60°**, ~**5:4** near 120° · without scapular rotation → deltoid active insufficiency → abduction limited to **60–70°**
Scapular plane	**30° anterior** to coronal plane
Abduction needs ER	clears greater tuberosity from acromion → **IR contracture → cannot abduct >120°**

Osteology

Structure	Key numbers
Humeral head	Spheroidal in 90% · diameter **43 mm** · **inclination 130°** to shaft · tuberosities = cuff attachments
Glenoid	**Pear-shaped** · upward tilt **5°** · covers only **25–30% of head** (TO)
Coracoid	**"Lighthouse"** of deltopectoral approach · coracobrachialis, pec minor, short head biceps
Acromion	3 ossification centres: **pre- (tip), meso- (mid), meta-acromion (base)**
Acromiohumeral interval	**7–8 mm**

Glenoid & humeral version

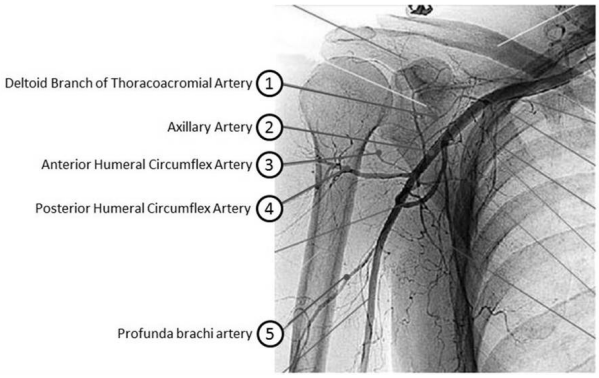
	Glenoid	Humeral head
Orthobullets	**5° retroversion** avg (scapular body axis); **7° retro → 10° ante**	**~20° retroversion** (transepicondylar axis)
TO	**Near 0°**, more often **slight retroversion**; <10° either way	**Retroverted 26° ± 11°**
TO (bone-loss lecture)	**~7° retroversion** to scapular plane · **~5° superior inclination**	**~30° retroversion** · neck–shaft **130°** · diameter **~43 mm**

Humeral retroversion	**20°** (Orthobullets) · **26° ± 11°** (TO) – read the options
Humeral retroversion (TO bone-loss lecture)	**30°** – read the options

False (TO)	**"both glenoid and humeral head anteverted"**

Humeral head blood supply

Main supply	**posterior humeral circumflex artery** (current literature) → injury most likely → **AVN**
Why PHCA?	**AHCA disrupted in ~80%** of proximal humerus fractures yet AVN infrequent
Ascending branch of AHCA	runs **parallel to lateral aspect of LHB** in bicipital groove → **arcuate artery** = intraosseous continuation into head; do not injure when plating



Arteriogram: 4 = posterior humeral circumflex artery

✗ Arteriogram distractor	Label
Deltoid branch thoracoacromial	1
Axillary	2
Anterior humeral circumflex	3
Profunda brachii	5

Labral anatomy & blood supply

Composition	**fibrocartilage** (TO)
Function	**50% of socket depth** · cavity-compression · ↑ depth, ↑ articular surface area, **centralises head**, **seal → intra-articular pressure** (TO)
Embryology (TO)	labrum forms **~8th week**; LHB + capsule insertion defined by **9th week**
Blood supply	**suprascapular** · **circumflex scapular** (TO) / anterior humeral circumflex (Orthobullets) · **posterior humeral circumflex**; from **capsule + periosteal vessels**, not underlying bone
Poorest blood supply	**anterosuperior labrum** (TO) ✗ Wrong options: posterosuperior
Biceps attachment (TO)	to labrum, **mainly posterior**
Weak links	anterior labrum anchors **IGHL → Bankart** · superior labrum anchors **biceps → SLAP**

– – Active space – –

2 Mechanism

Static vs dynamic stabilisers

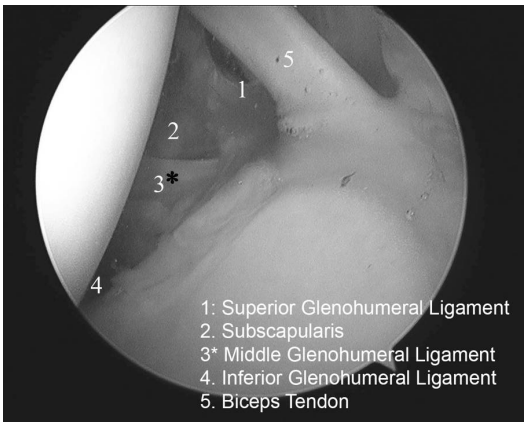
Static	Dynamic
Glenohumeral ligaments	**Rotator cuff** (main)
Glenoid labrum	Rotator interval
Articular congruity + version	**Long head biceps** (humeral head depressor)
Negative intra-articular pressure (release → head subluxes **inferiorly**)	Periscapular muscles

Bony articulation **inherently unstable**

Posterior capsule thin (**<1 mm**), **no ligaments**

Glenohumeral ligaments – function by arm position (IGHL, MGHL, SGHL, CHL)

Arm position	Anterior restraint	Inferior restraint	Posterior restraint
0° (side) + adduction		**SGHL / CHL**	
45° abduction + ER	**MGHL**		**MGHL**
90° abduction + ER	**Anterior band IGHL**	**Anterior band IGHL**	**Posterior band IGHL**
90° flexion, adduction, IR	Anterior band IGHL		**SGHL / CHL**



Arthroscopic view: 1 SGHL, 2 subscapularis, 3 MGHL, 4 IGHL, 5 biceps

Anterior static stability (TO) **anterior band IGHL = main restraint** (90° abduction + ER) · **MGHL** (45° abduction + ER) · **SGHL** (arm at side)

Hammock (TO) at **90° abduction** IGHL = hammock: **anteroinferior band = anterior** support, **posteroinferior band = posterior** support

Abduction + ER main stabiliser (TO) **IGHL**

>45° abduction, anterior + posterior stress (TO) **anterior inferior GHL**
✗ Wrong options: SGHL · CHL · anterosuperior labrum

– – Active space – –

SGHL anterosuperior labrum → humerus · **inferior translation at 0° abduction / adduction** (neutral rotation) → **greatest stress = inferiorly translated, arm adducted** · restrains **ER of adducted arm** · prevents **anteroinferior translation of LHB** (biceps pulley)

MGHL **anterior (and posterior) translation in midrange ~45° (45–60°) abduction + ER** · from labrum, crosses **subscapularis obliquely (~45°)** → **just medial to lesser tuberosity** · do not confuse with subscapularis

Anterior band IGHL **anterior/inferior translation at 90° abduction + maximal ER** (**late cocking**) · anchors anterior labrum → **Bankart weak link**

Posterior band IGHL **posterior subluxation at 90° flexion + IR** (forward flexed, adducted) · restraint in **posterior instability** (Kim test) → must repair · **tightness → internal impingement** + ↑ shear on superior labrum (SLAP)

Superior band IGHL **most important static stabiliser** (Orthobullets); **100% ↑ strain with SLAP lesion**

CHL coracoid → **rotator cable** · limits **posterior translation in flexion, adduction, IR** · limits **inferior translation + ER in adduction**

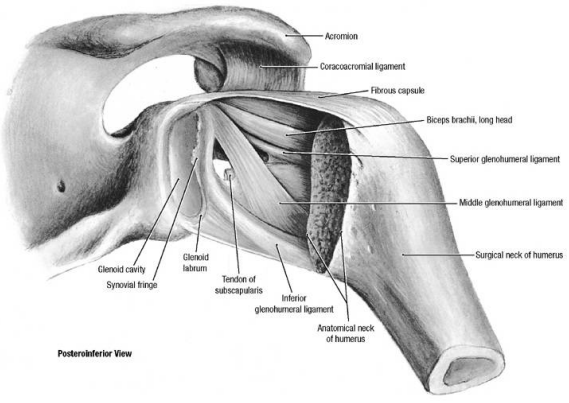
Primary restraint to inferior translation **CHL** (OB; Bigliani: inferior capsule + IGHL also contribute) · **SGHL/CHL** at 0° (Orthobullets, OB) – read the options

Ligament anatomy (TO)

Ligament	Origin → insertion (TO)	Note
CHL	**Proximal ⅓ of horizontal coracoid** → variable: superficial fibres **greater tuberosity**; only **15–50%** to lesser tuberosity	Blends with supraspinatus + subscapularis
SGHL (Flood ligament)	**Superior glenoid tubercle** → **fovea capitis** just above lesser tuberosity	Key **stabiliser of LHB** in groove
MGHL	Just beneath/medial to SGHL → **just medial to lesser tuberosity**	Blends with subscapularis
IGHL	Anterior + posterior bands + **axillary pouch** (reinforced by **fasciculus obliquus**) → humeral neck like a **collar** under articular margin	Hammock

Foramina (TO) **Weitbrecht** = between SGHL + MGHL · **Rouvière** = between MGHL + IGHL

IGHL anterior band origin **2–4 o'clock** (one TO) · **3–5 o'clock** (another TO); posterior band **7–9 o'clock** – read the options



Glenohumeral ligaments (Netter)

– – Active space – –

Stabilisers by abduction, neutral rotation (TO)

Abduction	Anterior	Posterior	Inferior
Rest	Supraspinatus	Posterior capsule	**Superior capsule, CHL, SGHL**, supraspinatus, biceps
0–45°	Subscapularis, **MGHL**, IGHL	Posterior capsule, IGHL	IGHL, MGHL
45–90°	**IGHL**, posteroinferior capsule, biceps (IR), cuff, deltoid	Posterior capsule, IGHL, biceps (ER)	IGHL, MGHL
>90°	**IGHL, axillary pouch**, subscapularis, infraspinatus	Posterior capsule, IGHL	–

Glenoid inclination also listed among stabilisers in the TO table

Rotator interval contents

Definition triangular space between **upper border of subscapularis** + **anterior margin of supraspinatus** (TO); bridged by capsule

Boundary	Structure
Base (medial)	**Coracoid** (lateral base)
Superior	Anterior edge **supraspinatus**
Inferior	Superior border **subscapularis**
Apex (lateral)	**Transverse humeral ligament**

Contents **CHL + SGHL** + LHB + thin capsule; CHL + SGHL = most constant structures of fibrous capsule
✗ Wrong options: MGHL

Layers (TO) 4 layers incl. **superficial CHL** · supraspinatus + subscapularis fibres · **deep CHL**

Arthroscopy area between **biceps + subscapularis tendon** = **rotator interval**; biceps deep to interval

Contracture **adhesive capsulitis** → release · **laxity** → **sulcus sign** (inferior laxity in adduction) → **RI closure / CHL imbrication** improves inferior stability (Harryman)

Dynamic stabilisers – concavity compression & rotator cuff

Rotator cuff **main dynamic stabiliser**; **compresses humeral head into glenoid**; works most in **midrange**, not extremes

Concavity compression **most important midrange stabilising mechanism**; depends on **depth of concavity + magnitude of compressive force**
✗ Wrong options: isometric ligaments · capsule tension · deltoid

✗ Distractor	Actual role
SGHL, MGHL	**Static** stabilisers
Teres major	Adducts + IR; not a significant stabiliser
Deltoid	Abducts; not main stabiliser

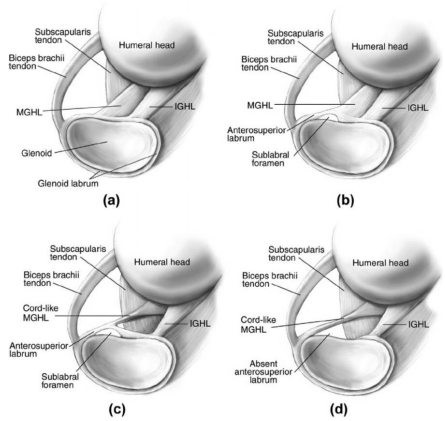
Free body analysis of deltoid

Arm 90° abduction A (deltoid lever) = 3 cm, B = 30 cm → (A × D) − (B × 0.5W) = 0 → **D = 5W**

4 Diagnosis & Classification

Normal variants (Buford complex, cord-like MGHL) – do not repair

Variant	Frequency	Description
Normal (labrum on rim + flat/broad MGHL)	Most common	Cord-like MGHL quoted in **86%** (Orthobullets)
Sublabral recess (sulcus) (TO)	**Most common** variant	**11–1 o'clock** at biceps–labral complex; smooth, **<2 mm**; follows rim **medially**; **does not extend posterior to biceps anchor**
Sublabral foramen (hole) (TO)	**~12%** (Orthobullets) · **11%** (TO)	Anterosuperior labrum detached **1–3 o'clock**, anterior to biceps anchor; smooth, **<1.5 mm**
Sublabral foramen + cord-like MGHL		
Buford complex	**~1.5%**	**Cord-like MGHL + absent anterosuperior labrum** (MGHL attaches to base of biceps anchor)
Meniscoid labrum	1%	



Anterosuperior labral variants: (a) normal, (b) sublabral foramen, (c) foramen + cord-like MGHL, (d) Buford

Buford (TO) **normal variant**, 1.5% · cord-like **MGHL** (✗ Wrong options: SGHL) + absent **anterosuperior** (✗ Wrong options: anteroinferior) labrum · not abnormal · **should not be repaired** → ↓ ROM

Repairing Buford to glenoid **painful loss of ER** (and elevation) · **loss of ER with arm at side** > at 90° abduction (MGHL tighter at side)

Sublabral foramen vs tear (TO) true anterosuperior tear **propagates superiorly into biceps anchor or inferiorly into IGHL** attachment

Tubercle of Assaki (TO) ridge of subchondral bone at **centre of glenoid** with focal **cartilage thinning** → do not mistake for cartilage defect

Subacromial pseudospur (TO) prominence of acromial angle at **CA ligament** attachment → mimics caudal osteophyte; also at deltoid tendon attachment → mimics enthesophyte

– – Active space – –

Key radiology findings

Sublabral recess — **best seen with arthrography** (TO)

AHI — **7–8 mm**; normal on X-ray but ↓ on supine MRI (arm weight removed) → **multiple-tendon cuff tear**

HAGL on MRI — tear of **humeral attachment of IGHL**

Acromial morphology

Type	Shape
I	**Flat**
II	**Curved**
III	**Hooked**

HAGL lesion

Mechanism — **abduction + hyperextension + ER** (e.g. 15-year-old wrestler)

HAGL — humeral avulsion of IGHL; **much less common than Bankart** (anteroinferior labral tear)

✗ Distractor	Why
Lesser tuberosity avulsion	Very rare; **12–13-year-olds**
Subscapularis tear	Same mechanism, **older** patients

5 Management

Arthroscopic vs open repair – external rotation preservation

Advantage of arthroscopic — **preservation of ER** (spares subscapularis; no anterior capsular incision)

Not advantages — ↓ healing time · outpatient (both) · faster return (recovery identical) · ↓ recurrence (arthroscopic **slightly higher or equal**)

Shoulder arthrodesis position

Plane	Angle
Abduction	**15–20°**
Forward flexion	**20–25°**
Internal rotation	**40–50°**

Other sources — **30-30-30** (Orthobullets arthrodesis note) · ~40° abd / 35° FF / 20° IR (OB) – read the options

X vs Y

Ligament	Position tested	Restrains
SGHL	**Arm at side (0°/adducted)**	**Inferior** translation, ER
CHL	Adducted · flexion + adduction + IR	**Inferior** + ER · **posterior**
MGHL	**45° abduction + ER**	**Anterior** (± posterior)
AIGHL	**90° abduction + max ER**	**Anterior / inferior**
PIGHL	**90° flexion + IR** (adducted)	**Posterior**

	Sublabral recess	Sublabral foramen	Buford complex
Position	**11–1 o'clock** (biceps anchor)	**1–3 o'clock** (anterosuperior)	Anterosuperior
Labrum	Superior labrum not attached to cartilage	Detached from rim	**Absent**
MGHL	Normal	Normal / cord-like	**Cord-like**
Width	<2 mm	<1.5 mm	–
Repair?	No	No	**No (→ loss of ER)**

How it's asked

"Most important static stabiliser at 45° abduction + ER?"	→ **MGHL** (TO)
"Main stabiliser in abduction + ER?"	→ **IGHL** (TO)
"At 90° abduction, main stabiliser against anterior dislocation?"	→ **anterior-inferior GHL** (TO)
">45° abduction, restraint to anterior + posterior stress?"	→ **anterior inferior GHL** (TO)
"Primary static restraint to anterior translation at 90° abduction?"	→ **IGHL complex** ✗ Wrong options: bony shape · negative pressure
"SGHL under greatest stress?"	→ **inferiorly translated, arm adducted (5°)**
"Asterisk structure (crosses subscapularis) – function?"	→ **prevents anterior translation at 45° abduction** (MGHL)
"Labelled ligament restraining anterior translation at 45° abduction + ER?"	→ **MGHL (D)**
"Primary restraint to inferior translation?"	→ **coracohumeral ligament** ✗ Wrong options: MGHL · subscapularis · CAL
"90° abduction, full ER – resists anterior translation?"	→ **anterior band of IGHL complex**
"Arrowed structure, posterior portal – function?"	→ **resists anterior + posterior translation at 45° abduction** (MGHL)
"Posterior instability, + Kim test – structure to repair?"	→ **posterior band IGHL** ✗ Wrong options: AIGHL · CHL

– – Active space – –

"Besides biceps, rotator interval contains?"	→ **CHL + SGHL** ✗ Wrong options: MGHL
"Arthroscopic area between biceps and subscapularis?"	→ **rotator cuff interval**
"Dynamic stabiliser compressing head into glenoid?"	→ **rotator cuff muscles** ✗ Wrong options: teres major · deltoid
"Most important midrange stabilising mechanism?"	→ **concavity compression**
"Buford complex?"	→ **normal variant: cord-like MGHL + absent anterosuperior labrum** (TO)
"Repairing Buford complex causes?"	→ **loss of external rotation**
"Probe on cord-like MGHL repaired to glenoid — result?"	→ **loss of ER with arm at side**
"Asterisk, posterior portal, anterior structure crossing subscapularis?"	→ **MGHL**
"15-year-old wrestler, abduction-hyperextension-ER, MRI?"	→ **tear of humeral insertion of IGHL (HAGL)** ✗ Wrong options: lesser tuberosity avulsion · Bankart
"AVN after proximal humerus fracture — artery?"	→ **posterior humeral circumflex (4)**
"Advantage of arthroscopic anterior stabilisation?"	→ **preservation of external rotation**
"False about glenoid labrum?"	→ **posterosuperior labrum has poorest blood supply** (TO) (anterosuperior is poorest)
"GH : ST motion ratio?"	→ **2:1** (TO)
"False about glenoid/humeral version?"	→ **both anteverted** (TO)

INTERNAL IMPINGEMENT

– Shoulder & Elbow · 20 / 60 –

1 Basics

Epidemiology

Who major cause of shoulder pain in **throwing + overhead athletes**; very common in **elite baseball pitchers** · also swimming, volleyball, javelin

Definition repetitive contact of **undersurface (articular side) of posterior rotator cuff** + greater tuberosity against **posterosuperior glenoid/labrum**

Impinged structure **posterior rotator cuff**
✗ Wrong options: anterior cuff · GH ligaments · biceps

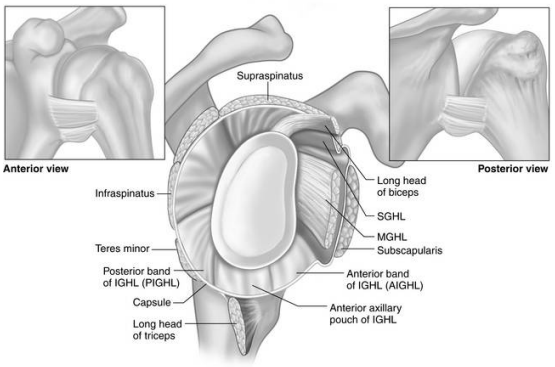
Contact may be **physiologic or pathologic** in throwers

Anatomy – glenohumeral stability & cuff force couples

Restraint	Structures
Static	**GH ligaments**, labrum, articular congruity + version, negative intra-articular pressure
Dynamic	Rotator cuff, biceps, periscapular muscles

Force couple plane	Balance
Coronal	**Inferior cuff (IS, TM, subscapularis)** vs superior moment of **deltoid**
Transverse	**Subscapularis (anterior)** vs **IS + TM (posterior)**

Cuff role dynamic stability, centres head → stable fulcrum; goal of cuff repair = restore equilibrium



Glenohumeral ligaments – posterior band of IGHL

– – Active space – –

Ligament	Role in internal impingement
Posterior band IGHL (PIGHL) + posterior capsule	**Tight + thickened** ✓ (target of sleeper stretch)
Anterior band IGHL	**Lax** (✗ Wrong options: tight)
Anterior axillary pouch, MGHL, SGHL	Not implicated
Superior band IGHL	Most strain in **SLAP** lesions

2 Mechanism

Pathomechanics – late cocking, abduction–external rotation & posterosuperior contact

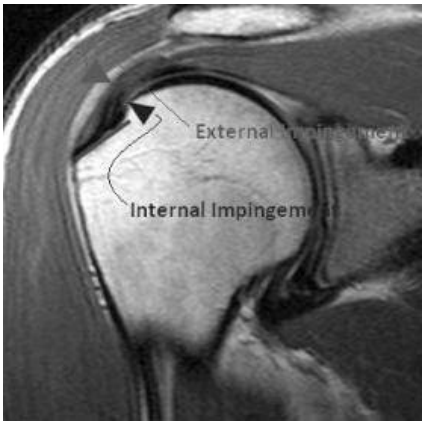
Position — **abduction + maximal ER**
✗ Wrong options: adduction/IR · abduction/IR · flexion/ER

Phase — **late cocking** (+ early acceleration); max ER
✗ Wrong options: wind-up · early cocking · deceleration · follow-through

Contact — **posterior articular supraspinatus** (± anterior infraspinatus) between greater tuberosity + **posterosuperior glenoid**

Peel-back — posterosuperior labrum peeled back by biceps → SLAP

Internal vs external — internal = **articular side** · subacromial ("external") = **bursal side**



Internal vs external impingement

Posterior capsular contracture & GIRD – stretching

Most clearly associated cause — **posterior capsular contracture** (✓ primary process)
✗ Wrong options: anterior capsular laxity (variable, less direct) · CA arch stenosis (subacromial) · Bennett lesion (no causal link)

Contracture effect — blocks normal anteroinferior translation in late cocking → **obligatory posterosuperior translation** → posterosuperior labral + articular cuff pathology

Etiology — tight **posterior band IGHL** + **anterior micro-instability** · multifactorial: altered biomechanics, subtle instability, posterior contracture, scapular dyskinesia

Primary etiology — **posterior capsular contracture** (OB, Orthobullets) · **GH instability** commonly primary (one OB explanation) – read the options

Anterior labrum — may become **thinned** (obligate translation → anterior micro-instability)

Associated conditions — **GIRD**, SLAP tears, **SICK scapula** + dyskinesia, biceps tendinitis, anterior instability

3 Presentation

Symptoms diffuse **posterior shoulder** pain (posterior deltoid), deep-seated · worse with throwing, esp. **late cocking + early acceleration** · ↓ velocity

Inspection proximal humeral retroversion · tender along **infraspinatus**

ROM **↑ ER + ↓ IR** · IR loss **>20° at 90°** (Orthobullets) · stabilise scapula · total arc **preserved** (Orthobullets, OB) · total arc **↓** (one OB stem) – read the stem

Strength rotator cuff weakness often (may be 5/5)

Isolated finding ↓ IR with **no other positive provocative test**
✗ Wrong options: sulcus sign · Speed's · ↓ ER · ↓ abduction

Test	Technique	Positive
Apprehension test (internal impingement)	Max ER + abduction + extension	**Posterior** pain, relieved in neutral
Whipple	Forward flexion + adduction + scapular retraction, resist	Pain → **partial supraspinatus tear**
Sulcus sign	Downward pull, arm adducted	**Inferior instability** (superior GH laxity)
Speed's	90° FF, supinated, resist downward	**Biceps tendinitis**

4 Diagnosis & Classification

Classification **none formal**

Key radiology findings

X-ray complete shoulder series; usually normal · **Bennett lesion** on AP



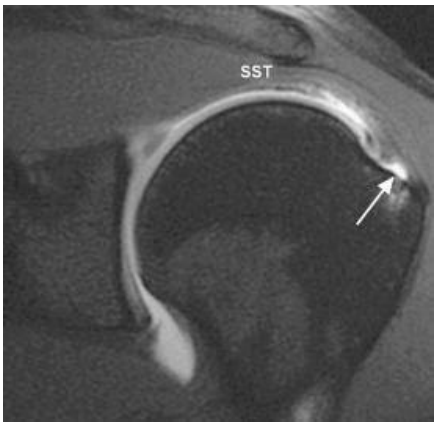
Bennett lesion

Bennett lesion **mineralisation / exostosis of posteroinferior glenoid** = **traction spur** of posteroinferior capsule (PIGHL) · formed in **deceleration + follow-through** · **baseball pitchers** · **22%** of MLB pitchers, **none needed surgery**
✗ Wrong options: football, swimmers, rowers

MRI / MRA **PASTA** (partial articular supraspinatus–infraspinatus tendon avulsion), fraying, tear · signal at **greater tuberosity** (posterolateral head edema, **cysts**) + **posterosuperior labrum** · ↑ anterior capsular volume · thickened **PIGHL**

ABER view reproduces impingement position; dynamic humeral + glenoid sides

– – Active space – –



PASTA lesion

Associated lesions & diagnosis (articular-sided cuff tears, posterosuperior labral fraying)

Lesion (spectrum)	Detail
Articular-sided partial cuff tear	**Supraspinatus–infraspinatus interval** (✗ Wrong options: bursal = external)
Posterior + superior labral lesion	**SLAP with posterior extension**; posterosuperior fraying
Bennett lesion	Posterior capsule/glenoid hypertrophy + scarring (calcified)
Chondral injury	**Posterior glenoid**
Greater tuberosity	Cystic changes
Capsule	**Lax anterior**, **tight posterior**

Not part of it	subscapularis tear · bursal-sided tear · Bankart · Hill-Sachs · tight anterior capsule · lax posterior capsule · full-thickness tear (needs significant trauma)
Bennett + ↓ arc + deep pain + posterosuperior labral fraying	**internal impingement** ✗ Wrong options: SLAP alone · subacromial = Hawkins + · calcific tendinitis
Javelin thrower, failed PT, SLAP with posterior extension	**internal impingement** ✗ Wrong options: EDS/MDI · traumatic anterior instability · osteonecrosis

Asymptomatic MRI findings in pitchers

Common asymptomatic findings	partial cuff tear · SLAP / posterior-superior labral tear · **↑ humeral retroversion** · thick posterior capsule · posterolateral head cysts
Asymptomatic pitcher	**continue pitching** ✗ Wrong options: rest · posterior release · cuff repair · rotational osteotomy
GIRD definition	**IR deficit 15° + total arc deficit ≥10°** (OB) · **>20° IR loss** (Orthobullets) · **>25°** (Orthobullets GIRD note) – read the options

5 Management

Internal impingement

- First line → **stop throwing + PT (posterior capsule stretching)**
- Failed extended PT + PASTA <50% / Bennett / peel-back → **arthroscopic debridement**
- Failed PT + partial >50% / full-thickness / unstable labrum → **repair (arthroscopic or mini-open)**
- Persistent posterior contracture / anterior instability → **add posterior release or anterior plication**

Nonoperative

Regimen	break from throwing until pain-free → supervised return with proper mechanics · **sleeper + cross-body stretches**, cuff strength balancing, scapular stabilisation, kinetic chain · **6 months** stretching
Stretch target	**posteroinferior capsule** · **posterior band IGHL** ✗ Wrong options: superior, anterior, anteroinferior, inferior
Prevention ✓	**posterior capsular stretching** ↓ shoulder + elbow injury risk ✗ Wrong options: cuff strengthening · periscapular strengthening (less direct) · SLAP / partial cuff repair
Outcome	correlates with **compliance**; surgery results **unpredictable** → only after extended failed PT

Operative criteria

Procedure	Indication (after failed nonop)
Arthroscopic debridement (cuff ± labrum)	**Partial tear <50%** · **Bennett lesion** · **peel-back labral lesion**
Cuff ± labral repair (arthroscopic / mini-open)	**Partial >50%** or full-thickness · **unstable labral tear**
Posterior capsule release vs anterior stabilisation	Persistent posterior contracture or anterior instability (adjunct)

Techniques

Debridement	EUA + diagnostic arthroscopy (intra-articular + subacromial) · shaver · **accelerated rehab**, shorter immobilisation
Cuff repair	arthroscopic addresses labrum too · bursectomy · **no acromioplasty** if no bursal pathology · footprint abrasion · **portal of Wilmington** · partial: **in situ trans-tendinous** (pulley; functionally shortens tendon) or complete the tear → single/double row
Posterior release	cautery / shaver · **axillary nerve risk** at inferior border of infraspinatus
Anterior stabilisation	**capsular plication** most common

6 Complications

Progression to full-thickness tear	small risk after debridement alone
Return to play	**delayed / worse after cuff repair** in throwers

– – Active space – –

Axillary nerve injury posterior release, **inferior border of infraspinatus**

X vs Y

	Internal impingement	Subacromial (external) impingement
Cuff side	**Articular** (posterior SS/IS)	**Bursal**
Pain	**Posterior**, ABER, late cocking	**Anterolateral**, IR in abduction
Test	Posterior apprehension pain, ↓ IR	**Hawkins / Neer** +
Cause	Posterior capsular contracture, micro-instability	Hooked acromion, CA arch stenosis

	Bennett lesion	Hill-Sachs
Site	**Posteroinferior glenoid**	Posterosuperior humeral head
Nature	Traction spur (PIGHL)	Impaction fracture
Context	Pitchers	Anterior dislocation

How it's asked

"Posterosuperior labrum impinges on which structure?"	→ **posterior rotator cuff**
"Posteroinferior glenoid mineralisation — which athletes?"	→ **baseball pitchers**
"Phase of internal impingement?"	→ **late cocking**
"Arm position?"	→ **abduction + external rotation**
"Structure pathologically thickened (figure)?"	→ **posterior IGHL + posterior capsule** ✗ Wrong options: cuff = thinned · acromion · biceps · anterior labrum = thinned
"30° IR deficit — stretch which capsule?"	→ **posteroinferior**
"Pitcher, ↓ velocity, normal MRA, normal Hawkins — exam finding?"	→ **↓ internal rotation** ✗ Wrong options: sulcus · Speed's
"Sleeper stretch addresses?"	→ **posterior band of IGHL**
"Proven to ↓ shoulder and elbow injury?"	→ **posterior capsular stretching**
"Most clearly associated with internal impingement?"	→ **posterior capsular contracture** ✗ Wrong options: Bennett · anterior laxity
"25° IR deficit — tight structure (figure)?"	→ **A = posterior band IGHL** ✗ Wrong options: AIGHL, MGHL, SGHL, axillary pouch
"16-year-old pitcher, MRI posterolateral head edema + undersurface tear?"	→ **internal impingement**
"Characteristic lesion?"	→ **articular-sided rotator cuff tear**

"Javelin, failed PT, arthroscopy SLAP with posterior extension — association?"	→ **internal impingement**
"Late-cocking pain, ER 100°/IR 45° — MRI?"	→ **articular-sided supraspinatus tear** ✗ Wrong options: bursal · Bankart · Hill-Sachs
"Clinical + radiographic findings?"	→ **↑ ER/↓ IR + articular-side posterosuperior fraying + posterior capsule/glenoid scarring**
"Bennett lesion + ↓ arc + posterosuperior labral fraying — diagnosis?"	→ **internal impingement**
"Asymptomatic pro pitcher, MRI partial cuff + labral tear — advice?"	→ **continue pitching**

– – Active space – –

LATERAL EPICONDYLITIS (TENNIS ELBOW)

– Shoulder & Elbow · 21 / 60 –

1 Basics

Epidemiology

Frequency	**most common cause of elbow pain** · **1–3% of adults/y** · **dominant arm**
Demographics	**45–64 y** · ♂ **=** ♀ · up to **50% of tennis players** · labourers with **heavy tools**, repetitive gripping/lifting
Risk factors (TO)	**smoking**, **obesity**, repetitive wrist extension
Tennis risk factors	**poor swing technique (backhand)**, **heavy racket**, **incorrect grip size**, **high string tension**
Association	**radial tunnel syndrome** coexists: **5%** (Orthobullets, associated conditions) · **up to 15%** (Orthobullets, missed-diagnosis complication) – read the options

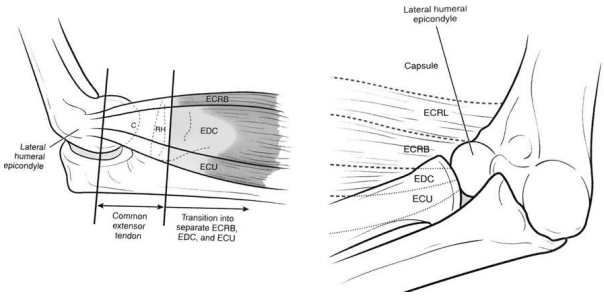
ECRB anatomy – origin, insertion & innervation

Most common tendon (TO) **ECRB**; **ECRB > ECRL** (TO); also ED, EDM, ECU, supinator can be involved
✗ Wrong options: ECRL · supinator · ED · ECU

Muscle	Origin	Insertion	Key fact
ECRB (extensor carpi radialis brevis)	**Lateral epicondyle** (anterior aspect); **13 × 7 mm diamond** superior to centre of epicondyle	**Base of 3rd MC**	**Deep branch radial n. (fibres of sixth + seventh cervical nerves, C6–C7)**; tendon just superficial to capsule; **deep + posterior to ECRL**
ECRL	**Lateral supracondylar ridge** (just below BR)	**Dorsal base 2nd MC**	**First fleshy fibres proximal to common extensor tendon**
Anconeus	**Same attachment site as ECRB**	–	–
ED, EDM, ECU	Lateral epicondyle (common extensor origin)	–	–

✗ Distractor	Belongs to
Largest cross-sectional area of elbow flexors	**Brachialis**
Lateral border of cubital fossa; greatest mechanical advantage of elbow flexors	**Brachioradialis**
Inserts on Lister's tubercle	**No muscle**

– – Active space – –



Common extensor origin: ECRB, EDC, ECU, ECRL

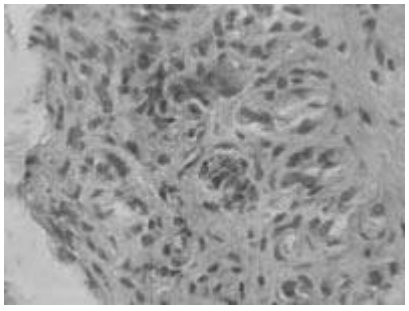
PIN	enters **supinator just distal to radial head** → radial tunnel syndrome
LUCL	**lateral epicondyle → supinator crest of ulna**

2 Mechanism

Overload	**eccentric overload** at common extensor origin → **microtear of ECRB origin** (± ECRL, ECU)
Triggers	**repetitive wrist extension + forearm pronation** · tenodesis effect to optimise grip → ECRB overuse · **backhand** in tennis · hammering
Underuse (TO)	**stress shielding** / pain-inhibited **underuse** of ECRB → weakening → **TRUE cause** ✗ Wrong options: "underuse cannot cause it"
Shear (TO)	shearing forces → **fibrocartilaginous ECRB enthesis** → weaker bone junction ✗ Wrong options: tensile

Histology – angiofibroblastic tendinosis

Lesion	**angiofibroblastic hyperplasia / dysplasia / tendinosis** = **fibroblast hypertrophy** (dense fibroblasts) + **disorganised collagen** + **vascular hyperplasia**
Inflammation	**NO inflammatory cells** → "**tendinosis**", tendonitis/epicondylitis = **misnomers**; **failed reparative process** (TO) ✗ Wrong options: macrophages · neutrophils
Also present	**normal tendon histology**, **hypercellularity** (TO), nerve sensitisation
✗ Histology distractors	abundant inflammatory cells · neutrophils around microvessels · cellular atypia · chondroblastic proliferation · primarily calcium deposition · osteoblastic rimming · reactive hyperaemia · cystic degeneration



Angiofibroblastic hyperplasia

Nirschl (OB)	1,213 patients, 88 operated; immature fibroblastic + vascular infiltration of ECRB origin; **97% improved** after excision + repair

– – Active space – –

Stages of lateral epicondylitis (TO)

Stage	Features
1	**Acute inflammatory** response; may resolve completely
2	**Angiofibroblastic hyperplasia** (fibroblasts, vascular hyperplasia, disorganised collagen), hypercellularity → **tendinosis**; **most common stage at presentation**
3	**Structural failure** → partial or complete **rupture**
4	Stage 2/3 + **fibrosis**, soft matrix calcification, **hard osseous calcification**

3 Presentation

Symptoms lateral elbow pain with **resisted wrist extension**, **gripping**; **↓ grip strength**

Tenderness **point tenderness at ECRB origin**, few mm **distal to tip** of lateral epicondyle

Neuro exam differentiates from **entrapment syndromes**

Provocative test	Manoeuvre
Cozen (TO)	**Resisted wrist extension, elbow extended** (± forearm pronated) → lateral pain
Maudsley	**Resisted extension of 3rd (long) finger** → lateral epicondyle pain
Passive stretch	**Maximal wrist flexion** / passive wrist flexion in pronation
✗ Wrong option: radial tunnel	**Resisted supination** (arm + wrist extended); tenderness **3–4 cm distal + anterior** to epicondyle

Rule out **LUCL insufficiency** (**chair test, push-up test**) · OCD / arthritis (↓ ROM) · **posterolateral plica** (pain/clicking on passive flexion in pronation + supination)

Differential of lateral elbow pain in athletes

Differential posterolateral plica · **PLRI** · **radial tunnel** · occult fracture · **cervical radiculopathy** · **capitellar OCD** · triceps tendinitis · radiocapitellar OA · **shingles**

Tennis player, continuous elbow pain (TO) source gives no key; Orthobullets causes = **incorrect grip size**, poor (backhand) technique, heavy racket, high string tension

4 Diagnosis & Classification

Diagnosis **clinical** (symptoms + exam)

Key radiology findings

X-ray (AP + lateral) usually **normal**; **calcification near epicondyle (up to ½)**; rarely changes management

MRI **not needed**; **↑ signal at ECRB origin (up to 90%)**, thickening/thinning, oedema · **does not correlate with severity** → does not dictate management · use when diagnosis unclear / refractory

US **thickened, hypoechoic ECRB**; **most useful tool in experienced hands** (operator-dependent)



MRI: increased signal at ECRB origin

5 Management

Lateral epicondylitis

- First line → **activity modification + eccentric PT / OT ± counterforce strap** *(NSAIDs, ice; injections second line)*
- Persistent → **injection (PRP better at 12 mo than steroid)**
- Failed ≥12 mo nonop + clear diagnosis → **ECRB release + debridement (open = arthroscopic)**

Nonoperative treatment – eccentric therapy, steroid & PRP injections

Natural history	resolves with time; recovery **6–12 mo**; **no consistent difference between nonop treatment modalities and placebo** / observation
Success	**80–90%** at 1 y, only **2–4%** need surgery (Orthobullets, OB) · **up to 95%** (Orthobullets prognosis, TO) · **>95%** (OB) – read the options
Eccentric exercise	**first line**; **eccentric > concentric** (TO); ↓ pain (grip gain unclear); beneficial effect on **tendon biology**
Occupational therapy (OB)	activity modification + **eccentric strengthening** + pain management = best initial
PT at 1 y	**PT > steroid injection**; **steroid (± PT) = worst outcomes**
NSAIDs	short-term benefit only; not an inflammatory condition; ↑ **GI side effects**

Injection	Evidence
Corticosteroid	**Short-term (2–6 wk)** ↓ pain + ↑ grip vs placebo (TO); **not beyond 6 wk** · RCT vs placebo: **equivalent** pain, grip, PROMs · **worst at 1 y** · **skin/muscle atrophy**, **tendon attenuation + rupture** · second line / not recommended
PRP	Orthobullets: **better than steroid** for long-term pain, healing, function · OB: **↓ pain + ↑ PROMs at 12 mo vs steroid**, **no ↑ strength/function**, no faster return to work, not better at 6 wk (steroid better short term) · TO (Cochrane): **no patient-important benefit vs placebo** – read the options
PRP type	**Leukocyte-rich** promising (✗ Wrong options: leukocyte-poor best); heterogeneous preparations
Others	Botulinum toxin, autologous blood, hyaluronate, acupuncture, ionto/phonophoresis, **nitroglycerin patch**

– – Active space – –

Steroid mechanism	analgesic on **neuropeptides (CGRP, substance P)** ↑ in tendinopathy; **suppresses fibroblast collagen + ECM** production
ESWT	**no difference vs placebo at 6 mo**; + stretching = stretching alone (TO)
Orthoses (TO)	**inelastic, nonarticular, proximal forearm strap** → may ↓ pain + ↑ grip at **3 wk**; bracing ≤6 wk may help ADLs · conflicting: not better than sham; may be inferior short term to steroid + topical NSAIDs
Tennis modifications	slower surface, more **flexible racquet**, **lower string tension**, **larger grip**

Operative criteria

Criterion	Detail
Indication	Failed **prolonged nonop (12 mo)** · **clear diagnosis** (isolated LE) · intra-articular pathology
Contraindication	**Inadequate nonop trial** · **noncompliance** with nonop
Poor-outcome / surgery predictors	**Depression, anxiety, poor coping (greatest)** · ipsilateral **radial tunnel** · **prior injection** · **workers' compensation**

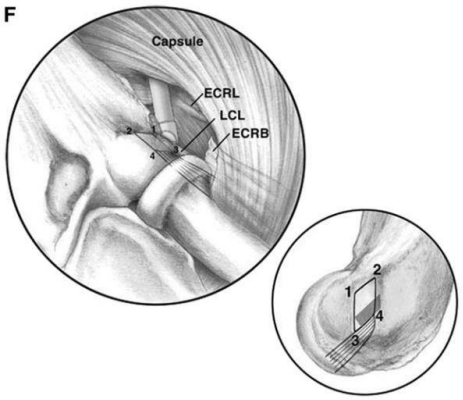
ECRB release & debridement

Open	incision over common extensor origin → **lift ECRL off ECRB** → **Nirschl scratch test** → excise degenerate tissue → **decorticate** epicondyle → **watertight capsule repair** → side-to-side closure → reattach (anchor / tunnel) ✗ Wrong options: synovial fistula
Arthroscopic	visualises + treats intra-articular pathology · resect lateral capsule **anteriorly** · release ECRB where muscle begins · decorticate · **do not pass mid-radial head**
Outcome	**open = arthroscopic** (satisfaction, return to work, complications) · **90–100% improve**, **up to 40% persistent pain**

6 Complications

Iatrogenic LUCL injury after debridement

Cause	debridement **posterior to equator of radial head** → **LUCL** injury → **PLRI**
Safe vs unsafe zones	anterior + superficial resection safe · **too distal → PIN** at risk
Presentation	lateral pain, **clicking/catching**, can't push up from chair / do push-ups → **chair push-up test +**
PLRI mechanism	instability with **elbow extension + supination + valgus**
Chronic iatrogenic PLRI (3 mo)	**LUCL reconstruction with palmaris longus or gracilis allograft** ✗ Wrong options: primary repair (chronic) · revision ECRB debridement · MCL procedures (milking manoeuvre/thrower)



Arthroscopic ECRB release zone relative to LCL

Complication	Detail
Persistent pain	**Up to 40%**
LUCL injury → PLRI	Beyond radial head equator
Missed radial tunnel	**Up to 15%**
Radial nerve / PIN injury	Distal dissection
HO	↓ with **thorough irrigation after decortication**
Stiffness	Esp. with intra-articular procedure
Synovial fistula	Capsule not repaired watertight
Infection, missed PLRI	–

X vs Y

	Lateral epicondylitis	Radial tunnel syndrome	PLRI
Tenderness	**ECRB origin**, just distal to epicondyle	**3–4 cm distal + anterior** (supinator)	Lateral
Painful test	**Resisted wrist extension** (elbow extended), long-finger extension	**Resisted supination**, resisted 3rd-finger extension	**Chair / push-up test**, clicking
Structure	ECRB	PIN	**LUCL**

How it's asked

"Most common tendon in tennis elbow?"	→ **ECRB** (TO)
"Microtear of which muscle?"	→ **ECRB** (TO) ✗ Wrong options: ED, ECU, ECRL
"Backhand pain, resisted middle-finger extension painful – attachment?"	→ **ECRB** (TO) ✗ Wrong options: BR, EDC, ECRL
"Carpenter, pain hammering – tendon?"	→ **ECRB** (TO; key not provided by source)
"Hammer pain, resisted wrist extension elbow extended – attachment?"	→ **ECRB** ✗ Wrong options: distal biceps, BR, ECRL, supinator

– – Active space – –

"Shares origin with the angiofibroblastic tendon?"	→ **anconeus** ✗ Wrong options: BR, annular ligament, FCU, PL
"Origin + insertion of implicated muscle?"	→ **lateral epicondyle, 3rd metacarpal** ✗ Wrong options: supracondylar ridge / 2nd MC = ECRL
"Labelled extensor origin (figure)?"	→ **B = ECRB** ✗ Wrong options: A ECRL, C ECU, D EDM, E EDC
"True about structure involved, golfer?"	→ **innervated by C6–C7 fibres** ✗ Wrong options: largest flexor CSA = brachialis · 2nd MC · cubital fossa border = BR · first fleshy fibres = ECRL
"True about histopathology?"	→ **inflammatory cells not seen** (TO)
"NOT true about lateral epicondylitis?"	→ **abundant macrophages** (TO) (underuse, hypercellularity, paucity of inflammatory cells = true)
"Region with asterisks on histology?"	→ **angiofibroblastic dysplasia**
"Histology of resected tissue?"	→ **angiofibroblastic tendinosis** ✗ Wrong options: chondroblastic · inflammation · calcium · no normal tendon
"3 mo pain, NSAIDs + band failed — next?"	→ **eccentric exercises** (TO) ✗ Wrong options: PRP, steroid, ESWT
"Squash player 3 mo — next step?"	→ **eccentric exercises** ✗ Wrong options: PRP, steroid, ultrasonic aspiration, surgery
"Golfer, short duration, X-rays normal — next?"	→ **eccentric PT program** ✗ Wrong options: steroid · radial tunnel injection · MRI · percutaneous tenotomy
"At 1 year, PT vs steroid?"	→ **PT more improvement than steroid**
"Steroid vs PRP vs placebo?"	→ **no consistent difference from placebo**
"PRP vs prior steroid — expected outcome?"	→ **↓ pain at 12 months** ✗ Wrong options: equal PROMs · faster return · better at 6 wk · better strength
"Most appropriate initial management (pathophysiology-based)?"	→ **OT: activity modification + eccentric strengthening + pain management** ✗ Wrong options: steroid, NSAIDs, surgery, PRP
"Post-arthroscopic release clicking, chair push-up pain at 3 mo — surgery?"	→ **LUCL reconstruction with palmaris / gracilis allograft** ✗ Wrong options: primary repair · MCL

"Excessive debridement beyond which point → instability?"	→ **C – equator of radial head** (posteriorly) ✗ Wrong options: distal = PIN
"Continuous elbow pain in tennis player?"	→ key not provided; risk factors = **incorrect grip size**, poor technique, heavy racket, high string tension

– – Active space – –

– – Active space – –

LATERAL ULNAR COLLATERAL LIGAMENT INJURY (PLRI)

– Shoulder & Elbow · 22 / 60 –

1 Basics

Primary elbow stabilisers (coronoid)

Primary constraint (TO) release alone → laxity · **secondary** – release alone insufficient, ↑ laxity only after primary divided

Primary constraints (TO)	Secondary constraints (TO)
Ulnohumeral articulation (coronoid)	**Radiohumeral articulation** (radial head)
Anterior MCL (AMCL)	Common **flexor-pronator** tendon
Lateral collateral ligament complex (LCLC)	Common **extensor** tendon
	Capsule

Coronoid (TO) **main stabiliser** of elbow; transection affects stability most · **loss of ≥50% → instability**
✗ Wrong options: anterior band MCL · LCL · radial head

Main osseous stability (TO) trochlea ↔ ulnar olecranon articulation; LCL + MCL blend into capsule

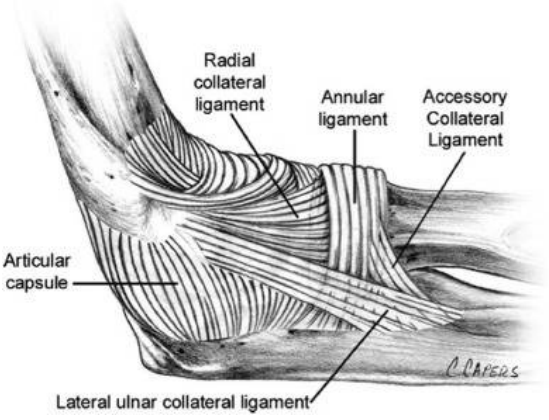
LUCL anatomy & failure site (humeral avulsion)

LCL complex (4) **radial collateral (LRCL)** · **annular** · **accessory lateral collateral** · **lateral ulnar collateral (LUCL)**

LUCL **primary stabiliser to varus + external rotation** (supination) stress; **most robust** of LCL complex; prevents ER subluxation of ulna on humerus

Origin → insertion **lateral epicondyle → tubercle of supinator crest (crista supinatoris)** of ulna

Failure site **humeral origin (lateral epicondyle)**, just anterior + distal to it; common extensor origin may also avulse; **radioulnar articulation intact**
✗ Wrong options: ulnar attachment



Lateral ligament complex: RCL, annular, accessory, LUCL

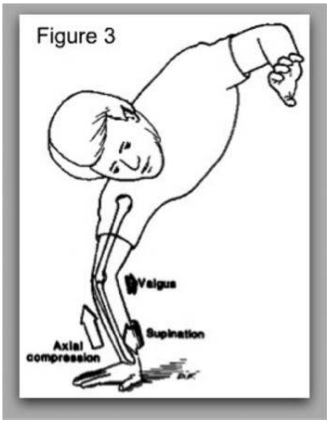
✗ Distractor (labelled diagrams)	Role
Annular ligament	Stabilises radial head in pronation/supination; does not need repair; not part of LUCL recon
Radial collateral / accessory collateral	Less important for PLRI
Orbicular ligament	Much less important laterally
Anterior band MCL	**Primary valgus** restraint (medial)
Posterior band MCL	Disrupted in **VPMRI**
Transverse ligament of Cooper	Coronoid → olecranon; **no stability role**
Posterior capsule	Contributes but does not cause PLRI
Anconeus	Does not need repair

✗ Location distractor	Site
Supinator crest	**LUCL insertion**
Radial tuberosity	Biceps insertion
Medial epicondyle	**MUCL origin**
Sublime tubercle	**MUCL insertion**

2 Mechanism

Definition PLRI = **posterolateral subluxation/dislocation of radiocapitellar + ulnohumeral joints**; ulna **supinates (ER) away from humerus**, radial head subluxes **posteriorly on capitellum** in **supination + extension**

Traumatic (most common) **elbow dislocation**: FOOSH → **supination + axial load + valgus** (posterolateral) + extension → LCL complex fails first → **anterior capsule / coronoid** → semilunar notch externally rotates off trochlea



Mechanism: axial compression, valgus, supination

Iatrogenic lateral elbow surgery (**lateral epicondylitis / ECRB debridement**, arthroscopic or open) going **posterior to equator of radial head** → excessive resection → **stay anterior to equator**

Chronic attenuation **cubitus varus malunion** → lateral thrust + abnormal **triceps vector** stretch LUCL

– – Active space – –

Radial head fractures usually **valgus posterolateral rotatory load** → concomitant LUCL injury common (avulsed from epicondyle)

O'Driscoll stages (circle of Horii)

Stage	Disruption	Result
1	**LUCL** ± partial/total LCL complex	**PLRI** (pain with varus stress)
2	+ **anterior capsule**	Incomplete posterolateral dislocation
3a	All soft tissue incl. **posterior MCL**, **except anterior bundle**	Dislocates posteriorly pivoting on **anterior bundle**
3b	**Complete MCL**	Full dislocation

Sequence **lateral → medial**

3 Presentation

Symptoms **pain** (vague, poorly localised) · **mechanical: clicking, catching, snapping, clunking, locking** with extension · **frank dislocation is not the most common presentation**

Classic **pushing off armrests to rise from chair** (axial load + supination + valgus) · difficulty with **triceps dips**, **push-ups**

History prior **elbow dislocation** ("never felt right"), or prior **tennis elbow surgery**

Causes of chronic lateral instability (TO) **neglected trauma** (fracture fixed, **LUCL not addressed**) · **steroid injections for lateral epicondylitis** (ligament attenuation) · chronic varus load

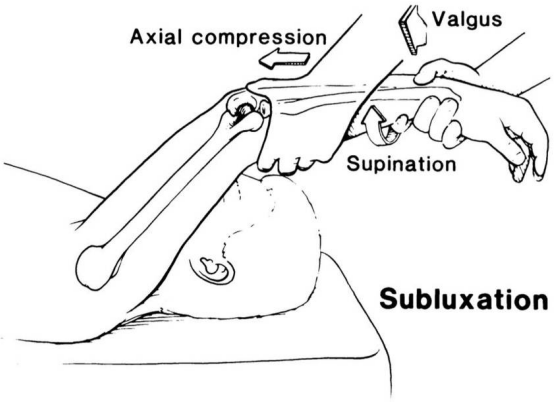
Symptoms (TO) apprehension/weakness **reaching out** (varus load) · painful clicking/snapping · apprehension on **axial load in supination + extension** (chair armrests) · unremitting lateral pain + swelling

Exam often benign; **tenderness over LUCL**, **varus instability** (no firm endpoint); full ROM

Diagnosis – lateral pivot-shift, chair push-up & clinical signs

Lateral pivot-shift (O'Driscoll) supine, arm **overhead** (shoulder fully ER), forearm **supinated**, **valgus + axial load**, elbow from **full extension → flexion** (to **40°** – Orthobullets; to 90° – OB)

Pivot-shift findings subluxed in extension → **reduces with flexion (triceps tension)** with **clunk**; max rotatory/posterior displacement **~40° flexion**; **skin dimple** proximal to radial head · awake = **apprehension**, anaesthetised = **clunk** (more reliable) · analogous to ITB in ACL-deficient knee



Lateral pivot-shift: supination, valgus, axial compression

Test	Technique / positive
Posterolateral rotatory drawer	Supine, overhead, supinated; finger under radial head → **posterior force + ER → radial head subluxes** behind capitellum
Apprehension	Overhead, supinated, valgus while flexing
Chair rise (chair push-up)	Pain/apprehension pushing up with forearms supinated (**= positive pivot shift**)
Table-top relocation	
Floor / prone push-up	**Can't push up with forearm supinated**

✗ Wrong option: valgus (MUCL) tests	Technique
Valgus stress	Valgus at **30° flexion**, medial pain
Milking manoeuvre	Elbow 90°, supinated, thumb pulled posteriorly
Moving valgus stress	Ranged **70–120°**, supinated, valgus
Symptom	**Pain late cocking / acceleration** = valgus instability

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral) rule out fractures, confirm **concentric reduction**; often **normal / little value** in PLRI; may show **impaction defect of posterolateral capitellum**

Fluoroscopy / stress views during pivot-shift → **radial head subluxation** (radiocapitellar + ulnohumeral)

X-ray clues (TO) **"drop sign"** on lateral (~**4 mm ulnohumeral gap**) · radiocapitellar gapping on AP · small avulsion/ossification at lateral epicondyle · **varus stress view under IITV widens radiocapitellar joint** · MRI confirmatory

Drawer at high flexion (TO) positive at **80–90°** flexion = **posterolateral capsule laxity** too, not just LCL · pivot shift dimple + clunk at **~40°**

MRI identifies **acute LUCL avulsion**; poor in recurrent/attenuated LUCL (**oblique course**); detects LUCL pathology in only **50%**

– – Active space – –

Elbow instability types

Instability	Cause / association
PLRI	**LUCL** – after dislocation; lateral pivot shift
Posteromedial rotatory (VPMRI)	**Varus** + **anteromedial coronoid facet** fracture
Valgus	**MUCL** – throwers, overuse
Varus	Rare; **lateral gapping**
Longitudinal forearm	**Essex-Lopresti**

	VPMRI	VPLRI
Radial head	**No** fracture	**Fracture**
Coronoid	**>15%, anteromedial facet**	**<15%, tip**
MCL	**Posterior band** ruptured; anterior band intact (on AM facet)	**Anterior band** ruptured
LCL	LCL complex (LUCL) avulsion	LCL complex (LUCL) avulsion
Exam	Valgus stress, moving valgus, milking	**Varus stress, chair rise, lateral pivot shift**

5 Management

LUCL injury / PLRI

- Acute simple dislocation, stable after reduction → splint 5–7 d → **early ROM**
- Unstable after reduction / entrapment / complex → **ORIF + LUCL repair**
- Mild chronic, low-demand → **brace + extensor strengthening, avoid varus**
- Chronic PLRI, failed nonop → **LUCL reconstruction with graft**

Acute dislocation – splinting position

Splint posterior splint **5–7 days** · **90° flexion** (Orthobullets) · **within stable arc, no specific angle** (OB) – read the options

Ligaments disrupted	Forearm position
LCL disrupted, MCL intact (usual)	**Full pronation** (tightens lateral structures)
LCL + MCL	**Neutral**
Isolated MCL	Supination – **not in PLRI** (LCL always torn)

Then early active ROM; full pronation/supination from 90° → full flexion; ↑ **extension by 30°/week in full pronation**; **full supination in extension after 6 wk**

✗ **Wrong options:** sling only, no immobilisation · splint at 45°/90° in supination

Operative criteria

Procedure	Indication
Open reduction + fracture fixation + LUCL repair	**Osteochondral fragment / soft-tissue entrapment** blocks concentric reduction · **complex dislocation** (fractures) · **acute instability** (e.g. unstable 0–60° after reduction)
LUCL reconstruction with graft	**Chronic PLRI** (tissue **attenuated**, not amenable to repair)
Bracing, extensor strengthening, avoid gravity varus	**Mild chronic PLRI, low-demand**
MCL reconstruction / hinged ex-fix	Only if lateral restoration fails – **rare**

Chronic ✗ traps direct repair, isolated plication (less reliable) · **thermal shrinkage** (long-term effect unclear) · continued bracing after failed nonop · ECRB debridement · RCL reconstruction · MUCL (Tommy John, docking) reconstruction

Most common lateral ligament needing repair after posterolateral dislocation **LUCL** (reliably torn, off humerus)

Reconstruction & postoperative protocol (avoid varus, pronation, splinting position)

Approach **posterior midline** or **Kocher**

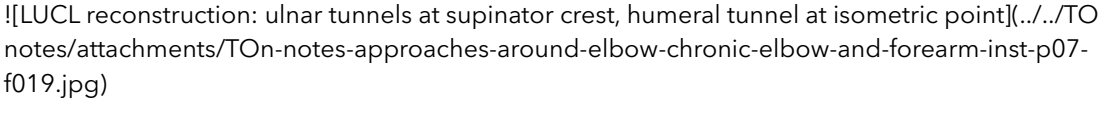
Graft **palmaris longus** (most common), gracilis, triceps fascia; auto- or allograft

Configuration tunnel at **supinator crest → "Y" tunnel** in humerus, tied over lateral column; graft must cover **> posterior 25% of radial head** (sling) · **+ capsuloligamentous plication**

Tensioning **neutral rotation + 45° flexion** (Orthobullets)

Tensioning (TO) **45–60° flexion + full pronation** (reduces residual posterolateral subluxation) – read the options

TO reconstruction details supine, Kocher or **EDC split** · graft **palmaris longus** (may be short), **half FCR**, **gracilis** (best diameter + length) · ulnar tunnels: one at **proximal margin of radial head**, a few mm posterior to joint, second **~2 cm distal over supinator crest** · humeral tunnel **just anterior + distal to isometric point** (anteroinferior lateral epicondyle) → exits posterolateral cortex, **cortical button** · graft **extracapsular** · **box-loop / sling** technique when both collaterals need reconstruction



Fixation **docking** (humeral Y tunnels) · interference screws both sides · overlay (sutures over bone)

– – Active space – –



Docking LUCL reconstruction: Y tunnel humerus, supinator crest tunnel

Coronoid	**large fragment → screw from dorsal ulna** (ACL-type guide) · **small → excise + suture plication of anterior capsule** to tip
Post-op	**locked hinge brace**; **avoid varus** and **shoulder abduction** (gravity varus) **6–12 wk**; ROM with extension block → full extension + supination by **6–8 wk**
Forearm in rehab	**full pronation** during ROM / extension exercises until 6 wk (least stress on lateral repair); forearm most stable in pronation
Post-op avoid	**varus + supination** ✗ Wrong options: valgus + pronation

Ligament compromised	Extension exercises with forearm
LUCL (lateral)	**Pronation**
MUCL (medial)	**Supination**
Both	**Neutral**

Radial head arthroplasty + lateral repair	extension exercises in **pronation** ✗ Wrong options: avoid motion 6 wk → stiffness

Iatrogenic injury after lateral epicondylitis surgery

Presentation	after ECRB release/debridement: ↑ lateral pain, **catching/locking**, painful rising from chair, **+ pivot shift**
Injured	**LUCL** ✗ Wrong options: annular · orbicular · LRCL · anterior MCL
Prevention	keep debridement **anterior to equator of radial head**

6 Complications

Recurrent instability	**3–8%** (Orthobullets) · **<10%** after repair (OB) ✗ Wrong options: >10%
Cutaneous nerve injury	↓ with **posterior midline** approach
Others	infection · **↓ ROM**

X vs Y

	PLRI (LUCL)	Valgus instability (MUCL)
Patient	After dislocation / lateral surgery	**Thrower**
Symptom	**Clicking pushing off chair**	**Pain late cocking / acceleration**
Test	**Lateral pivot shift**, drawer, chair rise	**Milking**, moving valgus, valgus at 30°
Splint/rehab forearm	**Pronation**	**Supination**
Post-op avoid	**Varus, shoulder abduction**	Valgus

How it's asked

"Ligament D injury – characteristic symptom + sign?"	→ **painful clicking pushing off chair armrests; lateral pivot shift** ✗ Wrong options: late cocking · milking
"Volleyball player, old dislocation, catching rising from chair, clunk in supinated extension, normal imaging?"	→ **posterolateral rotatory instability** ✗ Wrong options: posteromedial · valgus · longitudinal
"Apprehension flexing supinated elbow with axial + valgus, posterolateral capitellum defect – disrupted structure?"	→ **LUCL** ✗ Wrong options: posterior band MCL · ligament of Cooper
"Chronic PLRI, + chair rise – manoeuvre to assess degree?"	→ **radiocapitellar reduction as supinated overhead elbow flexes from extension with valgus** ✗ Wrong options: subluxation on flexion · milking · moving valgus
"Supine, extension → flexion with valgus, supination, axial load – tests?"	→ **lateral ulnar collateral ligament**
"Positive pivot shift after dislocation – labelled ligament?"	→ **LUCL (C)** ✗ Wrong options: annular · anterior band UCL
"Axial load + valgus + supination in extension – injured ligament?"	→ **LUCL (E)** ✗ Wrong options: annular A
"Primary stabiliser to varus + ER, deficient after dislocation?"	→ **LUCL (E)** ✗ Wrong options: RCL · accessory
"Posterolateral dislocation – structure most likely needing repair?"	→ **LUCL (1)** ✗ Wrong options: annular · anconeus
"Most common rupture site of injured ligament?"	→ **humeral origin, lateral epicondyle (A)** ✗ Wrong options: supinator crest · sublime tubercle
"Old open dislocation, clicking with extension, can't do dips/push-ups – reconstruction?"	→ **LUCL reconstruction with palmaris graft** ✗ Wrong options: Tommy John · RCL

– – Active space – –

"Chronic PLRI failed nonop — treatment?"	→ **free tendon graft reconstruction + capsuloligamentous plication** ✗ Wrong options: direct repair · thermal shrinkage
"12 wk after dislocation, instability rising from chair — minimum procedure?"	→ **LCL reconstruction** ✗ Wrong options: LCL repair · hinged ex-fix
"Post-op told to avoid varus — ligament reconstructed?"	→ **LUCL (E)**
"Unstable 0–60° after reduction, repair planned — true?"	→ **avoid shoulder abduction post-op** ✗ Wrong options: ulnar-side avulsion · recurrence >10% · anteromedial coronoid
"Radial head arthroplasty + lateral repair — extension exercises?"	→ **with forearm pronated** ✗ Wrong options: supinated = MUCL · neutral = both
"Simple posterolateral dislocation reduced, no fracture — immobilise?"	→ **splint within stable arc in full pronation**
"After tennis elbow surgery, catching + PLRI — injured ligament?"	→ **lateral ulnar collateral** ✗ Wrong options: annular · orbicular
"ECRB debridement, + pivot shift — labelled structure injured?"	→ **LUCL (E)**
"Transection affecting elbow stability most?"	→ **coronoid** (TO)
"Posterolateral rotatory drawer positive at 80–90° flexion implies?"	→ **posterolateral capsule laxity as well as LCL** (TO)
"LUCL graft tensioned in?"	→ **45–60° flexion + full pronation** (TO)

LITTLE LEAGUE ELBOW

– Shoulder & Elbow · 23 / 60 –

1 Basics

Medial epicondyle apophysitis – pathogenesis & diagnosis

Definition — spectrum of **medial elbow** injuries in **adolescent pitchers**: **medial epicondyle stress fracture / apophysitis** · **UCL injury** · **flexor-pronator mass strain**

Most common cause of medial elbow pain in adolescent pitcher — **Little League elbow (medial apophysitis)** (TO stem; answer not given by TO source – OB: apophysitis far commoner than other throwing elbow conditions)
✗ Wrong options: golfer's elbow · olecranon bursitis · capitellar OCD

Age rule — **younger / open physis → apophysitis or avulsion** > UCL sprain; medial epicondyle apophysis **fuses at 16–18 y**

Pathogenesis — **repetitive valgus loading → tension overload** of medial structures; **repetitive flexor-pronator contraction stresses chondro-osseous origin → apophysitis**

✗ Pathogenesis distractor	Condition
Acute fragmentation of **entire capitellar ossific nucleus**	**Panner disease**
Microtraumatic **vascular insufficiency of capitellum** (compression + rotation)	**Capitellar OCD**
Rupture anterior band of anterior bundle UCL	UCL tear
Repetitive triceps contraction	Olecranon (posterior)

MCL (UCL) anatomy & biomechanics (TO)

MUCL components (TO) — **anterior oblique ligament (AOL / anterior bundle)** = strongest, **primary valgus restraint** · **posterior oblique / posterior bundle** · **transverse** ligament

Anterior bundle (TO) — **medial epicondyle → sublime tubercle**; primary valgus stabiliser **30–120°** (when throwing valgus is highest)

Bands of AOL (TO) — **reciprocal**: **anterior band tight in extension**, **posterior band tight in flexion**

Band (TO)	Primary range
Anterior band	Up to **90°** (one TO) · **20–120°** (another TO) – read the options
Posterior band	**>90°**, taut **90–120°**
POL	**>120°**

Other valgus stabilisers (TO) — **radial head** + **flexor-pronator mass** (dynamic)

Degeneration in pitcher's elbow (TO) — **posterior band of anterior bundle** (max force in **acceleration at 90–100°** flexion)
✗ Wrong options: anterior band · posterior bundle · anteromedial olecranon

– – Active space – –

Posteromedial olecranon (TO) repetitive microtrauma → **osteophytes at posteromedial** olecranon
✗ Wrong options: anteromedial

LCL complex (TO) LCL, annular, LUCL

2 Mechanism

Risk factors – pitch counts

Risk factor	Threshold
Pitches per game	**>80**
Months of competitive pitching	**>8 months/year**
Fastball speed	**>85 mph**
Pitching with **arm fatigue / pain**	Yes
Showcases	Yes
Breaking pitches (curveball, slider) before puberty	Yes (✗ Wrong options: changeup)

Not risk factors playing another position after pitching · **2 games/week** (allowed) · **delayed ossification** of medial epicondyle apophysis · 65–75 mph · 6–7 months/year

Age (y)	Max pitches / game	Max games / week
8–10	50	2
11–12	65	2
13–14	**75**	2
15–16	90	2
17–18	105	2

3 Presentation

Triad **medial elbow pain** in throwing arm · **↓ throwing speed (velocity), accuracy, distance**

Exam tenderness **medial epicondyle** (inferior aspect) · **pain with valgus stress** · **valgus instability = more severe** (check through flexion → extension)

Valgus tests for MUCL (TO)

Test (TO)	Technique	Accuracy
Valgus stress test	Elbow **20–30° flexion** (**unlocks olecranon**) + **forearm supinated**, humerus ER, valgus	Sens **66%**, spec **60%**
Milking manoeuvre	Pull **thumb** → valgus; **forearm supinated**, **elbow 90°**	
Moving valgus stress (O'Driscoll)	Upright, shoulder **90° abduction + max ER**, elbow max flexion → extend to 30° under constant valgus; **pain 70–120°** (late cocking → early acceleration)	Sens **100%**, spec **75%**

AOL integrity test (TO) **valgus stress at 20° flexion**
✗ Wrong options: full extension · varus · milking in pronation

– – Active space – –



Milking manoeuvre: thumb pulled, forearm supinated, elbow 90°

4 Diagnosis & Classification

Key radiology findings

X-ray AP + lateral often normal · **physeal widening** of medial epicondyle · **fragmentation / avulsion**; **compare contralateral** elbow



Medial epicondyle physeal widening and fragmentation

MRI more sensitive; **oedema of medial epicondyle apophysis**; differentiates **bony/physeal vs ligament** injury; confirms **UCL insufficiency**



MRI: oedema of medial epicondyle apophysis

– – Active space – –

First step — **rule out MCL injury**: **medial widening on stress X-ray / dynamic US**, or valgus instability on exam → **no widening + tender + physeal widening/ oedema = apophysitis**

✗ Diagnosis distractor	Clue
Displaced medial epicondyle avulsion	Displaced fragment → ORIF
MUCL tear	Medial widening on stress / instability
Valgus extension overload	**Olecranon (posteromedial) osteophytes**
Ulnar neuritis	Typical throwing condition but much less common than apophysitis

5 Management

Adolescent thrower, medial elbow pain

- Apophysitis / stress fracture / strain → **rest + activity mod + PT**
  (↓ innings, minimal immobilisation, coach + parent education)
- Displaced medial epicondyle avulsion → **ORIF (cannulated screws)**
- UCL disruption + insufficiency → **UCL reconstruction (palmaris)**

Treatment – rest & activity modification

Mainstay — **rest, activity modification, ↓ innings pitched per week, PT** (also for acute medial pain + valgus pain after a pitch)

PT aims — strengthen **dynamic stabilisers** (flexor-pronator), shoulder + elbow ROM, **core**; gradual return with proper mechanics + safe pitching rules

Immobilisation — **minimal** (maintain ROM); no long-arm cast 6–8 wk → **stiffness**

Avoid fragment excision — preserve medial epicondyle (**ligament + soft-tissue attachments**, stability)

No UCL reconstruction in Little League elbow – ligament **competent**

✗ Wrong option: arthroscopy / microfracture — for **capitellar osteochondral lesions**, not medial apophysitis

Capitellar OCD in young thrower

Lateral elbow tenderness, nondisplaced capitellar OCD — **cessation of throwing** (heals if activity curtailed)

Cast — only for **noncompliant**; **≤6 weeks**

Surgery — **loose bodies or cartilage flaps**
✗ Wrong options: excision/drilling for intact lesion · steroid injection

Operative criteria

– – Active space – –

Procedure	Indication
ORIF medial epicondyle	**Displaced medial epicondyle avulsion fracture**
UCL reconstruction (palmaris longus autograft)	**UCL disruption + insufficiency**

MCL (UCL) injury – reconstruction (TO)

UCL reconstruction indications (TO)
Acute complete rupture in **competitive thrower** wishing to remain active
High-grade distal tear
Chronic pain/instability after **≥4–6 wk** supervised nonop

Midsubstance tear + degeneration, wants to play (TO) — **MCL reconstruction**
✗ Wrong options: repair · brace/physio · arthroscopy

Nonop (TO) — partial tears; **"active" rest 2–3 months** (no throwing, no valgus stress) + **flexor-pronator strengthening**; also for athletes not returning to sport and asymptomatic in daily life

Paediatric UCL reconstruction — similar to adult; **palmaris longus autograft**

6 Complications

List — **ulnar nerve neuropathy** · continued pain + instability · loss of motion · **inability to return to same level**

X vs Y

	Little League elbow (medial)	Capitellar OCD (lateral)
Force	**Tension** (valgus, flexor-pronator pull)	**Compression + rotation** (radiocapitellar)
Pathology	Apophysitis / avulsion / stress fracture	**Microtraumatic vascular insufficiency** of capitellum
Tenderness	**Medial epicondyle**	**Lateral** elbow
Treatment	Rest + activity mod + PT	**Stop throwing**; surgery for loose body / cartilage flap

How it's asked

"Medial-sided elbow pain in adolescent pitcher most commonly due to?" → **Little Leaguer's elbow** (TO; answer not given by source)

– – Active space – –

"10-year-old pitcher: medial pain, ↓ effectiveness, ↓ distance — pathogenesis?"	→ **repetitive flexor-pronator contraction stressing chondro-osseous origin → apophysitis** ✗ Wrong options: capitellar fragmentation · UCL rupture
"12-year-old, tender medial, no widening on dynamic US — cause?"	→ **medial apophysitis** ✗ Wrong options: MUCL tear · valgus extension overload
"13-year-old, 85 pitches/game, 7 months/y, 75 mph — contributor?"	→ **pitches per game** (max 75 at 13)
"14-year-old, ↓ velocity, physeal widening — risk factor?"	→ **averaging 85 pitches/game** ✗ Wrong options: changeup · 65 mph · 6 months · delayed ossification
"6-year-old pitcher, lateral tenderness, nondisplaced capitellar lesion — initial?"	→ **cessation of throwing** ✗ Wrong options: cast 3 months · excision · drilling
"14-year-old acute medial pain after a pitch, MRI — initial treatment?"	→ **rest + activity modification** ✗ Wrong options: cast 8 wk · fragment excision · ORIF · reconstruction
"11-year-old, 3 straight days pitching, + moving valgus — initial?"	→ **rest + physical therapy** ✗ Wrong options: ORIF screws · reconstruction · cast ≥6 wk
"Degenerative changes in pitcher's elbow most common in?"	→ **posterior band of anterior bundle of UCL** (TO)
"Test for integrity of anterior oblique band of MUCL?"	→ **valgus stress at 20° flexion** (TO)
"Elbow position for valgus stress test?"	→ **20–30° flexion + supination** (TO)
"Athlete, MCL midsubstance tear with degeneration, wants to play?"	→ **MCL reconstruction** (TO)

– – Active space – –

LITTLE LEAGUER'S SHOULDER

– Shoulder & Elbow · 24 / 60 –

1 Basics

Definition

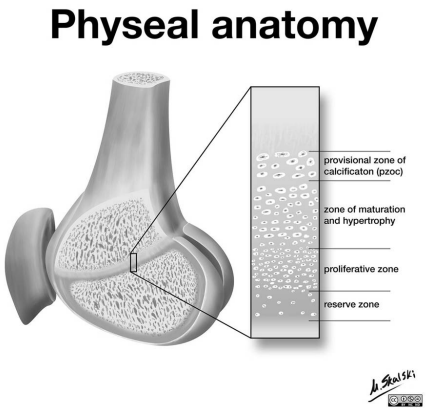
Little Leaguer's shoulder (LLS)	**proximal humeral epiphysiolysis** = overuse **Salter-Harris type I** stress injury of proximal humeral physis
✗ Wrong option: SH II	LLS does **not** produce a Salter-Harris II fracture

Epidemiology

Who	**skeletally immature overhead athletes**: **youth baseball pitchers** (dominant arm); occasionally **tennis**, football, racket sports
Sex	♂ > ♀
Age	**11–16 y** (Orthobullets, OB) · **11–14 y**, peak proximal physeal growth (OB)
Throwing	**10%** of paediatric shoulder pain is throwing-related

Physeal zone affected

Zone	**hypertrophic zone** = **weakest** part of growth plate
✗ Zone distractors	metaphysis (not a physeal zone) · provisional zone of calcification · **proliferative** · reserve



Physeal zones — LLS injures the hypertrophic zone

- - Active space - -

2 Mechanism

Forces repetitive **rotational (torsional) + distraction** stresses (± shear) on unossified physeal cartilage; **rotation > distraction** in importance

Throwing phase	Physeal stress
Late cocking (max ER, just before acceleration)	**Extreme rotatory torque ~400% greater** than physis tolerates → **commonest cause**
Deceleration	Forward arm motion vs cuff → **excessive eccentric** physeal stress
✗ Wrong option: early cocking	Not the peak

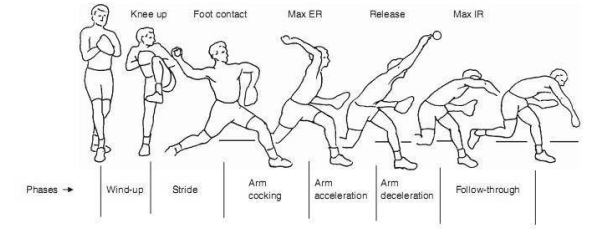


Fig. 1. Pitching phases and key events (adapted from Fleisig et al.,[12] with permission). **ER** = external rotation; **IR** = internal rotation; **max** = maximum.

Pitching phases — late cocking at max ER

Risk factors

Most important **number of pitches thrown** + **frequency of pitching** (innings, throwing days, pitches/innings per week–month–year, **multiple teams**, year-round play)

Breaking pitches **implicated** (Orthobullets) · **refuted / not worse than fastballs** (OB) — read the options; "% curveballs = most important factor" is wrong

Mechanics / body size poor throwing kinematics part of accepted theory (OB) · **no clear relationship** with weight/height or mechanics (another OB)

3 Presentation

History **↓ pitch velocity**, **↓ accuracy / control**; recent ↑ pitching (camp → season, two "select" teams)

Pain diffuse arm/shoulder pain **with throwing**, **anterolateral/lateral**; worse in **late cocking / deceleration**; resolves with rest → may progress to ADL / rest pain

Exam **point tenderness lateral proximal humerus over physis** · pain with rotation · **GIRD** (e.g. 30° IR difference) · ROM often full · no atrophy, no neuro deficit

4 Diagnosis & Classification

Key radiology findings

Modality	Finding
X-ray views	**AP in external rotation**, scapular Y, axillary
Subtle cases	**AP in ER** (changes clearest **anterolateral physis**) + **contralateral comparison**
X-ray findings	**Widened (lateral) proximal humeral physis vs contralateral** · metaphyseal sclerosis · demineralisation/lucency · osteopenia · metaphyseal calcification · **fragmentation** · cystic change
Normal X-ray	**17%**
MRI	**Metaphyseal oedema adjacent to physis**; rules out **labral tear**, **partial articular-sided cuff tear** (rare concomitant)
MRI necessity	**Not needed** for diagnosis when X-ray classic



Widened proximal humeral physis vs normal side



MRI – oedema around proximal humeral physis

– – Active space – –

✗ View distractor	Used for
Distal humeral axial	Paediatric **medial epicondyle** fracture displacement
West Point axillary	**Bony Bankart**, glenoid bone loss
Zanca	**AC joint** (AP, 15° cephalic tilt, OB)
Swimmer's	Lateral **C-spine** when C7 not seen

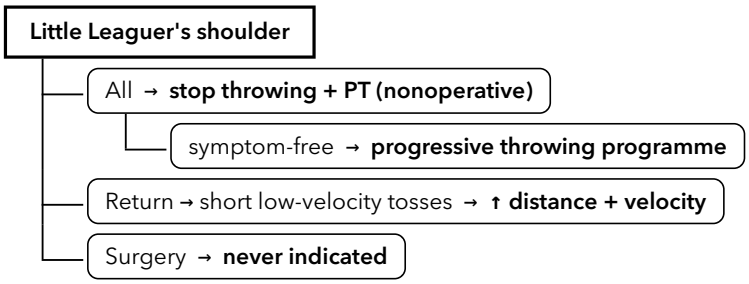
Diagnosis – physeal widening, imaging & throwing phase

Classic vignette — 11–16-yo pitcher, lateral shoulder pain with throwing, ↓ velocity, physeal tenderness, **widened physis → LLS**

15-yo, pain on throwing, ↓ IR, physeal widening (TO) — **Little League shoulder**
✗ Wrong options: adhesive capsulitis · pectoralis major tear · subacromial impingement

✗ Differential distractor	Why wrong
Septic arthritis	No infection
SLAP tear / internal impingement	Not this picture
External impingement	Subacromial bursitis, cuff tear
Chronic retracted supraspinatus tear	Age > 40
Fallen cortical fragment in empty cyst	**UBC** pathological fracture
Acromion tip stress fracture	LLS = **proximal humeral physeal** stress fracture
Deltoid strain	MRI oedema is **bony metaphyseal**

5 Management



Treatment – cessation of throwing & graded return

Mainstay — **cessation of throwing / pitching** until **completely asymptomatic → PT → progressive throwing programme**

Rest duration — **3 months** (Orthobullets; "minimum 3 months", OB) · **2–3 months** (OB) · **6–12 wk** (OB) · **6–8 wk** then progressive throwing (OB) · brief immobilisation + no pitching **for remainder of season** (OB) – read the options

PT — **rotator cuff strengthening**, **posterior capsule stretching**, core strengthening

Throwing programme start **only after symptom resolution**; short, low-velocity tosses → slowly ↑ distance + velocity

✗ Wrong management	Why
In situ screw fixation, ORIF (tension band), **CRPP**	Surgery not indicated; risks physis
Arthroscopic debridement / Bankart / labral repair	No labral pathology
Subacromial corticosteroid	Not impingement
Biopsy	Changes classic, not neoplastic
CT / MRI arthrogram as next step	Diagnosis already made
Continue playing with ↓ pitch count / breaking balls	Physis must heal
PT + continue pitching as tolerated	Risks growth arrest
Sling 4 wk + NWB	Unnecessary; ADLs allowed
Avoid throwing **until skeletal maturity**	Excessive

Prognosis **large majority return to pre-injury level**
✗ Wrong options: "unlikely to reach pre-injury level"

Prevention

Measures **proper mechanics (pitching coach) · discourage breaking balls until skeletal maturity · enforce pitch counts + rest days · avoid year-round pitching · ≤ 2 games/week**

Age (y)	Pitches/game	Max games/week
8–10	52	2
11–12	**68**	**2**
13–14	76	2
15–16	91	2
17–18	106	2

Recurrence risk factors (IR deficit, length of rest)

GIRD **3× more likely** recurrent symptoms; ↑ physeal injury risk

Inadequate rest **length of activity cessation** (e.g. only 1 month) = recurrence risk

> 2 games/week recurrence risk

Not recurrence risks pitch count within age limit (68 at 12 y) · avoiding breaking balls · pitching coach

6 Complications

Premature physeal closure **growth arrest** + **angular deformity** of proximal humerus

Cause **continuing to pitch** / not managed properly

– – Active space – –

X vs Y

	Late cocking	Deceleration
Arm	**Maximal external rotation**	Forward motion opposed by cuff
Physeal force	**Rotatory torque** (~400% > tolerance)	**Eccentric** stress
MCQ answer	**Commonest cause** of LLS	Also loads physis

	Orthobullets	OB explanations
Breaking balls	**Implicated**; discourage until maturity	**Refuted**; not worse than fastballs (avoidance still advised for recurrence)
Rest	**3 months**	6–8 wk · 6–12 wk · 2–3 months · remainder of season
Age	11–16	11–14 (also 11–16)

How it's asked

"15-yo, throwing pain, ↓ IR, physeal widening — diagnosis?"	→ **Little League shoulder** (TO)
"12-yo pitcher, lateral tenderness, AP vs contralateral — diagnosis?"	→ **Little Leaguer's shoulder** ✗ Wrong options: SLAP · internal/ external impingement · septic
"View that helps in subtle cases?"	→ **AP shoulder in external rotation** ✗ Wrong options: West Point · Zanca · distal humeral axial · Swimmer's
"12-yo throwing 100 pitches, anterolateral pain — imaging?"	→ **widened lateral proximal humeral physis** ✗ Wrong options: SH II · acromion stress fracture
"Physeal zone affected?"	→ **hypertrophic zone** ✗ Wrong options: proliferative
"Throwing phase causing LLS?"	→ **late cocking** ✗ Wrong options: early cocking
"14-yo pitcher, ↓ velocity, normal X-ray, MRI metaphyseal oedema?"	→ **proximal humeral epiphysiolysis** ✗ Wrong options: partial articular cuff tear · SLAP
"13-yo, MRI-confirmed physeal stress injury — treatment?"	→ **brief immobilisation + no pitching for remainder of season** ✗ Wrong options: until skeletal maturity · screw fixation
"15-yo pitcher, 3 months pain — next step?"	→ **stop throwing 6–8 wk → progressive throwing** ✗ Wrong options: MRI arthrogram · CT · biopsy
"12-yo on two select teams, physeal widening — management?"	→ **rest from throwing** ✗ Wrong options: steroid · biopsy · labral repair

"Next best step?"	→ **stop throwing until asymptomatic + PT** ✗ Wrong options: MRI · CRPP · continue activity
"Lateral shoulder pain worse in late cocking — plan?"	→ **no throwing 2–3 months + PT + progressive return** ✗ Wrong options: just ↓ pitch count · sling + NWB
"Team record for complete games — advice and outcome?"	→ **stop pitching, gradual return; likely back to pre-injury level**
"True statement about LLS?"	→ **IR deficit ↑ recurrence** ✗ Wrong options: proliferative zone · early cocking · % curveballs most important
"Rested 1 month, now 68 pitches, 1 game/wk, coach, no breaking balls — recurrence risk?"	→ **length of activity cessation**

– – Active space – –

– – Active space – –

LUXATIO ERECTA (INFERIOR GLENOHUMERAL JOINT DISLOCATION)

– Shoulder & Elbow · 25 / 60 –

1 Basics

Epidemiology

Incidence	**very rare: ~0.5%** of all shoulder dislocations (anterior + posterior far more common)
Definition	**inferior** GH dislocation; head trapped **under coracoid + glenoid**

Anatomy – static glenohumeral ligaments

Ligament	Restraint
SGHL	**Inferior** translation at **0° abduction** (neutral rotation)
MGHL	Anterior + posterior translation in **mid-abduction (~45°)** + ER
Anterior band IGHL	**Anterior/inferior** translation at **90° abduction + max ER** (late cocking)
Posterior band IGHL	**Posterior** subluxation at **90° flexion + IR**
Superior band IGHL	"Most important static stabiliser" (Orthobullets)

2 Mechanism

Energy	typically **high-energy**
Mechanism	**hyperabduction** → proximal humerus levered on **acromion** → tears inferior capsule/labrum → head disengages **inferiorly**
Soft tissue	variable tearing of static GH ligaments

3 Presentation

Clinical presentation – arm locked in abduction

Hallmark	arm **fixed in abduction, overhead** (e.g. **140°**); **can't lower arm → inferior** dislocation ✗ Wrong options: anterior · posterior · superior · lateral
Clinical diagnosis	possible on exam alone (unique posture); X-rays still for concomitant injury



Luxatio erecta — arm locked overhead

Associated injury	Rate / note
Neurological	**Up to 60%** — highest NV injury of all shoulder dislocations
Vascular	**Up to 39%** (axillary artery)
Proximal humerus fracture	Esp. **greater tuberosity**
Rotator cuff tear	**Extremely common** (older + young)
Anterior capsule / labral tear	Common

NV exam **pre- AND post-reduction**: radial + brachial pulse, **axillary nerve** + distal neuro exam

4 Diagnosis & Classification

Key radiology findings

X-ray complete shoulder series → **inferior dislocation, arm fully abducted**

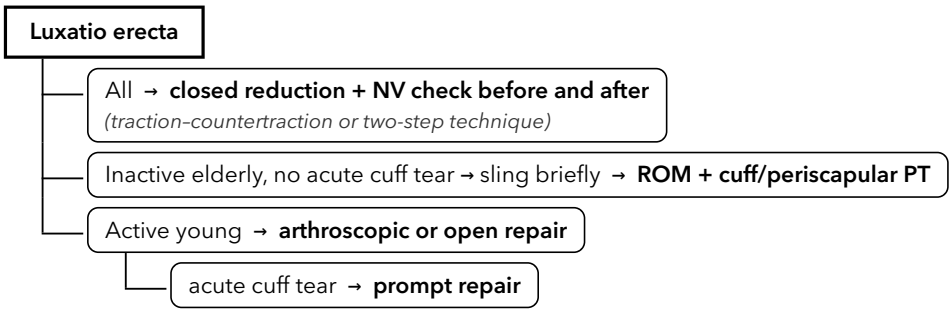


Inferior glenohumeral dislocation

MRI **after relocation** (soft-tissue injury common) → capsulolabral pathology, **cuff tears extremely common** · **prompt in young** to avoid missed cuff tear

– – Active space – –

5 Management



Closed reduction techniques

Traction–countertraction	as for anterior dislocation
Two-step technique	stand at head · **push humerus laterally + pull medial epicondyle superiorly** → head rotates **inferior → anterior** around glenoid (arm moves from abduction to **adduction against chest**) → then any anterior-dislocation reduction
Post-reduction	brief immobiliser → ROM (if cuff intact) → periscapular + cuff strengthening

Operative criteria

Option	Indication
Closed reduction + immobilisation	**Inactive elderly**; no acute traumatic cuff tear
Arthroscopic / open repair (repair vs reconstruction)	**Active younger** patients
Prompt cuff repair	**Acute traumatic cuff tear** (delay → retraction → irreparable)

Arthroscopy advantage addresses **capsulolabral damage + traumatic cuff tear** together

6 Complications

Complication	Key fact
Axillary nerve palsy	**Most common nerve palsy**; may resolve with reduction · persists → **EMG at 6–12 wk** (prognosis)
Brachial plexopathy	High energy + head displacement; usually resolves after reduction + observation
Axillary artery thrombosis	**May occur late**
Rotator cuff tear	Common in old + young → prompt MRI in young

X vs Y

	Luxatio erecta (inferior)	Anterior / posterior dislocation
Frequency	**0.5%**	**Much more common**
Arm fixed in abduction/ flexion	**Yes** (overhead)	**No**
NV injury	**Highest** (neuro ≤60%, vascular ≤39%)	Lower
Reduction	Two-step → convert to anterior	Standard techniques

How it's asked

"Shoulder locked at 140° abduction, can't lower arm — direction?"	→ **inferior** (luxatio erecta)
"Dislocation with highest neurovascular injury rate?"	→ **luxatio erecta**
"Most common nerve palsy?"	→ **axillary nerve**
"Persistent axillary palsy — when EMG?"	→ **6–12 weeks**
"Young patient after reduction — next?"	→ **MRI** (rotator cuff tear common)
"Late vascular complication?"	→ **axillary artery thrombosis**
"Two-step reduction converts inferior to?"	→ **anterior** dislocation

– – Active space – –

MEDIAL EPICONDYLITIS (GOLFER'S ELBOW)

– Shoulder & Elbow · 26 / 60 –

1 Basics

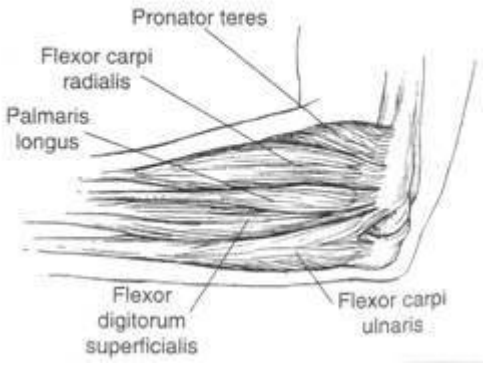
Epidemiology

Frequency	**5–10× less common than lateral epicondylitis**
Demographics	♂ = ♀ · **dominant arm 75%**
Age	30s–60s, peak **30s–40s** (Orthobullets) · **4th–6th decades** / 40–60 y, peak working years (OB) – read the options
Sports	repetitive **wrist flexion + forearm pronation** at ball release: **golf**, pitching, javelin, **bowling**, weightlifting, racquet sports
Tennis risks	late ball strike (racquet head behind elbow) · poor forehand mechanics · no vibration dampeners
Occupation	lifting **>20 kg**, forceful grip, vibration (plumbers, carpenters, construction)

Anatomy – common flexor tendon (CFT)

CFT — **3 cm** long · anterior medial epicondyle + anterior bundle MCL · fibres parallel to MCL · ulnar head PT confluent with anteromedial capsule

Flexor-pronator mass	Nerve
Pronator teres	Median
FCR	Median
FDS	Median
Palmaris longus	Median
FCU	**Ulnar**



Flexor-pronator origin at the medial epicondyle

Muscles affected — classically **PT > FCR** · newer studies: **all CFT muscles except palmaris longus** · 5 tendons form one common tendon
✗ Wrong options: number involved matters

Valgus role flexor-pronators offload the **anterior band of MUCL** (primary static valgus restraint; deep to PT + FCR) · dynamic valgus support in **early acceleration** + wrist flexion at release

2 Mechanism

Pathology repetitive **eccentric overload** → microtrauma at flexor-pronator origin

Stages peritendinous inflammation → **angiofibroblastic hyperplasia** (as in lateral epicondylitis) → breakdown / fibrosis / calcification · **inflammation uncommon**

Associated conditions

Ulnar neuropathy local inflammation involves ulnar nerve

UCL insufficiency rule out, esp. throwers

Occupational cases (84%) CTS · lateral epicondylitis · cuff tendinitis

3 Presentation

History insidious medial pain · worse with wrist/forearm motion + **gripping** · **late cocking / early acceleration** · ± ulnar digit paraesthesia · ↓ grip strength · rarely acute blow → CFT avulsion

Tenderness **5–10 mm distal + anterior to medial epicondyle** (PT/FCR origin)

Chronic flexion contracture

Diagnosis – provocative testing

Test	Condition
Resisted forearm pronation + wrist flexion (fist clenched)	**Medial epicondylitis**
Resisted supination + wrist extension	Lateral epicondylitis
Moving (dynamic) valgus stress	MCL injury
Milking manoeuvre	MCL origin apprehension/ instability/pain
Pinch grip	AIN palsy
Elbow flexion test (max flexion, pronation, wrist extension 30–60 s), Tinel, 2PD	Ulnar neuritis

UCL vs flexor-pronator pain UCL pain **posterior + distal** to medial epicondyle

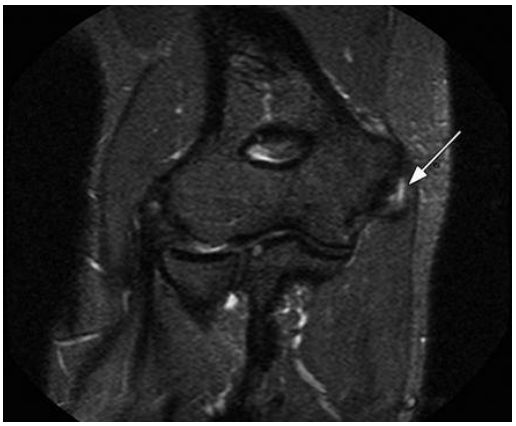
4 Diagnosis & Classification

Key radiology findings

X-ray usually normal · **25% calcification** of CFT or UCL · posteromedial osteophytes · stress views for valgus instability

– – Active space – –

US	**>90%** sensitivity/specificity/PPV/NPV · dynamic · hypoechoic/anechoic degeneration
MRI	standard of care · **↑ T2 signal** in CFT (tendinosis/disruption) + peritendinous oedema · use for UCL/osteochondral injury, loose bodies, flexor-pronator rupture, unclear source



MRI: T2 hyperintensity at the CFT origin

EMG/NCS	only if ulnar nerve symptoms (not routine)
Differential	MCL injury · cubital tunnel · fracture · cervical radiculopathy · triceps tendinitis · **herpes zoster**

5 Management

Medial epicondylitis

- First line → **rest, ice, NSAIDs, activity modification, PT, counterforce brace**
  - stop throwing 6–12 wk
- Refractory to above → **peritendinous CFT corticosteroid injection**
- Fails prolonged nonop, compliant, clear diagnosis → **open debridement ± reattachment**
  - ulnar symptoms → **+ cubital tunnel release / transposition**

Nonoperative treatment – injection

Prolonged nonop	appropriate: surgery **less predictable than lateral** epicondylitis
Modalities	counterforce brace / kinesiology tape · therapeutic US beneficial · **ESWT**: no definitive recommendation, equivocal (Orthobullets, OB) · ESWT "~90% pain relief" (one OB explanation) – read the options
Next step after 2 mo brace + NSAIDs + PT + rest	**corticosteroid into CFT peritendinous area** (short-term benefit) ✗ Wrong options: arthroscopic debridement · cubital tunnel injection · EMG without ulnar symptoms · acupuncture, not shown to help
Steroid complications	skin **depigmentation** (dark skin) · subcutaneous **atrophy** · tendon weakening · **ulnar nerve injury** · avoid multiple injections

Operative criteria

– – Active space – –

Criterion	Value
Failed nonop in **compliant** patient	**Up to 6 mo** (Orthobullets) · **4–6 mo** · **at least 6 mo** (OB)
Symptoms	Severe, affecting quality of life
Diagnosis	Clear
Exception	**Elite athlete with definite tendon tear** → earlier

Surgical outcomes & ulnar nerve involvement

Procedure — **open medial approach** · **PT–FCR interval** · excise pathologic tissue + side-to-side repair · ± epicondyle microfracture · **reattach flexor-pronator** if proximal origin involved · ± cubital tunnel release/transposition
✗ Wrong options: arthroscopic

Reattachment — debridement **with or without** reattachment → equivalent outcomes (Arevalo; complications 4.3%, RTS/RTW 81–100%)

Outcome — **pain and function both improve** · good/excellent **80%** (Orthobullets, < lateral) · **80–90%** (OB) · **87%** at 7 y (Gabel & Morrey)

Worst prognostic factor — **degree of ulnar neuropathy** (only parameter correlating with outcome; mild/none 24/25 good · moderate–severe 3/5 fair/poor)
✗ Wrong options: prolonged nonop · no reattachment · number of tendons · ESWT

Key exam finding — **pronation weakness at 90°**

Rehab — sling **1–2 wk** (avoid wrist volar flexion) · ROM after 2 wk · strengthening **6–8 wk** · **RTS 3–6 mo**

6 Complications

MABC neuropathy — avulsion/transection → if seen intra-op, **transpose into brachialis**

Others — ulnar nerve injury · infection

Prognosis — harder to treat + less studied than lateral epicondylitis

X vs Y

	Medial epicondylitis	Lateral epicondylitis
Frequency	5–10× less common	More common
Origin	Flexor-pronator (**PT, FCR**)	–
Provocation	**Resisted pronation + wrist flexion**	Resisted supination + wrist extension
Surgical success	**~80%**, less predictable than lateral	–
Associated nerve	**Ulnar**	–

	Medial epicondylitis	UCL injury
Tenderness	5–10 mm distal + anterior to epicondyle	**Posterior + distal** to epicondyle
Test	Resisted pronation + flexion	**Milking**, moving valgus stress

– – Active space – –

How it's asked

"Provocative test positive in medial epicondylitis?"	**→ resisted forearm pronation + wrist flexion with clenched fist** ✗ Wrong options: supination/extension = lateral · dynamic valgus/milking = MCL · pinch grip = AIN
"Masters tennis player, 2 mo PT, brace, NSAIDs, rest fail – next?"	**→ corticosteroid injection into CFT peritendinous area** ✗ Wrong options: arthroscopy · cubital tunnel injection · acupuncture · EMG
"Bowler-carpenter, 7 mo failed nonop, open debridement + reattachment – 1-y outcome?"	**→ pain and functional outcome scores both expected to improve**
"Surgical golfer's elbow – factor linked to inferior outcome?"	**→ degree of ulnar nerve involvement** ✗ Wrong options: prolonged nonop · no reattachment · number of tendons · ESWT
"Nerve injured at medial elbow surgery – intra-op fix?"	**→ MABC, bury in brachialis**

MEDIAL ULNAR COLLATERAL LIGAMENT INJURY

– Shoulder & Elbow · 27 / 60 –

1 Basics

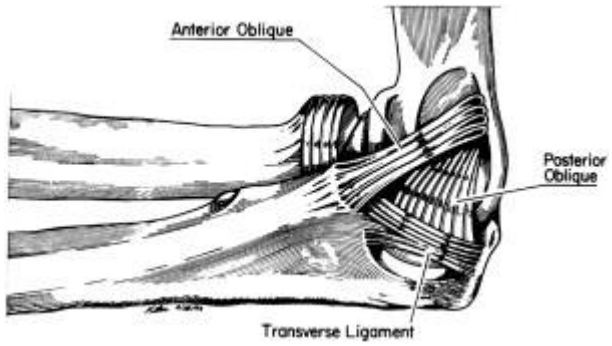
Epidemiology

Definition	attenuation / rupture of elbow UCL → **valgus instability** in **overhead throwers**
Who	originally **javelin** throwers → now mostly **baseball pitchers**; ↑ in high school + amateur pitchers; also water polo, wrestlers
Skeletal maturity	**uncommon in skeletally immature** → they get **little leaguer's elbow** (medial epicondyle) · isolated MUCL tear = **young skeletally mature** pitcher
Risk factors	**exceeding pitch count / innings limits** · short rest, no off-season · **higher pitch velocity** · poor mechanics · **kinetic chain deficits** (shoulder + core weakness, ↓ shoulder motion)

Anatomy & biomechanics (anterior bundle, isometry, posterior bundle)

Carrying angle	**6–11°** valgus (UCL note) · ♂ 5–10° / ♀ 10–15° (anatomy note) · ~11° ♂ / ~13° ♀ (exam note) – read the options
Stability	split evenly **osseous vs soft tissue**; UCL = primary valgus restraint **30–120° flexion** (+ flexor-pronator, capsule)

UCL component	Facts
Anterior oblique ligament (AOL) = anterior bundle	**Strongest, most important valgus stabiliser**; origin **anteroinferior medial epicondyle → sublime tubercle**, **2.8 mm distal** to ulnar articular margin; mean length **54 mm**; primary **30–120°**; greatest resistance at **~70°**
Posterior oblique ligament (POL) = posterior bundle	**Greatest change in tension** flexion → extension; **tight in flexion**; fan-like **capsular thickening**; inserts **sigmoid notch**; not subdivided into bands; secondary stabiliser at 30°
Transverse ligament	**No stability** – originates + inserts on **ulna** (same bone)



UCL complex – anterior oblique, posterior oblique, transverse ligament

Terminology trap	AOL = "anterior bundle" = older "anterior band"

– – Active space – –

AOL bands — **2 bands**: anterior + posterior (Orthobullets, most OB) · **3 bands**: anterior, central, posterior (2 OB explanations) · anterior, posterior, **oblique** (1 OB) – read the options

AOL band	Strain / tautness	Primary role
Anterior band	**Isometric** through ROM (✗ Wrong options: isokinetic); **taut 0–85°**; tightens at end-range **extension**; most vulnerable **in extension**	**Most important valgus restraint**; primary at **30°, 60°, 90°**
Posterior band	**↑ strain in flexion** (✗ Wrong options: isometric); **taut beyond 55°**; vulnerable **in flexion**	Primary at **120°**

Posterior bundle range — resists from **60° to full flexion** (one OB) · beyond **90°** (others)

Axis of rotation / isometric point — at UCL **origin = anteroinferior medial epicondyle**
✗ Wrong options: anterosuperior · posteroinferior · sublime tubercle = insertion

Flexion	Main valgus restraint
0–20° (extension)	**Bony: olecranon in fossa + radiocapitellar articulation**
~25°	Olecranon **unlocked** → UCL takes over
30–120°	**AOL (anterior band)**
>90° / max flexion	**POL**

Radial head — important **secondary** stabiliser in extension + flexion

Dynamic stabilisers — **flexor-pronator mass** (**FCU, FDS**) – dynamic, not static

2 Mechanism

Acute trauma — often with **elbow dislocation**

Overuse — **repetitive valgus microtrauma** → rupture of **anterior band** of AOL

Iatrogenic — **excessive olecranon osteophyte resection** puts MCL at risk

Associated — **traction ulnar neuritis** · **posteromedial olecranon impingement (VEO)** · elbow arthritis

Throwing phase of maximal valgus stress (late cocking/early acceleration)

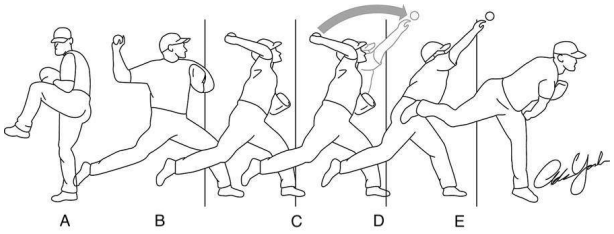
Max valgus — **late cocking → early acceleration**: forearm lags behind arm → valgus while elbow depends on **anterior band**; forces near **tensile limit** of UCL

Single best phase — **late cocking** (Orthobullets, several OB) · **early acceleration** (other OB: "highest in early acceleration") – choose **late cocking/early acceleration** if offered; else whichever of the two is listed

Forces valgus up to **64 N·m** · **500 N** compression at radiocapitellar joint · elbow **110° → 20°** · **3000°/s**

– – Active space – –

Phase	Key activity / pathology
Wind-up	Benign for elbow
Early cocking	Peak **deltoid**; cuff active
Late cocking	**Max shoulder ER ("layback")**, high torque → **max elbow valgus**; FDS + FCU protect
Early acceleration	Peak **triceps**; valgus + rapid extension
Late acceleration	Peak **pec major, latissimus, serratus anterior**
Ball release	Not a phase; end of acceleration
Deceleration	**Eccentric** contraction of all muscles, **highest torque**, "most harmful" → **SLAP, biceps, brachialis, teres minor** (✗ Wrong options: UCL)
Follow-through	Rebalance; elbow flexors prevent hyperextension; minimal UCL stress



Throwing phases – A wind-up, B early cocking, C late cocking, D late acceleration, E deceleration

Figure labels vary one figure: **C = late cocking** · another: **B = late cocking** (A early cocking, D deceleration) – read the figure

Valgus ↓ by **flexor-pronator contraction** · **forearm pronation** · **scapular protraction/retraction (trunk-scapular kinesis)** · **reduced velocity** · correct mechanics

Valgus ↑ by **↑ glenohumeral IR torque** · poor mechanics

3 Presentation

History acute **"pop"** + pain + can't throw · chronic: **loss of velocity ("zip")**, **loss of control/accuracy**

Pain medial/posteromedial in **late cocking + acceleration** · posteromedial in **deceleration** = **valgus extension overload**

Ulnar nerve paraesthesia ring + small (nerve **stretch** in late cocking) → check subluxation

Exam tenderness at/near **MCL origin** (posteromedial = VEO) · flexor-pronator integrity · **palmaris longus present?** (graft) · throwers may **lack full extension** · examine shoulder + kinetic chain

Throwing tolerance pain-free light throwing / several innings, no trauma → partial injury; complete rupture unlikely

– – Active space – –

4 Diagnosis & Classification

Diagnosis – moving valgus stress, milking manoeuvre & MRI T-sign

Test	Technique	Positive / accuracy
Valgus stress test	Elbow **20–30°** flexion (unlocks olecranon) + **humerus ER** + valgus	**50% sensitive**
Milking manoeuvre	**Pull thumb**, forearm **supinated**, elbow **90°**; supine/seated/standing	**Apprehension, instability or pain at MCL origin**
Moving valgus stress test	Milking position; valgus while ranging full arc (or valgus at full flexion → **quickly extend**)	Pain/apprehension **70–120°**; **100% sensitive, 75% specific**; **most strongly confirms**



Milking manoeuvre – valgus via thumb with forearm supinated

Milking (TO) — **shoulder ER, elbow flexed >90°**, pull thumb (valgus) → medial pain · **moving valgus**: constant valgus while flexion → extension, pain **70–120°** (TO)

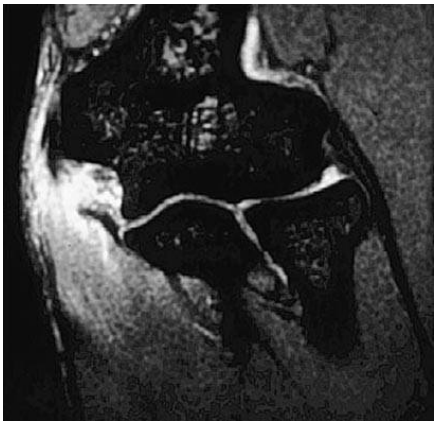
Chronic medial instability (TO) — repeated valgus (javelin, gymnasts) attenuates MCL · difficulty throwing · **ulnar neuritis** (weak grip, ulnar numbness) from traction/medial overload · **valgus stress view under IITV opens ulnohumeral joint** · small medial epicondyle avulsion/ossification on X-ray

Pain in mid-arc (70–120°) — **anterior bundle MCL** · **pain at terminal extension** = **posterior compartment / olecranon osteophytes**

✗ Exam distractor	Indicates
Pain with **resisted wrist flexion** (± pronation)	**Medial epicondylitis / flexor-pronator strain**
Froment's sign	Ulnar neuropathy
Hawkins	Shoulder impingement
Lateral pivot shift, **chair test**, **push-up sign**, PLRI test	**PLRI (LUCL)**
Painful **snapping** medially	**Ulnar nerve or triceps subluxation**

Key radiology findings

Modality	Fact
X-ray AP + lateral	Often **normal**; loose bodies, **UCL calcification**, **posteromedial osteophyte** (VEO)
Oblique views	Olecranon
Stress X-rays (gravity/ manual), both elbows	**Medial opening >3 mm = diagnostic**
MRI	**Gold standard / mainstay**; thickened ligament (chronic), calcification, midsubstance tear, proximal/distal avulsion, partial **undersurface** tear
Plain MRI	**57% sensitive, 100% specific**
MR arthrogram	**Diagnostic**, more accurate: **92% sensitive, 100% specific** (✗ Wrong options: "lacks specificity")
"Capsular T-sign"	**Contrast extravasates** around distal (undersurface) tear at sublime tubercle
Dynamic US	Laxity under valgus; operator dependent



MR arthrogram – capsular T-sign at sublime tubercle

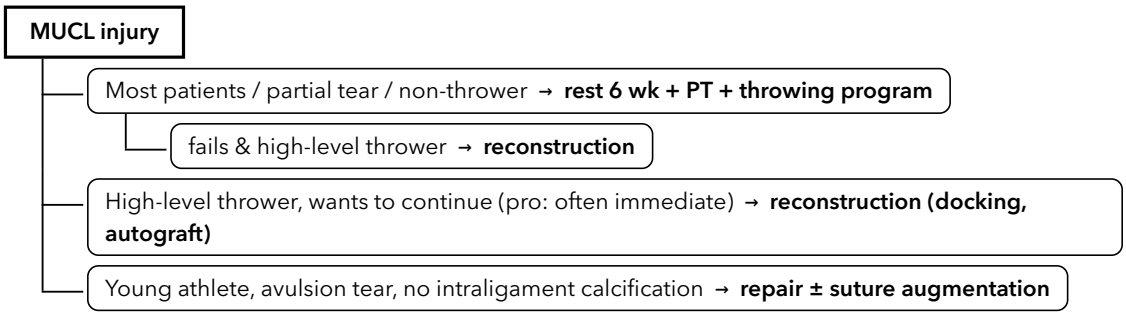
MRI labels (water polo Q) 1 = **flexor-pronator origin** · 2 = **UCL** · 3 = **RCL**

Differential diagnosis

Diagnosis	Discriminator
MUCL injury	**Moving valgus +**, milking +, loss of velocity
Medial epicondylitis / flexor-pronator strain	Pain on **resisted wrist flexion/pronation**
Ulnar neuropathy (isolated)	Doesn't explain valgus instability
Valgus extension overload	**Pain at terminal extension**; posteromedial osteophyte
Little leaguer's elbow (medial epicondyle stress fracture)	**Skeletally immature**

– – Active space – –

5 Management



Non-thrower partial injury – nonoperative care

First line	most patients + **partial tears**: **rest / activity avoidance 6 wk** + NSAIDs → **flexor-pronator strengthening** (secondary dynamic stabilisers) + mechanics, peri-scapular + core → **progressive throwing program**
Outcome	**42%** return to pre-injury sport at avg **24 wk** · most heal without surgery in non-high-level athletes
Adolescent acute partial (14-y basketball)	**activity avoidance 6 wk** ✗ Wrong options: stretching (full ROM) · arthroscopy (unless loose body) · static ex-fix (brace OK) · rheumatology work-up
Persistent at 6–12 wk	advanced imaging (MRI, MR arthrogram, US, stress X-rays)

Operative criteria

Procedure	Indication
UCL reconstruction (Tommy John)	**High-level throwers wanting to continue** · **failed nonop partial tears** willing to rehab · **midsubstance** tears · attritional tears with **calcification** (excision leaves void)
UCL repair	**Young athletes, avulsion-type tears** (proximal/distal), no intraligament calcification; ± high-strength suture augmentation
✗ Wrong option: repair	**Midsubstance tear** (reconstruction superior)

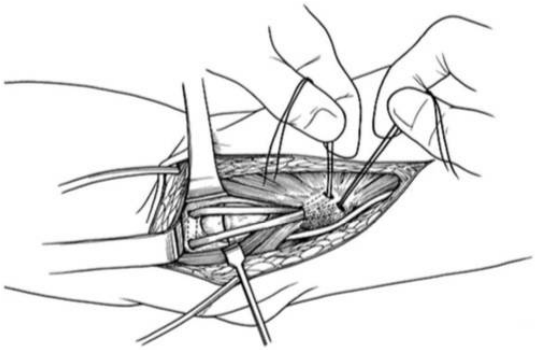
Repair history	early repairs poor → replaced by reconstruction; newer augmented repair promising

Reconstruction technique (docking vs Jobe) & return to play

Goal	anatomic reconstruction of **anterior bundle** (medial epicondyle → sublime tubercle); **none stronger than native**
Approach	**flexor-pronator muscle-splitting** (✓ ↓ morbidity) ✗ Wrong options: **detachment** (original Jobe; flexor-pronator avulsion risk) · **medial epicondylectomy** in throwers

Graft **ipsilateral palmaris longus autograft** most common/standard · gracilis autograft or allograft · PL vs gracilis both successful (PL possibly ↑ high-level return; gracilis does not ↓ laxity) · revision **autograft 1.6% vs allograft 2.6%**

Technique	Humerus	Ulna	Notes
Original Jobe (1986)	Tunnels	Tunnel	**Flexor-pronator tenotomy**, **figure-of-8**, **routine submuscular ulnar nerve transposition → abandoned**
Modified Jobe	**2 connected tunnels, "Y"**	Single tunnel (2 angled holes, sublime tubercle)	**Muscle-splitting**, no routine transposition; graft sutured **figure-of-8**
Docking	**Single socket**; sutures tied over bony bridge	**Two converging tunnels**	**Best**: **↑ return to sport, ↓ complications**, less nerve handling, **better graft tensioning**
Hybrid interference screw (DANE TJ)	Docking socket	**Single longitudinal socket + interference screw**	↓ iatrogenic fracture; docking + ulnar screw = **95% native strength**
Cortical button (Endobutton)	–	Suspensory	Mostly **revision**



Docking technique

Docking vs Jobe **~15% more excellent results** · **3× fewer complications** · **3–5% less ulnar neuropathy** · RTP **>90%** vs **80–85%** (figure-of-8)

Biomechanics docking + cortical button **stronger** than figure-of-8 + interference screw

TO technique table all use **sublime tubercle** ulnar tunnels · **Jobe**: complete flexor-pronator take-down (historical gold standard), **figure-of-8 3-ply** · **modified Jobe**: flexor-pronator split, figure-of-8 3-ply · **docking**: graft ends in **blind humeral tunnel**, tied over bone bridge, **2-ply → easier but weaker** (TO) – conflicts with "docking stronger" above · TO lists ulnar nerve transposition as **obligatory** in all three – read the options

– – Active space – –

Calcified attritional UCL in high-level thrower — **reconstruction (modified Jobe)**
✗ Wrong options: original Jobe · repair ± augmentation · arthroscopic debridement + repair

Return to play	Rate
Modern reconstruction	**~90%** (Orthobullets) · **85–95%** (OB) – answer **80–95%**
All athletes (systematic review)	**86%**
Collegiate	**95%**
Surgery overall good results	84–94%
Revision reconstruction	**Worse**

Rehab timeline — early wrist/elbow/shoulder ROM + core/hip · strengthening **4–6 wk** · **no valgus stress until 4 mo** · throwing program **4 mo** · **competition 9–12 mo**

Ulnar nerve management & postoperative neuritis

Pre-op ulnar status	At UCL reconstruction
No symptoms	**No routine decompression/transposition** (in situ release at most)
Symptoms (cubital tunnel)	Decompression ± transposition
Subluxation	**Transposition**

Routine transposition — avoid → **↑ post-op ulnar neuropathy**, worse outcomes

Midsubstance tear + cubital tunnel symptoms + subluxation — **autograft reconstruction + ulnar nerve transposition**
✗ Wrong options: repair · reconstruction without nerve

Ulnar nerve in forearm — **between FCU and FDP**
✗ Wrong options: medial to brachial artery / most volar in carpal tunnel = median

Postoperative rehabilitation – forearm position

Repair	Elbow extension exercises with forearm in
MUCL reconstruction	**Supination**
LUCL repair (± radial head replacement)	**Pronation**
Combined medial + lateral	**Neutral**

6 Complications

Ulnar neurapraxia — **most common**; **3–26%** (Orthobullets, OB) · "<10%" (one OB) – read the options; **treat by observation** (most resolve weeks–months)

↓ Neurapraxia by — muscle-splitting · **more superior exiting proximal humeral tunnels** · selective transposition · docking

MABC injury — crosses **distal** incision

Fracture — ulna or medial epicondyle; **small bone bridge** → ORIF / larger fixation

Stiffness — **HO** → early therapy; HO excision ~**6 mo**

Others	failure to regain pre-injury level (esp. **revision**) · infection · flexor-pronator avulsion (detaching approach)
Not MUCL complications	PLRI / radial head + coronoid tip fractures (terrible triad, LCL)

X vs Y

	Anterior band (AOL)	Posterior band (AOL)	POL (posterior bundle)
Strain	**Isometric**	↑ in flexion	**Greatest change**
Taut	**0–85°**	**>55°**	**>90°** / max flexion
Role	**Primary valgus restraint**	Primary at 120°	Secondary; max flexion; cubital tunnel floor

	Original Jobe	Modified Jobe	Docking
Flexor-pronator	**Detached (tenotomy)**	**Split**	**Split**
Ulnar nerve	**Routine submuscular transposition**	Not routine	Minimal handling
Humeral fixation	Figure-of-8 tunnels	**Y-tunnels, figure-of-8**	**Single socket, tied over bridge**
Outcome	Abandoned	Good	**Best RTP, fewest complications**

How it's asked

"Primary valgus stabiliser in mid-flexion?"	→ **AOL of MUCL** ✗ Wrong options: ulnohumeral · radiocapitellar · POL · flexor-pronator
"All protect elbow from valgus EXCEPT?"	→ **↑ glenohumeral IR torque** ✗ Wrong options: flexor-pronator contraction · pronation · scapular motion · ↓ velocity
"Native MCL kinematics?"	→ **posterior bundle = greatest tension change** ✗ Wrong options: posterior bundle isometric
"Most static valgus stability?"	→ **anterior band of UCL** ✗ Wrong options: flexor/pronator = dynamic
"Most important static valgus stabiliser in thrower?"	→ **anterior band of anterior bundle**
"Structure B (posterior bundle) – true?"	→ **most change in tension flexion → extension** ✗ Wrong options: subdivided into bands
"Javelin thrower – ligament, arc, insertion?"	→ **AOL, 30–120°, sublime tubercle** ✗ Wrong options: POL / sigmoid notch

– – Active space – –

"Isometric component?"	→ **anterior band of anterior bundle** ✗ Wrong options: isokinetic
"UCL anatomy – true (figure)?"	→ **AOL greatest resistance at 70°** ✗ Wrong options: both bands isometric · POL tightest in extension
"Axis of rotation for repair?"	→ **anteroinferior medial epicondyle** ✗ Wrong options: sublime tubercle
"Greatest stress on MUCL – phase?"	→ **late cocking** ✗ Wrong options: wind-up · early cocking · ball release · follow-through
"Pitcher ruptures MUCL – phase?"	→ **early acceleration** (when late cocking not offered)
"Forces approaching tensile limit?"	→ **late cocking / early acceleration**
"Water polo, lost velocity – phase + MRI label?"	→ **late cocking (C) + UCL (2)**
"Milking-positive pitcher – phase reproducing symptoms?"	→ **late cocking (figure B)** ✗ Wrong options: deceleration = labrum/biceps
"Highest valgus load phase (pro pitcher pre-reconstruction)?"	→ **early acceleration**
"Medial pain, positive milking, lost velocity – phase?"	→ **late cocking** (figure C)
"Valgus via thumb at 90° + moving valgus pain – phase?"	→ **late cocking**
"Pitcher lost 10 mph, normal EMG – exam that best confirms?"	→ **moving valgus stress test** ✗ Wrong options: resisted wrist flexion · Froment · Hawkins · pivot shift
"MR arthrogram – diagnosis?"	→ **MCL rupture**
"Medial pain after 3 innings, milking test reproduces?"	→ **MCL injury** ✗ Wrong options: ulnar nerve / triceps subluxation = snapping
"Pitcher, pain in acceleration, lost velocity – structure?"	→ **anterior bundle UCL**
"Valgus through passive ROM painful 70–120°?"	→ **anterior bundle MCL** ✗ Wrong options: olecranon osteophytes = terminal extension
"Positive moving valgus, lateral X-ray – labelled structure?"	→ **anterior oblique band (C)**
"ER humerus + supinate + valgus, apprehension 70–120°?"	→ **MUCL** ✗ Wrong options: LUCL · annular · anterior capsule · triceps
"Skeletally mature 17-y pitcher, overused, ulnar paraesthesia, valgus pain?"	→ **attenuation of anterior band of AOL** ✗ Wrong options: medial epicondyle stress fracture · isolated ulnar neuropathy · flexor-pronator

"Capsular T-sign – true?"	→ **evaluate by pulling thumb, forearm supinated, elbow flexed** ✗ Wrong options: MRA lacks specificity · all isometric · 0–20° · deceleration
"Failed rest + interval program, UCL tear on MRI?"	→ **open reconstruction with ipsilateral palmaris longus** ✗ Wrong options: ECRB release · OATS · flexor-pronator reattachment · osteophyte excision
"Lowest complication + best outcome technique?"	→ **muscle-splitting + docking** ✗ Wrong options: routine transposition · detachment
"Best return to sport, minor leaguer?"	→ **docking + ulnar nerve in situ release** ✗ Wrong options: classic Jobe · submuscular transposition · epicondylectomy
"Calcified UCL, failed rest – surgery?"	→ **modified Jobe reconstruction** ✗ Wrong options: original Jobe · repair
"Jobe vs docking?"	→ **docking: higher RTP, fewer complications** ✗ Wrong options: 20% neuropathy regardless
"Pro pitcher, no ulnar symptoms – procedure?"	→ **autograft docking reconstruction** ✗ Wrong options: allograft · figure-of-8 · nerve decompression
"Return to preinjury throwing after docking?"	→ **80–95%**
"Midsubstance tear + cubital tunnel symptoms + subluxation?"	→ **autograft reconstruction + ulnar nerve transposition**
"Most common complication – nerve lies where?"	→ **between FCU and FDP** (ulnar)
"Surgical management true (no ulnar symptoms)?"	→ **routine ulnar transposition not indicated** ✗ Wrong options: repair midsubstance · detachment equivalent · docking worse
"Most likely post-op complication – management?"	→ **observation** (ulnar neurapraxia)
"14-y basketball, acute partial UCL, full ROM?"	→ **activity avoidance 6 wk** ✗ Wrong options: static ex-fix · arthroscopy
"Extension exercises after MUCL vs LUCL?"	→ **supination for MUCL · pronation for LUCL**

– – Active space – –

MULTIDIRECTIONAL SHOULDER INSTABILITY (MDI)

– Shoulder & Elbow · 28 / 60 –

1 Basics

Definition & epidemiology

MDI	symptomatic instability in **≥2 planes** (anterior, posterior, inferior) from **capsular redundancy**; **one plane must be inferior** (TO, OB)
Age	peaks **2nd–3rd decades**
Who	overhead athletes: **swimmers, volleyball, gymnasts, throwers**, dancers
Generalised laxity	in **45–75%** of MDI patients undergoing surgery
Prevalence (TO)	**7–10%** of instability cases · young hyperlax / connective tissue disorders (**Marfan, Ehlers-Danlos**) · essential lesion = **↑ capsular redundancy** (TO)

Torn loose / worn loose / born loose (TO)

Type (TO)	Mechanism	Example
Torn loose	Trauma	Traumatic dislocation (TUBS)
Worn loose	**Microtrauma / overuse**	Overhead athletes → **MDI**
Born loose	**Lax ligaments / capsule**	**Collagen disorders: Ehlers-Danlos, Marfan → MDI**

TUBS vs AMBRI (Matsen) (TO)

TUBS	AMBRI
Traumatic dislocation → recurrent instability	**A**traumatic (minimal trauma)
Unidirectional	**M**ultidirectional
Bankart lesion most common	**B**ilateral (frequently)
Surgery mainstay	**R**ehabilitation first
–	**I**nferior capsular shift for nonresponders

Stanmore (polar) type (TO)	Aetiology	Key feature
I	**Traumatic structural** (e.g. Bankart)	Structural damage → **repair**
II	**Atraumatic structural** (ligamentous laxity)	**Rehab first**; capsular shift if persistent
III	**Neurological / muscle patterning**	Seen with brachial plexus injuries

Stanmore vs TUBS/ AMBRI (TO)	physiotherapists use **Stanmore**; surgeons prefer **TUBS/AMBRI**

Pathoanatomy

Cause (TO) **inferior capsular redundancy**
✗ Wrong options: Bankart · Hill-Sachs · cuff tear

Hallmarks **patulous inferior capsule** (IGHL anterior + posterior bands) on MRI · **rotator interval deficiency**

Component (TO)	Change in MDI
Capsule	**Redundant, patulous**
Rotator interval	**Lax (gap >1 cm)**
Labrum	**Shallow / dysplastic**
Dynamic stabilisers	Weak cuff, poor coordination
Proprioception	**Reduced**

Add-on lesions (TO) from traumatic events: **Bankart** (anteroinferior labrum) · **Kim lesion** (posteroinferior labral avulsion) · glenoid erosion · Hill-Sachs, chondral, SLAP — **less often** than in unidirectional; rarely change treatment

Glenohumeral stability

Static restraints	Dynamic restraints
Glenohumeral ligaments	**Rotator cuff** — compresses head into glenoid (primary role)
Glenoid labrum	Biceps
Articular congruity + version	**Periscapular muscles**
Negative intra-articular pressure (release → head subluxes inferiorly)	

MDI = static deficiency → therapy builds **dynamic** stabilisers to compensate

3 Presentation

Symptoms **pain > instability** often · weakness · **paraesthesia / transient numbness** · crepitus · instability **during sleep** · atraumatic, self-reduced dislocations, never needed ED

Impingement signs <20 y think **MDI** (TO)

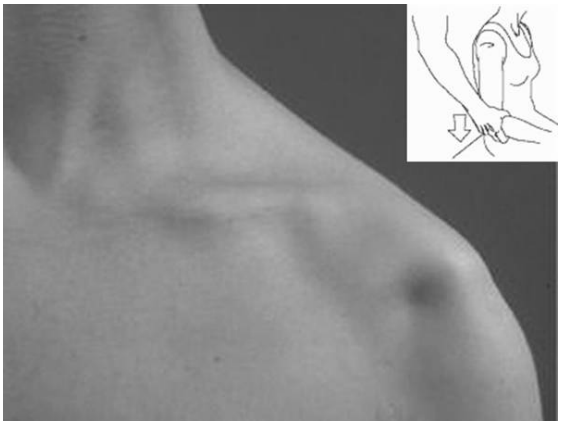
Presentation (TO) vague pain, fatigue, clicking, **"loose shoulder"** rather than frank dislocation · athletes: ↓ performance · numbness/tingling from **traction**

Voluntary dislocator "popping shoulders in and out" → **PT**; **contraindication to surgery**

– – Active space – –

Diagnosis – sulcus sign, generalised laxity & capsular redundancy

Test	MDI finding
Sulcus sign	**≥2+**; inferior traction at **0° and 45° abduction** (TO) → dimple below acromion; **most specific** for MDI (**2 cm → up to 97% specificity**)
ER with arm adducted (and at 90° abduction)	**↑ ER** = **incompetent rotator interval**
Anterior + posterior **load and shift**	**≥2+** both directions; less specific
Apprehension / relocation	Common in **both** traumatic + MDI
Jerk / Kim tests	Posterior traumatic **and** MDI
Neer / Hawkins	Impingement in **<20 y** → MDI



Sulcus sign – dimple below acromion with inferior traction

Sulcus grade (TO)	Acromiohumeral interval
1+	**<1 cm**
2+	**1–2 cm**
3+	**>2 cm**

2+/3+ sulcus + ↑ ER at side (TO) — **pathologic rotator interval** + patulous inferior capsule

Sulcus reflects — **patulous inferior capsule + rotator interval** (TO, Orthobullets) · "incompetent rotator interval" (OB)

Sulcus disappears in ER (TO) — **intact rotator interval**

Inferior laxity test (TO)	Positive
ER at side >85°	Inferior laxity
Gagey hyperabduction	**>105°** abduction → **inferior capsule (IGHL) laxity**
Beighton	Generalised hyperlaxity

Beighton score (generalised laxity)

Item (9 points)	Points
Palms to floor, knees straight	1
Genu recurvatum	2
Elbow hyperextension	2
Small-finger MCP hyperextension (passive >90°)	2
Thumb to ipsilateral forearm	2
Not included	**Excessive forearm supination**

Cut-offs generalised laxity **>4/9** (Orthobullets) · **>6** associated with **connective tissue disorder** (Marfan, Ehlers-Danlos; OB)



MCP hyperextension – generalised ligamentous laxity

✗ Exam distractors cross-body adduction (**AC joint**) · O'Brien (**SLAP**) · **axillary webbing** (**pec major rupture**) · Jobe's (**supraspinatus**)

4 Diagnosis & Classification

Key radiology findings

Modality	Finding
X-ray	Trauma series (**AP-IR, AP-ER, true AP, axillary, scapular Y**); **often normal**
MR arthrogram	Needed to assess **capsular volume**; **patulous inferior capsule** / capsular redundancy (e.g. posterior), **intact labrum**
MRI add-ons	Bankart (traumatic anterior) · Kim lesion (traumatic posterior) · glenoid bony erosion (chronic anterior)
Arthroscopy	**Drive-through sign**: scope passes easily between head + glenoid at **anterior band of IGHL** (also laxity)

– – Active space – –

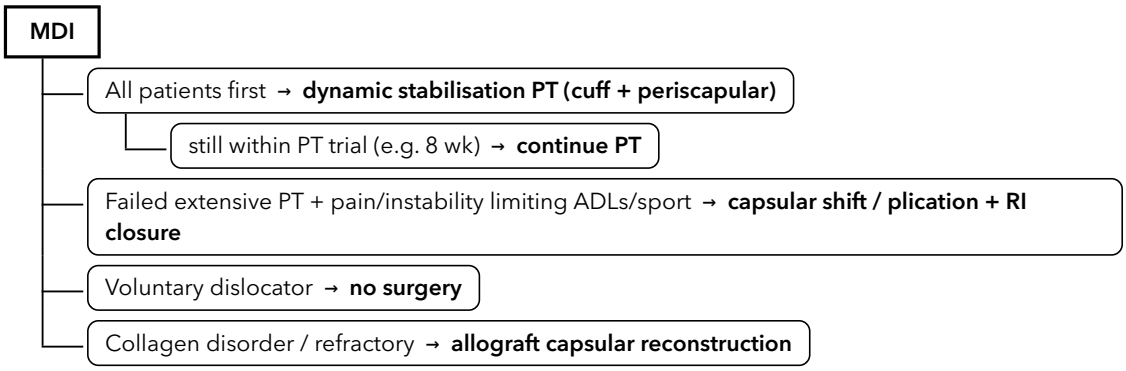
Diagnostic criteria

MDI requires	
Instability in **≥2 planes**	One **inferior**
Sulcus ≥2+	± ↑ ER at side
Load and shift ≥2+	Anterior + posterior

Differential diagnosis

Diagnosis	Clue
Anterior / posterior (traumatic) instability	TUBS, Bankart, **glenoid bone loss**, inverted pear
Long thoracic nerve palsy	Serratus weakness, **medial winging**
Buford complex	Normal variant: **absent anterosuperior labrum + cordlike MGHL** to base of biceps anchor
GIRD	IR deficit **25°** (OB) · **>20°** (TO, Orthobullets) vs contralateral → sleeper stretches
Little leaguer's shoulder	**Salter-Harris I** proximal humeral physis, excessive pitch counts
C-spine disease, brachial plexitis, TOS	Neurologic

5 Management



Initial treatment – prolonged rotator cuff & periscapular strengthening

First line — **all patients**, incl. bilateral, voluntary, collagen disorders: **dynamic stabilisation PT → rotator cuff + periscapular** (serratus anterior, rhomboids) strengthening, proprioception, humeral head control, scapulohumeral rhythm + **home exercises**

Early rehab — **closed kinetic chain exercises** → safe co-contraction of scapular + cuff muscles

Duration — **3–6 months** (Orthobullets, TO, OB) · **≥6 months** (several OB) · **6–12 months** (one OB) – read the options

Mandatory trial (TO instability lecture) — **6-month** PT (scapular stabilisation, cuff, deltoid, proprioception) before surgery · majority improve · **born loose > worn loose**

✗ Initial distractors **immobilisation / sling** (not effective; comfort only) · posterior capsular stretching (= GIRD) · bracing · steroid injections · labral repair · HAGL repair · thermal capsulorrhaphy · open/arthroscopic shift before PT trial

Outcome most patients satisfactory with nonop (pain + recurrence)

Operative criteria

Surgery for MDI	
Indications	**Failure of extensive PT** (sole indication, TO) · pain + instability interfering with ADLs / sport
Contraindication	**Voluntary dislocators**
Procedure of choice	**Inferior capsular shift** (open or arthroscopic)
Results	Good, but **less predictable** than traumatic anterior stabilisation

Surgery – capsular shift & rotator interval closure

Inferior capsular shift capsule shifted **superiorly** → addresses patulous inferior capsule

Capsular plication loose capsule to labrum, **balanced anterior + posterior** → ↓ capsular volume

Procedure (TO)	Indication	Mechanism
Inferior capsular shift (Neer & Foster, 1980)	Excess capsular redundancy	↓ capsular volume, centres head
Capsular plication (Snyder)	Lax capsule, **labrum intact**	Sequential tightening **posterior → anterior → RI closure**
Rotator interval closure	**RI gap >1 cm**	Sutured **superior → inferior**

Plication order (TO) 1 repair labrum if torn → 2 **posteroinferior** plication → 3 anterior plication → 4 **rotator interval closure** → 5 close portals

Rotator interval closure plicate **SGHL to MGHL** (anterosuperior capsule) → **eliminates sulcus sign** / inferior subluxation · greatest effect = **↓ ER with arm at side (0° abduction)** (~10°)
✗ Wrong options: restricts ER in cocking · IR at 90° · flexion/cross-body

Capsule rule tight **anterior** capsule → ↓ **ER** · tight **posterior** capsule → ↓ **IR**

Do not eliminate sulcus SLAP repair · Bankart repair (anteroinferior only) · supraspinatus repair · SAD

Open approach deltopectoral; subscapularis tenotomy vs split; also fix any labral pathology

Allograft capsular reconstruction rare; refractory, collagen disorders

– – Active space – –

Arthroscopic vs open similar outcomes; arthroscopic advantages

Post-op rehab	
0–4/6 wk	Sling / immobiliser
6–10 wk	ADLs, **45° limit** abduction + ER
10–16 wk	Gradual ROM
>16 wk	Strengthening
>10 months	**Contact sports** (only after normal strength + motion)

Recurrent instability with glenoid bone loss – bony augmentation

Clues to bone loss multiple recurrences · long history · instability in **mid-range (45–60° abduction + ER)**

Glenoid bone loss	Management
10%	Arthroscopic repair still OK
15% + labral tear, first-time (contact athlete)	**Surgical stabilisation** (✗ Wrong options: prolonged PT)
>20–25%	**Bony augmentation (Latarjet**: coracoid + conjoint tendon) – arthroscopic/open Bankart fail

✗ Bone loss distractors arthroscopic Bankart · subscapularis advancement · open capsular shift · hemiarthroplasty (no arthritis)

6 Complications

Thermal capsulorrhaphy – complications & failure

Status **historical / contraindicated**; abandoned for high complication + failure rates

Failure **high failure** for **anterior, posterior and MDI** at 2 + 5 y · recurrence peaks **2–3 wk** post-op (capsule weakest) · good 1-y result does not protect from later recurrence

Capsule revision finds capsule **thin + patulous** · **capsular necrosis / attenuation / insufficiency**
✗ Wrong options: thick fibrotic

Shrinkage depends on **total thermal energy + rate of delivery** · denatured tissue **does heal**
✗ Wrong options: fixed for peak temp

Revision after thermal failure **open revision** → capsular attenuation = **contraindication to arthroscopic revision**
✗ Wrong options: arthroscopic

Axillary nerve injury **heat-induced**; can't actively raise arm, full passive ROM, normal X-ray

Chondrolysis heat from **monopolar RF / laser** → progressive pain, ROM never regained, joint space loss

– – Active space – –



Glenohumeral chondrolysis after thermal capsulorrhaphy

✗ Chondrolysis distractor	Actually causes
Supraspinatus tear	Cuff tear arthropathy, superior head migration
Subscapularis dehiscence	Open deltopectoral → **anterior instability**
Anterior humeral circumflex coagulation	**Osteonecrosis** (main head supply)
Non-compliance with PT	Stiffness, pain (✗ Wrong options: chondrolysis)

Surgical complications

Recurrence	**most common** after arthroscopic or open capsulorrhaphy
Subscapularis deficiency	after **open anteroinferior shift**; **+ lift-off**, excessive ER; late **anterior subluxation** on axillary view
Loss of motion	over/asymmetric tightening → ↓ ER; treat **Z-lengthening subscapularis**; rare
Axillary nerve	iatrogenic neurapraxia → observe; **abduction + ER moves nerve away** from glenoid
Capsulorrhaphy arthropathy	**posterior glenoid wear**, posterior head subluxation, **glenoid retroversion**, **IR contracture**; **Putti-Platt, Magnuson-Stack**

X vs Y

	TUBS (traumatic)	AMBRI (MDI)
Onset	Trauma	**Atraumatic**
Direction	Unidirectional	**≥2 incl. inferior**
Lesion	**Bankart**, bone loss	**Capsular redundancy, RI deficiency**
Side	Unilateral	**Bilateral** often
Treatment	**Surgery**	**Rehab first** → inferior capsular shift

	MDI	GIRD / internal impingement
Exam	Sulcus, ↑ ER, load-shift both ways, Beighton +	**IR deficit ≥25°** vs other side
Treatment	**Cuff + periscapular strengthening**	**Posteroinferior capsular (sleeper) stretching**

– – Active space – –

How it's asked

"Cause of MDI?"	→ **inferior capsular redundancy** (TO) ✗ Wrong options: Bankart · Hill-Sachs · cuff tear
"Atraumatic recurrent dislocator, hyperextensible MCP – etiology?"	→ **capsular redundancy** ✗ Wrong options: inverted pear · LTN palsy · Bankart · Buford
"Connective tissue feature EXCEPT?"	→ **excessive forearm supination** (not Beighton)
"Gymnast, self-reduced dislocations, normal MRA – exam finding?"	→ **↑ ER in adduction** ✗ Wrong options: cross-body · O'Brien · axillary webbing · Jobe
"Most specific exam for MDI?"	→ **sulcus sign** ✗ Wrong options: apprehension · load-shift · jerk · Kim
"Voluntary popper, 2 cm bilateral sulcus – initial treatment?"	→ **cuff + periscapular strengthening** ✗ Wrong options: posterior stretching · plication · thermal · HAGL repair
"Swimmer, 2+ load-shift, 1.5 cm sulcus – next?"	→ **dynamic stabilisation therapy** ✗ Wrong options: bracing · labral repair
"Bilateral MDI volleyball, impingement signs?"	→ **physical therapy** ✗ Wrong options: injections · staged surgery
"17-y ♀ MDI – initial?"	→ **PT + home exercises** ✗ Wrong options: sling · thermal
"Swimmer, 2+ translation, elbow hyperextension, thumb to forearm?"	→ **PT cuff + scapulothoracic** ✗ Wrong options: open shift · thermal · sling
"13-y pitcher, Beighton signs, sulcus, 10° IR deficit?"	→ **MDI; periscapular training** ✗ Wrong options: GIRD sleeper stretches · little leaguer's shoulder
"Who gets prolonged PT rather than surgery?"	→ **17-y swimmer, Beighton 5 + sulcus** ✗ Wrong options: 15% bone loss dislocator · cuff tears
"Dancer, grade III sulcus, 8 wk PT done?"	→ **continue PT** ✗ Wrong options: shift · Bankart · Latarjet
"Rotator interval closure – true?"	→ **restricts ER at 0° abduction**
"Failed nonop MDI – successful surgery includes?"	→ **rotator interval closure** ✗ Wrong options: SLAP repair · capsular advancement into Hill-Sachs · IGHL repair
"Procedure that eliminates sulcus sign?"	→ **rotator interval closure** ✗ Wrong options: Bankart · SLAP · SAD

"Contraindication to arthroscopic revision?"	→ **capsular attenuation from thermal capsulorrhaphy** ✗ Wrong options: 10% bone loss · ALPSA · GLAD · SLAP + Bankart
"Can't actively elevate after thermal capsulorrhaphy, normal X-ray?"	→ **axillary nerve injury** ✗ Wrong options: chondrolysis
"Stiff painful shoulder 3 y after MDI scope, joint space loss?"	→ **monopolar RF thermal capsulorrhaphy** (chondrolysis)
"Thermal shrinkage — true?"	→ **high failure for anterior, posterior and MDI** ✗ Wrong options: thick fibrotic capsule · no healing
"6-y recurrent instability, mid-range apprehension, 3D-CT bone loss?"	→ **bony glenoid augmentation** ✗ Wrong options: arthroscopic Bankart · hemiarthroplasty
"Physiotherapist's classification of instability with polar types I–III?"	→ **Stanmore** (TO)
"Sulcus sign disappears with ER — implies?"	→ **intact rotator interval** (TO)

– – Active space – –

NEUROPATHIC (CHARCOT) JOINT OF SHOULDER

– Shoulder & Elbow · 29 / 60 –

1 Basics

Epidemiology

Incidence	**very rare** (~70 cases reported)
Demographics	mean age **~50** · ♂:♀ **2:1**
Sites	shoulder · elbow · foot & ankle
Syrinx → Charcot	**25% of syrinx patients** develop neuropathic arthropathy, **80% in upper limb** (Orthobullets, OB) · **25% of all Charcot joints** due to syringomyelia (Orthobullets associated conditions) – read the options

Syringomyelia – most common cause & MRI work-up

Cause	Note
Syringomyelia	**#1 cause of upper-limb (shoulder) Charcot**; **cervical syrinx = 75%** of Charcot shoulders; monoarticular (**shoulder > elbow**); Hatzis: syrinx in **5/6** shoulders
Arnold-Chiari malformation	**#1 cause of syringomyelia**
Hansen's disease (leprosy)	**2nd** most common cause in upper limb
Neurosyphilis (tabes dorsalis)	Usually **knee**; can be **polyarticular**; no longer the commonest
Diabetes	**#1 cause in foot & ankle**; reported in shoulder but less common; always rule out
Chronic alcoholism	Reported (Hatzis: 1/6)
Others	MS, ESRD, adhesive/TB arachnoiditis, meningomyelocele, traumatic SCI, familial sensory neuropathy
✗ Wrong option: **uremia**	**No documented association** (OB)

Pathophysiology

Syrinx	**fluid-filled cavity in spinal cord** → damages **decussating fibres of lateral spinothalamic tract** → **loss of pain + temperature**
Dissociated anaesthesia	pain/temperature lost · **proprioception + motor preserved** (early)
Syrinx enlarges	dorsal column + **anterior horn** → **areflexia**, weakness, atrophy
Neurotraumatic theory	↓ sensation/proprioception → **repetitive microtrauma** + abnormal joint pressures
Neurovascular theory	dysregulated reflexes → **↑ blood flow (hyperaemia)** → **↑ osteoclastic resorption**

Genes & lab markers

Molecular	**RANK/RANKL/OPG** pathway implicated
Labs	**ESR + WBC may be ↑** → hard to separate from osteomyelitis · can be **normal** (OB stem: CBC, ESR, CRP normal)
Histology	synovial hypertrophy · **detritic synovitis** (cartilage + bone fragments in synovium)

3 Presentation

History	trauma as inciting event in **30%**
Symptoms	swollen shoulder · **50% painless** · loss of function · instability · progressive **painless weakness** + ↓ ROM
Inspection	**swollen, warm, erythematous** (intermittent) → **mimics infection**
Motion	**loss of motion = most common finding (90%)** · crepitus · laxity/ mechanical instability
Neuro	↓ strength · **cape-like** sensory + temperature loss (back + arms) · asymmetric reflexes · **areflexia late**
Most common findings (OB)	**pain, loss of motion, instability**

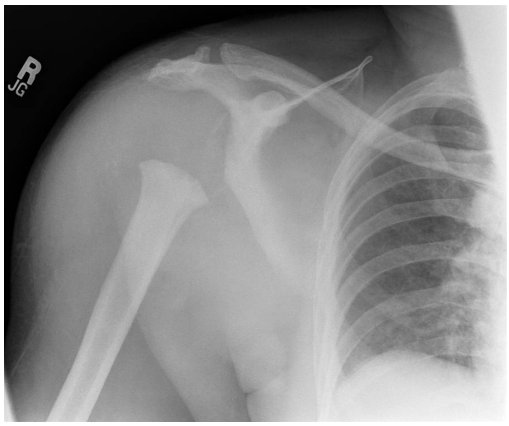
4 Diagnosis & Classification

Key radiology findings

X-ray (AP + scapular Y) — **gold standard / initial** diagnosis

Stage	X-ray finding
Early	Degenerative changes **mimic OA**
Late	**Superomedial flattening of humeral head**
Late	**Periarticular soft-tissue calcifications**
Late	**Glenoid sclerosis**
Late	**Extensive bone resorption** (humeral head "amputated"), fragmentation
Late	Destruction of **both sides** of joint → subluxation/dislocation

– – Active space – –



Charcot shoulder – resorption of humeral head, destruction both sides of joint

MRI cervical spine **next step in every Charcot shoulder** → syrinx (signal ≥ CSF on T1)



Cervical syrinx on MRI

CT if concern for osteomyelitis: intraosseous gas, cortical destruction, sequestra

Scan	Charcot	Osteomyelitis
Technetium bone scan	**Hot**	Hot
Indium-labelled WBC scan	**Cold (negative)**	**Hot (positive)**

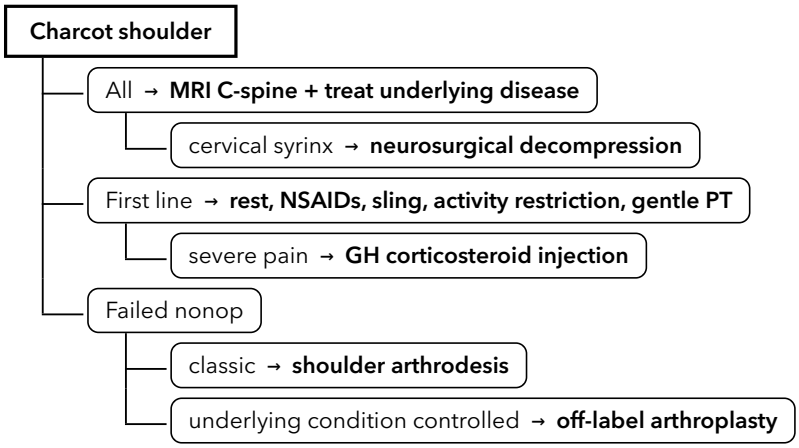
Best test to separate from osteomyelitis **indium WBC scan**

Differential diagnosis

✗ Distractor (OB)	Why not
AVN (loss of humeral head blood supply)	**Sclerosis + flattening** of head only
Metastasis (distant primary)	Variable; no obliteration of **both sides** of joint
Cartilage tumour	↑ calcification throughout lesion
Bone-forming tumour	↑ ossification throughout lesion
Osteomyelitis / septic joint	Same swelling, warmth, ↑ WBC/ESR, hot Tc scan → **indium scan hot**

Others synovial chondromatosis · soft-tissue sarcoma · tumoral calcinosis · Winchester syndrome · **Gorham's disease** · **Milwaukee shoulder**

5 Management



Nonoperative

First line rest, NSAIDs, **sling immobilisation** (slows ligamentous laxity), activity restriction, treat cause, gentle PT/passive stretching · **50% improve**

Corticosteroid injection severe pain · **glenohumeral** most effective · temporary **80%** pain ↓ (case reports)

Operative criteria

Procedure	Indication	Outcome / technique
Neurosurgical decompression	**Cervical syrinx present**	Slows progression, improves function + bone (**glenoid regrowth**); posterior fossa decompression, craniotomy, **syringoperitoneal shunt**, laminectomy
Arthrodesis	Severe pain, **failed nonop**; historically **only** option	**S-shaped incision** scapular spine → over acromion → anterolateral arm
Arthroplasty (hemi, TSA, rTSA, resurfacing)	Failed nonop **+ underlying condition treated**	**Off-label** (Charcot = **strict FDA contraindication**, loosening) · **70%** improved function at 5 y

Fusion position ("30-30-30") **20–30° abduction · 20–30° forward flexion · 20–30° internal rotation** → hand to mouth

No emergent ORIF / I&D not indicated (afebrile, normal labs) → **C-spine MRI first**

– – Active space – –

6 Complications

Complication	Risk factor
Infection	Surgery **without managing underlying condition**
Upper-limb DVT	Any surgery
Acromial stress fracture	**Reverse TSA** for Charcot
Progressive glenoid erosion	**Hemiarthroplasty**

Missed diagnosis often missed → **inappropriate surgery** (Hatzis)

X vs Y

	Charcot shoulder	Osteomyelitis	AVN humeral head
Pain	Often **painless** (50%)	Painful	Painful
Warm, red, swollen	Yes (intermittent)	Yes	No
X-ray	**Resorption + destruction both sides**, calcifications	Destruction, sequestra	**Sclerosis + flattening** (head only)
Indium WBC	**Cold**	**Hot**	–
Next step	**MRI C-spine**	Aspirate/CT	MRI shoulder

Charcot cause	Classic joint
Syringomyelia	**Shoulder** (upper limb)
Diabetes	**Foot & ankle**
Tabes dorsalis	**Knee** (polyarticular)

How it's asked

"Chronic shoulder pain, X-ray of destroyed/fragmented humerus – most common cause?"	→ **syringomyelia** ✗ Wrong options: diabetes · syphilis · alcoholism · uremia
"Most common cause of upper-extremity neuropathic arthropathy?"	→ **syringomyelia** ✗ Wrong options: Hansen's = 2nd · diabetes = foot · neurosyphilis = knee
"Painless progressive weakness, SCI, normal CBC/ESR/CRP, Charcot X-ray – early management?"	→ **cervical spine MRI** ✗ Wrong options: HIV test · repeat labs · emergent ORIF · emergent I&D
"Painless stiff shoulder, intermittent warmth, destroyed joint – work-up will show?"	→ **fluid-filled cavity in spinal cord** ✗ Wrong options: AVN · metastasis · cartilage/bone tumour
"Tract damaged by syrinx?"	→ **decussating fibres of lateral spinothalamic tract** (pain + temperature)

"Most common cause of syringomyelia?"	→ **Arnold-Chiari malformation**
"Test to differentiate Charcot from osteomyelitis?"	→ **indium WBC scan (cold in Charcot)**
"Syrinx found — treatment?"	→ **neurosurgical decompression**
"Shoulder fusion position?"	→ **20–30° abduction, flexion, IR each**
"Complication of rTSA in Charcot shoulder?"	→ **acromial stress fracture**

– – Active space – –

– – Active space – –

OLECRANON STRESS FRACTURE

– Shoulder & Elbow · 30 / 60 –

1 Basics

Epidemiology & location

Frequency	**rare**; predominantly **throwing athletes** (**baseball pitchers**), other high-level athletes; overuse injury
Location	predominantly the **olecranon epiphyseal plate** (persistent apophysis); line propagates **transverse or oblique**
Age	adolescent pitchers; may persist into **late adolescence** (**delayed closure of olecranon epiphysis**, 16–20 y; other elbow physes closed)
Associated throwing injuries	**UCL tear** · **medial epicondyle avulsion fracture**

2 Mechanism

Theory (3 accepted)	Note
Posteromedial olecranon impingement	–
Triceps traction force	Repetitive forceful triceps contraction
Valgus extension overload	Part of **valgus-extension overload syndrome**

Torg & Moyer	triceps contraction + **varus and valgus torques → olecranon epiphysis separates from epiphyseal plate**

3 Presentation

Pain	**posteromedial olecranon pain** of throwing arm; **worse with throwing** (after several innings), **better with rest**
Exam	mild swelling, **localised tenderness over proximal olecranon** (medial, distal to epicondyle); full ROM; no mechanical symptoms
Valgus stress test	**instability → associated UCL injury**; **normal** in isolated stress fracture
Late picture	acute worsening + **inability to throw** after failed nonop

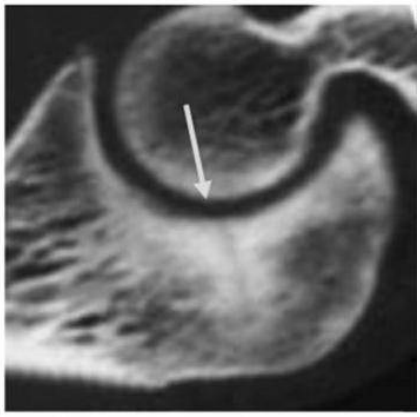
4 Diagnosis & Classification

Key radiology findings

Modality	Detail
X-ray	AP, lateral, oblique; optional **valgus stress** + **contralateral comparison**
X-ray findings	**Physeal widening, delayed fusion, fragmentation**; medial joint space widening if UCL injured
CT	Early confirmation; **best on lateral (sagittal) views**; **originates at articular surface → runs dorsal-proximal**
MRI	Early confirmation; **suspected UCL tear** → coronal **T2 fat-sat**: **T-sign** = UCL tear



Lateral X-ray – olecranon stress fracture through persistent physis



CT – fracture from articular surface running dorsal-proximal

Differential

✗ Distractor	Why wrong (11-yo pitcher, posteromedial tenderness, normal valgus stress)
Olecranon bursitis	No swelling/infection
OCD capitellum	No clicking / mechanical symptoms
UCL insufficiency	**Valgus stress normal**
Medial epicondylitis	Tenderness distal to epicondyle, over olecranon

– – Active space – –

5 Management

- **Olecranon stress fracture**
  - First line → **NSAIDs + rest ± splint (4–6 wk), no throwing up to 3 months**
    - delayed union / failed rest → **ORIF**
  - Undisplaced → **single large compression (cannulated) screw**
  - Displaced → **treat as olecranon fracture (screw ± TBW or plate)**

Nonoperative

Regimen short-term **NSAIDs**, **rest ± temporary splint**; **4–6 wk rest**, **removal from sport up to 3 months**; progressive ROM; **avoid valgus loading (throwing)**

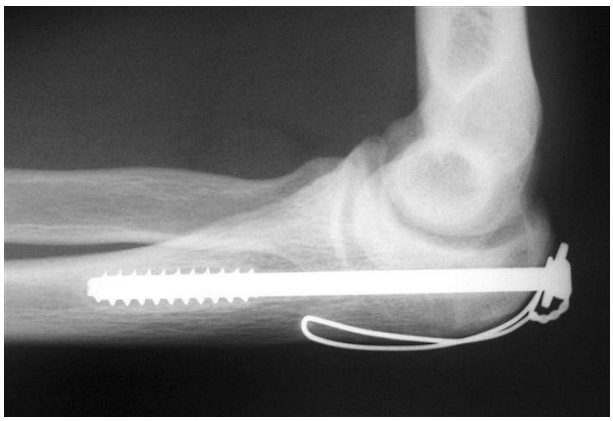
Adjuncts **electrical bone stimulation**; **vitamin D + calcium**

Operative criteria – compression screw fixation

Indication	Fixation
Delayed union / failed rest (e.g. **9 months**, or **two 6-wk rest periods**)	**Large compression screw** (intramedullary **cannulated**)
Undisplaced	**Single large screw** (e.g. **7.0 mm**)
Displaced	Screw **± tension band wire**, or plate

Plate fixation option once displaced (OB) · **not indicated** for this injury (another OB) – read the options

✗ Next-step traps elbow arthroscopy · open osteophyte removal · ulnar nerve transposition · **cast + strengthening** after failed rest · physiotherapy · **long-arm cast** (stiffness; only reasonable briefly at onset) · **hinged-elbow brace** (no healing benefit)



Compression screw + tension band construct, healed

Outcomes

Rettig (7.0 mm screw, 5 pitchers) union mean **15.4 wk**; return to throwing mean **29.4 wk**

Charlton & Chandler (16–20 y, delayed epiphyseal closure) UCL intact in all; ORIF with **TBW, screw ± autograft** → all **asymptomatic at 32 months** despite **> 30 months** pre-op pain

Surgery reliable results when rest fails

X vs Y

	Olecranon stress fracture	UCL insufficiency
Pain	**Posteromedial olecranon**	–
Valgus stress test	**Normal** (unless UCL also injured)	**Positive / instability**
Imaging	Physeal widening, CT fracture line	Medial joint widening; MRI **T-sign**

	Undisplaced	Displaced
Fixation	**Single large compression screw**	Screw ± TBW or plate (as olecranon fracture)

How it's asked

"19-yo pitcher, posterior elbow pain 9 months despite nonop, acute worsening — next?"	→ **internal fixation with compression screw** ✗ Wrong options: arthroscopy · osteophyte removal · ulnar transposition · cast
"11-yo pitcher, posteromedial pain, tender over proximal olecranon, normal valgus stress — diagnosis?"	→ **olecranon stress fracture** ✗ Wrong options: UCL insufficiency · OCD · medial epicondylitis · bursitis
"15-yo pitcher, two 6-wk rest periods failed, other physes closed — next?"	→ **cannulated screw fixation** ✗ Wrong options: plate · hinged brace · long-arm cast · physio
"Mechanism theories?"	→ **posteromedial impingement, triceps traction, valgus extension overload**
"First-line treatment?"	→ **NSAIDs + rest ± brief splint, no valgus loading**
"Indication for ORIF?"	→ **delayed union**

– – Active space – –

OS ACROMIALE

– Shoulder & Elbow · 31 / 60 –

1 Basics

Definition

Os acromiale	**unfused secondary ossification centre** of the acromion (**failure of fusion of anterior acromial apophysis**, OB) → impingement, cuff tendonitis

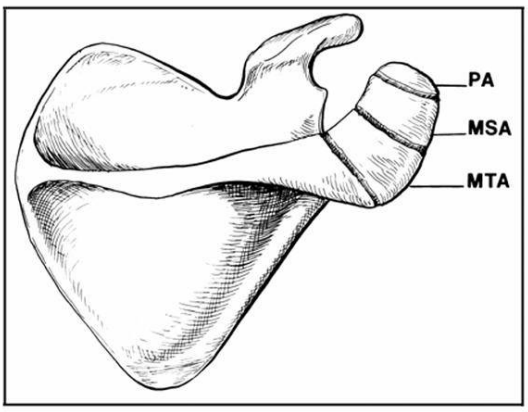
Epidemiology

Prevalence	**8%** (Orthobullets, OB) · **1–15%** (another OB) – read the options
Bilateral	**60%** (Orthobullets) · **33%** (Sammarco skeletons, OB) – read the options
Race	**Black > White** (**13.2% vs 5.8%**)
Sex	♂ **>** ♀ (**8.5% vs 4.9%**)
Usually incidental	found on imaging for other reasons, often **trauma**

Anatomy – failed fusion of meso- and meta-acromion

Ossification centre	Position	Origin of
Pre-acromion	**Tip**	**Anterior deltoid + coracoacromial ligament**
Meso-acromion	**Mid**	**Middle deltoid**
Meta-acromion	**Base**	**Posterior deltoid**
Basi-acromion	(4th centre, OB)	–

Number of centres	**3** (Orthobullets, OB) · **4: pre-, meso-, meta-, basi-** (another OB) – read the options
Fusion age	ossify medially toward clavicular facet **by 18 y** (OB) · **15–18 y** (Hurst, OB)
Most common site	**junction of meso- and meta-acromion** = commonest **symptomatic** form ✗ Wrong options: meso–pre · meta–pre
Blood supply	**acromial branch of thoracoacromial artery** → preserve at surgery



Ossification centres: pre- (PA), meso- (MSA), meta-acromion (MTA)

Pathological mechanism

✓ **Mechanism** **failure of fusion between ossification centres**

✗ Mechanism distractor	Condition
Failure of mesenchymal segmentation	**Tarsal coalition**
Defect in chondrocyte proliferation	**Achondroplasia**
Glycine substitution, abnormal collagen cross-linking	**Osteogenesis imperfecta**
Failure of cartilaginous differentiation	**Congenital radioulnar synostosis**

Associated conditions

Impingement + rotator cuff disease **meso-acromion** → cuff tendonitis + **full-thickness tears in 50%**

3 Presentation

Impingement pain anterior fragment **flexes / displaces downward** with deltoid contraction + arm elevation → ↓ subacromial space

Synchondrosis pain **motion at nonunion site**

Symptoms **superior shoulder pain**, **overhead activity**, **lying on affected side**; tenderness **anterolateral / superior acromion**

Trigger **trauma** can make a previously silent os symptomatic

Diagnostic injection **1% lidocaine** into site when pain source unclear (Spiegl)

4 Diagnosis & Classification

Key radiology findings

Modality	Use
Axillary lateral X-ray	**Best view** for diagnosis
CT	Nonunion site; **degenerative change (cysts, sclerosis, hypertrophy)**
Axial MRI	Shows os (superior pain + overhead vignette)

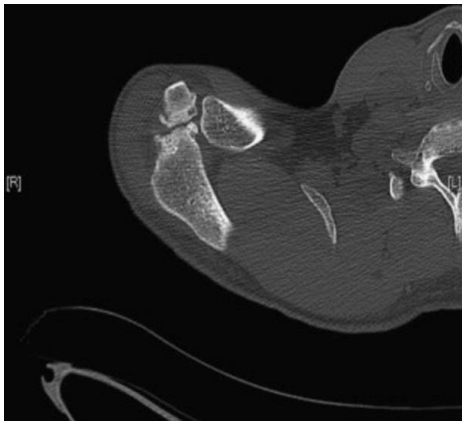
Os vs fracture os: **rounded edges + corticalisation of entire fragment** · fracture: acute

Fused os acromiale may be mistaken for a free ossicle (Sammarco)

– – Active space – –



Axillary lateral showing os acromiale



CT – unfused acromial fragment

✗ **Diagnosis distractor**	
Acute acromial fracture	Rounded corticated edges = os
Type IV AC separation	Posterior clavicle
Deltoid avulsion	–
AC OA / AC separation / PTRCT / SLAP	MRI shows os

5 Management

Os acromiale

- Incidental / mild symptoms → **observe, NSAIDs, PT, subacromial steroid**
  - fails 6–12 months → **surgery**
- Symptomatic os + impingement → **two-stage fusion**
- Impingement ± cuff tear, os incidental + nontender → **ASD + acromioplasty**
- Small symptomatic pre-acromion / failed ORIF → **fragment excision**

Nonoperative

Regimen	observation, **NSAIDs**, activity modification, therapy, **subacromial corticosteroid → majority**
Duration	surgery only after **6–12 months** of nonop

Operative criteria – treatment of symptomatic os acromiale (fusion)

Procedure	Indication	Detail
Two-stage fusion	**Symptomatic os + impingement**, failed nonop	**Stage 1: fuse ± bone graft** (rigid fixation: **TBW, sutures, cannulated screws**) → **Stage 2: acromioplasty**
Arthroscopic SAD + acromioplasty	**Impingement ± cuff tear, os incidental + nontender**	–
Fragment excision (open/arthroscopic)	**Symptomatic pre-acromion, small fragment · salvage after failed ORIF**	Arthroscopic: less periosteal + deltoid detachment; **better results with pre-acromion**

Direct excision of large fragment — **deltoid dysfunction** → prefer fusion

Warner (15 shoulders) — **cannulated screws + tension band + autograft** → union in **all but one**; 8 had cuff tears

Harris — commonest concurrent procedure **cuff repair (59%)**, then **distal clavicle excision (25%)**; all techniques improve outcomes

Abboud — meso-os + cuff pathology satisfactory in **4 of 8**; **80%** satisfactory overall vs **22%** in **workers' compensation**

44-yo, failed PT after cuff repair, symptomatic os + impingement — ✗ Wrong options: observation · more PT · ASD alone (os not incidental) · excision alone

6 Complications

Deltoid weakness — after **fragment excision**

Persistent pain / weakness

Prognosis — **poorer outcomes after cuff repair** with **meso-os acromiale**; os may worsen outcome of unrelated shoulder surgery (Hurst)

X vs Y

	Two-stage fusion	ASD + acromioplasty	Excision
Os status	**Symptomatic, tender** + impingement	**Incidental, nontender**	Small **pre-acromion** / failed ORIF
Deltoid	Preserved	Preserved	**Risk of dysfunction** (large fragments)

How it's asked

"27-yo, shoulder pain, X-ray shows os – cause?" — → **failure of fusion meso- and meta-acromion**
✗ Wrong options: meso–pre · acute fracture · type IV AC · deltoid avulsion

– – Active space – –

"23-yo, superior shoulder pain overhead + lying on side, axial MRI?"	→ **os acromiale** ✗ Wrong options: AC OA · PTRCT · SLAP
"Most common location?"	→ **junction of meso- and meta-acromion**
"Pathological mechanism?"	→ **failure of fusion between ossification centres** ✗ Wrong options: segmentation = coalition · glycine = OI
"44-yo, failed PT after cuff repair, symptomatic os with impingement – treatment?"	→ **two-stage fusion** ✗ Wrong options: ASD · excision only
"Os incidental + nontender, impingement ± cuff tear?"	→ **arthroscopic SAD + acromioplasty**
"Best view?"	→ **axillary lateral**
"Complication of excision?"	→ **deltoid weakness**

OSTEOCHONDRITIS DISSECANS OF ELBOW

– Shoulder & Elbow · 32 / 60 –

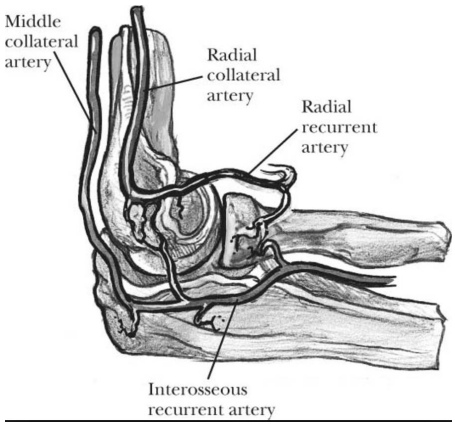
1 Basics

Presentation & population (capitellum, gymnasts/throwers)

Definition	localised injury → **separation of articular cartilage + subchondral bone** of the **capitellum**
Population	**gymnasts + throwing athletes** (pitchers) · repetitive overhead or **upper-limb weight bearing** · also cheerleaders ✗ Wrong options: swimmers/divers · linemen · rugby · cyclists
Age	**>10 y** (adolescent) (Orthobullets, OB) · "typically **>13 y**" (one OB explanation)
Sex	**boys > girls**
Site	**capitellum** of **dominant** arm (most common) · occasionally **radial head** (OB) ✗ Wrong options: "capitellum or lateral trochlea"
Juvenile vs adult	**juvenile OCD better prognosis**

Anatomy – blood supply

Capitellum	supplied by **2 end arteries**: **radial recurrent** + **interosseous recurrent** → vulnerable to ischaemia



Capitellar blood supply – radial recurrent + interosseous recurrent arteries

Pathophysiology

Mechanism	**repetitive compression** (lateral compression + axial loading) of **immature capitellum** → **vascular insufficiency** + **repetitive microtrauma** → osteochondral necrosis → separation
Throwers	**repetitive valgus overload of radiocapitellar joint**, **late cocking** phase
Panner's + OCD	likely a **continuum** (OB)

– – Active space – –

3 Presentation

Pain insidious, **activity-related lateral elbow pain**, dominant arm · posterolateral pain with throwing

Sign	Timing
Loss of extension (mild, ~15°)	**Early** sign = typical early presentation
Catching / locking / grinding / clicking / popping	**Late** – loose bodies; **poorer prognosis**
Lateral tenderness, mild **effusion**, ± crepitus	Any
Painful click on passive rotation, painful pronation/supination	Unstable lesion

Instability testing normal
✗ Wrong options: MUCL laxity

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral; slightly flexed-elbow view) confirms diagnosis: **well-defined subchondral lesion** in capitellum · radiolucency, **flattening**, sclerosis, fragmentation, loose bodies · initial films may be **negative**



Capitellar OCD – flattening and lucency

MRI best for **size, extent of oedema, cartilage status** · OCD: **separation of cartilage from capitellum + chondral fissuring** · stable: **intact cartilage + subchondral signal change**, T2 no signal around lesion

MR arthrogram contrast **outlining margins** of lesion = **unstable** fragment → **poor** prognostic sign
✗ Wrong options: good

Os supratrochleare dorsale normal variant, incidental
✗ Wrong options: prognostic

Classification – radiographic & arthroscopic

– – Active space – –

Type	Findings	Image
I	**Intact cartilage**; bony stability may or may not be present	
II	**Cartilage fracture** with bony collapse or displacement	
III	**Loose bodies** in joint	

Stability criteria (Takahara)

Stable (heals with rest) – all of	Unstable (surgery better) – any of
Open capitellar physis	**Closed** capitellar physis
Localised flattening or radiolucency of subchondral bone	**Fragmentation**
Good elbow motion	**Restricted motion ≥20°**

5 Management

Capitellar OCD

- Type I stable → stop activity 3–6 wk → **graded return over 6–12 wk**
- Unstable type I / stable type II → **arthroscopic microfracture or drilling**
- Large, incompletely displaced → **arthroscopic reduction + fixation**
- Unstable type II / type III → **arthroscopic debridement + loose body excision**
- Large type II–III engaging radial head → **OATS**
  - uncontained → **size-matched fresh allograft**

Stable lesions – activity cessation

First line — **cessation of offending activity** ± immobilisation · **>90% success**

Duration — **cessation of activity 3–6 weeks**, followed by slow progression back to activities over next **6–12 weeks** (Orthobullets, OB) · "**minimum 6 wk**" / "**6 wk**" (other OB explanations) – read the options

– – Active space – –

Return to play resumption when **asymptomatic** + examination shows **restoration of painless range of motion**
✗ Wrong options: wait for complete radiographic healing (residual X-ray abnormality persists even when asymptomatic)

No participation until symptoms resolve + motion recovers
✗ Wrong options: continue until worse

Avoid casting 6–8 wk too long → stiffness

Avoid immediate throwing programme · no surgery before trial of nonop (stable, no loose body)

Nonop healing >50% of stable lesions heal with activity restriction; **smaller lesions without cyst-like change** heal better

Operative criteria

Procedure	Indication	Post-op
Arthroscopic microfracture / drilling (extra- or transarticular)	**Unstable type I** · **stable type II** · symptomatic lesion with **intact articular surface** (in situ drilling)	Protected ROM; strengthen 2 mo; throw/WB **4–6 mo**
Fixation (arthroscopic reduction)	**Large**, **incompletely displaced** (partial separation)	As above; results variable
Arthroscopic debridement + loose body excision	**Unstable type II** · **type III** · mechanical symptoms	Early ROM ± brace; no throwing/WB **3 mo**
OATS (autograft/allograft)	**Large type II–III engaging radial head**; uncontained → fresh allograft	Strengthen 3 mo; throwing 5–7 mo
Lateral column osteotomy	**Investigational**, salvage (offloads lateral compression)	–

Surgery overall failed extended nonop · mechanical symptoms · loose bodies on imaging · unstable lesions

OATS (single plug) **93%** graft incorporation, high return to sport, low donor morbidity · large unstable lesions: **fixation/reconstruction > simple excision**

Microfracture among the most commonly performed + studied

Prognosis – skeletal maturity

Best predictor **skeletal age / open physes**
✗ Wrong options: symptom duration · gender · activity-related symptoms · anatomic variant

Healing most heal in **6–18 months** of nonop (open physes)

Poor nonop healing **radial head enlargement** · skeletal age difference · closed physis

Long term **~50%** residual ADL symptoms (Takahara: 50% nonop, 46% fragment removal) · **up to 50% arthritis**

6 Complications

Complications **stiffness** · pain · **failure to return to sport** · **arthritis**

X vs Y

	OCD of capitellum	Panner's disease
Age	**>10 y** (adolescent)	**<10 y** (first decade)
Lesion	**Focal, well-defined subchondral** lesion	**Entire capitellar ossific nucleus** fragmented; irregular epiphysis
MRI	Cartilage separation + chondral fissuring	–
Loose bodies	Common	Rare
Course	Variable; arthritis up to 50%	**Benign, self-limiting**, normal regrowth, no sequelae
Surgery	May be needed	**Contraindicated**

Distractor (X-ray / stem)	Diagnosis
Calcification at common extensor origin	**Lateral epicondylitis**
Widened medial epicondyle physis	**Little Leaguer's elbow**
Incarcerated medial epicondyle fragment	Medial epicondyle fracture
Olecranon osteophyte, posterior pain in extension	**Valgus extension overload**
Medial pain + instability	UCL (Tommy John)
After dislocation	PLRI
Posterior pain in extension	Olecranon stress fracture
Lateral pain, rarely effusion/mechanical	Plica

How it's asked

"13-y pitcher, lateral pain, effusion, painful click on rotation?"	→ **OCD** ✗ Wrong options: Tommy John · PLRI · stress fracture · plica
"Population for capitellar OCD?"	→ **gymnasts + throwing athletes**
"Characteristic of elbow OCD?"	→ **MRI: cartilage separation from capitellum + chondral fissuring** ✗ Wrong options: whole ossific nucleus · normal regrowth · <10 y · MUCL laxity
"Typical early presentation?"	→ **loss of extension** ✗ Wrong options: catching/locking = late · boys <10 = Panner · MR arthrogram outline = poor sign
"14-y gymnast, can't fully straighten, popping – which X-ray?"	→ **capitellar flattening + lucency** ✗ Wrong options: extensor calcification · physeal widening · olecranon osteophyte
"12-y gymnast, 15° extension loss, 3 × 7 mm stable lesion, no loose bodies?"	→ **no gymnastics until symptoms resolve + motion recovers**

– – Active space – –

"Little League pitcher, catching, lateral pain — initial management?"	→ **rest; resume when asymptomatic + painless full ROM** ✗ Wrong options: wait for X-ray healing · drilling · osteotomy
"11-y pitcher, lacks terminal extension, first visit?"	→ **stop pitching 3–6 wk + slow return** ✗ Wrong options: medial epicondyle ORIF · OATS · debridement
"Type I OCD on MRI, full ROM — initial treatment?"	→ **cessation 3–6 wk then progression over 6–12 wk** ✗ Wrong options: casting 6–8 wk · throwing programme
"Intact cartilage + subchondral signal change?"	→ **cessation of activity for 6 wk** ✗ Wrong options: microfracture · fixation · OATS
"Unstable type I or stable type II?"	→ **microfracture / drilling**
"Large lesion engaging radial head?"	→ **OATS**
"16-y cheerleader, closed physes — largest prognostic factor?"	→ **skeletal age**

– – Active space – –

PECTORALIS MAJOR RUPTURE

– Shoulder & Elbow · 33 / 60 –

1 Basics

Epidemiology

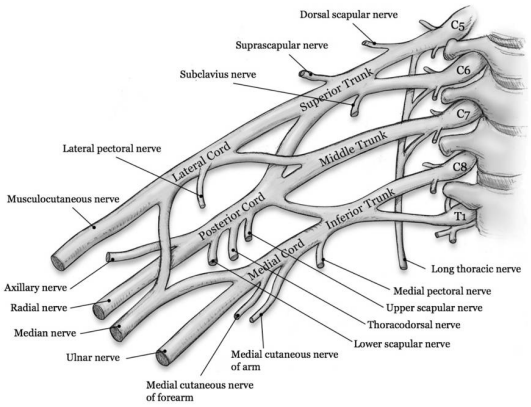
Incidence	rare (**<1/100,000/y**) but rising · **75%** of reported cases since 1990
Demographics	**almost exclusively ♂, 20–40 y** · **weightlifters**, **bench press** · almost no reports in ♀ (TO) ✗ Wrong options: equal ♂/♀ · old gym-goers
Risk	**anabolic steroids** / supplements
Other sports (TO)	rugby, football, wrestling, water skiing, boxing
Most common site	**tendinous avulsion at humeral insertion (65%)** > **musculotendinous junction (27%)** (TO) · intramuscular rare (direct trauma, stab) ✗ Wrong options: "MTJ most common" · muscle belly
Most common portion	**sternocostal (sternal) head** (✓ TO "all except" answer) · clavicular head often intact

Anatomy – sternal head & pectoral nerve innervation

	Clavicular head	Sternocostal head
Origin	**Medial half of clavicle** (+ proximal sternum)	Sternum, **costal cartilages 1–6** (TO), external oblique aponeurosis
Size	Smaller; may look confluent with deltoid	**Larger, >80% of volume**
Tendon layer at insertion	**Anterior**	**Posterior** (lower sternal fibres)
Nerve	**Lateral pectoral (C5–C7)**	**Lateral + medial pectoral** (dual, C5–T1)
Rupture	Fails last	**Fails first**, often isolated

Innervation	**lateral pectoral (C5–C7)** = clavicular + upper sternocostal · **medial pectoral (C8–T1)** = lower sternocostal + pectoralis minor (TO, Orthobullets)
Insertion	**lateral lip of bicipital groove** (humeral shaft lateral to groove), anterior + posterior layers (TO)
Action	**adduction + internal rotation** · forward elevation/flexion lesser (chiefly clavicular head) (TO) · not predominant forward flexion = **anterior deltoid + coracobrachialis**
Thoracoappendicular muscles	pec major, pec minor, subclavius, serratus anterior

– – Active space – –



Brachial plexus: lateral pectoral (lateral cord) and medial pectoral (medial cord)

✗ Nerve distractor	Supplies
Suprascapular (C5–C6)	Supraspinatus, infraspinatus
Thoracodorsal (C6–C8)	Latissimus dorsi
Medial brachial cutaneous (T1)	Sensation medial arm
Upper/lower subscapular	Subscapularis
Axillary / musculocutaneous	Deltoid, teres minor / biceps, brachialis

2 Mechanism

Mechanism – eccentric bench press

Mechanism — excessive tension on **maximally eccentrically contracted** muscle · **downward (deceleration) phase of bench press**
✗ Wrong options: upward · max elevation · bar on chest

Position — **final 30° of humeral extension** (inferior sternal fibres at maximal stretch) · extension + abduction + ER · TO: **sudden forceful abduction + ER of contracted muscle**

Failure sequence — **inferior sternocostal fibres** (insert more posteriorly) → superior sternocostal → **clavicular head last**

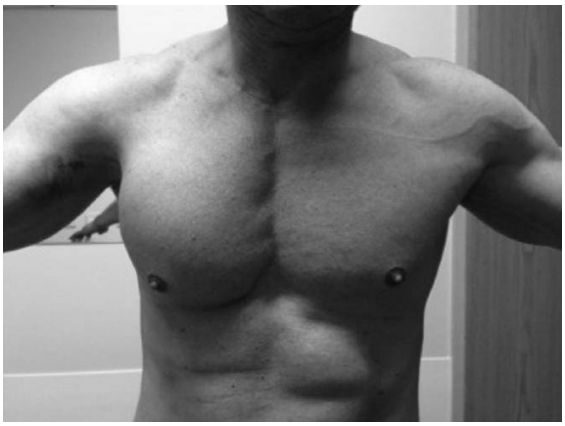
3 Presentation

History — **sudden pop / tearing** with resisted adduction + IR · pain, weakness

Inspection — **swelling + ecchymosis anterolateral chest wall and/or proximal medial arm** · **loss of anterior axillary fold**, muscle retracted medially · palpable defect accentuated by **resisted adduction** · **"dropped nipple" sign**

Ecchymosis location — **anterior brachium/upper arm → humeral avulsion** · **chest wall** → mid-substance / MTJ tear (less benefit from surgery)

Weakness — **adduction + internal rotation** (± forward flexion); ER, abduction normal; apprehension negative



Loss of axillary fold, chest asymmetry, upper arm ecchymosis

✗ Exam-sign distractor	Lesion
Popeye	Proximal LHB tear
Medial winging	Long thoracic nerve / serratus
Belly-press	Subscapularis
ER lag	Infraspinatus

✗ Wrong option: axillary fold subscapularis and biceps injuries do **not** change the axillary fold

4 Diagnosis & Classification

Diagnosis – MRI

X-ray trauma series (true AP, scapular Y, axillary) · usually **normal** · lost pec shadow or bony avulsion

MRI **investigation of choice / gold standard** · complete vs partial · tear site + retraction (muscle belly vs bone–tendon junction) · needs **dedicated chest-wall sequence** (standard shoulder MRI misses it) · **T2 acute**, **T1 chronic** · intact clavicular head may mask partial sternal tear

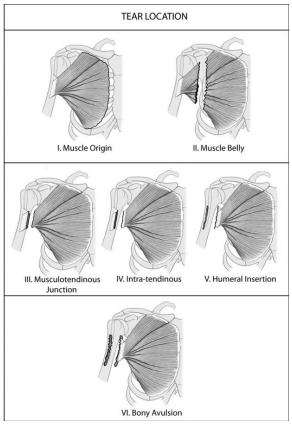
EMG no role without nerve injury

– – Active space – –

Modified Tietjen classification

Type	Description
I	Contusion / sprain
II	Partial tear
III	**Complete** tear, by location ↓
III-A	Muscle origin
III-B	Muscle belly
III-C	**Musculotendinous junction**
III-D	Intratendinous
III-E	**Tendon avulsion off humerus (no bone)**
III-F	Bony avulsion off humerus

Tietjen (TO) sprain · partial · complete (origin, belly, MTJ, tendon) + added **bony flake avulsion** + **intratendinous**



Modified Tietjen tear locations

5 Management

Pectoralis major rupture

- Low-demand, elderly, sedentary → **sling, ice, NSAIDs, PT**
  - muscle belly / low-grade partial / chest-wall tear → **nonop**
- Acute complete avulsion or MTJ, young/active → **open primary repair**
- Chronic, can't mobilise → **allograft reconstruction**
  - primary repair still possible years later

Operative criteria

Situation	Treatment
Avulsion from humeral insertion (complete pecto-humeral tear)	**Primary repair** (✓ surgery indication)
Acute tear, young athlete / labourer	**Primary acute repair** = gold standard
MTJ tear	Repair (Orthobullets, OB) · therapy may be considered (one OB) · may need augmentation/reconstruction (one OB) – read the options
Muscle belly / partial / chest-wall ecchymosis	**Nonop**
Chest asymmetry alone, no deficit	not an indication
Chronic, inadequate excursion	**Reconstruction (allograft)**

Treatment – acute repair, chronic allograft reconstruction & nonoperative deficits

Primary repair — **open, deltopectoral** · reattach to humerus lateral to bicipital groove · **suture anchors or buttons** (TO) · transosseous with cortical trough · cortical button · anchors (PEAK/all-suture) · all comparably excellent; some evidence **button + trough > anchors**

Acute vs delayed — acute easier, less scar, avoids later reconstruction/allograft
✗ Wrong options: delayed repair after failed nonop

Best long-term result (sternocostal tear) — **open repair of sternocostal portion**
✗ Wrong options: PT/NSAIDs · steroid injection · LHB repair · clavicular repair

Repair outcome — reliable strength, return to sport, satisfaction; improves regardless of tear site · most can be repaired primarily, **even chronic**
✗ Wrong options: majority need allograft

Chronic reconstruction — release adhesions superficial + deep (**protect medial + lateral pectoral nerves**) ± fascial release · graft: **Achilles allograft (most common)** (Orthobullets) · **semitendinosus weave** (OB) · gracilis weave · inferior to primary repair, better than nonop

Chronic weakness 2 y after PT, young labourer — **allograft reconstruction (semitendinosus weave)**
✗ Wrong options: latissimus transfer · electrical stimulation · arthrodesis · arthroscopic repair

Nonop deficits — **weakness of adduction + IR** (esp. isokinetic adduction, peak torque) · cosmetic defect · delayed recovery · poor satisfaction · ↓ return to sport

Nonop protocol — sling in adduction + IR · immediate PROM · AAROM/AROM over 6 wk · strengthening/unrestricted at **2–3 mo**

Tendon transfer — not for acute tear (pec major is a **donor** for subscapularis insufficiency)

– – Active space – –

6 Complications

Complication	Fact
Persistent pain	**Most common**
Re-rupture	**5–7%**; fails at **suture–tendon interface**
Residual weakness	–
Cosmetic deformity	–
Nerve injury in reconstruction	Medial + lateral pectoral during deep release

X vs Y

	Pec major tear	LHB rupture	Subscapularis tear
Sign	**Lost anterior axillary fold**, dropped nipple	**Popeye**	**Belly-press**, ↑ passive ER
Weakness	**Adduction + IR**	–	–
Axillary fold	**Lost**	Normal	Normal

	Sternocostal head	Clavicular head
Torn	**Most often, first**	Last, often intact
Nerve	Lateral + medial pectoral	Lateral pectoral

How it's asked

"Pec major tear – all false EXCEPT?"	→ **sternal head most commonly ruptured** (TO) ✗ Wrong options: old gym-goers · MTJ most common · equal sexes
"Bench press pop, upper medial arm pain, lost anterior fold?"	→ **pectoralis major tear** (TO) ✗ Wrong options: LHB · axillary fold tear · cuff
"Weightlifter, pop, MRI retracted muscle – nerve?"	→ **lateral + medial pectoral** ✗ Wrong options: axillary · musculocutaneous · subscapular · suprascapular
"Labelled plexus – innervation of most commonly torn portion?"	→ **A + E (lateral + medial pectoral)**
"True about this injury?"	→ **sternal head torn more than clavicular** ✗ Wrong options: ♀ · allograft needed · flexion loss · belly
"Bench press phase at greatest risk?"	→ **downward deceleration**
"38 y weightlifter, MRI arrow?"	→ **pectoralis major tear** ✗ Wrong options: biceps · pec minor · subscapularis

"When is surgery indicated?"	→ **avulsion from humeral insertion** ✗ Wrong options: never · after failed PT · asymmetry only · chest-wall ecchymosis
"30 y lifter, lost axillary contour, weak IR — definitive?"	→ **primary acute surgical repair**
"Bodybuilder, axillary + anterior brachium ecchymosis, MRI — best long-term?"	→ **open repair of sternocostal portion**
"Young electrician, lost fold worse with adduction, normal X-ray?"	→ **open repair of pec major avulsion** ✗ Wrong options: subscap repair · LHB tenodesis · brace
"Chronic tear 2 y after PT, weak — treatment?"	→ **allograft reconstruction (semitendinosus weave)**
"Nonop treatment leaves weakness of?"	→ **adduction + internal rotation**
"Repair to red label; preop sign?"	→ **lost axillary fold + chest asymmetry + arm ecchymosis (Figure C)** ✗ Wrong options: Popeye · winging · belly-press · ER lag
"42 y lifter, 3 days, MRI typical tear — next step?"	→ **primary repair** ✗ Wrong options: EMG · sling + PT · reconstruction · tendon transfer

– – Active space – –

PHASES OF THROWING

– Shoulder & Elbow · 34 / 60 –

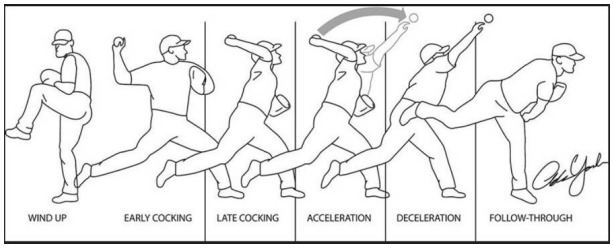
1 Basics

The phases

Number of phases — **5**: wind-up · cocking (early + late) · acceleration · deceleration · follow-through (Orthobullets, TO) · **4**: wind-up · cocking · acceleration · deceleration/follow-through (one TO) – read the options

Six phases (TO biomechanics lecture) — wind-up · **stride** · cocking (early + late) · acceleration · deceleration · follow-through – read the options

Timing — whole throw **~2 s**; wind-up + acceleration ≈ **75% (1.5 s)** (Orthobullets) · **wind-up + cocking = 80% (~1500 ms)**, acceleration **2% (50 ms)**, deceleration **18% (350 ms)** (TO) – read the options



Phases – wind-up, early cocking, late cocking, acceleration, deceleration, follow-through

Phase	Mechanics	Peak muscle activity	Associated pathology
Wind-up (500–1000 ms, TO)	Body rotates, hips + shoulders 90° to target; "controlled fall"; minimal shoulder force	**Rotator cuff inactive**	–
Early cocking	Arm loading; max valgus not yet reached	**Deltoid**	–
Late cocking (500–1000 ms, TO)	**Max shoulder ER** ("loaded", apprehension position), **high torque**; anterior capsule coiled	**Supraspinatus, infraspinatus, teres minor** (rotator cuff); subscapularis late cocking → early acceleration	**MUCL injury** (**max elbow valgus**) · **internal impingement** · **GIRD** · **SLAP (peel-back)**
Acceleration (**50 ms**, shortest, TO)	Max ER → **ball release (~ear level)**; elastic capsular recoil + IR muscles	Early **triceps** → late **pec major, latissimus dorsi, serratus anterior**	Enormous elbow valgus at release (TO)
Deceleration (350 ms, TO)	Eccentric braking; centre of gravity over plantar foot	**Eccentric contraction of all muscles**; **highest torque**	**Most harmful**: **SLAP**, **biceps**, **brachialis**, **teres minor**
Follow-through	Body rebalances, stops forward motion	Muscles **return to resting**	–

Most harmful phase — **deceleration** (Orthobullets, TO) · "**late cocking + deceleration** put greatest force on shoulder" (TO)

Rotator cuff greatest stress **deceleration** (eccentric slowing of arm)

Minimal throughout short head of biceps + coracobrachialis

Forces & kinematics (TO)

Parameter (TO)	Value
Shoulder ER	**>160°**
IR angular velocity	**>6000°/s**
IR torque	**>60 N·m**
Anterior joint force, end of cocking	**>350 N**; **AIGHL + anteroinferior capsule** under greatest strain
Distraction force, deceleration	**>500 N**

Phase detail (TO biomechanics lecture)	Value
Wind-up	Initial movement → **maximum knee lift** ("balance point") · weight on stance leg · hides ball
Stride	Max knee height → **stride-foot contact** · foot **~15° outward**, **~10 cm** from midline · **+3.0 N** anterior force per cm stride · **+2.1 N** per degree foot angle
Pelvic rotation	**400–700°/s**
Cocking energy	**~80%** of kinetic energy from lower limbs + trunk · ends at **maximum ER (MER)**
Early cocking	**90° abduction + 50° ER** (semi-cocked)
Late cocking	Abduction **90–95°**, **ER 165–175°** (MER 150–180°, kinetic-chain lecture) · subscapularis eccentric control
Acceleration	MER → release, **42–58 ms** · IR **7000–9000°/s** · elbow **120° flexion → ~25°** at release · trunk flexion 32–55°
Deceleration	Release → max IR · posterior shear **~400 N**, inferior shear **~300 N**, compression **>1000 N** · **teres minor = peak eccentric**
Follow-through	Horizontal adduction up to **60°** · lowest load → **lowest injury risk**

IR velocity conflict **>6000°/s** (TO forces table) · **7000–9000°/s** (TO biomechanics lecture) – read the options

Most injury-prone phase **deceleration** (Orthobullets, TO) · **acceleration** (TO kinetic-chain lecture: highest torque) – read the options

Elbow in cocking (TO) valgus torque from pelvic/torso rotation + shoulder ER → medial tension · **varus counter-torque by forearm flexor-pronators** protects UCL · **triceps eccentric → isometric** resists centripetal flexion torque · acceleration → **valgus extension overload**, posterior impingement

IR muscles in acceleration (TO) subscapularis, pec major, latissimus dorsi, teres major

Deceleration brakes (TO) eccentric **ER cuff**, **scapular stabilisers**, **posterior deltoid**; elbow flexors limit extension; spine = force attenuator

Scapular kinematics & muscle activation

Scapula + humerus must work **in concert** for GH stability

– – Active space – –

Scapular rotation must **rotate in cocking + acceleration** → elevate / clear acromion → prevents cuff impingement

Retraction / protraction **maximal retraction in late cocking** (acromion starts to elevate) · **protraction in acceleration + ball release** (**serratus anterior, long thoracic nerve**)

✗ Scapula distractors max retraction at ball release · protracts in late cocking · fixed during throw (must be dynamic) · no effect on concavity-compression
✗ Wrong options: – centres head → ↑ concavity-compression

Kibler – 5 scapular roles stable GH articulation · retraction + protraction · **acromial elevation** · base for muscle attachment · **link in kinetic chain**

Kinetic chain (TO) **~½ of kinetic energy** transferred through **scapulothoracic** joint

Kinetic chain & return to throwing (TO)

Links (TO) feet → lower limbs → hip/pelvis → trunk → scapulothoracic → shoulder → elbow → wrist → hand · **proximal-to-distal sequencing**: each distal link fires as proximal velocity peaks

Hip + core (TO) generate **~50%** of throwing energy · "if the hip is weak, the shoulder will suffer" · ↓ hip IR linked to shoulder injury

Dysfunction (TO)	Example
Anatomical disruption	**GIRD** or ↓ hip IR → shoulder overload
Force-distribution error	Over-reliance on arm (tennis / pitcher's shoulder)
Inefficient motor pattern	Scapular dyskinesis → ↓ velocity, impingement

Hip–shoulder separation (TO) rotational angle hips vs torso in stride/wind-up · **30–55°** (peak **67°**) · stores elastic energy (rubber band)

Catch-up phenomenon (TO) stride deficiency → compensatory shoulder load · stride faults → lumbar stress fracture, disc degeneration

Kinetic chain tests (TO) one-leg stability (pelvic drop, valgus collapse) · **Y-balance** (anterior, posteromedial, posterolateral reach) · **prone hip IR** · core/balance

Thrower's paradox (TO) capsule loose enough for ER yet stable enough to avoid subluxation → improve hip/thoracic ROM rather than more GH rotation

Return-to-throwing criteria (TO)	
Pain-free, no swelling	✓
ROM + strength	**Within 10%** of contralateral
Scapular kinematics	Normal
Plyometric progression	Asymptomatic
Interval throwing	With workload monitoring

SICK scapula (TO) **S**capular malposition · **I**nferior medial border prominence · **C**oracoid pain + malposition · dys**K**inesis → anterior shoulder pain, coracoid tenderness, AC pain, ↓ forward flexion

Thrower's shoulder – static/dynamic stability (TO)

Thrower's shoulder (TO) overuse + **microinstability** + adaptive GIRD · triad: **concavity-compression, cam effect, peel-back**

Bony values (TO) humeral retroversion **~30°** · glenoid retroversion **~7°** · articular offset **~2 mm** · ABER → ↑ congruence + contact area, ↓ pressure

Cuff muscle (TO)	Role
Supraspinatus	Initiates abduction (**0–15°**)
Infraspinatus	ER — **60%** of ER force
Teres minor	ER — **45%** of ER force
Subscapularis	IR; prevents anterior translation

Force couple (TO)	Muscles	Failure
Transverse (AP)	Subscapularis ↔ infraspinatus/teres minor	Tear → instability (isolated SS tear = minimal)
Coronal	Deltoid (up) ↔ cuff depressors	Superior migration
Scapular rotation	Trapezius (upper, middle, lower) + serratus anterior	Dyskinesis

Anterosuperior escape (TO) loss of subscapularis/infraspinatus → deltoid overpull → superior migration · **sourcil sign** (sclerosis under acromion) · coracoacromial impingement

Tubercle of Assaki (TO) thinnest central glenoid cartilage → concavity compression · labrum = **suction cup**, anti-shear bumper, GHL attachment

Thrower finding (TO)	Mechanism
Posterior ache, late cocking	Internal impingement: **GT abuts posterosuperior glenoid** → cuff entrapment
↑ ER	Anterior capsular laxity (**pseudo-laxity**)
↓ IR (GIRD)	Posterior capsule tightness / fibrosis
Type II SLAP	**Peel-back**: hyper-ER → torsion on biceps anchor

Cam effect (TO) tight posterior capsule → head migrates **posterosuperiorly** → ↓ anterior capsular tension (pseudo-laxity) → **↑ ER** but instability + impingement

Pectoralis minor tightness (TO) **>3 cm** side-to-side asymmetry = abnormal

Dynamic medial elbow stabilisers (FDS, FCU)

Local dynamic elbow stability **FDS + FCU** in **cocking + acceleration** → dynamic compression; **protect UCL**
✗ Wrong options: FDP · ECRL/ECRB · FCR

Implication strengthen forearm flexors to protect UCL

2 Mechanism

Phase of maximal valgus stress – late cocking/early acceleration

Max elbow valgus **late cocking** (TO, Orthobullets, OB) → **MUCL** injury; MCQs combine as **late cocking/early acceleration**

– – Active space – –

Medial elbow pain + valgus stress test	reproduced in **late cocking** ✗ Wrong options: early cocking (max valgus not yet) · acceleration/deceleration (more force at shoulder)
Varus instability	**LUCL** ✗ Wrong options: valgus

Throwing athlete adaptations (TO)

Shoulder	**↑ ER range** (anterior capsule/IGHL stretch) → anterior instability + secondary impingement
IR:ER strength	normal **3:2**; exaggerated in throwers → ER weakness ↑ injury risk
Bony	↑ **humeral retrotorsion** (↓ anteversion) · ↑ **glenoid retroversion**
Elbow	breakdown of medial stabilisers (MCL, capsule, flexors) → **↑ carrying angle**; anterior capsular strain, posterior impingement, flexor strain → **fixed flexion deformity**

3 Presentation

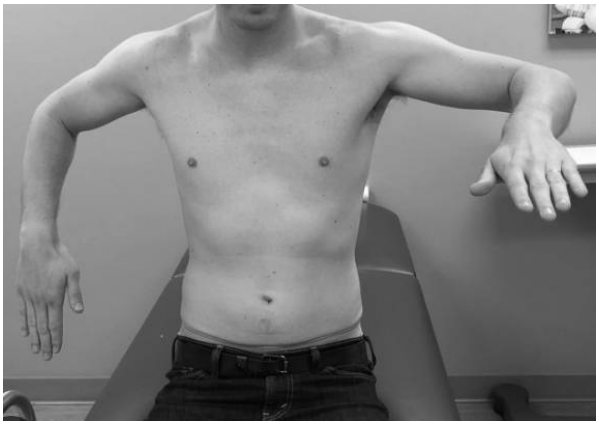
Internal impingement & GIRD (TO)

Internal impingement (TO)	late cocking (**ABER**): posterior cuff undersurface pinched between **greater tuberosity/humeral head and posterosuperior glenoid → posterosuperior labral tear** (hallmark) + **articular-sided (PASTA) supraspinatus** tear
GIRD (TO)	adaptive **loss of IR** in throwing shoulder; caused by **late cocking** (repetitive cocking)
GIRD pathology (TO)	**posteroinferior capsule + posterior band IGHL contracture** (thicker posterior capsule) ± posterior cuff tightness → ↑ GH contact pressure + impingement
GIRD symptoms (TO)	stiffness · prolonged warm-up · **loss of velocity ("dead arm")** · vague **posterior** pain · late cocking provocative
GIRD associations (TO)	posterior labral tears · posterosuperior PASTA tears · **SLAP (peel-back in ABER)** · scapular dyskinesia · **UCL tears**
Sulcus sign in GIRD (TO)	**~⅔ of pitchers**; rotator interval laxity (**CHL + SGHL**)
Cuff tears in throwers (TO)	mostly **degenerative partial/full** from impingement (**internal > external**) · minority **acute traumatic** from **deceleration** overload

4 Diagnosis & Classification

Diagnostic criteria – GIRD

GIRD	
Definition	IR loss **>20°** / **≥20°** vs contralateral (TO) · **25°** (OB, MDI topic) – read the options
Measurement	**Supine, 90° abduction + 90° elbow flexion**; max passive IR = point **just before scapula lifts**; goniometer; also TRM
Posterior tightness (Tyler)	Lateral decubitus, 90° abduction, neutral rotation, scapula stabilised → adduct; **1 cm loss ≈ 5° IR loss**
Posterior impingement sign (Meister)	Max passive **ABER (late cocking) → deep posterior pain** = posterosuperior labral / PASTA



Left GIRD – IR deficit on 90/90 exam

5 Management

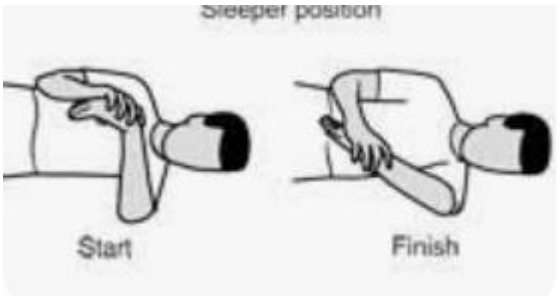
Throwing shoulder with GIRD

- First line → **posterior capsular stretching + scapular strengthening**
  - sleeper stretch, cross-body stretch, passive prone stretch
- Severe GIRD + ↓ TRM, failed extensive nonop → **arthroscopic posteroinferior capsular release**

Throwing athlete rehabilitation (TO)

Mainstay (TO)	**posterior capsular stretching** + **scapular** strengthening
Sleeper stretch (TO)	lie on **affected side**, shoulder + elbow **90°**, scapula stabilised → other arm pushes forearm to floor (**IR**); **30 s × 5**

– – Active space – –



Sleeper stretch

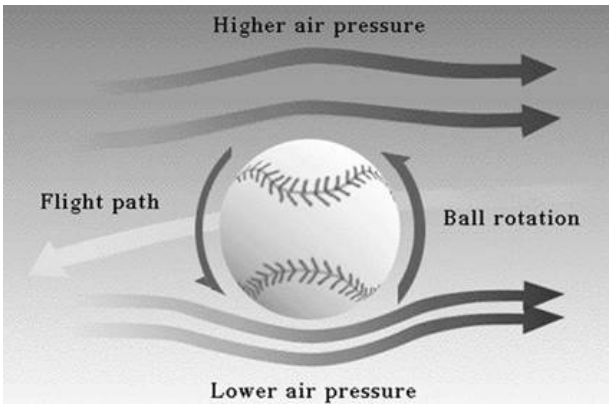
Cross-body stretch (TO)	standing/sitting, shoulder flexed 90° → other hand pulls arm across chest (**horizontal adduction**); **30 s × 5**
Not appropriate for GIRD (TO)	**latissimus dorsi stretch** (✓ sleeper, ✓ cross-body, ✓ passive prone shoulder stretch)
Stretching benefits (TO)	↑ IR · ↑ TRM · ↑ **acromiohumeral distance**
Surgery (TO)	only after exhausted nonop; SLAP repair + articular cuff tear treatment **controversial** (variable RTP); **arthroscopic posteroinferior capsular release** for severe GIRD

Follow-through & injury prevention (TO)

Long follow-through (TO)	**decreases** chance of muscle injury ✗ Wrong options: increases injury · ↑ fatigue · no relation
Why	dissipates deceleration forces; forward torso rotation reacquires energy → ↓ burden on serratus + scapular stabilisers

Ball-flight physics (Magnus effect) (TO)

Magnus effect (TO)	**spinning ball** drags boundary layer of air → **high pressure** where boundary layer opposes passing air · **low pressure** where they move together → **lift (Magnus) force** high → low → **curved flight path**
Sports (TO)	**tennis**, cricket, soccer, baseball, golf ✗ Wrong options: sprints · swimming · basketball



Magnus effect – spin creates pressure differential and curved flight

X vs Y

	Late cocking	Deceleration
Shoulder position	**Max ER**, ABER	Rapid IR, arm slowing
Muscles	Cuff (supra, infra, teres minor) peak	**Eccentric all**; cuff greatest stress
Elbow	**Max valgus → MUCL**	Elbow flexors (brachialis)
Shoulder pathology	**Internal impingement, GIRD, SLAP peel-back**	**SLAP, biceps, teres minor**
Label	High torque	**Highest torque, most harmful**

	Early cocking	Acceleration
Peak muscle	**Deltoid**	**Triceps → pec major, lat, serratus**
Scapula	–	**Protracts** (serratus)
Duration	Part of 500–1000 ms	**50 ms – shortest**

How it's asked

"Scapula during throwing – true?"	→ **must rotate in cocking + acceleration to prevent cuff impingement** ✗ Wrong options: max retraction at release · protracts in late cocking · fixed · no concavity-compression effect
"Peak muscle in early cocking (figure)?"	→ **deltoid** ✗ Wrong options: subscapularis = late cocking/early acceleration · supraspinatus/teres minor = late cocking
"Most important dynamic medial elbow stabilisers in cocking + acceleration?"	→ **FDS + FCU**
"Phase of maximum valgus at elbow in bowling?"	→ **late cocking** (TO) ✗ Wrong options: acceleration · deceleration
"Phase causing GIRD?"	→ **late cocking** (TO)
"Medial epicondyle tenderness + valgus stress test – phase reproducing pain?"	→ **late cocking**
"Valgus instability after seasons of pitching – structure + phase?"	→ **MUCL; late cocking (C)** ✗ Wrong options: LUCL = varus
"Young athlete, IR deficit on exam – exercise NOT appropriate?"	→ **latissimus dorsi stretch** (TO) ✗ Wrong options: sleeper · cross-body · passive prone
"Magnus effect important in?"	→ **tennis** (TO) ✗ Wrong options: sprints · swimming · basketball
"Long follow-through in bowling?"	→ **decreases muscle injury** (TO)

– – Active space – –

"Most harmful / highest torque phase?"	→ **deceleration**
"Shortest phase?"	→ **acceleration (50 ms)** (TO)
"Peak eccentric muscle during deceleration?"	→ **teres minor** (TO)
"Share of throwing energy from hip + core?"	→ **~50%** (TO)

POSTERIOR LABRAL TEAR

– Shoulder & Elbow · 35 / 60 –

1 Basics

Epidemiology

Definition	**reverse Bankart** lesion / attenuation of posterior capsulolabral complex
Frequency	relatively **rare** vs anterior labral pathology
Athletes	**weightlifters (bench press)** · **football linemen (blocking)** · **wrestlers** · swimmers · gymnasts · **baseball batters**

Anatomy

Posterior labrum	**fibrocartilage** · cavity-compression · **50% of socket depth** · **posterior stability** · **anchors PIGHL**
Static restraints to posterior translation (flexion, adduction, IR)	**PIGHL**, augmented by **SGHL**

✗ Distractor	Actual restraint role
Supraspinatus	**Dynamic**; prevents **inferior** instability
MGHL	Static; anterior + posterior at **45° abduction**
Subscapularis	**Dynamic**; posterior translation in **ER**
AIGHL	Static; **anterior** at **90° abduction + ER**

Most important factors in posterior instability (Bäcker)	**capsular lesions + IGHL rupture**

2 Mechanism

Mechanism & provocative position (flexion–adduction–internal rotation)

Most common mechanism	**repetitive microtrauma** to posterior capsulolabral complex
Provocative position	**posteriorly directed force** with arm **flexed, ADducted, internally rotated** (net posterior vector)
✗ Wrong option: flexion + ABduction + ER	anterior instability → **anteroinferior** labral tear
Batter's shoulder	**acute posterior subluxation** on **adducted lead shoulder** while swinging a bat → acute posterior labral tear ✗ Wrong options: recurrent anterior events · anabolic steroids
Posterior capsular tightening	→ **GIRD** ✗ Wrong options: cause of batter's shoulder

– – Active space – –

Glenoid retroversion	**~17° → 6× more likely** posterior instability vs ~7°
Kim lesion	**incomplete, sometimes concealed avulsion of posteroinferior labrum**
First-time posterior dislocation	**less likely to recur** than first-time anterior

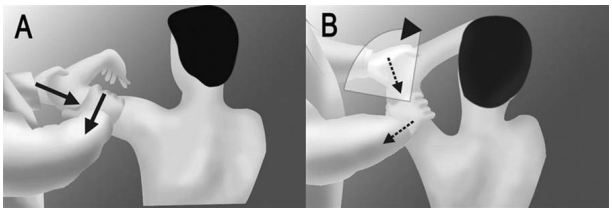
3 Presentation

Symptoms	**vague, deep, nonspecific posterior shoulder pain** (most common) · **↓ strength** (bench press, **batting power**) · clicking/popping · instability less common
Aggravated by	posterior force: **pushing heavy doors, bench press, push-ups, blocking, grappling**
Throwing pain	**late cocking** (Orthobullets) · late cocking/early acceleration = **anterior** instability (OB) – read the options
Exam	posterior joint-line tenderness; **full ROM**; large patients hard to examine

Diagnosis – jerk/Kim tests & MRI

Test	Technique	Positive / meaning
Posterior apprehension	**90° flexion + adduction**, stabilise scapula anteriorly, push humerus posteriorly	Pain / instability
Posterior load and shift	Arm at side; posterior force on proximal humerus	Graded translation (below)
Jerk (Jahnke) test	**90° abduction + full IR** (arm flexed, IR), **axial load while adducting across body** (push elbow posteriorly, scapula anteriorly)	**Clunk** = head subluxes off posterior glenoid → **posterior labral tear**
Kim test	**90° abduction + 45° forward flexion**, axial load on elbow, **posteroinferior** force	**Pain → posteroinferior labral tear** (most sensitive)
Push-pull test	(OB)	Pain from posterior labrum

Jerk/Kim accuracy — **highly sensitive + specific** (Orthobullets) · sensitivity/specificity **"under question"** (OB) – read the options



Kim test: abduction with forward flexion, axial + posteroinferior force

Load-and-shift grade	Translation
0	None
1	To **edge** of glenoid
2	**Over edge**, **spontaneously relocates**
3	Over edge, **does not relocate**

✗ Exam distractor	Indicates
Forward elevation in scapular plane	**Impingement**
ER + abduction	**Anterior apprehension**
Flexion + abduction	Nonspecific
O'Brien active compression	SLAP / labrum

4 Diagnosis & Classification

Key radiology findings

X-ray — **true AP + scapular Y + axillary**; **axillary required** (posterior dislocation missed on AP alone); often **normal**

Chronic X-ray (axillary) — **glenoid retroversion** · **posterior glenoid erosion / bone loss**

MRI — **study of choice**; **intra-articular contrast (MR arthrogram)** ↑ sensitivity · alternative **CT arthrogram**

MR arthrography for labrum — **sensitivity 89%, specificity 91%, accuracy 90%** (Bencardino); standard MRI often misses labral lesions → **next step after failed nonop + labral test**
✗ Wrong options: diagnostic arthroscopy · CT · stress views

SLAP on MRA — coronal oblique: deep cleft between superior labrum + glenoid extending below biceps anchor

Incidental — posterior labral tears common on MRI of **asymptomatic throwers** → treat only if history + exam fit

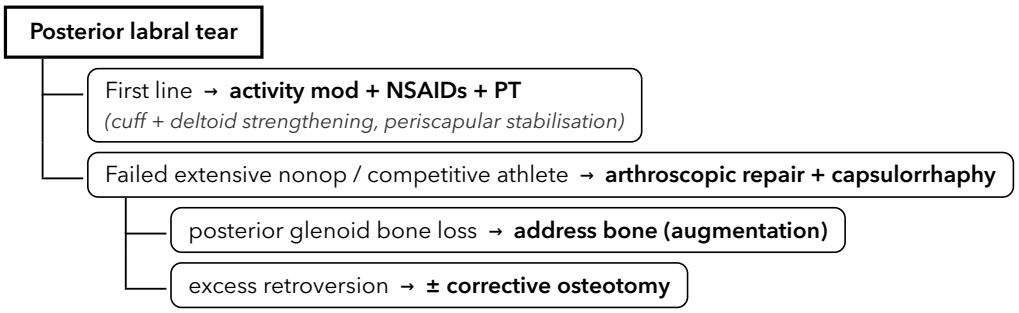


Axial MR arthrogram: posterior labral tear

✗ Diagnosis distractor (bodybuilder, MRI arthrogram)	
Pectoralis major rupture · supraspinatus PT tear · SLAP · LHB tendinitis	Not on MRI

– – Active space – –

5 Management



Treatment – therapy vs arthroscopic repair, recurrence & postoperative restrictions

Nonop	**first line**; cuff + deltoid strengthening, **periscapular stabilisation**
Thrower with anterior follow-through pain, end-of-game, full ROM, small posterior labral contrast	**rotator cuff deconditioning → PT with throwing program** ✗ Wrong options: labral repair · debridement · capsular release · IR stretching (ROM is full)
Operative indication	**failed extensive nonop** (or competitive athletes)
Arthroscopic > open	avoids extensive posterior dissection; **more reliable return to play**
Suture anchors + capsulorrhaphy	**fewer recurrences + revisions** than non-anchored repair
Probe posterior labrum	to rule out subtle **Kim lesion**
Post-op	**avoid adduction + internal rotation** (stress posterior repair) → failure risk ↑ if not avoided

Post-op restriction	Repair
Avoid adduction + IR	**Posterior labrum**
Avoid ER	**Subscapularis**, **anterior labrum**
No restriction (stiffness risk)	Calcific tendinitis decompression, distal clavicle resection

Outcome	generally good; **overhead throwers** return less reproducibly

Posterior glenoid bone loss

Posterior glenoid bone loss	Failure of isolated capsulolabral repair
11%	**10.4×**
15%	**24.4×**

"Sub-critical" bone loss	even small posterior loss → **recurrence after isolated arthroscopic repair** ✗ Wrong options: "not critical, no bony procedure"
Glenoid width vs version	**width more predictive** of recurrence than version
Obligate translation	tightening posterior tissues → anterior translation = **therapeutic** ✗ Wrong options: causes anterior instability
Corrective osteotomy	for excessive retroversion; **does not obviate bone block** when bone loss present

6 Complications

Complications – axillary nerve injury

Axillary nerve (posterior branch)	within **1 mm of inferior capsule + glenoid rim** → injured by **suture passage at posteroinferior glenoid**
Posterior branch gives	**superolateral brachial cutaneous** nerve + **nerve to teres minor** → **lateral arm burning/numbness** + **teres minor ER weakness**
Overtightening posterior capsule	**anterior subluxation** or **coracoid impingement**
Recurrence	↑ if **posterior bone loss** not addressed

X vs Y

	Posterior labral tear (reverse Bankart)	Anterior labral tear (Bankart)
Position	**Flexion + adduction + IR**, posterior force	**Abduction + ER**, anterior force
Who	Linemen, lifters, wrestlers, batters	–
Painful activity	**Bench press, push-ups, blocking**	Late cocking, spiking, overhead serve
Test	**Jerk, Kim, load-and-shift**	Apprehension, relocation
Ligament	**PIGHL** (+ SGHL)	**AIGHL**
Post-op avoid	**Adduction + IR**	**ER**

	Jerk test	Kim test
Position	90° abduction + IR, adduct across body with axial load	90° abduction + 45° flexion, axial + posteroinferior force
Positive	**Clunk**	**Pain**
Best for	**Posterior** labral tear	**Posteroinferior** labral tear

How it's asked

"Bodybuilder, vague deep pain, weak bench press, MR arthrogram – etiology?"	→ **posterior labral tear** ✗ Wrong options: pec major · SLAP · LHB tendinitis
"Old 'pop', never reduced, axial MRI posterior labrum – painful activity?"	→ **heavy bench press** ✗ Wrong options: late cocking · volleyball spike = anterior · curl = SLAP
"Wrestler, + Kim + jerk – structure augmenting PIGHL in flexion-adduction-IR?"	→ **superior glenohumeral ligament** ✗ Wrong options: MGHL · AIGHL · subscapularis
"Wrestler, posterior labral tear MRI – position increasing incidence?"	→ **flexion, ADduction, internal rotation**

– – Active space – –

"Baseball player, ↓ batting power, + Kim/jerk — etiology?"	→ **acute posterior subluxation while batting** ✗ Wrong options: posterior capsular tightening = GIRD
"Ex-pitcher, failed nonop, + O'Brien — next diagnostic step?"	→ **MRI-arthrography** ✗ Wrong options: diagnostic arthroscopy · CT
"Lineman, pain with blocking, axial MRI — manoeuvre reproducing pain?"	→ **flexion, adduction, internal rotation** ✗ Wrong options: ER + abduction = apprehension
"Flexed IR arm pushed posteriorly, girdle anteriorly, clunk — injured structure?"	→ **posterior labrum (C)** (jerk test)
"Quarterback, anterior pain in follow-through, end of game, small posterior labral contrast — initial?"	→ **PT with throwing program**
"Post-op: avoid adduction + IR — initial diagnosis?"	→ **posterior labral tear (Figure C)**
"NFL tackle, posterior glenoid bone loss on CT — true?"	→ **capsulolabral repair alone may lead to recurrence** ✗ Wrong options: obligate anterior instability · version more predictive than width · osteotomy obviates bone block
"Burning lateral arm pain after arthroscopic posterior labral repair — nerve?"	→ **axillary** ✗ Wrong options: suprascapular · radial

– – Active space – –

POSTERIOR SHOULDER INSTABILITY & DISLOCATION

– Shoulder & Elbow · 36 / 60 –

1 Basics

Epidemiology

Incidence	**2–5%** of unstable shoulders (Orthobullets) · **2–12%** of instability (OB) – read the options
Incidence (TO)	**2–4%** of all dislocations · causes: **epilepsy, electrocution, seizures, dashboard injuries** · **subacromial type commonest** (head posterior to glenoid, inferior to acromion)
Missed	**50%** of traumatic posterior dislocations in ED undiagnosed · Hawkins & Neer: 41 locked dislocations, **50% with fracture**, 20 missed, mean **1 y** to diagnosis
Prior dislocation	up to **50%** of patients evaluated for posterior instability report a prior posterior dislocation · frank dislocation in only **2–5%** of posterior labral tears
Risk factors	**ligamentous laxity** · **glenoid retroversion** / hypoplasia (less common) · not excessive glenoid **anteversion** = anterior instability
Seizures + electric shock	tetanic contraction pulls head out; posterior dislocation unlikely without such force · anterior **still more common** with seizures (Orthobullets, OB) · posterior "more common after seizure" / **as common as anterior in epilepsy** (OB) – read the options
Atraumatic (esp. posterior) dislocation	warrants **neurological workup**

Anatomy – posterior restraints

Restraint	Role
Posterior band IGHL	**Primary restraint in IR**; most important at **90° flexion + IR**
SGHL + CHL	Primary **static** stabiliser to posterior subluxation in **flexion, adduction, IR** · also inferior translation of adducted arm + ER
Subscapularis	**Primary dynamic restraint against posterior subluxation** (and in ER)
Labrum	Deepens glenoid **50%**

– – Active space – –

Restraint conflict posterior band IGHL (OB, Orthobullets) vs SGHL (OB, Orthobullets) for flexion + IR – read the position in the stem

✗ Restraint distractor	Role
Anterior band IGHL	Anterior/inferior at **90° abduction + max ER** (late cocking)
MGHL	Anterior + posterior at **~45° abduction** in ER
Anterior labrum	Anterior; IGHL anchors here → Bankart
Superior labrum	**Biceps anchor**

Dynamic stabilisers by direction posterior labral tear → **subscapularis** · anterior labral tear / HAGL → **infraspinatus + teres minor**

Tubercle of Assaki (TO)

Tubercle of Assaki (TO) ridge / **focal elevation of subchondral bone at centre of glenoid** · **normal variant** · **focal thinning of overlying cartilage** · not a cartilage defect
✗ Wrong options: "depression of humeral head cartilage"

2 Mechanism

High-risk position arm **flexed, adducted, internally rotated** + **axial load**
✗ Wrong options: abduction + ER = anterior · hyperabduction = **luxatio erecta**

Type	Mechanism	Presentation
Acute (dislocation)	Trauma (~50%), **seizure, electric shock**	Dramatic
Chronic (instability)	**Repetitive microtrauma** → labral tear, incomplete avulsion, erosion, **patulous posterior capsule**	Insidious, vague **pain > instability**

At-risk athletes **football linemen** (blocking), linebackers, **weightlifters/bench press**, push-ups, weighted dips, rugby, swimmers, overhead athletes · stretches posteroinferior GHL

Classification acute vs chronic · voluntary vs involuntary

Associated lesion	Association
Posterior band IGHL avulsion	Acute subluxation
Reverse Bankart	Detached posteroinferior capsulolabral complex
Reverse Hill-Sachs	**Locked**, hard-to-reduce dislocation
Posterior labral cyst / posterior glenoid rim fracture	Chronic reverse Bankart
Lesser tuberosity fracture	Acute posterior dislocation
Large capsular pouch	Chronic instability (MR arthrogram)

Posterior lesion (TO)	Description
Reverse bony Bankart	Avulsion fracture of posteroinferior glenoid
POLPSA	**Posterior labrocapsular periosteal sleeve avulsion**: capsule stripped with intact periosteum
Reverse HAGL	Posteroinferior capsule avulsed from **humerus**
Kim lesion	**Incomplete** posteroinferior tear between labrum + cartilage · **always probe** at arthroscopy

Reverse Hill-Sachs (TO) — **anteromedial** humeral head · Hill-Sachs = **posterosuperolateral**
✗ Wrong options: posterolateral

Cuff tears — **not** commonly associated with posterior dislocation

3 Presentation

Acute dislocation — **prominent posterior shoulder + coracoid** · **locked in IR + adduction** · comfortable in "sling position"

Most consistent sign — **lack of ER** (mechanical block: anterior head defect on posterior glenoid) · ER loss differential: adhesive capsulitis, GH OA

Chronic instability — may look normal · pain with **flexion, adduction, IR** · slipping when pass-blocking, pain bench pressing/push-ups

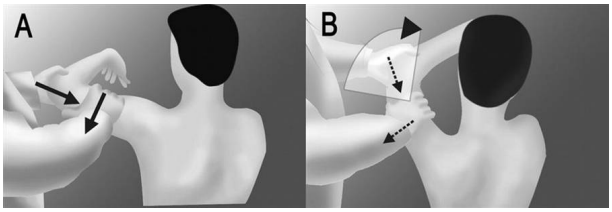
Recurrent posterior instability – Kim lesion/labral tear, jerk & Kim tests, restraints

Kim lesion (TO) — **incomplete + concealed avulsion of posteroinferior labrum** (superficial labrum intact, deep labrum detached) in posterior or multidirectional posteroinferior instability · usually **diagnosed at arthroscopy** · needs **arthroscopic repair**
✗ Wrong options: ALPSA = anterior labral periosteal sleeve avulsion · meniscal root avulsion

– – Active space – –

Posterior labral tear site **posteroinferior quadrant: 6–9 o'clock right, 3–6 o'clock left**

Test	Technique	Positive / use
Jerk	90° abduction/flexion + IR, elbow bent · **axial load** along humerus → adduct into forward flexion (bring back to extension → reduces)	**Clunk** ± pain · posterior labral tear in **mid-range** glenoid
Kim	Seated, 90° abduction → **flex to 45°** + axial load at elbow + posteroinferior force (Orthobullets) · arm elevated **125°** + IR, downward + posterior force (OB)	**Pain** (± clunk) · more sensitive for **posteroinferior** tears
Jerk + Kim	Combined	**97% sensitive** for posterior labral tear
Load & shift (TO)	Push head anterior/posterior with thumb	**90% sensitivity, 85% specificity**
Circumduction (TO)	Circumduction under load	Reproduces **clunk** → confirms instability
Posterior stress	Stabilise scapula, push posteriorly	Pain common, **not specific**
Load & shift	Supine, neutral rotation, 40–60° abduction + flexion	Grade translation



Kim test

Load & shift grade	Translation
1+	Apparent, not to rim
2+	**To glenolabral rim**
3+	**Over** rim
4+	Complete dislocation

✗ Test distractor	Assesses
Apprehension / relocation (abd + ER)	**Anterior** instability, Bankart
Hornblower; ER weakness at 90° abduction	**Teres minor** (axillary n.)
Jobe	Supraspinatus
Speed / Yergason	Biceps
O'Brien	SLAP
Belly-press, lift-off, IR lag	Subscapularis
Sulcus	Rotator interval
Pinch grip	AIN

✗ Lesion distractor	Feature
Bankart / Hill-Sachs	Anterior instability, apprehension
GIRD	Throwers, **tight posteroinferior capsule**, ↓ IR → posterior impingement, **anterior subluxation**
AC separation	Piano key sign
SLAP	Overhead pain, dead arm
PASTA	Partial articular supraspinatus avulsion
Pec major tear	↓ adduction + IR, lost axillary fold

4 Diagnosis & Classification

Locked posterior dislocation – diagnosis (axillary/Velpeau view, loss of ER, seizures, electrocution)

AP — **unreliable** · **"lightbulb" sign** (head internally rotated) · humeral head **overlaps glenoid**



Lightbulb sign on AP

Radiographic sign (TO)	Description
Vacant glenoid	Empty anterior glenoid fossa
Rim sign	Anterior rim–head distance **>6 mm**
Light bulb	Internally rotated head looks circular
Trough sign	Reverse Hill-Sachs (anterior head impaction line)
Moloney's arch	**Acute angle** of scapulohumeral arch

Apparently normal AP (TO) — posterior dislocation **often missed**

Axillary lateral — **best view**; mandatory (needs **20–30° abduction**) · **more sensitive than trans-scapular** · any suspected dislocation → full trauma series (AP, scapular Y, axillary)

Can't abduct (pain/spasm) — **Velpeau view**
✗ Wrong options: Zanca = AC/clavicle · Stryker notch = Hill-Sachs · MRI not first-line in trauma · EUA is for locked dislocation

– – Active space – –

Scapular Y humeral articular surface **posterior** to glenoid

CT chronic dislocation (**>2–3 wk**): size of **reverse Hill-Sachs** + bone loss · posterior glenoid bone loss (% diameter)

MRI / MR arthrogram chronic instability without dislocation · posterior labral tear, reverse Hill-Sachs, cuff · **Kim lesion**: incomplete avulsion/cyst with intact chondrolabral junction · **posteroinferior capsulolabral separation**



MR arthrogram: posterior labral tear

Next step pattern any "light bulb" / overlapping head / ER to neutral after seizure or fall → **axillary view**
✗ Wrong options: PT for frozen shoulder · sling · cuff repair · EMG · injection

Glenoid hypoplasia **posteroinferior glenoid deficiency**, enlarged distal clavicle, glenoid indentation · **usually bilateral** → X-ray other side · rarely syndromic · **PT first** · glenoid osteotomy variable · early OA
✗ Wrong options: echo / abdominal US / skeletal survey unless Holt-Oram, Apert stigmata

5 Management

Posterior shoulder instability

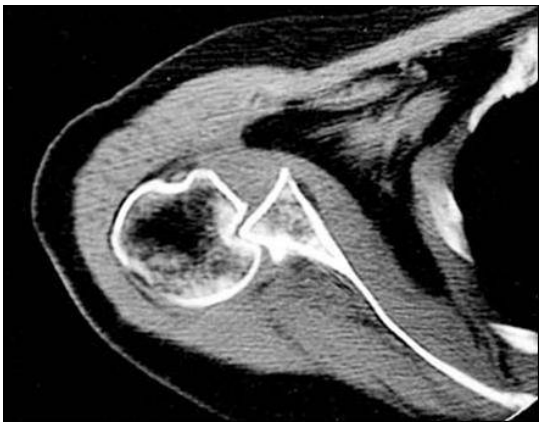
- Acute traumatic dislocation → **reduce, immobilise ER 4–6 wk, then PT**
- Chronic instability → **PT (cuff + periscapular)**
  - fails, pain with loading → **posterior labral repair**
- Chronic locked dislocation → **by duration + reverse Hill-Sachs size**
- Uncontrolled seizures → **neurology control first**

Chronic locked dislocation – reverse Hill-Sachs management (McLaughlin, arthroplasty)

Source	Closed reduction	McLaughlin / lesser tuberosity transfer	Hemiarthroplasty
Orthobullets	All acute (most reduce spontaneously) · disimpaction + graft if **<3 wk**	Chronic **<6 mo** + RHS **<40%**	**>6 mo**, RHS **>40%**, severe head arthritis, **head collapse on reduction**
Hawkins & Neer (OB)	**<6 wk + <20%**	**6 wk–6 mo + 20–45%**	**>6 mo or >45%** (glenoid normal)
OB (other)	**<3 wk + <25%** under GA; stable if IR to abdomen → 6 wk orthosis in slight extension + ER	Defect **<~40%**	Larger; **>50%** or several months
TO	**<3 wk + <20%** · **>3 wk / >20% → open reduction (posterior approach)**	**25–45%** McLaughlin / modified	**>45%** bone graft / arthroplasty

TSA	significant **glenoid arthritis** + an HA indication · glenoid intact → no glenoid component
Closed reduction after ~3 wk	exceedingly difficult · may **displace unrecognised surgical neck fracture**
Neglected >6 wk (TO)	**open reduction**; **closed reduction contraindicated**
McLaughlin	**subscapularis** into anteromedial defect · **modified (Neer/Hawkins)** = **lesser tuberosity + subscapularis** (used by most) · deltopectoral
Modified McLaughlin candidate	young man, **locked posterior dislocation after electric shock / seizure** ✗ Wrong options: AVN · RA = arthroplasty · subscap insufficiency = repair or pec transfer · anterior Hill-Sachs = graft/remplissage
6-wk locked dislocation, 38 y	**open reduction ± lesser tuberosity transfer** (CT first) ✗ Wrong options: PT · closed manipulation · cuff repair · HA
12-wk locked post-seizure, large RHS + lesser tuberosity fracture, 60 y	**hemiarthroplasty + subscapularis repair** · younger → **osteochondral allograft** + subscap repair
6 mo after motorcycle fall, 52 y	**hemiarthroplasty** most reliable ✗ Wrong options: PT · MUA · capsular release/plication
HA version	**anatomic version (20–30° retroversion)** · not excessive anteversion (re-dislocation, per source)
7 mo locked, 63F, light bulb	first **further radiographs (axillary)** → then HA
Open complications	stiffness · **AVN** · OA

– – Active space – –



CT: locked posterior dislocation with anteromedial reverse Hill-Sachs

Uncontrolled epilepsy, 9th dislocation, humeral + glenoid bone loss	**neurology + social work to control seizures first** · later: **posterior bone block** (iliac crest autograft or **distal tibial allograft**) ± modified McLaughlin / head grafting · HA ± glenoid reconstruction for chronic locked ✗ Wrong options: urgent HA · reconstruction before control · anterior procedure alone

Operative criteria

Procedure	Indication
Posterior labral repair (open/ arthroscopic)	**Recurrent instability despite PT** · pain loading arm in forward flexion (bench, blocking) · **negative Beighton**
Capsular shift + rotator interval closure	**Positive Beighton** (laxity) · shift less desirable in throwers · interval closure controversial
Posterior bone block	Posterior glenoid bone loss **≥11%** in contact athlete (**subcritical** by anterior standards)
Opening-wedge glenoid osteotomy	Excessive **congenital retroversion** (limited data)
Modified McLaughlin	Chronic locked, RHS <40%, <6 mo
HA / TSA	RHS >40%, >6 mo, collapse; TSA if glenoid arthritis

Treatment – therapy vs arthroscopic capsulolabral repair & postoperative protocol

Acute reduction	immobilise **10–20° ER, elbow at side, 4–6 wk** → PT (cuff + periscapular) + avoid high-risk position
After reduction, avoid	**cross-body adduction at 90° flexion + IR** (tensions SGHL + posterior IGHL)
Chronic instability first line	**PT** (cuff + periscapular; in-season athlete) · esp. **generalised laxity** · ± brief immobilisation, steroid injection
Failed rehab, positive jerk, grade 2 load-shift, lineman	**arthroscopic posterior capsulolabral repair** (arthroscopic ≈ open) ✗ Wrong options: thermal · open anterior capsulorrhaphy · injection · ER immobilisation for recurrent
Lineman, clunk, bench-press pain	**posterior labral tear → arthroscopic labral repair** ✗ Wrong options: thermal capsulorrhaphy · SLAP debridement/tenodesis
Thermal capsulorrhaphy	**contraindicated** · breaks collagen cross-links at **65–75 °C** · capsular necrosis, **chondral damage**, axillary nerve injury, recurrence

Technique **lateral decubitus** · **high lateral / posterosuperior portal** for anchor trajectory

Posterior repair (TO) open: **deltoid split → infraspinatus–teres minor interval** · anchors at **7, 8, 9 o'clock** (right shoulder)

Portal (TO)	Use
Kim's posterior portal — 2 cm lateral to posterolateral acromial corner	Working (anchors)
Mid-glenoid anterior	Viewing
Trans-cuff anterosuperior	Shuttle

Precautions (TO) **posterior circumflex humeral artery ~4 cm** · **suprascapular nerve ~3 cm** away · probe to rule out Kim lesion

Pathology (TO)	Procedure
Posterior Bankart	**Posterior labral repair**
No posterior Bankart	**Posterior capsulolabroplasty**
Posteroinferior instability	**Posterior capsular plication**
Anterior + posterior + inferior	**Combined labral repair + global capsular plication**

Results open **80–85% success at 5–7 y** · arthroscopic similar (shorter follow-up)

Post-op sling in **neutral rotation** (ER sling) or standard, up to 6 wk · **avoid flexion, adduction, IR** · full contact/heavy labour **6 mo**
✗ Wrong options: no elbow flexion = biceps tenodesis · avoid abd + ER = anterior/HAGL repair · PROM ≥6 wk = cuff repair · unrestricted ROM

Posterior glenoid bone loss **lower critical value than anterior**; **11%** of glenoid diameter ↑ recurrence after isolated labral repair → **bone block in contact athletes**
✗ Wrong options: modified McLaughlin = chronic dislocation · only with large RHS · isolated repair better · augmentation doesn't change recurrence

First-time traumatic dislocation, young athlete with recurrent subluxations **arthroscopic capsulolabral repair reasonable** · recurrence up to **90%** in young active · immobilisation (IR or ER) may not reduce recurrence · arthroscopic ≈ open with modern anchors if no glenoid loss/engaging lesion

6 Complications

Rank	Complication after labral repair
1	**Stiffness** (↓ **IR**)
2	**Recurrence** (7–50%) · ↑ with glenoid bone loss ≥11%, uncontrolled seizures
3	**DJD**
Other	Adhesive capsulitis · **overtightening → anterior subluxation / coracoid impingement** · **axillary / suprascapular** nerve

Not after posterior repair anterior instability not expected · acromial stress fracture not typical

– – Active space – –

X vs Y

	Posterior instability	Anterior instability
Position	**Flexion, adduction, IR**	Abduction + ER
Symptom	**Pain** > instability	Instability, apprehension
Labral lesion	**Reverse Bankart / Kim** (posteroinferior)	Bankart, ALPSA
Head lesion	**Reverse Hill-Sachs (anteromedial)**	Hill-Sachs (posterosuperolateral)
Tests	**Jerk, Kim**, load & shift	Apprehension, relocation
Dynamic stabiliser	**Subscapularis**	Infraspinatus + teres minor
Post-op avoid	**Flexion, adduction, IR**	Abduction + ER
Critical glenoid bone loss	**Lower (~11%)**	Higher

	Jerk test	Kim test
Positive	**Clunk**	**Pain**
Best for	Posterior (mid-range)	**Posteroinferior**

How it's asked

"63F, 7 mo after fall, ↓ ER, light-bulb AP — next?"	→ **further radiographs (axillary view)** ✗ Wrong options: PT for frozen shoulder · arthroplasty first · ORIF
"Seizure 1 wk ago, ER to neutral, sling — next?"	→ **axillary lateral radiograph**
"Posterior dislocations as common as anterior in?"	→ **epilepsy patients**
"Drunk, woke with pain, 'cuff tear' sling 6 wk — next?"	→ **axillary radiograph**
"Acute posterior dislocation — shoulder locked in?"	→ **internal rotation**
"Electrician, no axillary possible due to spasm?"	→ **Velpeau view** ✗ Wrong options: EUA · MRI · Zanca · Stryker notch
"Exam finding of fixed posterior dislocation?"	→ **lack of external rotation** ✗ Wrong options: apprehension · relocation · sulcus · jerk
"ICU head injury, ER −30°, overlap on AP?"	→ **posterior shoulder dislocation**
"Light-bulb sign — predisposing factor?"	→ **seizures** ✗ Wrong options: ↓ IR = GIRD · inflammatory · anteversion · anterior IGHL injury
"Epileptic teen, 9th dislocation, can't afford levetiracetam, bone loss on CT?"	→ **social work + neurology to control seizures**
"Reverse Hill-Sachs location?"	→ **anteromedial** humeral head (TO)
"NOT true about tubercle of Assaki?"	→ **depression of humeral head cartilage** (TO)

"Who benefits from modified McLaughlin?"	→ **locked posterior dislocation after electric shock**
"52 y, locked posterior dislocation 6 mo — most reliable?"	→ **hemiarthroplasty**
"Seizure 12 wk ago, no ER, large RHS + lesser tuberosity fracture?"	→ **HA + subscapularis repair** ✗ Wrong options: sling · closed reduction · labral repair · LT transfer
"38 y, posterior dislocation 6 wk old?"	→ **open reduction ± lesser tuberosity transfer**
"61 y, seizure 5 mo, half head lost?"	→ **HA in anatomic version** ✗ Wrong options: anteversion · TSA · closed reduction · LT transfer
"After surgery for posterior dislocation, most common complication besides stiffness?"	→ **instability (recurrence)**
"Kim's lesion is?"	→ **incomplete, concealed avulsion of posteroinferior labrum** (TO)
"Lineman, posterior pain after block in flexion + IR?"	→ **posterior labral tear** ✗ Wrong options: Bankart · SLAP · HAGL · AC
"Lineman subluxes blocking, positive jerk — pathology?"	→ **Kim lesion (reverse Bankart)** ✗ Wrong options: Hill-Sachs · GIRD
"Positive jerk + Kim test?"	→ **posterior labral tear** ✗ Wrong options: SLAP · ALPSA · PASTA
"After posterior dislocation, 'arm will pop out' — positive test?"	→ **Kim test** ✗ Wrong options: Hornblower · Jobe · apprehension · Speed
"Arthroscopy shows posteroinferior capsulolabral tear — preop exam?"	→ **painful clunk with arm elevated 125° + IR, downward/posterior force (Kim)**
"Basketball player, posterior labral tear on MR arthrogram — test?"	→ **jerk test** ✗ Wrong options: O'Brien · Speed · Neer · apprehension
"Lineman slipping when pass blocking, no apprehension?"	→ **posterior labral tear**
"Primary restraint to posterior displacement in flexion + IR?"	→ **posterior band IGHL** ✗ Wrong options: anterior band · MGHL · anterior/superior labrum
"Football player, axial load flexed + IR = pain + clunk?"	→ **jerk test** ✗ Wrong options: Hornblower · Speed · apprehension · O'Brien
"Powerlifter, pain bench + push-up, normal exam — MRI?"	→ **posteroinferior capsulolabral separation**

– – Active space – –

"Subscapularis is critical stabiliser in which lesion?"	→ **posterior labral tear (C)** ✗ Wrong options: SLAP · HAGL · spinoglenoid cyst · Bankart
"Lineman, Kim test pain — matching MRI?"	→ **posterior labral tear (B)**
"Mechanism of posterior labral tear + reverse Hill-Sachs?"	→ **axial load on flexed, adducted, IR shoulder**
"Quarterback after reduction of posterior dislocation — avoid?"	→ **cross-body adduction at 90° flexion + IR**
"Swimmer, clicking, glenoid hypoplasia on X-ray — next?"	→ **physical therapy** ✗ Wrong options: echo · abdominal US · skeletal survey · osteotomy
"Lineman, failed rehab, positive jerk, grade 2 load-shift — best?"	→ **arthroscopic posterior capsulolabral repair**
"16 y, first dislocation + 2 subluxations — true?"	→ **arthroscopic capsulolabral repair is reasonable**
"Lineman, bench pain, clunk — MRI + surgery?"	→ **posterior labral tear; arthroscopic labral repair**
"Pitcher, positive Kim, labral repair — likely complication?"	→ **decreased internal rotation (stiffness)** ✗ Wrong options: posterior instability = 2nd · DJD = 3rd
"Post-op protocol after posterior labral repair?"	→ **avoid flexion, adduction, IR**
"Rugby player, posterior tear + 11% glenoid loss, bone block — true?"	→ **subcritical bone loss is an indication in contact athletes**
"Rim sign threshold for posterior dislocation?"	→ **>6 mm** anterior rim–head distance (TO)
"Neglected posterior dislocation >6 weeks?"	→ **open reduction — closed reduction contraindicated** (TO)

QUADRILATERAL SPACE SYNDROME

– Shoulder & Elbow · 37 / 60 –

1 Basics

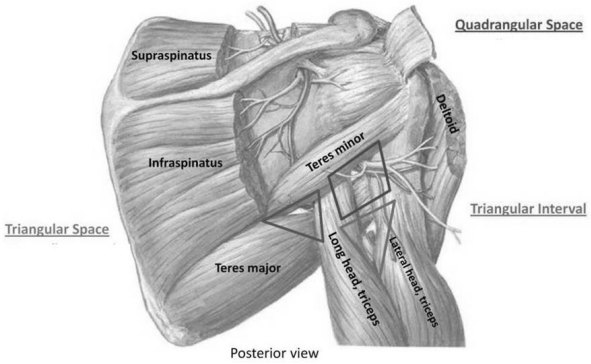
Epidemiology

Definition	compression of **axillary nerve + posterior humeral circumflex artery (PHCA)** in the quadrilateral (quadrangular) space → **posterolateral shoulder pain**
Incidence	**rare**; often **misdiagnosed as subacromial impingement**
Age	**20–40 y** (Orthobullets) · **20–35 y** (TO) – young adults
Side	**dominant shoulder** most commonly (TO)
Risk factors	**overhead athletes** (basketball, baseball pitchers), contact/throwing sports, **weightlifting** (muscle hypertrophy)

Anatomy – quadrilateral space

Boundary	Structure
Superior	**Teres minor** (inferior border)
Inferior	**Teres major** (superior border)
Medial	**Long head of triceps** (lateral border)
Lateral	**Surgical neck of humerus** (medial cortex)
Contents	**Axillary nerve** (C5 root, **posterior cord**) + **PHCA** (+ vein, TO)

Location posterior compartment of arm; **lateral to triangular space**, **medial to triangular interval** (with lateral + medial triangular spaces, TO)



Quadrangular space, triangular space and triangular interval

Axillary nerve supplies	**deltoid + teres minor** + sensation **lateral arm** (upper lateral brachial cutaneous)
Long head of triceps	**can have axillary innervation** (case report, OB)
Teres minor motor nerve	2 variants of fascial envelopment; nerve takes **highly angulated course at the fascial sling → stout fascial sling** may compress it (OB)

– – Active space – –

2 Mechanism

Etiology – causes of compression

Cause	Note
Fibrous bands	**Oblique fibrous bands = most common cause** (TO); compress nerve + artery in abduction/ER
Muscular hypertrophy	Weightlifting after a break (OB)
Paralabral cyst	**Most commonly inferior labral tears** (Orthobullets, TO) · **posterior labrum** ("occasionally", TO) – read the options
Trauma	Scapular fracture, shoulder dislocation
Masses	Benign or malignant

Position of max compression — **late cocking phase** of throwing = **abduction + external rotation** (TO)

Throwers — repetitive motion irritates nerve → dysfunction; usually **self-limited** (OB)

3 Presentation

Symptoms

Pain — **poorly localised posterior/lateral shoulder pain**; **worse at night**, overhead activity, late cocking/acceleration

Paraesthesia — **non-dermatomal**, lateral shoulder + **lateral arm**

Weakness — **external rotation** weakness, esp. **arm abducted 90° (throwing position)**; **5/5 ER with arm at side** (OB)

Aggravated by — active + resisted **abduction + ER**; forward flexion (OB)

Examination

Palpation — **point tenderness over quadrilateral space** (near teres minor insertion, TO)

Atrophy — **teres minor + deltoid** (✓ **deltoid atrophy**, TO)
✗ Wrong options: supraspinatus atrophy · positive apprehension · positive dynamic labral shear

Provocation (TO) — **arm abducted + externally rotated for 1 min** reproduces symptoms
✗ Wrong options: abducted + internally rotated

Detecting weakness (TO) — weakness/atrophy **difficult to detect unless significant deltoid involvement**

Neuro exam — usually **normal**; mild sensory change in axillary distribution

Typical negatives (OB) — full ROM, negative Hawkins/Neer, belly press/lift-off, O'Brien, normal radiographs, no cuff/labral tear on MRA

4 Diagnosis & Classification

Diagnosis – axillary nerve compression & teres minor atrophy

Diagnosis	**clinical** (point tenderness ± teres minor atrophy); MRI may show compression

Key radiology findings

Test	Finding
X-ray (AP, lateral, axillary)	**Usually normal**; rules out other pathology
MRI	**Fatty atrophy of teres minor** (✓ OB; T1 streaked white + atrophied) · deltoid fatty infiltration · compression of space · **inferior paralabral cyst** with labral tear
Arteriogram	**PHCA occlusion in abduction + ER, normal with arm at side** (TO)
EMG	**Confirms diagnosis**; axillary nerve involvement
Lidocaine block	Confirms diagnosis (see Management)

Selective teres minor atrophy on MRI	**highly suggestive** of QSS (TO)
Sanders	teres minor fatty atrophy in **3% of shoulder MRIs** (OB)
Next step to confirm (OB)	**subclavian arteriogram** with arm **adducted + abducted/ER**
MRI ✗ distractors	↓ intra-capsular volume (= **adhesive capsulitis** on arthrogram) · ↑ capsular volume · infraspinatus atrophy (suprascapular) · latissimus atrophy
Isolated teres minor atrophy (OB)	**common**, unclear aetiology (axillary nerve variants); **clinically distinct** from QSS

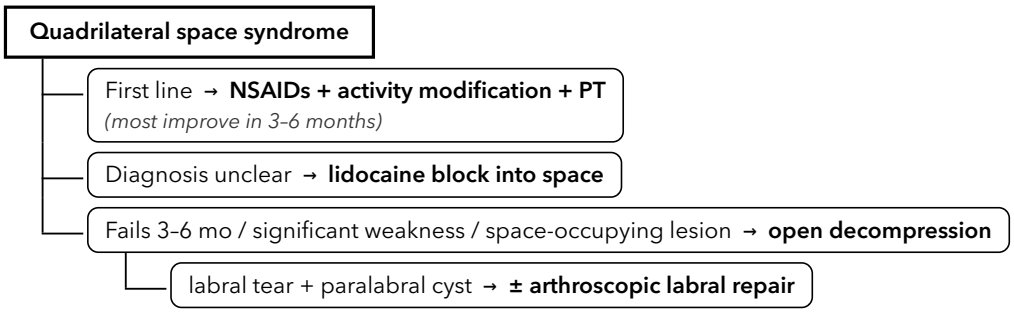


Fatty atrophy of teres minor on sagittal MRI

Brown classification (TO)

Type	Cause	Who
Vascular QSS	**Repetitive trauma to PHCA** in abduction + ER	**Professional/collegiate overhead athletes**
Neurogenic QSS	**Fixed anatomic anomalies – fibrous bands most common**	

5 Management



Nonoperative

Regimen	NSAIDs, activity restriction, physiotherapy (GH mobilisation + strengthening, **posterior capsule stretching**, massage)
TO regimen	activity modification, NSAIDs, **cortisone injections**, PT for **rotator cuff strengthening** – indicated initially for **all** patients
Outcome	**most improve with 3–6 months** nonoperative

Diagnostic lidocaine block

Technique	plain lidocaine into quadrilateral space; start **2–3 cm inferior to standard posterior arthroscopy portal**
Positive	**no point tenderness + pain-free full ROM** after injection

Operative criteria

Indication	Procedure
Failed nonop (3–6 mo)	Open release of quadrilateral space
Significant weakness + functional disability	Open release
Space-occupying lesion (paralabral cyst)	Decompression **± arthroscopic labral repair**
Isolated teres minor atrophy	**Release of fascial sling** to teres minor → ↓ pain, **Hornblower's** weakness resolves (OB)

Open quadrilateral space decompression

Approach	**lateral decubitus**; **posterior approach**, 3–4 cm incision over space; reflect posterior border of deltoid superolaterally; fat between teres minor + major
Technique	find axillary nerve using **humeral neck** as reference · **avoid cutting PHCA** · free fibrous bands (**lysis of fibrous bands**) · confirm release in **abduction + ER**
Post-op	sling for comfort · **early pendulums** (avoid new adhesions) · supervised full active ROM
Results	good results in overhead athletes after failed nonop (McAdams, 4 athletes, OB); usually symptom relief (TO)

6 Complications

Prognosis long-standing cases → **atrophy/weakness of teres minor + deltoid**

X vs Y

Lesion	Atrophy	Weakness
QSS (axillary n.)	**Teres minor** ± deltoid	**ER with arm abducted 90°**
Suprascapular notch	**Supraspinatus + infraspinatus**	Abduction + ER **at side**
Spinoglenoid notch (cyst)	**Infraspinatus only**	**Isolated ER at side**
Chronic cuff tear	Infraspinatus (retracted tear on MRI)	ER
Subscapularis	Rare (**upper + lower subscapular** nerves)	IR

QSS vs subacromial impingement QSS often **misdiagnosed as impingement**; in QSS **Hawkins + Neer negative**, MRI shows **teres minor atrophy**

✗ Wrong options (other diagnoses) Parsonage-Turner · long thoracic neuritis · internal impingement · external impingement · scapular dyskinesia

How it's asked

"Posterior shoulder pain, worse in abduction + ER, paralabral cyst in quadrilateral space — exam finding?"	→ **deltoid atrophy** (TO) ✗ Wrong options: supraspinatus atrophy · apprehension · dynamic labral shear
"True about quadrilateral space syndrome?"	→ **most common cause is oblique fibrous band** (TO) ✗ Wrong options: anterior pain · abducted + internally rotated · middle-aged diabetic female
"MRI in chronic QSS?"	→ **fatty atrophy of teres minor** ✗ Wrong options: infraspinatus · latissimus · loss of capsular volume
"Pitcher, pain + lateral arm paraesthesia, T1 fatty teres minor + deltoid — diagnosis?"	→ **quadrilateral space syndrome** ✗ Wrong options: Parsonage-Turner · long thoracic neuritis · internal/external impingement
"EMG shows QSS — muscle affected along with deltoid?"	→ **teres minor** ✗ Wrong options: teres major · supraspinatus · subscapularis
"Weightlifter, weakness of resisted ER at 90° abduction only, isolated teres minor atrophy?"	→ **QSS** ✗ Wrong options: suprascapular notch · spinoglenoid notch · scapular dyskinesia
"Next step to confirm QSS?"	→ **subclavian arteriogram, arm adducted + abducted/ER**

– – Active space – –

"Pitcher, weak ER at 90° abduction — muscle and aetiology?"	**→ teres minor; axillary nerve at quadrilateral space**
"Boundaries of quadrilateral space?"	**→ teres minor above, teres major below, long head triceps medial, surgical neck humerus lateral**

REVERSE SHOULDER ARTHROPLASTY

– Shoulder & Elbow · 38 / 60 –

1 Basics

Concept & history

Design	**convex glenoid hemisphere (glenosphere)** + **concave humeral cup** (semiconstrained)
History	early reverse designs failed (**COR too lateral** → torque → glenoid loosening) · **Grammont 1985** = first successful reverse (TO) · popular in Europe; North America since **1990**
Grammont's 2 innovations (OB)	**large glenoid hemisphere with no neck** + **small, almost horizontal humeral cup at 155°** (non-anatomic valgus) covering **< half the glenosphere**
Indications summary	**cuff tear arthropathy (CTA)**, comminuted **4-part proximal humerus fractures**, **failed arthroplasty**
TO indications	CTA · irreparable cuff + arthritis · fracture with **tuberosity nonunion** · failed TSA/HA · **RA with cuff involvement** · prerequisites: **functional deltoid + adequate bone stock** · contraindications: active infection, **deltoid paralysis**, poor bone stock, neurological disorder

2 Mechanism

Biomechanics – medialised/inferior centre of rotation & deltoid tensioning

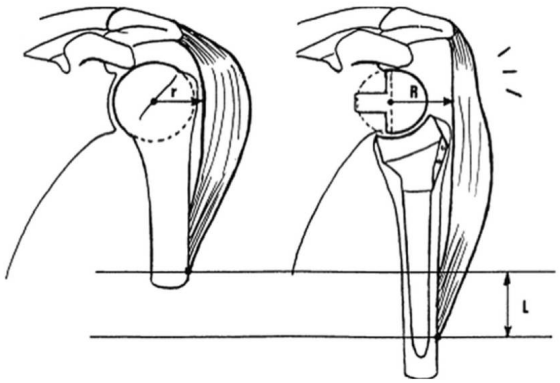
COR shift	**medial + inferior (distal)** (✓ TO, OB) ✗ Wrong options: superior · lateral · ↑ lateral humeral offset · posterior · proximal
Standard prosthesis without cuff (TO)	deltoid contraction → humeral head **subluxes superiorly**; reverse **corrects the abnormal vector** and re-creates a **fulcrum**
Medial COR	↑ **deltoid moment arm** (less force for same torque) · ↓ **torque/shear at glenoid interface** (✓ glenoid stability; force vector passes through component) · ↓ articular shear stress
Distal COR	**lengthens/tensions deltoid** (↑ resting length + tone) · humerus lowered relative to acromion · recruits **anterior + posterior deltoid** as abductors

COR shift (TO)	Effect
10 mm medial	↑ deltoid moment **~20%**
10 mm distal	↑ deltoid efficacy **~30%**

Grammont advantages (OB)	**medialisation of COR + lengthening deltoid + ↑ deltoid moment arm** ✗ Wrong options: lateralisation · shortening deltoid
Matched radius of curvature	glenosphere + humeral cup → **stable fixed fulcrum** for elevation; shear converted to **compression**

– – Active space – –

Motions gained	**forward elevation + abduction** ↑ · **IR, ER (at side or at 90°), adduction** not significantly improved → **latissimus dorsi transfer** for ER ✗ Wrong options: not normal
↑ deltoid moment (OB)	**place glenosphere more inferiorly** (✓) · anything moving COR medial/inferior or lengthening deltoid ✗ Wrong options: constrained poly (no effect) · short stem (no effect) · **smaller NSA (varus) shortens deltoid**



Grammont biomechanics: medial COR lengthens deltoid lever arm (R > r) and deltoid (L)

3 Presentation

Pseudoparalysis	**inability to actively elevate arm** with **free passive ROM**, **no neurological lesion**; irreparable cuff tear (± GH arthritis)
Pseudoparalysis without arthritis (TO)	loss of fulcrum when cuff fails → head translates superiorly → deltoid loses lever · **first line = rehabilitation** (recruit accessory muscles) → **RSA only after PT fails**
Anterosuperior escape	**incompetent coracoacromial arch** → humeral head escapes into subcutaneous tissue (esp. after **HA + acromioplasty**); **anterosuperior bulging** on attempted elevation
Failed HA (cuff failure)	pseudoparalysis, lateral deltoid pain, **no crepitus**, near-normal passive ROM · weakness of ER at side ✗ Wrong options: infection → passive ROM also ↓

4 Diagnosis & Classification

Key radiology findings

Investigation	Purpose / finding
True AP (Grashey)	Extent of arthritis, **superior migration of humerus**, acetabularisation of acromion
Axillary lateral	**Posterior glenoid wear**
Scapular-Y	
CT	**Glenoid version + bone stock** (if axillary inadequate), osteopenia
MRI	**Cuff integrity + fatty infiltration**

Anterosuperior migration on X-ray	cuff tear + **CA ligament deficiency** (OB)

Hamada ≥3 — **acetabularisation of acromion**
✗ Wrong options: not linked to post-op acromial fracture

RA glenoid — **central wear, glenoid medialisation**
✗ Wrong options: posterior wear = primary OA

– – Active space – –

5 Management

Massive irreparable cuff tear ± GH arthritis

- Balanced shoulder (isolated SST tear + arthritis) → **unconstrained TSA**
- Full active elevation retained → **no RSA**
- Pseudoparalysis, no arthritis → **PT first**
  - PT fails → **RSA**
- CTA / pseudoparalysis, elderly, working deltoid → **RSA**
  - also no active ER (abducted) → **RSA + latissimus dorsi transfer**

Failed hemiarthroplasty / aTSA

- Infection signs → **2-stage revision**
- Cuff failure / anterosuperior escape → **convert to RSA**
  (not RCR, aTSA or tendon transfer – poor results)
- Intractable pain / persistent infection → **resection arthroplasty**

Indications – cuff tear arthropathy, pseudoparalysis, failed arthroplasty or cuff repair, fracture

Indication	Detail
Cuff tear arthropathy	Severe GH arthritis + **superior escape** + massive cuff tear; **best results**
Pseudoparalysis	Irreparable cuff tear (± arthritis, TO)
Anterosuperior escape	Incompetent CA arch
Proximal humerus fracture in elderly	**3- or 4-part, age >70** · head-splitting · osteopenia (GT poor healing)
Cuff insufficiency "equivalent"	**Tuberosity nonunion/malunion** after trauma or arthroplasty
Failed arthroplasty	Prior HA or aTSA with **cuff insufficiency**; options exhausted
Failed cuff repairs	Irreparable tear after ≥2 repairs
Rheumatoid arthritis	**Only if glenoid bone stock sufficient**
Others (OB)	Proximal humeral tumours, B2 glenoid

Patient factors — **low demand** · **physiological age >70** ("exclusively >70 y", OKU 9) · **sufficient glenoid bone stock** · **working deltoid (intact axillary nerve)**

Not ideal — **40-y labourer** (glenoid loosening in young/active) · 40-y painful malunion · arthritis with **active overhead motion** · intact cuff → **aTSA** · 30-y locked posterior dislocation

– – Active space – –

Not indications (TO)	**massive cuff tear + arthritis with full active elevation** · **isolated supraspinatus tear + arthritis** → balanced → **unconstrained TSA**
Revision options — wrong (OB)	revision SST repair after 2 failed (doesn't address subscap) · **HA in subscap + SST insufficiency → instability + anterosuperior escape** · **latissimus transfer = younger, no arthritis** · cortisone/PT (doesn't restore function)
HA best results	marked glenoid bone loss, younger, **RA with forward elevation >90°**
HA failure reasons	**progressive glenoid degeneration + cuff deficiency** (✓ cuff failure; not advanced glenoid wear when pseudoparalytic without crepitus)
Instability after HA	elderly fall → **rotator cuff injury** → **convert to RSA** ✗ Wrong options: labrum
Humeral stem loosening	most commonly linked to **infection**
Sirveaux (OB)	80 shoulders: **96% no/minimal pain**, 3 revisions; ↑ forward elevation at 44 mo

Proximal humerus fracture – RSA vs alternatives

Option	Elderly 3/4-part fracture
Nonoperative	**Majority** of geriatric fractures
rTSA	**Best surgical option** in osteoporotic elderly; **better outcomes + lower reoperation than HA**; doesn't rely on tuberosity healing
HA	Outcome **depends on tuberosity healing** (cuff function)
ORIF	High demand + expected tuberosity healing; elderly → **= nonop function, ↑ complications**
IMN	2/3-part; ≈ ORIF; **high failure in osteoporosis**
PHR	**Last resort** — no bone stock; failed revisions, tumour

Fracture RSA – greater tuberosity healing

GT fixation	↑ **forward elevation + ER** (✓ OB); **anatomic GT repair** → ↑ ER, function, satisfaction vs resection/malunion · not mandatory (deltoid still powers elevation)
No effect on outcome	press-fit vs cemented stem · inlay stem · LTO (LT already fractured) · suprascapular decompression
Humeral height in fracture	calcar/tuberosity; if calcar missing → **pectoralis major insertion 5.6 cm** below top of humeral head

Contraindications – deltoid dysfunction & axillary neuropathy

Contraindication	Note
Deltoid denervation / axillary nerve dysfunction	**Absolute**; separate permanent vs temporary
Global deltoid deficiency	Absolute · **partial = relative** (reasonable results)
C5–C6 root avulsion · posterior cord palsy	→ complete deltoid palsy
Acromion deficiency	Contraindication (Orthobullets) · **relative** (OB)
Glenoid osteoporosis / insufficient bone stock	
Active infection (glenoid osteomyelitis)	
Complete brachial plexus injury · prior arthrodesis	
Severely impaired deltoid · isolated SST tear · full active elevation (TO)	

NOT contraindications
RA (sufficient glenoid stock; excellent short/midterm results) · **pseudoparalysis** (= indication) · **prior infection** (if not active) · **prior partial acromioplasty** · **absent GH joint-space narrowing** · **age >70** · **incompetent CA arch** (= indication)

Deltoid wasting + scapular dysrhythmia
not a candidate
✗ Wrong options: "candidate due to CTA" · "not candidate due to age/ massive tear"

Operative criteria

Factor	Requirement
Deltoid	**Functioning** (intact axillary nerve)
Age	**Physiological >70**, low demand
Glenoid	**Sufficient bone stock**
Cuff	Irreparable / deficient (pseudoparalysis)
Failed nonop	PT, injections
Fracture	**3/4-part, >70 y**, osteoporosis

– – Active space – –

Approach

	Deltopectoral	Anterosuperior (superolateral)
Deltoid	**Preserved**	**Anterior deltoid split/ violated**
Subscapularis	**Take-down needed**	**Preserved**
Glenoid	**Inferior pole exposed** (✓ baseplate position)	↑ exposure; **risk excess height / superior tilt**
Instability	Extensive capsular release → ↑ instability (TO: ↑ vs superolateral)	↓ (anterior stabilisers kept)
Axillary nerve	**↓ risk**	**↑ risk** (distal branches)
Notching	Not associated	**↑ notching** (insufficiently inferior baseplate)
Other	Extensile; **LD transfer possible**; poor posterior glenoid access; stiffness (immobilise for subscap)	Easier humeral axial prep + **GT fixation**

Technique

Humerus osteotomy **0–30° retroversion (typically 20°)**; more retroversion → ↑ post-op ER · LHB tenotomy/tenodesis · ream + broach as TSA
✗ Wrong options: no effect on notching

Glenoid excise labrum, **circumferential capsular release**, expose **inferior glenoid** subperiosteally · central guidewire by best bone stock · **baseplate as inferior as possible + inferior tilt** (↓ loosening + notching) · navigation

Screws **superior → coracoid base**, **inferior → scapular body** · unicortical screws to coracoid may ↓ acromial fracture; 2 screws may suffice

TO technique deltopectoral · humeral cut **30° retroversion, 45° inclination** · concave poly liner · **metaglene baseplate with 10–15° inferior tilt** → glenosphere · **4 parts**: stem, poly liner, metaglene, glenosphere · inferior tilt ↓ notching + ↑ longevity

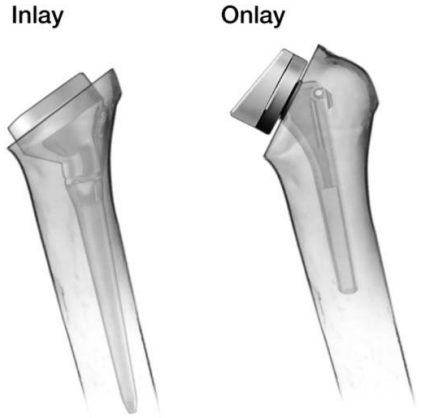
Glenosphere size **36 mm** females/smaller · **40 mm** large men

Newer prostheses (OB) **locking peripheral screws + 15° inferior tilt** · NSA **155° → 143° or 135°** → ↓ notching, ↓ instability, ↓ need for subscap repair

Humeral component design (inlay vs onlay)

	Inlay (Grammont)	Onlay
Stem	Straight, horizontal inlay tray	Curved
Bone	**More metaphyseal reaming → ↑ GT fracture risk**; ↑ bony contact	**Preserves proximal bone**
Cut		**More varus cut** → preserves GT, **minimises cuff damage**
Offset		**↑ humeral offset / lateralisation**
ROM		**↑ adduction, extension, ER**
Conversion		**Convertible to/from HA + aTSA**
Scapular spine fracture	**↓ risk** (✓ prevention, OB)	**↑ risk** (humeral lateralisation)

TO trap "onlay has more **valgus** cut and more cuff damage" = **FALSE** (onlay = **varus** cut, less damage)



Inlay vs onlay humeral components

Latissimus dorsi transfer with RSA

Indication **loss of active elevation + ER** → RSA restores elevation, **LD restores ER**; primary or revision (✓ **failed HA, no active ER with arm abducted**)

✗ Wrong options deltoid dysfunction (RSA contraindicated) · brachial plexus injury/ arthrodesis · CTA with **active ER >50°** (posterior cuff functioning → RSA or CTA head alone) · AVN + partial infraspinatus tear (nonop → resurfacing) · OA with 10° retroversion (TSA, ream anterior glenoid)

Rehabilitation

Early sling; passive / active-assisted only

Sling duration **3 wk if subscap NOT repaired** · **6 wk if repaired**; limit passive ER + active IR

Avoid **pushing out of chair** (extension + adduction + IR) · TO: avoid these **8–12 wk**

Early ROM not ↑ dislocation vs 6 wk immobilisation (OB)

Subscap re-tear **anterior instability → early exploration + repair**

– – Active space – –

Outcomes

Best indication	**CTA** (failed arthroplasty = **highest complications**)
Deterioration	radiographic after **6 y**, clinical after **8 y**
Survivorship	**~90% at 10 y**

Surgeon learning curve & postoperative surveillance

Learning curve	early complications ↓ after **18–45 cases** (✓ 40; ✗ Wrong options: 10, 75, 100, 200)
Asymptomatic notching	**observe, routine follow-up with serial X-rays** (✓) · (✗ Wrong options: aspirate for C. acnes · sling for acromial fracture · revise baseplate) · IR to **L5 = excellent** for RSA

6 Complications

Overview – which is "most common"

Claim	Source
Scapular notching = most common (radiographic) complication	OB, TO, Orthobullets incidence
Dislocation = most common complication	OB (2 questions) – read the options
Instability/dislocation = most common cause of **early failure / early revision**	Orthobullets, TO, OB
Glenoid loosening = most common **mechanism of failure / long-term** complication	Orthobullets, TO
Glenoid loosening/fracture = most common after **revision RSA**	OB

TO ranking (early)	**notching > instability > acromion/scapular spine fracture > infection > axillary neurapraxia**; overall complication rate **~15%**
Complications ∝ indication	highest for **failed shoulder arthroplasty** (dislocation, loosening, ↓ function); low for CTA / failed RCR

Scapular notching – superior baseplate, neck-shaft angle & lateralisation

Definition	erosion of **axillary border of inferior scapular neck** (scapular pillar) from **medial rim of humeral cup in adduction**; unique to RSA
Incidence	**44–96%** Grammont (Orthobullets) · **51–96%** (OB) · **68%** (Lévigne, OB) · **62%** (Lévigne, 326 pts, OB); appears early (**1–2 y**), variable progression
TO notching	**40–90%** · humeral poly impinges on scapular neck in adduction · risk: medialised COR, superior placement/tilt · prevention: **inferior tilt 10–15°**, inferior baseplate, **larger glenosphere (↑ lateral offset)** – mnemonic **"lateralise & inferiorise"** · Nerot–Sirveaux: 1 scapular neck only → 2 up to inferior screw → 3 beyond inferior screw → 4 baseplate loosening (TO)
TO complications	instability (soft-tissue tensioning) · infection (thin soft-tissue cover) · **weak ER from teres minor deficiency** → combine **LD transfer with RSA if teres minor deficient** · **acromial stress fracture** (deltoid over-tensioning)

Most common technical error — **superior placement of glenoid component** (✓) → **inferior** scapular notching
✗ Wrong options: "superior notching"

↑ Notching	↓ Notching (protective)
Superior baseplate placement (upper half)	**Inferior placement / inferior overhang** – **greatest reduction**, least adverse effect
Superior tilt of glenosphere	**Inferior tilt** (more even force, impingement-free arc)
155° NSA (Grammont; medialises humeral component)	**Varus NSA (135°)** – **most important factor** (Gutiérrez)
Medialised COR / medialised glenosphere	**Lateralised offset** (bone/metal augment) – **protective** (✓ TO "all except")
Superior glenoid erosion (E2) → base plate in valgus	**Larger glenosphere** (OB)
Anterosuperior / superolateral approach	↓ **poly depth** (↑ ROM, ↓ notching)
↓ **scapular neck length** · CTA with narrow AHD · **infraspinatus FI grade 3/4** · high BMI · ↑ activity · duration	

Gutiérrez (sawbones) — **NSA** = largest effect on **inferior impingement** → then glenosphere position → lateral offset → glenosphere size · **COR offset** = largest effect on **ROM** · **15° inferior tilt** → most uniform compression, least micromotion

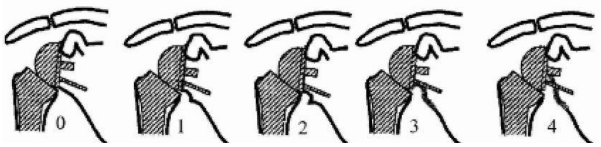
No effect on notching — humeral retroversion · stem length · horizontal humeral cut · retentive liner · cemented stem · locking screws · over-reaming

Source conflicts — **inferior/posterior tilt**: protective (Orthobullets, TO, most OB) · **minimal effect, may ↑ notching** (OB7681; low position = biggest effect) · **glenosphere size**: larger ↓ notching (OB) · **no effect** (OB214573) – read the options

Consequences — ↓ **ROM, strength, Constant scores** (✓ with preop superior erosion, Lévigne) · "not linked to detrimental outcomes" (same OB item) · particulate debris → osteolysis → **implant failure**; association with **baseplate failure** · rarely needs revision · not ↑ dislocation · not ↑ aseptic loosening

Baseplate lateralisation — ↓ notching but ↑ **baseplate loosening** (more stress)

Sirveaux grade	Extent
0	No notch
1	**Limited to scapular pillar**
2	**Contact with inferior screw** of baseplate
3	**Beyond inferior screw**
4	**Under baseplate, approaching central peg**



Sirveaux grades 0–4 of scapular notching

– – Active space – –

Instability/dislocation – risk factors & positions to avoid

Incidence **2–3.4%** (Orthobullets) · **3.4% avg, 2–14%** (TO) · **2–4%** (OB) · **2–3%** (OB) · **1.5–31%** (OB) – read the options

Direction (TO) **exclusively anterior / anterolateral**

Position **extension + adduction + internal rotation** (pushing out of chair, tucking shirt) · TO keys: **adduction + extension** and **adduction + IR** · not abduction + ER (native shoulder; deltoid "wrap effect" stabilises RSA)

✗ Position distractors flexion + abduction (RCR) · flexion + IR + adduction (**posterior labral repair**) · ER + abduction (**anterior labral repair**) · IR at 90° abduction (face/mouth hygiene)

Risk factor	Note
Irreparable / inadequate subscapularis	**Strongest** (Orthobullets); ↑ with medialised prosthesis (OB) · **less of an issue with newer implants** (OB4801) – read the options
Proximal humeral bone loss / loss of tuberosities	
Failed prior arthroplasty / revision	**Most significant** in revision vignette
Proximal humeral nonunion / malunion	Tuberosity failure
Fixed preop GH dislocation	
Male sex · **BMI >30 / obesity**	(✗ Wrong options: female)
Previous shoulder surgery (open, extensive capsulotomy)	Not arthroscopic / mini-open biceps tenodesis
Axillary nerve injury · Grammont-style implant	
Deltopectoral (vs superolateral) · subscap **FI > Goutallier 3** (TO)	
Inadequate humeral length (↓ compression) → shallow socket	**Main biomechanical** parameters (TO)
✗ Wrong option: **rheumatoid arthritis**	Not a risk factor

Stability determinants (TO) soft-tissue balance (humeral length), glenosphere size, humeral joint-line inclination, humeral version, metaglene position

Prevention **lateralise baseplate/glenosphere** (✓) · ↑ glenosphere size · **eccentric baseplate** · lateralise/distalise humeral component · **constrained implant** ✗ Wrong options: valgus stem (medialises humerus, ↓ deltoid wrapping) · subscap tenotomy without repair

Treatment **immediate closed reduction + sling** (OB) · CR can be tried but **redislocation usual → open reduction + correct the surgical error** (TO) – read the options

✗ Dislocation treatment distractors EMG (deltoid weakness) · I&D + poly exchange (**acute infection**) · I&D + antibiotic spacer (**chronic infection**) · ORIF (periprosthetic fracture)

RSA dislocation (OB algorithm)

- Closed reduction successful → **immobilise in abduction 4–6 wk**
- CR fails → **open reduction**
  - soft-tissue laxity → **thicker PE ± metal spacer ± retentive PE ± glenosphere change**
  - adequate tension → **immobilise in abduction 4–6 wk**
- Recurrent → **work-up: full-length humerus X-ray, CT, WBC, ESR/CRP**

Recurrent RSA instability – cause

- ↑ ESR/CRP + positive aspirate → **<3 mo: I&D + PE exchange · >3 mo: 1- or 2-stage revision**
- Humeral shortening
  - <15 mm → **thicker PE ± metal spacer ± retentive PE**
  - >15 mm → **humeral revision ± proximal humeral allograft**
- Humeral medialisation
  - <15 mm → **larger glenosphere ± ↑ lateral offset**
  - >15 mm → **baseplate revision with augment / lateralising graft**
- Acromial stress fracture → **immobilise in abduction 4–6 wk**

Acromial & scapular spine stress fractures

Incidence — **4%** (Orthobullets, OB) · **4–5%** / **~5%** (OB) · **4–8%** (OB) · **10.2%** with **lateralised glenosphere** (Levy, TO)

Mechanism — ↑ deltoid tension/lever arm → stress through acromion (deltoid origin) → **stress/fatigue fracture**; propagates from **superior or posterior screw** at base of acromion/scapular spine; **2–48 mo** post-op (TO) · OB: "most often occur during implantation" — read the options

✓ **Glenoid** — **universally stable glenoid implant** (TO)
✗ Wrong options: "fall on abducted shoulder"
✗ Wrong options: loose glenoid

Risk factor	Source
Female sex (up to 100% of series) · age 70–80	Orthobullets, OB, TO
Osteoporosis	✓ OB
Medialised preop COR (✓ OB) → **RA** (central wear)	Orthobullets, OB
Superior baseplate screw (✓ TO) · posterior screw	TO, OB
Lateralised glenosphere / glenoid · superior glenoid placement	TO, OB
Onlay humeral component (humeral lateralisation)	OB
↑ NSA (larger deltoid moment arm)	OB
Acromial thinning (end-stage CTA acetabularisation) · inflammatory arthritis · revision · massive cuff tear · prior surgery	TO, OB

– – Active space – –

Deltoid length conflict — **less change in deltoid length post-op** = risk (OB215540) · **distalisation / over-tensioning** = risk; superior baseplate ↓ risk (OB217851, OB218963) – read the options

Not risk factors — male · **Hamada ≥3** · RSA for fracture · age <50 · **inferior** baseplate screw

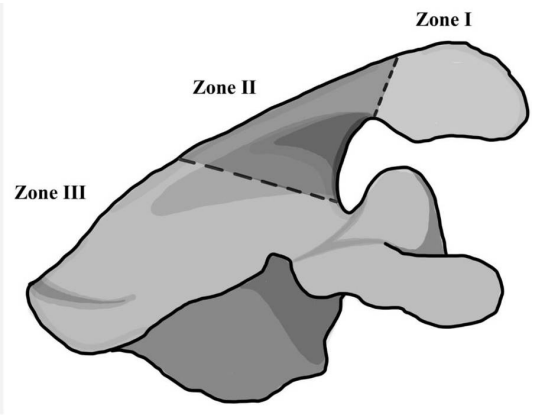
Presentation — acute **superior shoulder pain** + loss of function, often **atraumatic** or after fall; **missed on X-ray in up to 20% → CT**

Outcome — **inferior to RSA without fracture**; **elevation** most limited · better than preop · chronic pain **not** usual · **↑ revision**
✗ Wrong options: ER

Treatment — **nonop: sling (abduction pillow) 6 wk** · nonop union only **40–50%** · **ORIF / tension band** → ↑ union · TO: types I–II nonop (late ORIF if symptomatic nonunion); **type III → ORIF** if displaced, loss of function, nonunion
✗ Wrong options: nonunion <5%

Rare sequel — new instability (glenosphere orientation change, ↓ deltoid tension)

Levy type (Orthobullets)	Location	Periprosthetic scapular fracture (TO)
1	Acromion **anterior** to posterolateral corner	**Anterior acromion**
2	**Mid acromion** (base → posterolateral corner)	Acromion **posterior to AC joint, anterior to spine**
3	**Acromial base**	**Scapular spine**



Levy zones of acromial fracture

Periprosthetic fracture after a fall

Fall 8 mo post-RSA → acromial fracture — ✓ **osteoporosis is a risk factor**
✗ Wrong options: incidence 15% (≈5%) · chronic pain likely · nonunion <5% nonop · equal revision rate

Infection & complications after revision

Incidence — **1–2%** deep infection; large **subacromial dead space**

Organisms — **C. acnes + staphylococci** · C. acnes → **vancomycin + clindamycin**

Risk factors — **younger age (<65) + male = greatest** (✓ 45-y male RSA) · **primary reverse > aTSA** · **arthroplasty for trauma** · failed arthroplasty
✗ Wrong options: BMI/obesity, race, ASA, smoking, RA, diabetes

Treatment	**2-stage revision = gold standard** · acute → I&D + poly exchange · chronic → I&D + antibiotic spacer
Revision RSA	complication rate **11–36%** after failed arthroplasty; **glenoid loosening/ fracture ↑ (~25% at 5 y)** (✓) – high metal : small glenoid bone ratio; **periprosthetic glenoid** fractures commonest · also dislocation/recurrent instability ✗ Wrong options: humerus
Glenoid loosening treatment	**staged**: fill glenoid with **autograft** + **hemiarthroplasty** → await incorporation → new glenosphere

Nerve injury during glenoid preparation

Most commonly injured	**axillary nerve** (✓ label B) ✗ Wrong options: radial (A), musculocutaneous (C), median (D), ulnar (E)
Neurapraxia	**0.5–1%**; risks **anterosuperior approach** (deltoid retraction/levering) + **humeral lengthening** · direct laceration during glenoid implantation · **usually transient → observe**
Lädermann	subclinical EMG changes (mainly axillary) at 3.6 wk; **8 of 9 resolved <6 mo**; arm lengthening implicated

Baseplate failure – central peg error

Scenario	RSA + **iliac crest graft behind baseplate** → early baseplate migration → **central peg error** (peg in graft, not native glenoid; peg too short) (✓) ✗ Wrong options: tilt · dissociation · small glenosphere (larger = instability option)
Glenosphere dissociation	rare; intra-op failure to engage glenosphere on baseplate

X vs Y

	Anatomic TSA	Hemiarthroplasty	Reverse TSA
Cuff	**Must be functional**	Tuberosity healing needed	**Not needed** (deltoid)
Anterosuperior escape	**Contraindicated**	Performs poorly	✓ **Treatment**
Fracture elderly		Old standard; ↑ revision	**Best**
Infection risk	Lower		**Higher** (dead space)
Scapular spine fracture			**Highest (~5%)**

	Grammont 155°	135° (varus) / lateralised
Humeral component	Medialised	↑ lateral offset
Notching	↑	↓
Instability	↑ (↓ deltoid wrap)	↓
Deltoid length		Varus shortens deltoid (OB)
Acromial stress		Lateralisation ↑ (TO, OB)

– – Active space – –

Technical change	Notching	Dislocation	Acromial fracture	Baseplate loosening
Inferior baseplate	↓		↑ (distalisation, OB)	not ↑
Superior baseplate	↑	no direct effect	↓ (OB)	
Lateralised glenosphere	↓	↓	↑	↑
Larger glenosphere	↓	↓		
Superior screw			↑	

How it's asked

"Which is NOT an indication for RTSA?"	→ **massive cuff tear + arthritis with full active elevation** (TO) ✗ Wrong options: CTA · pseudoparalysis without arthritis · CTA + pseudoparalysis
"Most appropriate RSA candidate?"	→ **75-y man, painful arthritis + massive irreparable cuff tear** ✗ Wrong options: 40-y labourer · active overhead motion · failed HA from glenoid wear
"76-y, high-riding head, FF 80°, normal belly press?"	→ **RSA** ✗ Wrong options: TSA · RCR · arthrodesis
"71-y massive tear + arthritis + FF 40° vs 45-y failed RCRs vs intact cuff?"	→ **71-y, FF 40°**
"75-y, 2 failed RCRs, FE 60°, +lift-off, stage 4 FI?"	→ **reverse TSA** ✗ Wrong options: HA → anterosuperior escape · LD transfer
"Failed aTSA 3 y, FE 80°, deltoid intact — best function?"	→ **revision to RSA** ✗ Wrong options: tendon transfer · CTA hemi
"HA + acromioplasty, can't elevate >40°, anterosuperior bulge?"	→ **convert to RSA** ✗ Wrong options: TSA contraindicated
"HA 6 y, cuff atrophy, no infection?"	→ **convert to RSA** ✗ Wrong options: 2-stage · resection · aTSA · RCR
"HA dislocation after fall, can't elevate, anterosuperior escape?"	→ **rotator cuff; convert to RSA** ✗ Wrong options: labrum
"76-y osteoporotic 3-part fracture, low demand, surgery chosen?"	→ **rTSA** ✗ Wrong options: ORIF · IMN · HA · PHR
"Pseudoparalytic HA, no crepitus — reason for failure?"	→ **rotator cuff failure** ✗ Wrong options: glenoid wear · infection · acromial fracture

"Deltoid denervation is a contraindication to?"	→ **reverse TSA** ✗ Wrong options: arthrodesis · ACDF
"Deltoid wasting, can't flex/ER — RSA candidate?"	→ **no — deltoid dysfunction**
"Contraindication to RSA?"	→ **axillary neuropathy** ✗ Wrong options: prior infection · pseudoparalysis · prior acromioplasty · no joint-space narrowing
"NOT a contraindication to RSA?"	→ **rheumatoid arthritis** ✗ Wrong options: C5–6 avulsion · acromial deficiency · glenoid osteomyelitis · posterior cord palsy
"Malunion + arthritis + IR lag — contraindication to best surgery?"	→ **global deltoid deficiency** ✗ Wrong options: RA with bone stock · age >70 · incompetent CA arch
"Biomechanical advantage of RSA over standard arthroplasty / hemiarthroplasty?"	→ **COR more medial** (TO) ✗ Wrong options: superior · lateral · ↑ lateral offset
"Early reverse designs failed — why?"	→ **COR too lateral** → glenoid torque + loosening
"Grammont advantages?"	→ **medial COR, lengthened deltoid, ↑ deltoid moment arm**
"Directional COR change improving kinematics?"	→ **medial**
"Technique increasing deltoid moment?"	→ **glenosphere more inferior** ✗ Wrong options: smaller NSA · constrained poly · short stem
"Motion gaining most besides abduction?"	→ **forward elevation** ✗ Wrong options: ER · IR · adduction
"Scapular notching risk — all EXCEPT?"	→ **lateralised offset** (TO) ✗ Wrong options: superior placement · superior tilt · superior glenoid erosion
"Technical error causing scapular notching?"	→ **superior glenoid component placement**
"Superior baseplate → which complication?"	→ **inferior scapular notching**
"Prosthetic factor most reducing notching (Grammont)?"	→ **inferior positioning of glenoid component** ✗ Wrong options: inferior/posterior tilt · cemented stem · locking screws
"Notching 2 y post-op — implicated factor + sequela?"	→ **preop superior glenoid erosion; ↓ ROM, strength, Constant**
"Superior tilt on X-ray — most at risk of?"	→ **scapular notching** ✗ Wrong options: acromion fracture · baseplate failure · dissociation
"Modification minimising notching?"	→ **inferior baseplate** ✗ Wrong options: smaller glenosphere · retentive liner · horizontal cut

– – Active space – –

"Notching at 4 y – what happened at surgery?"	→ **glenosphere too superior** ✗ Wrong options: too much inferior tilt · shallow poly · over-lateralisation
"Modification MOST increasing notching risk?"	→ **155° NSA humeral component** ✗ Wrong options: inferior baseplate · larger glenosphere · lateral offset · retroversion
"155° prosthesis, baseplate upper half – future concern?"	→ **scapular notching**
"Prevent most common complication intra-op (OB = dislocation)?"	→ **lateralised baseplate** ✗ Wrong options: 4 bicortical screws · ↑ stem angle · drain · subscap tenotomy
"Baseplate in upper 50% – most at risk?"	→ **scapular notching** ✗ Wrong options: dislocation · baseplate loosening
"Step preventing notching seen at 2 y?"	→ **varus humeral NSA** ✗ Wrong options: smaller glenosphere · superior glenosphere · medialising
"Most common instability position in RTSA?"	→ **adduction + extension** (TO)
"Most common mechanism of RTSA dislocation?"	→ **adduction + internal rotation** (TO)
"Position to avoid after RSA?"	→ **extension, internal rotation, adduction** (pushing out of chair) ✗ Wrong options: ER + abduction · flexion + IR + adduction
"Dislocation risk factors EXCEPT?"	→ **rheumatoid arthritis** ✗ Wrong options: malunion · bone loss · failed arthroplasty · fixed dislocation
"Medialised reverse in pseudoparalysis – true?"	→ **instability ↑ with irreparable subscapularis** ✗ Wrong options: contraindicated in pseudoparalysis · LD transfer allows deltoid dysfunction · superior + lateral COR
"Revision RSA for failed aTSA – most common cause of early revision?"	→ **instability** ✗ Wrong options: notching · infection · acromion fracture
"Dislocation 2 wk after revision lateralised RSA – biggest risk?"	→ **revision arthroplasty** ✗ Wrong options: early ROM · female · mini-open tenodesis
"Management of most common post-op complication after RTSA?"	→ **immediate closed reduction + sling** ✗ Wrong options: EMG · I&D + poly exchange · spacer

"True about scapular fracture after RSA?"	→ **superior baseplate screw is a risk factor** (TO) ✗ Wrong options: loose glenoid · fall on abducted arm · medialised glenosphere
"Acromial fracture after fall at 3 mo, nonop — long-term counselling?"	→ **inferior to RSA without fracture** ✗ Wrong options: ER most limited · persistent pain likely
"Which procedure has highest risk of scapular spine fracture?"	→ **reverse TSA** (~5%)
"Onlay RSA, atraumatic scapular spine fracture at 1 y — prevention?"	→ **inlay humeral implant** ✗ Wrong options: ↑ NSA · avoid inferior screw · thinner liner
"RA patient, RSA — elevated risk of?"	→ **acromial stress fracture** ✗ Wrong options: dislocation · glenoid loosening · poly wear
"Atraumatic acromial fracture at 6 mo — risk factor?"	→ **medialised preop COR** ✗ Wrong options: male · Hamada ≥3 · RSA for fracture · age <50
"Acromial fracture after fall — true?"	→ **osteoporosis is a risk factor** ✗ Wrong options: 15% incidence · nonunion <5%
"Complications most frequent when RSA done for?"	→ **failed shoulder arthroplasty**
"After revision RSA — significantly ↑ risk of?"	→ **glenoid component loosening** (~25% at 5 y)
"Most likely to get infection after shoulder arthroplasty?"	→ **45-y male after reverse TSA** ✗ Wrong options: diabetic · obese · RA
"Nerve injured during glenoid preparation?"	→ **axillary nerve (B)**
"Baseplate on iliac crest graft migrates at 2 wk?"	→ **central peg error**
"RSA + latissimus dorsi transfer — patient?"	→ **failed HA, no active ER with arm abducted** ✗ Wrong options: CTA with ER >50° · brachial plexus injury
"Fracture RSA — improves FE + ER?"	→ **greater tuberosity fixation** ✗ Wrong options: press-fit · inlay · LTO
"Surgeon learning curve for RSA?"	→ **18–45 cases** (✓ curve at 40)
"Asymptomatic notching at 9 mo?"	→ **routine 1-year follow-up with X-rays** ✗ Wrong options: aspiration · revision
"Humeral prosthesis in RSA — all true EXCEPT?"	→ **onlay has more valgus cut + more cuff damage** (TO) (✓ inlay ↑ GT fracture, ✓ onlay ↑ ER, ✓ onlay convertible)
"10 mm medialisation of COR in RSA increases deltoid moment by?"	→ **~20%** (TO)

– – Active space – –

"Mnemonic to minimise scapular notching?" → **lateralise & inferiorise** (TO)

– – Active space – –

REVISION TOTAL ELBOW ARTHROPLASTY

– Shoulder & Elbow · 39 / 60 –

1 Basics

Epidemiology

Lifetime revision rate	**~13%** · much **higher than TKA/THA**
TEA survival	**92% (5 y)**, **81% (10 y)**, **71% (15 y)**, **61% (20 y)**
Shift in indication	originally **RA**; with anti-rheumatics (DMARDs) → now mostly **post-traumatic** (distal humerus fractures)
Risk factors for TEA failure	**smoking** · **medical comorbidities** (RA, hypothyroidism, uncontrolled DM, iron-deficiency anaemia, obesity) · **non-compliance with activity restrictions** · **non-constrained** (dislocation) or **highly constrained** (loosening) systems

Failure modes & revision rates by indication

Mode of failure	Rate	Key facts
Aseptic loosening	**15%**	↑ with **cementation + highly constrained** designs; **anterior impingement** of coronoid into anterior humeral flange in max flexion · **most common in OA**
Component failure	**15%**	Mostly **bushing wear** (linked/semiconstrained); **younger** patients or **pre-existing deformity**
Infection	**~8% → ~3%** since DMARDs (Orthobullets) · **3–5%** (OB) – read the options	**Higher than other joint arthroplasties**
Periprosthetic fracture	**5%**	Intra-op (preparation, reaming, implantation) · post-op (non-compliance, ground-level falls)
Instability	–	**Most common failure of unconstrained (unlinked) TEA** – late ligament insufficiency or malposition

Revision rate by indication	**traumatic < inflammatory arthritis < OA** (lowest → highest)
OA patients	**aseptic loosening = most common complication** (higher demand) ✗ Wrong options: infection · fracture
RA patients	↑ **tendon + ligament insufficiency**
Design	unconstrained/unlinked · **semiconstrained/linked** (e.g. **Coonrad-Morrey**) = **best longevity + function** · constrained
Polyethylene / bushing wear	linked + semiconstrained · not the main failure of unconstrained · **component fracture uncommon**
Gill & Morrey (OB)	RA, Coonrad-Morrey, 10–15 y: **97%** no/mild pain · arc **28–131°** · **92.4% survival** · 14% reoperation · 86% good/excellent

– – Active space – –



Aseptic loosening of TEA

Infection – organisms, risk factors & staged revision

Organism	Fact
S. aureus (GPC clusters, catalase +, **coagulase +**)	**Most common (~40%)**
S. epidermidis (GPC clusters, catalase +, **coagulase –**)	**2nd (30–35%)**; **worst outcomes / lowest reimplantation** – **biofilm / encapsulation → most difficult to eradicate**
Cutibacterium (P.) acnes (anaerobic aerotolerant GP bacillus)	**Most common in TSA**, not common in TEA
GPC chains, α-haemolytic, optochin-sensitive	S. pneumoniae (pneumonia)
GPC chains, β-haemolytic, bacitracin-sensitive	**Group A strep** → necrotising soft-tissue infection

Infection risk factors **RA on steroids** · **hypothyroidism** (independent) · **diabetes** · **thin soft-tissue envelope** · **prior elbow infection** · **prior elbow surgery** · **psychiatric illness**

Not risk factors female gender · Nordic heritage · **isolated +RF with –anti-CCP** · HbA1c <6.0%
✗ Wrong options: RA

Genes & lab markers

Test	Value / meaning
TSH ↑ + free T4 ↓	**Hypothyroidism** = independent PJI risk (TSH normal 0.5–5.0; free T4 normal 0.7–1.9)
ESR, CRP, WBC	Usually ↑ in PJI (WBC may be normal)
IL-6, alpha-defensin	**Not studied** in elbow PJI
Synovial WBC	**No accepted cut-off** for elbow
Aspiration culture	**+ = chronic infection**; **very high PPV, very low NPV → negative aspirate does not rule out**
Intra-op histology	**Most effective** diagnostic test; **specificity 93%**, **NPV 90%**, **sensitivity 51%**
Intra-op cultures	**Negative in 10%** of infected TEA; chronic **~15%** vs acute **3%**

No definitive test diagnosis needs **high clinical suspicion**

– – Active space – –

3 Presentation

Symptoms **squeaking / crepitus** (bushing wear) · pain · swelling · stiffness

Red flags for infection **pain at night or rest** · **progressive loss of motion without cause** · **early unexplained loosening** · erythema, **sinus tract**, persistent wound drainage

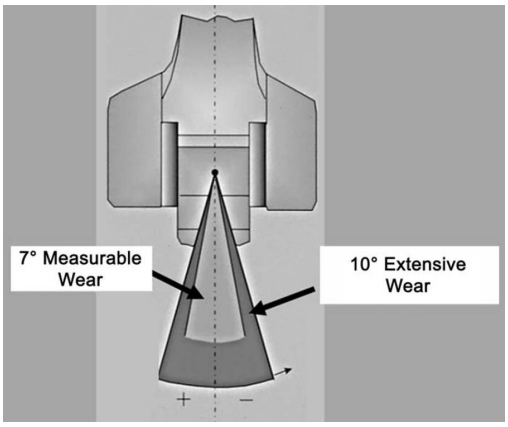
Exam diffuse tenderness, swelling, **deformity** (significant bushing wear), painful limited ROM

4 Diagnosis & Classification

Key radiology findings

X-ray **AP + lateral of humerus, elbow, forearm** → loosening, periprosthetic fracture, **bushing wear**

Bushing wear angle angle between ulnar and humeral components · **>10° = mild–moderate wear** (Orthobullets text) · figure labels **7° measurable, 10° extensive** — read the options



Bushing wear angle

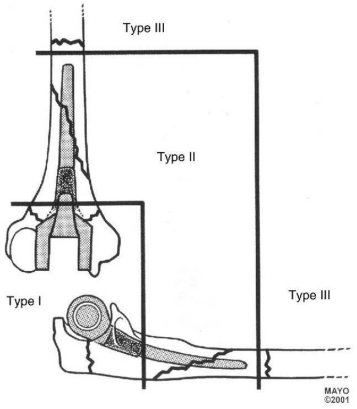
CT periprosthetic **osteolysis / loosening**; **cement mantle intact or broken** in fracture

MRI **metal subtraction + contrast** → abscess / soft-tissue infection

– – Active space – –

Mayo (O'Driscoll & Morrey) classification of periprosthetic fracture

Type	Location / features	Treatment
I	**Periarticular** – humeral condyle or olecranon; osteolysis + muscle distraction	**Undisplaced** → immobilisation / soft-tissue repair (**fibrous union OK**) · **Displaced** → ORIF with heavy non-absorbable sutures or **tension band** · intact extensor + well-fixed stem → **fragment excision**
II1	Along stem, **well-fixed implant**	**ORIF + component retention ± strut allograft** (unicortical screws + cerclage cables)
II2	Along stem, **loose implant, good bone stock**	**Revision with long-stem** ± strut allograft + impaction grafting (± locking plate / cerclage)
II3	Along stem, **loose implant, severe bone loss**	Revision with **extensive allograft**; often **resection arthroplasty**
III	**Distal to prosthesis**	Well-fixed → **humerus: immobilise**, **ulna: ORIF (plate)** · loose → treat as **II2** · CT/X-ray to confirm implant + cement mantle



Mayo classification of periprosthetic TEA fracture

5 Management

Failed TEA

- Bushing wear, stable implants, no infection → **I&D + bushing exchange (retain)**
- Acute PJI (<90 d) → **I&D + component retention (63%)**
- Chronic PJI → **explant + 2-stage revision (90%)**
  - unfit / severe bone loss / poor soft tissue → **resection arthroplasty**
- Aseptic loosening / Mayo II2 → **single-stage long-stem revision**

Operative criteria

Procedure	Indication	Outcome
Nonop (immobilise 2–4 wk → brace)	**Type I** with stable prosthesis · **Type III humerus** with stable implant (→ Sarmiento) · unfit for surgery	–
I&D + bushing exchange, retain	**Squeaking/crepitus**, stable implants, no infection · **acute PJI <90 d**	Bushing exchange **75% good at 5 y** · acute PJI **63%** eradicated · chronic only **31%**
ORIF, retain ± excision ± strut	**Type I olecranon** · **II1** · **III ulna**	**80–90%** no complications
Single-stage revision ± ORIF/ allograft	**Type II2** · **aseptic loosening**	Chronic infection only **66%** eradicated
Explant + 2-stage revision	**Infected TEA** · Mayo II2	Chronic infection **90%** eradicated
Resection arthroplasty	**Salvage**: resistant PJI, can't tolerate multiple surgeries, **severe bone loss**, poor soft tissue · **II3** not reconstructable	**71%** eradication; **lowest MEPS** function

Chronic P. acnes (6 mo), well-fixed, good bone stock, healthy — **2-stage revision**
✗ Wrong options: IV antibiotics alone · arthroscopic I&D · I&D + poly exchange · single stage

Technique pearls

Bushing exchange — prior approach · ulnar nerve **explore/decompress if symptomatic**, else identify + protect · **uncouple** components → inspect integrity, loosening, rotation → exchange bushing

ORIF — humerus: **posterior paratricipital** muscle-sparing · ulna: posterior + subcutaneous border extension · **radial nerve crosses lateral IM septum 14 cm proximal** to lateral epicondyle · Type I with extensor disruption → **tension band, K-wires dorsal to implant** engaging cement + figure-of-8

Single-stage revision — remove all loose implant + cement · **long stem in all**, **2 cortical widths beyond fracture** · **cement stem BEFORE struts** (extravasation under strut → no incorporation, nerve irritation) · humerus: **longer posterior + shorter anterior strut** · ulna: dorsal strut under **anconeus** or **FCU**
✗ Wrong options: flexion impingement

2-stage (infection) — **posterior triceps-sparing** · humeral removal: burr + flexible osteotomes, slap hammer; stuck → **posterior humeral cortical split** · well-fixed ulna → **extended olecranon osteotomy** (repair with cables) · **antibiotic spacers/ beads** in both canals; no cement in subcutaneous tissue · ± static ex-fix · **≥6 wk IV culture-specific antibiotics** → reimplant when markers normalise, **long stem + antibiotic cement**

2-stage (fracture) — stage I: union with **ICBG + plate** · stage II: **longer stem + impaction graft**

Resection arthroplasty — thick flaps; nerves in unpredictable places; remove all cement; stabilise with heavy sutures/wires if needed

– – Active space – –



Resection arthroplasty

6 Complications

Complication	Rate / risk factors	Treatment
Persistent deep infection	**~10% after 2-stage**, **~30% after single stage** · **polymicrobial**, **S. epidermidis**, **RA on steroids**, **extensive bone loss**	**Resection arthroplasty**
Ulnar nerve palsy	**Cement extravasation**, **not visualising nerve**	–
Radial nerve palsy	**Cement extravasation**, **long plates** impinging	–
Symptomatic hardware	**ORIF of periprosthetic ulna** fractures	Remove after union

Prognosis depends on **cause of failure**, **comorbidities**, remaining **bone stock**

X vs Y

	Unconstrained (unlinked)	Semiconstrained (linked)	Highly constrained
Main failure	**Instability**	**Bushing / poly wear**, loosening	**Aseptic loosening**
Longevity	–	**Best** (Coonrad-Morrey)	–

	Single-stage revision	Two-stage revision	Resection arthroplasty
Chronic infection eradication	**66%**	**90%**	**71%**
Persistent infection	~30%	~10%	–

How it's asked

"Chronic P. acnes TEA, 6 mo, well-fixed, good bone, healthy – treatment?"	→ **two-stage revision arthroplasty**
"RA, cemented semiconstrained TEA 8 y, ↑ ESR/CRP, lysis – hardest organism to eradicate?"	→ **Staphylococcus epidermidis** ✗ Wrong options: S. viridans, E. coli, Vibrio, C. difficile

"History item most linked to TEA PJI (TSH 10.6, free T4 0.23)?"	→ **TSH and free T4 levels** = hypothyroidism ✗ Wrong options: female · Nordic · HbA1c 5.9% · +RF with −anti-CCP
"Draining TEA at 12 wk — organism with worst reimplantation prognosis?"	→ **gram-positive cocci in clusters, catalase-positive, coagulase-negative** = S. epidermidis ✗ Wrong options: coagulase + = S. aureus, most common
"Most common failure of unconstrained TEA?"	→ **instability** ✗ Wrong options: poly/bushing wear · humeral loosening · component fracture
"TEA revision rates lowest → highest?"	→ **traumatic < inflammatory arthritis < osteoarthritis**
"TEA for OA — most common complication (figure)?"	→ **aseptic loosening (Figure A)** ✗ Wrong options: fracture · infection · normal X-ray
"Most effective test for TEA infection?"	→ **intra-op histology** ✗ Wrong options: negative aspirate excludes

– – Active space – –

ROTATOR CUFF ARTHROPATHY

– Shoulder & Elbow · 40 / 60 –

1 Basics

Epidemiology

Demographics	♀ **>** ♂ · **7th decade** · **dominant** shoulder
Risk factors	**rotator cuff tear** · **RA** · **crystal-induced arthropathy** · **haemorrhagic shoulder** (haemophilia, elderly on anticoagulants)
History	term **cuff tear arthropathy** coined by **Neer (1983)**

Defining features

3 common features (all patients)	**non-functioning rotator cuff** · **GH degenerative changes** · **superior migration of humeral head**
Full picture	+ **subchondral osteoporosis** · **humeral head collapse** · **femoralisation** of head · **acetabularisation** of coracoacromial arch
Why diagnosis matters	definitive treatment depends on whether **cuff functions**

2 Mechanism

Pathogenesis – loss of force couples/concavity compression & anterosuperior escape

Concavity compression	cuff compresses head into glenoid → concentric rotation → **primary mechanism centring head against upward pull of deltoid**
Force coupling	muscle balance centres the net force vector on glenoid → stable **fulcrum** for deltoid + periscapular muscles · loss → **instability, eccentric wear, CTA**
Supraspinatus + infraspinatus	depress head caudally; dynamic stabilisers
Pathway	cuff deficiency → **loss of dynamic compression** → abnormal joint forces → **superior migration** → **mechanical cartilage wear** → **acetabularisation of acromion**

✗ Concept trap	Truth
Absent supraspinatus → centring impossible	**Subscapularis + infraspinatus** can still centre head
Cuff repair restores stability after concavity lost	**False** – cannot fully restore once **effective glenoid concavity lost**
Intact biceps prevents migration	**False** – biceps is **not a head depressor**
Cuff vector optimally depresses head	**False** – not optimally oriented

Mechanical factors	loss of concavity (compression effect), ↓ ROM + function, head migration, instability ± recurrent dislocation

Nutritional factors — **loss of watertight joint space** → ↓ joint fluid → **cartilage atrophy** (↓ water + GAG) + **subchondral collapse** (disuse osteoporosis)

Crystal-induced — synovial degradation proteins destroy cuff + cartilage → end-stage **calcium phosphate crystal** deposits

Pseudoparalysis mechanism

Pseudoparalysis — inability to **actively forward flex/elevate** with **full passive ROM** · cuff **can't DEPRESS** head → deltoid pulls humerus up → **anterosuperior subluxation / escape** · deltoid lever arm ↓
✗ Wrong options: can't elevate

Injection test — **no change** in active elevation after GH local anaesthetic → **pain is NOT the cause** → mechanical (RSA)

✗ Pseudoparalysis distractor	Why wrong
Posteroinferior / inferior subluxation	Deltoid pulls **anterosuperiorly**
Deltoid atrophy	Not the classic chronic-tear mechanism
Suprascapular nerve palsy	Weak supra + infraspinatus but **cuff still checkreins** head → no pseudoparalysis
C5 radiculopathy	Weak deltoid + cuff, **paraesthesias**, no pseudoparalysis
Fibroblastic capsular thickening	**Adhesive capsulitis** (passive ROM lost)

✗ Pathogenesis distractor	Diagnosis
Haemosiderin, synovial hyperplasia, cytokines	**Haemophilic arthropathy** (can cause massive cuff tears)
GPC proliferation, PMNs	**Septic arthritis** (P. acnes = **gram + bacillus**, may have **normal markers**)
Microcirculation occlusion, osteocyte death	**AVN** (subchondral lucency/collapse, no superior migration)
T-cell autoimmune	**RA** (**central** glenoid erosion, periarticular osteopenia)

3 Presentation

Symptoms — pain incl. **night pain** · subjective weakness + stiffness

Inspection — **supra/infraspinatus atrophy** · **anterosuperior escape** (head prominence anteriorly on elevation) · **subcutaneous effusion** (capsular fluid leak, "geyser")

ROM — ↓ active ± passive · **crepitus** GH/subacromial · **pseudoparalysis**

Acute-on-chronic — low-energy injury → **acute decompensation** of compensated chronic tear → pain + disability

Test	Finding	Structure
ER lag sign	Can't hold passive ER, elbow 90°	**Massive infraspinatus** tear
Hornblower	Can't ER / hold ER at **90° abduction + 90° elbow flexion**	**Teres minor** dysfunction
Belly press	Normal → subscapularis intact	Subscapularis

- - Active space - -

4 Diagnosis & Classification

Key radiology findings

Views AP, axillary, **Grashey (true AP)**

X-ray **↓ acromiohumeral interval** · **acromial acetabularisation** · **femoralisation** of head · **asymmetric superior glenoid wear** · **lack of osteophytes** · **osteopenia** · **"snowcap" sign** (subchondral sclerosis) · **anterosuperior escape**



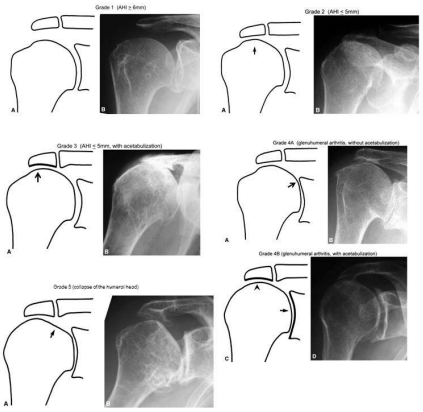
Cuff tear arthropathy, high-riding head

MRI irreparable tear: **massive fatty infiltration** + **severe retraction** · **not needed** if X-ray already shows anterosuperior escape

Labs normal WBC/ESR/CRP (vs septic, RA)

Hamada classification

Grade	Findings
1	**AHI ≥6 mm**
2	**AHI ≤5 mm**
3	AHI ≤5 mm + **acetabularisation of acromion**
4A	**GH arthritis without acetabularisation** (AHI <7 mm)
4B	**GH arthritis with acetabularisation** (AHI ≤5 mm)
5	**Humeral head collapse**

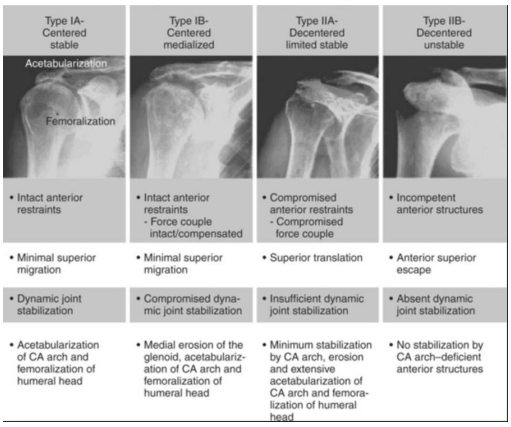


Hamada classification

Hamada versions stage 2+ AHI **≤5 mm**, stage 5 **head collapse** (Orthobullets) · AHI **<7 mm**, stage 5 **osteonecrosis** (TO) — read the options

Seebauer classification

Type	Pattern	Features
IA	**Centred, stable**	Intact anterior restraints; minimal superior migration; dynamic stabilisation; **femoralisation + acetabularisation**
IB	**Centred, medialised**	Intact/compensated anterior restraints; compromised stabilisation; **medial glenoid erosion**
IIA	**Decentred, limited stability**	Compromised anterior restraints; **superior translation**; minimal CA-arch stabilisation
IIB	**Decentred, unstable**	Incompetent anterior restraints; **anterosuperior escape**; no dynamic or CA-arch stabilisation



Seebauer classification

5 Management

Rotator cuff arthropathy

- Function preserved / first visit → **NSAIDs, steroid injection, PT (deltoid, scapular, cuff)**
- Pain only, intact CA arch + deltoid, young active → **hemiarthroplasty**
- Pseudoparalysis / elderly >70 / anterosuperior escape → **reverse TSA**
  - + ER pseudoparalysis (Hornblower +, ER lag) → **+ latissimus dorsi transfer**
  - deltoid / axillary nerve dysfunction → **no RSA**
- Infection / poor soft tissue → **resection arthroplasty (salvage)**

Nonoperative care when function is preserved

Indication **first-line**; esp. **good active elevation** (e.g. 150°), first presentation

Regimen activity modification · **NSAIDs** · **subacromial steroid injection** · PT (**deltoid**, scapular, cuff strengthening) · not straight to surgery

Acute decompensation rest, PT, injection often settle the painful phase

– – Active space – –

Operative criteria

Procedure	Indications / facts
Arthroscopic debridement	**Controversial**, unpredictable; **preserve CA arch** (✗ Wrong options: acromioplasty · CA ligament release)
Hemiarthroplasty	**Anterior deltoid preserved** + **CA arch intact** (else **anterosuperior / subcutaneous escape**) · **younger, active** · relieves pain (good–excellent **~75%**) but **not function** (elevation **40–70°**)
Reverse TSA	**Pseudoparalytic CTA** · **elderly (>70)**, low demand · **anterosuperior escape** · needs **functioning deltoid (axillary nerve)** + **good bone stock** · improves **pain + function**
Latissimus dorsi transfer	**ER pseudoparalysis** · with RSA
Pectoralis major transfer	**IR deficiency / subscapularis insufficiency** (± aTSA for massive anterior tear + arthritis)
Resection arthroplasty	**Salvage** – chronic osteomyelitis, infection, poor soft tissue
Glenoid resurfacing / anatomic TSA	**Contraindicated** – superior glenoid **shear / edge loading** ("**rocking horse**") → **glenoid loosening** (#1 cause of aTSA revision)
Cuff repair	Only if no end-stage changes, active, **minimal fatty infiltration**; **not with supraspinatus atrophy**
SCR / tendon transfer	Younger, irreparable tear, must follow strict rehab
Ream-and-run	**Young labourer, GH arthritis, intact cuff** (✗ Wrong options: CTA)

Reverse arthroplasty for pseudoparalysis & outcomes

Biomechanics moves **centre of rotation inferior + medial → ↑ deltoid mechanical advantage / lever arm →** deltoid powers elevation

Outcome **improves pain and range of motion even in setting of preoperative pseudoparalysis** · low loosening with modern constructs · durability > hemi

Can't elevate after successful local anaesthetic → RSA
✗ Wrong options: biceps tenolysis, open debridement, hemi (pain-reducing only)

Young vs old younger: cuff repair, **SCR**, or hemi · **older, less active: RSA**

Implant options convertible platform stem, central post or screw baseplate, **short cementless stem** (osteopenia) all OK · **severe osteoporosis / low deltoid tuberosity index → cemented stem** · **metal-backed polyethylene glenoid (aTSA) = contraindicated**

RSA with latissimus/teres major transfer for loss of external rotation

Indication for LDT with RSA **pseudoparalysis of ER** (deltoid restores elevation but **not ER**) from infraspinatus + teres minor dysfunction

ER dysfunction criteria **+ Hornblower** · **ER lag >15° vs contralateral** · **teres minor fatty degeneration ≥ stage 2** (Goutallier / Fuchs)

Not LDT indications negative Hornblower · <5° ER lag · **IR pseudoparalysis** · supraspinatus retraction medial to glenoid · isolated supraspinatus tear
✗ Wrong options: RSA either

RSA contraindications **axillary nerve neurotmesis** · **severe deltoid dysfunction** (e.g. stroke)

Post-traumatic arthritis, normal motor — TSA, no transfer

Pectoralis major transfer for subscapularis insufficiency – nerve risk

Technique — upper part or whole pec major tendon → near **lesser tuberosity**; route **deep to conjoint tendon, superficial to musculocutaneous nerve**

Nerve at risk — **musculocutaneous** (tension neurapraxia) → ↓ **elbow flexion** (biceps, brachialis)

✗ Wrong option: weakness	Nerve
Elbow extension	Radial
Shoulder ER	Suprascapular (infraspinatus), axillary (teres minor)
Shoulder adduction	Medial/lateral pectoral, thoracodorsal, lower subscapular
Shoulder abduction	Suprascapular, axillary

Traumatic subscapularis tear mechanism

Mechanism — fall, grasping to stop fall → **forced external rotation** against resistance
✗ Wrong options: IR · extension · abduction · flexion

6 Complications

Implant choice & complications (superior glenoid wear, metal-backed glenoid)

Scapular notching — humeral poly abuts **inferior glenoid in adduction** → bone loss, osteolysis, loosening, **lower outcome scores**

Notching	Factor
↑ Risk	**CTA with superior glenoid wear** → glenosphere too **superior** / **superior tilt** · **anterosuperior (deltoid-splitting) approach**
↓ Risk	**Lateralised humeral implant** with **lower neck-shaft angle 132.5°** (vs Grammont **155°**) · **~10° inferior tilt** · **inferior-offset glenosphere** with **~3 mm inferior overhang** · deltopectoral approach



Scapular notching after RSA

– – Active space – –

Acromial / scapular spine stress fracture mechanism = **deltoid tensioning by implant design** · **~4–5%**, mean **9 mo** · ↑ **onlay humeral component**, **lateralisation**, **osteoporosis**, **female**, medialised pre-op COR · **nonop union 40–50%**
✗ Wrong options: bisphosphonate mechanism · C. acnes · superior tilt (= notching)

Complication	Note
Instability	**More common after hemi**, rare after RSA
Infection	C. acnes → indolent infection
Neurovascular injury	Musculocutaneous with pec major transfer
Deltoid dysfunction	–
Glenoid loosening	aTSA in CTA (rocking horse)

X vs Y

	Hemiarthroplasty	Reverse TSA	Anatomic TSA
Patient	Younger, active	**Elderly >70**, low demand	Not in CTA
Needs	**Intact CA arch + anterior deltoid**	**Functioning deltoid (axillary n.)**, bone stock	Functioning **supraspinatus + subscapularis**
Pain	Relieved (~75%)	Relieved	–
Function	**Poor (40–70°)**	**Improved**, even pseudoparalysis	Glenoid loosening

	CTA	RA	AVN	Septic (P. acnes)
Glenoid wear	**Superior**, asymmetric	**Central** erosion	–	–
X-ray	↓ **AHI**, acetabularisation, **no osteophytes**, snowcap	**Periarticular osteopenia**	Subchondral lucency / **head collapse**, no superior migration	–
Labs	Normal	↑ inflammatory markers	–	**May be normal**

How it's asked

"Remote cuff repair, normal labs, high-riding head – pathogenesis?"	→ **loss of dynamic compression → abnormal forces → mechanical cartilage wear** ✗ Wrong options: haemosiderin · GPC · AVN · T-cell
"Untreated progression of cuff tear on MRI (figure)?"	→ **cuff tear arthropathy (Figure E)** ✗ Wrong options: chondrosarcoma · OA · dislocation · fracture
"True about CTA before RSA (Hamada 4a)?"	→ **concavity compression is primary centring mechanism vs deltoid**

"Full passive ROM, can't flex, no change after injection — pathomechanics?"	→ **anterosuperior subluxation of humeral head** ✗ Wrong options: posteroinferior · deltoid atrophy · SSN · failure to elevate
"Pseudoparalysis after ladder fall, acetabularisation — process?"	→ **inability of cuff to depress head** ✗ Wrong options: capsulitis, SSN, C5, inferior subluxation
"75 y, subcutaneous fluid, 80° active / 170° passive, CTA X-ray — next?"	→ **reverse TSA** ✗ Wrong options: biopsy · aspiration · anatomic TSA · syrinx MRI
"75 y, high-riding head, severe arthritis — treatment?"	→ **reverse shoulder arthroplasty** ✗ Wrong options: resurfacing · hemi · TSA · debridement
"30° active elevation, AHI 0 mm — best function?"	→ **reverse shoulder arthroplasty**
"Can't raise arm even after injection relieved pain?"	→ **reverse TSA**
"82 y fall, acute-on-chronic cuff tear, failed 4 mo — surgery?"	→ **reverse TSA** ✗ Wrong options: cuff repair · ream-and-run · hemi · aTSA
"72 y, walks + housework, CTA — implant (figure)?"	→ **reverse TSA (Figure F)**
"True about best option for CTA, 70° elevation?"	→ **improves pain + ROM even with pseudoparalysis** ✗ Wrong options: needs CA arch · pain only · high loosening
"Pain on top of shoulder 9 mo after RSA, CT fracture — mechanism?"	→ **deltoid tensioning by implant design** ✗ Wrong options: bisphosphonates · ↓ lateralisation · C. acnes · superior tilt
"RSA + latissimus transfer most appropriate?"	→ **70 F, pseudoparesis of elevation + ER, acetabularisation**
"Pseudoparalysis + Hornblower +, belly press normal — treatment?"	→ **RSA + latissimus dorsi transfer** ✗ Wrong options: aTSA + pec transfer = anterior tears
"Indication for concurrent latissimus transfer?"	→ **pseudoparalysis of external rotation** ✗ Wrong options: axillary neurotmesis · negative Hornblower · IR
"82 F, 1 mo pain, elevates to 150° — management?"	→ **trial of anti-inflammatory medication or cortisone injection and/or deltoid strengthening**
"Iatrogenic nerve injury after pec major transfer — weak movement?"	→ **elbow flexion** (musculocutaneous)
"Traumatic subscapularis tear mechanism?"	→ **forced external rotation**

– – Active space – –

"Risk factor for scapular notching?"	**→ CTA with superior glenoid wear** ✗ Wrong options: lateralised 132.5° · 10° inferior tilt · inferior offset · deltopectoral
"Implant feature contraindicated in CTA?"	**→ metal-backed polyethylene glenoid component** ✗ Wrong options: convertible stem · central post · medialised COR · short cementless stem

ROTATOR CUFF TEARS

– Shoulder & Elbow · 41 / 60 –

1 Basics

Epidemiology

FT tear prevalence	**28% >60 y · 65% >70 y** (Orthobullets, OB) · ~1/3 >60 y, >60% at 70 y (OB) · ~30% >60 y (OB) – read the options
Risk factors	**age** · **smoking** · **hypercholesterolaemia** · **family history**
Burden	leading cause of shoulder disability; ~**250,000 RCR/y** (USA); **2nd most common reason manual labourers miss work**
Age pattern	mostly **>40 y**; degenerative/attrition in elderly · acute trauma in young

Genes & lab markers

Inherited / metabolic	**family history** + **hypercholesterolaemia** = risk factors (no gene named in sources)
Indolent post-RCR infection labs (OB)	WBC borderline (11.0) · **ESR ↑ (40)** · **CRP mildly ↑ (1.5)** · Gram stain negative · culture positive at **day 7–21 → Cutibacterium acnes**

Natural history – asymptomatic prevalence & tear progression

Fact	Value
Asymptomatic tears (FT + PT) **>60 y** on MRI/US	**30–55%** (Tempelhof US **30%** · Sher MRI **54%** = 28% FT + 26% PT)
Asymptomatic ≥60 y (Orthobullets)	MRI **55%** · US **23%** of asymptomatic
Asymptomatic → symptomatic	**50% in 2–3 y** (Orthobullets) · **30–50% in 2–3 y** (OB) – read the options
Symptomatic FT tear enlarges	**50% at 2 y** (Orthobullets) · ~**50% within 2–3 y** (OB) · **20–50% at ~2 y, >50% at 5 y** (OB)
Asymptomatic tear enlarges	~**1/3** within 2–3 y (OB)

Progression rules	**larger initial tears progress faster** (**linear correlation size ↔ rate**) · **FT > PT** (faster, more atrophy) · ↑ pain = marker of progression · ↑ fatty atrophy / ↓ muscle quality ↔ ↑ pain
✗ Progression traps	FT tears don't progress · symptoms unrelated to size · pain independent of muscle quality · **all** asymptomatic tears become symptomatic (subset never painful) · "10–15% FT enlarge by 2 y" · "20% symptomatic in 3 y" · PT tears progress faster
Incidental FT tear, minimal symptoms (72 y, 7 mm)	**observation** ✗ Wrong options: repair · debridement · tenotomy
Atraumatic FT tear + PT	**subjective functional improvement** (↑ PROMs) · ~**75% avoid surgery at 2 y** · ↑ ROM (forward elevation) · minimal pain long-term · PT aim = strengthen surrounding muscles, **not tendon healing**

– – Active space – –

Anatomy – footprint, rotator cable, biceps pulley, innervation & CA ligament

Function — **dynamic stability** by **balancing force couples** → stable fulcrum; **coronal**: inferior cuff (IS, TM, SSc) vs deltoid's superior moment · **transverse**: SSc (anterior) vs IS + TM (posterior); goal of treatment = restore equilibrium

Footprint	Value
Supraspinatus ML width	**12.7 mm** (Orthobullets) · **12.1–12.7 mm** (Dugas, Ruotolo; OB) · **14–16 mm** (OB) · ~12–13 mm (OB) – read the options
Infraspinatus · teres minor · subscapularis	13.4 · 13.9 · **17.9 mm** (SSc widest)
AP dimension	**20 mm** (SS + anterior IS)
Max insertional length × width	23 × 16 mm (OB)
Articular cartilage → medial footprint	**1.6–1.9 mm**
Tendon thickness	10–12 mm
Usual tear site	**15 mm posterior to biceps**, near **SS/IS junction**

Footprint ↔ % tear — **6–7 mm** exposed ≈ **50%** PT tear

Supraspinatus (TO) — **most commonly ruptured tendon in the body** · compresses head + initiates abduction · loss → deltoid elevates head → **anterosuperior escape → cuff arthropathy**

Insertion facet (TO)	Muscle
Superior facet GT	Supraspinatus
Middle facet GT	Infraspinatus
Inferior facet GT	Teres minor (origin lateral border scapula)
Lesser tuberosity	Subscapularis

Footprint (TO) — **AP 25 mm × ML 15 mm**, thickness **~12 mm** – read the options vs table above · **critical zone ~1.5 cm proximal to insertion** = poor vascularity, commonest tear site (TO)

5 histologic layers — **articular side = half the strength of bursal side → most tears articular-sided** · I CHL fibres (1 mm) · II dense parallel fibres (3–5 mm) · III loose 45° bundles (3 mm) · IV loose CT + CHL · V capsule (2 mm)

Bursal vs articular tissue — bursal = tendon bundles → greater deformation + tensile strength · articular = tendon + ligament + capsule complex

Blood supply — subscapular, suprascapular, humeral circumflex aa (layers II–III); **bursal > articular** (hypovascular); critical zone next to lateral SS insertion

Rotator interval — capsule + **LHB** + **SGHL** + **CHL**; gap between SS leading edge and SSc superior border

Rotator crescent — thin sheet = distal SS + IS insertions (asterisk on arthroscopy)

Rotator cable — **thickened CHL** near its **avascular zone**, **perpendicular to SS fibres**, spans SS + IS insertions · **>2.5× thicker** than crescent · "**suspension bridge**" · cable disruption → destabilises tendon, tear propagates

Biceps pulley (medial sling) — **SGHL + CHL** (rotator interval tissue) · SSc = **anterior wall of bicipital groove** → SSc tear tears pulley → **medial LHB subluxation**

CA ligament extra-articular · suspension function · restrains anterior translation (with CHL) · contains head in cuff arthropathy · resection → **↑ GH translation** (anterior/inferior)

✗ CA resection distractors ↑ passive IR · ↑ axillary recess volume · ↓ AC joint forces · ↓ LHB resting tension

Nerve	Muscles
Upper + lower subscapular	**Subscapularis** (lower also **teres major**)
Suprascapular	Supraspinatus + infraspinatus
Axillary	**Teres minor** + deltoid
Thoracodorsal (middle subscapular)	Latissimus dorsi
Musculocutaneous	Biceps, brachialis, coracobrachialis
Lateral pectoral · medial pectoral	Pec major (± minor) · pec major + minor
Long thoracic · dorsal scapular	Serratus anterior · rhomboids + levator scapulae

Biceps subluxation on MRI → innervation of injured muscle? **subscapular nerves** (upper + lower) – SSc tear

Sagittal MRI marked muscle **infraspinatus** (TO, FNB 2022/2023) · **teres minor** (OB, sagittal T1 = muscle quality)

Lateral subacromial portal view through FT tear glenoid, labrum, **biceps tendon** visible

MGHL anterior restraint at **45° abduction**; **deep to SSc** (can be mistaken for it); **absent MGHL** → suspect **SSc tear** (MGHL translates medially with retracted SSc)

Posterior lever push **IR arm + posterior stress on head** → tensions SSc → shows **SSc insertion on lesser tuberosity**

2 Mechanism

Etiology

Mechanism	Features
Chronic degenerative (intrinsic = primary aetiology)	Older; **SIT**; large tears extend to superior SSc
Chronic impingement (extrinsic)	Starts **bursal surface** or intratendinous
Acute avulsion	**SSc in young** (hyperabduction/ER, fall) · **SIT >40 y with dislocation**
Iatrogenic	Failed repair, esp **SSc after open anterior surgery** (open Bankart)

Disease continuum subacromial impingement → subcoracoid impingement → calcific tendonitis → **RCT** → cuff arthropathy

Associated AC pathology · proximal biceps subluxation/tendonitis · **internal impingement** (throwers; PT tears)

– – Active space – –

PT tear causes (OB)	age-related metabolic/vascular degeneration · subacromial impingement · **anterior instability** · internal impingement · repetitive microtrauma · acute trauma less often · **articular side = poorer blood supply**
Articular vs bursal origin	articular = **intrinsic** degeneration · bursal = **extrinsic** (acromial spurs, AC arthritis); degenerative tears "start bursal side" (one OB explanation) vs articular commoner (Orthobullets, TO) – read the options

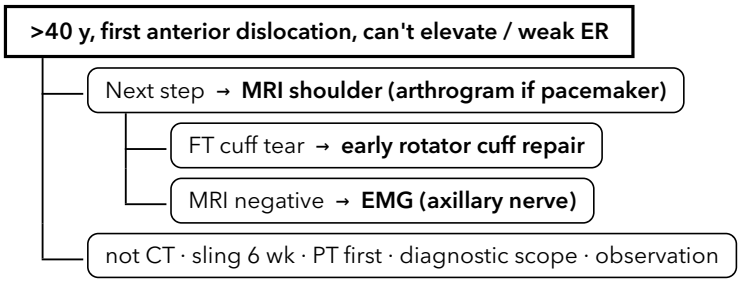
Throwing athletes – mechanism & management

Phase	**deceleration** (label E): cuff = principal decelerator → **eccentric tensile failure**
Site	**articular side of posterior SS or IS**; **posterior pain after ball release**
Overhead-athlete cuff dysfunction	primary impingement · primary tensile overload · secondary impingement/tensile overload from GH instability; microtrauma (overhead) vs macrotrauma (contact)
FT tear in thrower	**must be repaired** ✗ Wrong options: debrided
Adolescent pitcher, undersurface PT tear, never treated	**cessation of all throwing 6 wk** → cuff + periscapular strengthening → slow supervised return, pitching mechanics ✗ Wrong options: ↓ pitch count · continued play · arthroscopic/mini-open repair

Cuff tear after shoulder dislocation in older patients

Rule	**>40 y** anterior dislocation → **rotator cuff tear** (pre-existing degeneration, ↓ elasticity) » recurrent instability; often **missed**
Which tendon	**supraspinatus** commonest (TO, OB) · **SIT** (SS, **IS**, TM) > SSc · SSc tears more with **posterior** dislocation or young hyperabduction/ER
Incidence	**35–86%** >40 y (OB) · **54%**, rising to **100% >70 y** (Simank) · **15%** >40 y, **up to 40% >60 y** (OB) – read the options
Bimodal instability	peak <40 y (Bankart, HAGL) · 2nd peak >60 y (cuff tears, fractures); Hill-Sachs common, usually not addressed
Clue (TO)	pain lifting in **scapular plane (scaption)** after 6 wk immobiliser → **US/MRI** for cuff ✗ Wrong options: LHB entrapment · loose body
Axillary nerve	**7–10%**; far less common than cuff tear; usually too mild for gross weakness; **lateral arm sensation intact** → nerve palsy/radiculopathy unlikely

(✗ Wrong options: **Distractors** – posterior Bankart (posterior dislocation)) · HAGL (only ~11% of first dislocations >35 y) · SLAP (throwers, no link to anterior dislocation) · SSN injury (uncommon)



Arthrogram	dye leaks **GH → subacromial space** (geyser) = cuff tear; normal = dye contained in capsule

Treatment	**rotator cuff repair**; improved pain + function vs nonop; delay → atrophy + retraction ✗ Wrong options: hemiarthroplasty · ORIF · TSA · RTSA · coracoid transfer · labral repair · PT alone
Most likely to need surgery (52 y)	**supraspinatus** ✗ Wrong options: labrum · GH ligaments · humeral head · biceps
Massive but completely repairable	**acute tear after dislocation, 45 y** (no retraction/atrophy yet) · (✗ Wrong options: RA (attrition, steroids) · prior repair (revision) · SS fossa atrophy · superior humeral migration (→ arthropathy)) · chronic massive repairs retear **~50%**

3 Presentation

Pain	insidious, deltoid region, **overhead** activity · **night pain** = poor predictor for nonop
Weakness	**active ROM lost, passive ROM preserved** · acute pain + weakness after trauma

Physical examination – external rotation lag sign

Muscle	Weakness	Test	Technique / positive
Supraspinatus	Elevation in Jobe position	**Jobe / empty can**, **drop arm**	Abducted, IR, **30° forward in scapular plane, thumb down** → resist pressure · drop test: lowers slowly → sudden drop
Infraspinatus	ER at **0° abduction**	**ER lag sign**	**Elbow 90°, passive ER → can't actively maintain**
Teres minor	ER at **90° abd + 90° ER**	**Hornblower**	90° abd, 90° elbow flexion, 90° ER → **arm falls into IR**
Subscapularis	IR at 0° abduction	**Belly press**, **lift-off**, **bear hug** (upper border), **IR lag**, **↑ passive ER**	Belly press: **elbow drops back** · lift-off: can't lift hand off back

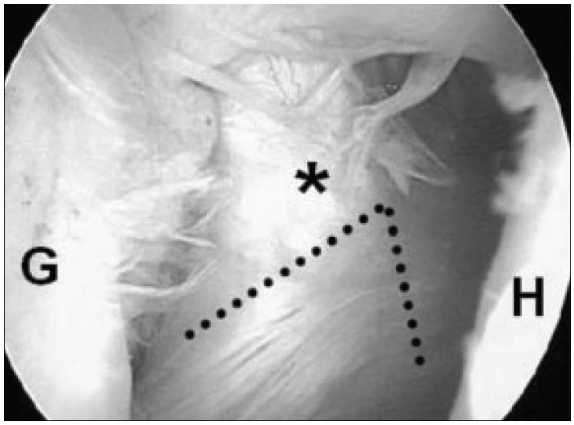
✗ Wrong options (non-cuff tests)	Yergason (resisted supination, elbow 90°) = biceps · Jerk = posterior labrum · Wright = **TOS** · relocation = anterior instability · Hawkins/Neer = impingement · O'Brien = SLAP

Subscapularis tears – belly-press/lift-off tests, biceps instability

Who	**young**, acute avulsion, **hyperabduction/ER** (fall, wave, door handle) · iatrogenic after **open Bankart** · degenerative = superior portion; largest cuff muscle, prevalence under-recognised
Symptoms	**anterior shoulder pain** (mimics **subcoracoid impingement**)
Signs	**↑ passive ER** (e.g. 90° vs 45°, 100° vs 70°) · **↓ active IR** · belly press, lift-off, bear hug, IR lag
✗ ROM traps	↑ passive flexion/extension/abduction · ↑ active IR · ↑ passive IR
MRI	SSc discontinuity off **lesser tuberosity** + **LHB out of groove (medial subluxation/dislocation)** = near-pathognomonic
LHB dislocated medially → also injured?	**subscapularis**; rare lesser tuberosity injury ✗ Wrong options: MGHL · Bankart (not biceps stabilisers)

– – Active space – –

Biceps dislocation routes	beneath SSc · within SSc · extra-articular
Arthroscopic findings (all of the above)	uncovered lesser tuberosity · retraction to glenoid · avulsed SGHL · medial biceps subluxation
Godenèche (218 scopes)	pathologic LHB with intact SGHL/CHL in only **11%**; even then **medial sheath wall torn in 92%** → SGHL/CHL lesion = LHB nearly always unstable
Comma sign	**SGHL + CHL** complex attached to **superolateral corner of retracted SSc** (Lo & Burkhart) · trace it **inferiorly** → SSc; chronic SSc retracts medially, scarred to deltoid fascia



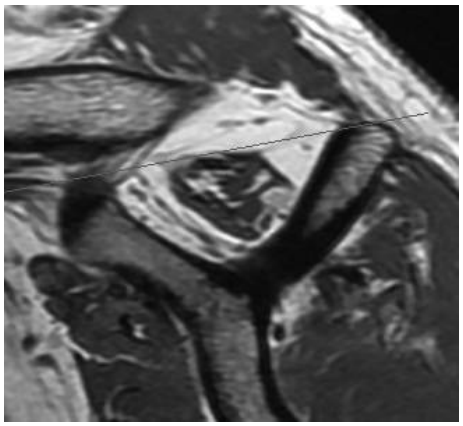
Comma sign: SGHL/CHL complex (asterisk) on superolateral SSc border

4 Diagnosis & Classification

Key radiology findings

Modality	Finding / fact
X-ray	True AP, AP IR/ER, axillary, **outlet view** (acromion); GT cysts, calcific tendonitis, CHL calcification, **type III acromion**, **AHI <7 mm** = proximal migration (chronic RCT)
Arthrogram	When **MRI contraindicated** (pacemaker); dye **GH → subacromial** = tear; MR arthrogram ↑ sens/spec
MRI	**Diagnostic standard**; size, shape, retraction; **fatty atrophy best on sagittal**; **medial biceps subluxation = SSc tear**; humeral head cysts in almost all chronic RCT
ABER MRI	↓ effacement of articular surface → contrast enters defect → **↑ sensitivity for PT (PASTA) tears**
Ultrasound	Dynamic, cheap, confirms injections; **user dependent**; = MRI accuracy for cuff

Tangent sign	sagittal oblique: line **scapular spine top → coracoid top**; SS belly **below line = positive → SS fatty atrophy → irreparable** / poorer outcome after repair (TO) ✗ Wrong options: SSc atrophy · retraction medial to glenoid · biceps dislocation



Tangent sign on sagittal MRI

PASTA on coronal T2 **articular-sided SS tear**; intra-articular contrast delineates

Lesion	Imaging / definition
PASTA	Partial articular supraspinatus tendon avulsion
ALPSA	Labroligamentous complex **medial + inferior**, rolled under **intact periosteum**
GLAD	Nondisplaced anteroinferior labral tear + **glenoid cartilage** avulsion
HAGL	IGHL avulsed from **humeral neck**
Bankart	Anterior scapular periosteum **disrupted**
SLAP	Superior labrum anterior → posterior to biceps root

Diagnostic criteria – irreparable cuff

Marker	Threshold
Size	**>5 cm** (Cofield) or **≥2 tendons**
Fatty atrophy	**Goutallier 3–4**
AHI	**<6 mm** (TO) · **<5–7 mm** (OB) · <7 mm = proximal migration (Orthobullets)
Tangent sign	**Positive**
Retraction	Medial to glenoid
Clinical	**Chronic pseudoparalysis** · **profound ER weakness** · long tear duration
Pattern	**Posterosuperior (SS + IS) > anterosuperior (SS + SSc)**

Irreparability (TO)	Threshold
Retraction	**To glenoid (Patte III)** · **>3 cm** (TO RSA lecture)
Fatty infiltration	**>50% (Goutallier III)** · **≥ grade 2** (TO RSA lecture)
AHI	**<5 mm** (TO cuff lecture) · **<7 mm** (TO RSA lecture)
Size	Gap **>5 cm**, **≥2 tendons**
Function	**Absent active ER / positive ER lag**

– – Active space – –

Patte retraction (TO, coronal MRI)	
I	Minimal / none
II	To humeral head
III	**To glenoid → irreparable**

Forsyth sign (TO) **sclerosis under acromion** = subacromial impingement

Impingement (TO)	Age	Tear side	Tear
Internal	**<40** throwers	**Articular**	Partial + **SLAP**
External	**>50**	**Bursal**	Degenerative partial/full

Zaslav internal rotation resistance test (TO) simulates throwing, resisted IR → **posterior pain = internal impingement / articular-sided tear**

Pseudoparalysis can't actively elevate **>90°** (some authors **>45°**, TO), **full passive**; **no improvement after lidocaine** · **SSc + SS complete tear → 80% pseudoparalysis**

Anatomic classification

Type	Features
SIT (SS, IS, TM)	**Majority**; subacromial impingement; degenerative older or **dislocation >40 y**
Subscapularis	More prevalent than thought; **subcoracoid impingement**; acute avulsion young (hyperabduction/ER) or **iatrogenic failed repair**

Cuff tear size

Size	cm
Small	0–1
Medium	1–3
Large	3–5
Massive	**>5** or **≥2 tendons** (Orthobullets) · ">2 tendons" (one OB)

Cuff tear shape

Shape	Repair
Crescent	Little retraction, mobile ML → direct to bone, minimal tension
U-shape	Apex at/medial to glenoid → **margin convergence** side-to-side first (↓ strain on lateral margin)
L-shape	One leaf more mobile → margin convergence
Massive immobile	U or longitudinal → **interval slide**

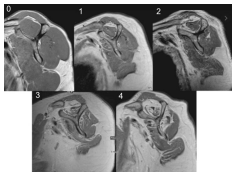
Ellman classification of partial-thickness tears

Grade	Depth
I	**<3 mm (<25%)**
II	**3–6 mm (25–50%)**
III	**>6 mm (>50%)**
Location	**A** articular · **B** bursal · **C** intratendinous

Ellman depth (TO) I **<⅓** · II **⅓–½** · III **>½** of ~12 mm thickness (vs 25% / 50% above) — read the options · **PASTA (Snyder)** · **PAINT = partial articular intratendinous tear (Conway)** (TO)

Snyder also classifies PT tears by location + depth

Goutallier classification of fatty atrophy

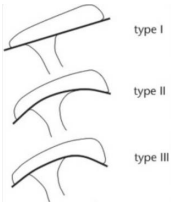
Grade	Fat vs muscle	MRI
0	Normal	
1	**Some fatty streaks**	
2	More muscle than fat	
3	Equal fat and muscle	
4	**More fat than muscle**	

Lafosse classification of subscapularis tears (TO)

Type	Lesion	Surgical approach
I	**Partial**, superior third (articular)	Debridement or single-row
II	Complete, superior third	Single-row
III	Complete, **superior two-thirds**	Double-row / footprint restoration
IV	Complete, **entire tendon**, head centred (concentric)	**Extensive 3-sided release** + robust repair
V	Complete + **head subluxation** (eccentric; coracoid impingement)	Often open repair or **pec major transfer**

– – Active space – –

Acromial morphology & impingement (TO)

Bigliani type	Shape	Note
I	Flat	
II	Curved	**Most common type**
III	**Hooked**	↑ anterior + inferior extension → **most associated with impingement + RCT**
IV	Convex (upturned), 3%	no impingement

Parameter	View	↑ RCT risk
Acromion type	**Outlet**	Type III
Acromial slope	**Outlet**	Normal 23–45°; **small angle** (steep)
Acromial tilt	**Outlet**	**↑ tilt** (keyed); **>35°** (normal 29–34°) · ">8.5 mm" (other TO copy) · "lower tilt angle → ↑ tears" (one TO text) – read the options
Lateral acromial angle	True AP (TO) · oblique coronal MRI (Banas, TO)	Normal 64–99°; **<70°** → high likelihood FT tear
Acromion index	**True AP** (✗ Wrong options: outlet)	**>0.7** (deltoid vector more superior)
Critical shoulder angle	–	**>35°**
Acromiohumeral interval	**Outlet**	**<7 mm**

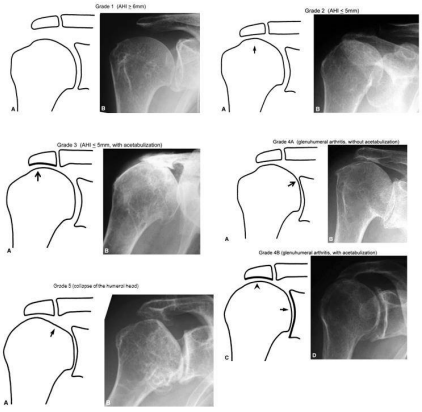
Direction summary ↑ RCT with **type III**, **↑ AI**, **↑ tilt**, **↑ CSA**, **↓ AHD**, **↓ LAA**

Bursal tears invariably associated with severe degenerative acromial change (friction/rubbing)

Cuff tear arthropathy – radiographic signs (TO)

Sourcil (eyebrow) sign — **sclerosis/erosion of acromion undersurface**; superiorly migrated head abuts acromion → **rotator cuff tear arthropathy** (anterosuperior escape) · sens/spec/PPV/NPV all **poor**
✗ Wrong options: calcific tendinitis · SS tendinitis · adhesive capsulitis · SLAP · instability

Hamada stage	Finding
1	AHI **>6 mm** (TO) · ≥6 mm (OB figure)
2	AHI **<7 mm** (TO) · **≤5 mm** (OB figure) – read the options
3	+ **acetabularisation** of acromion
4a	GH arthritis **without** acetabularisation
4b	GH arthritis **with** acetabularisation
5	Humeral head **osteonecrosis** (TO) · **collapse** (OB figure)



Hamada grades of cuff tear arthropathy

5 Management

Rotator cuff tear

- Asymptomatic / incidental → **observation**
- Atraumatic FT or PT → **PT + NSAIDs + subacromial CSI, 3–6 mo**
  - fails → **arthroscopic debridement or repair**
- Acute traumatic FT (young; >40 after dislocation) → **early repair**
  (do not delay for PT; better within 3 wk – 3/4 mo)
- Massive irreparable → **partial repair / SCR / transfer / RTSA**

– – Active space – –

Operative criteria

Indication	Procedure
Low-grade PASTA (<50%, <5–7 mm footprint) + impingement, failed nonop	**SAD + cuff debridement**
Bursal >3 mm (>25%)	Release, debride degenerative tissue + **repair**
Articular >50% (>6 mm (OB) · >7 mm (Orthobullets, OB) exposed footprint)	**Completion + repair**
PASTA, young, acute, **≥25% healthy bursal tissue**	**In situ repair** (bursal tissue left intact)
Intrasubstance tear, intact footprint	**Trans-tendon repair**
Acute (<3 mo) traumatic tear, <60 y	Repair (relative indication)
FT or PT failing nonop	Repair (relative)
Massive irreparable, intact SSc, no arthritis	SCR / LD / lower trapezius transfer
Massive + GH arthritis, **intact deltoid**	**RTSA**
Relative CI to repair	Advanced fatty infiltration/atrophy · significant GH arthritis

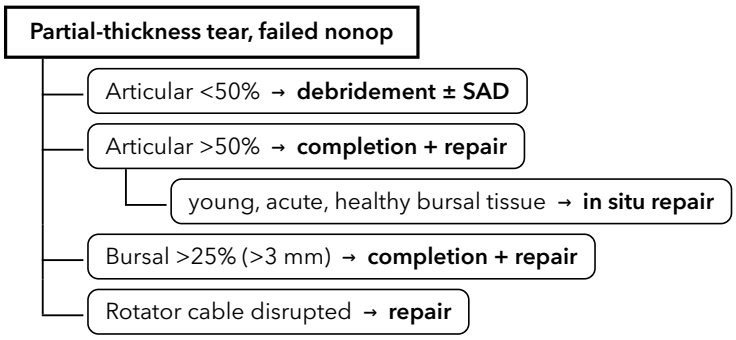
Operative vs nonoperative indications (acute traumatic vs atraumatic tears)

Nonop regimen	PT + NSAIDs + subacromial CSI = **first line**; avoid overhead; aggressive **cuff + scapular stabiliser** strengthening **3–6 mo**
Partial tear, improving, never had formal PT (70 y)	**formal PT** ✗ Wrong options: repair · debridement · RTSA · repeat MRI unless unexplained progression
PASTA, full strength, 6 wk (carpenter)	**physical therapy** first ✗ Wrong options: PRP · repair · SLAP repair · SAD
Failed 12 wk PT + injection, PASTA	**repair or debridement** ✗ Wrong options: repeat injection · continued PT · capsular release (full passive ROM) · rotator interval release (for adhesive capsulitis or retracted SSc)
Low-grade PASTA + impingement, 5/5 strength, failed nonop	**SAD + debridement** ✗ Wrong options: SR/DR repair · labral repair · tenodesis
Subacute FT SS tear, weakness, type 1 acromion, no fatty change (47 y)	**rotator cuff repair alone** ✗ Wrong options: repeat CSI · porcine SIS xenograft · + acromioplasty · + remplissage/tenodesis/DCE
AAOS points	exercise + NSAIDs for PT tears · **routine acromioplasty not required** · **non-cross-linked porcine SIS xenograft not to be used** · workers' comp → less favourable outcome
Purpose of repair	restore muscle function; then ↓ pain, prevent irreversible atrophy
Massive tear SS + SSc, 40 y, 5 mo, no atrophy	**rotator cuff repair**; attempt repair before salvage; surgery **within 6 mo** → better outcome ✗ Wrong options: debridement/SAD/DCE · pec major transfer · 6 wk more PT · SCR

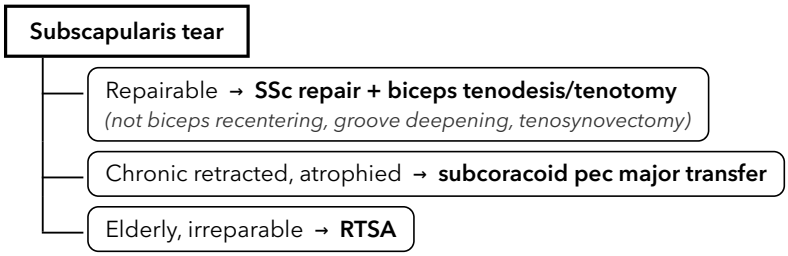
Traumatic FT tear, young — outcome driver	**time to surgical repair** (repair within **3–4 mo**; acute repair **within 3 wk** best) · (✗ Wrong options: BMI · age · male gender) · higher preop ASES → ↑ VAS improvement only
Atraumatic FT tear — strongest predictor of surgery	**patient expectation of PT** (also higher activity, non-smoker) · pain not related to anatomic severity · pain ↔ comorbidities, lower education, race

Partial-thickness tears – PASTA, bursal vs articular, debridement vs completion & repair

PASTA (TO)	**articular side**; far more common than bursal or intratendinous
Articular vs bursal	articular **more common** · bursal **more symptomatic** (subacromial bursitis) · **bursal = higher risk of progression** · articular fares better → higher % tolerated before repair
Repair thresholds	**bursal >25% (>3 mm)** · **articular >50%** ✗ Wrong options: bursal <25% · articular >25%
Articular >50% tear but poor lateral tissue	**complete the tear → repair** ✗ Wrong options: fascial graft · allograft · trans-tendon · biopsy
Bursal tear 4 mm deep, 1.5 cm AP, softening	**convert to FT + repair with suture anchors** ✗ Wrong options: debride + acromioplasty · abort · incidental
Footprint exposed 3 mm, bursal intact	**debridement + SAD** (<5 mm exposure ≈ <50% thickness; ✗ Wrong options: completion · transtendinous repair · tenotomy · synovectomy)
6 mm ML width tear	**50%** of SS tendon
Rotator cable involved	**bursal 4 mm tear involving anterior cable → repair** · intact cable 3–4 mm → less likely to need repair · articular <7 mm → debridement · SSc upper-border fraying without retraction → no repair (but often-missed tear)
High-grade PASTA (>50%), failed 9 mo PT + PRP (tennis, 51 y)	**tear completion + repair to footprint** ✗ Wrong options: isolated DCE (negative cross-body) · more nonop · SAD ± tenodesis · trans-tendon + DCE



Subscapularis repair – biceps, 3-sided release (TO)



– – Active space – –

Biceps with SSc tear unstable → **tenodesis or tenotomy** (recentering unpredictable; prevents recurrent biceps pain/failure) · tenodesis **low in groove**; address biceps **first** (rubs/fails repair)

SSc tear after open Bankart (6 mo) **subscapularis repair** + biceps tenotomy/tenodesis
✗ Wrong options: PT · tenolysis → cosmetic deformity · pec major (tendon repairable)

Which lesion → RCR + biceps tenodesis? **SSc tear with dislocated biceps**; no significant GH arthritis
✗ Wrong options: Hill-Sachs · GT fracture · bony Bankart · Bankart

Chronic retracted atrophied SSc (55 y carpenter) **pectoralis major transfer** under conjoined tendon · Achilles allograft alternative for degenerated SSc
✗ Wrong options: fusion = salvage · CSI · arthroscopic/open repair

Lafosse IV – release from coracoid base (TO) **coracohumeral ligament**
✗ Wrong options: CA ligament · transverse humeral ligament · conjoined origin

3-sided release (TO)	Structure
Anterior (subcoracoid space)	**CHL** + adhesions to **base of coracoid** – key for lateral excursion
Superior	Rotator interval; free **comma tissue** from coracoid base
Posterior (intra-articular)	Capsulotomy from **capsule + MGHL** to **6 o'clock**

Repair steps (TO) biceps first → **70° scope, posterior portal** for lesser tuberosity → traction stitch at comma via accessory anterolateral portal → anchors **inferior first → superior** → tie in **internal rotation** · III/IV → modified double-row

Arthroscopic vs open SSc repair arthroscopic superior motion + pain (Orthobullets)

Repair technique – single vs double row, transosseous-equivalent & failure mode

Mini-open deltoid-splitting; ↓ deltoid avulsion vs open; immediate PROM; active ~6 wk · arthroscopic = equivalent results

Arthroscopic vs open (medium tear) **no difference in retear rate** or function · arthroscopic → earlier rehab (no deltoid to protect) · mini-open → worse early pain/function only

Interval slides **anterior** = release SS from rotator interval (incise CHL) · **posterior** = release SS from IS, then margin convergence

Double-row (TOE) medial **mattress** + lateral **simple** → **more anatomic footprint**, larger contact area

DR vs SR	Finding
Retear	**DR lower** (esp tears/retraction **>3 cm**); SR higher (Millett meta-analysis, 7 level I RCTs)
Healing rate	DR higher
Biomechanics	DR **higher ultimate tensile load**; = peak-to-peak elongation, stiffness, conditioning elongation
Function, pain, time to heal (short/mid term)	**No difference** (esp <3 cm)
Long term	DR may improve function (one RCT)
Cost	DR more (more anchors)
Blood loss	No difference

Lowest retear risk — **Goutallier 1 + DR TOE**
✗ Wrong options: Goutallier 3–4 · SR

Two DR constructs (B, C) — equivalent retear; SR (A) possibly higher; **A and B = equivalent outcome scores at 1 y**

Vascularity — **anchor drilling** (holes in GT) + **peribursal tissue** → main supply to repair; ↑ with exercise

✗ Wrong options (false repair "truths") — motion restriction ↑ blood flow · DR better clinical results · SAD ↑ healing · non-healing always → poor outcome (**~half don't heal**, outcomes preserved)

GT trough (cancellous) — **not stronger** repair (animal models)

Failure mode — **suture pull-out through tendon = tissue failure** (most common, early) → gap formation → poor tendon–bone healing

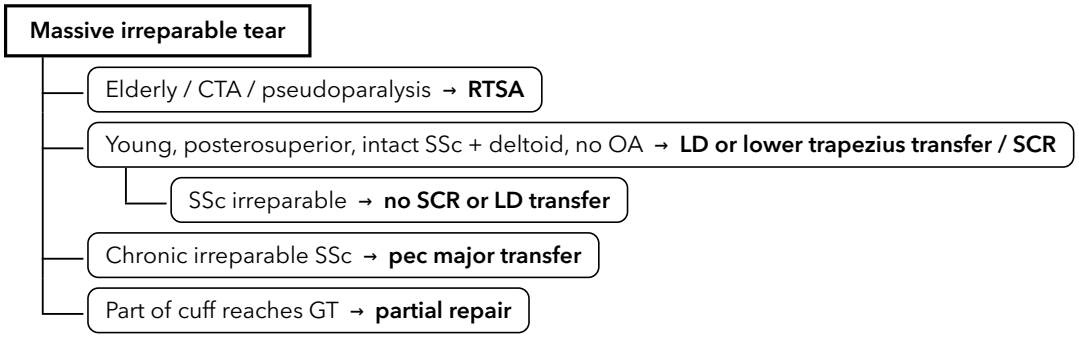
Less common failure	Note
Anchor breakage	Biocomposite fatigues less than metal
Anchor pull-out	Poor BMD, improper placement
Eyelet abrasion suture rupture	Metal anchors (sharper)
Knot failure	
Infection	~1.9%

Causes of failed RCR — failure to heal (**19–94%**: poor tissue, vascularity, bone) · surgical (deltoid, infection, foreign body, stiffness, nerve) · diagnostic (missed interval, LHB, SSc) · technical (excess tension, anchor placement); arthroscopic RCR complication rate ~10%

Traumatic retear 3 wk post-op, 1 cm, <65 y — **revision RCR** (healing >95%)
✗ Wrong options: PT · CSI · sling · debridement (for chronic irreparable)

Revision RCR vs RTSA — RTSA favoured by older age, **static proximal humeral migration**; also aetiology of retear, tissue/MRI quality

Massive/irreparable tears – partial repair, SCR & tendon transfers



Partial repair — SS can't reach GT, posterior cuff can → **partial repair** (balances coronal + axial force couples) · pain options: acromioplasty + CA release, **reverse acromioplasty (tuberoplasty)**, biceps tenotomy · ✗ RTSA in 48 y · other young options: SCR, transfers, **subacromial balloon spacer** (none superior)

Pseudoparesis vs pseudoparalysis — **active FE >90° (e.g. 100°) after lidocaine injection → pseudoparesis** (pain) · no change → pseudoparalysis · arthritis/retraction occur in both

– – Active space – –

Young (35 y), pseudoparalysis, AHD <6 mm, Goutallier 3, minimal arthritis — least preferred (TO)

RSA (✓ LD, ✓ lower trapezius, ✓ SCR)

SCR (TO, Mihata)	Detail
Mechanism	Passive constraint to **superior head translation**; space-occupier + tether; restores force couples
Graft	**Folded fascia lata 6–8 mm** · acellular dermal **3–4 mm**; anchors **superior glenoid rim → medial GT**
Indications	Failed nonop massive irreparable **SS + IS**, **Hamada 1–2**, **intact/reparable SSc**, **functional deltoid**; intolerable pain; pseudoparalysis
Contraindications	**Irreparable SSc** (subscapular nerves) · **Hamada ≥3** · GH OA
Best candidate	**59 y, SS + IS, normal X-ray**

SCR caveat irreparability confirmed only after full diagnostic arthroscopy (Hartzler & Burkhart); short-term data only, graft failure risk

Transfer	Indication	Key facts
Pectoralis major (subcoracoid, **under conjoined**)	**Chronic irreparable SSc**	Replicates SSc vector; restores IR; **4–6 wk rigid immobilisation**; not posterosuperior
Latissimus dorsi (± teres major)	**Irreparable posterosuperior (SS + IS)**, **young labourer**, Goutallier ≥3	To cuff/SSc/GT; brace **6 wk, 45° abd, 30° ER**; restores ER + elevation
Lower trapezius (± **Achilles allograft**)	Irreparable posterosuperior, middle-aged labourer	More anatomic; **superior outcomes vs LD** (Orthobullets); restores ER

Transfer counselling pain relief more reliable than function (Omid & Lee)

TO age rule **<60 y → tendon transfer / SCR** · **>70 y → RSA** · posterosuperior → **LD**, anterosuperior → **pec major**

Gerber transfer principles (TO) donor **expendable** · similar **excursion + tension** · matching **line of pull** · replaces at least one lost function

Transfer detail (TO)	Facts
Pec major	Deltopectoral; **sternal head** under conjoined → **lesser tuberosity** · pectoral nerves **~12 cm** from insertion · thoracoacromial (pectoral branch) · line of pull differs → **anterior humeral translation** risk
Latissimus dorsi	**Gerber 1988** · origin T7–L5, iliac crest, ribs 10–12 → intertubercular groove · passed between deltoid + cuff → GT · **thoracodorsal nerve (C6–C8)** · **subscapularis intact = prerequisite** · contraindicated: SSc tear, pseudoparalysis, deltoid dysfunction, stiffness
Lower trapezius (Elhassan)	Origin **C7–T12** → medial scapular spine · line of pull **matches infraspinatus**, synergistic → **now preferred** · **spinal accessory nerve 2 cm medial** to scapular border

SCR (TO) **Mihata 2013**, **TFL autograft = gold standard** (donor morbidity) · dermal allograft costly · graft superior glenoid → GT · tension to restore **AHI ~8 mm**

LHB-based options (TO) — **biceps augmentation / anterior cable reconstruction** (LHB tenotomised distally, transposed to SS footprint, glenoid attachment kept) · **biceps rerouting (Kim)**: LHB intact proximally, moved to posterior groove at footprint, cinch anchors → **all-inside, no donor morbidity**, internal splint + head depressor

Subacromial spacer balloon (InSpace) (TO) — biodegradable saline balloon, **resorbs ~1 y** · **low-demand, pain-dominant, motion preserved** · **not for pseudoparalysis or GH arthritis** · overinflation → deltoid tension · underinflation → **escape**

Rehab after transfer / SCR (TO)	Protocol
0–4 wk	Immobilisation (USI brace)
3 days–6 wk	Passive ROM (30° flexion, 80° IR, ER to neutral)
6 wk–3 mo	Active ROM all planes
>3 mo	Strengthening

LD transfer – best candidate — **young labourer, massive posterosuperior tear + SS atrophy/fatty infiltration** · **some active elevation (90–100°)** · intact SSc + deltoid, no OA (e.g. 50 y, FE 100°, ER −10°)

LD transfer – worse outcomes — **positive lift-off / SSc deficiency** (strongest; Gerber, Irlenbusch) · **deltoid weakness/detachment** · **nonsynergistic** (out-of-phase) action · stage 3–4 fatty degeneration, poor tendon · **female** · preop elevation **<90°** · **complete loss of ER** · superior escape · salvage after failed repair (vs primary)

Not a worse-outcome factor — **absence of CA ligament** · other prior surgery

LD relative CI — SSc deficiency · deltoid deficiency · pseudoparalysis · advanced GH arthritis · CTA (stays painful) · >80 y pseudoparalytic · acute tear in young (repair first) · thoracodorsal palsy · disability claimant

Belly press elbow drops back — SSc dysfunction → **↓ LD transfer benefit**
✗ Wrong options: ↑ passive IR · drop sign = SS · Hornblower = TM · axillary fold ecchymosis = pec major rupture

Posterosuperior deficiency + intact belly press → transfer? — **latissimus dorsi**
✗ Wrong options: pec major = SSc · pec minor · conjoined = instability · LHB

LD nerves at risk — **radial nerve** deep (anterior) to LD tendon, **~3 cm medial to insertion → IR arm** at tenotomy; injury → **EPL (extensor pollicis longus)** + wrist/digit extensors · posterior branch **axillary** (deep fascia of posterior deltoid, tendon passage)

Lower trapezius best for — **45 y manual labourer, irreparable posterosuperior, loss of ER**
✗ Wrong options: pitcher SLAP · irreparable SSc → pec major · GH OA → TSA · CTA → RTSA

Retear 2 mo after massive repair (rough PT), ER lag — **revision repair SS/IS**; failure: intrinsic (age, size, tissue, systemic disease, smoking) + extrinsic (construct, aggressive rehab)
✗ Wrong options: SSc revision · LD first · interval release · anatomic TSA

RTSA mechanism — **lengthens deltoid lever arm**; COR **medial + inferior**; convex glenoid + concave humerus; ↓ shear · survival **90–95% at 10 y** · most reliable for 65 y painter with SS + IS atrophy, no OA
✗ Wrong options: dermal augmentation · SCR · partial repair · trapezius transfer

– – Active space – –

Other grafts (Orthobullets) xenograft (bovine dermis/intestine; mixed) · allograft (human skin/fascia; good short term) · synthetic (foreign body reaction) · **lateral acromionectomy = contraindicated** (historic)

Concomitant procedures (acromioplasty, distal clavicle excision, suprascapular nerve release)

Add-on to RCR	Effect
Acromioplasty / SAD	No difference in PROs, symptomatic retear, ROM, strength, pain · **lower long-term reoperation** (Level 1, esp **type 2–3** acromion; OB) · "routine acromioplasty not recommended" (AAOS, OB) – read the options
Distal clavicle excision	**No effect on 2-y/long-term outcomes**, ROM, revision
SSN release	**No difference** in pain or function; SSN dysfunction in **8–27%** of massive tears (traction from retraction → fatty infiltration)
LHB tenotomy vs tenodesis	Negligible difference
CA ligament release	↑ anterior/inferior GH translation

No long-term effect either mini-open vs arthroscopic · starting PT at 6 wk · SR vs DR

Negative predictors of RCR outcome smoking · severe obesity · tear size · fatty infiltration · **age 70+** · pre-existing arthropathy · **workers' comp**

Postoperative rehabilitation – early vs delayed motion

Healing tendon-to-GT healing **8–12 wk** = rate-limiting step · protect ≥**6 wk**, ~**8 wk** for large 2-tendon tear

Protocol **early PROM** (prevents adhesive capsulitis) → **AROM at 6 wk** → strengthening **≥2–3 mo** · no active strengthening at 1 mo
✗ Wrong options: isometric ER · eccentric SS · resistance · deltoid (can't isolate from cuff)

EMG **passive forward flexion = minimal cuff activation** · active, active-assisted, isometric, **pulley-assisted** all activate cuff (avoid early)

Early (<6 wk) vs delayed (6 wk sling)	Result
Functional outcomes	**Equivalent**
Terminal ROM / stiffness at 1 y	**No difference**
Retear / healing	No difference (Cuff & Pupello, 68 pts) · **delayed may ↑ healing**
Short-term PROs	Not better with early; early may ↑ pain

PT not available first 6 wk **won't affect ROM at 1 y**
✗ Wrong options: peer-to-peer · home band strengthening · strength testing · prophylactic CSI

Scapular plane forward elevation, prone retraction, 2 mo post-RCR (TO)

lower trapezius (LT > MT + SA) · SA activity ↑ with elevation angle; **LT ↑ steeply at end range** · deficit = endurance/control/timing, not weakness → low–moderate repetitive resistance

Trapezius part (TO)	Exercise
Upper	Shrug ± retraction; overhead shrug
Middle	Overhead retraction against wall
Lower	**Prone scapular setting** (retract + depress, lift arm off table)
Lower + middle	Prone scapular retraction, arm to ~120°

Return to work

full duty 6–10 mo · light duty after progressive strengthening (earliest **8–10 wk**) · labourers later + lower outcomes than desk jobs; most return to full · PROM PT starts 2–6 wk

Analgesia

NSAIDs improve pain at 1 wk (Orthobullets)

6 Complications

Healing & retear risk factors (age, fatty infiltration, workers' comp, injections)

Retear rate

10–90% · massive chronic ~**50%**
✗ Wrong options: <5%

Factor	Effect
Age >65	**Highest risk of non-healing** (✗ Wrong options: protective); >70 → inferior outcomes
Goutallier grade (3–4)	**Greatest retear risk** (vs age, rehab, technique, gender)
Tear **>5 cm**, multiple tendons, retraction **medial to glenoid**	↑ retear
Diabetes · smoking · severe obesity	↑ retear
Poor post-op compliance	↑ retear
Concomitant AC / biceps procedure	↑ retear (Orthobullets)
Workers' comp / work-related	**↓ functional improvement + ↓ satisfaction**, later RTW
Frequent CSI	**↑ revision surgery** (frequency-dependent)
Not risk factors	Female gender · DR technique · delayed rehab · hand dominance · BMI/age/sex (young traumatic tear)

Timing of surgery

repair **<3 mo** from acute tear → better · operative time >45 min → not better (longer = overnight stay, SSI)

– – Active space – –

Complications – infection, positioning & regional anaesthesia

Suprascapular nerve (TO)	on **undersurface of SS + IS**; **~2.9 cm from superior glenoid rim**, **1.8 cm from scapular spine → no sharp dissection >1 cm medial to glenoid rim** · tethered at supraspinous fossa + notch → **tension on lateral SS advancement** (and medial retraction) · injured during **posterosuperior cuff mobilisation** in massive RCR, posteriorly directed glenoid-neck sutures (anterior labral repair), glenoid osteotomy ✗ Wrong options: thoracoacromial a. · axillary n. · PCHA
Infection	**<1%** (Orthobullets) · ~**1.9%** (OB) · **0.006–3.4%** (OB) – read the options; S. aureus, strep, **C. acnes** (formerly *P. acnes*)
C. acnes	**delayed/indolent** (10 wk), healed portals, mild ↑ ESR/CRP, Gram negative → anaerobe, cultures positive **7–21 days** ✗ Wrong options: Pseudomonas (water-borne, not shoulder)
Pre-op steroid injection	**within 1 month → highest infection risk** (**65–75% ↑**) · 1–3 mo ↑ · **>3 mo = no ↑ risk** → delay surgery ≥3 mo · not dose, location, number, steroid type
6 injections, surgery 3 mo later	**↑ revision, no ↑ infection**
Pneumothorax	after **interscalene/supraclavicular block** or arthroscopy · risk = **smoking/ COPD** (hyperinflated lungs, high pleural dome) · dyspnoea also from block **diaphragm paralysis** → CXR ✗ Wrong options: DM · revision · obesity · SSc repair
Beach chair	**LFCN palsy** up to **2%**; resolves <6 mo; padding, seatbelt, slight reverse Trendelenburg; **BMI >35** risk ✗ Wrong options: femoral · obturator · pudendal (perineal post/hip scope) · sciatic
Others	**deltoid detachment** (open) · AC pain · axillary nerve injury · stiffness (early PT/ROM doesn't reduce at 1 y)

X vs Y

	Articular-sided (PASTA)	Bursal-sided
Frequency	**Far more common**	Less
Cause	Intrinsic degeneration, weaker hypovascular side	Extrinsic (acromial spur, AC arthritis)
Symptoms	Less	**More** (bursitis)
Progression	Lower	**Higher**
Repair threshold	**>50%** (>6–7 mm footprint)	**>25%** (>3 mm)

– – Active space – –

	Supraspinatus/SIT tear	Subscapularis tear
Patient	Older, degenerative; >40 y dislocation	Young, hyperabduction/ER; post-open Bankart
Pain	Lateral, overhead, night	**Anterior**
Signs	Jobe, drop arm, ER lag	**↑ passive ER**, belly press, lift-off, bear hug, IR lag
MRI	Tear 15 mm behind biceps	**Medial LHB subluxation**
Impingement	Subacromial	**Subcoracoid**
Nerve	Suprascapular	**Upper + lower subscapular**

	Pseudoparalysis	Pseudoparesis
Active elevation	**<90°**, full passive	Limited by pain
After lidocaine	**No change**	**Improves >90°**

	SCR	LD transfer	Pec major transfer	RTSA
Tear	SS + IS	SS + IS	**SSc**	Any massive + CTA
Patient	Younger	**Young labourer**	Younger	**Elderly**, low demand
Needs	Intact SSc, deltoid, Hamada 1–2	Intact SSc + deltoid, some elevation	Under conjoined	**Intact deltoid**

	Dislocation <40 y	Dislocation >40 y
Main lesion	Bankart, HAGL	**Cuff tear** (SS, SIT)
Problem	Recurrent instability	**Weakness**; redislocation low
Sling 6 wk		Unnecessary

How it's asked

"Sagittal MRI, orange arrow – muscle?"	→ **infraspinatus** (TO)
"Sagittal T1 arrow – muscle?"	→ **teres minor**
"Thick CHL band perpendicular to SS near avascular zone?"	→ **rotator cable** ✗ Wrong options: crescent · rotator interval · MGHL
"CA ligament resection results in?"	→ **↑ GH translation**
"SS footprint ML width?"	→ **14-16mm** (OB) · 12.7 mm (Orthobullets) – read the options
"6 mm ML articular tear = what %?"	→ **50%**
"Comma sign / biceps pulley ligaments?"	→ **CHL + SGHL** ✗ Wrong options: MGHL · CA · IGHL
"Lateral subacromial portal through FT tear – arrow?"	→ **biceps tendon**

– – Active space – –

"Medial LHB subluxation — nerve to injured muscle?"	→ **upper + lower subscapular** ✗ Wrong options: suprascapular · axillary · thoracodorsal
"Posterior lever push shows?"	→ **SSc insertion (structure #5)**
"Posterior lever push + which finding supports SSc tear?"	→ **absent MGHL**
"Lafosse IV — release from coracoid base?"	→ **coracohumeral ligament** (TO)
"SSc tear arthroscopic findings?"	→ **all of the above** (uncovered LT, retraction, avulsed SGHL, biceps subluxation)
"Abnormal belly press after motorcycle fall?"	→ **subscapularis tear**
"Repaired SSc — pre-op abnormal test?"	→ **lift-off** ✗ Wrong options: jerk · Wright · Jobe · Hornblower
"Truck driver held door, MRI SSc tear — test?"	→ **belly press**
"Can't lift hand away from back — diagnosis?"	→ **subscapularis rupture**
"SSc tear + biceps out of groove, failed 3 mo nonop?"	→ **SSc repair + biceps tenodesis**
"Biceps treatment with SSc rupture?"	→ **tenodesis or tenotomy** ✗ Wrong options: recentering
"Chronic retracted atrophied SSc, 55 y?"	→ **pectoralis major transfer**
"Unstable LHB dislocated medially — also injured?"	→ **subscapularis**
"Weakness 6 mo after open Bankart, MRI SSc rupture?"	→ **subscapularis repair**
"Which image → RCR + biceps tenodesis?"	→ **SSc tear with dislocated biceps (A)**
"SSc tear exam finding?"	→ **↑ passive ER** ✗ Wrong options: ↑ active IR
"Medial biceps subluxation — provocative test?"	→ **bear hug (D)**
"Surfer, SSc tear — injured pulley ligaments?"	→ **CHL + SGHL**
"PASTA occurs on?"	→ **articular side** (TO)
"Coronal T2, carpenter fell from ladder — lesion?"	→ **PASTA** ✗ Wrong options: ALPSA, HAGL, SLAP, GLAD
"Bursal tear 4 mm deep, softening?"	→ **convert to FT + suture-anchor repair**
"MRI + US undersurface SS defect?"	→ **articular-sided SS tear**
"Poor tissue, >6 mm footprint PT tear?"	→ **complete tear + repair**
"PASTA, 3 mm footprint exposed?"	→ **debridement + SAD**
"True statement about PT tears?"	→ **bursal-sided progress more**
"Which PT tear needs repair over debridement?"	→ **bursal 4 mm involving anterior cable**
"High-grade PASTA failed 9 mo?"	→ **tear completion + repair**

"Lateral portal sees intra-articular structures in pitcher?"	→ **rotator cuff tendon tear**
"Phase of eccentric cuff failure?"	→ **deceleration (E)**
"16 y pitcher, undersurface tear, untreated?"	→ **stop all throwing 6 wk**
"50 y, pain after 6 wk immobiliser post-dislocation?"	→ **rotator cuff tear** (TO)
"65 y, pacemaker, arthrogram dye in subacromial space?"	→ **rotator cuff repair**
"Can't abduct 10 days after dislocation — confirm with?"	→ **MRI shoulder** ✗ Wrong options: EMG · CTA
"60 y, can't elevate + excess ER after dislocation — next?"	→ **MRI** (EMG only if negative)
"8 wk after dislocation, can't elevate — cause?"	→ **rotator cuff tear** ✗ Wrong options: axillary · spinal accessory
"52 y, 3/5 empty can + ER, intact sensation — next?"	→ **MRI shoulder**
"45 y, SS tear 6 wk after dislocation?"	→ **rotator cuff repair**
"Massive tear most amenable to complete repair?"	→ **45 y with anterior dislocation**
"70 y, 12 wk weak after dislocation, sensation intact?"	→ **MRI shoulder; rotator cuff tear**
"Expected injury after dislocation at 52 y?"	→ **rotator cuff tear (B)**
"65 y dislocation — structure most likely injured?"	→ **infraspinatus** (SIT) ✗ Wrong options: axillary · SSN · SSc
"Structure most likely needing surgery after dislocation at 52 y?"	→ **supraspinatus**
"Asymptomatic tears >60 y?"	→ **30–55%**
"72 y, incidental 7 mm FT tear on lipoma MRI?"	→ **observation**
"True about progression?"	→ **larger tears progress faster**
"Nonop chronic tears — true?"	→ **linear correlation size ↔ progression**
"SS tear most commonly located?"	→ **15 mm posterior to biceps**
"Atraumatic FT tear + PT — likely outcome?"	→ **subjective functional improvement**
"PASTA, carpenter, 6 wk, full strength?"	→ **physical therapy**
"47 y subacute crescent SS tear, type 1 acromion?"	→ **rotator cuff repair** ✗ Wrong options: + acromioplasty
"Low-grade PASTA + impingement after failed nonop?"	→ **SAD + cuff debridement**
"Strongest predictor of surgery, atraumatic FT tear?"	→ **patient expectation of PT**
"40 y skier, massive SS + SSc tear, 5 mo, no atrophy?"	→ **rotator cuff repair**
"70 y partial bursal IS tear improving with home exercise?"	→ **formal PT**
"PASTA failed 12 wk PT + CSI?"	→ **repair or debridement**
"Traumatic FT tear — factor most affecting outcome?"	→ **time to surgical repair**

– – Active space – –

"True about RCR biology?"	→ **bone anchor drilling ↑ vascularity**
"Repair failure at 2 mo, biocomposite DR?"	→ **suture pull-out from tissue**
"Fall 3 wk after repair, 1 cm retear, 52 y?"	→ **revision RCR**
"Early failure mechanism?"	→ **tissue failure**
"DR biomechanical advantage?"	→ **higher ultimate tensile load**
"DR vs SR – true?"	→ **↓ retear with DR** ✗ Wrong options: function · pain · healing time
"Arthroscopic vs open, 2.8 cm tear?"	→ **no difference in retear rate**
"Lowest retear risk?"	→ **Goutallier 1 + DR TOE**
"Constructs A (SR) vs B (DR) at 1 y?"	→ **equivalent outcome scores**
"DR vs SR clinically?"	→ **similar short-term function**
"Figure C (DR) vs B (SR)?"	→ **improved healing + lower retear**
"Workers' comp outcomes?"	→ **less functional improvement + lower satisfaction**
"73 y smoker – highest non-healing risk?"	→ **73 years of age (>65)** ✗ Wrong options: smoking · 4 wk timing · workers' comp · dominance
"Goutallier 4, DR, delayed rehab – retear risk?"	→ **Goutallier grade**
"CSI variable linked to infection?"	→ **injection within 1 month**
"Work-related cuff injury outcome?"	→ **poorer outcomes**
"6 CSIs, surgery in 3 mo?"	→ **↑ revision, not infection**
"Labourer – return to full duty?"	→ **6–10 months**
"Prone scapular-plane elevation exercise targets?"	→ **lower trapezius** (TO)
"Large tear, mini-open transosseous – rehab?"	→ **early PROM, AROM at 6 wk**
"No PT visits for 4 wk post-op?"	→ **won't affect ROM at 1 y**
"Early PROM vs 6 wk immobilisation (RCTs)?"	→ **equivalent functional outcomes**
"Minimal cuff EMG activity?"	→ **passive forward flexion** ✗ Wrong options: pulley-assisted
"1 mo after 2-tendon DR repair?"	→ **continue PROM, no strengthening**
"Early vs delayed ROM – true?"	→ **no difference in terminal ROM**
"SCR contraindication?"	→ **irreparable subscapularis** (TO)
"35 y pseudoparalysis, AHD <6 mm – least preferred?"	→ **reverse shoulder arthroplasty** (TO)
"Positive tangent sign predicts?"	→ **irreparable SS from fatty atrophy** (TO)
"Optimal LD transfer candidate?"	→ **36 y labourer, massive tear + SS atrophy**

"NOT a worse-outcome factor after LD transfer?"	→ **absence of CA ligament**
"Best LD indication by ROM?"	→ **50 y, FE 100°, ER −10°**
"47 y landscaper, ER lag + Hornblower, intact SSc?"	→ **latissimus dorsi transfer**
"Retear 2 mo post-op, ER lag?"	→ **revision SS/IS repair**
"50 y irreparable — best isolated LD candidate?"	→ **man with active elevation 90°**
"SS won't reach GT, posterior cuff does?"	→ **partial repair**
"Pre-op factor → poor LD result?"	→ **positive lift-off**
"Nerve 3 cm medial to LD insertion — lost muscle?"	→ **EPL** (radial)
"Patch-like lateral pain, ER weak, belly press intact — transfer?"	→ **latissimus dorsi**
"LD transfer indicated?"	→ **30 y carpenter, massive tear + fatty infiltration (A)**
"Finding that ↓ thoracodorsal-innervated transfer benefit?"	→ **elbow drops back on belly press**
"Pseudoparesis over pseudoparalysis?"	→ **FE 100° after injection**
"Best SCR candidate?"	→ **59 y, SS + IS, normal X-ray (A)**
"Lower trapezius transfer candidate?"	→ **45 y labourer, irreparable posterosuperior, loss of ER**
"SCR contraindicated if muscle innervated by label D torn?"	→ **upper subscapular nerve (D)**
"65 y painter, painless pseudoparalysis — mechanism of best procedure?"	→ **lengthening deltoid lever arm** (RTSA)
"No effect on 2-y RCR outcome?"	→ **concomitant distal clavicle resection**
"SSN release with massive repair?"	→ **no difference in function or pain**
"RCR + acromioplasty vs without?"	→ **lower long-term reoperation** ✗ Wrong options: ROM · PROs · strength · retear
"Mobilising posterosuperior cuff — structure at risk?"	→ **suprascapular nerve** (TO)
"10 wk post-RCR, cultures positive day 12?"	→ **Cutibacterium acnes**
"Pneumothorax after interscalene — risk factor?"	→ **35-pack-year smoking**
"Beach chair lower-limb nerve palsy?"	→ **lateral femoral cutaneous**
"Infraspinatus (label 10) tear — exam?"	→ **can't maintain passive ER, elbow 90° (ER lag)** ✗ Wrong options: Hornblower = TM · belly press = SSc · Jobe = SS · Yergason = biceps
"Acromial parameter ↑ RCT risk?"	→ **increased acromial tilt** (TO) ✗ Wrong options: type 2 · ↑ LAA · ↑ AHD

– – Active space – –

"Most common acromion type for impingement?"	→ **type III** (TO) (type II = commonest overall)
"Acromion index NOT measured on outlet view?"	→ **acromion index** (true AP) (TO)
"Sourcil sign?"	→ **rotator cuff tear arthropathy** (TO)
"Most commonly ruptured tendon in the body?"	→ **supraspinatus** (TO)
"Gold standard graft for SCR?"	→ **tensor fascia lata autograft (Mihata)** (TO)

SCAPULAR WINGING

– Shoulder & Elbow · 42 / 60 –

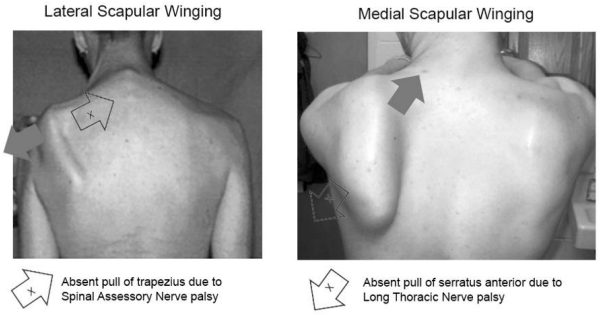
1 Basics

Definition & types

Definition dysfunction of **scapular stabilising muscles** → imbalance + abnormal scapular motion; rare, debilitating

Named by direction of the **superomedial corner** of the scapula

	Medial winging	Lateral winging
Muscle	**Serratus anterior**	**Trapezius** (rarely rhomboids)
Nerve	**Long thoracic (C5, C6, C7)**	**Spinal accessory (CN XI)**
Unopposed pull	**Medialising retraction (rhomboids)** + **elevation (trapezius)**	**Lateralising protraction** (serratus, pec major + minor)
Who	**Young athletes**; **more common**	**Iatrogenic** (neck surgery)
Commonest cause	**Repetitive stretch/traction** (>50%)	**Iatrogenic** (**up to 70%**)



Lateral (trapezius/CN XI) vs medial (serratus/LTN) winging

Anatomy – scapulothoracic stabilisers

Scapulothoracic articulation **not a true joint**; attached via **AC joint** ligaments; **suction** by serratus anterior + subscapularis holds scapula to thorax

– – Active space – –

Scapula attachment for **17 muscles**

Muscle	Origin → insertion	Action	Nerve
Serratus anterior	**Ribs 1–8** (Orthobullets) · top **8–9 ribs** (OB); 7–10 slips → anteromedial (vertebral) border	**Protraction** (+ upward rotation, stabilises vertebral border)	**Long thoracic (C5–7)**
Trapezius	Superior nuchal line, EOP, nuchal ligament, **C7–T12** spinous → lateral ⅓ clavicle, acromion, spine	**Upward rotation + elevation**	**Spinal accessory (CN XI)**
Rhomboid major	**T2–T5** spinous → medial border below minor	**Retraction** (+ elevation, OB)	**Dorsal scapular**
Rhomboid minor	**C7–T1** spinous → medial border at base of spine	Retraction	**Dorsal scapular**
Levator scapulae	**C1–C4 transverse processes** → medial border at spine level	Elevation + downward rotation	**C3–C4 cervical plexus** + dorsal scapular

Serratus parts upper: downward rotation (glenoid down), stabilises superior scapula · middle: protraction · lower: **upward rotation + abduction**

Serratus blood supply long thoracic artery (superior) · **thoracodorsal artery** branches (inferior)

Trapezius parts upper: upward rotation + elevation · middle: **retraction + adduction** · lower: downward rotation + depression

Long thoracic nerve

Roots **ventral rami C5, C6, C7 · C4 contribution 13% · no C7 in 8%**
✗ Wrong options: C8–T1 · C4–6 · C6–8

Course **first nerve to branch** off brachial plexus · posterior to axillary vessels + plexus · **superficial to serratus**, branch to each slip

Vulnerability **long course, small diameter, little connective tissue** · easily injured in **axillary dissection**

Crow's foot where LTN crosses the dominant inferior **serratus branch of thoracodorsal artery**

Spinal accessory nerve

CN XI spinal + cranial parts → accessory trunk → **jugular foramen** → crosses **lateral to IJV** → **SCM** → **posterior triangle** → trapezius · also ventral rami **C2–C4** (OB)

Vulnerability **superficial course in posterior triangle**

Dorsal scapular nerve – rhomboid & levator

Medial scapular muscles innervated by **dorsal scapular nerve** (TO)
✗ Wrong options: median · ulnar · musculocutaneous · superior/inferior subscapular · thoracodorsal

Dorsal scapular nerve (TO) **anterior ramus of C5 → levator scapulae + rhomboids** · OB: **C4 + C5 (mostly C5)**; less commonly causes winging

Medial-border origins (TO) posterior surface: **levator scapulae, rhomboid major + minor** (deep extrinsic) · anterior surface: **teres major** (TO) · **teres minor = lateral border** (TO)

Attachments (TO) levator → **transverse processes of upper cervical** · rhomboids → **spinous processes of upper thoracic**

Fossa muscles (TO) supraspinatus, infraspinatus, subscapularis from their fossae

Nerve (TO, OB)	Origin	Muscles	Winging?
Dorsal scapular	**C5 anterior ramus**	**Levator, rhomboids**	Less common
Suprascapular	**Superior trunk** (C5–6)	Supraspinatus, infraspinatus	No
Upper (superior) + lower (inferior) subscapular	**Posterior cord** (C5–6)	Subscapularis, **teres major**	No
Thoracodorsal	**Posterior cord** (C6–8)	**Latissimus dorsi**	No
Axillary	**Posterior cord**	Deltoid, teres minor	No
Long thoracic	C5–7	Serratus	**Medial**
Spinal accessory	CN XI	Trapezius	**Lateral**

2 Mechanism

Medial winging – causes

Cause	Detail
Traction (>50%)	**Repetitive stretch = most common** (gradual) — head tilted away during overhead activity, **throwing, swimmers, weightlifters, volleyball**, prolonged abduction · acute high-energy (motorcycle)
Compression – acute	Blunt chest/head/neck trauma (contact sports, **MMA blows**), sudden shoulder depression (fall)
Compression – chronic	**Scalenes** · **subcoracoid** (coracoid–1st/2nd rib; carrying loads) · inflamed bursae · anteroinferior scapular border
Iatrogenic (10%)	**Radical mastectomy / axillary node dissection**, thoracic surgery (**flail chest fixation**), **chest tube**
Penetrating	Chest wall
Neuralgic amyotrophy (Parsonage-Turner)	**Isolated LTN palsy in 22%**; no trauma, antecedent immune/inflammatory insult
Mechanical	**Serratus avulsion** · displaced **inferior pole scapula fracture**
Muscle inhibition	Underlying GH pathology (cuff, labrum — Kibler dyskinesia), fatigue from repetitive tensile use

Most common cause conflict **repetitive traction** (Orthobullets, OB) · **Parsonage-Turner = most often associated** (OB6211) · Parsonage-Turner **"very rarely"** (OB7315) — read the options; nerve = **long thoracic** either way

Vastamäki (OB) 92 isolated serratus palsies: **exertion 40%**, trauma 24%, surgery/anaesthesia 11% · **infection → better recovery**, **surgery/anaesthesia → poorer**

– – Active space – –

Lateral winging – causes

Iatrogenic (most common)	**cervical lymph node biopsy**, **radical neck dissection**, posterior cervical surgery, carotid endarterectomy
Traumatic	traction (**sudden lateral neck flexion**, MVA), blunt (**deep tissue massage**), penetrating neck, fall in elderly

3 Presentation

Medial winging – long thoracic nerve/serratus anterior

Symptoms	**vague, nonspecific shoulder girdle pain + fatigue** (base of neck, scapula, deltoid) · spasms · weak **lifting away from body/overhead** · discomfort sitting against chair · subjective instability (no stable platform) · ± traction plexus symptoms
Deformity	**inferomedial scapula elevates, protrudes posteriorly + medially**; scapula superior + medial with **inferior pole rotated medially** · **worse with forward flexion / wall push-up**
Mechanism of deformity	**compensatory overpull of rhomboids (major + minor) + levator scapulae** (✓ OB) · unopposed **trapezius** pulls medially (antagonist)
Exam	abduction often **≤90°** · weak **forward flexion + abduction** · wasting of anterior scalene triangle (SCM atrophy)
Scapular compression test	examiner pushes scapula anteriorly + stabilises chest during flexion → **positive = ↑ motion** with support
Scapular assistance test (OB)	upward pressure on **inferomedial angle** during elevation → **↓ pain + ↑ arc** = positive
✗ Exam distractors	ER lag/hornblower (teres minor) · **belly press** (subscapularis) · **apprehension** (anterior instability) · **Wright's** (TOS)

Lateral winging – spinal accessory nerve/trapezius & Eden-Lange transfer

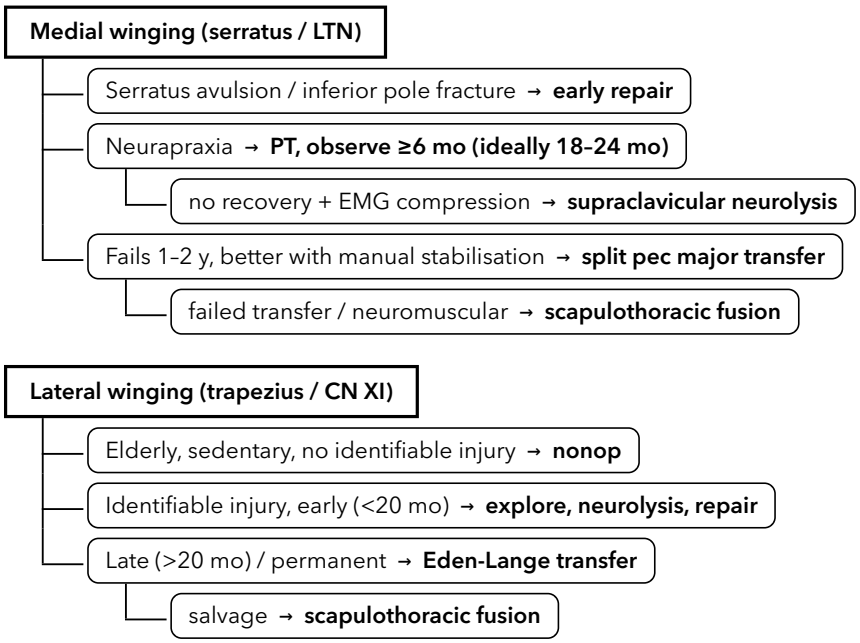
Symptoms	**similar to medial** + **impingement** (CA arch translates inferiorly as scapula depresses)
Deformity	**superomedial scapula drops + protrudes posterior and lateral**; **inferior angle rotates laterally** · **worse with abduction + resisted ER**
Exam	**drooping/depressed shoulder girdle** · **trapezius atrophy** (flattened superior trapezius) · weak flexion + abduction · **manual scapular stabilisation ↓ pain**

4 Diagnosis & Classification

Key radiology findings

EMG medial	LTN palsy vs **mechanical** (serratus avulsion); **serial every 3 mo** to follow recovery
EMG lateral	isolated CN XI vs wider neurological injury; **repeat every 3–6 mo**
EMG → which nerve	medial winging + abnormal EMG = **long thoracic (C5–7)**; trapezius denervation = **spinal accessory**

5 Management



Nonoperative – medial

Regimen	observation, PT (**serratus anterior strengthening**, stretching), avoid heavy lifting · modified thoracolumbar brace (**poor compliance, little benefit**)
Duration	**≥6 months**, ideally **18 mo–2 y**
Recovery	majority resolve **by 2 y** (Orthobullets; **within 24 mo**, Martin, OB) · **within 1 y** (OB) · **~25% need surgery** (OB)

Operative criteria

Procedure	Indication
Early serratus repair	**Mechanical avulsion** / inferior pole fracture + symptomatic winging
LTN neurolysis	Failed nonop **≥6 mo** + EMG **compression** (distal latency, denervation)
Split pec major transfer	Failed nonop **1–2 y** + **relief with manual scapular stabilisation**
Surgery (OB)	Symptomatic serratus winging **>1 y + EMG denervation** → fusion, fascial sling, muscle transfer
Eden-Lange	CN XI injury **diagnosed late (>20 mo)**
CN XI exploration/ repair	**Identifiable injury diagnosed early** (<20 mo)
Scapulothoracic fusion	**Diffuse neuromuscular disorders**, **failed muscle transfer**; not first choice; goal **pain relief**

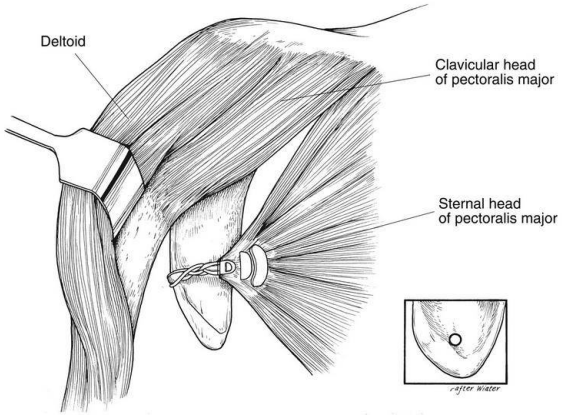
LTN neurolysis

Technique	**supraclavicular decompression** where nerve traverses scalenes
Outcome	**98%** pain relief + winging resolution after failed nonop · **better strength (flexion, abduction) than muscle transfer**

– – Active space – –

Split pectoralis major transfer

Technique	**sternal head** of pec major → **inferior (inferomedial) border/pole of scapula**; direct (± bone block) or with **hamstring/allograft** extension · **most effective** for medial winging · dynamic stabiliser
Predictor of success	**symptom relief with preop manual scapular stabilisation**
Outcome	persistent **abduction weakness** common
Complications	transfer failure at scapula · **breast asymmetry** · infection · adhesive capsulitis



Split pectoralis major transfer

Nerve transfer – medial

Donors	**lateral branch of thoracodorsal nerve → LTN** · **medial pectoral + sural graft → LTN**; preserves muscle biomechanics · nerve/muscle transfers **mixed results** (OB)

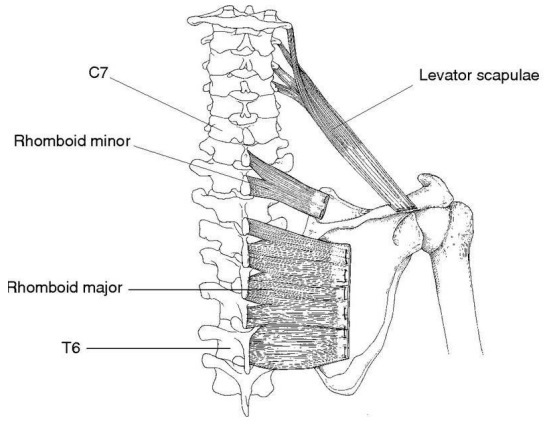
Lateral – nonoperative

Role	**controversial** (most injuries = iatrogenic direct → surgery) · nonop for **elderly, sedentary, no identifiable injury** · **less effective than for serratus palsy**
Poor prognosis with nonop	**can't raise arm above shoulder** at presentation · **dominant arm**
Delay	marked delay in recognition common

Eden-Lange transfer

Classic	**levator scapulae → lateral scapular spine** + **rhomboid minor + major → infraspinatus fossa** (lateralises: medial border → lateral)
Triple tendon transfer	all three to **scapular spine** (Rm medial → RM → LS lateral) to reconstruct trapezius
Results	**reliably restores ROM + function** · **better than scapulothoracic fusion** (OB)

(✗ Wrong options: **Distractors** – split pec major (= serratus/medial)) · long head triceps / infraspinatus transfers (rarely used) · latissimus transfer · levator only



Eden-Lange: levator, rhomboid minor and major transferred laterally

Scapulothoracic fusion

Technique	anterior scapula to posterior ribs (wire cables, plates/screws)
Outcome	**~20° gain in abduction** · **82% satisfied at 5 y**
Complications	nonunion · **pleural effusion** · adhesive capsulitis · symptomatic hardware

X vs Y

	Medial winging	**Lateral winging**
Scapula	Inferior pole **medial**, scapula elevated	**Drooping**, inferior angle **lateral**, depressed
Worse with	**Forward flexion / wall push-up**	**Abduction + resisted ER**
Atrophy	–	**Trapezius**
Nonop	**Effective** (most recover)	**Less effective**
Muscle transfer	**Split pec major (sternal head)**	**Eden-Lange (levator + rhomboids)**
Nerve transfer	Thoracodorsal lateral branch → LTN	

Nerve injury	**Result**
Long thoracic	**Medial winging**
Spinal accessory	**Lateral winging**
Thoracodorsal	Latissimus weakness, no winging
Suprascapular	Supra/infraspinatus wasting, no winging
Upper/lower subscapular	Subscapularis/teres major, no winging
Axillary	Deltoid/teres minor, no winging
C8–T1	No winging

How it's asked

"Long thoracic nerve injury causes?"	→ **medial scapular winging** ✗ Wrong options: lateral · teres minor/ infraspinatus atrophy
"Swimmer, medial winging, abnormal EMG – roots?"	→ **C5, C6, C7**

– – Active space – –

"MMA fighter, blows to torso, medial winging – deficient muscle?"	→ **serratus anterior (D)** ✗ Wrong options: trapezius · teres major · latissimus · infraspinatus
"Medial winging + abnormal EMG – nerve?"	→ **long thoracic** ✗ Wrong options: suprascapular · axillary · thoracodorsal
"Chest wall stabilisation for flail chest, inferior angle translates medially?"	→ **long thoracic nerve** ✗ Wrong options: spinal accessory · C8–T1 · subscapular
"Quarterback, medial winging, PT – additional finding?"	→ **inferomedial scapular support ↓ pain, ↑ ROM (scapular assistance test)**
"Most common condition causing winged scapula – nerve?"	→ **long thoracic**
"Clinical photo of medial winging – nerve/muscle?"	→ **long thoracic / serratus anterior** ✗ Wrong options: dorsal scapular/ trapezius
"Nerve to muscle not functioning in medial winging?"	→ **long thoracic** ✗ Wrong options: dorsal scapular · thoracodorsal · lower subscapular
"Swimmer, wall push-up winging, EMG – nerve on plexus diagram?"	→ **A = long thoracic** ✗ Wrong options: medial pectoral · thoracodorsal
"True of wall push-up winging?"	→ **compensatory overpull of rhomboids + levator scapulae** ✗ Wrong options: C8–T1 · Eden-Lange · node biopsy · retraction deficit
"Winging after neck exploration, trapezius denervated?"	→ **lateral winging, spinal accessory injury**
"72-y fall, lateral winging – nerve?"	→ **cranial nerve XI** ✗ Wrong options: XII · C3–5
"Cervical node biopsy, winging at rest + at 90° flexion?"	→ **spinal accessory**
"Posterior cervical surgery, lateral winging – best restores function?"	→ **rhomboids + levator transfer (Eden-Lange)** ✗ Wrong options: split pec · fusion
"Node biopsy 2 y ago, trapezius denervated – surgery?"	→ **lateral transfer of levator + rhomboid minor + major** ✗ Wrong options: pec transfer · levator only · latissimus
"75-y after ENT node biopsy, winging, untreated – nerve?"	→ **spinal accessory**
"Medial scapular muscles innervated by?"	→ **dorsal scapular nerve** (TO) ✗ Wrong options: subscapular · thoracodorsal · musculocutaneous

– – Active space – –

SCAPULOTHORACIC DYSKINESIS

– Shoulder & Elbow · 43 / 60 –

1 Basics

Definition **abnormal scapular motion** during coordinated scapulohumeral movement → impingement + dysfunction · rare

Who athletes, **baseball pitchers** · sequela of prior shoulder injury disrupting **scapular stabiliser activation**

Incidence (TO) in **61–100%** of athletes with shoulder injury (throwers: baseball, cricket, volleyball, javelin, tennis) · often a **symptom, not a diagnosis** → treat underlying cause (TO)

Normal scapula at rest (TO)	Value
Internal rotation	**30–45°**
Upward rotation	5–10° (20° in another TO lecture)
Anterior tilt	10–15°
Vertebral border	**2–3 cm** from spine
Scapular plane	30–45° anterior to frontal plane

During elevation (TO) upward rotation **~60°** (with 120° humeral) · **posterior tilt 20°** · ER 15–20° · rhythm **2:1**

Muscle (TO)	Scapular action
Serratus anterior	Upward rotation + **posterior tilt**
Lower trapezius	Depression + upward rotation
Upper trapezius	Elevation + upward rotation
Rhomboids, levator scapulae	Downward rotation + retraction
Pectoralis minor	**Anterior tilt** + IR

Scapular roles (TO) stable base · kinetic-chain link · **shock absorber in deceleration** · glenoid orientation · elevation via upward rotation + posterior tilt

Aetiology

Causes (multifactorial) neurologic injury · thoracic **kyphosis** · **periscapular muscle fatigue** · **poor throwing mechanics** · pain (shoulder, neck)

Mechanisms (TO) weak **serratus / lower trapezius** + tight **pec minor / levator** · GH instability · kyphosis · kinetic-chain disruption · posterior capsule tightness (anterior tilt) · nerve: **long thoracic → serratus (winging)**, **spinal accessory → trapezius**

Associated lesions (TO) internal impingement · SLAP · GIRD · subacromial impingement

– – Active space – –

2 Mechanism

Pathoanatomy scapulothoracic **power imbalance → scapular protraction** → altered GH mechanics → ↑ stress on **anterior capsule + posterosuperior labrum**

Injury risk labrum · rotator cuff (**articular-surface tears**) · capsule

Thrower's cascade **posteroinferior capsular tightness** + scapular dyskinesis → articular-side cuff tear + **posterosuperior labral tear**

SICK scapula

SICK muscular **fatigue syndrome** of throwers

Letter	Meaning
S	**Scapular malposition**
I	**Inferior medial border prominence**
C	**Coracoid pain + malposition** (tender coracoid = **pectoralis minor contracture**)
K	dys**K**inesis of scapular movement

GIRD **loss of internal rotation** (e.g. 15°) = posterior capsule tightness

3 Presentation

Symptoms shoulder pain/dysfunction **worse with arm elevation** · **loss of throwing velocity** · posterosuperior ± anterior pain

Exam **scapulothoracic crepitus** · scapula **lower + protracted** · **symptoms relieved by scapular stabilisation** (diagnostic)

Associated coracoid tenderness · IR deficit · positive relocation test



Wall push-up – protracted, winged scapula

4 Diagnosis & Classification

Diagnosis **clinical**

Kibler type (TO)	Pattern	Muscle	Pathology
I	**Inferomedial border** prominence ("wing")	Weak lower trap / serratus	**Labral tear**, impingement
II	**Medial border** prominence	Weak rhomboids	**GH instability**
III	**Superior translation, shrug**	Overactive upper trap, weak lower trap	**Rotator cuff**
IV	Normal symmetric	–	Normal

Test (TO)	Positive
Scapular assistance (SAT)	↓ pain / ↑ ROM with manual upward rotation + posterior tilt
Scapular retraction (SRT)	↑ supraspinatus strength with manual retraction · **most sensitive for dynamic dyskinesis** (TO)
Lateral scapular slide	**>1.5 cm** side difference
Wall push-up	Medial border prominence (serratus weakness)

Key radiology findings

X-ray (complete shoulder series) **usually unremarkable**

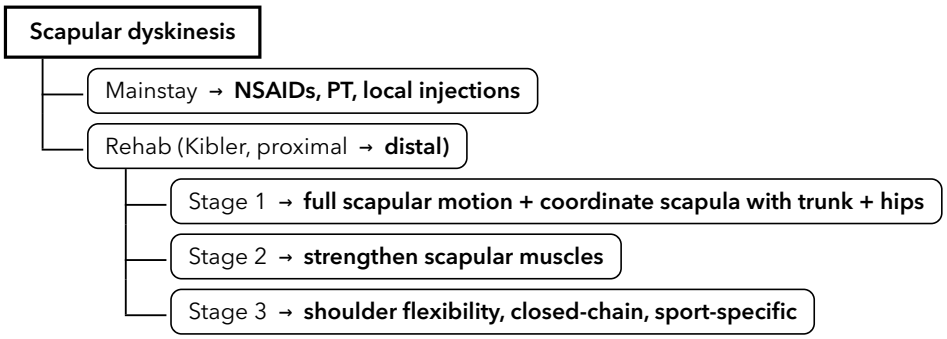
MRI often no intra-articular lesion



Scapular asymmetry with arm abduction

– – Active space – –

5 Management



Rehabilitation – kinetic chain, pectoralis minor & posterior capsule stretching

Principle — **proximal → distal** (kinetic chain); progress by **function, not timetable**; acute → recovery → maintenance phases

Initial stage — **coordination of scapular motion with trunk + hip movements**
✗ Wrong options: isometric · isokinetic · closed chain · axial loading (later)

PT emphasis — **core** strengthening · scapular stabilisers (**serratus anterior, trapezius**) · rotator cuff · proper mechanics in throwers

SICK / GIRD initial regimen — **stretch posterior capsule** (IR deficit) **+ pectoralis minor** (coracoid tenderness)

Rehab (TO)	Technique
Stretch **pec minor**	Doorway / corner
Stretch levator	Contralateral flexion + rotation
Stretch posterior capsule	Cross-body adduction
Strengthen **serratus**	**Push-up plus**, dynamic hugs
Strengthen **lower trapezius**	**Prone Y raises**, wall slides
Rhomboids	Prone retraction, rows
Kinetic chain	**Closed chain first** → open-chain overhead; "proximal stability for distal mobility"

Return to play (TO) — full pain-free ROM, no winging · strength **≥90%** of other side · no kinetic-chain deficit · **negative SAT/SRT**

✗ Regimen distractor	Why wrong
Posterior capsule + **subscapularis strengthening**	Misses pec minor (coracoid pain)
Posterior capsule + **plyometrics**	Misses pec minor (coracoid pain)
Anterior capsule stretching (any)	IR deficit = **posterior** capsule tightness

How it's asked

"Pitcher, posterior pain, scapular dyskinesis, normal MRI – emphasise first in rehab?" → **coordination of scapular motion with trunk + hip movements**
✗ Wrong options: isometric · isokinetic · closed chain · axial loading

"Pitcher, 15° IR loss, coracoid tenderness, + relocation, normal MRI — initial regimen?"	→ **stretch posterior capsule + pectoralis minor** ✗ Wrong options: subscapularis strengthening · plyometrics · anterior capsule stretching
"SICK scapula — coracoid pain due to?"	→ **pectoralis minor contracture**
"Diagnostic exam finding?"	→ **symptoms relieved by scapular stabilisation**
"Scapular position in dyskinesis?"	→ **lower + protracted**
"Kibler type III dyskinesis?"	→ **superior translation / shrug** (TO)
"Lateral scapular slide test abnormal if?"	→ **>1.5 cm side difference** (TO)

– – Active space – –

– – Active space – –

SHOULDER ARTHRODESIS

– Shoulder & Elbow · 44 / 60 –

1 Basics

Goal

Aim	**stable base** for the upper limb → optimise **hand + elbow** function
Ideal patient	**flail shoulder** (deltoid + cuff paralysis) with **functional hand, wrist + elbow**; young, esp. **labourer**

Anatomy

GH fusion surface	small → decorticate **GH articular surfaces + humeral head–acromion undersurface** to ↑ fusion area
Fixation strength	only **glenoid fossa + base of coracoid** strong enough for screws

3 Presentation

Symptoms	**specific to underlying condition**; symptomatic GH dysfunction

4 Diagnosis & Classification

Key radiology findings

Test	Purpose
X-ray AP, lateral, axillary	Bone stock for fusion, deformity
CT	**Glenoid bone loss** (esp. failed arthroplasty)
EMG	When **neurological status of scapular muscles** is ill-defined

5 Management

Indications – deltoid and rotator cuff paralysis

Indication	Note
Paralytic disorders	**Deltoid + cuff paralysis** (post-traumatic), **brachial plexus palsy**, paralysis in infancy
Irreparable deltoid + cuff deficiency with arthropathy	
Failed TSA / failed revision arthroplasty	Salvage
Tumour resection	Bone deficiency
Chronic infection	Painful ankylosis
Recurrent instability	Failed previous repairs (severe, refractory)

Rühmann (OB) 43 cases: paralysis **74%**, OA/humeral head necrosis 35%, infection 16%, instability

Contraindications

List **ipsilateral elbow arthrodesis** · **contralateral shoulder arthrodesis** · **no functional scapulothoracic motion** · **trapezius, levator scapulae or serratus anterior paralysis** · **Charcot** in acute inflammatory stage (**Eichenholtz 0–2**) · **elderly** · **progressive neurological disease**

Operative criteria – who benefits vs alternatives

Patient	Best option
Flail shoulder (deltoid + cuff paralysis), hand works	**Arthrodesis** ✓
No cuff function	No **TSA** (contraindicated)
No deltoid function	No **reverse TSA** (contraindicated)
Chronic lateral winging	Eden-Lange (levator + rhomboids)
Irreparable cuff tear, overhead motion	Latissimus dorsi transfer
Young AVN	Hemiarthroplasty
Recurrent MDI after arthroscopy	Therapy, revision arthroscopic/open stabilisation
Infected TSA removed (70-y)	Treat infection → **2nd-stage TSA / HA / reverse**
Cleidocranial dysostosis	**Scapulothoracic** (✗ Wrong options: glenohumeral) arthrodesis

Fusion position

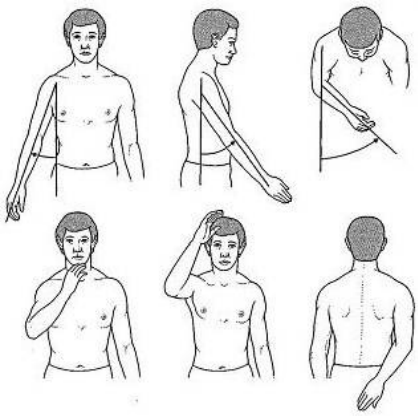
Goal **hand to mouth** (feeding) + **groin** (hygiene)

Source	Abduction	Forward flexion	Internal rotation
Orthobullets "30-30-30"	**20–30°**	**20–30°**	**20–30°**
OB typical	**~40°**	**~35°**	**~20°**
Rühmann range (OB)	20–60°	20–40° (anteversion)	0–50°; **no ER**
Orthobullets GH anatomy note	15–20°	20–25°	40–50°

TO (arthrodesis as final bail-out after failed RSA/TSA) **20° abduction · 30° forward flexion · 40° IR** ("20-30-40") → hand to mouth – read the options

Positions conflict 30-30-30 (Orthobullets) · 40-35-20 (OB) – read the options

– – Active space – –



Arthrodesis position: reach mouth, head, back

Technique

Approach	**S-shaped incision**: scapular spine → over acromion → anterolateral arm
Steps	resect cuff from proximal humerus · **tenodese biceps** · decorticate glenoid, humeral head + acromial undersurface · position 30-30-30
Fixation	**10-hole 4.5 mm pelvic reconstruction plate** along **scapular spine → acromion → humeral shaft** · **compression screws across GH into glenoid fossa** · screw into **base of coracoid**

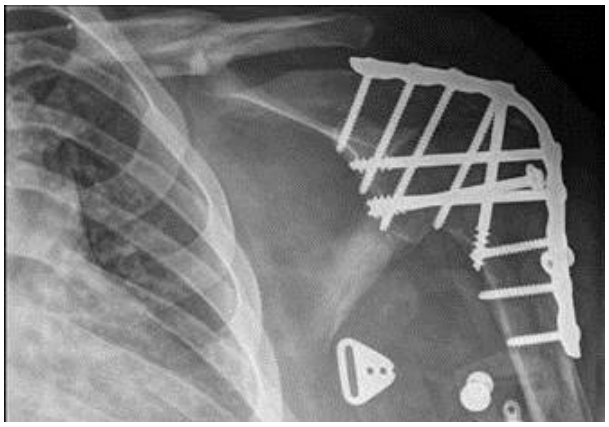


Plate fixation of shoulder arthrodesis

Postoperative care

0–6 wk	**thermoplastic orthosis** from day 1
6 wk	**sling** if no radiographic loosening
3 mo	**mobilisation + thoracoscapular strengthening** if no loosening
Recovery	**6–12 months**; marked ↑ pain + function in selected patients

6 Complications

List	**infection** · **nonunion** · **malposition** · **prominent hardware** (pain) · **humeral shaft / periarticular fracture**

How it's asked

– – Active space – –

"45-y, deltoid + cuff paralysis 5 y after motorcycle injury, hand/wrist work?"	→ **shoulder arthrodesis** ✗ Wrong options: Eden-Lange · TSA · reverse TSA · latissimus transfer
"Who benefits most from GH arthrodesis?"	→ **30-y labourer with deltoid + cuff paralysis** ✗ Wrong options: AVN · MDI · infected TSA removed · cleidocranial dysostosis
"Position of shoulder fusion?"	→ **30° abduction, 30° flexion, 30° IR** – hand to mouth (OB: ~40/35/20)

– – Active space – –

SHOULDER ARTHROSCOPY: INDICATIONS & APPROACH

– Shoulder & Elbow · 45 / 60 –

1 Basics

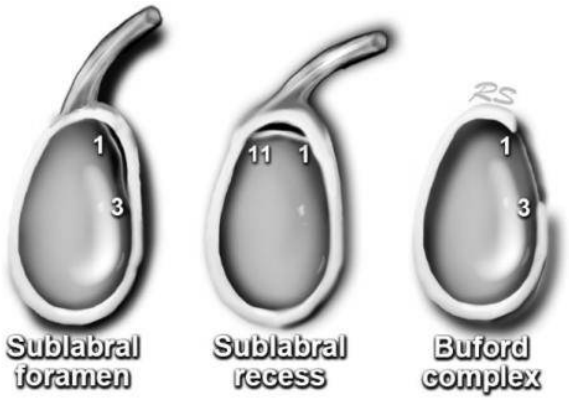
Indications

Indications	diagnostic · loose bodies · **cuff repair/debridement** · labral/SLAP/ instability repair · **subacromial decompression** · AC joint / **distal clavicle resection** · **suprascapular nerve release** · capsular release (contracture) · synovectomy · **biceps tenotomy/tenodesis**

Diagnostic scope – normal anatomy & variants

Diagnostic scope	**30° scope via posterior portal**: biceps, SS/IS/TM, **rotator interval**, MGHL/ IGHL, **subscapularis recess (loose bodies)**, anterior labrum, glenoid, humeral head
Rotator interval	**biceps tendon + superior edge of subscapularis + glenoid**
Anterosuperior labrum + MGHL	**widest anatomic variability**

Variant	Labrum	MGHL
Attached labrum	Attached	**Broad – most common**
Sublabral hole (foramen)	Sublabral hole	**Cord-like**
Buford complex	**Absent** anterosuperior labrum	**Cord-like**



Sublabral foramen, sublabral recess, Buford complex

Normal bare areas	**central glenoid** + **posterior humeral head** ✗ Wrong options: pathology

5 Management

Positioning

	Beach chair	Lateral decubitus
Orientation	Upright, **anatomic**	Glenoid parallel to floor (non-anatomic)
Traction	**None → ↓ neurapraxia**	**10–13 lb** (avoid over-distraction) → **traction neurapraxia** (NV + soft tissue)
Arm	Mobile; arm holder possible	**30–60° abduction + 20–30° forward flexion**
Conversion to open	**Easy** (deltopectoral; no re-prep)	Must reposition/redrape
Bleeding	**↓ venous pressure + bleeding**	–
Cerebral perfusion	**↓ (hypoperfusion, CDEs, stroke, death)**	**Better**
Hypotension / bradycardia	**↑ risk**	No ↑
Posterior portals	**Difficult** (head in the way)	Easy
Labrum	Less capsular distortion	**Traction accentuates labral tears**
Anaesthesia	**Regional + sedation possible**	Not ideal if regional not tolerated
5 o'clock portal	**Safer** (Gelber)	**Axillary + MCN closer**

Lateral position (TO) **vacuum beanbag + kidney rest**; **10–13 lb** traction; **30–60° abduction, 20–30° forward flexion**

Lateral modification (TO) tilt patient **20–30° posteriorly** → glenoid parallel to floor → **less traction** (↓ plexus neurapraxia), **labral tears accentuated** (pulled away from bed), **better access to inferior third** of labrum/capsule

Beach chair advantages (TO) faster positioning · **↓ neurapraxia (no traction)** · less capsular distortion · arm mobility · easy open conversion

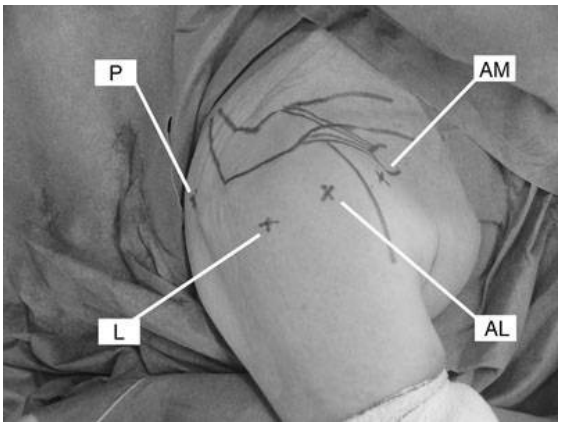
Beach chair disadvantages (TO) difficult posterior portals · **↓ cerebral perfusion with hypotensive anaesthesia → stroke, death**

✗ Wrong options: **EXCEPT (TO)** – "beach chair → **more** brachial neurapraxia" is **false**

Primary portals

Portal	Location	Function	Passes / risk
Posterior	**2 cm inferior, 1 cm medial** to posterolateral acromion; aim at **coracoid tip**	**Primary viewing**; **first portal**	Between **infraspinatus (SSN)** + **teres minor (axillary)** or through IS
Anterior	**Lateral to coracoid**, anterior to AC joint; spinal needle under vision	Viewing + SAD	Between **pec major (pectoral nn.)** + **deltoid (axillary)**
Lateral	**1–2 cm distal** to lateral acromion	**SAD**	Through **deltoid (axillary)**

– – Active space – –



Portal sites: posterior (P), lateral (L), anterolateral (AL), anteromedial (AM)

Secondary portals

Portal	Location	Function	Risk
Anteroinferior (5 o'clock)	Slightly inferior to coracoid	**Anterior labral repair anchors**	**Axillary (> 7 o'clock risk, Meyer)** + MCN
Posteroinferior (7 o'clock)	Under vision, spinal needle	**Posterior labral repair anchors**	**Axillary nerve** (low posterior)
Neviaser (supraspinatus)	**Just medial to lateral acromion**	Anterior glenoid view, **SLAP repair**	Through **supraspinatus (SSN)**; SSN if into **suprascapular notch**
Port of Wilmington (posterolateral)	**Just anterior to posterolateral acromion**	**Posterior SLAP + cuff lesions**	–

Trans-subscapularis portal — **avoid** in either position (Gelber)

Placement — secondary portals under direct vision from posterior portal with **spinal needle**

6 Complications

Portal-related nerve injury

Structure	Anatomy	At risk when
Axillary nerve	Quadrangular space → deep surface deltoid, **~7 cm below acromion**	Posterior portal **too inferior** (**7 o'clock**) → pain, weakness, ↓ **lateral arm/shoulder** sensation
Suprascapular nerve	Supraspinatus → infraspinatus fossa	Posterior portal **too medial**; Neviaser into notch
Cephalic vein	Deltopectoral groove	Anterior portal **too lateral** – **structure most at risk** (Lo, Meyer)
Musculocutaneous nerve	Enters muscle **2–8 cm distal to coracoid tip**	Anterior portal **too inferior**
Phrenic nerve	–	**Interscalene block**

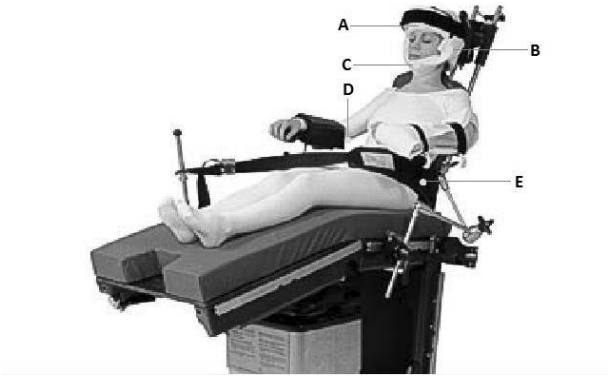
Lo (outside-in) — all NV structures **> 20 mm** from all portals

No neural structure at risk lateral superior portal · posterior superior portal

– – Active space – –

Positioning complications (beach chair, lateral decubitus, obesity)

Pressure point (beach chair)	Nerve	Deficit
A – forehead (tight face mask)	**Supraorbital** (terminal **CN V**)	Paraesthesia **forehead + anterior scalp**
B – mastoid process (mask straps)	**Great auricular (C2–C3)**	Paraesthesia **ear, posterior auricular area, angle of mandible**
C – chin strap	–	**Skin breakdown lip/chin**
D – contralateral elbow	**Ulnar**	Ulnar neurapraxia
E – lateral abdominal belt / ASIS	**LFCN**	Burning/numbness **anterolateral thigh**, above knee, **no weakness**



Beach chair pressure points A–E

Neck position	Complication
Hyperextension (± rotation)	**Stroke** + **CN XII (hypoglossal)** palsy — traction + compression by angle of mandible (✗ Wrong options: face-mask compression)
Hyperflexion	**Spinal cord ischaemia → paraplegia**

Not reported with beach chair CN III (ptosis, mydriasis, diplopia) · **AIN** (FPL + index/long FDP) · **femoral** (quadriceps, anterior thigh, medial leg)

Cerebral hypoperfusion

CN VI palsy post-op **lateral rectus** paralysis (can't abduct eye) → cerebral hypoperfusion in beach chair

↓ Risk **lateral decubitus** · **regional anaesthesia + sedation** (fewer CDEs) · **SBP > 90 mmHg**

Increases or no effect (not protective) **general anaesthesia (↑ CDE)** · **SBP < 70 mmHg** (↑) · open vs arthroscopic · metal vs bioabsorbable anchors

Stroke also reported after lateral decubitus (risk small in both)

LFCN palsy (obesity)

Path LFCN crosses near **ASIS** → compressed by belt / waist flexion / pannus

– – Active space – –

Risk factors	**obesity (BMI > 30; highest-risk option BMI > 35)** · **prolonged operative time** · **inadequate ASIS padding**
Not risk factors	age > 50 · type of surgery (open/arthroscopic/arthroplasty) · **diabetes** · foam between pannus + thighs (= **protective**)
Prevention	hips **centred at bed break**, **hip flexion ~45°** (less waist flexion), **padding under belt** + between pannus and thighs
Holtzman (397 pts)	LFCN palsy **5.5%**, mostly **transient**; weight + BMI significant
Levy protocol (BMI > 30)	LFCN **5 → 1** case (OB) · **none, even BMI > 40** (another OB)
Obese patients	↑ nerve palsies; post-op ↑ **DVT, infection, wound breakdown, hardware failure, mortality**

X vs Y

	5 o'clock (anteroinferior)	7 o'clock (posteroinferior)
Use	**Anterior** labral anchors	**Posterior** labral anchors
Nerve	**Axillary (more risk, Meyer)** + MCN	**Axillary**

	Sublabral foramen	Buford complex
Labrum	**Sublabral hole**	**Absent** anterosuperiorly
MGHL	Cord-like	**Cord-like**

How it's asked

"Lateral arm numbness + weakness after arthroscopy – portal?"	→ **posterior inferior (7 o'clock)** ✗ Wrong options: Neviaser = SSN · anterior = MCN · lateral/posterior superior = none
"All true about positioning EXCEPT?"	→ **beach chair has more brachial neurapraxia** (TO)
"Lateral decubitus traction?"	→ **10–13 lb**, 30–60° abduction, 20–30° flexion (TO)
"Beach chair pad at mastoid – deficit?"	→ **great auricular: ear, posterior auricular, angle of mandible** ✗ Wrong options: CN III · CN XII · AIN · femoral
"PACU, can't gaze left after beach chair – prevention?"	→ **lateral decubitus position** ✗ Wrong options: GA · SBP < 70 · open · metal anchors
"Obese, burning outer thigh above knee, no weakness – pad where?"	→ **lateral abdominal belt / ASIS (LFCN)**
"Anterolateral thigh numbness after beach chair – biggest risk factor?"	→ **BMI > 35** ✗ Wrong options: age · diabetes · surgery type
"Hypoglossal palsy after beach chair – cause?"	→ **neck hyperextension**

"Absent anterosuperior labrum + cord-like MGHL?"	→ **Buford complex**
"Cephalic vein at risk — portal error?"	→ **anterior portal too lateral**

– – Active space – –

– – Active space – –

SHOULDER EXAM

– Shoulder & Elbow · 46 / 60 –

1 Basics

Exam components

Sequence inspection · palpation · ROM · neurovascular · provocative tests · always **compare both sides**

Inspection skin, scars, symmetry, swelling, atrophy, hypertrophy, **scapular winging**

Palpation C-spine spinous processes, SC joint, clavicle, **AC joint**, acromion, coracoid, scapular spine · paraspinals, periscapular, deltoid, **greater tuberosity (cuff insertion)**, **LHB in groove**

Range of motion

Plane	Normal
Forward elevation	**180°**
Abduction (scapula stabilised)	**90°**
ER at side	**80°**
ER at 90° abduction	Compare sides
IR to vertebral level	**T4–T8**
IR at 90° abduction	Compare sides

ROM rule active vs passive, both sides; normal relative to contralateral

Neurovascular

Root	Motion	Muscle	Reflex
C4	Scapular stabilisation (winging)	Upper serratus anterior	–
C5	Shoulder abduction · elbow flexion palm up	Deltoid · biceps	**Biceps**
C6	Elbow flexion thumb up · wrist extension	Brachioradialis · ECRL	**Brachioradialis**
C7	Elbow extension · wrist flexion	Triceps · FCR	**Triceps**
C8	**Finger flexion, grip, thumb extension**	**FDS**	–
T1	Finger abduction	Interossei (ulnar)	–

Sensation C4–T1 dermatomes · **pulses** brachial, radial, ulnar

Anterior band of IGHL – restraint in abduction/external rotation

– – Active space – –

Ligament	Restrains
Anterior band IGHL	**Anterior translation at 90° abduction + full ER** (moves anteriorly)
MGHL	Anterior translation at **45° abduction**
SGHL + CHL	**Inferior translation of adducted arm** + ER in adduction
Posterior band IGHL	**Posterior translation in IR**

4 Diagnosis & Classification

Impingement & biceps tests (Hawkins, Speed)

Impingement syndrome (TO) anterior pain in **60–120° abduction** (painful arc) · greater tuberosity slides under acromion, bursa smooths it · causes: **subacromial bursitis**, **hooked acromion**, cuff pathology (supraspinatus tendinitis, calcific tendinitis, partial tear), **GT fracture** (TO)

Test	Technique	Positive / meaning
Neer sign	Stabilise scapula, **passive forced forward elevation**	Anterior pain (GT under acromion **70–110°**) · needs **full ROM** · sens **75–86%**, spec **50%** · also + in stiffness, OA, instability, bone lesions
Neer test	Repeat Neer after **subacromial lidocaine**	**Marked pain relief** = subacromial impingement
Hawkins-Kennedy	**Flex 90°, elbow 90°, forcible IR** (GT under CA ligament)	Pain · **sensitive, not specific** · sens **75–92%**, spec **45%** · = **subacromial bursitis** (TO)
Scapular assistance (TO)	Push **inferomedial scapular angle** during elevation (upward rotation + posterior tilt)	**Relief** of impingement pain + ↑ arc
Internal impingement	90° abduction + max ER + extension (late cocking)	**Posterior** throwing pain · relieved by relocation / elbow in front of body



Hawkins-Kennedy test

– – Active space – –

Neer test injectate — **5 mL 1% lidocaine** (TO) · **4 cc 1% lidocaine + 4 cc 0.5% bupivacaine + 2 cc steroid** (Orthobullets) – read the options

Biceps test	Technique	Positive
Speed	Resisted forward elevation, **elbow extended, forearm supinated**	Pain in **bicipital groove**
Yergason	Elbow 90°, pronated → **resisted supination**	Groove pain = biceps tendon pathology
Popeye	Bulge in arm	**LHB proximal rupture**
Groove tenderness	Palpation	Inflamed LHB or **SLAP**

Speed + / empty can – — **bicipital tendinitis** (TO)
✗ Wrong options: supraspinatus tendinitis · SLAP · impingement

Rotator cuff strength & lag tests (lift-off, hornblower, ER lag)

Muscle	Tests
Subscapularis	**Lift-off**, **belly-press / Napoleon**, **bear hug**, **IR lag**, ↑ passive ER
Supraspinatus	**Jobe empty can**, **drop arm (Codman)**
Infraspinatus	**ER lag sign** (specific), drop sign (Hertel), ER resistance stress
Teres minor	**Hornblower (Patte)**, ER strength at 90° abduction

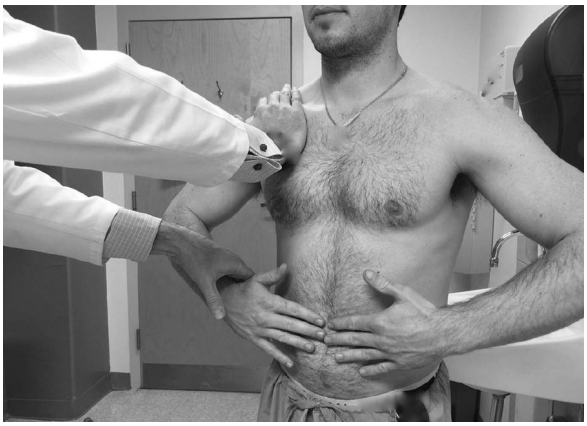
Lift-off (Gerber) (TO) — dorsum of hand on lumbar spine (IR) → can't lift off = **subscapularis tear** · specific for **IR deficit** (TO) · best for **inferior subscapularis** · more accurate if hand reaches opposite scapula · confounded by other muscles
✗ Wrong options: abduction · ER deficit · anterior instability



Lift-off test

Belly-press (Gerber) / Napoleon (TO) — palm presses abdomen, elbow in front of trunk · weak → **elbow drops behind trunk** (shoulder extension) = belly-press · **wrist flexes** = **Napoleon sign** · best for **superior subscapularis**

– – Active space – –



Belly-press test

Bear hug (Burkhart & De Beer) (TO)	palm on **opposite shoulder**, elbow in front, examiner pulls hand away **anteriorly** · can't resist = positive · **most sensitive**: detects tears **<50%** (upper third) missed by lift-off/belly-press (TO) · positive if **≥20% weaker** than other side (Orthobullets)
Most sensitive + specific subscap test	**IR lag sign** (Orthobullets, OB) · **bear hug** most sensitive (TO) – read the options
IR lag sign	behind patient, elbow 90°, shoulder 20° elevation + 20° extension, lift hand off lumbar spine to near-max IR, release wrist → lag
Not subscap testing	**isolated IR at side** (pec major + latissimus contribute)
Subscapularis insertion	**lesser tuberosity → positive bear hug after trauma = lesser tuberosity avulsion** ✗ Wrong options: GT fracture = drop arm · superior migration = CTA · dislocation can't be positioned · AC separation
Jobe empty can (TO)	**90° abduction in scapular plane (30° forward), IR thumb down**, resist downward push · **weakness = supraspinatus tear** (TO) · **pain only** = subacromial bursitis/irritation, tear only with true drop arm (Orthobullets)
Drop arm (Codman) (TO)	arm elevated overhead → lower slowly · sudden pain or uncontrolled lowering = **supraspinatus tear** · passive 90° scapular plane → drop = **most specific for full-thickness tear (98%)** (Orthobullets)
ER resistance stress (TO)	arms at side, elbows 90°, **ER 60°**, resist examiner's IR push → pain/ weakness = **infraspinatus + teres minor**
ER lag sign (TO)	seated, elbow 90°, shoulder **20° elevation in scapular plane**, full ER → hold · angular drop = **infraspinatus tear** (specific) ✗ Wrong options: subscapularis · teres minor · adhesive capsulitis
Drop sign (TO)	same as ERLS but **90° elevation** in scapular plane · infraspinatus · ERLS slightly more sensitive + specific · not the Orthobullets "drop sign" (= drop arm, supraspinatus)
Hornblower (Patte) (TO)	arm **90° abduction in scapular plane**, elbow 90° → attempt ER against resistance · can't = hornblower posture · **most accurate for teres minor** · TO option: "test for **infraspinatus and teres minor**" · Orthobullets: hold 90° abd + 90° ER → **arm falls into IR** · also + in **posterior supraspinatus** tears

– – Active space – –



Positive hornblower's sign

Teres minor **axillary nerve**, ER

✗ Distractor	Tests
Jobe	Supraspinatus weakness ± impingement
Belly-press, IR lag	Subscapularis
Hawkins	Impingement
"Jones test"	not supraspinatus test

Lesser tuberosity fracture – subscapularis testing

Bear hug after ATV accident can't hold palm on opposite shoulder vs anterior pull → **lesser tuberosity fracture** on X-ray

Labral tests (Porcelli)

Test	Technique	Positive
O'Brien (active compression)	**90° flexion, elbow extended, 10–15° adduction, pronated (thumb down / full IR)** → resist downward force · repeat supinated/ER	**Deep GH pain pronated, not supinated = SLAP** · **superficial AC pain = AC joint**
Crank	Abducted, passive rotation + axial load	Click/pain = labrum
Modified dynamic labral shear (TO)	Standing, **abduct 120°** scapular plane, elbow 90°, ER → lower arm keeping ER	**Painful click posterior joint line 120°→90°** = SLAP
Uppercut (TO)	Neutral rotation, elbow 90°, supinated fist → punch to chin vs resistance	Pain = **SLAP**
Porcelli (TO)	See below	**Posterior labral tear**

Porcelli test (TO) Morey 2016 · **posterior labral tear** · **sens 100%, spec 99%**, inter-examiner reliability **>0.8** · outperformed jerk, Kim, load-and-shift · **not yet validated**
✗ Wrong options: SLAP · subscapularis · PASTA

Porcelli technique (TO) part 1: **90° flexion, 10–15° adduction, max IR (thumb down)**, examiner behind stabilises scapula, resisted elevation → note pain/strength · part 2: same, examiner's **thumb on posterior joint line** pushing anteriorly (prevents posterior subluxation) · **pain ↓/gone in part 2 = positive** · pain persists = **AC joint** pathology

Instability tests

– – Active space – –

Direction	Test	Technique / positive
Anterior	**Apprehension** (Jobe, TO)	Supine, **90° abduction + full ER** (behind coronal plane, elbow 90°) → apprehension
Anterior	**Relocation**	Posterior force on head → apprehension relieved
Anterior	**Anterior release**	Release hand in abd/ER → subluxation · **"3 in 1"** (apprehension + relocation also +)
Anterior	Anterior load & shift / drawer	Supine, **40–60° abd** (TO drawer: 60°, neutral rotation, axial load) · grade = laxity, pathologic only if symptomatic
Posterior	**Jerk (Jahnke)**	Sitting/standing, **flex 90°, IR, adducted**, axial load at elbow → posterior jerk; abduct → reduces (TO) · clunk or pain
Posterior	**Kim**	Seated, 90° abduction → flex to 45° + axial load + posteroinferior force · **pain**
Posterior	**Posterior stress**	Flexion, adduction, IR + posterior force → pain + instability
Posterior	Posterior load & shift / drawer	Increased translation vs other side
Posterior	**Loss of ER**	Locked in IR = posterior subluxation/dislocation
Inferior / MDI	**Sulcus**	Pull arm down (TO: at **0° and 45° abduction**) · persists at **2+/3+ with ER at side = pathologic rotator interval**

Translation grade (drawer / load & shift)	Finding
1+	**To glenoid rim**
2+	**Over rim, reduces spontaneously**
3+	**Dislocates / locks out**, no spontaneous reduction

Sulcus grade	Acromiohumeral interval
1+	**<1 cm**
2+	**1–2 cm**
3+	**>2 cm**

AC joint & other tests

Test	Meaning
Cross-body adduction	AC joint pain
AC tenderness	AC pathology
Wright's	**TOS**: abduct + ER, head turned away → **radial pulse lost**
Medial winging (wall push at 90° flexion)	**Serratus anterior / long thoracic nerve**
Lateral winging	**Trapezius / spinal accessory (CN XI)**
Axillary webbing defect	**Pectoralis major rupture**

– – Active space – –

X vs Y

	Lift-off	Belly-press	Bear hug
Subscap portion	**Inferior**	**Superior**	Upper third, partial <50%
Position	Hand on lumbar spine	Palm on abdomen	Palm on opposite shoulder
Positive	Can't lift off	Elbow drops back / wrist flexes (Napoleon)	Can't resist anterior pull

	ER lag sign	Drop sign (TO)	Hornblower
Elevation	**20°** scapular plane	**90°** scapular plane	**90°** abduction
Muscle	**Infraspinatus**	Infraspinatus	**Teres minor**

	Neer	Hawkins
Manoeuvre	Passive forced elevation, scapula fixed	90° flexion + **forced IR**
Sens / spec	75–86% / 50%	75–92% / 45%

	O'Brien – SLAP	O'Brien – AC
Pain site	**Deep** in GH joint	**Superficial** over AC
Forearm	Pronated, not supinated	Pronated, not supinated

How it's asked

"Full-thickness teres minor tear – positive test?"	→ **Hornblower's sign** ✗ Wrong options: Jobe · belly-press · IR lag · Hawkins
"90° abduction + full ER – ligament resisting anterior translation?"	→ **anterior band IGHL** ✗ Wrong options: CHL · SGHL · MGHL · posterior band
"ATV crash, positive bear hug – X-ray?"	→ **lesser tuberosity fracture**
"Correct statement for test in figure (arm in hornblower posture)?"	→ **Hornblower's test for infraspinatus and teres minor** (TO) ✗ Wrong options: Jones · O'Brien · IR lag
"Muscle tested in figure (hand behind back)?"	→ **subscapularis** (lift-off) (TO)
"Lift-off test is specific for?"	→ **internal rotation deficit** (TO)
"ER lag sign suggests?"	→ **infraspinatus tear** (TO)
"Anterior pain on IR, positive lift-off – muscle?"	→ **subscapularis** (TO)
"Diagram: 90° flexion + forced IR?"	→ **Hawkins-Kennedy test** (TO) ✗ Wrong options: Neer · empty can · ER lag

"Positive Hawkins sign indicates?"	→ **subacromial bursitis** (TO) ✗ Wrong options: cuff tear · subscap tear · brachial plexus
"Positive Speed + negative empty can?"	→ **bicipital tendinitis** (TO)
"Porcelli test detects?"	→ **posterior labral tear** (TO)

– – Active space – –

– – Active space – –

SHOULDER HEMIARTHROPLASTY

– Shoulder & Elbow · 47 / 60 –

1 Basics

Definition

Hemiarthroplasty (HA)	humeral articular surface replaced with **stemmed humeral component**; glenoid left native
Most common indication	GH arthritis with **inadequate glenoid bone stock** for TSA
Mechanics	relies on **rotator cuff compression** for motion + strength; **CA arch** = barrier to proximal migration
Use	less common since RSA; still for young patients/laborers, CTA, osteonecrosis, complex proximal humerus fractures

4 Diagnosis & Classification

Key radiology findings

Modality	Use
True (Grashey) AP	**30–40° oblique** to coronal plane; extent of DJD, fracture pattern
Axillary	**Posterior glenoid wear**; fracture displacement
CT	**Glenoid version + glenoid bone stock**; unclear fracture pattern
MRI	**Rotator cuff** status

↓ Acromiohumeral distance superior head migration → **cuff deficiency** → evaluate cuff (MRI / CT arthrogram)



Axillary view – posterior glenoid wear

5 Management

Indications & limitations (cuff integrity, osteonecrosis, CTA head, ream-and-run)

Indication	Detail
Primary arthritis	**Cuff deficient** · glenoid stock inadequate · **high glenoid loosening risk (young, active laborers)**
Rotator cuff arthropathy	**HA > rTSA if forward flexion > 90°**
Osteonecrosis	**Without glenoid involvement**
Proximal humerus fracture	3-part with poor bone · **4-part** · **head-split** · major articular destruction

Contraindication	Note
Infection	–
Neuropathic (Charcot) joint	e.g. syrinx
Unmotivated patient	–
CA ligament deficiency	CA arch + cuff deficient → **superior (anterosuperior) escape**
RA · capsulorrhaphy arthropathy	Listed as contraindications (one OB)

RA / capsulorrhaphy arthropathy contraindication (OB) · RA results "satisfactory in most, ↓ ROM" (TO) · ream-and-run works in capsulorrhaphy arthropathy (OB) – read the options

Outcomes by diagnosis

Best results (TO) **humeral head AVN / osteonecrosis**: consistent pain relief in **90%**, near-normal ROM

Less good (TO) **RA, OA, glenoid dysplasia, post-traumatic arthrosis**: satisfactory in most, **↓ ROM**

Most influential factor on function **status of the rotator cuff**
✗ Wrong options: CA ligament · glenoid wear · inferior osteophyte · AC arthritis

Irreparable cuff **TSA contraindicated** → HA (limited function) or **RSA**

Fractures **excellent pain relief** in majority; outcome **inversely proportional to age + time from injury to surgery**

– – Active space – –

HA vs TSA for GH OA

Statement	Truth
HA outcomes with eccentric glenoid wear	**Inferior to TSA** ✓
HA function overall	Inferior to TSA
HA revision rate	**Equal or greater** than TSA (mid–long term); higher with eccentric wear
HA outcomes eccentric vs concentric	**Worse with eccentric** wear
Glenoid	**Progressive erosion** may occur

Extended (CTA) head

Theoretic advantage **metal-to-bone articulation with the acromion** → greater arc of smooth metal contact with glenoid + acromion

Evidence unproven; results **comparable to standard HA**; **no other biomechanical advantage**
✗ Wrong options: superior stability · fixed fulcrum · ↑ deltoid moment arm · ↑ offset

Ream-and-run

Definition HA + **concentric glenoid reaming** (**non-prosthetic glenoid arthroplasty**) with **nubbed spherical reamer** → single smooth concavity

Advantage avoids **prosthetic glenoid failure** → **activity beyond that recommended for TSA**

Ideal patient **young, active, motivated** (e.g. 42-yo weightlifter)

Works in OA, **capsulorrhaphy arthropathy**, **post-traumatic arthritis**
✗ Wrong options: contraindicated

Rehab **therapy is critical**
✗ Wrong options: "TSA needs more therapy"

Outcomes excellent 2-year functional + radiographic results

Technique

Approach **deltopectoral**

Head cut start at **medial insertion line of supraspinatus**

Humeral stem version

Version	Result
30° retroversion ✓ (broach + stem, TO)	**Ideal**; lateral fin **slightly posterior to bicipital groove**
Excess anteversion	**Anterior dislocation**
Excess retroversion (e.g. 45°, TO)	**Posterior dislocation** (✗ Wrong options: anterior instability)
✗ Wrong options: neutral · 10° anteversion · neutral–10° retroversion	Wrong (TO)

45° retroversion (TO) FNB 2024 item, answer not provided by source; options: anterior instability · limit IR · best function · shaft fracture

Height, head size & fixation

Greater tuberosity height **7–8 mm below top of head** (Orthobullets) · **6–8 mm** (OB) — read the options

Why maintain **cuff + biceps tension**; recreate **medial calcar** contour

Height landmark top of head → **upper border of pectoralis major = 56 mm**; cement prosthesis proud

Head size contralateral X-ray or **measure resected head**; too large → **overstuffing**

Fixation **cemented = standard of care** (better QoL, ROM, strength vs uncemented)

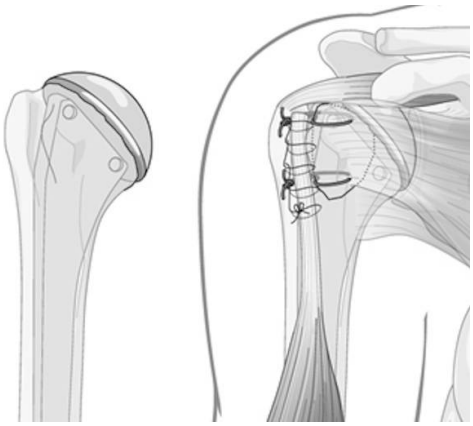
Tuberosity malposition after fracture hemiarthroplasty

Importance **tuberosity migration = commonest cause of failure** in fracture HA; non-anatomic placement → variable outcomes, **rotation deficit**

Technique secure tuberosities **to each other + to shaft**; **autograft from resected head** → ↓ pull-off, ↑ healing

Malposition	Consequence
GT too proximal (above head)	**GT impingement on acromion** in abduction
GT too distal	**Over-tensioned cuff**, stiffness, ↓ motion
LT too lateral (close to GT)	**↑ Subscapularis tension → ER deficit**

✗ GT position traps does **not** change nonunion risk · does **not** cause overstuffing



Tuberosity fixation around hemiarthroplasty

Rehab

Fracture HA **early passive motion 6–8 wk** until healed → **strengthening once tuberosity healed**

– – Active space – –

6 Complications

Complication	Detail	Consequence / treatment
Progressive glenoid arthrosis	Young, active	**Conversion to TSA**
Tuberosity displacement / malunion	Commonest in fracture HA	Reposition + bone graft
Overstuffing	Head too large	→ **stiffness**, **accelerated glenoid arthritis**
Subcutaneous (anterosuperior) escape	**Cuff + CA arch deficient**	**RSA > anatomic TSA**



Anterosuperior escape after hemiarthroplasty

Failed hemiarthroplasty – glenoid erosion & conversion to TSA/RSA

Painful hemiarthroplasty

- Infection excluded? (WBC 1900, 20% PMN, cultures −ve 3 wk) → **aseptic**
- Intact cuff + glenoid wear → **keep stem, revise head + add glenoid (TSA)**
- Pseudoparalysis + superior escape → **convert to RSA**

Glenoid arthritis, intact cuff, well-fixed stem	**modular**: retain stem, **revise head + prosthetic glenoid** ✗ Wrong options: MRI · two-stage spacer · explant stem · RSA
Cuff failure + superior escape (pseudoparalysis)	**conversion to RSA** ✗ Wrong options: MRI (cuff failure implied) · cuff repair + TSA · fusion if glenoid stock good · revision HA · PT
Why not TSA for cuff-deficient failed HA	poor function, **continued superior migration**, **glenoid loosening (rocking-horse)**
Cuff arthropathy after HA	bony erosion, superior head migration, **acromion + AC joint erosion**

X vs Y

	Hemiarthroplasty	Anatomic TSA	Reverse TSA
Glenoid	Native	Prosthetic	Prosthetic (reverse)
Cuff needed	**Yes**	**Yes**	No
Best for	**AVN, 4-part fracture**, young/laborer, poor glenoid stock	OA, **eccentric glenoid wear**	Cuff deficiency, **superior escape**

How it's asked

"Best results of hemiarthroplasty?"	→ **humeral head AVN** (TO)
"Recommended stem position?"	→ **30° retroversion** (TO)
"Lesser tuberosity placed too close to GT — problem?"	→ **ER deficit** (subscap over-tension)
"GT above superior head after fracture HA — risk?"	→ **tuberosity impingement** ✗ Wrong options: nonunion · overstuffing
"Function after HA or TSA most affected by?"	→ **rotator cuff integrity**
"Extended CTA head advantage?"	→ **metal-to-bone articulation with acromion**
"66-yo — hemiarthroplasty most indicated for?"	→ **humeral head osteonecrosis** ✗ Wrong options: Charcot · CA ligament deficiency · RA · capsulorrhaphy arthropathy
"42-yo with GH OA — HA vs TSA?"	→ **HA inferior with eccentric glenoid wear**
"42-yo weightlifter — ream-and-run?"	→ **allows activity beyond TSA limits**
"Painful HA 10 y, intact cuff, aseptic aspirate — next?"	→ **revise head + prosthetic glenoid, retain stem**
"Pseudoparalysis + superior escape after fracture HA?"	→ **convert to RSA**
"Contraindication to HA?"	→ **CA ligament deficiency** (superior escape)

– – Active space – –

SHOULDER IMAGING

– Shoulder & Elbow · 48 / 60 –

1 Basics

MRI physics – field strength

Field strength stronger magnet → ↑ intrinsic **signal-to-noise ratio**

3 T vs 1.5 T **3 T = 9× proton energy** of 1.5 T → **1.5 T ≈ 11% of 3 T**
✗ Wrong options: 33% · 66%

Sequence	TR / TE	Bright	Dark	Shoulder use
T1	**Short TR, short TE**	**Fat**	Fluid, bone, ligament, marrow, fibrocartilage	**MR arthrogram**; **Hill-Sachs**
T2	**Long TR, long TE**	**Fluid** (inflammation), marrow	Bone, ligament, muscle, fibrocartilage	**Cuff pathology** – full-thickness tear
STIR	Fat saturation (suppression)	Fluid + oedema ↑	Fat ↓	**Oedema vs fatty infiltration** of cuff muscles

CT vs MRI

	CT	MRI
Best for	**Cortical + trabecular bone**, **bony defects**	**Soft tissue**, **bone contusion / trabecular microfracture**
Cost	**↑ Radiation**	–
Artefact	**No magnification artefact** (unlike X-ray)	–

4 Diagnosis & Classification

Radiographic views (Zanca, West Point)

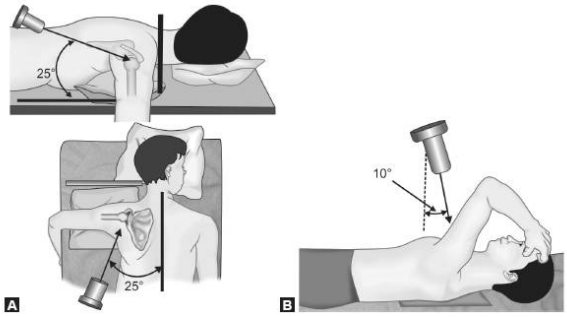
Complete trauma series	Focus
Shoulder AP	GH joint space, DJD
Grashey (true AP)	GH joint space, DJD, **proximal migration of humerus**
AP in ER / AP in IR	**Hill-Sachs**
Axillary lateral	**Anterior + posterior dislocation**, acromion
Velpeau	Axillary modification **if can't abduct arm**
Scapular Y lateral	**Anterior + posterior dislocation**

Additional view	Focus
Supraspinatus outlet	**Acromion morphology** – type I flat, II curved, **III hooked** (→ impingement + cuff pathology)
Zanca	**AC joint** – AC disease, **distal clavicle osteolysis**, distally projecting osteophytes
Stryker notch	**Hill-Sachs**
West Point axillary	**Anteroinferior glenoid – bony Bankart**, proximal humerus fracture
Garth	Anteroinferior glenoid – **bony Bankart**
Hobbs	**Anterior + posterior SC dislocation**
Serendipity	**Anterior + posterior SC dislocation**

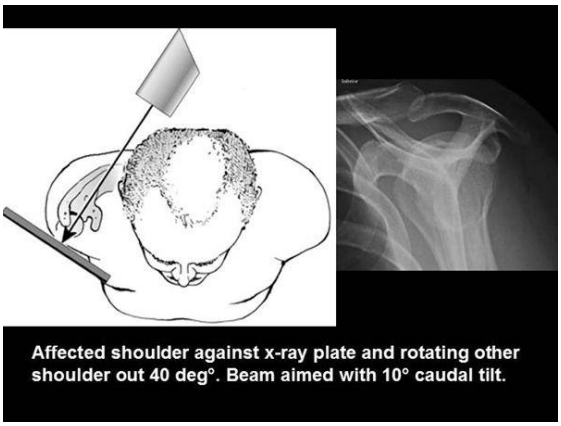
Zanca technique (TO) beam angled **10–15° superiorly (cephalad)** + **kV ↓ to 50%** of standard

Bony Bankart (TO) **West Point axillary** = popular view for **glenoid bone defect**
✗ Wrong options: IR/Stryker = Hill-Sachs · serendipity · Greenspan

Hill-Sachs (TO) **internal rotation view + Stryker notch view**



West Point prone axillary (A) and Stryker notch (B) positioning



Supraspinatus outlet view technique

Acromion morphology (outlet view)

Type	Shape	Significance
I	**Flat**	–
II	**Curved**	–
III	**Hooked**	**Impingement + cuff pathology**

– – Active space – –

X-ray views to remember (TO)

View	Use (TO)
Axillary	Lateral view of shoulder
West Point axillary	**Bankart lesion**
IR view + Stryker notch	**Hill-Sachs**
Zanca	**AC joint**
Serendipity	**SC joint** dislocation
Greenspan	**Radial head + capitellum** fractures
Jones (AP) elbow	Reduction of **supracondylar humerus** fracture
Oblique + PA wrist in ulnar deviation	**Scaphoid** fracture
Brewerton	**Metacarpal head** fractures
Robert's	**Thumb CMC** joint
Carpal tunnel view	**Hook of hamate** fracture
Ball catcher's	**RA hand erosions**
Judet (obturator + iliac oblique)	**Acetabular** fractures
Frog leg	Intra-op lateral hip
Von Rosen	**CDH**
Axial / skyline / Merchant	**Patellar subluxation**
Mortise (AP ankle in 15° IR)	Ankle injuries
Canale	**Talar neck** fractures
Harris (hindfoot)	**Calcaneus** fractures
Broden	Intra-op **calcaneal** fracture assessment
Swimmer's	Lateral C-spine with shoulder pulled down → **C6, C7, T1**
Ferguson	**20° caudocephalic AP lumbar** → L5 transverse process vs sacrum (**far-out syndrome**)

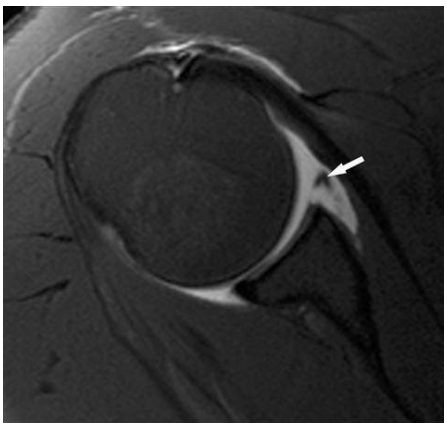
Key radiology findings – CT

CT plane	Best for
Axial	**Reverse Hill-Sachs**
Coronal	**Fractures**
Sagittal	**Anteroinferior glenoid insufficiency** (bone loss)
3D reconstruction	**Glenoid version** for TSA

Key radiology findings – MRI & MR arthrogram

ABER position sequence beyond coronal/sagittal/axial · hand **behind head** · **tensions anteroinferior GHL + labrum, relaxes capsule → Bankart**, **partial/full-thickness cuff tears**, **internal impingement**
✗ Wrong options: true 90–90

MR arthrogram percutaneous **dilute gadolinium** into joint → **labral + ligament** pathology: **Bankart**, **SLAP**, **GLAD**, **ALPSA**, **HAGL**



MR arthrogram, Bankart lesion

X vs Y

Lesion	Best X-ray view	Best cross-sectional
Hill-Sachs	**AP in IR / ER**, **Stryker notch**	T1 MRI
Reverse Hill-Sachs	–	**Axial CT**
Bony Bankart	**West Point**, **Garth**	**Sagittal CT** (glenoid insufficiency)
Soft-tissue Bankart / SLAP / HAGL	–	**MR arthrogram**, ABER
AC joint / DCO	**Zanca**	–
SC dislocation	**Serendipity**, **Hobbs**	–
Cuff tear	Grashey (proximal migration)	**T2 MRI**; STIR (fatty vs oedema)

How it's asked

"1.5 T proton energy as % of 3.0 T?"	→ **11%** (3 T = 9×)
"Best view for AC joint?"	→ **Zanca view** (TO) ✗ Wrong options: serendipity · Greenspan · Brewerton
"Best view for bony Bankart lesion?"	→ **West Point axillary view** (TO) ✗ Wrong options: IR + Stryker notch · serendipity · Greenspan
"Hill-Sachs best seen on?"	→ **IR view / Stryker notch** (TO)
"SC joint dislocation view?"	→ **serendipity** (or Hobbs)
"Can't abduct arm for axillary view?"	→ **Velpeau**
"Glenoid version before TSA?"	→ **3D CT**
"Reverse Hill-Sachs?"	→ **axial CT**
"MRI position for internal impingement / anteroinferior labrum?"	→ **ABER**

– – Active space – –

"Distinguish cuff muscle oedema vs fatty infiltration?" → **STIR**

SHOULDER PERIPROSTHETIC FRACTURE

– Shoulder & Elbow · 49 / 60 –

1 Basics

Epidemiology

Incidence — **0.6–3%** of shoulder arthroplasty (Orthobullets, OB) · **1.5–3%** in stemmed anatomic TSA (OB) · age **50–85**

Share of complications — **20% of all complications** (intra-op fractures, Orthobullets) · overall arthroplasty complication rate **11%**; post-op periprosthetic fractures = **6.6%** (reverse) vs **1.2%** (anatomic) of complications

Timing — **most are intra-operative**, at the index implant (anatomic + reverse); stemless: few post-op reports

Location — **adjacent to stem tip** · **greater tuberosity** (esp. **revision** TSA)

Risk factors — **osteoporosis** · **aggressive reaming** · endosteal notching · incorrect reaming entry (ideal **9 mm lateral** to anatomic centre of rotation) · **press-fit > cemented** stems · revision · soft-tissue contracture · instability history · ♀ · post-traumatic arthritis · ↑ comorbidity · osteonecrosis (**steroid** > alcohol, post-traumatic)

2 Mechanism

Intra-operative

Revision TSA — highest risk: greater exposure, subacromial/cuff scarring → ↑ retraction + torque

Intra-op step (revision)	Share of intra-op fractures
Removing previous implant	**Up to 81%**
Reaming / broaching	Up to 31%
Trialing / insertion	Up to 19%
Exposure (forceful retractor)	Up to 15%

Implant removal — metaphyseal-filling → GT fracture (cement removal, interface disengagement) · diaphyseal-engaging → GT fracture (**metaphyseal stress shielding**) · prevent: clear lateral soft tissue, implant-specific tools, **controlled corticotomy**

Primary TSA — lower risk · **endosteal notching from reaming** · **excessive humeral ER during exposure** · cortical breach, esp. osteopenic bone

Post-operative

Fall (FOOSH) — **stem tip = stress riser** → most post-op fractures **at/adjacent to stem tip** · fatigue fracture through stress riser

– – Active space – –

Atraumatic	**loosening** → implant "rattling" weakens cortex
Nonunion risk ↑	prosthesis **disrupts endosteal blood supply** · **stem distracts fragments** (no end-to-end healing)
Nonunion factors	osteoporosis · **RA** · ♀ · **displacement >2 mm** · **cement at fracture site**

3 Presentation

History	fall on outstretched hand · **prodromal / worsening pain** = mechanical loosening
Exam	**varus/valgus deformity**, ecchymosis, shortening, diffuse tenderness · can't move elbow/shoulder (**pain**) · document full NV exam
Radial nerve	check status; palsy common with distal humerus fracture, **mostly neurapraxia** ✗ Wrong options: surgery for it

4 Diagnosis & Classification

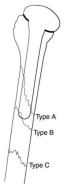
Key radiology findings

X-ray	orthogonal humerus: **AP, axillary lateral, Grashey, scapular Y** · fracture at **stem tip**, usually **oblique** · decide on **stem stability** + **zones of lucency** (no ingrowth / osteolysis)
CT (metal suppression)	concern for **loose stem** (pattern, prodromal pain) or comminution · morphology, bone stock, cuff muscle quality, glenoid version (if revising)



Loose humeral stem – axillary lateral

Wright and Cofield classification

Type	Fracture	Diagram
A	Centred near **stem tip**, extends **proximally** ("proximal to stem tip" – OB)	
B	Centred near **stem tip**, extends **distally** (at level of tip)	
C	**Distal to stem tip** (into distal metaphysis)	

Wright series (Mayo) 499 arthroplasties → **9 fractures** (1 A, 5 B, 3 C) · 3 splinted → **failed to heal**

– – Active space – –

5 Management

Post-operative periprosthetic humerus fracture

- Loose stem (any type) → **revision arthroplasty + supplementary fixation**
  (poor stock → long stem bypassing 2–3 cortical diameters; good stock → shorter stem + fixation)
- Stable stem, type A/B → **ORIF: hybrid locking plate + cerclage cables**
  (long spiral/oblique A/B → bracing possible, union <50%)
- Type C → **fracture brace (nonop)**
  - fails / nonunion → **ORIF ± allograft strut**

Intra-operative fracture

- Type A → **stem spanning fracture by 2 cortical diameters**
- Type B / C → **proximally porous-coated long stem, 2–3 cortical diameters**
- Type C unsuitable for long stem → **ORIF**

Operative criteria

Situation	Treatment
Long oblique / spiral A or B, stable stem · **type C**	Immobilisation (coaptation → **functional brace**); **union <50%**
Intra-op type A	Longer stem spanning **2 cortical diameters** ± supplementary fixation
Intra-op type B or C	**Proximally porous-coated long stem**, **2–3 cortical diameters**; distal cement
Post-op transverse A/B, stable stem · C failed nonop · can't tolerate nonop	**ORIF**
Loose prosthesis, any type	**Revision arthroplasty + supplementary fixation**
Poor bone stock (loose)	**Long stem** bypass ≥2–3 cortical diameters ± allograft
Good bone stock (loose)	**Short stem** (ends proximal to original) + fixation; tip doesn't distract fracture

Source nuance long spiral **type B**, stable stem: bracing possible (Orthobullets) · **ORIF hybrid plate + cables** best for active patient (OB) – read the stem

Stable-stem fractures – ORIF with plate/cerclage & radial nerve risk

Best for stable stem A/B **ORIF hybrid locking plate + cerclage cables**
✗ Wrong options: revision (stem well fixed) · **lag screw + neutralisation** · nonop near stem tip → nonunion

– – Active space – –



Hybrid locking plate and cerclage cables

Construct	plate placed **eccentrically** → locking screws skive past stem (bicortical) · **cables proximal** to stem tip · **screws distal** · usually **4.5 mm LC-DCP** · alternatives: dual plates, **3.5 mm attachment plates** · deficient cortex → **cortical strut allograft** + cables (2 hemicylinders = "sarcophagus")
Approach	**anterolateral** (extend deltopectoral: coracoid → DP groove → lateral biceps) · biceps–brachialis interval, **split brachialis** (medial = musculocutaneous, lateral = radial) · **posterolateral** for distal fracture with stable stem
GT / calcar fracture	suture or cable fixation
No nonop with sling / hanging arm cast	bracing is the nonop method; active patient wants fast return → ORIF
No APC (allograft-prosthetic composite)	not with good bone quality + no bone loss
Evidence	Kurowicki: 7 type B ORIF without revision → **all united** · Andersen: ORIF 17/17, revision 18/19 united · stable IM stem + supplementary fixation > loose stem

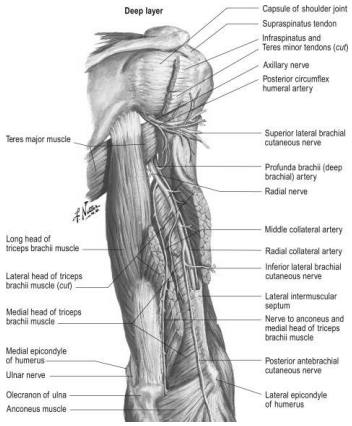
Radial nerve and cerclage

Landmark	Distance to radial nerve at spiral groove
Coracoid	~13 cm
Distal lesser tuberosity	~10 cm
Superior latissimus insertion	~8 cm
Inferior latissimus insertion	**~4 cm**

Cable 4 cm below inferior LD insertion → iatrogenic radial nerve injury
✗ Wrong options: median (medial septum) · axillary (proximal) · subscapularis · brachial artery (superficial, anteromedial)

Safe zone	cables **just inferior / proximal to inferior edge of LD insertion**

– – Active space – –



Radial nerve in spiral groove

Long stem technique

Porous-coated long stem	proximal coating = metaphyseal stability · **pack cement distally**, **avoid cement extrusion into fracture** · strut + cables if deficient · **olecranon fossa** may prevent it

6 Complications

Complications	**nonunion** · delayed union · **GH stiffness** · **radial nerve injury** (fracture or **iatrogenic cerclage**)
Prognosis	high union when treated appropriately; union **2–8 months** · **significant ↓ shoulder ROM**

X vs Y

	Stable stem	**Loose stem**
Clues	No lucency, no prodromal pain	Lucent zones, **prodromal pain**, "rattling"
Type A/B	**ORIF hybrid plate + cables**	**Revision long stem** + fixation
Type C	Brace → ORIF if fails	Revision

How it's asked

"Fall, stable rTSA stem, type B fracture – best treatment?"	→ **ORIF hybrid locking plate + cerclage cables** ✗ Wrong options: revision cemented/ cementless long stem · lag screw · nonop
"Cable placed 4 cm inferior to latissimus insertion – complication?"	→ **iatrogenic radial nerve injury** ✗ Wrong options: median · axillary · brachial artery
"Active 69-year-old, T-score −0.45, long spiral B, stable stem – treatment?"	→ **ORIF plate + screws/cerclage cables** ✗ Wrong options: sling · hanging cast · revision · APC

– – Active space – –

"Loose stem with any fracture type?"	→ **revision arthroplasty + supplementary fixation**
"Intra-op fracture — stem should bypass by?"	→ **2 cortical diameters** (2–3 for type B/C)
"Why high nonunion near stem tip?"	→ **disrupted endosteal blood supply** + stem distracts fragments
"Most common step causing intra-op fracture in revision?"	→ **implant removal** (up to 81%)
"Type C fracture, stable stem?"	→ **fracture brace** ✗ Wrong options: sling

SHOULDER PROSTHETIC JOINT INFECTION

– Shoulder & Elbow · 50 / 60 –

1 Basics

Epidemiology

Significance — **major cause of revision within first 2 years** after shoulder arthroplasty

Incidence — **~4% primary TSA + RSA** (Orthobullets, OB) · 3–4% · 0–5% · up to 4–5% (other OB explanations) – read the options · **~15% revision** TSA/RSA

aTSA vs rTSA — **no difference** in primary shoulders without prior surgery · ↑ in primary **rTSA after ≥1 prior non-arthroplasty operation** (Florschütz)

Risk factors & prevention (smoking, steroid injection, nasal decolonisation)

Risk factor	Note
Male	**2.5×** PJI rate · ↑ C. acnes colonisation
Arthroplasty for fracture / trauma	**3×** vs elective
Revision surgery	~15%
Intra-articular steroid injection within 3 months	Injection **4 months** (or 15 months) before = **not** a risk factor
Smoking	↑ infection + fracture + loosening
Young age	–
Postoperative haematoma	–
Coagulopathy, **ESRD**, **lupus**, **RA**, uncontrolled comorbidities	Medical comorbidities

Pre-op smoker — **smoking cessation programme next** (smoking = suggested contraindication to TSA/RSA)
✗ Wrong options: PT trial (pseudoparalysis CTA) · tramadol (= 1st line knee OA) · CBC/ESR/CRP (infection unlikely) · rheumatology referral

Complication risk (patient) — younger age · ↑ BMI · RA · diabetes · smoking

– – Active space – –

C. acnes colonisation correlates **male · native hair at surgical site · humeral loosening**
✗ Wrong options: age >70 · CKD · female · glenoid lucent lines

✓ Validated prevention	Detail
MRSA (methicillin-resistant Staphylococcus aureus) screening + nasal decolonisation	If nares colonized → application of **5% povidone-iodine solution to each nostril** pre-operatively (**10 s/nostril, 1 h pre-op**) · or **2% mupirocin BID × 5 days**
IV antibiotic prophylaxis	Within **30 min** of incision, continue **24 h**
Penicillin allergy prophylaxis	**Clindamycin for C. acnes** (Orthobullets, OB) · **vancomycin, not clindamycin** (OB) – read the options
Skin prep	**Chlorhexidine (ChloraPrep)** · hydrogen peroxide prep
Hair removal	**Electric clippers**
Intra-op	Frequent **saline irrigation** · **dilute povidone-iodine** before closure
Closure / dressing	Dermabond · **silver-impregnated** dressings

Not validated	Note
Axillary hair shaving	No meaningful ↓
Sterile helmets/hoods	May ↑ infection
Laminar flow	Not validated

2 Mechanism

C. acnes – microbiology, prolonged culture & antibiotic choice

Most common organism **Cutibacterium acnes** (formerly Propionibacterium acnes) · others: S. aureus, S. epidermidis

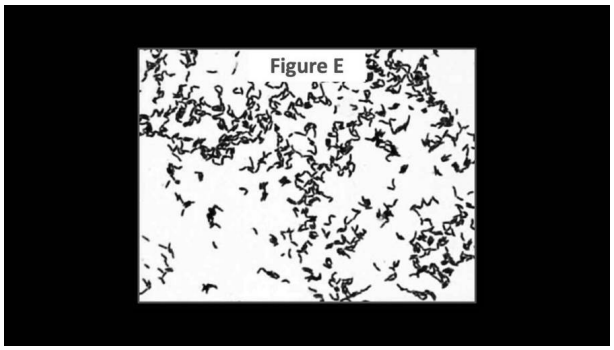
Description **gram-positive, anaerobic rod** (OB) · **gram-positive rod, facultative anaerobe** (OB) · "gram-positive, facultative, aerotolerant, anaerobic rod" (Orthobullets) · **ferments lactose → propionic acid** · non-spore-forming

Habitat **axilla**, dermal **sebaceous glands**; commensal skin flora

Biofilm forms in **18–90 h** on implant + synovium (>> planktonic) → **aspiration only 17% sensitive**

Behaviour indolent, low-grade, low virulence · usually **no swelling, erythema, fever, pus** · inflammatory markers **often normal**

Culture mean incubation **7–21 days** · usually negative in first **3–5 days** · may take **up to 3 weeks**



C. acnes – gram-positive rods

✗ Organism distractor	Description
S. aureus	GP cocci, facultative anaerobe, **coagulase +** · commonest in **hip/knee** PJI
S. epidermidis	GP cocci, facultative anaerobe, **coagulase –**
E. coli	GN rod, facultative anaerobe · rare in shoulder
Klebsiella	GN **encapsulated** rod
N. gonorrhoeae	GN diplococci
C. diphtheriae	GP **club-shaped**, nonmotile, noncapsulated bacillus
Streptococci · fungi	Rare · exceedingly rare

Biofilm & timing

Biofilm (all bacteria) on implant **within 4 weeks** · **15% cells + 85% polysaccharide (glycocalyx)** → adherence + immune protection · no safe removal → **explant if >4 wk**

Source **direct** (sinus tract, wound dehiscence) · **haematogenous** (dental work, gallbladder) in longstanding infection-free joint

3 Presentation

Symptoms **persistent pain (most common)** · **draining sinus (2nd)** · systemic (fever, chills, sweats) · **stiffness**

Sinus tract to joint **diagnostic**

Typical C. acnes stem months–years after arthroplasty, **gradually ↑ pain**, normal WBC/ESR/CRP, humeral lucency

4 Diagnosis & Classification

Classification – time of onset

	Acute	Chronic
Timing	**<3–6 wk** (CDC **<90 days**)	**>3–6 wk** (CDC **>90 days**)
Biology	Confined to joint; no bone–implant invasion; less biofilm	**Biofilm**; invades **prosthesis–bone interface**
Treatment	**I&D + retention (DAIR)**	**2-stage revision**

– – Active space – –

DAIR window — **<4 wk** (Orthobullets treatment) · **within 5–6 wk** (OB) · <3–6 wk (Orthobullets classification) – read the options

Key radiology findings

X-ray — normal early · osteopenia · **periprosthetic lucencies** · pseudosubluxation of head · effusion · **endosteal scalloping** · bony resorption

Humeral vs glenoid loosening — **glenoid loosening = most common cause of TSA failure**, not specific for infection · **humeral loosening → suspect PJI**



Glenoid radiolucency after TSA

US / MRI — fluid collections (US: loculated away from GH joint)

Bone scan — not routinely (efficacy not shown)

PET — adjunct; **98% sensitivity + 98% specificity**

Genes & lab markers

Marker	Fact
WBC	Not sensitive/specific; **normal in C. acnes**
CRP	Peaks **2–3 d** post-op → normal **14–21 d**; may be normal in C. acnes
ESR	Peaks **5–7 d** → normal **90 d**; may be normal
ESR/CRP overall	**Insufficient sensitivity/specificity** for PJI
Alpha-defensin	Investigational
Synovial WBC	**>50,000**
Crystals	**Don't exclude** infection
Gram stain	Often **negative**; + in ~75%

Diagnostic work-up – aspiration

Synovial tissue culture — **gold standard** (+ ~80%) · **≥4 specimens** at surgery · **aerobic + anaerobic + broth** media

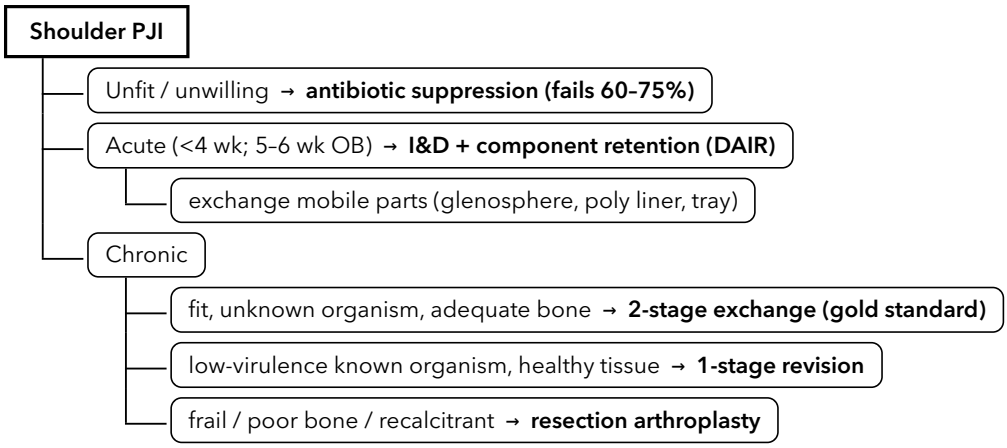
Hold cultures — **14–28 days** (Orthobullets) · **14–21 days / 2–3 weeks** (OB) – read the options · fungal **4 wk** · mycobacterial **8 wk**

Aspiration — **low sensitivity + low NPV · dry tap common → does not rule out · + S. aureus culture = high specificity + PPV**

Fluoroscopic / US guidance — ensures intra-articular placement · **no ↑ PPV/NPV or yield**, operator-dependent, ↑ cost

Antibiotics	stop **2 weeks before aspiration** ✗ Wrong options: current antibiotics don't affect yield
No growth at 5 days	**observe cultures additional 2 weeks** before revision ✗ Wrong options: revision · arthroscopic I&D

5 Management



Antibiotics

C. acnes drug of choice	**penicillin** (low MIC) · penicillin G, amoxicillin ✗ Wrong options: TMP-SMX · meropenem · vancomycin (not 1st line)
Penicillin allergy	**ceftriaxone or clindamycin**
Duration	prolonged IV; guided by cultures + ID consult

Mechanism of action	Antibiotic
Binds transpeptidase (PBP) → inhibits cell wall synthesis	**Penicillin** ✓
Reversibly binds **50S**	Macrolides
Irreversibly binds **30S**	Aminoglycosides
Blocks **30S–50S initiation complex**	**Linezolid**
Binds cell wall, disrupts peptidoglycan **cross-linking**	**Vancomycin**

– – Active space – –

Operative criteria

Option	Indications	Outcome
Antibiotic suppression	Severely ill, unwilling, multiple comorbidities	**Fails 60–75%**
I&D + retention	**Acute** infection	**50% failure**
Single-stage revision	Uncommon; **low-virulence** organism with good sensitivity; healthy patient + soft tissue	Up to **90%**; ↓ stay, cost, antibiotic time · consider **RSA** (cuff sacrificed)
2-stage exchange	**Gold standard**; medically fit; **unknown organism**; adequate bone stock; eradication before stage 2	**60–90%**; some keep spacer
Resection arthroplasty	Frail/complex, **insufficient bone stock**, recalcitrant	Poor function; **pain relief >50%**; **tuberosity preservation** predicts better result
Arthrodesis	Rare (bone stock poor)	Pain ↓
Amputation	**Life-threatening** infection	–

Spacer antibiotic-PMMA; local antibiotics, **pseudocapsule**, maintains soft-tissue tension, ↓ pain, allows PT · stem from **Steinmann pin + chest tube**



Resection arthroplasty

Resection **antero-superior subluxation** if tuberosities + cuff removed

Arthrodesis S-shaped incision · resect cuff, biceps tenodesis, decorticate glenoid/head/acromion undersurface · **10-hole 4.5 mm pelvic recon plate** · **30° abduction, 30° flexion, 30° IR** · complications: infection, nonunion, malposition, prominent hardware, humeral shaft fracture

Approach **deltopectoral** for all

6 Complications

Complications **failure to eradicate** (inadequate clearance after suppression, I&D, 1-stage) · recurrent infection (2-stage) · humerus/glenoid fracture · poor bone stock

X vs Y

	Shoulder PJI	Hip/knee PJI
Commonest organism	**C. acnes**	**S. aureus / S. epidermidis**
Inflammatory markers	**Often normal** (C. acnes)	↑ with S. aureus / MRSA
Aspiration value	Less clear: low sensitivity, dry taps	Clearer
Gold-standard treatment	2-stage	2-stage

How it's asked

"rTSA, drainage at 4 wk, I&D + poly exchange, cultures negative at 10 d – MOA of best antibiotic?"	→ **binds transpeptidase, inhibits cell wall synthesis (penicillin)** ✗ Wrong options: 50S macrolide · 30S aminoglycoside · linezolid · vancomycin
"Normal labs, painful aTSA, aspiration – hold cultures 2–3 wk for?"	→ **gram-positive, anaerobic bacteria** ✗ Wrong options: GP aerobic cocci · GN rods · diplococci · club-shaped
"18-mo TSA, lucent lines, normal labs, no growth at 5 d – next step?"	→ **observe cultures additional 2 weeks** ✗ Wrong options: arthroscopic I&D · revision
"Independent correlate of P. acnes colonisation?"	→ **native hair at surgical site** ✗ Wrong options: age >70 · CKD · female · glenoid lucent lines
"aTSA 2 mo ago, infection confirmed – organism characteristics?"	→ **gram-positive rod, facultative anaerobe** ✗ Wrong options: coagulase + / – cocci · GN rods
"C. acnes grown – antibiotic of choice?"	→ **penicillin** ✗ Wrong options: TMP-SMX · clindamycin = allergy only · meropenem · vancomycin
"Smoker with CTA planned for arthroplasty – next step?"	→ **smoking cessation programme**
"rTSA infection – NOT a risk factor?"	→ **steroid injection 4 months before surgery** (✓ risk: coagulopathy, fracture, male, smoking)
"Validated PJI prevention before rTSA?"	→ **5% povidone-iodine to each nostril if MRSA-colonised** ✗ Wrong options: axillary shaving · helmets · laminar flow · clindamycin for allergy
"rTSA 2 y, ↑ ESR/CRP – most common organism (Gram picture)?"	→ **gram-positive rod (C. acnes)** ✗ Wrong options: staph · strep · E. coli · fungus

– – Active space – –

"Shoulder aspiration — true statement?"	**→ positive S. aureus culture = high specificity** ✗ Wrong options: high sensitivity/NPV · dry tap rules out · fluoroscopy ↑ PPV/NPV · antibiotics don't matter
"Gold standard for chronic shoulder PJI?"	**→ 2-stage exchange**
"Acute PJI within weeks of surgery?"	**→ I&D + component retention (DAIR)**

SLAP LESION

– Shoulder & Elbow · 51 / 60 –

1 Basics

Epidemiology

Incidence	uncommon: **<5%** of shoulder injuries · yet a common cause of pain + weakness in throwers
Who	**dominant shoulder of overhead/throwing athletes** · older patients after traumatic falls
Risk factor	**GIRD**
Associations	internal impingement · GIRD · **articular-sided** cuff tears · (subtle) instability · scapular dyskinesis · paralabral cyst · sublabral clefts, Buford complex
History (TO)	**Andrews (1985)**: superior labral tears with **partial cuff tears in pitchers** · **Snyder (1990)** coined "**SLAP lesion**"
Age (TO)	usually **<40 y** athletes/throwers · isolated SLAP uncommon; often with **articular-sided supraspinatus tear**

Anatomy – biceps anchor, Buford complex & vascularity

LHB attachment	**~50% to supraglenoid tubercle, ~50% to labrum** (predominantly posterior labrum) · tubercle **>6 mm medial** to glenoid surface
Posterior / posterior-dominant attachment	**~70%** of shoulders (✗ Wrong options: 15% · 30% · 100%) · most common: **posterior to 12 o'clock** (✗ Wrong options: "anchored anteriorly")
Biceps origin (Habermeyer, TO)	**posterosuperior labrum 48% (commonest)** · **supraglenoid tubercle only 20%** · both **28%** · predominantly posterosuperior labrum, **10–2 o'clock**

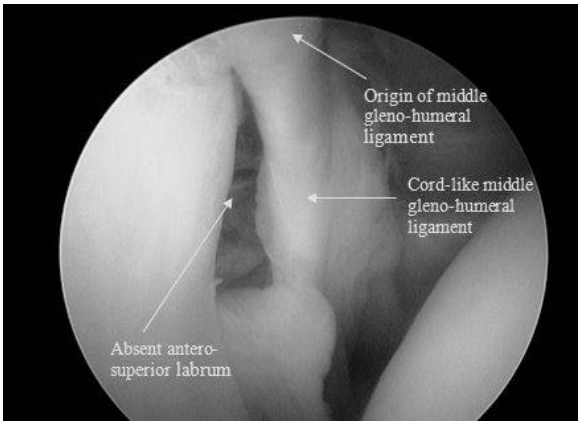
Tuoheti type	Biceps attachment	Share
I	Completely posterior	I + II **>70%**
II	Predominantly posterior	
III	Anterior + posterior	25%
IV	Anterior	5%

Biceps function	torsional + horizontal stability in **late cocking** · **poorest blood supply** in intra-articular course
Labrum	fibrocartilage; **chock block** to subluxation · superior labrum attaches more **medially** + anchors biceps = **weak link**

– – Active space – –

Labral blood supply — **suprascapular, circumflex scapular, posterior humeral circumflex** arteries via **capsule + periosteum** · **anterosuperior labrum = poorest** → degenerative tears, poor healing
✗ Wrong options: bone · single source

Variant	Feature	Pitfall
Sublabral recess / foramen	± thickened MGHL	Mistaken for tear on MRI
Buford complex	**Cord-like MGHL + absent anterosuperior labrum**; **normal variant**, **1.5%**	**Do not repair → significant loss of ER** + pain
Meniscoid labrum	**1%**	–
Superior MGHL attachment	On glenoid	–



Buford complex – arthroscopic view

Arthroscopy tell — erythema + tearing **under** labrum = tear (vs normal recess)

2 Mechanism

Mechanisms — **repetitive overhead activity** · FOOSH with **tensed biceps** · traction on arm

Thrower pathoanatomy — tight **posterior IGHL** → GH contact shifts **posterosuperiorly** → ↑ shear on superior labrum → **"peel-back"**

Consequence — SLAP **↑ strain on anterior band IGHL** → ↓ stability

Disabled throwing shoulder (Burkhart 2003) (TO) — repetitive throwing → posterior capsular stretch → **posterior capsular fibrosis/contracture** → **GIRD (>25°)** → head shifted **posterosuperiorly** in ABER → **internal impingement** → **SLAP + PASTA**

Internal impingement complex (TO) — **SLAP + articular-sided supraspinatus tear**

3 Presentation

History — "pop" during overhead activity or trauma · **lag** between injury + symptoms

Symptoms — **vague deep shoulder pain** · popping/clicking · weakness, fatigue, ↓ performance

Exam bicipital groove tenderness · check **IR + total arc** vs contralateral · SS ± IS **atrophy** (suprascapular neuropathy from cyst) · apprehension positive in **85%**

Tests **no single specific test**

Test group	Tests
Biceps provocation	**Speed's**, **Yergason's**, Kim biceps load
SLAP	**O'Brien (active compression)**, **crank** (full abduction, load + rotate), **dynamic labral shear**, Kibler anterior slide

SLAP test (TO)	Manoeuvre	Positive
O'Brien	90° flexion, **10° adduction**, thumb down → resist	Deep pain in IR not ER = SLAP · superficial = AC
Biceps load I	**90° abduction**, ER, supination → resisted elbow flexion	TO: **pain relief = positive**
Biceps load II	Supine, **120° abduction**, ER, supination → resisted elbow flexion	**Pain ↑** = positive
Uppercut	Elbow 90°, supinated, punch to chin vs resistance	Pain = SLAP
Modified dynamic labral shear	Standing, 120° abduction scapular plane, ER → lower to 90°	Posterior click/pain · **most specific / confirmatory test** (TO)

Speed / Yergason (TO) may be positive in early biceps tendinitis or evolving SLAP

Related topics (distal clavicle excision, biceps dislocation)

Arthroscopic vs open DCE advantage = **evaluate GH joint** · failed open DCE: **15/20 had SLAP** (9/15 good–excellent after arthroscopic treatment)
✗ Wrong options: fewer complications · infection · time

Arthroscopic DCE sacrifices **inferior** AC ligament, **preserves superior** AC ligament

Medial LHB dislocation **subscapularis tear**
✗ Wrong options: supraspinatus (LHB rupture) · transverse ligament · shallow groove · type I SLAP; type II SLAP can cause instability

4 Diagnosis & Classification

Diagnosis – MR arthrogram

X-ray usually normal

Study of choice **MR arthrogram** · **gold standard = arthroscopy**
✗ Wrong options: US · EMG · labs · PT first when suspicious

MRI sign T2 signal between superior labrum + glenoid, **lateral to glenoid rim, posterior to biceps**

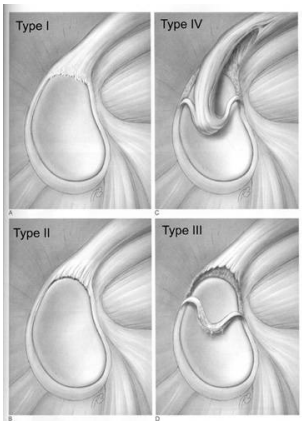
Accuracy sens **~50%**, spec **~90%**, ↑ with arthrogram (Orthobullets) · linear high signal on oblique coronal T2: sens **41%**, spec **86%**, accuracy 63% (OB) – read the options

– – Active space – –

Paralabral ganglion cyst	**highly specific for labral tear**; usually **spinoglenoid notch** ✗ Wrong options: suprascapular notch
Overdiagnosis	concern (↑ SLAP repairs in ABOS case logs)
Arthroscopic peel-back test	labrum peels back at **90° abduction + ER**
Arthroscopic SLAP criteria (TO)	**superior labral excursion >5 mm** · sublabral granulation/fraying · **peel-back sign → confirms type II**
Plain MRI (TO)	limited for typing · **MR arthrography = investigation of choice** · arthroscopy = gold standard for classification

SLAP classification

Type	Orthobullets description (share)	Clock face (TO)	TO description
I	Labral + biceps **fraying, anchor intact** (11%)	11–1	Fraying superior labrum
II	Fraying + **detached biceps anchor** (**41%**, most common)	11–1	Detachment of biceps anchor
III	**Bucket handle, intact anchor** (33%); biceps separates from fragment	11–1	**Bucket handle, intact biceps**
IV	**Bucket handle + detached anchor** (15%); biceps stays with fragment	11–1	Bucket handle with **biceps extension**
V	II + **anteroinferior extension (Bankart)**	**11–5**	Bankart + SLAP
VI	II + **unstable flap**	11–1	Flap tear of bucket-handle SLAP
VII	II + **MGHL injury**	**11–3**	Extension into MGHL
VIII	II + **posterior extension**	**11–7**	Posterior extension
IX	**Circumferential**	**11–11**	Pan-labral
X	II + posteroinferior extension (**reverse Bankart**) (Orthobullets)	**11–1+**	SLAP + **noncontiguous** posterior labral tear (TO)



SLAP types I–IV

Systems	**Snyder** = I–IV · **Maffet** added V–VII · high inter/intra-observer variability
Type X wording	"posteroinferior extension / reverse Bankart" (Orthobullets) · "noncontiguous posterior labral tear" (TO) – read the options

Classification & variants (type III, Andrews lesion) (TO)

Dynamic labral shear + tear 11–1 o'clock, intact biceps (TO)	**type III** SLAP ✗ Wrong options: type II = biceps anchor detached · Bankart
Andrews lesion (TO)	**anterior + superior labral detachment without extension posterior to biceps** footplate · **throwers** ✗ Wrong options: SLAP II · posterosuperior · anterior + posterior · posterior IGHL avulsion

5 Management

SLAP lesion

- First line → **rest + PT + NSAIDs (scapula, cuff; treat GIRD)**
- Failed nonop, type I / III / IV <1/3 biceps → **arthroscopic debridement**
- Failed nonop, type II, young overhead athlete → **SLAP repair**
  - >40 y (failure >36 y) → **debridement + biceps tenodesis/tenotomy**
- Type IV >1/3 biceps → **biceps tenodesis/tenotomy ± labral repair/debridement**

Treatment – nonoperative vs repair vs tenodesis (age, type IV) & return to sport

Initial ✓	**rest + rehab for scapular mechanics + cuff strengthening** (+ NSAIDs; address GIRD) ✗ Wrong options: aggressive **concentric biceps** strengthening (aggravates) · surgery first
Labral tear in thrower	initial: **posterior capsular stretching + cuff strengthening**

Operative criteria

Situation (failed nonop, severe symptoms)	Procedure
Type I, III, IV <1/3 biceps	**Arthroscopic debridement** (III → debridement ± repair of fragment)
Type II, young overhead athlete	**Labral repair** (most amenable to repair)
Type II, >40 y · biceps fraying **>50%**	**Debridement + biceps tenotomy/tenodesis** (✗ Wrong options: repair (stiffness))
Type IV with >1/3 (>30%) biceps	**Biceps tenodesis** ± labral repair/debridement
Type V (II + Bankart) · II + posterior extension	**SLAP + Bankart repair** (active patient)
Paralabral cyst + SLAP II	**Cyst decompression + SLAP repair**

– – Active space – –

Age thresholds tenodesis consensus **>40 y** (Orthobullets, OB) · repair failure **>36 y** (Orthobullets, OB) · stiffness risk **>35 y** (OB) — read the options

TO treatment by type	Plan
I	Exhaust nonop → **debridement** if mechanical symptoms or **paralabral cyst**
II	**Low-demand → nonop** · high-demand athlete → **arthroscopic repair**
III	**Resect** bucket-handle fragment
IV	Biceps **<50% → leave** · **>50% → tenodesis / tenotomy**

Type IV biceps threshold **>1/3 (>30%)** (Orthobullets) · **>50%** (TO) — read the options

Tenodesis vs tenotomy (TO) **<40 y → tenodesis · >40 y → tenotomy** · similar long-term outcomes · tenotomy cosmetic deformity **~20%**

Tenodesis in young viable even **<25 y** athletes (recent studies)

Repair failure factor **age >36**
✗ Wrong options: recreational swimming · male · sedentary job · type II

Return to sport

Group / treatment	Rate
Same level after repair (overall)	**60–70%**
Return to prior play after repair	**50–60%**
Non-pitchers returning to baseball after repair	**65–85%**
Nonop: return to play	**~40%**
Nonop: return to prior play	**20–30%**

Pitchers vs non-pitchers **non-pitchers return at higher rates** after repair · RTP similar for repair vs tenodesis · elite throwers: variable, unpredictable

Repair technique

Steps debride flaps · **decompress cysts** · accessory/percutaneous portals for anchors near rim · **sutures anterior to biceps → stiffness**

Tenodesis all-arthroscopic or mini-open; tenotomise near labrum

SLAP repair portal (TO)	Use
Posterior	Viewing
Anterior (above coracoid)	Instrumentation
Anterolateral (acromial edge)	**Anchor insertion**
Neviaser (superior)	**Suture shuttling** to superior labrum

Repair steps (TO) confirm tear → debride + prepare bed → **anchor posterior to biceps** (posterosuperior labrum = main attachment) → shuttle via Neviaser → tie · usually **1–2 anchors**

Rehab (TO) sling **3 wk** · passive ROM after 3 wk (avoid ER stress) · strengthening **6–8 wk** · overhead sport **4–6 months** (Orthobullets ~6 months) — read the options

Postoperative rehabilitation after SLAP repair

Period	Allowed	Avoid
Wk 0–4	**Passive + active-assisted flexion in scapular plane**; passive **0–90° FE**	**Passive ER at 90° abduction** (~4 wk), extremes of abduction + ER, **resisted biceps**
Wk 4–6	Active ROM, isometrics; **isotonic cuff** 4–6 wk	**Resisted elbow flexion until 6 wk**
Wk 6–12 (to 8)	Functional + light strengthening	–
Wk 8–16 / 12+	**Sport-specific** strengthening	–
~6 months	Return to sport	–

Limit after repair **abduction + ER** – max tension on repair
✗ Wrong options: IR

✗ Rehab traps immediate full active ROM · **6 wk sling, no motion** (stiffness) · cuff strengthening by wk 2 · eccentric biceps at wk 2 · latissimus pull-downs

6 Complications

Stiffness

Most common after SLAP repair (~78%, Orthobullets) · risk: incidental SLAP repaired in **>45 y** during cuff repair · **post-surgical adhesive capsulitis**

Stiffness rate (TO) **up to 75%** · prevented by rehab compliance

Treatment early pendulum/AAROM from wk 1 → **capsular release** if fails >6 months PT

Suprascapular nerve – anchor placement & paralabral cyst

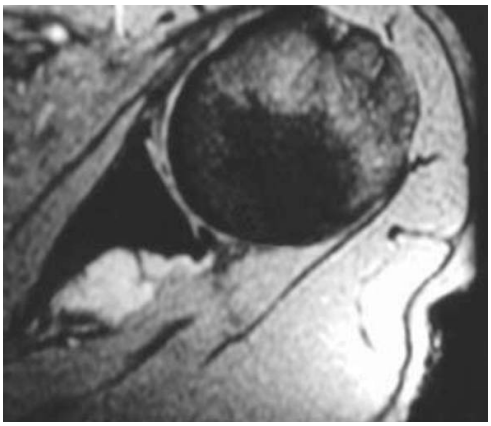
Mechanism deep drilling / anchors from **anterior or anterosuperior portal** · instrumentation **medial to glenoid** · overdrilling glenoid

Distance SSN **1.5–2.0 cm** from glenoid cortex at scapular spine level

Origin SSN from **superior (upper) trunk**
✗ Wrong options: middle trunk (no branches) · C5–7 roots = long thoracic · posterior/medial cord

✗ At risk distractors axillary nerve, thoracoacromial artery, thoracodorsal nerve, superior thoracic artery

– – Active space – –



Spinoglenoid paralabral cyst

Cyst location	Muscles affected
Suprascapular notch	**SS + IS**
Spinoglenoid notch	**IS only**

Cyst management no structural lesion → rehab · with labral tear → **decompression + labral repair**

Failed repair

Causes chondral injury, hardware failure, **anchor pullout** · age **>36** → tenodesis better

X vs Y

	Buford complex	Sublabral foramen	SLAP II
Labrum	**Absent** anterosuperior	Present; recess/foramen beneath	Detached at biceps anchor
MGHL	**Cord-like**	± thickened	–
Key point	**Do not repair** (↓ ER)	MRI pitfall – not a tear	Repair / tenodesis

	SLAP III	SLAP IV	Andrews (TO)
Labrum	Bucket handle	Bucket handle	Anterosuperior detachment
Biceps	**Intact**	**Torn into**	Not posterior to footplate

Distractor lesion	Key feature
HAGL	Humeral avulsion of anterior IGHL
ALPSA	Capsulolabral periosteal sleeve displaced **medially** on glenoid neck (axial)
Bankart	Anteroinferior labrum, after anterior dislocation
Hill-Sachs	Anterior instability; engaging defects → surgery
Reverse Hill-Sachs	Posterior instability; surgery if **>20%** of articular surface

How it's asked

"MRI tear 11–1 o'clock, intact biceps, positive dynamic labral shear?"	→ **SLAP type III** (TO)
"Andrews lesion?"	→ **anterior + superior labral tear** (TO)
"Buford complex?"	→ **normal variant: cord-like MGHL + absent anterosuperior labrum**
"% posterior/posterior-dominant biceps attachment?"	→ **70%**
"True about the superior labrum (SLAP on MRI)?"	→ **anterosuperior portion most susceptible to vascular insufficiency** ✗ Wrong options: anterior anchor · single artery · suprascapular notch cyst · limit IR after repair
"FOOSH, overhead athlete, MRI superior labral signal?"	→ **SLAP tear** ✗ Wrong options: HAGL · ALPSA · Bankart
"Pitcher, O'Brien + Jobe relocation, total arc 180° – next?"	→ **MR arthrogram**
"47-year-old, type II SLAP, biceps fraying >50%, failed 8 months nonop?"	→ **debridement + biceps tenotomy/ tenodesis** ✗ Wrong options: SLAP repair
"Factor associated with SLAP repair failure?"	→ **age >36**
"Third-baseman, type II SLAP – counsel?"	→ **better return to same level than pitchers**
"38-year-old, O'Brien/Speed/Yergason +, initial treatment?"	→ **rest + rehab for scapular mechanics + cuff**
"Finding favouring tenodesis over repair?"	→ **type IV with 40% biceps involvement**
"Rehab after type II SLAP repair?"	→ **limited passive 4 wk → active to 8 wk → sport-specific to 12–16 wk**
"Avoid at 2 weeks after SLAP repair?"	→ **passive ER at 90° abduction**
"Immediate rehab after SLAP repair?"	→ **passive + active-assisted flexion in scapular plane**
"Anchor drilling from anterosuperior portal – structure at risk?"	→ **suprascapular nerve**
"SLAP + cyst on MRI – treatment?"	→ **supraglenoid cyst decompression + SLAP repair**
"Instrument passed medial to glenoid – nerve origin?"	→ **superior trunk**
"Arthroscopic DCE advantage?"	→ **evaluation of GH joint**
"Medial biceps dislocation cause?"	→ **subscapularis tear**
"Who coined the term SLAP?"	→ **Snyder (1990)** (TO)
"Arthroscopic sign confirming type II SLAP?"	→ **peel-back** (TO)

– – Active space – –

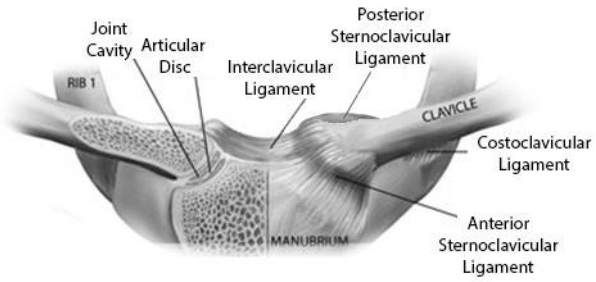
STERNOCLAVICULAR JOINT

– Shoulder & Elbow · 52 / 60 –

1 Basics

Joint anatomy

Type	**diarthrodial saddle** joint, **incongruous (~50% contact)**; **fibrocartilage**; contains **intra-articular disc** ✗ Wrong options: hyaline/synovial cartilage
Motion	arm elevation to **90° → 30° rotation** at SC joint



SC joint: disc, anterior/posterior SC, costoclavicular, interclavicular ligaments

Stabilising ligaments

Ligament	Function
Posterior SC capsular ligament	**Primary restraint to anterior–posterior stability**
Anterior SC ligament	Primary restraint to **superior displacement** of medial clavicle
Costoclavicular (rhomboid) – anterior fasciculus	Resists **superior rotation + lateral displacement**
Costoclavicular – posterior fasciculus	Resists **inferior rotation + medial displacement**
Intra-articular disc ligament	**Prevents medial displacement**; secondary restraint to superior displacement

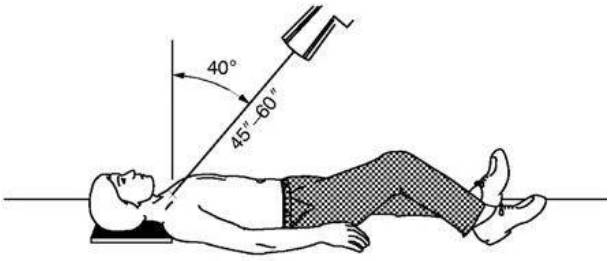
Medial clavicle anatomy

First to ossify	**clavicle**, **5th–6th week of gestation**
Last to fuse	**medial clavicular epiphysis**, closes **20–25 y** (last physis)
Before 25 y	most "SC dislocations" are **medial clavicular physeal fractures** → distinguish

4 Diagnosis & Classification

Key radiology findings

Serendipity view	**40° cephalic tilt** of sternum + clavicles; best plain radiograph



Serendipity view: 40° cephalic tilt

CT **most sensitive + specific**

5 Management

Instability after medial clavicle resection – figure-of-eight reconstruction

Cause medial clavicle resection (e.g. for **SC degenerative joint disease**) compromising **costoclavicular ligament → medial clavicular instability**: ↑ pain, **medial bump**, digit tingling

Treatment **semitendinosus figure-of-eight graft** through **clavicle + manubrium** → best **peak load-to-failure**

(✗ Wrong options: **Distractors** – subclavius tendon transfer) · medial clavicular osteotomy · further medial resection · SC arthrodesis

X vs Y

	SC dislocation	Medial clavicle physeal fracture
Age	Adult (physis closed)	**<25 y** (physis closes 20–25)
Frequency <25 y	Less common	**Most injuries**

How it's asked

"Instability after medial clavicle resection (medial bump, tingling) – procedure?"	→ **semitendinosus figure-of-eight graft** ✗ Wrong options: subclavius transfer · further resection · arthrodesis
"Primary restraint to AP translation of SC joint?"	→ **posterior SC capsular ligament**
"Ligament preventing medial displacement of clavicle?"	→ **intra-articular disc ligament**
"Best plain view of SC joint?"	→ **serendipity view (40° cephalic tilt)**; **CT** most sensitive
"Last physis to close?"	→ **medial clavicle, 20–25 y**

– – Active space – –

SUBACROMIAL IMPINGEMENT

– Shoulder & Elbow · 53 / 60 –

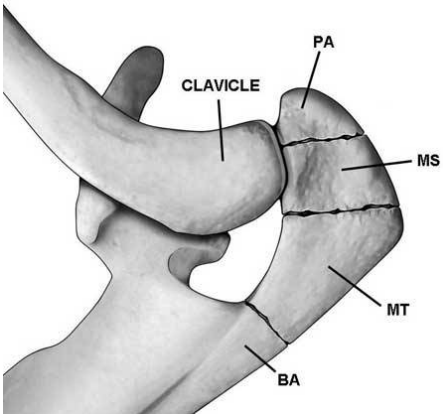
1 Basics

Epidemiology

Frequency	**most common cause of shoulder pain** · **44–65%** of shoulder disorders · with biceps tendinitis, among commonest causes

Anatomy – acromion ossification

Ossification centres	**pre-acromion** (tip) · **meso-acromion** (mid) · **meta-acromion** (base)
Os acromiale	failure of centres to fuse



Acromial ossification centres

Pathophysiology – metalloproteases

Extrinsic compression	cuff between humeral head and **anterior acromion**, **CA ligament**, **AC joint**
Intrinsic degeneration	supraspinatus attrition → can't balance head → **superior migration** → narrowed subacromial space
Bursitis	abutment → inflamed subacromial bursa → anti-inflammatory treatment rationale

Genes & lab markers

Marker	Condition
Metalloproteases (MMP) ↑	**Subacromial bursitis / impingement** ✓
TNF, **COX-1 + COX-2** ↑	Subacromial bursitis
TSH (hypothyroid), **HbA1c** (diabetes)	Wrong – tests for **adhesive capsulitis**
WBC	Wrong – tests for **septic bursitis**
ALP	Wrong – tests for osteoblastic tumours, Paget, hyperparathyroidism
AFP, PSA, CEA, CA-125	Wrong – tumour markers

Histology (tendinopathy) **disorganised collagen** · **mucoid degeneration** · inflammatory cells · bursa: high MMP + cytokines

2 Mechanism

Continuum (Neer) **impingement + bursitis (stage 1)** → tendinitis/fibrosis → partial → full-thickness → **massive tear** → **cuff tear arthropathy**

Associated conditions **hooked (type III) acromion** · **os acromiale** · **posterior capsular contracture** · **scapular dyskinesia** · tuberosity malunion · instability

3 Presentation

Pain **insidious** · **overhead activity** + lifting away from body · **night pain** = **poor predictor of nonop success**

Strength **usually normal**; full passive ROM

Test	Technique	Positive / meaning
Neer sign	Passive FF **>90°**	Pain
Neer test	Subacromial **lidocaine** then repeat	**Pain relieved**
Hawkins	**IR + passive FF to 90°**	Pain
Jobe (empty can)	Resisted pronation + FF 90°	**Supraspinatus** pathology
Painful arc	Abduction in scapular plane	Pain **60–120°**
Yocum	Hand on opposite shoulder, raise elbow	**Sensitive, nonspecific**
Internal impingement	ABER	Pain
Speed's	–	Biceps tendinitis

O'Brien pain not relieved in neutral equivocal → **not labral**; don't order MRA first

4 Diagnosis & Classification

Key radiology findings

View	Shows
True AP	**Acromiohumeral interval** – normal **7–14 mm**
30° caudal tilt	**Subacromial spur**
Supraspinatus outlet	**Acromial morphology** (Bigliani)
Axillary lateral	**Os acromiale** (best)

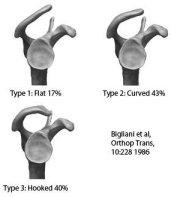
X-ray findings proximal humeral migration (CTA) · traction osteophytes · **CA ligament calcification** · GT cysts · **type III hooked acromion** · os acromiale

MRI degree of cuff pathology; subacromial/subdeltoid bursitis · CT arthrogram + US also image cuff accurately

MRI indications acute injury · failed nonop · **weakness** / suspected large tear
✗ Wrong options: first with benign exam

– – Active space – –

Bigliani classification of acromion morphology

Type	Shape	Diagram
I	**Flat**	
II	**Curved**	
III	**Hooked** → associated with impingement	

Reliability **poor inter-observer**

Acromial morphology – os acromiale

Types pre- · **meso-** · meta- os acromiale (named by unfused centre)

MRI (axial + sagittal) unfused segment ± **edema** in skeletally mature adult → **meso os acromiale**
✗ Wrong options: type I/III acromion



Os acromiale on outlet view

5 Management

Subacromial impingement (no cuff tear)

- First line → **PT (cuff + periscapular) + NSAIDs ± subacromial steroid**
  (scapular dyskinesia → integrated kinetic-chain rehab)
- Failed ≥4–6 months nonop → **subacromial decompression ± acromioplasty**
- Os acromiale → stage 1 fuse with graft → **stage 2 acromioplasty**

Nonoperative management – physical therapy & PRP

Mainstay **PT** (aggressive cuff strengthening + periscapular stabilisation), NSAIDs, subacromial injection

Benign exam (full strength, Jobe/Speed painful) **PT** before imaging
✗ Wrong options: AC injection · ASD · MRA · MRI

Outcome — **most improve with PT alone** (even with type III acromion) · but **tendinitis tends to recur**
✗ Wrong options: **"most fail nonop"**

PT vs surgery — PT **as effective** as subacromial decompression ± debridement (tendinitis, partial tears) · ASD + PT not better at 2 y

PRP — **no effect** on pain, function, ROM at **1 y** vs therapy alone · not harmful · does not prevent arthropathy · meta-analysis: insufficient evidence

Operative criteria

Procedure	Indication
Subacromial decompression / acromioplasty	Impingement **failed ≥4–6 months** nonop
Two-stage os acromiale	Fuse with bone graft → later acromioplasty (✗ Wrong options: direct excision → deltoid dysfunction)
Avoid acromioplasty + CA ligament release	**Massive irreparable cuff tear** (preserve CA arch → avoid anterosuperior escape)

Technique – acromioplasty

Step 1 — **anterior acromionectomy** (arthroscopic: limited by **anterior deltoid origin**, keep intact)

Step 2 — **anteroinferior acromioplasty** smooths undersurface (rasp open; shaver/burr arthroscopic) · open → meticulous **deltoid repair** to bone

Subacromial decompression outcomes (workers' compensation)

Lower satisfaction — **workers' compensation** (also longer return to work) · **anxiety + depression**
✗ Wrong options: dominant arm · male · smoker · age <60

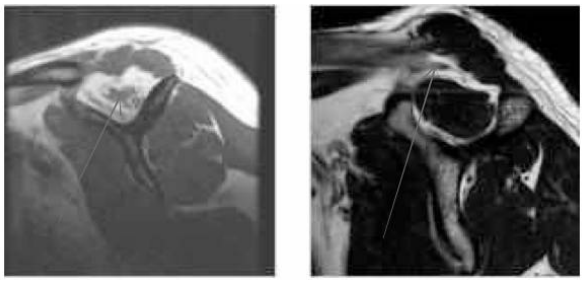
Massive cuff tears – irreparability & SCR contraindications (TO)

Irreparable (TO) — can't reach tuberosity despite release (size, retraction, atrophy, fatty infiltration)

Irreparability marker (TO)	Value
Massive tear	**>5 cm** (Cofield) or **≥2 tendons**
Pattern	**Posterosuperior (SS + IS)** > anterosuperior (SS + subscap)
Fatty atrophy	**Goutallier III–IV**
Acromiohumeral distance	**<6 mm**
Function	**Chronic pseudoparalysis** (can't elevate **>90°**; some say >45°)
MRI	**Positive tangent sign**

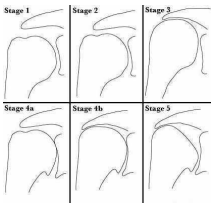
Tangent sign (TO) — sagittal oblique; line **top of scapular spine → top of coracoid** · SS belly **below line = positive → fatty atrophy, irreparable supraspinatus** / poor repair outcome
✗ Wrong options: subscapularis · retraction medial to glenoid · biceps dislocation

– – Active space – –



Tangent sign

Hamada classification of cuff tear arthropathy (TO)

Stage	Finding (TO wording)	Diagram
1	AHI **>6 mm**	
2	AHI <7 mm	
3	AHI <7 mm + **acetabularisation** of acromion	
4a	AHI <7 mm + **GH arthritis**, no acetabularisation	
4b	AHI <7 mm + acetabularisation + GH arthritis	
5	AHI <7 mm + **humeral head osteonecrosis**	

Hamada versions AHI **<7 mm**, stage 5 **osteonecrosis** (TO) · stage 2+ AHI **≤5 mm**, stage 5 **head collapse** (Orthobullets) – read the options

Superior capsular reconstruction (TO)

Concept (TO) passive restraint to **superior head translation** → restores force couples, ↓ subacromial impingement, ↓ pain

Graft (TO) folded **fascia lata 6–8 mm** or **acellular dermal 3–4 mm** · anchors on **superior glenoid rim** + **medial tuberosity**

SCR (TO)	Criteria
Indications	Failed nonop **massive irreparable** tear (SS + IS) · **Hamada 1–2** · **intact/reparable subscapularis** · **functional deltoid** · intolerable pain · pseudoparalysis
Contraindications	**Irreparable / torn subscapularis** ✓ · **Hamada ≥3** · GH OA

✗ Contraindication distractors (TO) Hamada I · pseudoparalysis · massive tear failed conservative treatment (= indications)

Young patient, massive irreparable posterosuperior tear (TO)

Option (TO)	Indication
Latissimus dorsi transfer	Retracted posterosuperior tear, **Goutallier ≥3** SS + IS; not if subscap insufficient or deltoid atrophy
Lower trapezius transfer	**More anatomical** for posterosuperior; promising
Pectoralis major transfer	**Anterosuperior** (SS + subscap) – most common
SCR	Young, no significant arthritis
Reverse TSA	**Older** patients with cuff arthropathy → **least preferred in 35-year-old** ✓

Tendon transfer prerequisites (TO) young, active, no significant GH arthritis, **intact subscapularis**

6 Complications

Deltoid dysfunction failed repair after open acromioplasty · **excessive acromionectomy** arthroscopically · **direct excision of os acromiale**

Anterosuperior escape acromioplasty + CA ligament release in **massive irreparable** tear

X vs Y

	Subacromial impingement	Adhesive capsulitis	Septic bursitis
ROM	**Full passive**	↓	–
Marker	**MMP, TNF, COX**	TSH, HbA1c (associations)	**WBC**

	Latissimus transfer (TO)	Pectoralis major transfer (TO)
Tear	**Posterosuperior**	**Anterosuperior**
Requires	Intact subscapularis, deltoid	–

How it's asked

"Insidious overhead pain, full ROM, MRI bursitis – elevated marker?"	→ **metalloproteases** ✗ Wrong options: TSH · HbA1c · WBC · ALP
"Markers contributing to subacromial bursitis pain?"	→ **metalloproteases** ✗ Wrong options: AFP, PSA, CEA, CA-125
"Partial SS tear, lidocaine relieved 90%, asks for PRP?"	→ **no effect on pain, function, ROM at 1 year**
"Type III acromion, Hawkins + empty can +, 5/5 – true about nonop?"	→ **most improve with PT alone** ✗ Wrong options: most fail · ASD better at 2 y · PRP better · never recurs

– – Active space – –

"Golfer, Jobe + Speed's painful, full strength — next?"	→ **PT for cuff strengthening** ✗ Wrong options: ASD · MRI · MRA · AC injection
"Lower satisfaction after ASD?"	→ **workers' compensation**
"Axial + sagittal MRI unfused acromion with edema?"	→ **meso os acromiale**
"Contraindication to SCR?"	→ **irreparable subscapularis tear** (TO)
"35-year-old, pseudoparalysis, AHD <6 mm, Goutallier 3 — least preferred?"	→ **reverse shoulder arthroplasty** (TO)
"Positive tangent sign predicts?"	→ **irreparable supraspinatus (fatty atrophy)** (TO)
"Best view for os acromiale?"	→ **axillary lateral**
"Night pain significance?"	→ **poor predictor of nonop success**
"Massive irreparable tear — avoid?"	→ **acromioplasty + CA ligament release** (anterosuperior escape)

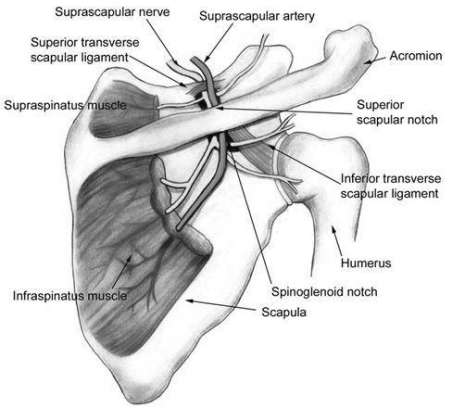
SUPRASCAPULAR NEUROPATHY

– Shoulder & Elbow · 54 / 60 –

1 Basics

Anatomy – suprascapular nerve course

Origin	**C5–C6**, **superior (upper) trunk** of brachial plexus → across **posterior triangle of neck** → superior border of scapula
Course	under **superior transverse scapular (suprascapular) ligament** at **suprascapular notch** → supraspinous fossa (**supraspinatus**) → around lateral border of scapular spine through **spinoglenoid notch** (under **spinoglenoid ligament**) → infraspinous fossa (**infraspinatus**)
Suprascapular ligament	from medial base of coracoid over notch · **artery ABOVE, nerve BELOW**
Function	**motor only** (SS + IS) · **no sensory** → no sensory deficit
Transglenoid fixation risk	nerve lies **1.5 cm medial** to glenoid labrum at spinoglenoid notch



Suprascapular nerve – sites of compression

Site	Level	Muscles weak
Suprascapular notch	**Proximal**	**Supraspinatus + infraspinatus**
Spinoglenoid notch	**Distal** (SS branch already given off)	**Infraspinatus only**

TO wording	"infraglenoid notch" compression → infraspinatus only (= spinoglenoid notch)
Atraumatic compression sites	**suprascapular + spinoglenoid notches** (fascial bands, ganglion cysts) ✗ Wrong options: scalenus anterior · cervical rib · conjoined tendon · subcoracoid

Epidemiology & associations

Who	**overhead athletes**: **volleyball** (classic), baseball pitchers/catchers, tennis · also weightlifters, rowers, linebackers
Associated	**SLAP tears** · posterior labral tears · internal impingement

– – Active space – –

2 Mechanism

Suprascapular notch – causes

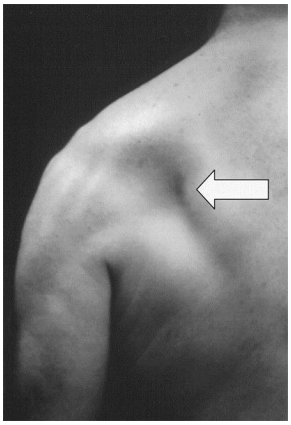
Causes	**ganglion cyst** (often with labral tear) · **transverse scapular ligament** entrapment/thickening · **fracture callus**

Spinoglenoid notch – causes

Causes	**posterior labral tear → cyst** · spinoglenoid ligament · spinoglenoid notch ganglion · **traction** (**45% of volleyball players**) · transglenoid fixation
Cyst mechanism	labral tear = **one-way valve** → synovial fluid exits joint, can't drain → paralabral cyst
Cyst frequency	ganglion cysts at **spinoglenoid notch** > **less commonly suprascapular notch** · cyst origin: transverse scapular ligament, scapular fibrous tissue, or GH joint
Labral source	**SLAP** most often, posterior labrum less (OB) · **posterior labral tear** (Orthobullets, OB) – read the options; posterior SLAP / **posterosuperior** labral tear fits both

3 Presentation

Symptoms	**deep, diffuse, posterolateral shoulder pain** · weakness, fatigue with overhead activity · ↓ velocity · no numbness
Suprascapular notch	tender notch · **supraspinatus weak**: abduction 90° + 30° FF + IR (**Jobe +**) · **infraspinatus weak**: **ER with elbow at side** · atrophy above + below spine
Spinoglenoid notch	**isolated weak ER with arm at side** · **atrophy below scapular spine** (posterior shoulder) · **supraspinatus normal** (empty can negative) · O'Brien + (associated SLAP)



Isolated infraspinatus atrophy

Most specific exam finding (spinoglenoid cyst)	**weakness of ER** ✗ Wrong options: impingement sign · apprehension · active compression · abduction weakness
ER in elevation	isolated IS palsy: **full ER with arm elevated** · combined SS + IS: **can't ER in forward elevation**

Distinguishing shoulder tests

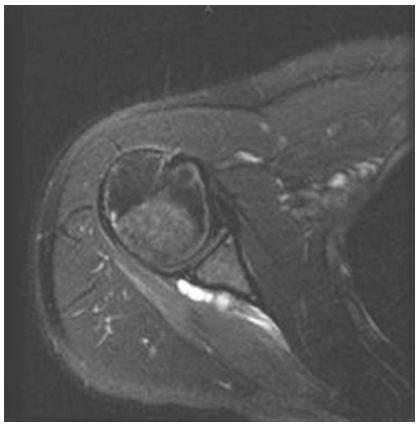
Test	Muscle / lesion	In SSN entrapment
Jobe / empty can	**Supraspinatus**	Abnormal at **suprascapular notch** ✓
ER with arm at side	**Infraspinatus**	Abnormal at **both** sites ✓
Hornblower (90° abduction + 90° ER, falls into IR)	**Teres minor** (axillary)	Normal
Belly press · bear hug · lift-off	**Subscapularis**	Normal
Anterior load and shift (supine, 40–60° abduction + FF)	Anterior instability	Normal
Kim test (seated, 90° abduction → 45° FF + axial + posteroinferior force)	**Posterior instability**	Normal
Jerk test	**Posterior labral tear**	May be + (cyst source)

4 Diagnosis & Classification

Key radiology findings

MRI identify **compressive mass / cyst** + associated **labral lesion** · cyst: well-defined, **low T1, high T2** · denervation: ↓ bulk + **fatty infiltration** (SS ± IS)

Paralabral cyst most commonly with **labral tears**



Spinoglenoid notch cyst



Suprascapular notch cyst

– – Active space – –

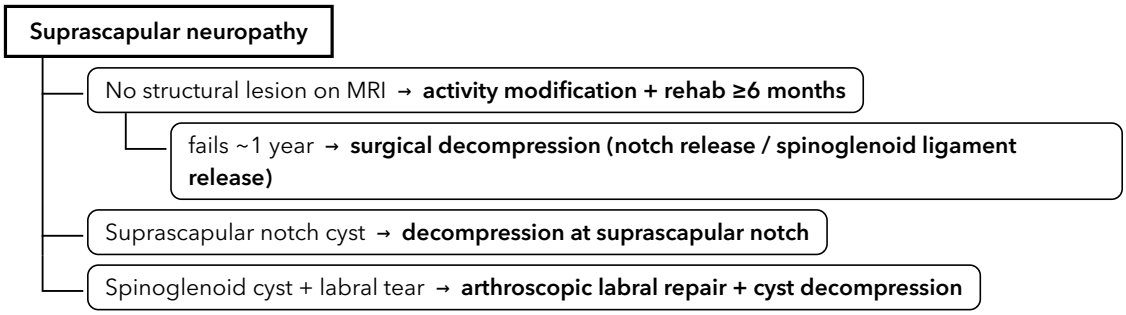
Diagnosis – EMG & compression sites

EMG/NCV	**diagnostic**; with MRI = investigations of choice · localises compression
Isolated ER weakness + atrophy, MRA normal (no cyst, no tear)	**EMG/NCV first** → found notch entrapment with early SS denervation ✗ Wrong options: cervical MRI first · subacromial steroid · notch release before EMG · CRP/ESR
C5 radiculopathy	weakness out of proportion; not when specific SSN pattern

Lesion (distractor)	Nerve / site	Deficit
Quadrangular (quadrilateral) space	**Axillary + PHCA**	**Teres minor** atrophy, deltoid (forward elevation, abduction)
Triangular interval	**Radial nerve + profunda brachii**	Elbow + wrist extension
Subscapular recess	Anterior to subscap, below coracoid	Loose bodies
Upper/lower subscapular nerves	Subscapularis	IR (lift-off)
Long thoracic (C5–7 roots)	Serratus anterior	Not affected by spinoglenoid cyst

Brachial plexus figure	suprascapular = **A (upper trunk)** · dorsal scapular · medial pectoral · lower + upper subscapular = wrong options

5 Management



Operative criteria

Situation	Treatment
No structural lesion	**Nonop ≥6 months** (posterior capsule stretching for spinoglenoid)
No lesion + failed **~1 year** nonop	Suprascapular notch release · or **spinoglenoid ligament release** (posterior approach)
Structural lesion (cyst) at suprascapular notch	**Nerve decompression at notch**
Labral tear + spinoglenoid cyst	**Arthroscopic labral repair + cyst decompression** ✓

Treatment – labral repair with cyst decompression

Young competitive athlete, posterior labral tear + spinoglenoid cyst	**cyst decompression + posterior labral repair** ✗ Wrong options: 6 wk rest · PT · suprascapular notch decompression · cuff repair + acromioplasty
Catcher, posterosuperior tear + large cyst, severe ER weakness	**arthroscopic repair + decompression** ✗ Wrong options: aspiration + steroid · rest · acromioplasty · rehab
Labral tear key	address tear (cause of cyst) · **no isolated cyst decompression** when tear present
Labral repair alone	not enough (doesn't decompress cyst, atrophy may progress — OB) · "good outcomes with labral repair alone; needle aspiration also used" (one OB explanation) — read the options; best answer = **repair + decompression**
Failed 6 wk PT with cyst + IS atrophy	operate ✗ Wrong options: another 6 wk PT · cervical MRI + EMG · tenodesis
Mechanical compression	surgical decompression usually required

6 Complications

Iatrogenic injury

Arthroscopic SSN release at suprascapular notch → injured nerve → weak **abduction, forward elevation + ER**, wasting **above + below scapular spine** = suprascapular notch level
✗ Wrong options: spinoglenoid = IS only · axillary · spinal accessory · dorsal scapular

SLAP repair	anchors/instruments **medial to glenoid** → SSN injury

X vs Y

	Suprascapular notch	**Spinoglenoid notch**
Muscles	**SS + IS**	**IS only**
Weakness	**Abduction (Jobe) + ER**	**ER at side** only
Atrophy	Above + below spine	**Below spine**
Usual cause	Cyst, **transverse scapular ligament**, callus	**Paralabral cyst (SLAP/posterior labrum)**, traction
Treatment (lesion)	Decompress at notch	**Labral repair + cyst decompression**

	Spinoglenoid cyst	**Quadrilateral space syndrome**
Nerve	Suprascapular	**Axillary**
Atrophy	**Infraspinatus**	**Teres minor** (± deltoid)
Test	ER at side weak	Hornblower; forward elevation/abduction weak

– – Active space – –

How it's asked

"Burning pain, SS + IS atrophy, weak abduction + ER – site?"	→ **suprascapular notch** (TO)
"Volleyball, isolated IS atrophy, axial MRI cyst – compression?"	→ **suprascapular nerve in spinoglenoid notch** ✗ Wrong options: quadrangular space · triangular space
"Volleyball, MRI cyst in suprascapular notch – weakness?"	→ **abduction + external rotation**
"SLAP + cyst on sagittal MRI – extra exam finding?"	→ **weak external rotation**
"Weak ER at side, atrophy below spine, O'Brien + – MRI?"	→ **SLAP tear + spinoglenoid notch cyst**
"Isolated IS atrophy on clinical photo + MRI – cyst location?"	→ **spinoglenoid notch** ✗ Wrong options: quadrangular space · subscapular recess
"Isolated ER weakness, normal flexion/abduction, 44 y?"	→ **spinoglenoid notch cyst** ✗ Wrong options: CN XI · Parsonage-Turner
"Normal empty can + belly press, weak ER – cause?"	→ **SSN entrapment at spinoglenoid notch**
"Suprascapular notch cyst – weakness on exam?"	→ **ER with arm at side** ✗ Wrong options: scapular protraction · IR
"Volleyball, posterosuperior pain – most specific exam finding?"	→ **weakness of external rotation**
"Weak ER at side, normal lift-off – nerve + site?"	→ **suprascapular nerve, spinoglenoid notch**
"Pitcher, O'Brien +, ER weak, empty can 5/5 – muscle at risk?"	→ **infraspinatus**
"Posterior labral tear + paralabral cyst – structure affected (MRI figure)?"	→ **infraspinatus** ✗ Wrong options: supraspinatus · subscapularis
"Spinoglenoid cyst – exam?"	→ **weak ER with arm adducted** ✗ Wrong options: Jobe · belly press · Hornblower
"Linebacker, weak ER at side, spinoglenoid cyst – arthroscopic finding?"	→ **SLAP tear** ✗ Wrong options: Bankart · PASTA · Hill-Sachs
"Cyst in suprascapular notch on MRI – which muscles?"	→ **supraspinatus + infraspinatus**
"Spinoglenoid cyst on MRI – which muscle weak (figure)?"	→ **infraspinatus (B) only**
"Weak ER at side, full ER when elevated – MRI?"	→ **spinoglenoid cyst + IS fatty atrophy**

"Weak forward elevation + ER at 0° abduction — commonest compression site (figure)?"	→ **suprascapular nerve (A)**
"Rower, IS atrophy, negative empty can — cause?"	→ **spinoglenoid notch cyst** ✗ Wrong options: Bankart · subscapularis tear
"Volleyball, suprascapular notch cyst — abnormal test?"	→ **Jobe's test** ✗ Wrong options: load-and-shift · bear hug · Hornblower · Kim
"Weightlifter, weak ER, posterior labral tear + cyst — treatment?"	→ **spinoglenoid cyst decompression + posterior labral repair**
"Catcher, cyst + posterosuperior labral tear — management?"	→ **arthroscopic repair + decompression**
"Jerk test +, labral tear + spinoglenoid cyst — procedure?"	→ **labral repair + cyst decompression** ✗ Wrong options: isolated decompression
"Posterosuperior tear, cyst, mild IS atrophy, failed 6 wk PT — next?"	→ **labral repair + cyst decompression** ✗ Wrong options: repair only · cervical MRI
"MRI cyst, abnormal EMG — nerve?"	→ **suprascapular**
"Tennis player, paralabral cyst, ER weakness — cause?"	→ **suprascapular nerve compression** ✗ Wrong options: C5 radiculopathy · IS tear
"Superior labral cyst — clinical finding?"	→ **atrophy of posterior shoulder** ✗ Wrong options: any sensory deficit
"Weak ER, no tear or cyst on MRA — next?"	→ **EMG/NCV**
"Atraumatic SSN compression sites?"	→ **suprascapular + spinoglenoid notches**
"After arthroscopic notch release: weak abduction + ER, wasting above and below spine?"	→ **suprascapular nerve at suprascapular notch**

– – Active space – –

THORACIC OUTLET SYNDROME

– Shoulder & Elbow · 55 / 60 –

1 Basics

Epidemiology

Definition compression of **brachial plexus ± subclavian vessels** between neck and axilla (first rib–clavicle); heterogeneous, potentially disabling

Incidence **1–2%** of population

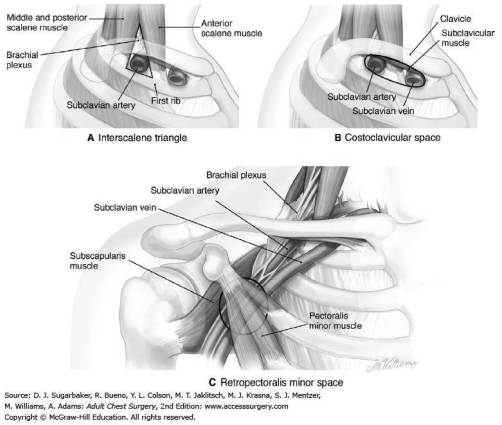
Demographics ♀ > ♂ **(3:1)**; **thin, long neck, drooping shoulders**; age **20–60**

Type	Frequency	Note
Neurogenic	**95% (most common)**	Lower plexus
Venous	**4%**	Paget-Schroetter; athletic ♂ > ♀
Arterial	**<1%**	Aneurysm, emboli; athletic ♂ > ♀

Anatomy – thoracic outlet spaces

Boundaries (OB) **clavicle, first rib, subclavius, costoclavicular ligament, anterior scalene**

Space	Borders	Contents	Test
Interscalene triangle (proximal)	**Ant: anterior scalene · Post: middle scalene · Inf: 1st rib**	**Plexus trunks + subclavian artery** (✗ Wrong options: vein (runs **beneath anterior scalene**))	**Adson**, supraclavicular pressure
Costoclavicular space (middle)	Ant: clavicle + **subclavius** · Post: 1st rib + scalenes · Med: **costoclavicular ligament** · Lat: upper scapular border	**Divisions + subclavian artery + vein**	**Costoclavicular manoeuvre**
Retropectoralis minor (subcoracoid; distal)	Sup: **coracoid** · Ant: **pectoralis minor** · Post: **ribs 2–4**	**Cords + axillary artery + vein**	**Wright**



Three spaces of the thoracic outlet

Most affected (OB) **subclavian artery, vein + lower trunk (C8–T1)**

2 Mechanism

Pathogenesis **anatomic predisposition + superimposed neck trauma** (acute or repetitive)

Split **soft tissue 70% · osseous 30%**

Soft tissue	Detail
Scalene abnormalities	**Anterior scalene hypertrophy** · plexus **through** anterior scalene · **middle scalene anterior insertion** on 1st rib
Scalenus minimus	Accessory; **30–50%** of TOS; C-transverse process → 1st rib **between subclavian artery + T1 root**
Fibromuscular bands	↑ stiffness, ↓ compliance; congenital band → lower trunk
Costoclavicular ligament	Abnormal insertion → **Paget-Schroetter**
Tumours	**Pancoast** (1–3% lung cancers; **lacks cough/haemoptysis/dyspnoea**), neuroblastoma, plexus schwannoma
Pectoralis minor	Abnormal

Osseous	Detail
Cervical rib	**<1%** population; from **C7**
Prominent C7 transverse process	
Clavicle / 1st rib	Displaced fracture, **hypertrophic callus**, malunion
AC / SC injury	Dislocation
Tumours	1st-rib metastasis (**breast, prostate, kidney**), osteoid osteoma

Cervical rib types

Type	Description
1	**Complete**, articulates with 1st rib or manubrium
2	Incomplete, **free bulbous tip**
3	Incomplete, **fibrous band** distally
4	**Short bar** (mm) beyond C7 transverse process

Overuse repetitive overhead lifting, **hyperabduction**; **weightlifting, rowing, swimming**

Vascular damage aneurysm, thrombosis, emboli, limb-threatening ischaemia

Paget-Schroetter syndrome

Definition **venous TOS** in **well-developed young athletes → upper-limb DVT**

Cause intermittent **subclavian vein** obstruction in **costoclavicular space** by **abnormal costoclavicular ligament** or **anterior scalene hypertrophy**

– – Active space – –

3 Presentation

Neurogenic

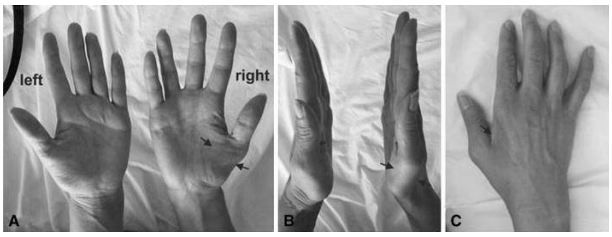
Pain	neck, **trapezius (92%)**, chest, shoulder, arm
Paraesthesia	**98%**; **non-radicular**, **wide plexus distribution** · **lower plexus (C8–T1) or combined (C5–T1) in 90% → ring + small fingers**, medial forearm
Other	weakness, numbness, **heaviness with overhead activity** · activity-related + **night symptoms** (↓ pressure → returning sensation felt as pain) · uni- or bilateral
Posture	rounded shoulders, ↑ kyphosis, depressed/downward-rotated scapula

Gilliatt-Sumner hand (TO)

Definition (TO)	rare, severe sign of **true neurogenic TOS**; **chronic lower-trunk (C8–T1)** compression by **cervical rib or congenital fibrous band**
Pattern	**preferential APB atrophy, relative hypothenar sparing** (TO, ✓) · atrophy of **APB + hypothenar + interossei** (Orthobullets, TO Elvey list) – read the options
Why APB (TO)	**T1-innervated** muscles hit harder (median + ulnar both get lower-trunk fibres)
Red flag (TO)	thenar wasting usually = CTS, but in TOS **+ medial forearm sensory change** + milder ulnar intrinsic involvement

(✗ Wrong options: **Distractors (TO)** – symmetric interossei/hypothenar wasting sparing thenar) · claw hand from ulnar compression at humerotricipital groove · thumb/index C6 sensory loss

Diagnosis (TO)	EDX (neurogenic changes) + MRI/US
Treatment (TO)	**surgical decompression** → stops progression of atrophy + pain; partial motor recovery possible
Differential (TO)	**CTS, ALS (motor neuron disease), Hirayama disease**



Gilliatt-Sumner hand: thenar (APB) wasting

Vascular

Venous	**episodic cyanosis + swelling**, distended veins, deep arm ache, **heaviness after activity**
Arterial	**unilateral Raynaud-type**: coolness + **pallor → cyanosis → erythema**, worse in cold; **hand** predominantly; ulcers/microemboli (rare)

Provocative tests & neurogenic TOS

All provocative tests high false-positive rate

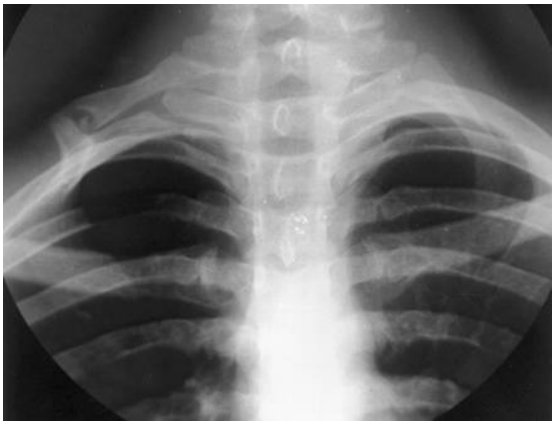
Test	Technique	Positive	Space
Supraclavicular pressure	Squeeze upper trapezius + anterior scalene **30 s**, arm at side	Pain/paraesthesia	Interscalene
Adson	Arm slightly abducted + ER (TO: abducted + **extended**), elbow extended; **neck extended + rotated TOWARD tested side**; inhale + hold	↓/absent radial pulse ± symptoms; **51% of normals** ↓ pulse; TO: **low sensitivity + PPV** (modified: tilt to opposite side / symptom aggravation)	**Interscalene**
Costoclavicular (military / "attention")	Shoulders retracted + depressed, chest out; shoulder extended ~30° **1 min**	↓ pulse ± symptoms	**Costoclavicular**
Wright (hyperabduction)	Neck rotated **away**; shoulder ER + **max abduction above head 1 min** (TO: 1–2 min)	↓ pulse (**7% of normals**) ± symptoms	**Retropectoralis minor**
Roos / EAST	Orthobullets: **abduct 90° + elbow 90°**, open/close hands **3 min** · TO: arms **abducted, ER, elevated overhead**, open/close fingers **1–3 min**	**Reproduction of symptoms**; can't complete 3 min; relief on dropping arms	**Entire outlet**
Cyriax release	Examiner behind, elevates shoulder girdle passively **3 min**	Symptoms reproduced (unloading)	Unloading
Halstead (TO)	Downward traction on arm, **neck hyperextended + rotated to opposite side**	**Radial pulse disappears**	
ULTT of Elvey (TO)	Abduct 90°, elbow extended, **wrist dorsiflexed**, then **ear to contralateral shoulder**	Ipsilateral symptoms = **neurogenic TOS**	Plexus stretch
✗ Wrong option: **Spurling (TO)**	Neck extension + lateral flexion + **axial compression**	**Radicular** pain	**Cervical radiculopathy**

– – Active space – –

Key radiology findings

Test	Use / finding
CXR + C-spine X-ray	**Cervical rib**, prominent C7 TP, low shoulder girdle, **Pancoast** (or normal)
CT	Osseous lesions, rib/clavicle malunion
MRI	Soft-tissue anomalies
EMG/NCV	Historically equivocal (normal until permanent damage) · early: **↓ NCV in medial antebrachial cutaneous + median motor to APB** (C8–T1)
Doppler US	Subclavian vein obstruction/thrombosis — **95% sensitive, 92% specific** for venous TOS
CTA/MRA · arteriography	**Embolic disease / aneurysm**
Venography	Suspected **subclavian/axillary vein thrombosis**

Normal neuro exam + EMG/NCS does **not** exclude TOS (✓ cervical rib vignette)



Bilateral cervical ribs

Diagnosis – cervical rib

Classic vignette **bilateral ring + small finger tingling**, worse **overhead**, normal neuro exam + EMG/NCS, cervical ribs on X-ray → **TOS**
✗ Wrong options: C5–6 / C6–7 disc · cubital tunnel · radial tunnel

5 Management

Thoracic outlet syndrome

- Neurogenic → **PT, activity modification, NSAIDs**
  - fails 6 mo / progressive atrophy → **decompression**
- Scalene contracture → **US-guided scalene block (lidocaine / Botox)**
- Vascular → **by lesion (see table)**

Nonoperative

First line	avoid provocative activity (overhead, ± change job) · NSAIDs, muscle relaxants · **PT**: core/back + shoulder-girdle strengthening, **posture**, relaxation · TENS
Worse results	**obese** · **workers' compensation** · **double crush** (carpal/cubital tunnel)
Anterior scalene block	**US-guided lidocaine or botulinum toxin**; successful block → **14% higher** good surgical outcome
Leffert (OB)	reserve surgery for **intolerable** symptoms (complications significant)

Operative criteria

Indication	
Failed nonop 6 months	Decompression
Progressive atrophy / worsening neuro deficit	Decompression
Embolic events · stenosis with pain + insufficiency · **subclavian aneurysm** · thrombosis with critical ischaemia	Vascular intervention

Decompression procedures

Procedure	**Indication**
1st rib resection + anterior + middle scalenectomy + neurolysis	**Most common**; **95% good outcomes**; combined transaxillary + supraclavicular
Isolated scalenectomy	**Upper plexus** symptoms · **no bony anomaly** · muscular/obese · **recurrence after 1st rib resection**
Isolated pec minor tenotomy	Symptoms reproduced at **retropectoralis minor** space
Cervical rib resection · band release · costoclavicular ligament resection · clavicle malunion ORIF	Per cause

Vascular intervention

Lesion	**Treatment**
Acute embolism	**IV heparin** ± embolectomy ± thrombectomy ± **tPA** + anticoagulation
Small-vessel embolism	**tPA + anticoagulation**
Large/proximal embolism	**Embolectomy + anticoagulation**
Mild stenosis	**Endovascular stent**
Subclavian aneurysm / severe stenosis or thrombosis + critical ischaemia	**Resection** ± repair / saphenous / arterial / synthetic graft
Chronic emboli + critical ischaemia	**Bypass**

– – Active space – –

Approaches

	Transaxillary	Supraclavicular	Posterior
Use	**Most common**; **best 1st rib exposure**	**Upper plexus**, scalenes, 1st rib neck, vessels	**Recurrent TOS**, **prior neck surgery**
Best for	1st rib, cervical rib, bands, **lower plexus (C7–T1)** neurolysis	**Isolated scalenectomy + arterial reconstruction**	Proximal plexus
Pros	**Cosmetic scar**, no NV retraction for rib		
Cons	**Plexus injury risk**	Poor 1st rib view (needs plexus/vessel retraction)	Extensive dissection → shoulder dysfunction; **LTN, dorsal scapular, accessory nerve** risk

6 Complications

Pneumothorax **one of the most common** after 1st rib resection

X vs Y

	Neurogenic TOS	Cervical radiculopathy	Cubital / carpal tunnel
Distribution	**Non-radicular, wide (plexus)**, C8–T1	**Radicular**	**Clear dermatomal/peripheral** distribution
Provocation	Overhead, Roos, Adson, Wright	**Spurling**	
EMG	Often normal early; MABC + APB motor ↓		

	Gilliatt-Sumner (TOS)	CTS
Thenar (APB) wasting	**Yes** (preferential)	Yes
Medial forearm sensation	**Abnormal**	Normal
Lesion	**Lower trunk C8–T1**	Median at wrist

How it's asked

"35-y, bilateral ring/small tingling overhead, normal EMG, X-ray shown?"

→ **thoracic outlet syndrome (cervical rib)**
✗ Wrong options: C5–6/C6–7 disc · cubital tunnel · radial tunnel

"Arms abducted, ER, overhead; open-close fingers 1–3 min — test?"

→ **Roos test** (TO)
✗ Wrong options: Adson · Wright · Spurling

"Gilliatt-Sumner hand in true neurogenic TOS?"	→ **preferential APB atrophy, relative hypothenar sparing** (TO) ✗ Wrong options: symmetric interossei sparing thenar · claw from ulnar at humerotricipital groove · C6 thumb/ index loss
"Shoulder ER + max abduction above head 1 min, neck turned away — test and space?"	→ **Wright test; retropectoralis minor space** ✗ Wrong options: Adson = interscalene · costoclavicular
"Most common complication of 1st rib resection?"	→ **pneumothorax**

– – Active space – –

TOTAL ELBOW ARTHROPLASTY

– Shoulder & Elbow · 56 / 60 –

1 Basics

Epidemiology & forms of elbow arthroplasty

Trend — **TEA for trauma** (elderly distal humerus fractures) = one of the **fastest-growing** indications; historically for **RA**

Forms — **TEA** · **hemi** (radiocapitellar, distal humeral) · **ulnohumeral distraction / interposition arthroplasty** · **olecranon fossa debridement** · radial head arthroplasty

Implant design (linked, semiconstrained, unlinked) & survivorship

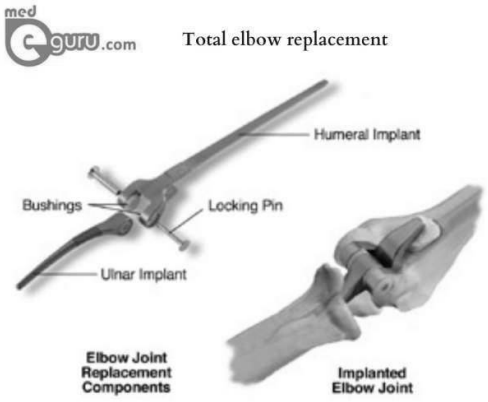
	Unconstrained (unlinked, resurfacing)	Semiconstrained (linked)	Constrained
Example	**Ewald capitello-condylar**	**Coonrad-Morrey**	**Dee**
Hinge	None	**"Sloppy hinge": 7° varus-valgus + 7° axial rotation**	**Rigid hinge**
Needs	**Competent collaterals + soft-tissue envelope**, good bone, **precise alignment**	**collaterals not needed**; radial head not needed	–
Use	–	**Most commonly used**	**No longer used**
Results	No proven superiority	**Best results** of all designs	**Highest loosening**
Commonest failure	**Instability** (5–10% dislocation)	Instability (Orthobullets, TO) · bushing wear (OB)	**Aseptic loosening**, then **bushing wear**

Semiconstrained laxity — ↓ stress at **bone–cement interface → ↓ aseptic loosening**, at the cost of **bushing wear**

Constrained — no coronal/rotational flexibility → **high shear stress → early loosening**; theoretically most stable

Unconstrained suits — humeroulnar disease with **intact collaterals + radiocapitellar joint**: OA, post-traumatic arthritis, acute intra-articular distal humerus fracture, distal humerus malunion, bony ankylosis

Unconstrained relative contraindication — **late-stage RA** (capsuloligamentous instability + bony erosion); RA → **linked**
✗ Wrong options: unlinked: tendon injury → instability



Semiconstrained (linked) TEA: humeral + ulnar implants, bushings, locking pin

Design pearls

Stems humeral + ulnar stems → better fixation, ↓ loosening

Anterior humeral flange extracortical; resists **posteriorly directed + rotational** forces; designs without flange → **early humeral loosening**; improper flange placement → humeral loosening (TO)

Radial head **not needed for stability** in linked designs; often debrided/resected in RA

Elbow arthroplasty type (TO)	Description	Use
Debridement arthroplasty	Remove osteophytes, preserve surfaces	Painful stiff degenerative elbow, **young high-demand**
Interposition arthroplasty	**Triceps fascia or fascia lata** between surfaces	When implant contraindicated (**paediatric / poor bone stock**)
Implant arthroplasty (TEA)	Metal + polyethylene	End-stage arthritis, trauma, RA

Fully constrained examples (TO) metal-on-metal hinges **GSB-1, Stanmore, Dee, Mazas** → high loosening/ breakage · semiconstrained: **Coonrad-Morrey, Mayo, Pritchard-Walker** · semiconstrained survival **90–92% at 10–15 y** (TO)

Ideal elbow implant (TO) preserves **epicondyles + olecranon** · maintains carrying angle (valgus in extension → varus in flexion) · functional arc **30–130°** · minimal bone sacrifice

Selection (TO) **anterior medial ligament complex = 70% of stability** · good bone + intact ligaments → **unconstrained** · poor bone / weak muscles → **semiconstrained** · severe destruction + instability → **fully constrained**

Coonrad-Morrey (TO) **7° varus-valgus + rotatory laxity** · **triangular humeral stem** with flattened base + **anterior flange** (torque) · long stems · HMW polyethylene bushing between titanium components

4 Diagnosis & Classification

Key radiology findings

Imaging	Purpose
AP + lateral elbow	Bone stock, **medial + lateral columns intact**, **canal diameter**
C-spine flexion–extension	**RA** patients (pre-op)
CT	Bone stock / fracture
True AP in full extension (post-op)	**Bushing wear**: humeral yoke–ulnar component angle **> 7° = wear**, **> 10° = extensive/complete** (TO) · **> 10°** concerning (Orthobullets)
Gruen zones (4)	Zone 1 periarticular, zones 2–4 = stem thirds; **diffuse lucency ≥ 3 zones**, **focal ≤ 2 zones**

Semiconstrained laxity (TO) **7° total = 3.5° each direction**; **> 7° = partial wear**; **> 10° = complete wear**



Bushing wear — varus-valgus tilt of ulnar component on AP

Cement mantle quality (Morrey, TO)

Quality	Radiolucent zone	Cement beyond stem tip
Adequate	**< 1 mm**	Yes
Marginal	≥ 2 mm	Yes
Marginal	< 2 mm	No
Inadequate	**≥ 2 mm**	**No**

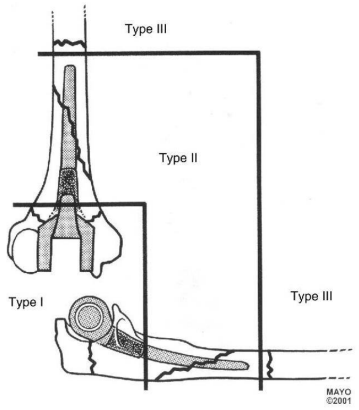
Humeral bone loss grading

Grade	Humeral bone stock
I	Articular surface damage only; **intact subchondral bone**
II	**Absent trochlea**; intact medial + lateral epicondyles
III	**Absent medial or lateral epicondyle**
IV	**Loss of both epicondyles** or resection **proximal to olecranon fossa**

Mayo (O'Driscoll & Morrey) periprosthetic fracture classification

Type	Characteristics	Treatment
I (**commonest**, TO)	**Periarticular – humeral condyle or olecranon**; osteolysis around hinge + muscle distraction	Undisplaced → **immobilise / soft-tissue repair** (fibrous union OK) · displaced → **ORIF heavy nonabsorbable suture or TBW**
II1	Along stem	**ORIF, retain components ± strut allograft**
II2	Along stem, **loose implant, good bone stock**	**Revision long-stem ± strut allograft + impaction grafting** (± locking plate / cerclage)
II3	Along stem, **loose implant, severe bone loss**	**Revision + extensive allograft**; often **resection arthroplasty**
III	**Distal to prosthesis tip**	Well-fixed → **immobilise humerus, ORIF ulna** · loose → treat as **II2**

Basis modelled on **periprosthetic femoral** fracture classification; site + implant fixation + bone stock



Mayo periprosthetic elbow fracture types I–III

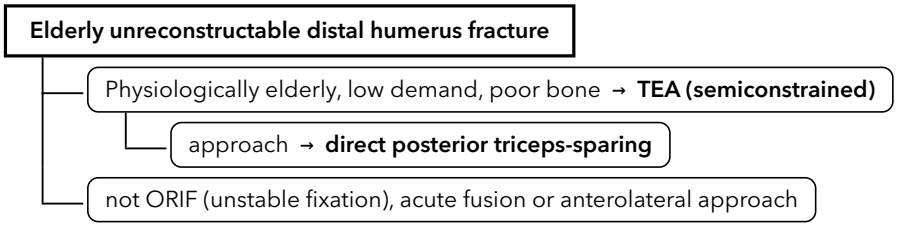
5 Management

Operative criteria – RA, elderly distal humerus fracture, failed interposition

Indication	Criteria
Rheumatoid arthritis	**10–20%** of RA have elbow disease; **Larsen 3–5** (Orthobullets) · grade III–IV (OB) + pain, functional loss, instability; ideally **> 65 y**; synovectomy for less advanced disease
Primary OA (advanced)	**> 65 y**; **mid-arc pain** from ulnotrochlear cartilage loss
Fracture	**Physiologically elderly (> 70 y)**: acute complex **unreconstructable intra-articular distal humerus** fracture, missed fracture-dislocation, **poor bone**
Post-traumatic OA (advanced) · chronic instability	–
Failed interposition arthroplasty	End-stage arthritis + instability → **semiconstrained TEA**

– – Active space – –

Contraindications	**absolute**: **active infection** (→ arthrodesis), **Charcot joint** · **relative**: poor neurological control (UE paralysis), **active patient < 65 y**, **olecranon osteotomy** · RA list (OB): active/**historical** infection, paralysis, won't follow instructions
Olecranon osteotomy	contraindication: preserve **extensor mechanism**; trans-olecranon approach **jeopardises ulnar component stability** → never for TEA
Not contraindications	central comminution · **ulnar nerve transposition** · radial head impaction · retrocapitellar comminution (radiocapitellar joint secondary)
TO indications	end-stage RA / post-traumatic arthritis · unreconstructable distal humerus fracture **>60 y** · **bilateral / fibrous ankylosis in poor position** · bone loss (tumour, trauma) · **haemophilic arthropathy**
TO contraindications	active/recent infection · poor soft-tissue envelope · **non-functional biceps/triceps** · poor compliance · flaccid paralysis · young vigorous post-traumatic elbow · **neuropathic joint** · **ipsilateral shoulder ankylosis**
Pre-op (TO)	AP + lateral for **humeral bow + canal diameter**, ulnar canal size/curvature · templating · ulnar nerve evaluation · **aspirate if warmth/redness (esp. RA)**



Fracture TEA vs ORIF	**faster recovery + more predictable function**; elderly accommodate limits, fewer complications ✗ Wrong options: slower recovery
Anterolateral approach	brachialis/brachioradialis (radial nerve between) → **no access for TEA**
Prior radial head resection in RA	**semiconstrained > unconstrained** (both better in native elbows)
Failed interposition, 60-yo sedentary	**semiconstrained TEA** ✗ Wrong options: unconstrained fails with ligament instability · revision interposition · Outerbridge-Kashiwagi · arthroscopic osteocapsular arthroplasty

Survivorship & outcomes

Indication	Outcome
RA	**Longest survivorship** of all indications (> fracture, flail elbow, OA, post-traumatic); **92.4% revision-free at 10 y**; **> 90% up to 15 y**
RA complications	**14%** (Orthobullets) · **~15%** (OB) – triceps avulsion, deep infection, periprosthetic fracture, aseptic loosening (✗ Wrong options: "< 5%")
Primary OA	**10-y survival 80–85%**; more active → **↑ aseptic loosening + stem fracture vs RA**
Overall	**> 80% at 10 y**
Post-traumatic	Functional ROM, satisfaction at 5 y; **complications 27–43%**, **re-operation 25%**

Longest survivorship combination	**RA + semiconstrained**

Linked TEA, OA vs RA OA ↑ **aseptic loosening**; **instability equal** (linked)

– – Active space – –

Surgical approach & triceps insufficiency

Approach	Technique	Key fact
Triceps-reflecting (Bryan-Morrey)	Triceps + **anconeus reflected medial → lateral**; reattached to ulna with **nonabsorbable suture through bone tunnels**	**Triceps failure**; **↓ extension power up to 30%**; **delayed ROM**
Triceps tongue	Fascial tongue raised from olecranon proximally; release collaterals	Fractures or TEA
Triceps-splitting	Longitudinal split in continuity with forearm fascia; or **V-shaped turndown** (insertion intact)	Allows **extensor lengthening**
Triceps-sparing (e.g. Alonso-Llames, para-olecranon)	Mobilise medial + lateral triceps borders; insertion retained	**Best for acute fracture TEA**; **earlier ROM**, ↑ final ROM; **↓ ulnar visualisation**
Triceps "on"	Midline incision; release/protect **ulnar nerve**; release flexor-pronator + MCL, common extensors + LCL; elevate triceps off posterior humerus; dislocate ulnohumeral joint	**No triceps release → no insufficiency**
Trans-olecranon	Osteotomy, triceps stays on olecranon	**ORIF only**, never TEA

Triceps-off vs triceps-sparing triceps-off: **delayed ROM**, ↑ triceps failure · **no difference**: cement mantle, ulnar nerve palsy · sparing: ↑ final ROM

Bryan-Morrey (1982) initial 49 patients no triceps failure; later series ↑ failure

TO approaches splitting/reflecting keep triceps attached to periosteum + forearm fascia (± olecranon wafer); **triceps-preserving (bilateral tricipital)** = exposure by ulnohumeral dislocation, **useful in severe RA**; no consensus on best approach

Post-op protection (triceps-reflecting) **restrict active extension** early; **6–8 wk** before **resisted extension**
✗ Wrong options: active flexion · supination · passive motion

Technique

Positioning **supine** (arm over chest, assistant; avoid ETT) or **lateral decubitus** over bolster (no assistant, ↓ manipulation)

Humerus resect **olecranon fossa**, **keep medial + lateral columns**; **bone wedge behind anterior flange**; orient to **posterior flat surface of distal humerus**

Ulna resect **olecranon tip** + **coronoid tip** (prevents impingement on flange → **axial pistoning + loosening**); orient **perpendicular to dorsal flat olecranon**; centre of **greater sigmoid notch**, rotation from **posterior olecranon cortex** (TO)

Alignment (TO) **implant–medullary canal parallelism** (Figgie); COR displacement limit **5 mm**; axis: **anterior humeral cortex** (AP), **roof of coronoid fossa** (depth), **condylar plane** (coronal rotation)

Pistoning (TO) check coronoid, protruding cement, HO on trial motion

Stability check (TO) joint line should separate **≤ 1–2 mm** on distraction at **90° flexion + pronation**; **UCL + extensor mass** contribute most

Cement modern cementing; semiconstrained commonest

– – Active space – –

Post-op care & lifelong lifting restrictions

Immobilisation	**all protocols acceptable** (immediate motion, or splint 60–90° for 2–5 d, 5–10 d, ≥ 10–14 d → active flexion + gravity-assisted extension); historically 4 wk; immediate motion safe
Evidence	**earlier ROM → fewer complications** · **prolonged immobilisation → higher implant survival** (Orthobullets); complications **lowest without immobilisation** in RA (OB)
Lifting limit	**5–10 lb** lifelong (Orthobullets, OB) · **≤ 5 lb** (OB) – read the options
TO post-op	elevate overnight, drains out next day → passive flexion–extension · collar and cuff · **no active extension 3 months** (triceps healing) · lift **≤2.2 kg (5 lb) for 3 months → lifelong ≈4.5 kg (10 lb)**
Avoid after TEA	impact sport (golf, tennis) · immediate aggressive triceps strengthening · **immobilising wrist/hand 4 wk** (mobilise immediately) · avoiding pronation/supination (doesn't stress ulnohumeral implant)
Patient selection	> 65 y, low demand, able to obey restrictions ("none do"), **medical optimisation**

6 Complications

Complication	Rate / fact
Aseptic loosening	**Radiographic 17%, clinical 6%** (Orthobullets) · clinically significant **7–15%** (TO) · **< 10%, usually > 10 y** (OB); **commonest failure of constrained**
Infection	**8%** (Orthobullets) · **3–8%** (TO) · **5–8%**, within first 5 y (OB); **commonest early failure**
Instability	**7–19%**; **commonest failure of semiconstrained** (Orthobullets, TO) and unconstrained
Bushing wear	Constrained + semiconstrained; isolated exchange **1–6%**, mean **7.9 y**; **commonest cause of revision / mechanical failure of semiconstrained** (OB)
Periprosthetic fracture	**5–30%** of primary TEA; **trauma** commonest cause
Component fracture	Humeral **0.65%** · **ulnar 1.2%** (ulna > humerus); usually **ulnar**, late
Triceps insufficiency	**2–8%**; complete disruption **< 2%**
Ulnar neuropathy	Paraesthesia **< 40%**, resolves **2–6 wk**; radial nerve less often
Wound healing	↑ with **long-term steroids**

Complications – instability & component loosening (TO)

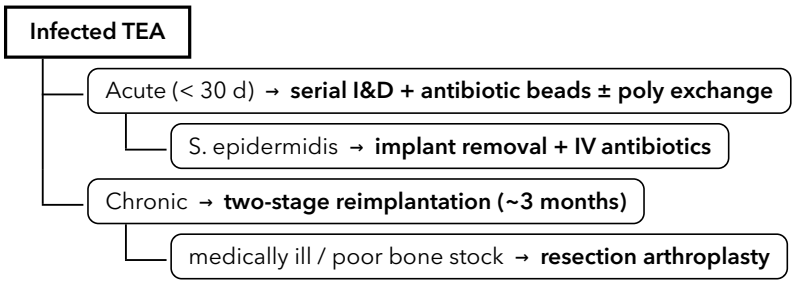
Unconstrained, 3 months post-op (TO)	**instability** ✗ Wrong options: ulnar loosening · infection · fracture
"True about TEA" (TO)	**loosening most commonly humeral component** ✓ ✗ Wrong options: instability commonest in semiconstrained (it is commonest in **unconstrained**) · loosening commonest in unconstrained · periprosthetic fracture humerus > ulna
Loosened component	**humeral** (TO answer; historically commonest revision cause, now **< 5%** in semiconstrained) · **ulnar** with current flanged designs (another TO) – read the options

Semiconstrained commonest failure	**instability** (Orthobullets, TO) · **bushing wear** (OB) · humeral loosening historically (TO) — read the options
Instability treatment (TO)	collateral reconstruction usually fails → **revision to semiconstrained hinge**
Constrained, commonest failure	**aseptic loosening** > bushing wear > infection (8%); instability = unconstrained/semiconstrained; ulnar neuropathy < instability
TO failure order	**instability > aseptic loosening > infection** · aseptic loosening = **most common late failure** · bushing wear → synovitis, foreign-body reaction (active males, post-traumatic) · infection > hip/knee (**S. aureus**, thin soft tissue) · periprosthetic fracture **6–22%** (Mayo classification; revise if loose) · heterotopic ossification limits arc

Morrey rating (TO)	Criteria
Good	No pain, **>90° flexion**, **>60° rotation**, no lucent lines
Fair	Mild pain, 50–90° flexion, 40° rotation, **<1 mm** lucency
Poor	Significant pain, **<50°** motion, **>2 mm** lucency

MEPS (TO)	Outcome
90–100	Excellent
75–89	Good
60–74	Fair
<60	Poor

Infection



Acute window	**< 30 days** (Orthobullets) · I&D + **polyethylene exchange within 6 wk** if components stable (OB) — read the options
Organisms (TO)	**S. aureus + S. epidermidis**; **S. epidermidis** = encapsulating, **biofilm** → persistent → **implant removal + IV antibiotics**
Frail 78-yo RA, WBC 25,000, S. epidermidis	**resection arthroplasty** ✗ Wrong options: lifelong suppression (only indolent e.g. **C. acnes**) · I&D + poly exchange (not acute) · one-stage (**50%** success only) · two-stage at 3 wk (wait **~3 months**)
Early failure, semiconstrained post-traumatic	**infection** ✗ Wrong options: loosening > 10 y · bushing wear = intermediate · component fracture = late · ulnar neuropathy = not implant failure

– – Active space – –

Triceps insufficiency

Bryan-Morrey, 4 wk, "pop" on resisted extension	**triceps repair failure** ✗ Wrong options: ulnar component fracture · axle disengagement · periprosthetic fracture (rare without major trauma, late)
Can't extend at 6 wk after soft-tissue repair	**Bryan-Morrey** ✗ Wrong options: triceps-sparing · triceps-on · trans-olecranon · Alonso-Llames
Triceps roles (TO)	**sole overhead extension** + thick **soft-tissue cover** (↓ wound problems, infection); prevention = meticulous approach + repair

Bone loss & salvage

Causes	multiple revisions, fractures, osteolysis
≤ 8 cm distal humeral loss	**longer prosthesis with extended anterior flange**; severe → **endoprosthesis (total humerus)**
Salvage	flail elbow, amputation, arthrodesis

X vs Y

	Triceps-reflecting (Bryan-Morrey)	Triceps-sparing
Triceps	Detached, reflected with anconeus, repaired	**Left on olecranon**
ROM start	**Delayed**	**Early**
Triceps failure	**Higher**; ↓ extension power up to 30%	Low
Exposure	Good	↓ Ulnar visualisation
Best for	–	**Acute distal humerus fracture**

	TEA for RA	TEA for primary OA
Survivorship	**Longest** (92.4% at 10 y)	80–85% at 10 y
Patient	Low demand	**More active → ↑ loosening, stem fracture**
Design	**Linked** (✗ Wrong options: unlinked)	Linked or unlinked

How it's asked

"Unreconstructable distal humerus fracture, 78-yo – contraindication to TEA?"	**→ olecranon osteotomy** ✗ Wrong options: ulnar transposition · central comminution
"85-yo nursing-home resident, comminuted distal humerus – treatment?"	**→ TEA via direct posterior triceps-sparing approach** ✗ Wrong options: ORIF · fusion · anterolateral
"82-yo RA, distal humerus fracture – shortest recovery, best function?"	**→ TEA**
"RA, prior synovectomy + radial head excision, Larsen III?"	**→ semiconstrained TEA** ✗ Wrong options: unconstrained · synovectomy

"Failed interposition arthroplasty, end-stage + instability?"	→ **semiconstrained TEA**
"TEA survival in RA?"	→ **good at 15 years**
"Unconstrained TEA — relative contraindication?"	→ **late-stage RA**
"Longest implant survivorship?"	→ **RA + semiconstrained**
"Constrained TEA — commonest failure?"	→ **aseptic loosening** ✗ Wrong options: bushing wear
"Linked TEA, OA vs RA?"	→ **OA ↑ aseptic loosening**
"Most accurate pre-op counselling?"	→ **semiconstrained = most favourable** ✗ Wrong options: no extension loss with reflecting · RA complications < 5% · fracture TEA slower
"Post-traumatic OA — true about designs?"	→ **semiconstrained varus-valgus laxity ↓ bone–cement stress** ✗ Wrong options: constrained best · semiconstrained needs collaterals
"Bryan-Morrey, pop + can't extend at 4 wk?"	→ **triceps repair failure**
"Triceps-reflecting — restrict which motion?"	→ **active extension**
"Can't extend at 6 wk, soft-tissue repair failed — approach?"	→ **Bryan-Morrey**
"Immobilisation after triceps-sparing TEA?"	→ **any protocol acceptable**
"Triceps-off vs triceps-sparing?"	→ **delayed ROM** with triceps-off
"Lifelong instruction after TEA?"	→ **limit load to ≤ 5 lb**
"Frail RA, chronic S. epidermidis TEA infection?"	→ **resection arthroplasty**
"Semiconstrained TEA — commonest early failure?"	→ **infection**
"True about TEA?"	→ **loosening most commonly humeral component** (TO)
"Unconstrained TEA at 3 months — commonest complication?"	→ **instability** (TO)
"Key stabiliser to assess when choosing an elbow implant?"	→ **anterior medial ligament complex (70% of stability)** (TO)
"Mayo Elbow Performance Score: excellent?"	→ **90–100** (TO)

– – Active space – –

TOTAL SHOULDER ARTHROPLASTY

– Shoulder & Elbow · 57 / 60 –

1 Basics

Principles

Definition replace humeral head + **resurface glenoid**; **cemented all-polyethylene glenoid = standard of care**

vs THA/TKA greater **ROM** · success depends on **soft tissues** (intact cuff, esp. subscapularis) · glenoid **less constrained → greater shear stress → mechanical loosening**

Requirements **intact + functional rotator cuff** · adequate **glenoid bone stock + version**

Type (TO)	Replaced	Use
Hemiarthroplasty	Humeral head only	**Poor glenoid stock** or non-functional cuff with good forward flexion · **AVN with intact glenoid** · **3/4-part non-reconstructible fracture**
TSA	Head + glenoid	**Intact functional cuff + adequate glenoid stock** · OA (commonest), RA if cuff functional
RSA	Reversed geometry (convex glenoid, concave humerus)	**Non-functional cuff**, CTA · RA with cuff deficiency

Outcomes **pain relief** = most predictable benefit (more than hemi) · preserved IR + ER · survival **93% at 10 y** (Orthobullets, OB) · ">90% at 10 y" (OB) · TSA/hemi for OA: good/excellent **95%** (Norris) · **worse for post-capsulorrhaphy arthropathy**

Humeral fixation cemented or press-fit (porous-coated); both good longevity; humeral loosening **uncommon without infection**

Humeral anatomy & implant design (version, stemless, platform stems)

Parameter	Native value
Humeral head retroversion	Mean **~20°** (transepicondylar axis) (OB) · Boileau & Walch mean **17.9°** (−6.7 to 47.5°) · range **0–55°** depending on method (TO)
Neck-shaft angle (inclination)	**45° ± 5°** (range **30–50°**); arthritic ≈ 50° (flatter) (TO)
Humeral head height	Top of head **8–10 mm above top of GT** (TO, native)
Lateral humeral offset (LHO)	**Lateral base of coracoid → lateral margin of greater tuberosity** (TO)
Humeral offset	Centre of rotation → humeral canal axis; medial **4–14 mm**, AP **−2 to 10 mm** (TO)
Head thickness	**70% of radius of curvature** (OB)
Radius of curvature	Glenoid **22–28 mm** · head **23–28 mm** (TO)
Glenoid version	**0–3° retroversion** (OB) · **1.5° retroversion** (10.5° retro to 9.5° ante) (TO) · ~0°, <10° either way (TO)
Glenoid inclination	Superior **~4°** (4.2°; −7 to 20°) (TO)

LHO (TO) **↓ LHO → ↓** lever arm of **deltoid + supraspinatus** → weak abduction · **↑ LHO → overstuffing → ↓** motion, **↑** poly wear · reconstruct within **4 mm** of normal

Parameter (TO arthroplasty lecture)	Value
Head–shaft angle	**135°** (130–150°) → sets osteotomy inclination
Humeral retroversion	**30°** (relative to elbow axis)
Scapular plane	**30°** to coronal plane (true AP)
Medial offset	**7 mm** (COR medial to shaft axis)
Posterior offset	**3 mm**
Head height	**0.5–1 cm above GT tip**
Lateral humeral offset	**5–6 cm** (coracoid → lateral humeral head border)
Normal glenoid version (CT)	**2° anteversion to 7° retroversion**

Deviation (TO) overstuffing, ↓ ROM, altered cuff tension → early wear

Retroversion axes (TO) proximal: articular surface plane / COR–articular centre / GT–articular centre · distal: **trochlear axis**, transepicondylar line, forearm

– – Active space – –

Restoring COR may prolong glenoid fixation + ↓ poly wear (TO)

Humeral implant	Key fact
Stemless	**Completely metaphyseal (biological) fixation** · **more bone preserving** · **no osteotomy needed for proximal humerus malunion** (not constrained by stem) · not for 3/4-part fractures (→ stemmed RSA in elderly)
Stemmed	Cemented or uncemented porous-coated
Short stems	↑ **varus malalignment** (metaphyseal-diaphyseal mismatch)
Large-diameter, high-fill stems	↑ **proximal stress shielding**
Platform (convertible) stem	Modular conversion TSA → rTSA without stem removal → ↓ **operative time**, ↓ blood loss, preserves humeral bone, ↓ cost

Subscapularis failure **independent of implant design**

2 Mechanism

Rocking-horse phenomenon & glenohumeral mismatch

Rocking horse **eccentric glenoid loading** → lifting of opposite unloaded rim → **glenoid loosening** · causes: **superior migration of head (cuff deficiency)** · abnormal glenoid version (ant/post translation)

Minimised by (TO) **glenoid radius of curvature larger than humeral head radius** (**2–4 mm**) → normal translation without rim loading
✗ Wrong options: perfect congruence · larger head radius

Mismatch (TO) recommended **<4 mm** (2–4 mm larger glenoid) · "mismatch <4 mm ↑ loosening" (another TO explanation) – read the options · matched within 2 mm: no ↑ loosening reported · **too much mismatch** → ↑ translation, ↑ poly wear, fracture

Modern glenoids cemented all-poly, radius **2–6 mm larger** than head (TO)

Glenoid conformity	Effect
Conforming	**More stable** · **rim stress** + radiolucencies
Nonconforming	↑ **polyethylene wear**
Overall	Neither superior

Osteolysis **rare** in TSA: larger particle size → harder for macrophages to mount RANK–RANKL reaction (OB)

3 Presentation

Indications & contraindications (cuff status, brachial plexus injury, RA, capsulorrhaphy arthropathy)

Indications pain (esp. **night**) + ↓ ADL · **glenoid chondral wear to bone** (primary OA, inflammatory arthritis esp. **RA**) · posterior head subluxation · OA after primary/recurrent dislocation · complicated proximal humerus fracture not fixable (TO)

TSA vs hemi TSA **preferred for OA + inflammatory arthritis** (better pain relief + forward elevation in RA) · **age <55: TSA > hemi** (outcomes, survival, satisfaction)

AAOS CPG (end-stage GH OA) **TSA with all-polyethylene cemented glenoid** · (✗ Wrong options: humeral head resurfacing · CTA head · **metal-backed glenoid (↑ revision/ loosening)**) · hemi + ream-and-run = option but TSA preferred

Contraindication	Note
Upper / preganglionic brachial plexus palsy	**Absolute** (TO FNB) · root avulsion = flail arm, poor recovery
Active infection	"**Only absolute** contraindication" (other TO explanation) – read the options
Irreparable rotator cuff tear	→ **hemi or rTSA**; **rocking horse** → glenoid loosening
Rotator cuff arthropathy (proximal migration on X-ray)	Glenoid resurfacing contraindicated
Insufficient glenoid bone stock	**Glenoid erosion to coracoid → favours hemiarthroplasty**
Deltoid dysfunction	e.g. can't abduct past **30°**
Relative (TO)	**Charcot**, severe plexopathy, **intractable instability**, prohibitive comorbidity, cuff + deltoid dysfunction
Isolated supraspinatus deficiency	**Relative** only (TO)

NOT a contraindication	Why
Non-retracted isolated supraspinatus tear / 2-cm full-thickness SS tear	Reparable at TSA
Passive ER <10°	Release at surgery
Eccentric posterior glenoid erosion / retroverted glenoid	Eccentric reaming
Poor glenoid bone stock	Build up with graft (TO)
Inflammatory arthritis	**Indication** if bone stock adequate
Subchondral cysts	Normal end-stage OA finding
Age (young)	TSA favoured over hemi

TO contraindication list non-functional cuff (**except isolated supraspinatus tear**) · inadequate glenoid stock · **active or previous infection** · **brachial plexus / axillary nerve palsy** → hemi or RSA instead

Cuff tears at TSA full-thickness in **5–10%** of TSA patients (Orthobullets) · **<5%** in primary OA (OB) – read the options · **+ impingement signs → pre-op MRI**

– – Active space – –

RA + good elevation (120°), no superior migration — **TSA cemented all-poly**
✗ Wrong options: rTSA · hemi · arthroscopic debridement = less advanced disease

Hemiarthroplasty indications (TO)	Hemiarthroplasty contraindications (TO)
Rough humeral surface, **intact glenoid cartilage**, enough glenoid arc	Recent sepsis
Insufficient bone for glenoid	**Neuropathic joint**
Fixed upward displacement (CTA, severe RA)	Paralytic disorder
History of **remote infection**	Cuff + deltoid deficiency
Heavy demands (occupation, sport, LL paresis)	Non-cooperative patient

Post-capsulorrhaphy arthropathy

Putti-Platt — subscapularis + anterior capsule divided; lateral stump → anterior glenoid neck; medial stump → tuberosity (**pants-over-vest**) → **limits ER** → incongruent contact loading → **post-capsulorrhaphy arthropathy**
✗ Wrong options: no coracoid transfer

Magnuson-Stack — middle subscapularis + piece of LT transferred **lateral to bicipital groove** (restrict ER) → **subscapularis lacerated** during biceps release at TSA → **↑ ER, anterior instability, + belly press**

Post-capsulorrhaphy arthropathy (TO) — tight anterior capsule → **posterior subluxation** + **multiple loose bodies** on X-ray

4 Diagnosis & Classification

Key radiology findings

View	Look for
True AP (Grashey)	Extent of arthritis · **superior migration** (= cuff deficiency) · Iannotti best-fit circle
Axillary	**Posterior glenoid wear** · head subluxation
CT	**Glenoid version + glenoid bone stock**
MRI	Rotator cuff (**Goutallier** fatty change)

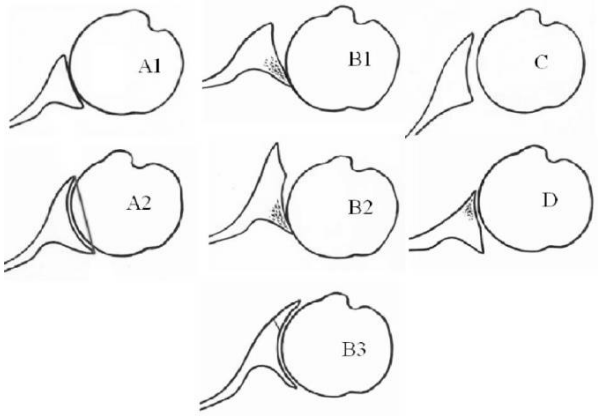
Cuff-deficiency signs — **acetabularization of acromion** → incompetent cuff → **rTSA** · disrupted **Shenton's line** of shoulder → cuff not intact

X-ray sign (TO)	Meaning
Goat's beard	**Inferior humeral osteophyte** → OA
Sourcil sign	**Acromial sclerosis** → cuff tear arthropathy
Multiple loose bodies	**Post-capsulorrhaphy arthropathy**
Posterior subluxation	**Posterior glenoid wear** (most typical OA pattern)

Infraspinatus atrophy on MRI — **independent risk factor for post-op superior cuff tear**

Walch classification of glenoid wear

Type	Description
A1	Concentric, centred head; minor central erosion
A2	Deeper central erosion; line joining rims transects head
B0	Pre-OA **posterior subluxation** of head
B1	Posterior narrowing, osteophytes, sclerosis, **no bone loss**
B2	**Biconcave**, **posterior rim erosion**, retroverted · max erosion **posteroinferior (7–8 o'clock)**
B3	**Monoconcave** posterior wear; head subluxation **>70%** or retroversion **>15°**
C1	**Retroversion >25°** regardless of erosion (dysplastic)
C2	Biconcave, posterior bone loss, posterior translation
D	**Anteversion** or **anterior** head subluxation (<40%)



Walch classification of glenoid wear

Commonest OA pattern (TO) — **posterior subluxation → posterior glenoid wear → ↑ retroversion**

B2 vs A1 — **B2 ↑ radiographic loosening** (TO)

5 Management

Implant choice

- **Glenohumeral arthritis**
  - Intact cuff + adequate glenoid → **aTSA (cemented all-poly glenoid)**
    - retroversion >15° → **augment or posterior bone graft**
  - Glenoid eroded to coracoid → **hemiarthroplasty**
  - Irreparable cuff / CTA / proximal migration → **rTSA (or hemi)**
  - Plexus palsy / active infection → **no TSA**

– – Active space – –

Approach

Deltopectoral	**cephalic vein** = interval landmark · detach **subscapularis + capsule** from anterior humerus → **dislocate anteriorly** · tight shoulder: release **upper half** (Orthobullets) · **upper 1/3** (OB) of **pectoralis major** tendon
Superior border of subscapularis (TO)	**biceps tendon** (just medial to pec major insertion → follow up to **rotator interval**) · biceps ruptured/dislocated → **base of coracoid** marks medial interval
Inferior border of subscapularis	landmark for **axillary nerve** (anterior → posterior under GH joint → quadrangular space)

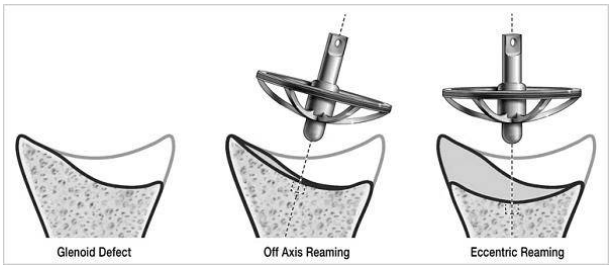
Palpate cuff insertion before head cut – ✓ protects cuff

Exposure (TO)	protect **cephalic vein** · release subscapularis **2 cm from lesser tuberosity**; **Z-lengthening** if contracted · dislocate in **abduction + ER** · **preserve CHL if cuff deficient** (adds stability)
Head osteotomy (TO)	**135°** inclination · **30° retroversion** (arm in 30° ER, cut anterior → posterior) · just proximal to cuff insertion · entry **posterior to bicipital groove, 1 cm medial to GT tip** · cemented or uncemented (**proximally coated stems preferred**)

Subscapularis management & failure (peel, tenotomy, LTO; postoperative restrictions)

Takedown options	**peel** (off bone) · **tenotomy** (trans-tendon) · partial tenotomy · **lesser tuberosity osteotomy (LTO)**
Which is best?	**no technique superior**: no difference in strength, ROM/function, biomechanical strength or failure rates ✗ Wrong options: LTO lower failure · peel more ROM · tenotomy higher failure
LTO	interest due to **bone-to-bone healing** (biomechanical strength); high union rate · **nonunion** → very poor outcomes, multiple revisions
Most commonly ruptured muscle after TSA (TO)	**subscapularis** · factors: **multiple previous operations**, **overstuffing**, **aggressive physio**, tendon length compromise (poor technique)

Glenoid morphology & retroversion correction (eccentric reaming, bone graft, augments)



Eccentric reaming of the high anterior side

Why correct	failure → glenoid malposition → ↓ contact area → ↑ contact pressure → **accelerated loosening**

Operative criteria – glenoid retroversion

Glenoid retroversion	Orthobullets / OB	TO
<10–15° (e.g. **9°, 11°, 13°**)	**Eccentric reaming of anterior (high) side**	**Up to 15°** → eccentric reaming
>15° (e.g. **19°, 21°**)	**Posterior bone graft** (Orthobullets) · **augmented glenoid or bone graft** (OB; augments ↓ complications)	**15–25° → glenoid augments** (reaming perforates anterior vault)
>25°	Allograft reconstruction of posterior defect	**Bone graft**
Insufficient bone (to coracoid)	No eccentric reaming → posterior bone graft or hemi	–
Cuff-deficient / acetabularised acromion	**rTSA** (reaming irrelevant)	–

15° threshold more reaming compromises **subchondral bone** + risks **glenoid vault penetration** · excessive **medialisation** weakens fixation + alters mechanics

Eccentric reaming not for **humeral anteversion/retroversion** (address with head cut + stem) · humeral head collapse · **superior inclination 18°** (exceeds correctable)

Glenoid deficiency build up with **iliac crest autograft** or **resected humeral head** · **never cement** to fill

Medial (central) glenoid wear (TO) hemiarthroplasty · **soft-tissue interposition (fascia lata / meniscus)** · **ream-and-run**

Persistent posterior subluxation (TO) **larger humeral head** · **↓ humeral retroversion** · **tighten posterior capsule** · immobilise in **ER** / delay ROM

Axial CT: retroversion + bone loss, glenoid to be resurfaced **bone grafting of the glenoid** (posterior)
✗ Wrong options: infection work-up · cuff repair · acromioplasty

Component positioning

Component	Target
Glenoid version	**Neutral to 10° retroversion** / "neutral or slight retroversion" (TO FNB) · **neutral** (5° ante to 10° retro acceptable) (TO) · **neutral to slight anteversion** (OB, another TO) · "recreate neutral" (Orthobullets) – read the options
Glenoid inclination	**Neutral**; superior/inferior inclination → highest failure; retroverted glenoid loosens more than anteverted (TO)
Humeral stem	**25–45° retroversion** (Orthobullets, TO) · **20–40°** (one TO explanation)
Glenoid retroverted	**↓ stem retroversion** → avoid **posterior dislocation**
Stem alignment	**avoid valgus**
Head height (implant)	Top of head **5–8 mm above GT** (Orthobullets, TO)
Head size	**avoid overstuffing** → ↑ joint reaction force + cuff tension

Accurate glenoid positioning (TO) **most critical** + most challenging step: limited bone stock, variable anatomy, few landmarks

– – Active space – –

Glenoid component design

Better	Worse	Note
Convex (curved) back	Flat back	Flat back ↑ loosening
Pegged	Keeled	Biomechanically superior; studies: no difference or ↑ loosening with keel
All-polyethylene	**Metal-backed**	Glenoid too small for metal + PE; metal-backed ↑ loosening/revision
Cemented (Orthobullets: ↓ loosening)	Uncemented	One TO states **uncemented better** – read the options

Central peg non-keeled designs: central peg usually **not cemented** → not a loosening risk factor

Glenoid implant (TO)	Key point
All-poly pegged	Cemented; **pegged in poor bone stock**
All-poly keeled	Good bone stock
Metal-backed	Porous, uncemented → **higher loosening**; less used
Hybrid	Central metal peg + poly pegs

Capsule & intra-operative fractures

Anterior capsular contracture (passive ER **<40°**) – **anterior release + Z-lengthening**

Stretched posterior capsule **plication** (volume reduction)

Intra-op GT fracture (minimal displacement) – standard stem + **suture fixation** + cancellous autograft

Intra-op shaft fracture remove → **long cemented stem + cerclage** · most common intra-op complication = **fracture**, humeral shaft **mid–distal diaphysis** (TO, Norris)

Rehabilitation & postoperative restrictions

Limiting factor early **protection of subscapularis repair** · restriction set by **strength of subscapularis repair**
✗ Wrong options: supraspinatus · dislocation · loosening · pec release · capsule · bone quality

Restrictions (≈6 wk) **no passive ER >30°** · **no active IR past plane of body** · **no pushing out of a chair** · passive/active-assisted only

Allowed early pendulums · passive FE to 90° · **passive IR to chest** · active elbow flexion

Progression ER isometrics → IR eccentric + isometric

Postoperative stiffness & opioid use

Stiff at 6 wk (FE 90°, ER 0°), X-ray normal, sling still on – **start PT for ROM now**
✗ Wrong options: more sling · aspiration · MRI · DVT duplex (DVT = swelling + pain)

Postoperative narcotic use	Factor
↓ use (protective)	**Increasing age**
↑ use	**Pre-op narcotic use** · unemployment
No correlation	Gender, smoking, alcohol

Orthopaedics highest narcotic-prescribing surgical specialty

Professionalism – impaired colleague

Attending smells of alcohol before TSA **notify the department chairman** before case (chain of command)
✗ Wrong options: medical board first · resident does case · discuss afterwards · refuse to participate

Substance abuse **leading cause of medical board sanctions**; physicians recover better than general population

6 Complications

Overview

Complication (TO)	Rate
Stiffness	66%
Glenoid loosening	**Commonest cause of failure (39%)**
Periprosthetic fracture (GT, shaft)	11%
Iatrogenic cuff tear (neck cut, overstuffing)	7.7%
Humeral loosening (infection, **RA, osteonecrosis**)	6.5%
Posterior instability	**6%**
Anterior instability (subscapularis failure)	**5%**
Neurovascular (arcuate artery)	4.8%
Infection	4.6%

Overall complication rate **~15%** · order: glenoid loosening > instability > cuff tear > periprosthetic fracture > infection > modular dissociation > deltoid dysfunction (TO)

Most common complication / failure **glenoid loosening** (TO, OB) · Orthobullets: newer evidence → **cuff failure may be more common** than glenoid loosening – read the options

– – Active space – –

Rocking-horse effect (TO) alternating anterior–posterior lift-off from eccentric loading = **most common cause of glenoid loosening**

Most common (TO)	Hemiarthroplasty	TSA	Reverse TSA
Complication	**Progressive glenoid erosion** (fracture cases: **tuberosity displacement**)	**Glenoid loosening**	**Scapular notching** > instability/dislocation
Cause of revision	Glenoid erosion	**Glenoid loosening**	Glenoid component loosening
Nerve	–	**Axillary (neurapraxia)**	Axillary

Glenoid loosening – most common complication & risk factors

Frequency most common cause of TSA failure: **~30% of primary OA revisions** (OB) · **39%** (TO) · reoperation **2.9% primary vs 28% revision** TSA

Commonest complication at 5 y **loosening of glenoid component**
✗ Wrong options: humeral loosening · acromial stress fracture · infection

✓ Risk factor	Not a risk factor
Glenoid morphology: **biconcave (Walch B2)**, **posterior glenoid wear**, excessive retroversion	**Age** (incl. >65)
Dysfunctional/deficient rotator cuff (rocking horse)	Inferior humeral head osteophytes
Insufficient glenoid bone stock	Subtle **anterior** subluxation (biconcave goes with **posterior**)
Excessive reaming removing subchondral bone	**Goutallier/cuff atrophy** (not correlated)
Flat-back glenoid	**Curved-back** (↓ loosening) · **pegged** component
Mismatch (see Mechanism)	Stemless humerus · diabetes (infection) · uncemented central peg
Soft-tissue instability · infection · aseptic osteolysis	✗ Wrong options: osteonecrosis · RA · OA (as cause of glenoid loosening)

Radiolucent lines presence **does not correlate with symptoms** or clinical failure (3 + 7 y) · **progression does correlate** (50% by 3–4 y)

Treatment glenoid removal (if bone stock) ± **bone graft** for later re-implantation · **convert to rTSA** (salvage) · glenoid morphology not fixable with aTSA → rTSA

Subscapularis failure

Commonest instability after TSA **anterior** (subscapularis + anterior capsule pull-off) (Orthobullets, TO) · TO complication list gives posterior 6% vs anterior 5% – read the options
✗ Wrong options: "pull-off of rotator cuff" (= supraspinatus)

Anterior dislocation after TSA **subscapularis rupture in all cases** (Wirth & Rockwood) · classic: **pushing up from chair** (fires subscapularis)

Mechanism fall/catching self · **forced IR** or **excessive ER** (beyond normal **40–60°**) · forced ER withdrawal (hot pan) → LTO failure · late rupture with sport (golf, tennis)

Sign of subscapularis insufficiency	Note
↑ Passive ER	e.g. 90° at side (abnormal at 72 y), 110° vs 80°
Lift-off / IR lag	Can't hold hand off back
Belly-press (abdominal compression)	Weak
Bear hug	**≥20% weaker** than other side
IR weakness · can't tuck shirt / reach back pocket	–
Anterior instability with ER	(✗ Wrong options: instability with IR = posterior)
Axillary X-ray: anterior subluxation	Head not centred



Anterior subluxation of humeral head after subscapularis failure

Confirm **ultrasound** · X-ray: displaced lesser tuberosity (failed LTO) · glenoid loosening/locked posterior dislocation **visible on X-ray** (normal X-ray → tendon rupture)

Consequence loss of **anterior–posterior force couple** / concavity-compression → instability, early loosening

Timing / patient	Treatment
Acute (<6 wk), young/active	**Early exploration + revision repair** (± augmentation)
>6 wk	Anterior capsular reconstruction · tendon transfer · **rTSA**
Chronic / older	**Revision to rTSA** (most reliable)

Subscapularis innervation **upper + lower subscapular nerves ← posterior cord**

✗ Plexus distractor	Branches
Lateral cord	**Lateral pectoral**, **musculocutaneous**
Medial cord	Medial pectoral, **medial brachial cutaneous (T1)**, MABC
Superior trunk	**Suprascapular**, nerve to subclavius
C5 root	**Dorsal scapular** (rhomboids, levator scapulae)

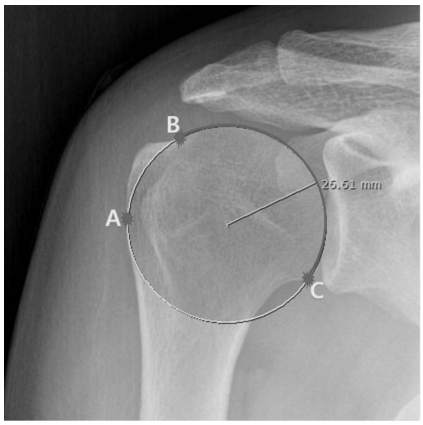
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Rotator cuff failure after TSA (overstuffing, falls, sizing)

Frequency **one of the most common complications**; common long-term secondary dysfunction · **common reason for conversion to rTSA**

Cause	Mechanism
Excessive bone removal at humeral neck osteotomy (cut too distal / excessive retroversion)	**Iatrogenic** cut/detachment of cuff insertion (✗ Wrong options: 30° retroversion = normal · 45° inclination · deltoid retraction = axillary nerve)
Overstuffing (oversized head, ↑ LHO)	**Attritional supraspinatus + subscapularis** tears; ↓ motion
Fall / trauma	**Traumatic cuff tear** (SS 2/5, normal X-ray → MRI)
Undiagnosed pre-op cuff tear	Improper soft-tissue balancing
Pre-op infraspinatus atrophy	↑ **superior cuff tear**
Degeneration, malposition	Late tear

Iannotti best-fit circle Grashey AP: 3 points = **lateral cortex below GT flare + lateral + medial ends of anatomic neck** · resurfaced head **outside circle = overstuffing → component sizing** error



Iannotti best-fit circle for humeral head sizing

Anterior + superior escape **SS + subscapularis** tears (Jobe + belly-press +; Hornblower −) ← **overstuffing**

High-riding humerus after fall **cuff failure → convert to rTSA**

Late pain at night, can't do overhead **positive empty can test** (SS tear) · repair if possible, else **rTSA**

Catastrophic glenoid fracture humeral head **directly on glenoid bone** (normally gap = poly thickness; poly is radiolucent)

Cuff tear → glenoid loosening (cuff deficiency predisposes)

Infection – C. acnes & work-up

Risk factor	Not
Male (more sebaceous glands, more C. acnes; more TSA for post-traumatic/ CTA)	Female
Age <65	Age >65
Reverse TSA (vs anatomic)	Anatomic TSA
Inflammatory or post-traumatic arthritis	Primary OA
Corticosteroid injection within 3 months (e.g. 2 months)	Injection 4 months before

Rate — C. acnes infection **1–2%** after primary TSA (Orthobullets) · **4.6%** (TO)

C. acnes (P. acnes) — **gram-positive, facultative, aerotolerant, anaerobic rod**; ferments lactose → propionic acid · **commonest cause of late/indolent infection + implant failure** · **>50% of chronic infections**; **S. epidermidis 2nd (15%)**

C. acnes facts — colonises **shoulder > hip/knee** · **men > women** bacterial burden · biofilm 18–90 h · presentation: early **pain + stiffness**, later swelling + redness · **usually no fever** (subtle)

Cultures — hold **14 days** (OB) · **10–14 days** in anaerobic bottles, mean detection **6 days** (Orthobullets TSA note) · **14–28 days** (Orthobullets PJI note) — read the options · 16S rRNA PCR · **sonicate** explanted implant

Sampling — **arthroscopic tissue culture 100% sensitive + specific** vs fluoroscopic aspiration **17% sensitive**, 100% specific · aspiration may be **normal** · imaging + in only 24%

New pain after pneumonia / ↑ WBC + ESR — **haematogenous → radiographs + aspiration with Gram stain + culture** · well-fixed implants → I&D may avoid full revision
✗ Wrong options: wait 2 wk for repeat labs · PT/NSAIDs/stretching · MRI · emergent revision

Timing	Treatment
Early (<6 wk)	**Open I&D**
Late (>6 wk) / grossly infected (frozen >10 WBC/HPF, GPC in clusters)	**Remove components + antibiotic spacer → 2-stage** reimplantation · IV antibiotics **6 wk** (penicillin G, ceftriaxone, clindamycin, vancomycin) then PO **2–6 mo**

Avoid in 2-stage stage 1 — bone grafting glenoid (won't heal with infection; do at stage 2) · resection arthroplasty (poor function) · head exchange/arthroscopic debridement (won't eradicate)

– – Active space – –

Neurovascular injury during approach (musculocutaneous, axillary, radial, arcuate artery)

Structure	At risk when	Deficit / fact
Axillary nerve (most common)	Inferior capsular release, inferior subscapularis, labrectomy; **excessive deltoid retraction**	Lateral shoulder numbness, weak abduction; **10–25 mm** below glenoid rim (↑ with abduction + ER); ≤3 mm from inferior capsule
Musculocutaneous (lateral cord)	**Retraction on conjoint tendon / strap muscles** (short head biceps + coracobrachialis)	**Weak forearm supination + elbow flexion** (biceps)
Radial nerve	**Latissimus dorsi tenotomy** for transfer (with rTSA)	–
Arcuate artery (terminal branch of **anterior humeral circumflex**, main GT supply)	**Elevating biceps tendon from groove** (runs lateral side of groove)	Bleeding
Anterior humeral circumflex vessels	Distal dissection at **inferior subscapularis** (peel)	–
Suprascapular nerve	≤1 cm from glenoid rim	–

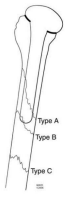
MCN distance from coracoid **~3 cm** (OB) · **2.0 ± 0.5 cm** (Apaydin; closest with arm **abducted + IR**) · **3.1 cm** (Boardman) – read the options · enters conjoint **5.8 cm** below coracoid · 0.5 cm from labrum at 90° abduction

Nerve injury rate **1–4%** of shoulder arthroplasty · 58% of iatrogenic injuries fail to recover (Carofino)

Axillary nerve + posterior humeral circumflex artery pass beneath GH joint through **quadrilateral space**

Latissimus transfer routine with **Grammont (medialised)** rTSA for ER deficit (+ Hornblower) · ↓ use with **lateralised** designs

Periprosthetic fracture – Wright & Cofield

Type	Location	Intra-op	Post-op	
A	Near **stem tip, extends proximally** (tuberosities)	Span with standard (2–3 cortical diameters) or long stem; transosseous sutures for tuberosity	**Nonop** if minimally displaced · significant overlap = loose stem → **long-stem revision**	
B	**Centred at stem tip**, extends distally	Span with long stem; cement distal canal (✗ Wrong options: cement escape); strut allograft + cerclage	**Revise to long-stem**	
C	**Distal to stem tip**	Long stem, or near olecranon fossa → plate ± cerclage, strut	**ORIF**; plate overlaps stem by **2 cortical diameters** (avoid stress riser)	

Acceptable alignment **≤20° flexion/extension · ≤30° varus/valgus · ≤20° rotation**

Humeral stem loosening more common in **RA + osteonecrosis → rule out infection** · uncommon with stemmed primary aTSA

X vs Y

	Anatomic TSA	Reverse TSA
Cuff	**Must be intact**	Deficient (CTA, pseudoparalysis, anterosuperior escape)
Design	Less constrained (shear → loosening)	Convex glenosphere + concave humerus; COR **inferior + medial** → longer deltoid lever
ER at side + IR behind back	**Better** (✓ reach behind back)	Worse
Return to activity	**Quicker**	Slower
Forward flexion / PROMs / revision / complications	**Similar**	Similar
Typical complication	Glenoid loosening, subscap failure	**Scapular notching**, instability, **acromial stress fracture (~4%)** (over-tensioning)
Dislocation	**Anterior** (subscapularis)	**Anterior**; mechanism **extension + IR + adduction** (rising from chair) (TO) · **hyperextension + ER** (another TO) – read the options

Post-op exam test	Muscle
Belly-press, lift-off, bear hug, IR lag, ↑ passive ER	**Subscapularis**
Jobe / empty can, drop arm / drop sign	Supraspinatus
ER lag sign	Infraspinatus
Hornblower	Teres minor
Yergason	Biceps
Loss of passive ER (complete)	**Locked posterior dislocation**

Late failure clue	Diagnosis
↑ Passive ER, + belly-press, **anterior** subluxation, early post-op	**Subscapularis failure**
High-riding humerus, + empty can	**Supraspinatus / cuff failure**
Head outside Iannotti circle, attritional SS tear	**Overstuffing (sizing)**
Progressive glenoid lucency, biconcave pre-op, normal labs/cultures	**Aseptic glenoid loosening**
Pain + stiffness, humeral lucency, normal or ↑ labs	**C. acnes infection**

How it's asked

"TSA absolute contraindication?" → **upper brachial plexus palsy** (TO)
✗ Wrong options: SS deficiency · poor glenoid bone stock · retroverted glenoid

– – Active space – –

"All-poly glenoid contraindicated in?"	→ **irreparable rotator cuff tear** ✗ Wrong options: RA · OA · osteoporosis · prior resurfacing
"Putti-Platt 20 y ago, painful stiff shoulder — diagnosis?"	→ **post-capsulorrhaphy arthropathy**
"Contraindication to TSA?"	→ **preganglionic brachial plexus injury** ✗ Wrong options: ER <10° · posterior erosion · 2-cm SS tear · inflammatory arthritis
"End-stage OA, AAOS CPG — next step?"	→ **TSA with all-polyethylene cemented glenoid** ✗ Wrong options: metal-backed · CTA head · resurfacing
"Most appropriate aTSA candidate?"	→ **72-y male, 9 months post TKA, debilitating pain, MRI shows intact rotator cuff** ✗ Wrong options: plexus palsy · can't abduct >30° · proximal migration · erosion to coracoid
"RA, FE 120°, no superior migration — most predictable?"	→ **TSA with cemented all-polyethylene glenoid**
"Favours hemi over TSA?"	→ **glenoid wear to level of coracoid** ✗ Wrong options: age · non-retracted SS tear · cysts
"aTSA vs rTSA counselling?"	→ **aTSA: increased ability to reach behind back** ✗ Wrong options: equal ER · better FF with rTSA · worse PROMs
"Primary OA, 7° retroversion — aTSA (Figure E) vs rTSA (Figure F)?"	→ **procedure shown in Figure E (aTSA) provides better post-operative range of motion** (similar complications + outcomes)
"Ideal glenoid component position?"	→ **neutral to 10° retroversion** (TO)
"Eccentrically ream the glenoid when?"	→ **retroversion 15°** (TO) · **13°** (OB) · **9°** (OB)
"Axial CT: retroversion + posterior bone loss, resurfacing planned?"	→ **bone grafting of glenoid**
"Walch B2 — maximum erosion?"	→ **posteroinferior (7–8 o'clock), area D**
"Glenoid augmentation needed for?"	→ **retroversion 21°** ✗ Wrong options: 11° · humeral retroversion · head collapse · acetabularization = rTSA
"Cause of glenoid loosening?"	→ **dysfunctional rotator cuff** (TO) ✗ Wrong options: osteonecrosis · RA · OA
"Rocking-horse effect minimised by?"	→ **glenoid radius larger than humeral head radius** (TO)

"Recommended mismatch?"	→ **<4 mm** (TO)
"Commonest complication 5 y after TSA?"	→ **glenoid component loosening** (TO)
"Pre-op feature most linked to early glenoid loosening?"	→ **biconcave glenoid / glenoid morphology** ✗ Wrong options: osteophytes · age · Shenton's line · cuff atrophy · pegged/ curved back · stemless · diabetes
"Aseptic glenoid loosening 5 y post — true?"	→ **pre-op posterior glenoid wear is a risk factor** ✗ Wrong options: 2nd to infection · lucent lines correlate · central peg uncemented
"Most likely mode of failure to warn about?"	→ **glenoid loosening** ✗ Wrong options: cuff tear · humeral loosening · fracture · infection
"Most common muscle ruptured after TSA?"	→ **subscapularis** (TO)
"Landmark for superior subscapularis border?"	→ **biceps tendon** (TO)
"Early rehab restricted to protect?"	→ **subscapularis tendon** ✗ Wrong options: supraspinatus · dislocation · loosening
"Greatest influence on early restrictions?"	→ **strength of the subscapularis repair**
"Can't lift hand off back, weak IR 8 mo after TSA?"	→ **subscapularis insufficiency** ✗ Wrong options: subscapular nerve palsy · biceps · brachial neuritis
"Rose from chair at 3 wk → anterior dislocation — mechanism?"	→ **subscapularis rupture**
"Fall at 4 wk, ER 80° passive, subscap tear — other complaint?"	→ **instability with external rotation** ✗ Wrong options: with IR = posterior · lateral numbness = axillary
"Fall at 4 wk, ER 90° at side, normal X-ray?"	→ **rupture of subscapularis repair** ✗ Wrong options: glenoid loosening · locked posterior dislocation
"Repaired tendon's nerves arise from?"	→ **posterior cord** (upper + lower subscapular)
"Belly-press weak, ER 110°, displaced LT on X-ray?"	→ **lesser tuberosity nonunion**
"Prior Magnuson-Stack — failed repair after TSA gives?"	→ **subscapularis, increased ER, anterior instability, positive belly-press**
"Movement to avoid immediately after aTSA?"	→ **passive ER to 60°** ✗ Wrong options: IR to chest · FE 90° · pendulums · elbow flexion
"Failed LTO after slip — exam?"	→ **increased passive external rotation** ✗ Wrong options: ER lag · drop arm · Hornblower · Yergason

– – Active space – –

"Superior subscapularis technique long term?"	→ **no technique is superior / no differences are observed between techniques**
"Click, 80° ER, axillary X-ray anterior subluxation?"	→ **subscapularis failure** ✗ Wrong options: infection · SS tear (empty can 5/5) · axillary neurapraxia
"Withdrew arm from hot pan, failed LTO – exam?"	→ **positive bear hug test** ✗ Wrong options: loss of passive ER = locked posterior dislocation
"Golfer 3 y post aTSA, lift-off lag, off-centre head?"	→ **subscapularis rupture**
"FALSE about TSA?"	→ **most common instability is pull-off of rotator cuff** (TO; it's subscapularis)
"Manoeuvre risking cuff injury?"	→ **excessive bone removal with humeral neck osteotomy**
"Fall, SS 2/5, normal X-rays 2 y after TSA?"	→ **rotator cuff tear**
"Anterosuperior escape, + Jobe + belly-press – cause + risk factor?"	→ **SS + subscapularis tear; overstuffing** ✗ Wrong options: osteolysis (rare)
"Pre-op infraspinatus atrophy → risk of?"	→ **superior rotator cuff tear** ✗ Wrong options: acromial stress fracture = rTSA
"High-riding humerus after fall, 15 y post TSA?"	→ **rotator cuff failure → rTSA**
"24 mo post, night pain, can't go overhead – also present?"	→ **positive empty can test**
"Head larger than best-fit circle, attritional SS tear?"	→ **component sizing** (overstuffing)
"P. acnes after arthroplasty – incorrect?"	→ **usually associated with fevers**
"Pain after pneumonia, ↑ WBC/ESR, TSA 5 y?"	→ **obtain radiographs + aspirate the shoulder**
"Late pain post pneumonia, mild ↑ labs, normal X-ray?"	→ **shoulder aspiration with Gram stain + culture**
"Late infection with lucencies – organism?"	→ **Propionibacterium acnes** (S. epidermidis 2nd)
"Frozen >10 WBC/HPF, GPC in clusters – treatment?"	→ **remove components + antibiotic spacer** ✗ Wrong options: graft glenoid now · resection · head exchange
"Draining sinus 3 mo post aTSA – risk factor?"	→ **corticosteroid injection 2 months before surgery** ✗ Wrong options: anatomic · female · >65 · OA
"Retraction on strap muscles – weakness?"	→ **forearm supination** (MCN)
"rTSA + latissimus transfer – nerve at risk at tenotomy?"	→ **radial**

"Arcuate artery injured during?"	→ **elevating biceps tendon from groove**
"Inferior border of subscapularis protects?"	→ **axillary nerve**
"Conjoint tendon over-retraction injures?"	→ **musculocutaneous nerve from lateral cord**
"Stem + glenoid position in TSA?"	→ **stem 25–45° retroversion + glenoid neutral or slight retroversion** (TO)
"Lateral humeral offset = GT to?"	→ **lateral base of coracoid process** (TO)
"Average humeral head version?"	→ **20 degrees retroversion** ✗ Wrong options: 40°/60° retroversion · anteversion
"Stemless vs stemmed — true?"	→ **stemless less likely to need osteotomy for malunion** ✗ Wrong options: fractures · ↓ subscap failure · stemmed more metaphyseal
"Retained platform stem at revision to rTSA?"	→ **lower overall operative time**
"Stiff at 6 wk in sling, X-ray normal?"	→ **PT for range-of-motion exercises**
"Factor with ↓ post-op narcotic use?"	→ **age** ✗ Wrong options: prior narcotics ↑
"Attending smells of alcohol before TSA?"	→ **notify the department chairman**
"Most common cause of glenoid loosening?"	→ **rocking-horse effect** (eccentric loading) (TO)
"Sourcil sign on shoulder X-ray?"	→ **acromial sclerosis in cuff tear arthropathy** (TO)

– – Active space – –

TRAUMATIC ANTERIOR SHOULDER INSTABILITY (TUBS)

– Shoulder & Elbow · 58 / 60 –

1 Basics

Epidemiology

TUBS	**T**raumatic **U**nilateral dislocation, **B**ankart lesion, requires **S**urgery
Frequency	one of the most common shoulder injuries; **1.7%/year** general population; shoulder = most commonly dislocated major joint
High-risk groups	**military**, **contact/collision athletes** (rugby, American football commonest)
Hyperlaxity	global hyperlaxity (Ehlers-Danlos, collagen disorders) → OR **2.68** for anterior instability · **3× recurrence** · fewer capsulolabral lesions · more often atraumatic instability
Dislocation direction (TO)	**anterior ~95%** · posterior ~5% · inferior (luxatio erecta) rare · one TO lecture: anterior **60–78%** of all dislocations (TO – verify: usual ~95%) – read the options
Bimodal age (TO)	young males **20–40 y** + elderly **>70 y**
Head vs glenoid (TO)	humeral head **~4× larger** than glenoid → stability sacrificed for mobility
Instability ≠ laxity (TO)	laxity = ↑ translation, **no symptoms** · instability = laxity **+ symptoms** · dislocation = complete loss of articulation
"Unstable painful shoulder" (Boyd) (TO)	pain without frank dislocation, labral tear present
Bone loss frequency in recurrent instability (TO)	glenoid bone loss **50–95%** · Hill-Sachs **~93%** · **bipolar 60%+**

Anatomy – static & dynamic restraints

Static	bony anatomy, capsule, GH ligaments, **labrum (+50% glenoid depth)**
Dynamic	rotator cuff, long head of biceps
Concavity-compression	cuff compresses head into glenoid concavity; **lost with anterior glenoid rim loss** → ↓ effective depth ✗ Wrong options: adhesion-cohesion · negative intra-articular pressure: both preserved in closed capsule

Ligament	Taut / restrains in	Direction
SGHL	**Arm at side (0° abduction)**	**Inferior**
MGHL	**45° abduction + ER**	Anterior (and posterior, OB)
Anterior band IGHL	**90° abduction + ER**	**Anterior – main anterior restraint**
Posterior band IGHL	90° abduction	Posterior

Main anterior stabiliser (TO)	**IGHL** ✗ Wrong options: MGHL · SGHL · anterior capsule · subscapularis

Main stabiliser above 45° abduction (TO)	**IGHL**
AIGHL glenoid attachment	usually to **labrum**; less commonly direct to glenoid neck
Negative intra-articular pressure (TO)	**~−42 cm H_2O** · labral tear → pressure leaks → loss of concavity-compression → recurrence
"X-factors" (TO)	**volition** + **proprioception** (besides static + dynamic stabilisers)
Labrum functional zone (TO)	labrum functional only up to **2 o'clock** → anchors placed **below the coracoid**

Pathoanatomy – Bankart lesion, IGHL & mechanism (abduction–external rotation)

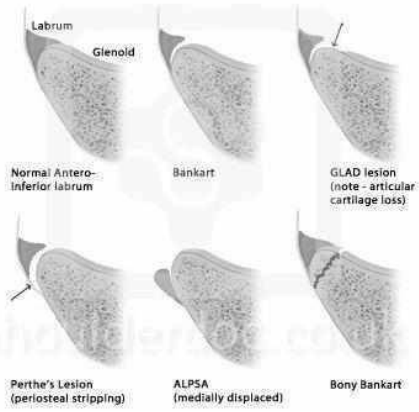
Bankart lesion — avulsion of **anteroinferior labrum + anterior band IGHL** from anteroinferior glenoid; labrum separated from the **bony glenoid**; "essential lesion" of traumatic anterior dislocation (TO)
✗ Wrong options: cartilage–labral junction only

Bankart frequency	Source
80–90% of TUBS	Orthobullets
85% (Rowe) · **87%** (Hintermann) · **97%** (Taylor 63/67) · 73.5% (Wolf)	OB studies

Most common surgical finding	**anteroinferior labral tear** ✗ Wrong options: anterosuperior · posterior · Hill-Sachs
Hintermann (212 dislocators)	Bankart 87% · ventral capsule insufficiency 79% · **Hill-Sachs 68%** · GHL insufficiency 55% · complete cuff tear 14% · posterior labral tear 12% · SLAP 7%

Lesion	Definition	Key point
Bankart	Labrum + IGHL off anteroinferior rim; **medial periosteal hinge undisrupted** (TO)	Most common
Perthes (TO)	Labrum **avulsed (not torn)**, attached by **intact but medially stripped periosteum**	May sit in **normal position** (neutral-arm MRI) → **hard to see arthroscopically**; stabiliser function lost → **recurrent dislocation**
ALPSA	"**Medialised Bankart**": labroligamentous complex **displaced medially (+ inferiorly)** on scapular neck, intact medial hinge, **fluid underneath**; anterior labrum absent on rim (TO)	**Heals medially** on glenoid neck · ↑ arthroscopic failure · common in **recurrent instability treated nonop**
GLAD	**Partial** anteroinferior labral tear + **adjacent glenoid articular cartilage defect**	↑ arthroscopic failure · debridement relieves pain (Neviaser)
HAGL	IGHL avulsed from **humeral** insertion	See HAGL block
Bony Bankart	Fracture of anteroinferior glenoid	Up to **49%** of recurrent dislocators
Kim lesion (TO)	Concealed tear at **posterior** glenoid chondrolabral junction + concealed deep labral detachment	**Posterior** instability; visible chondrolabral junction intact; probing → **labral fluctuance**; ± glenoid hypoplasia

– – Active space – –



Labral lesion spectrum – Bankart, GLAD, Perthes, ALPSA, bony Bankart

Association (TO)	Bankart, Perthes, HAGL, ALPSA, GLAD → all **recurrent anterior** instability · Kim → **posterior**
97% rule	**97%** of recurrent instability have **Bankart or ALPSA**
Mechanism	anteriorly directed force with arm **abducted + externally rotated** · posterior instability = **adduction + internal rotation**
Bankart vs position	AIGHL/anteroinferior labrum taut in **ER at 90° abduction** ✗ Wrong options: 45° = MGHL
Associated lesion in young dislocator	**Hill-Sachs** ✗ Wrong options: full-thickness cuff tear = elderly · reverse Hill-Sachs = posterior · coracoid fracture = rare · GT fracture = less common than Hill-Sachs
"Clunk" in 22-y basketballer	Bankart + **Hill-Sachs**

Combined anterior & posterior labral tears

Pan-labral tear	**270–360°** labral tear (anterior + inferior + posterior); anterior and/or posterior instability
Exam	positive **Jerk + Kim** (posterior) **+ apprehension + relocation** (anterior) → **270° tear** → anterior + posterior labral repair (arthroscopic)
✗ Wrong options (images)	bony Bankart / anterior Bankart (anterior repair only) · subscapularis tear (cuff repair) · Kim lesion (posterior only)
270° repair	**15% failure** (Mazzocca)
Military	high incidence of posterior and combined front-to-back tears → **unrecognised pan-labral tear** = cause of recurrence

HAGL lesion

HAGL	**humeral avulsion of the (inferior) glenohumeral ligament**, anterior band IGHL, **without subscapularis tear**; term coined by **Wolf (1995)**
RHAGL	reverse HAGL = posterior band IGHL → **posterior** instability
Frequency	**7–9%** of unstable shoulders (Wolf 9.3%: Bankart 73.5%, capsular laxity 17.2%) · Taylor **1/67** · Rhee 2% – **rare**
Patient	**slightly older** than Bankart (Orthobullets) · **older + more violent trauma** (OB) · female collegiate athletes · hyper-abduction + ER

MRI	best on **coronal** sequences: **J-shaped IGHL**, contrast **extravasating down humeral neck**; bony HAGL variant; "floating" AIGHL = HAGL + capsulolabral detachment
Associations	always with other lesions (Bankart, LHB tear, cuff tear); more severe anterior pain
Significance	"other end of the IGHL" → **missed HAGL = cause of failed Bankart repair** → higher recurrence if unrepaired
Treatment	**open repair = classic / gold standard** · arthroscopic possible but technically hard (**70° scope**, accessory portals) ✗ Wrong options: arthroscopic contraindicated classically

2 Mechanism

Hill-Sachs lesion

Hill-Sachs	**compression/impaction fracture of posterosuperior (posterolateral) humeral head** against **anteroinferior glenoid rim**; chondral impaction injury
Frequency	**80–100%** of traumatic dislocations · **25%** of traumatic subluxations · 68% (Hintermann) · 73% on X-ray (Edwards)

Most common associated intra-articular finding with Bankart
✗ Wrong options: cuff tear · SLAP · posterior labral tear · Buford

Arthroscopy	indentation on posterior head **beside the bare area** ✗ Wrong options: bare area = normal cartilage-free zone
Significance	clinically irrelevant **unless it engages**; **end-range (ABER) instability**, not mid-range (TO)
Glenoid bone loss (TO)	from anteroinferior rim fracture or **erosion** by humeral head; **6.8% loss per dislocation** (average)
Types of bone loss (TO)	glenoid **attritional (chronic)**: repeated dislocations shave anterior rim · glenoid **acute (bony Bankart)**: avulsed fragment with labrum · humeral **Hill-Sachs**: posterosuperior head impacted on anterior glenoid
History (TO)	Hill-Sachs described **1940** (Hill & Sachs); earliest impaction-fracture reference **1861** (Flower) · **Rowe (1978)**: bone loss → Bankart failure · **Burkhart & De Beer (2000)**: bone loss = main predictor of recurrence; **engaging lesion** concept
Reverse Hill-Sachs	anteromedial head; **posterior** dislocation; lesser tuberosity fracture = posterior dislocation

3 Presentation

History	traumatic event; ask **formal reduction vs spontaneous**; instability feeling; pain from subluxation/excess translation
Apprehension	supine, **90° abduction + 90° ER** → apprehension; positive in **mid-range abduction = glenoid bone loss**
Relocation	posterior force on humeral head → ↓ apprehension
Load and shift	grades 0–III (table below); **sulcus sign** with arm at side

– – Active space – –

Clues to significant bone loss (TO) apprehension at **lower degrees** of abduction/ER · **ease of dislocation** (sleep, minor movement) · catching/locking · **high-energy first episode** (bony Bankart) · failed Bankart

Dislocation sign (TO)	Finding
Dugas	Cannot touch opposite shoulder
Hamilton ruler	Straight edge from **acromion to lateral arm** lies flat (lost deltoid curvature)
Callaway	↑ **arm circumference** on affected side
Flattened deltoid	Loss of shoulder contour

Limb attitude (TO) anterior **abduction + ER** · posterior **adduction + IR** · inferior **arm locked overhead** · mnemonic **"A–E, P–I"**

Associated fractures (TO) **GT fracture → anterior** dislocation · **LT fracture → posterior** dislocation

Anterior test (TO)	Technique	Meaning
Apprehension (Jobe)	90° abduction + ER	Pain / apprehension
Relocation	Posterior pressure on head	Relief → confirms instability origin
Release	Remove hand	Pain / apprehension returns → confirms
Rowe	Edge of couch, push head forward	↑ sensitivity
Andrews	**130° abduction + ER**	Alternative to Jobe
Rockwood	Apprehension **maximal at 90° abduction**	IGHL involvement
Anterior drawer	Supine, 80–90° abduction + slight flexion + ER	Quantifies translation

Diagnostic trio (TO) **apprehension + relocation + release**

Diagnostic criteria – hyperlaxity

Criterion	Threshold
Beighton score	**>4** · **≥5/9** (TO)
Hyper-ER at side	**>85°**
Hyperabduction (Gagey)	**>105°**
Load–shift	**>2+ in ≥2 planes**

ISIS hyperlaxity anterior (ER >85°) or inferior (Gagey) laxity = 1 point

4 Diagnosis & Classification

Key radiology findings

Modality / view	Use
Trauma series: true AP (Grashey) + scapular Y + axillary	Required; Grashey = beam ~45° off AP, end-on GH joint; **Y = humeral head over Y intersection** (scapular spine + coracoid)
Axillary	**Best for detecting dislocation**; bone loss visible here = already significant; Velpeau if can't abduct
West Point	**Glenoid bone loss / bony Bankart**
Stryker notch	**Hill-Sachs** (supine, hand on head, **10° cephalic**); poor for Bankart
AP in ER	Hill-Sachs (ISIS item)
Bernageau profile	Glenoid profile, "cliff sign"; **less accurate than 3D imaging**
CT ± 3D	Bony injury, **% glenoid bone loss**; **3D CT = gold standard**
CT arthrogram	**MRI contraindicated** (pacemaker, cochlear implant); poorer soft tissue
MRI	**Best for labral tear**; validated for bone loss
MR arthrogram	**Study of choice for soft tissue**: sens **86–91%**, spec **86–96%** (MRI 44–100% / 66–95%)

X-ray view (TO)	Use
Garth (apical oblique)	**Both** Hill-Sachs + glenoid rim in one view
AP in IR	**Best AP view for Hill-Sachs**
Bernageau	Anterior glenoid contour: **blunted angle, "cliff sign"**
Axillary lateral	**Best for confirming direction** of dislocation

View mnemonic (TO) **West Point → glenoid · Stryker → Hill-Sachs · Garth → both**

CT vs MRI (TO) **CT = quantify** bone loss (**3D CT with humeral subtraction** best) · **MRI = qualify** (presence + soft tissue; poor correlation with size) → confirm on CT before surgery

CT indications (TO) recurrent dislocation · high-energy injury · failed Bankart · X-ray/MRI suggests bone loss · young athlete **<25 y** planning early repair

MRI clock face (TO) on axial MRI **coracoid = 12–2 o'clock** · **Bankart = 3–6 o'clock** · **Kim lesion = 6–9 o'clock** (posteroinferior)

MRI positioning – ABER view

ABER **abduction + ER → humerus in coronal plane, glenoid in axial plane**

Shows **anteroinferior labrum**, **anterior band IGHL**, articular surface of supraspinatus, posterosuperior labral peel-back (sens 73%, spec 100%)

– – Active space – –

Glenoid bone loss – assessment & critical thresholds

Method	Fact
3D CT en face	**Gold standard**; **highest interrater reliability**; predictive error **similar to MRI** (Rerko)
Best-fit circle (TO)	Circle fitted to **posterior + inferior** glenoid ≈ original glenoid; or **contralateral glenoid** reference
Bare-spot method	Bare spot = glenoid centre (Burkhart); GBL = (B − A) / 2B (A = bare spot→anterior edge, B = →posterior edge)
MRI	Tends to **overestimate** GBL (✗ Wrong options: underestimates); less bone–soft tissue contrast; cuts must be en face
MR arthrogram	Not a substitute for CT (poorer bony definition)
X-ray (Bernageau, West Point)	Significantly less accurate

Worked example (OB) bare spot 9 mm anterior, 12 mm posterior → (12 − 9)/24 = **12.5%** + non-engaging HS → **Bankart alone**

Worked example (OB) 6.9 mm lost of 28.3 mm → **24.4%** = critical → **coracoid transfer**

Inverted pear glenoid morphology with ↑ bone loss (**>25%**, TO/OB); **89% failure** after arthroscopic repair

Glenoid shape normally pear-shaped (narrow superior); anteroinferior loss → **inverted pear**

Bare spot (TO) central cartilage-free zone at **junction of lower ⅔ and upper ⅓** · glenoid **max AP width ≈ 25 mm** · reference for bone-loss measurement

Arthroscopic bare-spot formula (TO) written **(B − A) × 2 / B × 100%** (TO – verify: usual **(B − A) / 2B**)

Quantification method (TO)	Principle	Remark
PICO / surface area	Best-fit circle centred on bare spot → % area lost	Simple, reliable, **single CT**
Linear (bare spot)	Anterior vs posterior radius	Also arthroscopic
Glenoid index (ratio)	Involved vs **opposite** glenoid width	Needs **bilateral CT**
Osseous fragment length	Defect length **> glenoid radius → grafting**	Quick screening
Di Filippo	Multiplanar CT, both sides	Precise, complex
Sugaya	3D CT en face	Quantification

Most reliable + practical (TO) **PICO** and **glenoid index**

Saito severity (TO)	% glenoid loss	Meaning
Mild	**<15%**	Usually insignificant → Bankart
Moderate	**15–25%**	Borderline → arthroscopic engagement / on-off track
Severe	**>25%**	Bankart alone likely fails → bony procedure

Critical glenoid defect (TO) **>21%** → isolated Bankart fails ("universal failure", Saito / Itoi / Burkhart & De Beer) · **inverted pear = ≥25%** loss (TO)

Threshold	Meaning	Source
>20–25%	**Critical**; soft tissue alone fails (>2/3 failure) → bony procedure	Orthobullets, most OB
>25–30%	Osseous augmentation needed	OB (bone loss explanations)
20–30%	"Glenoid bone deficiency" → Latarjet; **>60% recurrence** with Bankart alone	OB
15–25%	Biomechanically unstable	OB
21% of glenoid length	Instability + ↓ ER after Bankart (Itoi)	OB
17.3%	Critical value (Shin: 43.9% vs 3.7% recurrence)	OB
>13.5%	**Subcritical** – critical may be this low; open Bankart ↑ recurrence	Orthobullets, OB
<15%	Soft-tissue stabilisation alone works	OB, TO

Bone-loss conflict critical **20–25%** (Orthobullets) · **25–30%** (some OB) · **13.5–17.3%** subcritical (recent studies) – read the options

Strongest predictor of failure after arthroscopic Bankart **glenoid bone loss**
✗ Wrong options: age at first dislocation · age at surgery · >3 anchors · on-track HS

Most predictive intra-op finding **anterior glenoid deficiency 35%**
✗ Wrong options: non-engaging HS · loose body · bursitis · 10% PASTA

Mechanism lost with rim loss **concavity-compression**

Hill-Sachs lesion & glenoid track (on/off-track)

Bipolar concept glenoid bone loss + Hill-Sachs together; bipolar > unipolar engagement risk

Engaging lesion (Burkhart & De Beer 2000) (TO) in **abduction + ER** the Hill-Sachs axis **parallels the anterior glenoid rim** → engages → dislocation · on/off-track (**Yamamoto, Itoi, Di Giacomo 2014**) = final evolution

Glenoid track 83% (TO) remaining **17% anterior margin = safety zone**

Hill-Sachs quantification (TO)	Significant if
Hill-Sachs quotient	Width × depth × length (cm) **>2.5**
Ratio method (true AP)	Depth **>20% of head radius**
Surface involvement	**>20%**

Franceschi grade (TO)	Hill-Sachs
I	**Cartilaginous** only
II	**Superficial bony scuffing**
III	**Deep "hatchet-type"** fracture

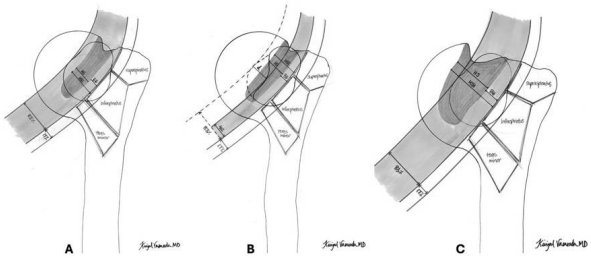
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Flatow & Warner head involvement (TO)	Significance
<20%	Clinically **insignificant**
20–40%	**Variable**
>40%	**Significant → surgery**

Glenoid track (TO) area of posterior humeral articular surface in contact with glenoid along **posterior end-range (ABD + ER)**; width = medial margin of track → medial margin of cuff footprint

Formula	Value
GT = 0.83D − d	D = inferior glenoid diameter, d = anterior bone loss width (Orthobullets, TO, most OB)
GT = **0.84(D) − ABL**	OB (glenoid-track formula question)
84% vs 83% (TO)	**84% cadaveric · 83% live at 90° abduction**
HSI = HS + BB	Hill-Sachs width + **bony bridge** to cuff insertion
HSI > GT	**Off-track = engaging** → recurrence
HSI < GT	**On-track = non-engaging**

Coefficient conflict **0.83** (Orthobullets, TO, most OB) · **0.84** (one OB; "HS >84% of glenoid width = off-track") – read the options



Glenoid track vs Hill-Sachs interval – on-track (A), off-track from glenoid loss (B) or large Hill-Sachs (C)

Medial extension **the more medial the HS extends, the more likely off-track** ✓

✗ Glenoid-track traps on-track engages · unipolar > bipolar risk · **GT only by CT** (MRI also usable) · HSI < GT engages

Dynamic arthroscopic exam (TO) assess engagement **after the Bankart repair**

Tyre & pothole glenoid track = tyre, HSI = pothole; off-track = tyre falls in (big HS or big glenoid loss)

Off-track predictive value 75% vs 8% recurrence (Shaha); **superior to >20% bone-loss cut-off**

Worked example	Calculation	Verdict
Kabaddi (TO): D 24, width 20, HSI 18	d = 4 → **16.6%**; GT = 19.92 − 4 = **15.92**	**Off-track + subcritical, elite contact → Latarjet**
Water polo (OB): D 29, d 2, HSI 19	GBL **6.9%**; GT = 24.07 − 2 = **22.07**	**On-track → isolated arthroscopic Bankart**
TO: 25% GBL, GT 18, HSI 21.3	Off-track + critical	**Latarjet**

Instability Severity Index Score (ISIS – Balg & Boileau)

– – Active space – –

Factor	Parameter	Points
Age at surgery	**<20 y**	**2**
Sport level	**Competitive**	**2**
Sport type	**Contact or forced overhead**	**1**
Hyperlaxity	Anterior or inferior	**1**
Hill-Sachs on AP in ER	Visible	**2**
Glenoid contour on AP	**Loss of contour**	**2**
Total		**10**

Score	Recurrence after arthroscopic Bankart	Advice
≤6	**~10%** acceptable (3 points → 5%)	Arthroscopic stabilisation
>6	**~70%**	**Open / Latarjet**

Latarjet trigger ISIS **>6** (Orthobullets summary, OB, TO) · ISIS **>4–6** (Orthobullets technique) – read the options

ISIS worked (OB) 18 ♀ competitive tennis (forced overhead), HS + glenoid contour loss = **9** (>70%) · competitive + contact + HS on ER AP, age 21, normal laxity (ER 80°, negative Gagey) = **5**

ISIS limitations (TO) Boileau, **JBJS 2007** · does **not quantify bone loss** · **ignores bipolar lesions**

Anteroposterior translation (load & shift)

Grade	Translation
0	Normal
1+	**Up to glenoid rim**
2+	**Over rim, spontaneously reduces**
3+	Over rim, **does not reduce**

Load & shift grade (TO)	Translation (% head diameter)
0	**<25%** (normal)
I	**25–50%**, rides up to rim
II	**>50%**, dislocates, relocates on release
III	Dislocates and **stays out** until pushed back

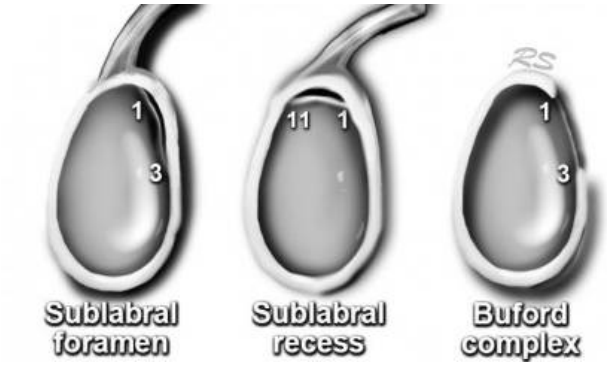
Sulcus sign

Grade	Acromiohumeral interval
1	**<1 cm**
2	**1–2 cm**
3	**>2 cm**

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Normal variants mistaken for pathology (Buford complex)

Variant (TO)	Location	Feature	Prevalence
Sublabral foramen (hole)	**1–3 o'clock** (anterosuperior)	Unattached anterosuperior labrum	**11%** (TO) · ~12% (Orthobullets)
Sublabral recess	**12 o'clock** (biceps anchor)	Doesn't extend to 1–3 o'clock; **mimics SLAP**	–
Buford complex	**1–3 o'clock**	**Absent anterosuperior labrum + thick cord-like MGHL**	**~1.5%**



Sublabral foramen, sublabral recess, Buford complex

Buford (TO) normal variant; **not associated with recurrent dislocation**; do not repair (→ loss of ER)

5 Management

First-time traumatic anterior dislocation

- Reduce under sedation / IA lidocaine → **traction-countertraction (commonest)**
  - fails 5–10% → **blocks: biceps tendon, capsule, fracture fragments**
- Low-risk → **sling ≤1 wk + PT (cuff + periscapular)**
- Age >40 + persistent weakness → **MRI shoulder (cuff tear)**
- In-season athlete, no bone loss → rehab, RTP in-season → **surgery end of season**
- Young (<20–25) contact / military + Bankart on MRI → **early arthroscopic Bankart**

Recurrent instability – glenoid side (3D CT)

- GBL <13.5–15% + on-track → **arthroscopic Bankart (≥3–4 anchors)**
- GBL <15% + off-track → **Bankart + remplissage**
- Subcritical 13.5–25% → contact athlete / high ISIS → **Latarjet**
  - non-contact, off-track → **Bankart + remplissage**
- GBL >20–25% / inverted pear → **Latarjet (coracoid transfer)**
- Failed Latarjet / massive loss → **ICBG (Eden-Hybinette) or distal tibia allograft**

Recurrent instability – humeral side (Hill-Sachs)

- On-track / non-engaging (<20%) → **nothing extra**
- Off-track / engaging 20–40% (>25%) → **remplissage**
- >40% → **humeral head allograft / ICBG; hemiarthroplasty if >40–45% + arthritis**

Glenoid side (TO bone-loss lecture)

- <10% → ignore → **Bankart**
- 10–25% → grey zone → **on/off-track**
- 25–40% → **Latarjet / ICBG**
- >40% → **ICBG / allograft reconstruction**

Humeral side (TO bone-loss lecture)

- <10% → ignore → **Bankart**
- 10–30% → **remplissage**
- 30–40% → **ICBG / Latarjet**
- >40% → **resurfacing / arthroplasty**

Operative criteria

Procedure	Indications
Arthroscopic Bankart ± capsular plication	**First-time** dislocation + MRI Bankart in athlete **<25 y** · high-demand athletes · **recurrent (>1) dislocation** after nonop · **GBL <20–25%** · on-track (± remplissage if off-track)
Open Bankart ± capsular shift	GBL <20–25% · **revision after failed arthroscopic Bankart** without GBL >20% · **HAGL** · acute glenoid fracture · **hyperlax needing formal capsular shift** · contact athlete returning to sport
Latarjet / Bristow	**GBL >20–25%** (inverted pear) · **subcritical >13.5%** in high-risk · **ISIS >6** · contact athletes / athletes of consequence (climbers, big-wave surfers) · Europe uses aggressively
ICBG / distal tibia / distal clavicle graft	GBL >20–25% · **revision of failed Latarjet** · **less graft resorption** than Latarjet
Remplissage + Bankart	**Off-track / engaging** HS (**>25–40%**) with **GBL <20–25%** · subcritical GBL (arthroscopic-alone recurrence ~20%)
Hill-Sachs bone graft	Engaging HS **>40%**
Tendon transfer (conjoint / LHB)	Capsulolabral deficiency; with labral repair if **GBL <15%**

Bipolar matrix (TO)	Hill-Sachs on-track	Hill-Sachs off-track
GBL <15%	**Bankart**	**Bankart + remplissage**
GBL 15–25%	**Latarjet** (Bankart + remplissage if lower demand)	**Latarjet**
GBL >25%	Latarjet or ICBG	Latarjet or ICBG

– – Active space – –

TO algorithm: GBL	Hill-Sachs	Treatment (TO)
<10%	None / non-engaging	**Bankart**
<10%	Engaging, head loss <30%	**Remplissage**
<10%	Engaging, head loss >30%	**Latarjet**
10–25%	None / non-engaging	**ISIS <6 → Bankart · >6 → Latarjet**
10–25%	Engaging, head loss <20%	**Remplissage**
10–25%	Engaging, head loss >20%	**Latarjet**
25–40%	Any	**Latarjet** + ICBG to humeral head if head loss >30%
>40%	Any	**ICBG / allograft**

Di Giacomo / Itoi / Burkhart 2014 (TO)	Hill-Sachs on-track	Hill-Sachs off-track
Glenoid defect <25%	**Arthroscopic Bankart**	**Bankart + remplissage**
Glenoid defect >25%	**Latarjet**	**Latarjet ± humeral procedure** (remplissage / bone graft)

TO recurrent-dislocation algorithm	Hill-Sachs	Procedure (TO)
<10%	Non-engaging	**Bankart**
<10%	Engaging	**Bankart + remplissage**
10–25%	Non-engaging	**Bankart ± capsular shift**
10–25%	Engaging	**Bankart + remplissage**
25–40%	Any	**Latarjet**
>40%	Any	**ICBG / allograft**

10–25% GBL + engaging HS	**Bankart + remplissage** (TO recurrent-dislocation lecture) · **Latarjet if head loss >20%** (other TO algorithm) – read the options
Bankart indication (TO)	**<20%** glenoid loss (recurrent lecture) · **<15%** (instability lecture) · "On-track → soft tissue safe; off-track → needs bone" (TO)
TO Hill-Sachs size rule	non-engaging (**usually <20%**) → nothing · engaging **20–40%** → remplissage · engaging **>40%** → bone graft (Eden-Hybinette)
TO soft-tissue zone	isolated Bankart if GBL **<15–20%** (stricter **13.5%** best-fit circle in high-demand athletes) + on-track
TO subcritical (15–20%)	**contact athlete (rugby, kabaddi, wrestling) → Latarjet even at 15%** (Bankart failure ~30%) · non-contact → Bankart + remplissage
TO bone grafting triggers	GBL >20–25% · **failed previous Bankart** · bipolar loss making an on-track HS off-track

Wrestler, 30 dislocations (incl. in sleep), Bankart + 40% HS (TO) → Latarjet
✗ Wrong options: Bankart · remplissage · McLaughlin

Recurrence risk – age, ISIS score & natural history

Age at first dislocation	Recurrence with nonop	Source
<20 y	**~90%** (80–90% teens; >90%)	Orthobullets, OB
14–18 y	**~90%**	TO
<25 y	**60–94%**; likely to re-dislocate	OB, TO
<30 y	↑ risk (significant)	OB
All comers	**33–67%**	OB

Most significant risk factor (TO, OB) **young age (<20)**; only consistent predictor regardless of immobilisation/exercise

Age cut-off conflict **<20** (Orthobullets, ISIS, OB) · **<25** (TO, some OB) · **<30** (OB immobilisation question) · 21–30 (Kralinger) – read the options

Age <20 recurrence (TO) **~70–90%** · up to **80%** in young athletes · **Beighton ≥5/9** = generalised laxity risk factor

Recurrence risk factors (TO)	
Modifiable	**Manual occupation**, overhead occupation, immobilisation after dislocation, **collision sport**, time to RTS, psychosocial (pain, **fear of reinjury**)
Non-modifiable	**GT fracture**, **Hill-Sachs**, **bony Bankart**, **axillary nerve palsy**, **age**, **gender**, **hypermobility**

Orthobullets risk factors for re-dislocation **age <20 (highest)**, **male**, **contact sport**, **hyperlaxity**, **GBL >20–25%**, **GT fracture**

Hill-Sachs & recurrence HS **not significantly associated** (TO) · HS = ISIS item / predictor after Bankart (OB) – read the options

Dominance **non-dominant less likely to recur** (dominant limb used in high-risk activity; TO) · **no difference** dominant vs non-dominant (OB) – read the options

Not top risk factors (OB) contralateral dislocation (seen in 25% of recurrent) · family history (34% recurrence risk) · patellar instability · dominance

Natural history (Hovelius, 25 y nonop) 43% never re-dislocated · 15% re-dislocated then stabilised · 27% had surgery · immobilisation length no effect

Recurrent after PT, dislocates in sleep → discuss surgical stabilisation
✗ Wrong options: steroid · NSAIDs · change therapist · 6 wk immobilisation

First-time dislocation – nonoperative vs early arthroscopic Bankart

First-time dislocators management controversial; **ASES: surgery for athletes 14–30 y at end of season** if **positive apprehension + bone loss**

Reduction techniques **traction-countertraction (commonest)** · **Kocher**: arm at side in ER → forward flex → IR · **Hippocratic**: heel in axilla · **Stimson**: prone, weight hung from arm · scapular manipulation

Kocher sequence (TO) **Traction → ER → Abduction → IR ("T-E-A-I")** (TO – verify: classic Kocher = traction → ER → **adduction** → IR) · **Hippocratic = obsolete** (TO) · reduce ideally **under anaesthesia + muscle relaxation** (TO)

– – Active space – –

Immobilisation (TO first-time lecture) traditional **3 wk adduction + IR** · current concept **1 wk in ER brace** (capsulolabral healing, ↓ stiffness) · **>3 wk → stiffness** esp. elderly

Rehab after first dislocation (TO)	Duration	Focus
Phase 1	**0–1 wk**	ER brace
Phase 2	1–3 wk	Pendulum + assisted ROM
Phase 3	3–6 wk	Active ROM, avoid extreme ER
Phase 4	**>6 wk**	Strengthening + proprioception

Return to sport after nonop (TO) **~3 months** if stable

Operative indications after first episode (TO) **GT / LT / glenoid rim fracture** · **bony Bankart** · **cuff tear (>40 y)** · **failed closed reduction** · high-demand athlete **<25 y + Bankart on MRI** · open reduction via **deltopectoral**

Immobilisation **no benefit >1 wk**; short (≤1 wk) = prolonged (≥3 wk) recurrence (41% vs 37%, Paterson) → **may stop sling at 1 wk**

ER immobilisation early studies ↓ recurrence <40 y (labrum reduced to glenoid) → **subsequent studies refuted**; IR 24.7% vs ER 30.8% (Liavaag, NS)

Immobilisation conflict no benefit / refuted (Orthobullets, OB) · **immobilisation decreases recurrence** (TO) – read the options

Bracing **no evidence it ↓ recurrence**; limits ABD + ER in season

Return to play (nonop) **faster than surgery**; 7–21 days (Orthobullets) · 2–3 wk (OB) · **3–4 wk** with full painless ROM + strength (OB) → can finish season

Nonop vs surgery recurrence nonop **higher**: 88% vs 14% (Arciero) · 77% vs 11% (Bottoni); RTP without recurrence next season 40% vs 90% (Dickens)

Delaying surgery further events → more labral damage, **more bone loss**, ↑ failure after Bankart; trend = **surgery after first dislocation**

Bony vs soft-tissue (first-time) Latarjet **equal or lower** recurrence than Bankart

Water polo, 3rd dislocation, GBL <7%, HSI 19 < GT 22.07 → isolated arthroscopic Bankart
✗ Wrong options: remplissage → ↓ ER · Latarjet / DTA · humeral OATS

Arthroscopic vs open Bankart repair

	Arthroscopic	Open
Re-dislocation	**Similar** (RCTs) · slightly higher in some · 11% vs 8% (Harris) · 15.1% vs 7.7% (Rollick)	Similar
ROM	**Slightly better** (less pain, ER preserved)	More pain, ↓ ROM
Return to work/sport	Equal (RCT)	**Equal or slightly better** (some studies)
Arthritis	**No difference**	No difference
Approach	Arthroscopic, ≥3 (preferably 4) anchors	**Deltopectoral**; subscapularis **transverse split or tenotomy**; **inferior capsule shifted superiorly**

Arthroscopic advantages cosmesis, ↓ morbidity, ↓ ER loss, diagnostic confirmation

Contraindications to arthroscopic (Stein) significant GBL · attenuated capsulolabral tissue · engaging HS · **HAGL** (open = gold standard)

Isolated open Bankart + capsular shift **recurrent after arthroscopic repair, minimal GBL (6%), on-track, contact athlete** ✓

✗ Open Bankart distractors 20% GBL rugby (→ Latarjet) · 5% GBL off-track (→ Bankart ± remplissage) · 45% HS (→ humeral graft) · 18% GBL wrestler (→ bony)

Technique **fully mobilise labrum** before repair · **≥3 anchors** (<3 → failure) · "drive-through sign" pre-repair · knotless anchors

Bankart goals (TO) reattach labrum · **retension IGHL** (superior capsular shift) · **"bumper" effect** on rim

Anchor positions (TO) on glenoid face at **3, 4, 5 o'clock** (right) · **2–3 anchors sufficient** (TO) · ≥3–4 anchors (Orthobullets, OB) – read the options

Balanced repair (Snyder) (TO) add **posterior capsular plication** if head not centred

Arthroscopic Bankart (TO) **gold standard** for anterior instability **without bone loss**

Position (TO)	Detail
Lateral decubitus (preferred)	**45° abduction + 15–20° forward flexion**, **20–30° posterior tilt**
Beach chair	Alternative; risk of **cerebral hypoperfusion**

Portal (TO)	Location	Use
Posterior	2 fingers down + medial to posterolateral acromion	Viewing
Anterior	Between coracoid + anterior acromial edge	Working: anchors, lasso
Anterosuperior	Just lateral to acromial edge	Shuttle

Modern trends (TO) **single anterior portal** Bankart (less time + morbidity, same stability) · **all-suture soft anchors** (preserve bone stock, less artefact) · **knotless** anchors (no suture abrasion) · biocomposite anchors

Rehab after stabilisation (TO)	Timing
Sling	**2–3 wk**
Passive ROM	From **3 wk** (avoid ER stress)
Active strengthening	**6–8 wk**
Return to sport	**4–6 months** (Latarjet later)

Too many anchors **postage-stamp fracture** through anchor holes

Recurrence after arthroscopic Bankart 13% (Ahmed) · 15.3% (Boileau) · 4% without bone defect vs **67%** with (Burkhart) · **89%** contact athletes with inverted pear

Bony Bankart – arthroscopic fixation

Small bony Bankart (<20%) **arthroscopic repair incorporating the fragment**; 95% RTS (Sugaya)
✗ Wrong options: ORIF · iliac crest · coracoid transfer

26-y accountant, 3rd dislocation, 10% bony defect, no prior surgery → arthroscopic bony Bankart repair
✗ Wrong options: ER immobilisation · remplissage · coracoid/iliac crest

– – Active space – –

ORIF large **acute** glenoid rim fractures only
✗ Wrong options: recurrent instability

Bony Bankart + deltoid weakness after reduction → bony Bankart stabilisation + observe axillary nerve (EMG changes in ~48%; surgery only if no recovery at **3–6 months**)

Fixation open: screws or suture (linear / bridge)

Imaging 3D CT > CT > MRI > X-ray for bone loss (MOON)

Unaddressed bony Bankart commonest cause of failed soft-tissue Bankart
✗ Wrong options: labral nonunion · HAGL · SLAP

Remplissage

Remplissage ("to fill") **infraspinatus + posterior capsule** sutured into Hill-Sachs (posterior capsulodesis + infraspinatus tenodesis) → **intra-articular HS becomes extra-articular** (TO) → can't engage rim in ER

Remplissage origin + range (TO) **Wolf** (2004 in one TO lecture · 2007 in another) · for **10–30%** defects (TO bone-loss lecture) · **20–40%** (TO recurrent lecture)

Humeral option (TO)	Indication	Key point
Disimpaction / elevation	**<3–4 wk old, <40%** surface	Cortical window → tamp → bone graft
Remplissage	10–30%	Infraspinatus tenodesis + posterior capsulodesis
Allograft plug (Kropf, Sekiya)	Moderate	Structural resurfacing
Rotational osteotomy (Weber 1969)	Large defect, **young**	**25° medial rotation** of head · **5.7%** recurrence
Resurfacing / hemiarthroplasty	**>40%** surface loss, elderly / arthritic	–

RSA / arthroplasty (TO) **elderly with >40%** humeral loss

Goal (TO) ✓ extra-articular HS
✗ Wrong options: more secure labrum · ↑ head diameter · ↑ glenoid track width

Effect shortens HSI → **off-track → on-track**; ↑ rotation before engagement (**does not increase glenoid track**)

Indications **off-track/engaging HS** + GBL **<20–25%** (<15% TO) · HS **>25%** (OB) / **25–40%** (Orthobullets) · chronic Bankart + engaging HS catching in **90/90 → arthroscopic Bankart + remplissage**
✗ Wrong options: remplissage alone · humeral graft alone

Technique arthroscopic, percutaneous / posterolateral portal; **pass sutures before anterior repair, tie after**; knotted or knotless; **double-pulley**

Post-op **avoid forward flexion + adduction** (tensions posterior capsule; abduction need not be avoided); brief ER immobiliser

Complications **ER loss 5–15°**; recurrence **5% at 2 y**; IR loss <2°

Outcomes vs **Latarjet (2 y): 1.4% vs 3.2% recurrence** despite greater bipolar loss · vs **Bankart alone: ↓ recurrence + revision** (MacDonald RCT) · Latarjet better ER

Moderate HS + labral tear after 2nd dislocation → arthroscopic labral repair + remplissage
✗ Wrong options: PT · labral repair alone · hemiarthroplasty – HS not >45% · thermal capsulorrhaphy

Connolly procedure — **infraspinatus + GT bone block** into HS · analogous **Neer-McLaughlin** = subscapularis + LT into **reverse HS** (posterior)

Humeral head allograft (OATS) (TO) — massive deep HS **>40%** where remplissage would severely limit ER

HS bone graft outcomes — small cohorts; early arthritis **16% at 4 y**; Constant 78.5–86.8; options: allograft, arthroplasty, rotational osteotomy

Latarjet/bone block – indications, mechanism & outcomes

Latarjet — transfer of **coracoid + conjoined tendon + CA ligament** to anteroinferior glenoid through **subscapularis split**; deltopectoral; open = arthroscopic outcomes (steep arthroscopic learning curve)
✗ Wrong options: detached

Triple effect	Mechanism
Bony	↑ glenoid surface/width, **lengthens glenoid track / articular arc**
Sling (TO)	**Conjoint tendon** (short head biceps + coracobrachialis) over lower subscapularis → **dynamic stability to anteroinferior capsule in ABD + ER**; brace across lower third of joint
Capsular	**CA ligament stump** repairs capsule

Triple effect question — **all three (bony + sling + CA ligament)** ✓

Triple effect = Patte (TO) — bone block + conjoint sling + **tenodesis effect** (conjoint + lower subscapularis reinforce capsule)

History (TO) — **Latarjet 1954**: **2–3 cm** coracoid transferred along long axis to anteroinferior glenoid neck · **Bristow (Helfet 1958)**: coracoid **tip** to glenoid neck; non-anatomic, relies on sling, poor bone coverage · gold standard for **>25%** loss (TO) · **25–40%** range (TO)

Latarjet technical point (TO)	Detail
Fixation	**2 screws parallel to glenoid face**, toward scapular spine
Safe zone	**<28° medial tilt** (axial) · **<29° superior inclination** (coronal)
Graft position	**2–5 o'clock** (right)
Nerve risk order (TO)	**Suprascapular > axillary > musculocutaneous**
Nonunion	**2–9%** – poor decortication / fixation
Recurrence	**<1%** if done correctly
Malposition	**Overmedialisation → persistent instability** · lateral overhang → arthritis

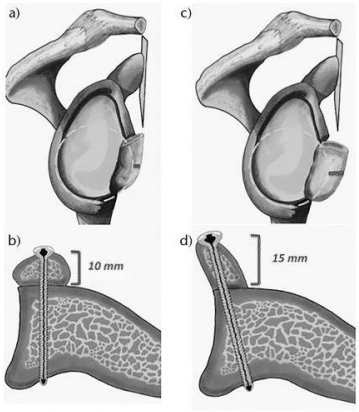
Arthroscopic Latarjet (Lafosse) (TO) — **5 stages**: exposure → coracoid prep → drilling/osteotomy → transfer → fixation · **91% excellent**, **80% grafts flush** on CT (Lafosse & Boyle 2010)

– – Active space – –

Sling effect (TO)

✓ conjoint tendon dynamic stability
✗ Wrong options: CA ligament transfer = capsular · bone surface = bony · LHB tenodesis = DAS

	Traditional Latarjet	Congruent arc	Bristow
Coracoid orientation	**Undersurface on glenoid**; lateral surface = new track	**Rotated 90°**; **inferior surface** = track, articulates with head	**Distal ~1 cm tip**, orientation unchanged
Track gain	Less (~10 mm)	**Most: ~15 mm**	Less
Note	**Less fracture-prone** at screw insertion	Better for **severe bone loss**	Less common; **MC nerve tethering** (TO)



Traditional (a–b, ~10 mm) vs congruent arc (c–d, ~15 mm) Latarjet

Graft position

flush, **no lateral overhang** (→ OA) · not **≥1 cm medial** or >1.5 cm migration (→ redislocation) · intra- vs extra-articular placement (intra-articular OA concern unproven)

Outcomes

recurrence **0–8%** (<10%) · good–excellent **>90%** · RTS same level **75–90%** · RTS any level 80–100%

Return-to-sport conflict

same level **75–90%** (OB) · 72.6% (Hurley) · 56–95% (Glogovac) · 49% contact athletes (Privitera) – read the options

Latarjet vs Bankart (bone loss) 4.9% vs 67% recurrence (Burkhart); Latarjet 2.7% (Rollick) with ↑ complications

Clinical scenario (OB)	Answer
Inverted pear + recurrent instability, failed arthroscopic Bankart	**Coracoid autograft (Latarjet)**
GBL **>20%**, rugby, 5 dislocations	**Open coracoid transfer**
GBL **30%** (best-fit circle), no prior surgery	**Open coracoid transfer** (✗ Wrong options: capsular shift = MDI · ORIF)
GBL **30%**, Beighton 2, HS	**Bony glenoid augmentation** (Latarjet / distal tibia)
GBL **33%**, lacrosse	Latarjet – **conjoint tendon sling**
Failed arthroscopic Bankart, **20% GBL**	**Latarjet – recurrence <10%**
23-y pro rugby, small HS, 15% GBL	**Latarjet** (✗ Wrong options: 33-y recreational with 10% → Bankart ± remplissage · linebacker large HS 6% → remplissage / humeral allograft · swimmer circumferential tear → repair)
≥10 dislocations, **24.4% GBL**, HS 15.4 mm	**Coracoid transfer**
Pole vaulter, off-track HS, significant GBL	**Coracoid transfer** (✗ Wrong options: remplissage + plication without labral repair · latissimus transfer with competent subscap)
Infantryman, **25% GBL**, off-track, failed repair	**Coracoid transfer** (✗ Wrong options: re-repair with ≥3 anchors · arthroplasty)
Chronic anterior dislocation + inverted pear	**Open reduction + glenoid augmentation** (if HS <20%)
Failed Bankart + **postage-stamp fracture**	**Bone block**
Failed arthroscopic Bankart, GBL >20–25% on CT	**Open capsulorrhaphy + bone block** (✗ Wrong options: arthrodesis)

Chronic dislocation (Sahajpal) glenoid defect **>20–30% → bony augmentation**; humeral defect engaging or **20–40% → address**; **>40% → hemiarthroplasty**

Postage-stamp fractures anterior rim fractures through anchor holes; male **≤25 y**, sport, **≥3 anchors**, **knot-tying anchors**, anchor-site osteolysis → **bone block**

Recurrence predictors after Bankart (OB) age <20, Hill-Sachs, glenoid lesion, hyperlaxity, off-track
✗ Wrong options: dominant side · SLAP · >5 preop dislocations · on-track

Distal tibia allograft (DTA) **true osteochondral graft**; **matches humeral head radius of curvature** + large articular surface (TO); functions mainly by **↑ glenoid track**; revision of failed Latarjet
✗ Wrong options: sling – differs from Latarjet

– – Active space – –

Iliac crest (Eden-Hybinette) (TO) autograft for large defects where **coracoid volume inadequate**; 89% healing at 1.4 y; screws or buttons; risks: hardware failure, subscapularis repair failure

Alternative graft (TO)	Source	Use
ICBG	**Inner table** of iliac wing	Matches glenoid concavity; **>30–40%** loss · Eden-Hybinette for **>40%**
Distal tibial allograft	**Lateral** articular surface	Cartilage match; massive defects
Distal clavicle autograft	Resurfacing	Smaller defects / backup
Allograft bone plug	Cancellous	Limited integration; selective

Revising a failed Latarjet hardware removal + ICBG or DTA; electrocautery/nerve check to protect MC nerve (scarred, overlaps axillary nerve)

Dynamic anterior stabilisation (DAS, TO) **LHB transferred to anterior glenoid through subscapularis split → sling effect**; for **chronic** anteroinferior instability with **poor labrum/GHL quality** and **limited bone loss**; also with SLAP, overhead athletes/throwers

DAS technique (TO) LHB tenotomy, whipstitch · split subscap in middle · drill **3 o'clock**, 1.5–2.0 cm deep · tenodesis screw · add labral repair (bumper effect)

DAS contraindications (TO) **significant bone defects** · LHB lesions · subscapularis lesion · prior LHB tenotomy/tenodesis; fills gap between soft-tissue and bone-block procedures
✗ Wrong options: alternative to Latarjet with bone loss

Conjoint tendon / LHB transfer (Orthobullets) conjoint detached from coracoid tip (CA ligament preserved) or LHB; socket in anterior glenoid + suspensory fixation through subscap split; low recurrence with labral repair if **GBL <15%**; biceps possibly better; no ROM loss

Latarjet in epileptics contraindicated with **uncontrolled seizures**

Seizure-related dislocation

Epileptic, >10 dislocations, uncontrolled seizures, GBL >20–25% → closed reduction + neurology referral
✗ Wrong options: no reduction attempt · reduction alone · stabilisation now · fusion

Rule **exhaust medical (seizure) management before surgery**; high recurrence even with bony augmentation

Pathology **large bony defects** (HS + bony Bankart) + degenerative change; muscle contraction + trauma; anterior = posterior frequency

Surgery once controlled **bony procedure** (Latarjet / bone graft); soft tissue alone fails

Failed stabilisation with subscapularis deficiency – tendon transfer

Signs **belly-press** weakness; post-open-repair subscap failure → **positive lift-off + excessive ER**

Irreparable retraction + **atrophy** on MRI

29-y, failed arthroscopic + open Bankart, irreparable subscap → tendon transfer (latissimus dorsi or pectoralis major)
✗ Wrong options: PT · Latarjet · Magnuson-Stack · RTSA – age

Latissimus vs pec major — latissimus **better replicates subscap line of pull / IR moment arm**, better scores; pec major historically to lesser tuberosity; pec major not after failed subscap repair post-arthroplasty

Historical & thermal procedures (Putti-Platt, capsulorrhaphy)

Procedure	Technique	Problem
Putti-Platt	**Lateral advancement of subscapularis + medial advancement of capsule** (imbrication)	**Loss of ER**, **arthrosis** with ↓ ROM (TO)
Magnuson-Stack	Subscapularis advanced lateral to bicipital groove (± to GT)	↓ ER, posterior glenoid loading
Boyd-Sisk	Biceps transferred laterally + posteriorly	**High recurrence**
Bristow	Coracoid tip to anterior rim	**MC nerve injury** commoner (TO)

Putti-Platt (TO) — ✓ arthrosis with compromised ROM · (✗ Wrong options: redislocation never reported · rotates HS by ↑ retroversion · MC nerve injury common (infrequent)) · Putti-Platt + Bristow → **IR contractures**

Post-capsulorrhaphy arthropathy — 1970s–80s open repair → pain, stiffness, **lack of ER**, **posterior glenoid wear + retroversion**, IR contracture

Thermal capsulorrhaphy (TO) — used in **shoulder instability**; monopolar RF, introduced **1996**
✗ Wrong options: adhesive capsulitis · dural tear · stiff elbow

Thermal mechanism (TO) — **65–80°C** shortens capsule; collagen denaturation, ↓ mechanical properties, cell necrosis → fibroblast repair; stretches if loaded early; **grid/stripe** technique aids healing; ↓ capsular proprioceptors

Thermal outcomes (TO) — volume ↓ **26–36%** (vs plication **19%**); best in mild, unidirectional anterior instability **with** suture repair; **isolated → unacceptably high failure**

Thermal complications (TO) — **failure (commonest)**, **chondrolysis**, axillary nerve injury, capsular attenuation; now contraindicated

6 Complications

Recurrence

Commonest cause — **unrecognised glenoid bone loss** with soft-tissue-only repair (80% of failures) · missed off-track HS · **<3–4 anchors** · missed HAGL · missed pan-labral tear (military) · seizures

Anchor conflict — **<4 anchors** = risk (Orthobullets) · **<3 anchors** = risk (OB) — read the options

Age at surgery — not related to failure (one OB) · independent predictor (Ahmed, OB) — read the options

Bankart repair complications

Stiffness — **loss of ER** (overtightening; worse with remplissage) → treat **Z-lengthening of subscapularis**; overtightening → **post-capsulorrhaphy arthropathy** (older patients)

Subscapularis failure (open) — positive lift-off, excessive ER

– – Active space – –

Hardware	anchor pull-out; postage-stamp fracture
Chondrolysis	**thermal capsulorrhaphy** or **intra-articular pain pumps** (both now contraindicated)

Neurovascular injury – axillary nerve & artery (dislocation, capsular shift)

Most common nerve after dislocation	**axillary** (course around surgical neck, close to joint); typically **neurapraxia** (stretch) ✗ Wrong options: neurotmesis = penetrating/laceration · brachial plexus · MC

Axillary nerve injury rate	Source
Up to 5% (transient neurapraxia)	Orthobullets, OB (older cohort)
42% clinical + EMG (Visser)	OB
~48% EMG changes	OB
Any nerve injury **21–65%**; neuro deficit 13.5% (Robinson); 11.6% (Atef)	OB

Rate conflict	**≤5%** clinical (Orthobullets) vs **42–48%** EMG (OB) – read the options
Axillary nerve	**C5–6** (upper trunk, posterior cord) → deltoid + teres minor; weakness esp. **30–100° abduction**; ↓ sensation **lateral proximal arm**
Timeline	neurodiagnostics after first **2–3 wk** (14–21 d if no recovery) · persistent weakness + sensory loss at **12 wk → EMG/NCS** · surgery (neurolysis, graft, transfer) if **no recovery 3–6 mo** ✗ Wrong options: observe more · repeat MRI · explore · triceps branch transfer before EDS
High-grade injury risk (Avis)	proximal humerus fracture, multiple nerves, neuropathic pain, **delayed reduction**
Surgical anatomy	axillary nerve anterior to subscapularis → drapes **inferolateral border → quadrilateral space**; **12 mm** (Orthobullets) · **12–16 mm** (OB) · 1–1.5 cm (Kang) from inferior glenoid / infraglenoid tubercle
Arthroscopic risk zone	**6 o'clock, inferior to infraglenoid tubercle** (also 4:30) during capsular shift/plication; plication **≤1 cm from rim safe**, >1 cm risk; anteroinferior portals riskiest ✗ Wrong options: spinoglenoid/suprascapular notch = suprascapular n. · 9 o'clock posterior · 12 o'clock biceps
Protective position	**abduction + external rotation** ↑ nerve–capsule distance ✗ Wrong options: adduction/IR – beach chair adduction + IR = risk
Open capsular shift	injury during **dissection of subscapularis off inferior capsule** / retractor on inferomedial subscap belly → **can't abduct, elbow flexion + supination intact = axillary (C5–6)** ✗ Wrong options: MC C5–7 · lower subscapular nerve
Axillary artery	**1–2 cm** from inferior glenoid · closest at **≥90° ER: 4–5 mm** (4.1 mm Stone) · ≥2 cm from coracoid · **greatest risk at inferior glenoid with arm in ER** ✗ Wrong options: thoracoacromial – at coracoid · MC – medial · radial – triangular interval · ulnar

Latarjet complications – musculocutaneous nerve, graft resorption & overhang arthritis

Complication rate up to **25%** (Orthobullets) · 15–30% (Gupta) · 30% (Griesser); higher than Bankart

Complication	Fact
MC nerve injury	**Most common nerve** (Orthobullets, most OB); **instrumentation / aggressive medial retraction around conjoint tendon**, coracoid harvest; pierces coracobrachialis ~**5.6 cm** distal to coracoid tip
Axillary nerve injury	**During graft fixation**; 12 mm from infraglenoid tubercle; 2nd commonest
Graft lysis/resorption	Up to **90%** some resorption **within 6 months**; **superior** graft; late pain + apprehension at ~9 mo
Lateral overhang	**GH OA** (up to **38%**); 20% new arthritis (Mizuno)
Stiffness	**ER loss** (✗ Wrong options: IR · elbow flexion)
Nonunion, hardware (screw pull-out)	
Axillary artery	4–5 mm at 90° ER
Subscapularis rupture	**Low** – split, not detached

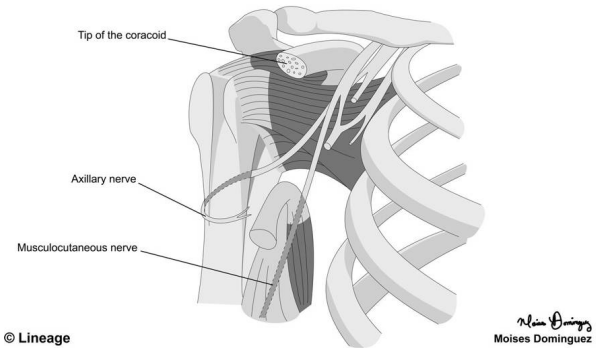
MC nerve **terminal branch of lateral cord**, **C5–7**; → coracobrachialis, biceps, brachialis; injury → **lateral forearm numbness (LACN)** + **↓ elbow flexion/ supination**

Latarjet nerve conflict **MC = most common** (Orthobullets, most OB) · **axillary = most common** (one OB; neuromonitoring: all clinical deficits axillary) – read the options

Nerve risk ranking (TO) **suprascapular > axillary > musculocutaneous** (TO Latarjet table) – read the options

Resorption conflict common, **up to 90%** (Orthobullets, OB) · "uncommon overall", mostly first 6 mo (another OB) – read the options

Nerve injury management mostly **traction/contusion neurapraxia → observe 3–6 wk**; delayed EMG if persistent; all resolved by 6 mo (neuromonitoring)



Axillary and musculocutaneous nerves relative to coracoid tip

After Latarjet MC + axillary nerves move medially, **MC overlaps axillary** → revision risk; axillary artery position unchanged

✗ Nerve distractors lateral winging (spinal accessory) · medial winging (long thoracic) · IR weakness (subscapular) · thumb extension weakness (PIN/radial) · median, ulnar, radial (paths far)

Sling post-Latarjet protects **graft healing**
✗ Wrong options: subscap detached

– – Active space – –

Older patients – rotator cuff tear after dislocation

Rule young → recurrence · **old → rotator cuff tear**

Age	Cuff tear after dislocation
>40 y	**30%** (Orthobullets, OB) · "30% at 30 y" (one OB)
>60 y	**80%**

Tendons "**SIT**": **supraspinatus** (commonest), infraspinatus, teres minor; recurrent instability in elderly → **subscapularis** tears
✗ Wrong options: subscap – bear hug

Cuff tear vs axillary palsy in elderly cuff tear **far more common**; **deltoid firing + full passive ROM + intact sensation** → cuff tear → **suprascapular nerve** muscles (SS/IS)
✗ Wrong options: dorsal scapular · lateral pectoral · MC · axillary

Persistent pain/weakness ≥2 wk after dislocation (>40 y) → MRI shoulder
✗ Wrong options: EMG · C-spine MRI · steroid injection – harms later repair · PT first · CT glenoid

Test for commonest pathology **drop arm test** (supraspinatus)
✗ Wrong options: O'Brien – SLAP · apprehension – young/bone loss · bear hug – subscap · Spurling – C-spine

MR arthrogram **most sensitive (and specific) for full- + partial-thickness cuff tears**; MRI ≈ US

Treatment surgical repair → better pain + satisfaction; full-thickness tears don't heal nonop; **GT fracture** with dislocation >50 y (↑ recurrence)

Terrible triad of the shoulder

Shoulder terrible triad (TO) **anterior dislocation + rotator cuff tear + brachial plexus (mostly axillary) nerve injury**; cuff usually **complete** tear

EXCEPT (TO) **proximal humeral fracture**

Late complications – chondrolysis, osteonecrosis & dislocation arthropathy

Osteonecrosis (skeletally immature) rare; after fracture-dislocation or stabilisation → **observation** (revascularises)
✗ Wrong options: arthroplasty · grafting · capsular release

Chondrolysis progressive pain + stiffness ~1 y after Bankart with **heat (RF) probe** → joint space loss
✗ Wrong options: frozen shoulder · subscap failure

Dislocation arthropathy **age at first dislocation** (>25 y, Hovelius) + high-energy primary dislocation → severe OA; "**goat's beard**" inferior osteophyte; ↓ ER
✗ Wrong options: current age · dominance · diabetes · sex

After Bristow-Latarjet mild arthropathy 35%, severe 14%; first dislocation >23 y = risk

X vs Y

	Bankart	Perthes	ALPSA	GLAD	HAGL
Labrum	Off rim	Off rim, may look **in place**	**Medially displaced**, healed on neck	Partial tear	Intact on glenoid
Periosteum	Hinge intact	**Stripped, intact**	Sleeve intact medially	–	–
Cartilage	–	–	–	**Defect**	–
Failure point	Glenoid	Glenoid	Glenoid neck	Chondrolabral	**Humerus**
Best repair	Arthroscopic	Arthroscopic	Arthroscopic (↑ failure)	Arthroscopic	**Open**

	Young dislocator (<25)	Older dislocator (>40)
Main lesion	**Bankart** + Hill-Sachs	**Rotator cuff tear** (30% >40, 80% >60)
Main problem	**Recurrence**	Weakness, pseudoparalysis
Next test	MRI/MRA, 3D CT for bone	**MRI shoulder**
Nerve	Axillary neurapraxia	Axillary / plexus (less common than cuff)

FEDS (TO)	Options
Frequency	Single / recurrent
Etiology	Traumatic / atraumatic
Direction	Anterior / posterior / inferior
Severity	Subluxation / dislocation

	West Point	Stryker notch	Garth
Shows (TO)	**Glenoid rim / bony Bankart**	**Hill-Sachs**	**Both**

	Remplissage + Bankart	Latarjet
Fixes	**Humeral side** (HSI ↓)	**Glenoid side** (GT ↑)
GBL	<15–25%	>20–25% or subcritical contact
Approach	Arthroscopic	Open (or arthroscopic)
ROM	**ER loss 5–15°**	ER loss; better ER than remplissage
Key risk	Stiffness	**MC nerve**, resorption, overhang OA

– – Active space – –

	Latarjet	Distal tibia allograft	Iliac crest (Eden-Hybinette)
Graft	Coracoid + conjoint + CA ligament	Osteochondral allograft	Tricortical autograft
Mechanism	**Triple effect**	**↑ glenoid track only**	Bony
Curvature	–	**Matches humeral head**	–
Use	Primary bony option	Massive loss, failed Latarjet	Large defect, coracoid inadequate, failed Latarjet

	MC nerve (Latarjet)	Axillary nerve
Cord / roots	Lateral cord, **C5–7**	Posterior cord, **C5–6**
When injured	Retraction / instrumentation **around conjoint tendon**	**Graft fixation**; capsular shift at 6 o'clock; dislocation
Deficit	**Lateral forearm numbness**, ↓ elbow flexion + **supination**	↓ abduction (deltoid), lateral arm numbness

	Anterior instability	Posterior instability
Position	**ABD + ER**	**ADD + IR**
Labral lesion	Bankart, ALPSA, Perthes, GLAD	**Kim lesion**, posterior labral tear
Humeral lesion	Hill-Sachs (posterosuperior)	**Reverse Hill-Sachs**, LT fracture
Ligament avulsion	HAGL	RHAGL
Humeral fill	Remplissage / Connolly (infraspinatus + GT)	**McLaughlin** (subscap + LT)

How it's asked

"Perthes lesion is?"	→ **associated with recurrent dislocation** (TO)
"Anteroinferior labral detachment with medially stripped intact periosteum?"	→ **Perthes** (TO) ✗ Wrong options: Bankart · GLAD · ALPSA
"Main anterior stabiliser of shoulder?"	→ **IGHL** (TO) ✗ Wrong options: MGHL · SGHL · anterior capsule
"Main stabiliser against anterior dislocation above 45° abduction?"	→ **IGHL** (TO) ✗ Wrong options: subscapularis
"MRI 2 months after dislocation – labrum displaced medially with periosteal sleeve?"	→ **ALPSA** (TO) ✗ Wrong options: Bankart · SLAP · HAGL
"Most common finding at surgery for traumatic anterior instability?"	→ **anteroinferior labral tear** ✗ Wrong options: Hill-Sachs
"17-y footballer, MRI after injury – arthroscopic finding?"	→ **Bankart lesion** ✗ Wrong options: cuff tear · SLAP · glenoid fracture · HAGL

"Labral tear + Hill-Sachs on MRA – position at injury?"	→ **abduction + external rotation**
"25-y basketballer, first dislocation – MRI shows?"	→ **anteroinferior labral tear** ✗ Wrong options: supraspinatus tear = >40 y
"Typical separation on MRI after anteroinferior dislocation?"	→ **anteroinferior labrum from bony glenoid** ✗ Wrong options: from cartilage · capsule from humerus
"Bony HAGL – when is this ligament most stressed?"	→ **anterior force at 90° abduction + ER** ✗ Wrong options: 0° inferior = SGHL · 45° = MGHL
"Structure torn, name, position of greatest contribution?"	→ **anteroinferior labrum, Bankart, ER at 90° abduction**
"Young dislocator with Bankart – most likely associated injury?"	→ **Hill-Sachs** ✗ Wrong options: cuff tear · reverse HS · coracoid · GT fracture
"Recurrent anterior + posterior instability, positive Jerk, Kim, apprehension – lesion needing both repairs?"	→ **270° pan-labral tear**
"Recurrent instability on MRA, surgeon recommends open treatment?"	→ **HAGL** ✗ Wrong options: GLAD · SLAP · ALPSA · PASTA
"Recurrent subluxation, coronal MRI – ligament injured?"	→ **inferior glenohumeral (HAGL)**
"Re-dislocation 1 month after arthroscopic Bankart – missed pathology?"	→ **HAGL**
"36-y diving catch, post-reduction MRI lesion?"	→ **HAGL** (older, violent trauma)
"Goal of remplissage?"	→ **make intra-articular Hill-Sachs extra-articular** (TO) ✗ Wrong options: ↑ glenoid track
"Rugby, 12% GBL, narrow off-track Hill-Sachs?"	→ **arthroscopic Bankart + remplissage** (TO) ✗ Wrong options: Eden-Hybinette · Latarjet
"Kabaddi, D 24 mm, width 20 mm, HSI 18 mm?"	→ **off-track; Latarjet** (TO) ✗ Wrong options: Bankart + remplissage – 16.6% subcritical, elite contact
"Associated intra-articular lesion with Bankart in rugby player?"	→ **Hill-Sachs** ✗ Wrong options: Buford – normal variant
"Best X-ray for Hill-Sachs?"	→ **Stryker notch** ✗ Wrong options: Grashey · Y · Zanca · axillary
"Ex-pro footballer, catching in 90/90, chronic Bankart + large HS?"	→ **arthroscopic Bankart + remplissage** ✗ Wrong options: remplissage alone

– – Active space – –

"Arrow on posterior head beside bare area after dislocation?"	→ **Hill-Sachs defect**
"Compressive lesion after anteroinferior dislocation – structure?"	→ **posterosuperior humeral head**
"12.5% GBL, non-engaging Hill-Sachs?"	→ **Bankart repair** ✗ Wrong options: remplissage · capsular shift alone · Latarjet · distal tibia
"Bankart + remplissage – post-op restriction?"	→ **avoid forward flexion + adduction**
"Glenoid track formula?"	→ **GT = 0.84(D) − ABL** (0.83D − d in most sources)
"Most increases glenoid track?"	→ **congruent arc Latarjet** (~15 mm) ✗ Wrong options: traditional · Bristow · remplissage
"True about glenoid track?"	→ **more medial HS → more likely off-track**
"Medium-large HS + labral tear after 2nd dislocation – best treatment?"	→ **arthroscopic labral repair + remplissage** ✗ Wrong options: thermal capsulorrhaphy
"Recurrence after arthroscopic repair in lineman – cause?"	→ **anteroinferior glenoid bone defect** (missed bony Bankart)
"Highest recurrence risk after arthroscopic Bankart?"	→ **18-y competitive tennis, HS + glenoid contour loss (ISIS 9)**
"Intra-op finding most predictive of recurrence?"	→ **35% anterior glenoid deficiency**
"Failed Bankart, glenoid rim loss – mechanism deficient?"	→ **concavity-compression**
"Water polo, 2 dislocations – greatest recurrence risk after labral repair?"	→ **inverted-pear glenoid** ✗ Wrong options: dominant arm · overhead athlete · age <25 · biceps extension
"Isolated labral repair would fail for which image?"	→ **anterior glenoid fracture / bone loss**
"Factor most linked to failure after arthroscopic Bankart, >10 dislocations?"	→ **glenoid bone loss** ✗ Wrong options: age at first dislocation · age at surgery
"Why 3D CT for bone loss?"	→ **highest inter-rater reliability** ✗ Wrong options: lowest predictive error · MRI underestimates
"Most significant risk factor for recurrence after first dislocation?"	→ **young age** (TO) ✗ Wrong options: bony Bankart · manual occupation · GT fracture
"Highest risk for recurrent instability?"	→ **age <20 at dislocation** ✗ Wrong options: contralateral · dominance · family history · patellar instability

"3 recurrences despite PT, last in sleep?"	→ **discuss surgical stabilisation**
"Competitive contact footballer, HS on ER AP, normal laxity – ISIS?"	→ **5 points**
"48-y first dislocation, 1 wk in sling, full strength?"	→ **stop immobilisation now – no added recurrence risk** ✗ Wrong options: ER/IR/neutral 2 more weeks
"Nonop first-time dislocator – true statement?"	→ **short immobilisation = prolonged** ✗ Wrong options: bracing helps · low risk · future events don't matter
"17-y footballer, first dislocation, small non-engaging HS?"	→ **nonop allows return this season** ✗ Wrong options: surgery faster · ER brace better
"Water polo, GBL <7%, HSI 19 < GT 22?"	→ **isolated arthroscopic Bankart**
"Advantage of arthroscopic over open stabilisation?"	→ **slightly better ROM** ✗ Wrong options: lower recurrence · better return to work/sport
"Arthroscopic vs open Bankart?"	→ **similar dislocation rates** ✗ Wrong options: more OA with open
"Isolated open Bankart most appropriate?"	→ **recurrent after arthroscopic repair, 6% GBL, on-track**
"Bony Bankart + deltoid weakness 3 days after reduction?"	→ **bony Bankart stabilisation** (observe axillary nerve)
"Accountant, 3rd dislocation, 10% glenoid defect on 3D CT?"	→ **arthroscopic bony Bankart repair**
"Acute dislocation, bony fragment 10% of glenoid?"	→ **arthroscopic repair incorporating fragment** ✗ Wrong options: ORIF · iliac crest · coracoid transfer
"25% glenoid loss, GT 18 mm, HSI 21.3 mm?"	→ **Latarjet** (TO)
"Wrestler, 30 dislocations, 40% posterolateral head loss?"	→ **Latarjet** (TO) ✗ Wrong options: McLaughlin
"Incorrect about DAS?"	→ **suitable arthroscopic alternative to Latarjet with bone loss** (TO)
"Latarjet sling effect?"	→ **conjoint tendon dynamic stability to anteroinferior capsule in ABER** (TO)
"Latarjet stabilises by?"	→ **bony support + conjoint sling + CA ligament capsular repair (1, 2 and 5)**
"Ideal patient for open reduction + glenoid augmentation?"	→ **chronic anterior dislocation + inverted-pear glenoid** ✗ Wrong options: large engaging HS · acute bony Bankart · cuff arthropathy

– – Active space – –

"Failed Bankart, inverted pear on CT?"	**→ coracoid autograft** ✗ Wrong options: Connolly · remplissage · humeral augmentation
"Rugby, 5 dislocations, >20% GBL?"	**→ open coracoid transfer**
"Failed arthroscopic Bankart, glenoid defect on CT?"	**→ open capsulorrhaphy + bone block**
"30% GBL, no prior surgery?"	**→ open coracoid transfer**
"Return to previous level after Latarjet?"	**→ 75–90%**
"33% GBL, bone transfer – true?"	**→ conjoint tendon sling** ✗ Wrong options: subscap detached · axillary = retraction nerve · medial graft → OA · traditional > congruent arc
"30% GBL, Beighton 2, HS?"	**→ bony glenoid augmentation**
"Failed Bankart, 20% GBL – true?"	**→ recurrence <10% after Latarjet** ✗ Wrong options: revision arthroscopic first · Latarjet = DTA mechanism
"Latarjet most appropriate in?"	**→ 23-y pro rugby, small HS, 15% GBL**
"19-y footballer, 24% GBL, HS 15.4 mm?"	**→ coracoid transfer**
"Pole vaulter, 5th dislocation, off-track, significant GBL?"	**→ coracoid transfer** ✗ Wrong options: latissimus transfer
"Infantryman, failed repair, 25% GBL, off-track?"	**→ coracoid transfer**
"Risk factor justifying bone block after failed Bankart?"	**→ postage-stamp fractures** ✗ Wrong options: SLAP · dominance · >5 dislocations · on-track
"Lateral forearm numbness after coracoid transfer – label?"	**→ musculocutaneous nerve** ✗ Wrong options: long thoracic · suprascapular · axillary · radial
"Nerve most at risk in Latarjet for 40% bony Bankart?"	**→ musculocutaneous**
"Lateral forearm numbness + supination weakness post-op – nerve and procedure?"	**→ musculocutaneous & Latarjet**
"Post-Latarjet neurapraxia finding?"	**→ lateral forearm numbness**
"True of Latarjet complications?"	**→ lateral graft overhang → GH OA** ✗ Wrong options: axillary = harvest nerve · IR stiffness · resorption uncommon
"Latarjet complication true?"	**→ lateral cord branch injury from aggressive retraction** ✗ Wrong options: medial graft → OA · resorption >2 y · subscap detached
"9 months post-Latarjet, new apprehension, CT superior graft loss?"	**→ coracoid graft resorption** ✗ Wrong options: nonunion · lateral placement
"Nerve most likely injured with anterior dislocation?"	**→ axillary**

"Most common neurological problem with simple dislocation?"	→ **neurapraxic axillary nerve injury**
"62-y, can't abduct 2 wk after dislocation, deltoid firing, sensation intact – nerve to injured structure?"	→ **suprascapular (label B)** – cuff tear
"Inferior capsular shift – arm position protecting the nerve?"	→ **abduction + external rotation**
"Neurapraxia after arthroscopic capsular shift in adduction + IR – site?"	→ **6 o'clock, inferior to infraglenoid tubercle**
"Structure at greatest risk at inferior glenoid with arm in ER during Latarjet?"	→ **axillary artery**
"Persistent abduction weakness + lateral arm numbness 3 months after dislocation, intact cuff?"	→ **electrodiagnostic studies**
"Can't abduct after open capsular shift, elbow flexion intact – nerve and roots?"	→ **axillary, C5–6**
"61-y, pain + weakness 6 wk after dislocation – next step?"	→ **MRI to evaluate rotator cuff** ✗ Wrong options: labrum · EMG · CT
"62-y, weakness 4 wk after dislocation?"	→ **MRI shoulder** ✗ Wrong options: injection · C-spine MRI · EMG · PT
"Provocative test for commonest pathology, 61-y after dislocation?"	→ **drop arm test**
"64-y, weakness 6 wk after dislocation, neck radiation?"	→ **shoulder MRI** ✗ Wrong options: EMG · C-spine MRI
"MR arthrogram after dislocation in 55-y – true?"	→ **most sensitive for partial-thickness cuff tears**
"Osteonecrosis after dislocation in skeletally immature – treatment?"	→ **observation**
"Pain + stiffness 1 y after Bankart with heat probe?"	→ **chondrolysis**
"Severe OA at 55 after single dislocation at 30?"	→ **age at time of dislocation**
"Epileptic, >10 dislocations, poor compliance, GBL?"	→ **closed reduction + neurology referral**
"Failed arthroscopic + open Bankart, belly-press weakness, atrophied retracted subscap?"	→ **tendon transfer** ✗ Wrong options: Latarjet · Magnuson-Stack · RTSA
"Arm position for humerus coronal + glenoid axial on MRI?"	→ **abduction + external rotation (ABER)**
"Terrible triad of shoulder EXCEPT"	→ **proximal humeral fracture** (TO)
"True of Putti-Platt?"	→ **arthrosis with compromised ROM** (TO) ✗ Wrong options: MC injury common · no redislocation
"Thermal capsulorrhaphy is used in?"	→ **shoulder instability** (TO)

– – Active space – –

"True about Buford complex EXCEPT"	→ **associated with recurrent dislocation** (TO)
"Critical glenoid defect above which isolated Bankart fails (Saito)?"	→ **>21%** (TO)
"Glenoid track as % of glenoid width?"	→ **83%** (TO)
"Inverted-pear glenoid corresponds to?"	→ **≥25% anterior bone loss** (TO)
"Dugas sign?"	→ **cannot touch opposite shoulder** (TO)
"Kocher reduction sequence?"	→ **traction → ER → abduction (adduction classically) → IR** (TO)
"ISIS above which Latarjet is advised?"	→ **>6** (TO)
"Latarjet triple effect described by?"	→ **Patte** (TO)

– – Active space – –

TRICEPS RUPTURE

– Shoulder & Elbow · 59 / 60 –

1 Basics

Epidemiology

Incidence rare · **0.8%** of tendon ruptures

Demographics ♂ **2:1** · age **30–50** · competitive **weightlifting** (bench press), bodybuilding, **football**

Risk factors & diagnosis (flake sign, MRI)

Risk factor	Yes / No
Anabolic steroids	✓ (established link)
Local steroid injection	✓
Fluoroquinolones	✓
Renal osteodystrophy / renal disease, **hyperparathyroidism**	✓
OI, RA, type I DM, Marfan	✓
Chronic olecranon bursitis, prior triceps surgery	✓
Creatine supplements	**not associated**

Anatomy – lateral expansion & anconeus

Head	Origin
Long	**Infraglenoid tuberosity**
Lateral	Posterior humerus between teres minor insertion and **above spiral groove**, lateral border, lateral IM septum
Medial	Posterior humerus **distal to spiral groove**, medial humerus, medial IM septum

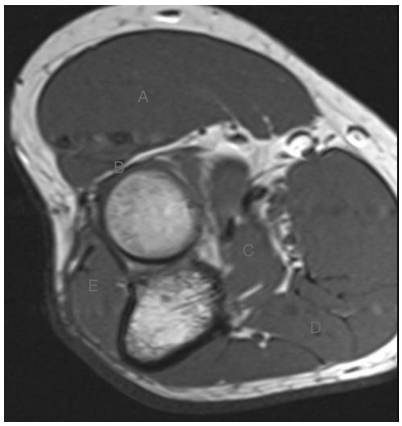
Triceps pennate · **only muscle of posterior arm compartment** · **radial nerve (C6–C8)** · primary elbow extensor

Insertion footprint **1.1 cm distal to olecranon tip** · width **1.9–4.2 cm** · triceps tendon proper (all 3 heads → olecranon) + **lateral triceps expansion** (posterior ulnar crest · **ECU fascia + anconeus deep fascia** · antebrachial fascia)

Anconeus **secondary elbow extensor**; terminal **15° extension** + supination · **compensates for torn triceps** → resisted extension may persist

Anconeus vs distractors brachioradialis, brachialis = flexion · LCL = varus stabiliser · pronator teres = pronation

– – Active space – –



Axial MRI: E = anconeus, A = brachioradialis, B = LCL, C = brachialis, D = pronator teres

2 Mechanism

Mechanism	**forceful eccentric contraction** (bench press, muscle-ups) · **FOOSH** · direct blow
Site	most often **osseous insertion** (medial or lateral head) · less often muscle belly / MTJ

3 Presentation

History	**painful pop**
Exam	posterior elbow pain, swelling, ecchymosis · **palpable defect** (not always)
Resisted extension	inability to extend against resistance **not always present**: **intact lateral expansion** or **anconeus** compensation · complete tear → can't extend **against gravity** (OB)
Modified Thompson squeeze	**prone**, elbow at table edge, forearm hanging → squeeze triceps → **no extension** = complete rupture of **triceps proper + lateral expansion**
Missed	**10/23** missed initially (van Riet) · **nearly 50%** misdiagnosed (Yeh)

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral)	**"flake sign"** on lateral = **pathognomonic** of triceps avulsion (olecranon tip fleck)



Flake sign: avulsed fleck at olecranon tip

MRI location + severity; confirms, classifies, guides treatment · **partial** = small fluid-filled defect in distal tendon · **complete** = **large fluid-filled gap**, paratricipital oedema · use when too painful to examine or partial suspected



MRI: distal triceps tear

✗ Flake-sign distractors distal biceps, brachialis, LUCL, MUCL tears · elbow arthritis · olecranon bursitis · anconeus avulsion (= **lateral epicondyle** fracture)

Descriptive classification

Feature	Options
Degree	Complete · partial · intact
Location	Muscle belly · MTJ · tendinous insertion · avulsion
Lateral expansion	Intact · torn

Formal system none

- - Active space - -

5 Management

Triceps rupture

- Partial, extends vs gravity / low-demand, poor health → **splint 30° flexion 4 wk**
- Acute complete → **primary repair (best within 3 wk)**
- Partial >50% + significant weakness → **primary repair**
- Chronic / delayed → **reconstruction ± tendon graft**

Operative criteria

Indication	Treatment
Acute complete tear (active, medically fit)	**Primary repair**
Partial >50% + significant weakness	**Primary repair**
Partial, extends against gravity	Splint
Low demand, poor health	Splint
Late (>3 wk)	Reconstruction, tendon graft

Approach posterolateral

Tear site	Technique
MTJ	**V-Y advancement** ± plantaris augmentation
Tendinous insertion	**Bunnell / Krackow** non-absorbable whipstitch → **transosseous tunnels**, periosteal flap, or **suture anchors**
Bony avulsion	Tension band · screw + washer

Tunnels vs anchors **no difference** in biomechanical strength or function · **transosseous: ↑ re-rupture + complications**

Post-op 30–45° flexion **2 wk** · active ROM at **4 wk** · no weightlifting **4–6 mo**

Outcome early diagnosis + repair → excellent; anatomic footprint repair

6 Complications

General elbow stiffness/weakness · **ulnar nerve injury** · repair failure

Repair-specific olecranon bursitis · flexion contracture · re-rupture

X vs Y

	Partial tear	Complete tear
Extension vs gravity	Preserved	Lost (unless anconeus / lateral expansion compensate)
Thompson squeeze	Extension present	**No extension**
MRI	**Small fluid-filled defect**	**Large fluid-filled gap**, paratricipital oedema
Treatment	Splint 30° × 4 wk · repair if **>50% + weak**	**Primary repair**

How it's asked

"Triceps rupture associated with all EXCEPT?"	→ **creatine supplements** ✗ Wrong options: anabolic steroids · local steroid · renal osteodystrophy · fluoroquinolones
"40 y weightlifter, pop on bench press, lateral X-ray fleck – MRI shows?"	→ **triceps tendon tear** ✗ Wrong options: distal biceps · brachialis · LUCL · MUCL
"44 y, FOOSH rollerblading, posterior pain, olecranon flake?"	→ **triceps rupture** ✗ Wrong options: arthritis · bursitis · LCL · anconeus avulsion
"Muscle-ups, positive Thompson squeeze — structure assisting resisted extension?"	→ **anconeus (E)**

– – Active space – –

VALGUS EXTENSION OVERLOAD (PITCHER'S ELBOW)

– Shoulder & Elbow · 60 / 60 –

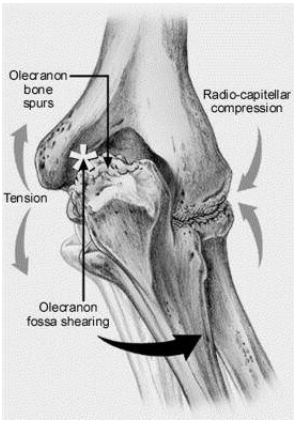
1 Basics

Epidemiology

Definition	**posteromedial elbow pain** from repetitive microtrauma in **throwing athletes**
Prevalence	up to **65%** of elite baseball players with symptomatic elbows
Who	**dominant arm** of throwers (pitchers, quarterbacks) · also swimmers, volleyball, gymnasts, racquet sports, golfers
Location	**posteromedial olecranon** (✓ label D) · posterior trochlea · olecranon fossa
Association	**cubital tunnel syndrome ~25%**

2 Mechanism

Forces of pitching	**medial tension (MCL)** · **lateral radiocapitellar compression** · **posterior extension overload** · **shear on medial olecranon tip + fossa**
Phases (OB)	**valgus stress in acceleration** (esp. with **UCL insufficiency**) + **extension impaction in deceleration**
Medial laxity	any cause → **compensatory ↑ stress posteromedially** → osteophytes / loose bodies
Sequence	repetitive impaction of olecranon into fossa → **chondromalacia/cartilage injury** → **osteophytes (posteromedial olecranon + humerus)** → **loose bodies** (fragmentation) · **capitellar osteochondral lesions** · **UCL attenuation** with repetitive strain

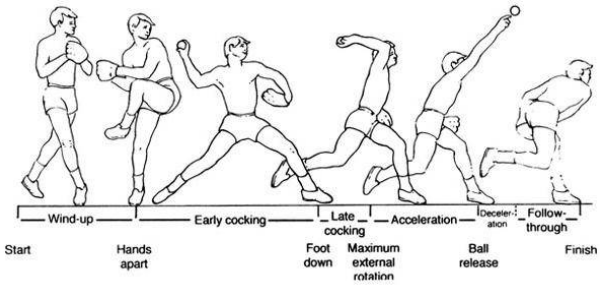


Valgus extension overload: medial tension, radiocapitellar compression, olecranon fossa shearing

3 Presentation

Diagnosis – posteromedial olecranon osteophytes, examination & throwing phase

Pain	**posteromedial**, at **full extension**; **deceleration / follow-through** (✓; after ball release) · rarely acceleration ✗ Wrong options: wind-up, early/late cocking
Function	**loss of terminal extension** (osteophytes block olecranon seating) · ↓ accuracy · pain after ~25 pitches
Exam	**tender posteromedial olecranon** · **flexion contracture** (e.g. 10°) · **crepitus** (loose bodies, synovitis) · **pain with forced extension**
Provocative tests	**arm bar test** (sustained extension) · **bounce test** (repeated terminal extension)
UCL assessment	check competency (failure → valgus instability); moving valgus stress may be painful **without instability** in VEO



Phases of pitching: VEO pain in deceleration/follow-through

4 Diagnosis & Classification

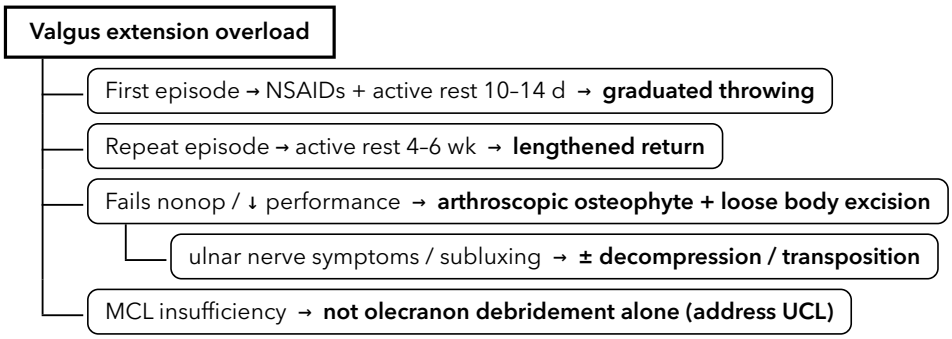
Key radiology findings

Modality	Finding
X-ray AP, lateral, oblique	**Posteromedial olecranon osteophyte** (AP key; medial olecranon) · loose bodies (capitellar fragmentation) · **MCL calcification** · **humeral hypertrophy** (↓ room in fossa)
CT	**Best osseous detail**; 3D for planning
MRI	Not needed for diagnosis · **associated MCL tear** (partial/complete) · capitellar OCD

Diagnosis	**clinical** (posteromedial pain ↑ with extension) + imaging of posteromedial osteophyte

– – Active space – –

5 Management



Nonoperative

First line NSAIDs, **stop throwing**, correct mechanics, ± steroid injection (**paucity of data**) · also **mid-season** or **end of career**

PT **flexor-pronator strengthening** (✓ next step) · shoulder, forearm + kinetic chain

Patient	Active rest	Then
First-time symptomatic	**10–14 days**	Graduated throwing
Repeat symptomatic	**4–6 weeks**	Lengthened gradual return

Operative criteria

Factor	Rule
Indication	**Persistent symptoms after failed (extended) nonop** · diminished performance
Relative contraindication	**MCL insufficiency** (for olecranon debridement alone)
Ulnar nerve release / transposition	**Significant ulnar symptoms** · nerve **subluxes** over medial epicondyle (transposition for symptomatic unstable nerve)

Arthroscopic osteophyte resection

Preferred arthroscopic (✓) or open **cheilectomy**; generally **excellent results** if no over-resection

Position lateral decubitus, prone or supine (**supine + articulating holder** if combined with **MCL reconstruction**)

Steps mark ulnar nerve course · diagnostic arthroscopy, remove loose bodies · **posterolateral viewing portal** · **direct posterior portal** (through triceps) · clear fibrous tissue (RF + shaver) · free fractured osteophyte (switch portals → remove via posterolateral) · **contour to native margin only** · **arthroscopic valgus stress test** (medial gapping = incompetent MCL)

Also chondromalacia debridement/drilling · lateral meniscoid lesion / posterolateral plica

Excessive olecranon resection – unmasking UCL insufficiency

Rule remove **osteophyte only, not native olecranon** → else **loss of bony restraint → ↑ MCL strain + valgus angulation**

Threshold	**≥9 mm** resection → **UCL rupture** under valgus (cadaveric; Orthobullets, OB) · **>3 mm** / past native margin → ↑ MCL stress + valgus instability (Orthobullets complications) – read the options
Result	**valgus elbow instability** (✓) · **MCL damaged** ✗ Wrong options: LUCL · RCL · annular · arcuate
Presentation	after debridement: **medial pain (late cocking)**, laxity + pain on valgus stress, **positive milking manoeuvre**, **negative arm bar**, new UCL tear on MRI
✗ Failure-cause distractors	accelerated return-to-throw · lateral instability (rare) · posterior bundle (dominant UCL structure = **anterior band of anterior bundle**) · anatomic variant · ulnar neuritis · capitellar OCD · olecranon stress fracture

6 Complications

Complication	Prevention
Valgus instability	Avoid over-resection (see thresholds)
Ulnar nerve injury	Mark nerve before medial portals · **"nick and spread"** · limit/remove **shaver suction** in posteromedial gutter

X vs Y

	VEO	UCL injury	Medial epicondylitis
Pain	**Posteromedial**	**Medial**	Medial, **5–10 mm distal to epicondyle**
Phase	**Deceleration / follow-through**	**Acceleration** (late cocking)	
Test	**Arm bar, bounce**; crepitus	**Valgus stress at 20–30°**, **milking manoeuvre**, moving valgus stress	**Resisted flexion/ pronation**
X-ray	**Posteromedial osteophyte**	No fracture, **no posteromedial osteophyte**	No VEO osteophytes
Contracture	**Yes**		Unlikely

– – Active space – –

Differential	Key feature
Acute UCL rupture (pop, ↓ velocity, MRA full-thickness)	**UCL reconstruction** in competitive thrower (✗ Wrong options: isolated ulnar transposition · osteophyte excision without osteophyte)
Partial UCL tear, non-thrower / low-risk sport	Nonop: activity modification + PT
MCL sprain (recreational thrower)	Medial pain **only when throwing**, ↓ speed + distance, no instability, normal X-ray
PLRI	Lateral; **pivot shift, chair rise**; LUCL; dislocation or lateral epicondylitis debridement
Capitellar OCD	Lateral pain, swelling, **catching/locking**
Ulnar nerve subluxation	Palpable in flexion; neuritis to hand
Pronator syndrome	**Volar forearm** pain on resisted pronation; not throwing-related
Medial epicondyle avulsion	**Skeletally immature**
Olecranon stress fracture	**Posterior** pain + swelling; lateral X-ray/MRI
✗ Wrong option: varus extension overload	Distractor — **valgus** EO (not varus) mimics MCL injury

How it's asked

"Typical location of osteophytes in VEO?"	→ **posteromedial olecranon (D)**
"Pitcher, posteromedial pain after ball release, normal milking — diagnosis + next step?"	→ **VEO; PT with flexor-pronator strengthening** ✗ Wrong options: UCL reconstruction · PRP · LUCL reconstruction · extension splint
"Pitcher, 10° flexion contracture, posteromedial tenderness, no instability?"	→ **valgus extension overload** ✗ Wrong options: medial epicondylitis · olecranon stress fracture · trochlear OCD · epicondyle avulsion
"Exam findings in VEO?"	→ **posteromedial pain, crepitus, flexion contracture** ✗ Wrong options: pivot shift · distal epicondyle tenderness · lateral locking
"Throwing phase most painful in VEO?"	→ **follow-through / deceleration** ✗ Wrong options: acceleration · late cocking
"Excessive normal olecranon resection → on return to play?"	→ **valgus elbow instability**
"Valgus instability 9 mo after osteophyte removal — structure?"	→ **medial collateral ligament** ✗ Wrong options: LUCL · annular
"Medial pain in late cocking + valgus laxity after olecranon debridement?"	→ **excessive olecranon resection**

"New UCL tear after VEO arthroscopy + accelerated rehab — cause?"	→ **excessive olecranon resection** ✗ Wrong options: accelerated protocol · posterior bundle
"Pitcher, pop, full-thickness UCL tear on MRA, wants to return?"	→ **UCL reconstruction** ✗ Wrong options: ulnar transposition · osteophyte excision
"Recreational thrower, medial pain only throwing, ↓ distance, normal X-ray?"	→ **MCL sprain** ✗ Wrong options: ulnar neuritis · pronator syndrome · medial epicondylitis · varus extension overload

– – Active space – –

– – Active space – –

BASIC ARTHROSCOPIC INSTRUMENTATION

– Shoulder & Elbow · TO notes –

1 Basics

History

Milestone (TO)	Name
Father of sports medicine	**Galen**
First successful arthroscopy (1918)	**Kenji Takagi** – TB knee, using a **cystoscope**
Therapeutic arthroscopy, **triangulation**	**Masaki Watanabe**
Brought to the West (1965 Tokyo Olympics)	**Robert Jackson** (Canada)
International Arthroscopy Association (**1974**)	Watanabe president · Jackson founding member
Arthroscopic camera	**McGinty and Johnson**

Operating room set-up

Tower (TO) monitor · camera control unit · light source · shaver control · pump

Monitor (TO) **≥20-inch** (usually 26-inch HD), at surgeon's eye level

Distension pressure (TO) **knee 30 mmHg** · **shoulder 40–50 mmHg** · 3–5 L bags hung high · **raising bag 1 ft → ~20 mmHg ↑**

Fluid (TO)	Pro	Con
Normal saline	Easily available; **routine use**	**Conductive** → not ideal with cautery
1.5% glycine	**Non-conductive → use with cautery**; less tissue toxicity	Slightly less clear view
Ringer lactate	Common in Western practice	Conductive
Distilled water	Occasional	**Hypotonic → tissue oedema**

4 Diagnosis & Classification

Arthroscope (telescope)

Parts (TO) eyepiece (to camera coupler) · light post (fibre-optic entry) · barrel · **objective lens system at distal tip** (thin lenses + spacers) → impact/ pressure misaligns lenses → **blurred image**

Joint (TO)	Diameter	Length
Knee / shoulder	**4 mm**	13.5 cm
Hip	4 mm	**18 cm**
Elbow / wrist / ankle	**2.7 mm**	6.7 cm
Small joints (hand / foot)	**1.9 mm**	6.7 cm

Movement (TO)	Description	Note
Pistoning	In–out	Depth
Angulation	Up / down / side	Orientation
Rotation	Twist along axis	**Most important** – changes field of view

Angle of inclination (TO)	Use
0°	Straight view (spine, endoscopy)
30°	**Standard: knee + shoulder**
70°	**Hip / periphery**; wider peripheral view but **central blind spot**

Field of view (TO) **apparent FOV depends on diameter · actual FOV = diameter + angle of inclination**

Sheath, light source & camera

Sheath (TO) scope through **sheath (cannula) + obturator (trocar)** · sheath larger than scope to allow saline flow (stated ~2 mm) · 4 mm scope → **6 mm** sheath · 2.7 → 2.9 mm · 1.9 → 2.2 mm

Sheath types (TO) **universal** = more lens exposed → ↑ damage · **hooded** = protective, **preferred**

Light source (TO) **300–350 W** · tungsten < halogen < xenon < **LED (best)** → **cooler, prevents chondral damage**

Fibre-optic cable (TO) **300 cm**, **4.8 mm** · core (fibres) → cladding → coating

Camera (TO) now **3-chip HD**

- - Active space - -

5 Management

Hand instruments

Instrument (TO)	Use / fact
Probe	"**Extension of surgeon's finger**" – palpate, pull, **measure** (tip **4 mm**, shaft 12 cm)
Punch (basket forceps)	Cuts tissue: straight / side-cutting / up-biting
Grasper	Tissue or foreign body
Retriever	Pulls sutures
Switching stick (Singer rod)	Portal exchange; **4 mm**
Cannula	Wash (rigid) · **passport (flexible)** → allows jig passage

Suture passer (TO)	Example	Mechanism
Direct	Champion (Stryker), Scorpion (Arthrex)	**Needle fires suture**, trap catches it – no assistant
Indirect – retrograde	**Lasso**	Loop **pulls suture back** through tissue
Indirect – antegrade	**Bird Beak**	Sharp jaws carry thread through tissue

Knot pusher (TO) single-hole · double-hole · **end-splitting = best** (max knot + loop security)

Powered & energy devices

Shaver / burr (TO) diameters **3.5, 4.5, 5.5 mm** · length 120 mm (knee), 180 mm (hip) · **shaver 1,000–3,000 rpm** · **burr 8,000–9,000 rpm** · blades: standard (smooth) vs **aggressive (toothed)** · **full-radius resector = most used**

Device (TO)	Use	Note
Cautery	Haemostasis	Conductive in saline
Laser (Nd:YAG)	Precision	Smooths frayed cartilage (chondroplasty)
Radiofrequency probe	Coagulation + **thermal shrinkage**	Tightening effect

Sutures & anchors

Suture (TO)	Composition	Feature
Ethibond	Braided **polyester**	Non-absorbable, moderate strength
FiberWire	**UHMWPE + polyester**	**4–5× strength of Ethibond**
OrthoCord	UHMWPE + **PDS core**	Profile ↓ as core degrades
FiberTape / TigerTape	Broad flat UHMWPE	↑ compression, may abrade tissue

– – Active space – –

Anchor material (TO)	Pro	Con
Metal (SS / titanium)	High pull-out, **visible on X-ray**	**Abrades cartilage**, permanent
PEEK	**Radiolucent, drillable**	Non-degradable, may migrate
Biocomposite (PLA + β-TCP)	Absorbable, **bone ingrowth**	Slow degradation
All-suture (JuggerKnot)	Minimal hardware, high pull-out	Technically demanding

Anchor technique (TO)	Use
Knotted	**Medial row** cuff repair
Knotless	**Lateral row** · **Bankart**

Anchor size (TO)	
Glenoid (Bankart)	**2.8 mm**
Cuff medial row	**4.5 mm**
Cuff lateral row	**5.5 mm**

Arthroscopic knots

Non-sliding (TO) **square knot + 5 half-hitches → best security** but cannot pass through cannula

Sliding knots (TO) advanced down cannula by pulling one limb · back every sliding knot with **3 reverse half-hitches on alternating posts**

Sliding knot (TO)	Example	Property
Non-locking	**Duncan loop**	Easy; loosens under back pressure
Locking – proximal	**Nicky's knot**	Secure, complex
Locking – middle	**SMC (Samsung Medical Center)**	**Most popular**, balanced security

Term (TO)	Definition
Knot security	Resistance to **slippage** under tension
Loop security	Resistance to **tissue back pressure**
Knot failure	Displacement **>3 mm**

Sterilisation

Item (TO)	Method
Camera, light cable, scope	**ETO or plasma** – **never autoclave** (lens + electronics)
Mechanical instruments, handpiece	**Autoclave** or **Cidex** (glutaraldehyde, **30 min**; TO states 4%) · Cidex + Steris common
Last resort	**Formalin chamber 24 h**

– – Active space – –

6 Complications

Most common complication of arthroscopy (TO)	**haemarthrosis** (esp. after **synovectomy** or **lateral release**)
Lens damage (TO)	universal sheath, impact on objective lens → blurred image
Chondral thermal damage (TO)	hot light sources → **LED** preferred

X vs Y

	Saline	**Glycine 1.5%**
Conductive	**Yes**	**No**
Use	Routine	**With electrocautery**

	Direct passer	**Indirect passer**
Mechanism	**Needle fires suture**	**Instrument passes through tissue** (lasso retrograde, bird beak antegrade)

	Knot security	**Loop security**
Resists	**Slippage** of knot	**Tissue back pressure** (loop enlarging)

How it's asked

"First successful arthroscopy?"	→ **Kenji Takagi, 1918** (TO)
"Introduced triangulation?"	→ **Masaki Watanabe** (TO)
"Most important arthroscope movement?"	→ **rotation** (TO)
"Standard scope angle for knee + shoulder?"	→ **30°** (TO)
"Fluid when using electrocautery?"	→ **1.5% glycine (non-conductive)** (TO) ✗ Wrong options: normal saline · Ringer lactate · distilled water
"Shoulder arthroscopy pump pressure?"	→ **40–50 mmHg** (TO)
"Item that must never be autoclaved?"	→ **camera / light cable / scope** (TO)
"Most popular locking sliding knot?"	→ **SMC knot** (TO)
"Most common complication of arthroscopy?"	→ **haemarthrosis** (TO)